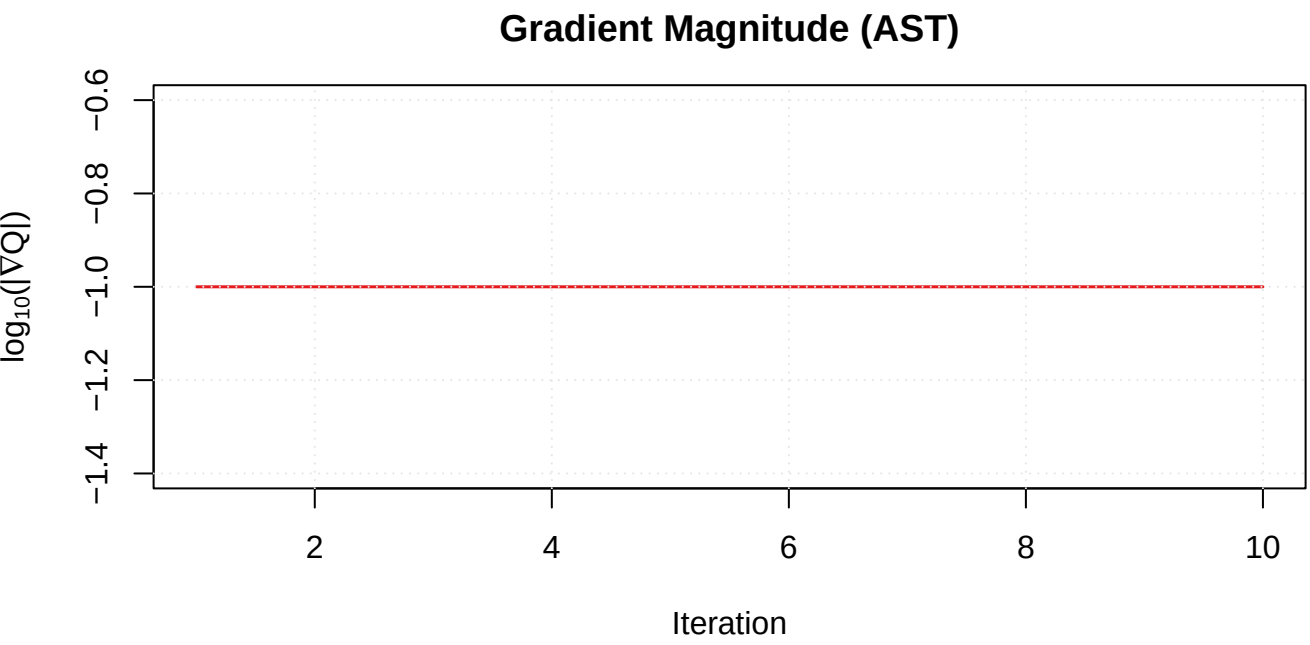
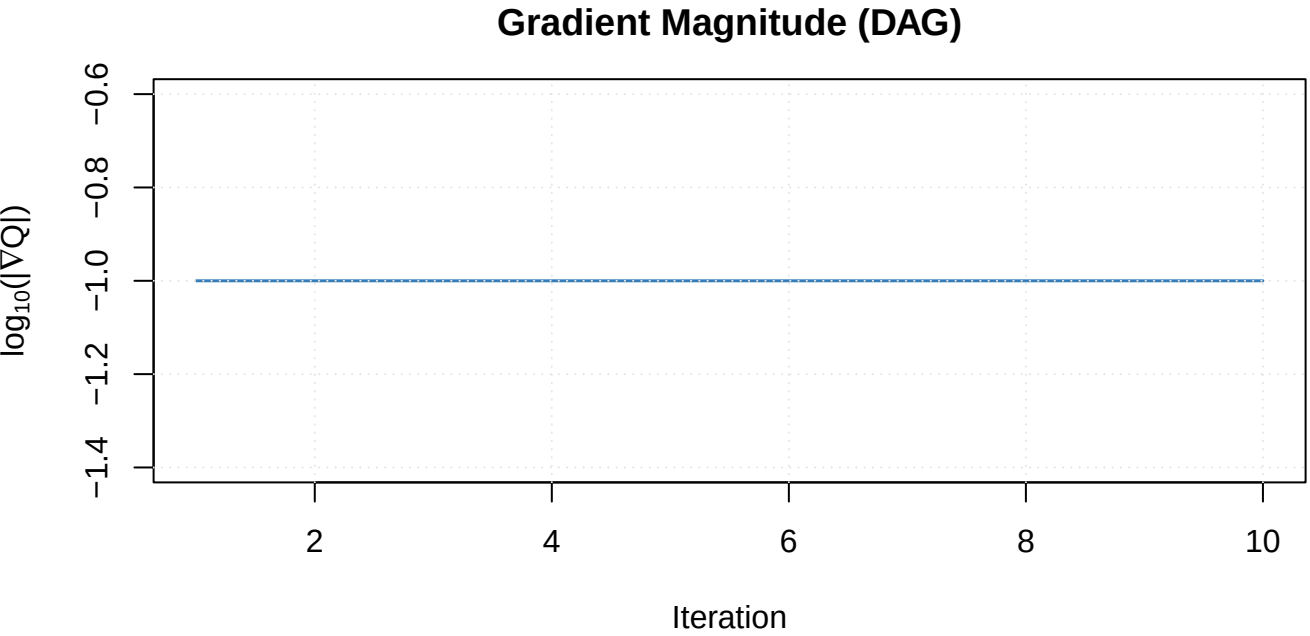
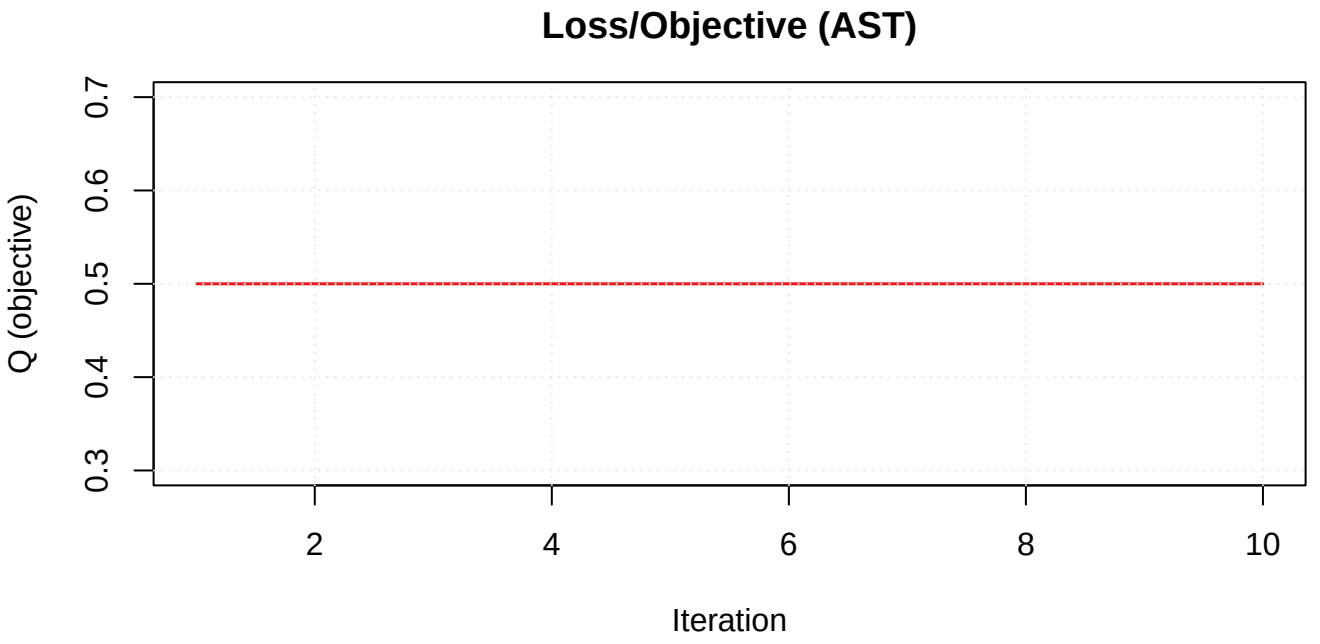
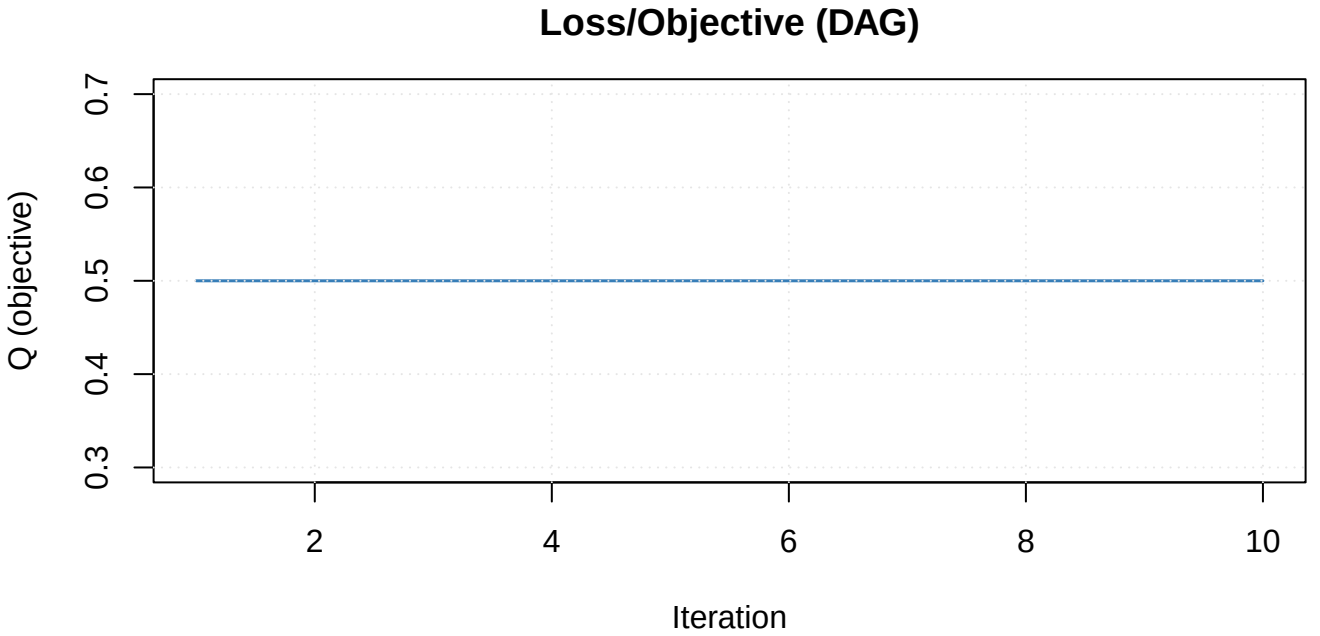
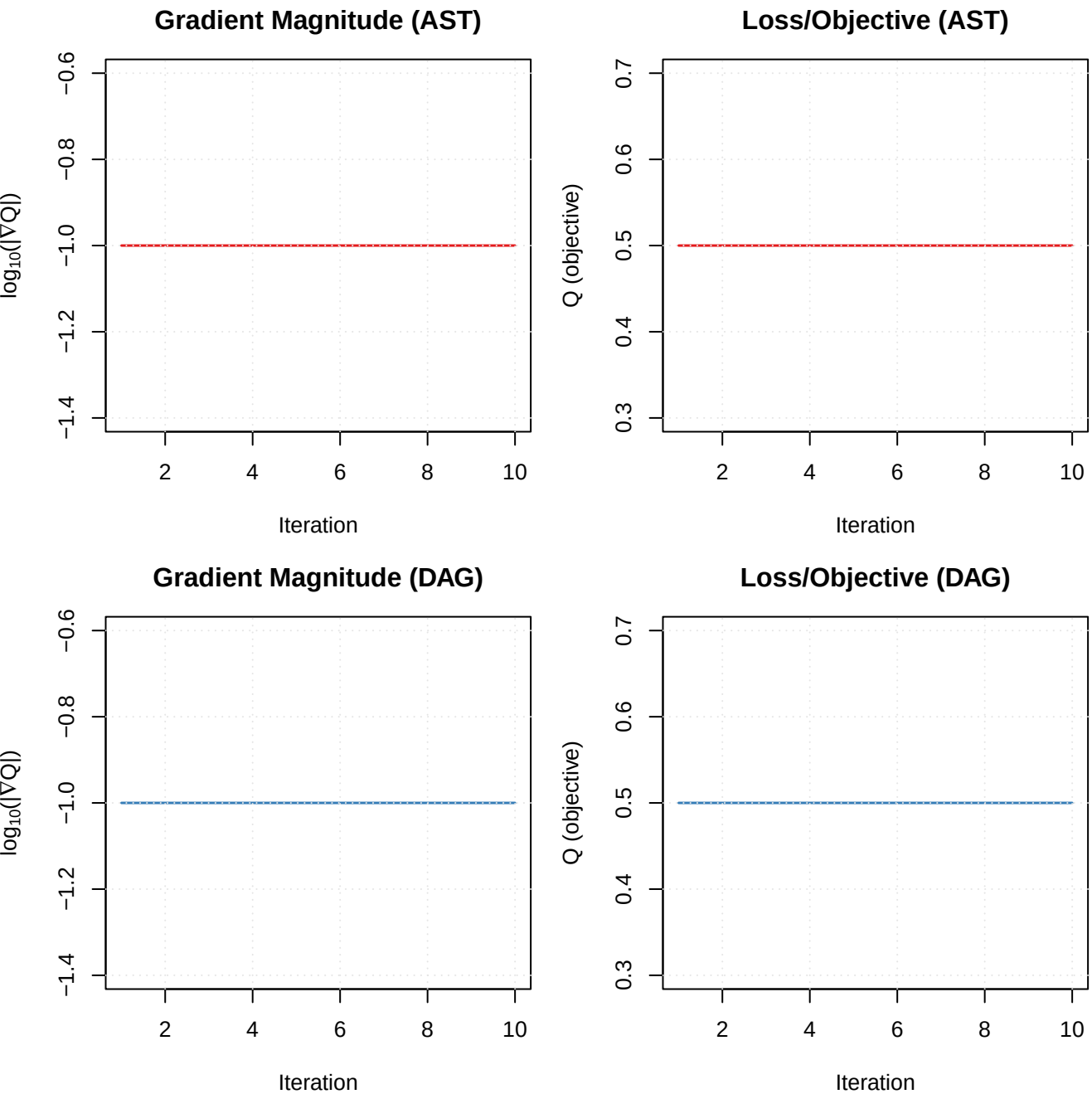
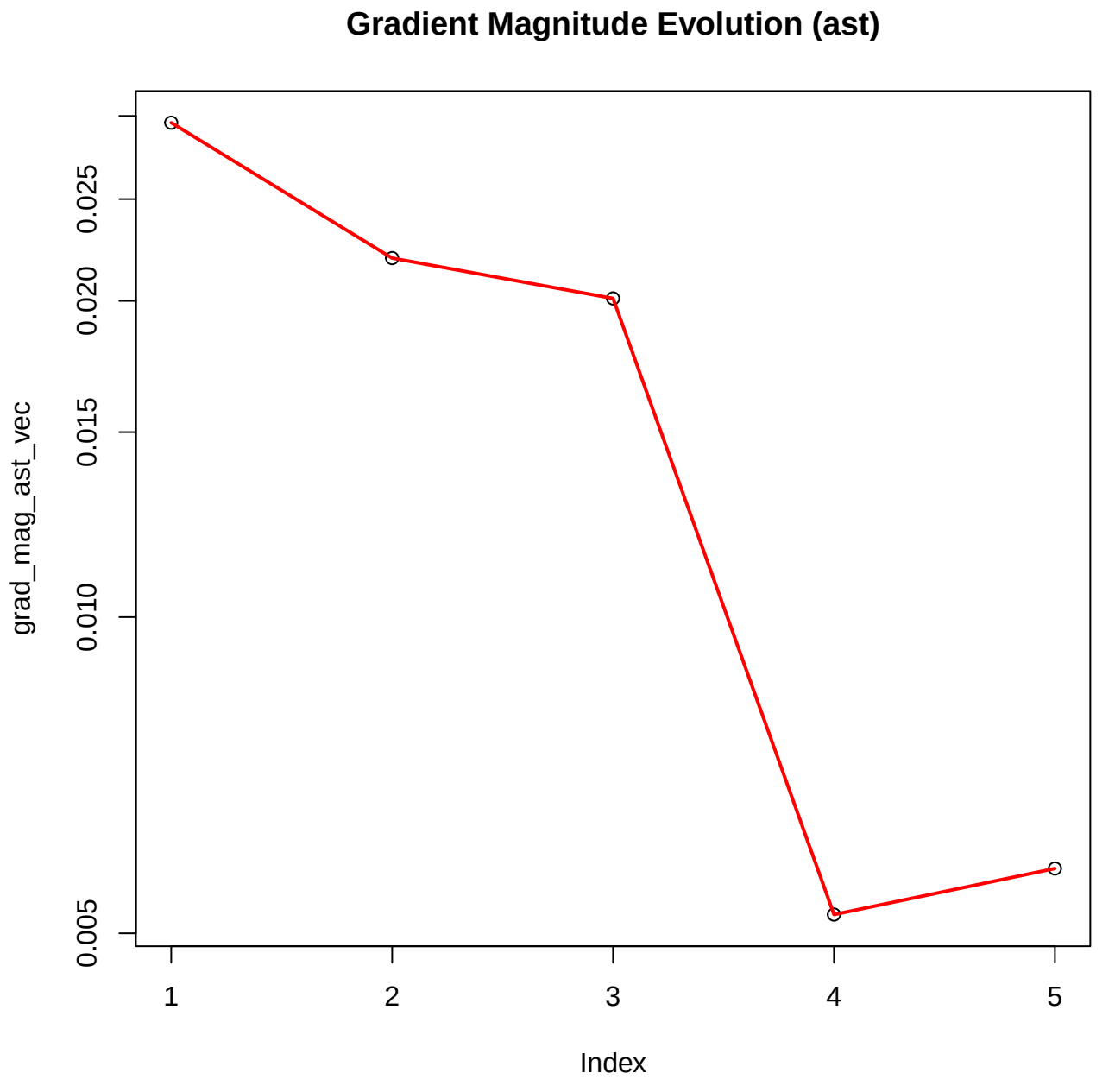

Gradient Magnitude (AST)
log10(|∇Q|)
−0.6
−0.8
−1.0
−1.2
−1.4
2
4
6
8
10
Iteration

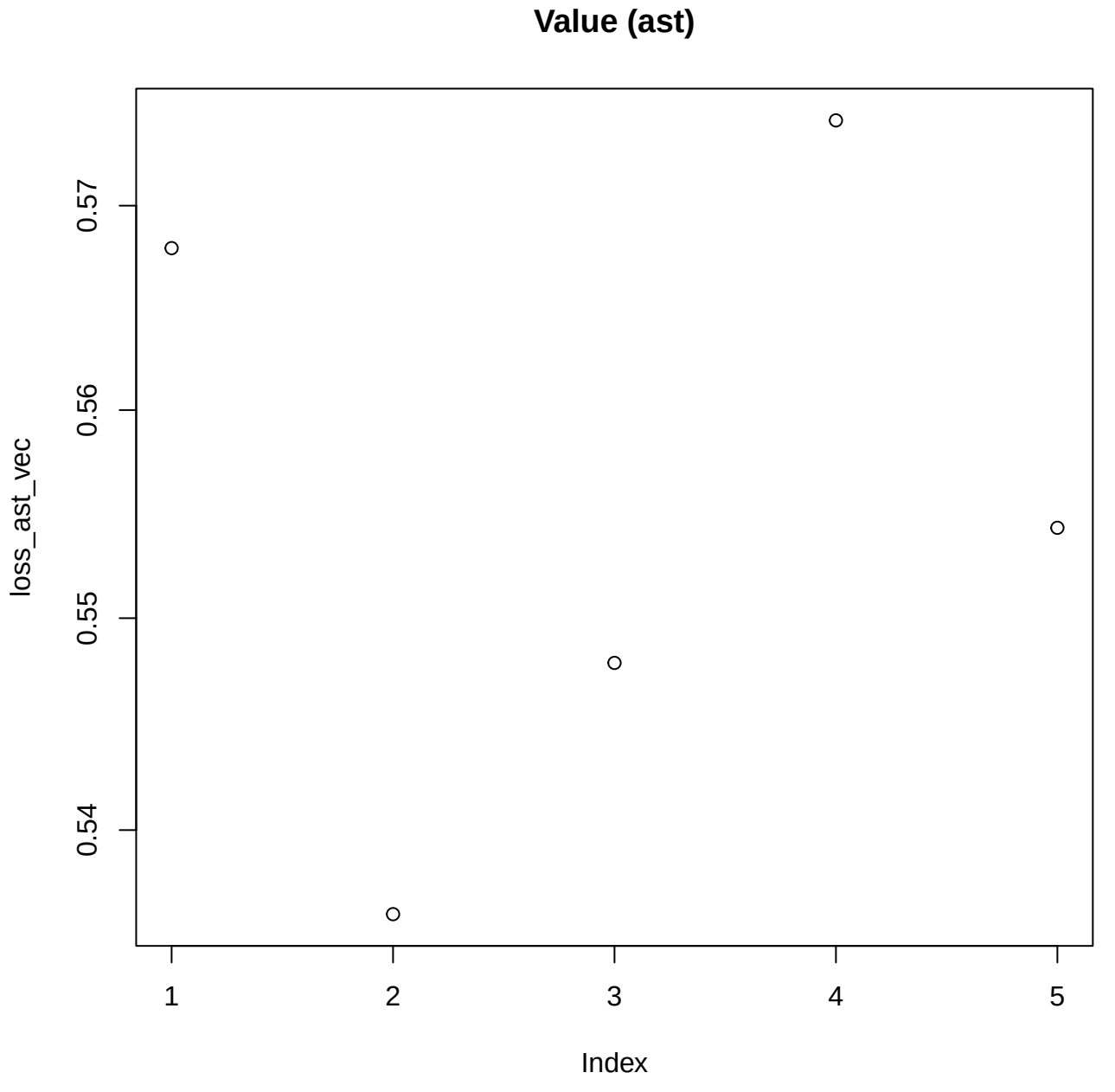

Gradient Magnitude (DAG)
log10(|∇Q|)
−0.6
−0.8
−1.0
−1.2
−1.4
2
4
6
8
10
Iteration

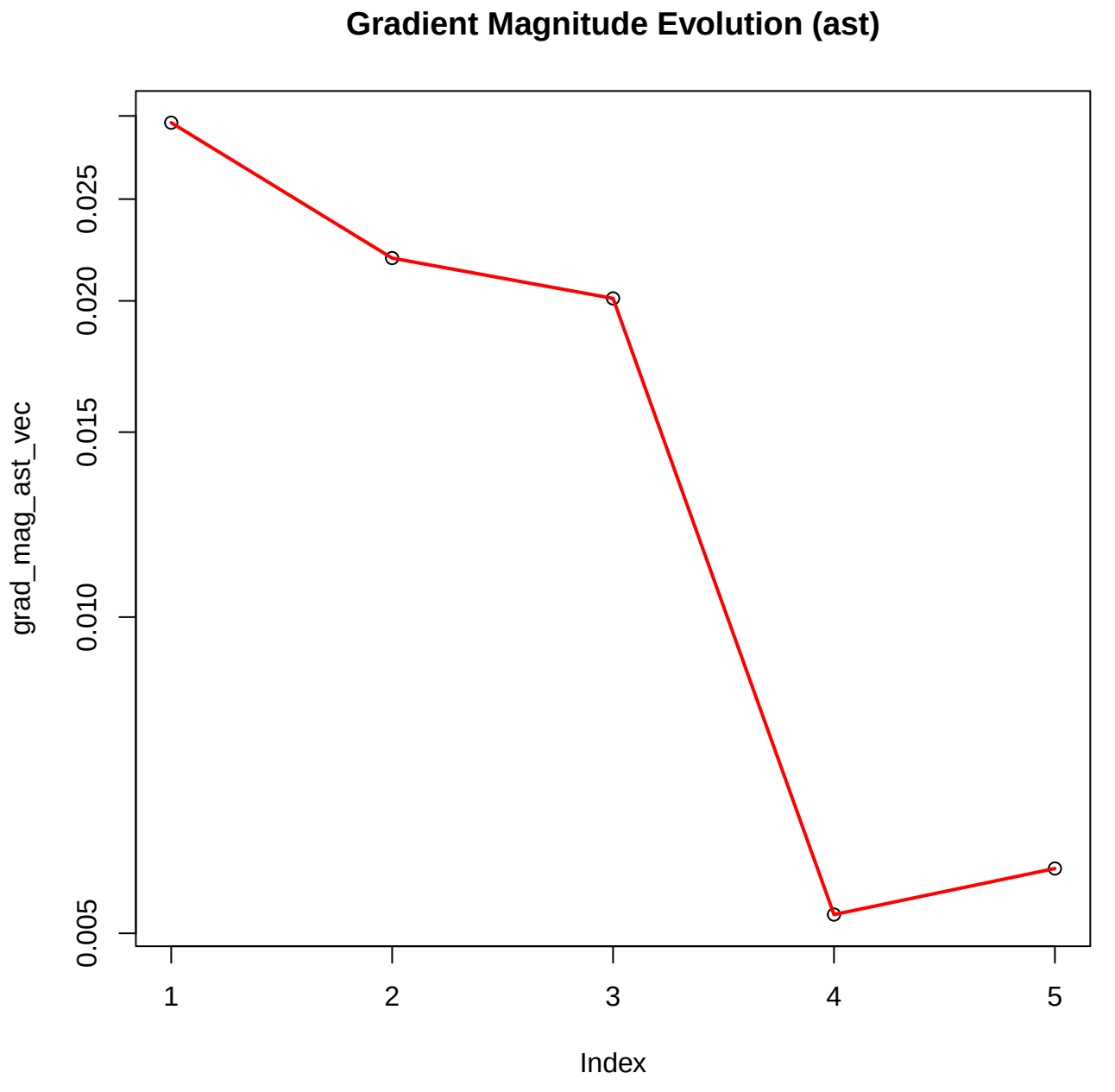

Loss/Objective (AST)
Q (objective)
0.3
0.4
0.5
0.6
0.7
2
4
6
8
10
Iteration

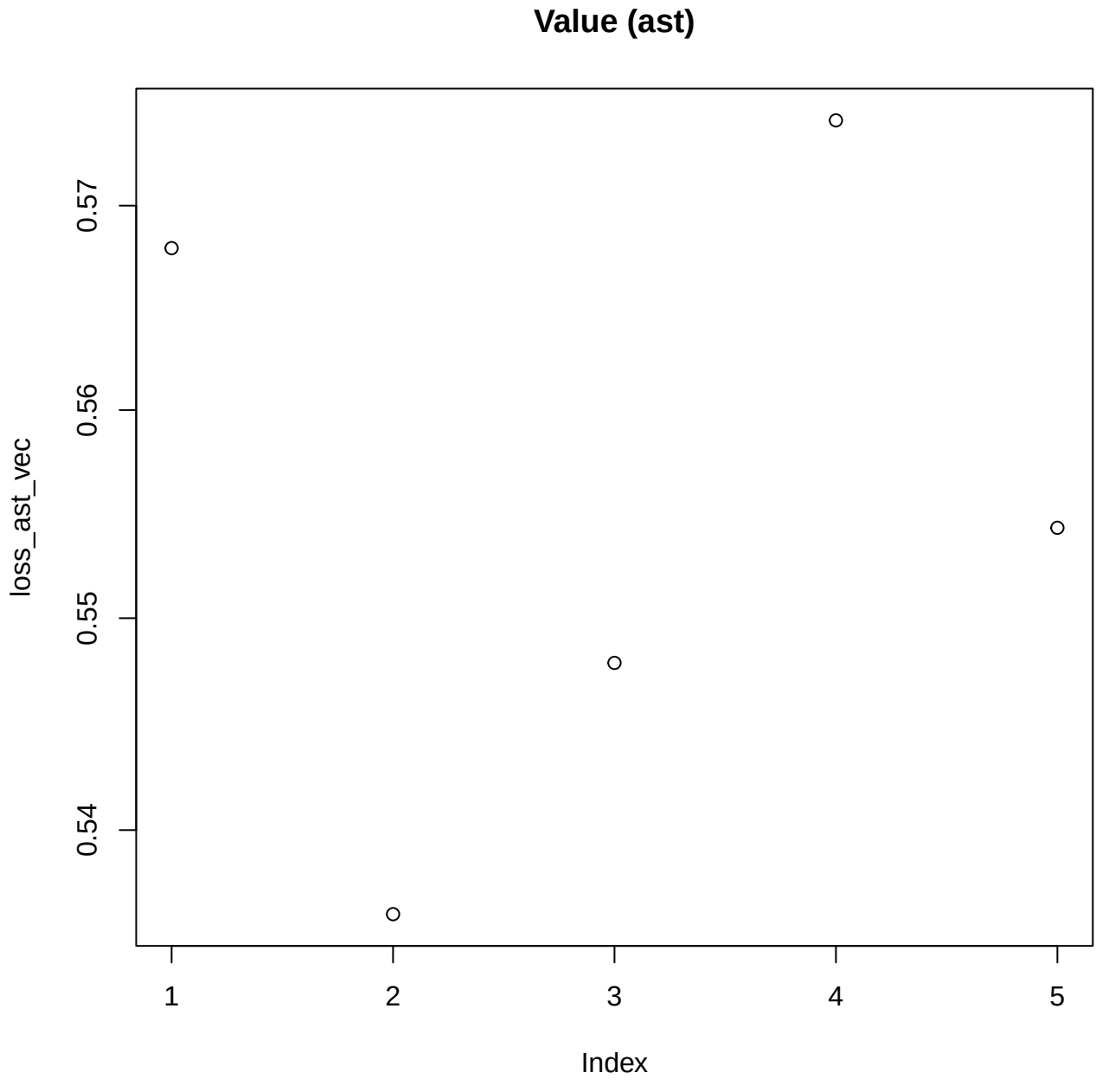

Loss/Objective (DAG)
Q (objective)
0.3
0.4
0.5
0.6
0.7
2
4
6
8
10
Iteration

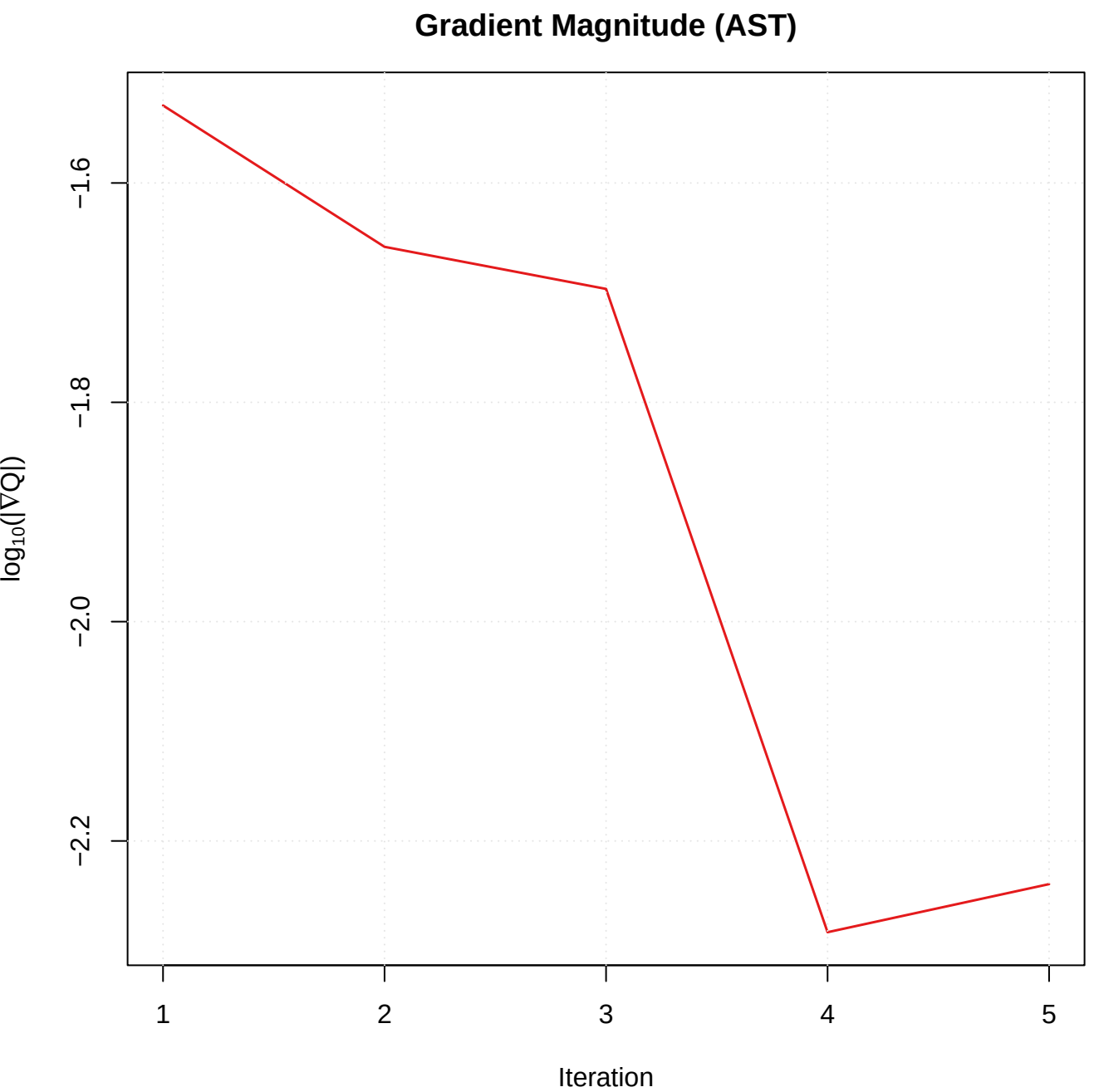

Gradient Magnitude (AST)
log10(|∇Q|)
−0.6
−0.8
−1.0
−1.2
−1.4
2
4
6
8
10
Iteration
Loss/Objective (AST)
Q (objective)
0.7
0.6
0.5
0.4
0.3
2
4
6
8
10
Iteration
Gradient Magnitude (DAG)
log10(|∇Q|)
−0.6
−0.8
−1.0
−1.2
−1.4
2
4
6
8
10
Iteration
Loss/Objective (DAG)
Q (objective)
0.7
0.6
0.5
0.4
0.3
2
4
6
8
10
Iteration

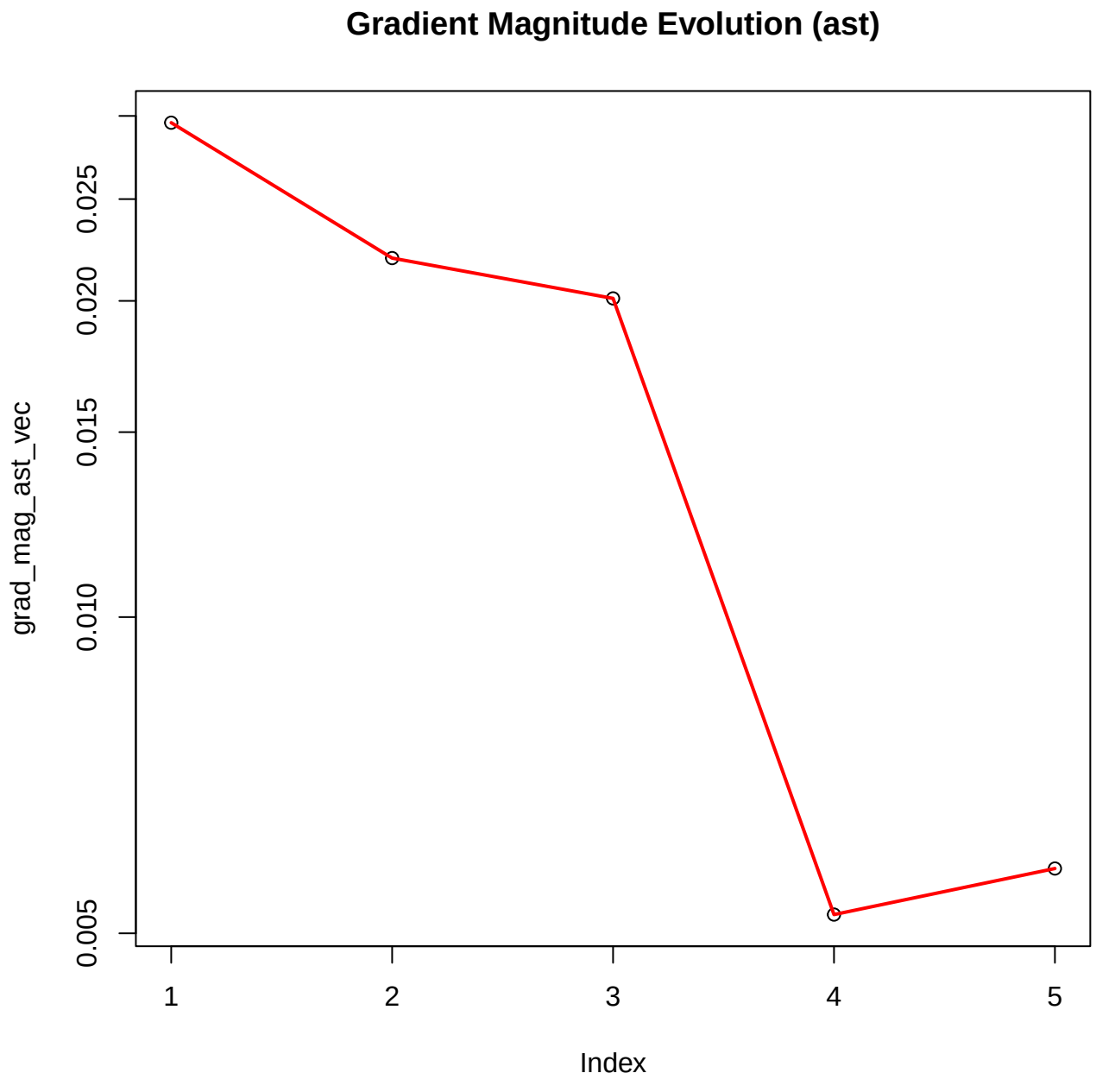

Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
0.005
0.010
0.015
0.020
0.025
1
2
3
4
5
Index

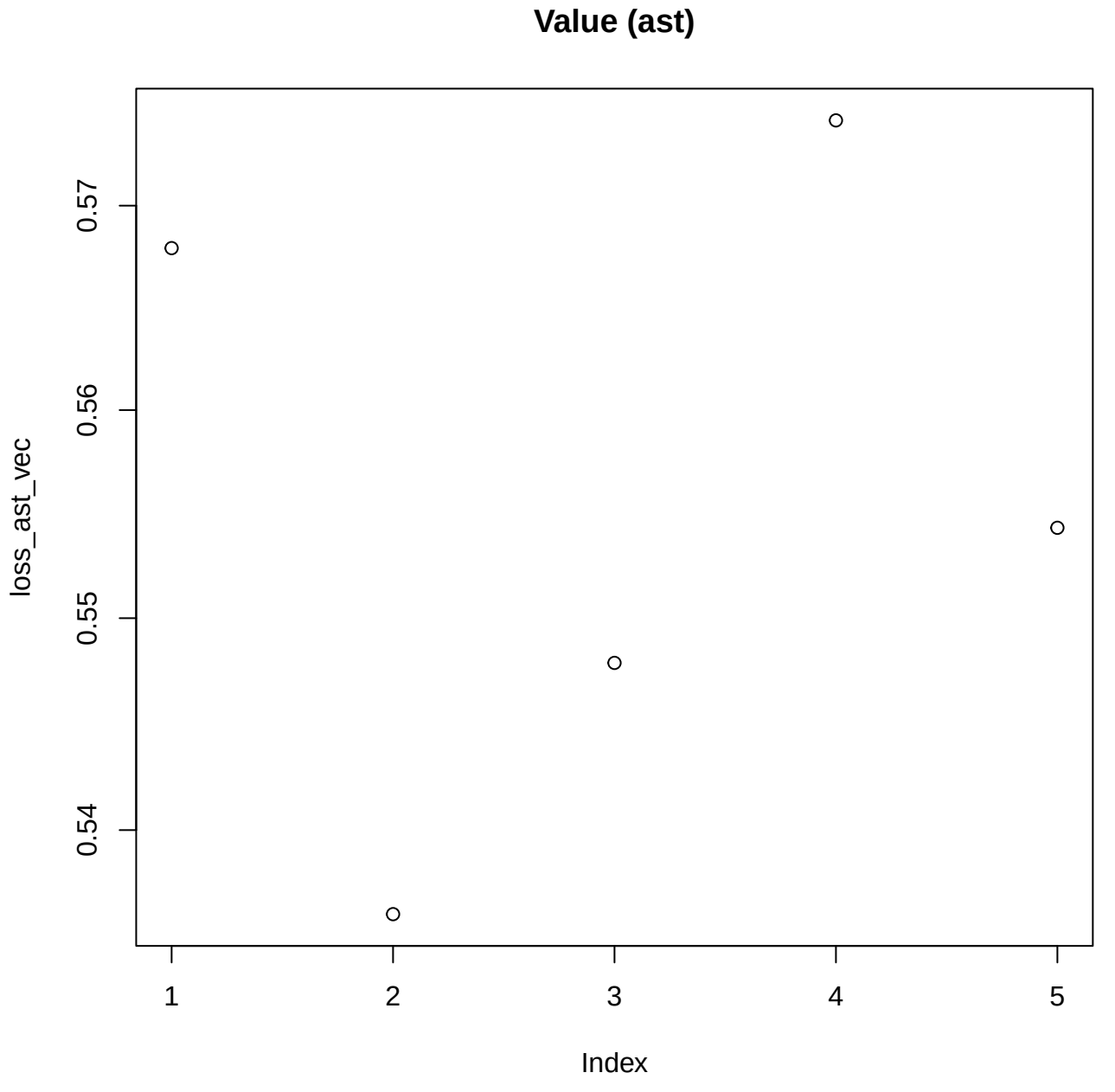

Value (ast)
loss_ast_vec
0.54
0.55
0.56
0.57
1
2
3
4
5
Index

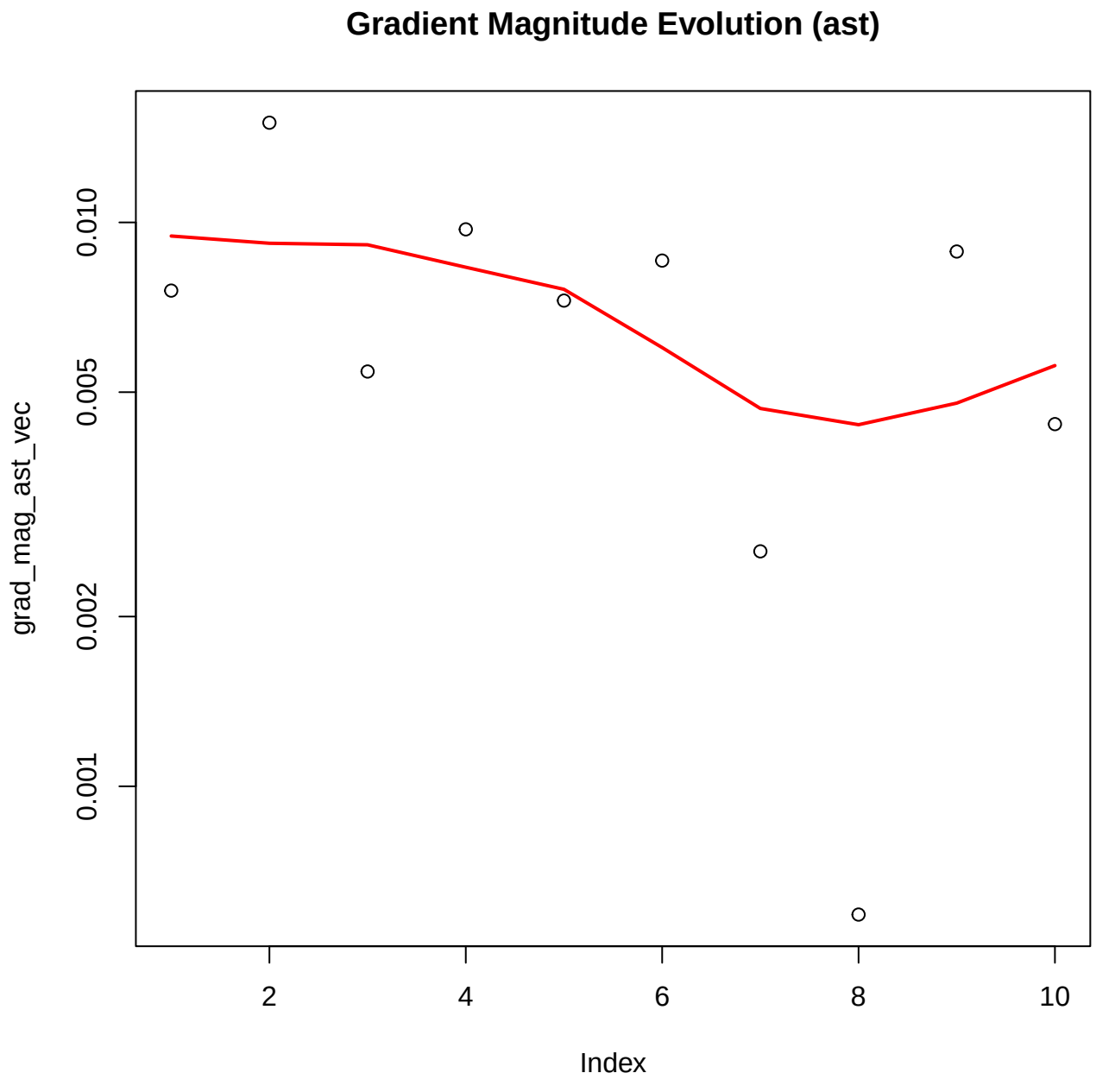

Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
0.005
0.010
0.015
0.020
0.025
1
2
3
4
5
Index

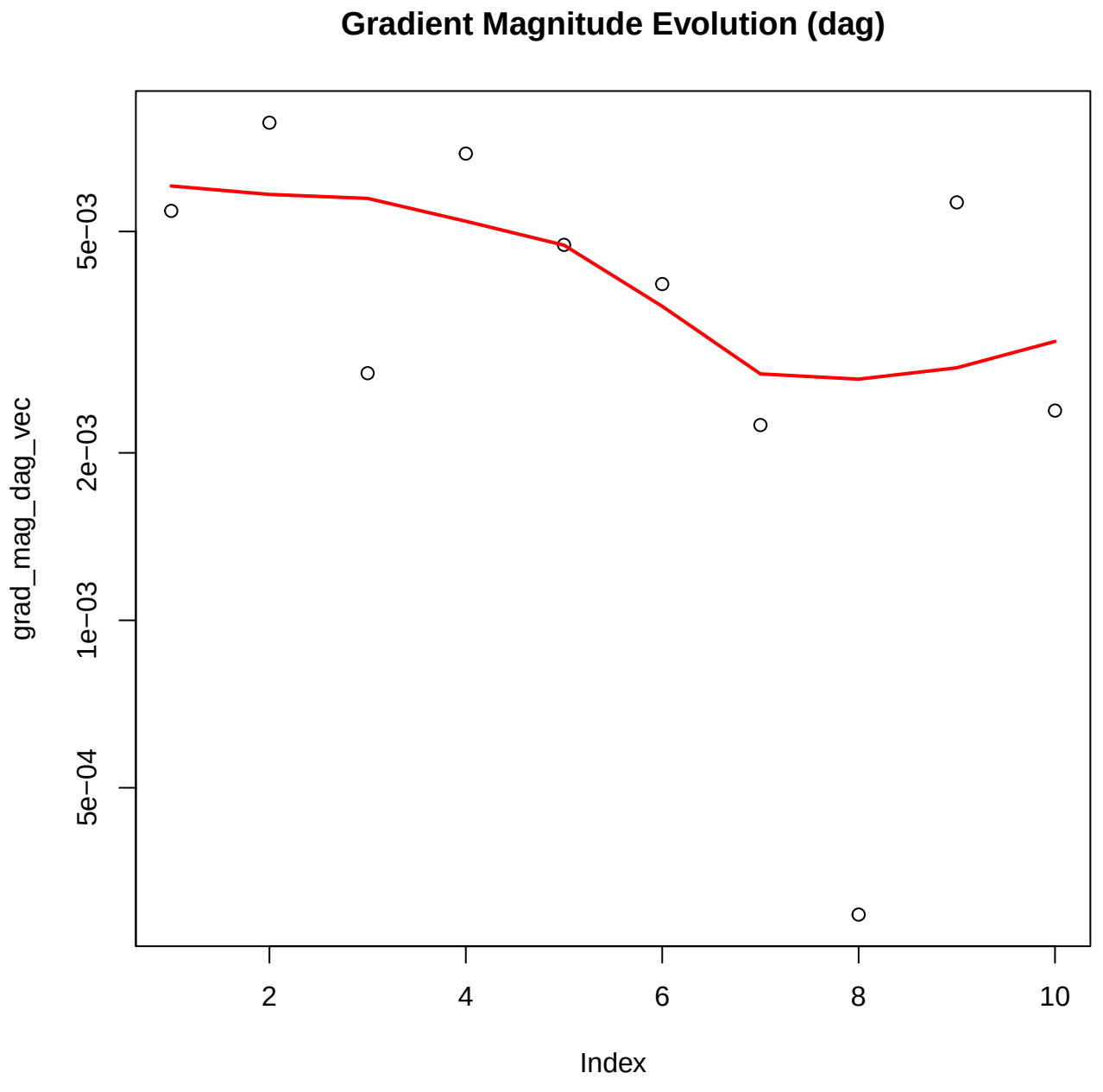

Value (ast)
loss_ast_vec
0.54
0.55
0.56
0.57
1
2
3
4
5
Index

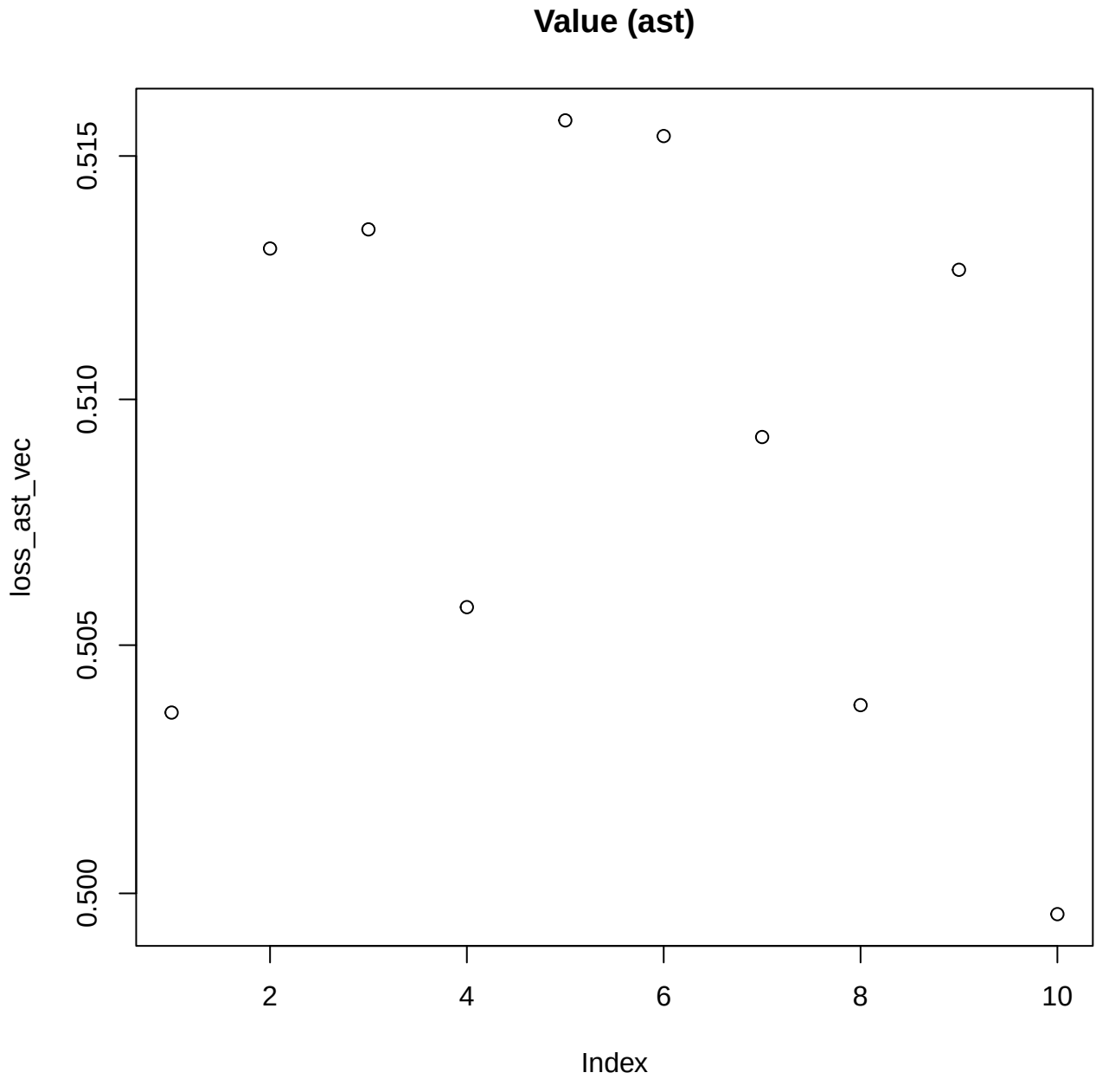

Gradient Magnitude (AST)
−1.6
−1.8
−2.0
−2.2
log10(|∇Q|)
1
2
3
4
5
Iteration

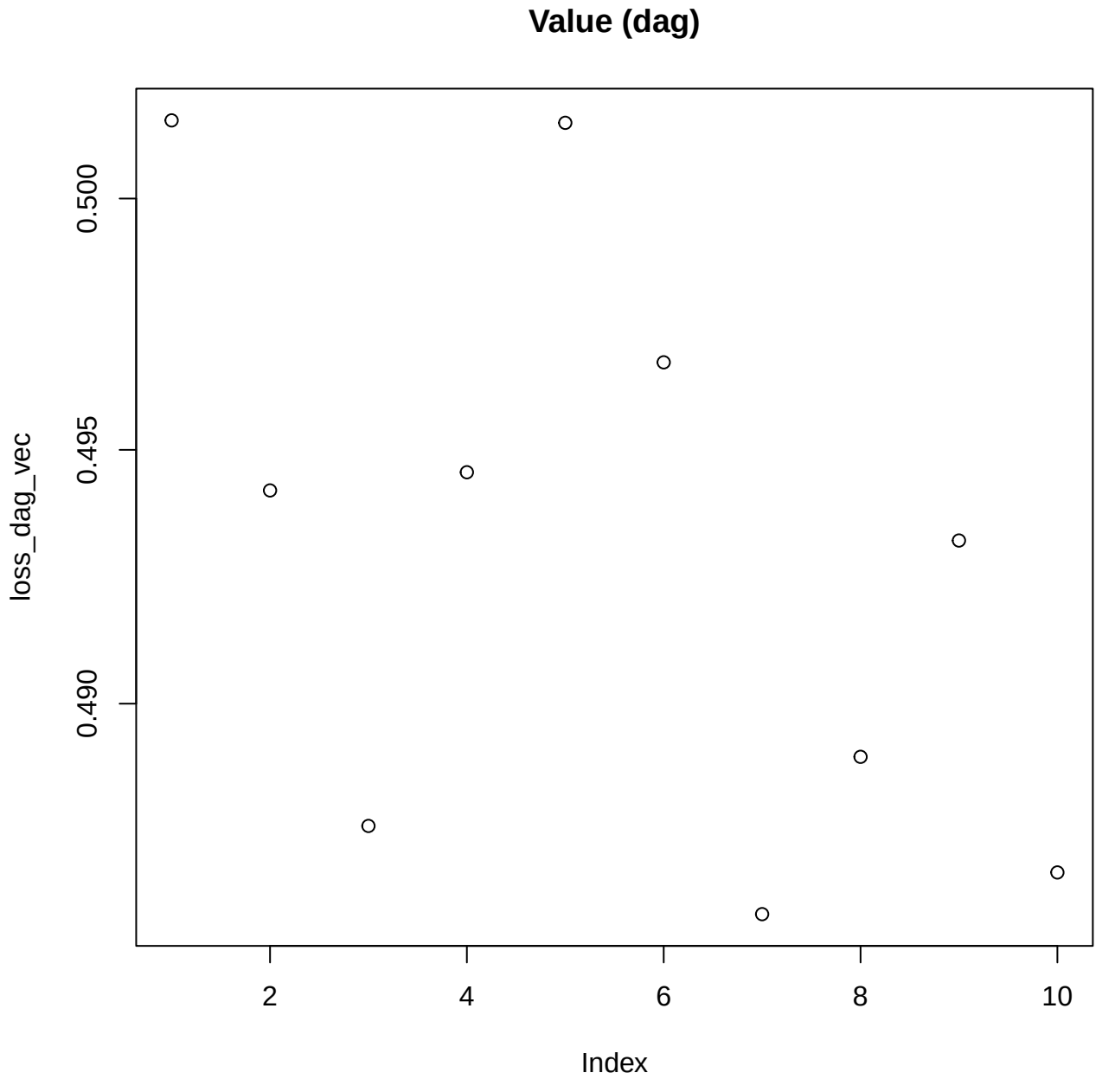

Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
0.005
0.010
0.015
0.020
0.025
1
2
3
4
5
Index

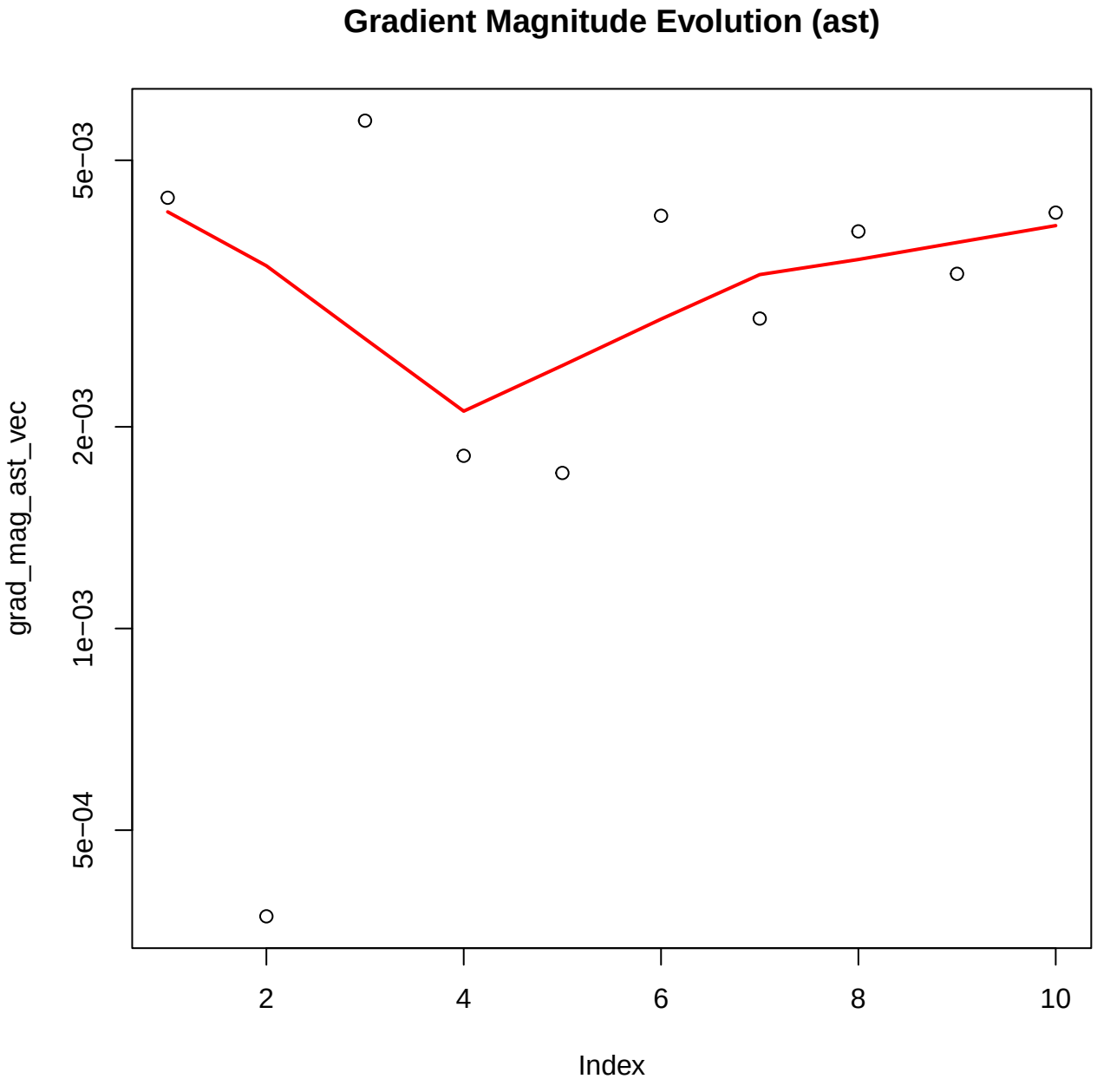

Value (ast)
loss_ast_vec
0.54
0.55
0.56
0.57
1
2
3
4
5
Index

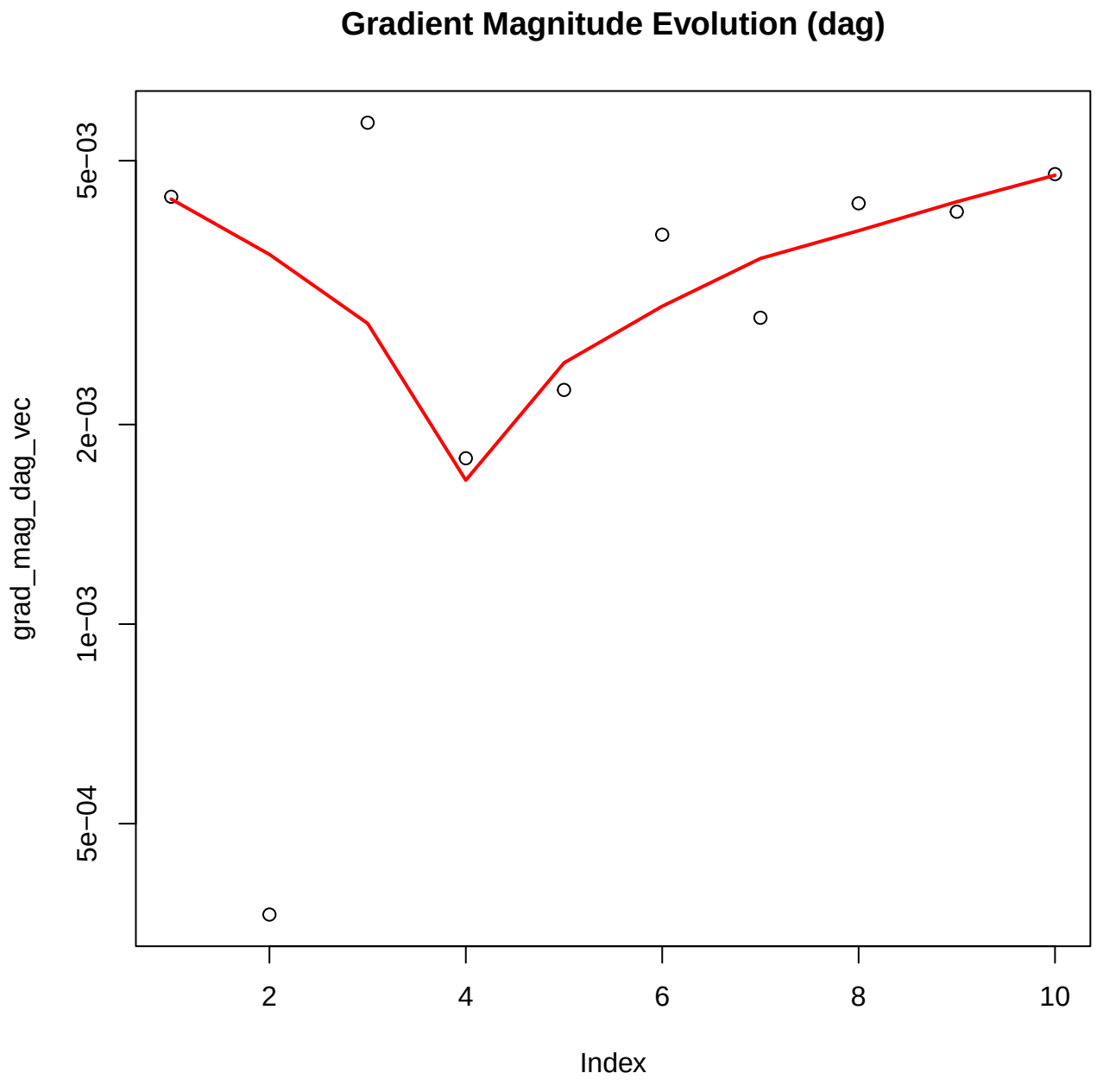

Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
0.001
0.002
0.005
0.010
2
4
6
8
10
Index

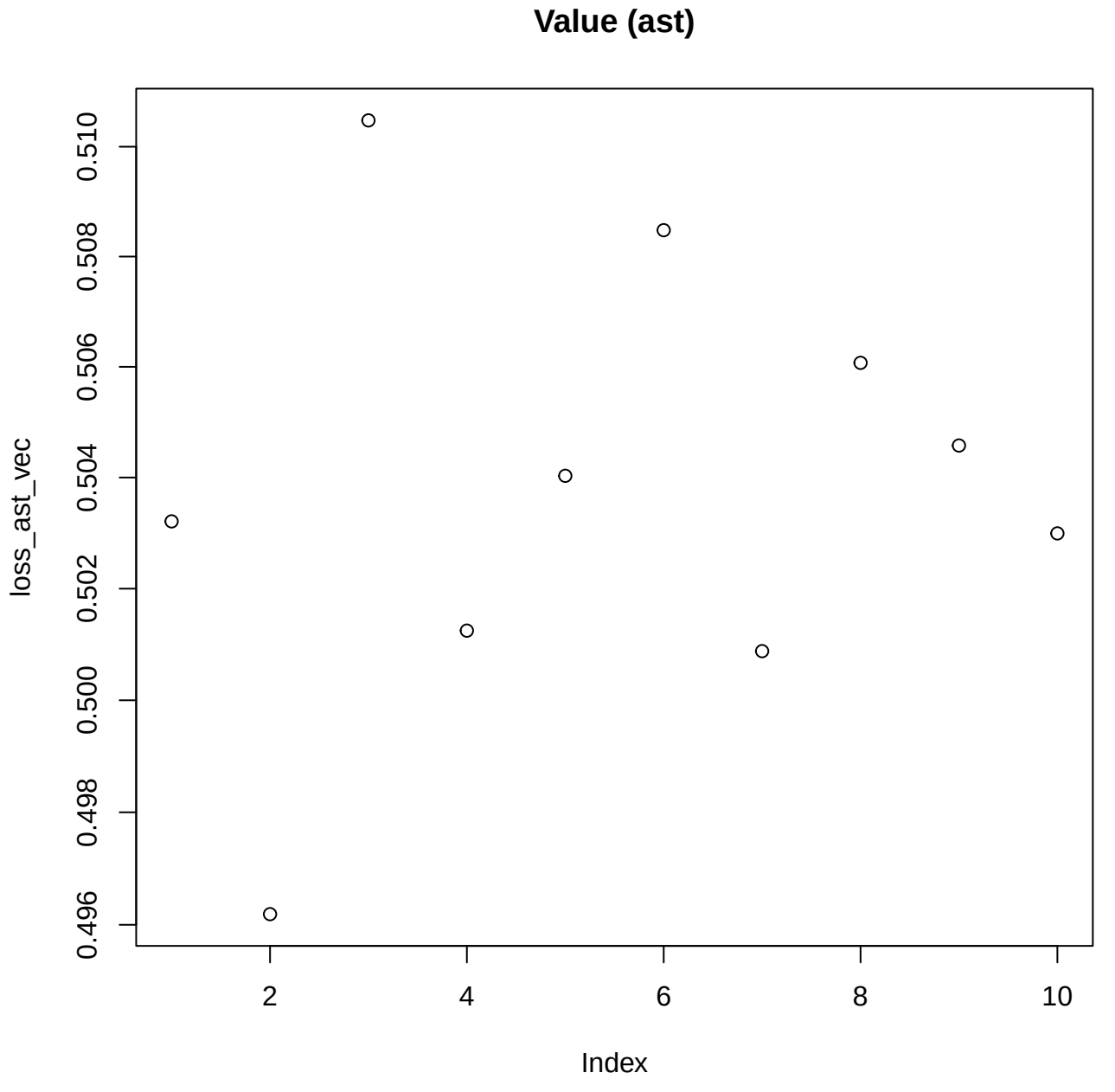

Gradient Magnitude Evolution (dag)
grad_mag_dag_vec
5e−03
2e−03
1e−03
5e−04
2
4
6
8
10
Index

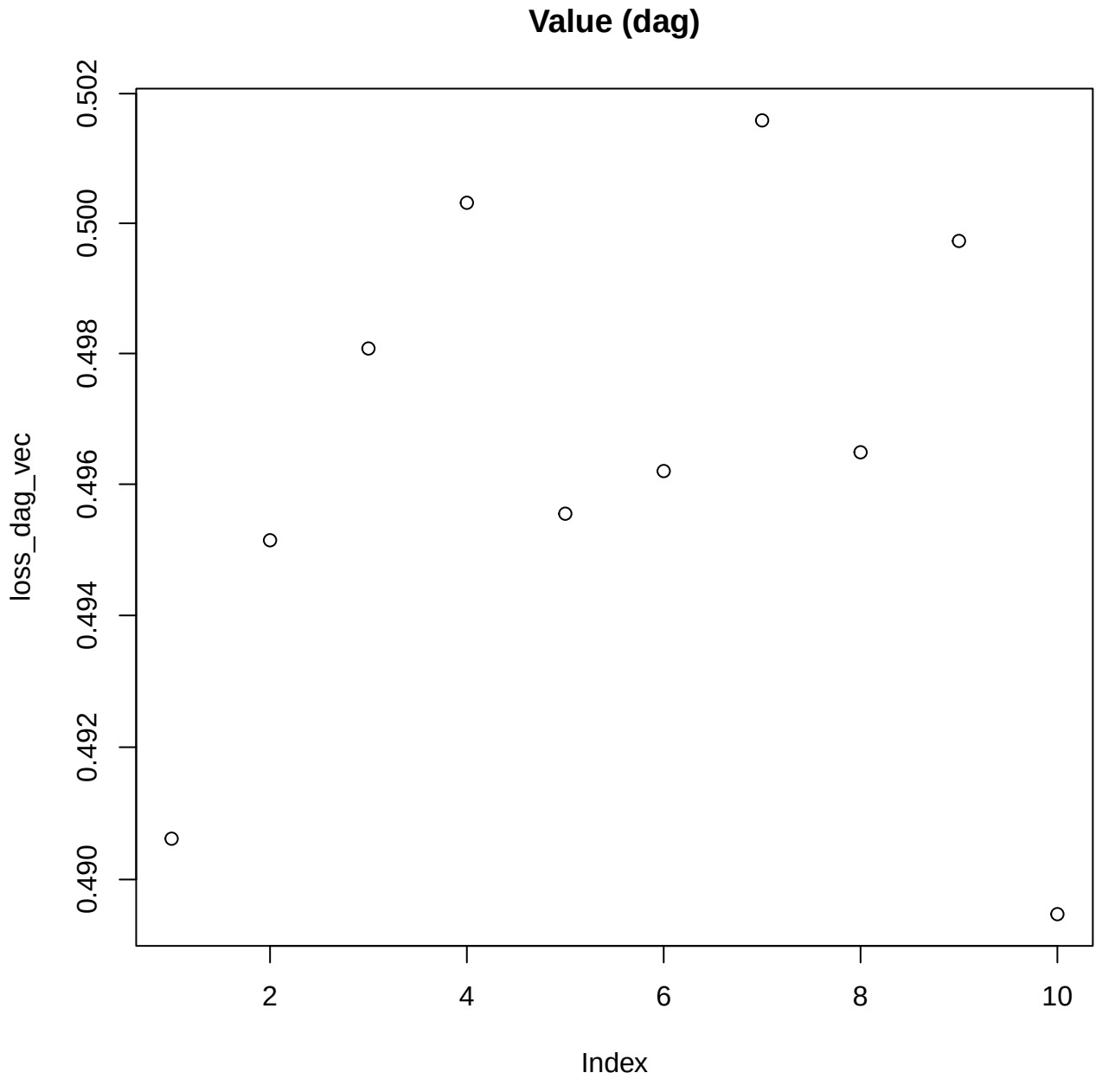

Value (ast)
loss_ast_vec
0.500
0.505
0.510
0.515
2
4
6
8
10
Index

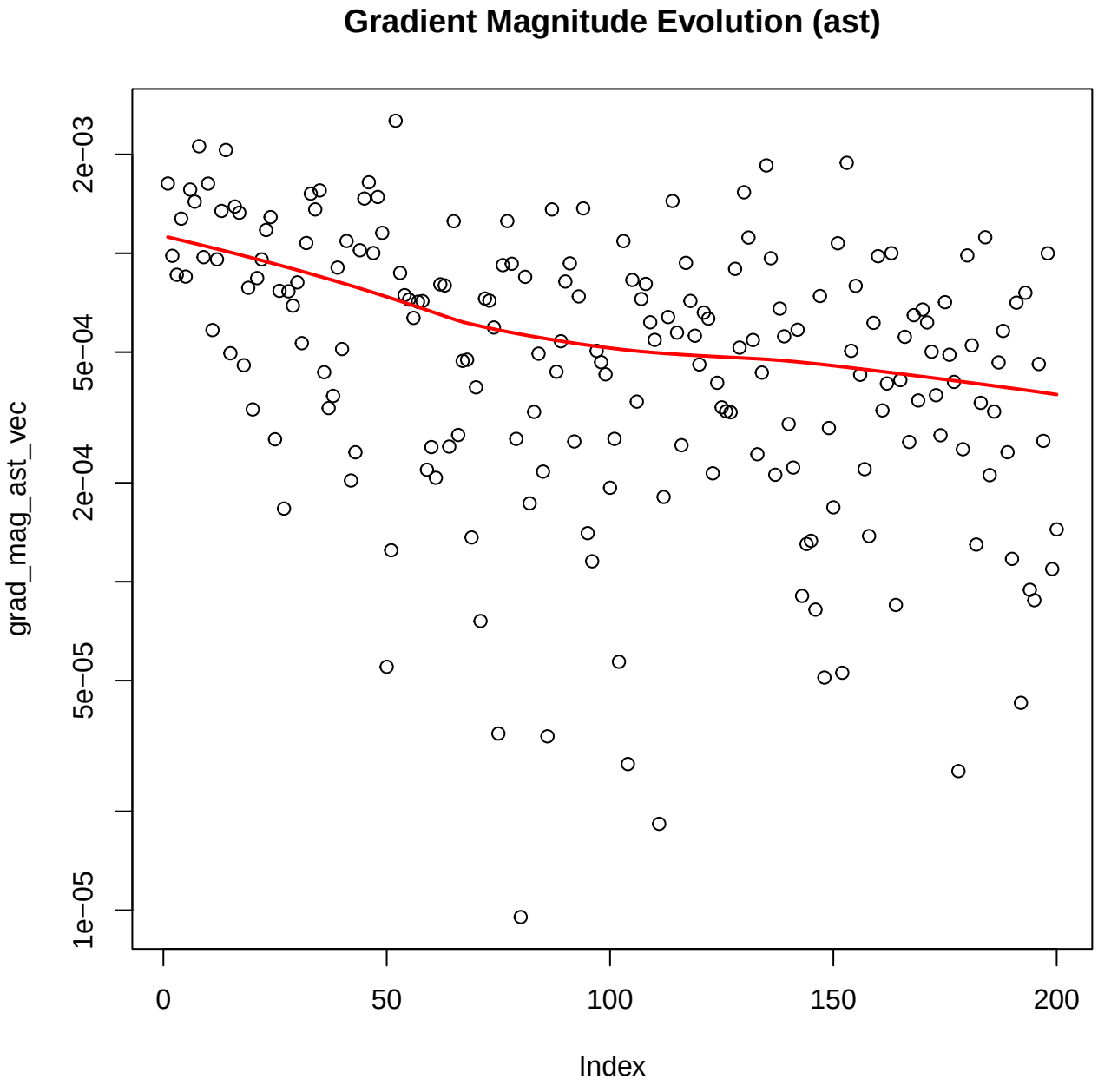

Value (dag)
loss_dag_vec
0.490
0.495
0.500
2
4
6
8
10
Index

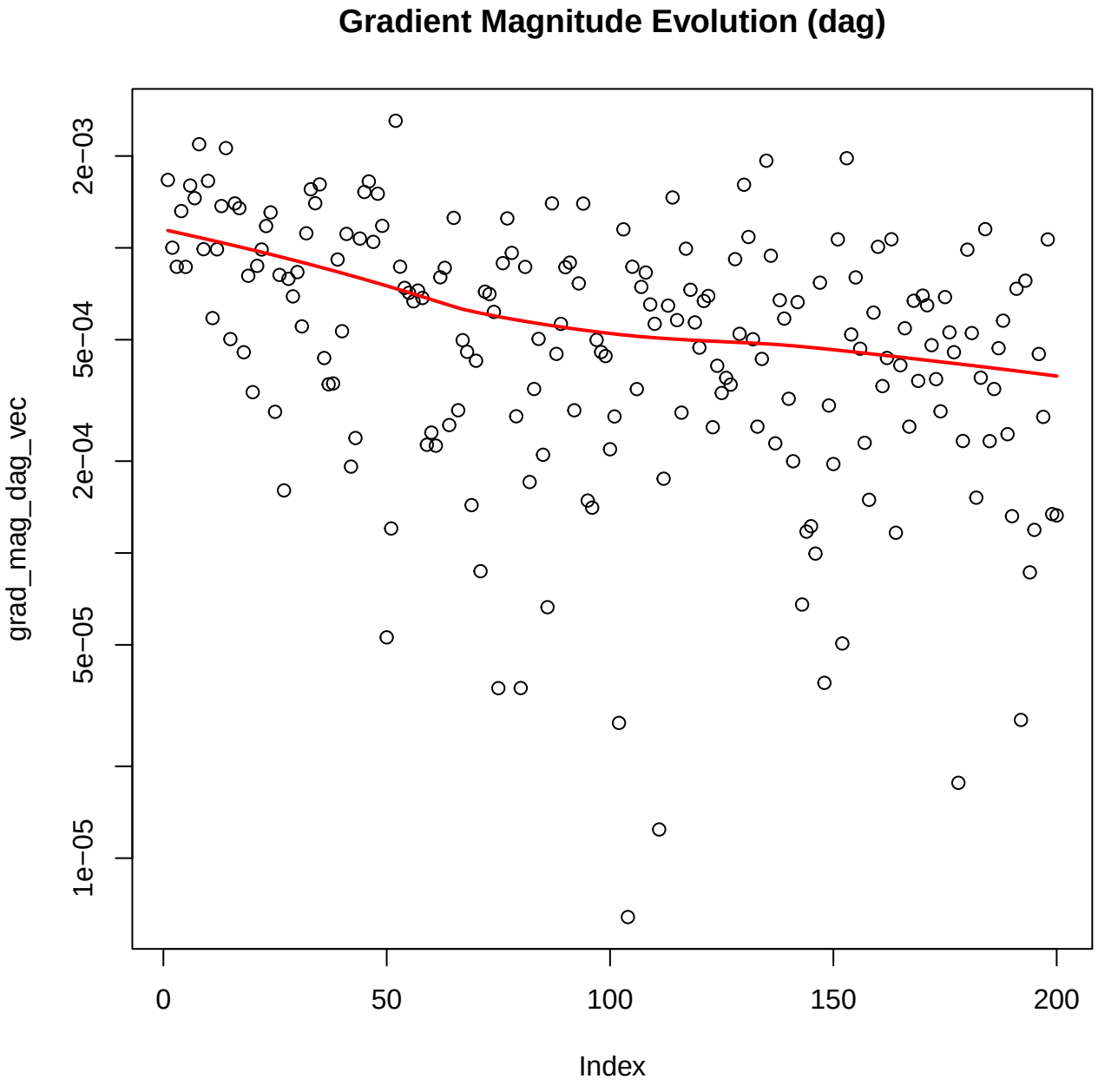

Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
5e−03
2e−03
1e−03
5e−04
2
4
6
8
10
Index

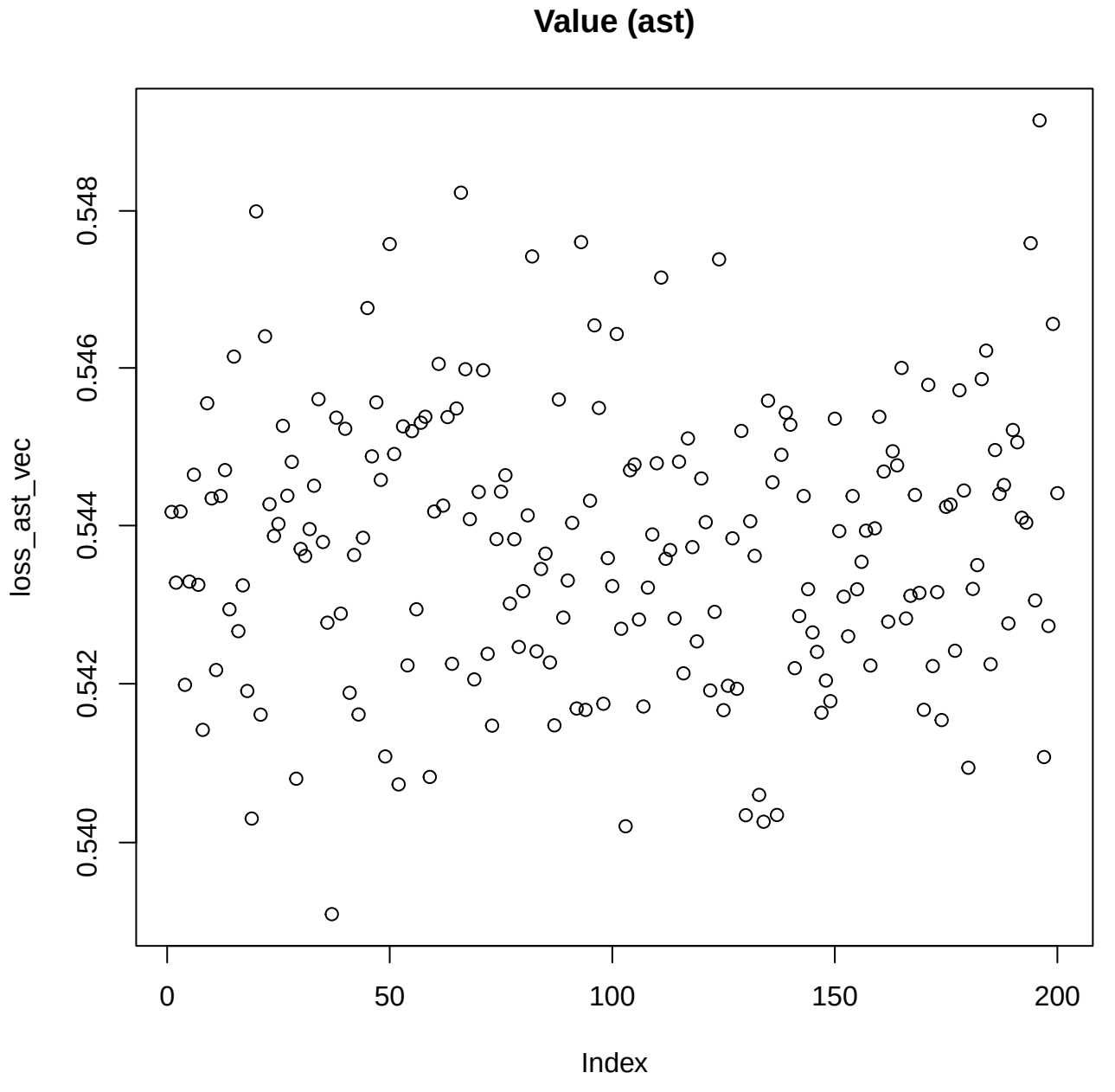

Gradient Magnitude Evolution (dag)
grad_mag_dag_vec
5e−03
2e−03
1e−03
5e−04
2
4
6
8
10
Index

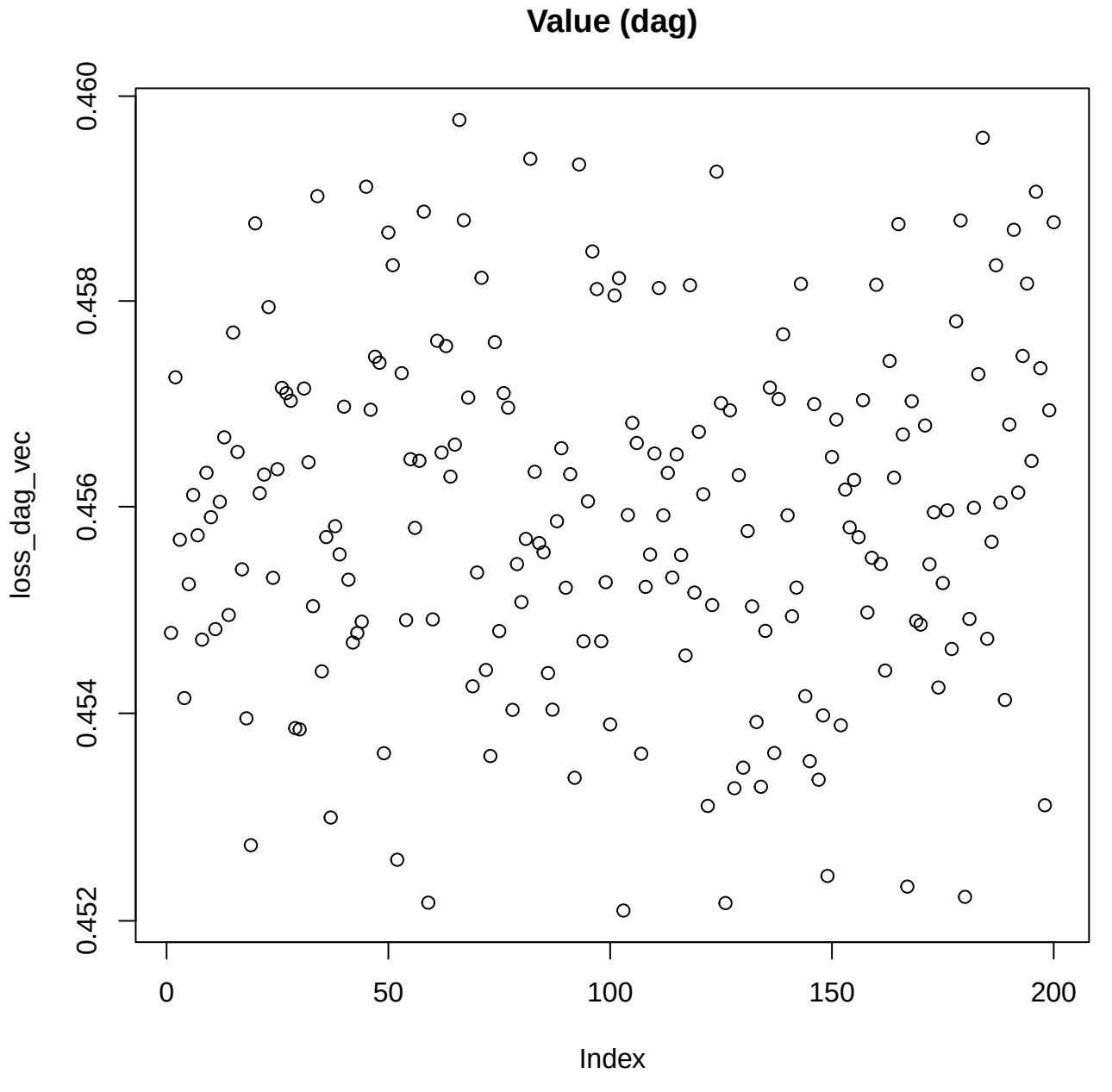

Value (ast)
loss_ast_vec
0.496
0.498
0.500
0.502
0.504
0.506
0.508
0.510
2
4
6
8
10
Index

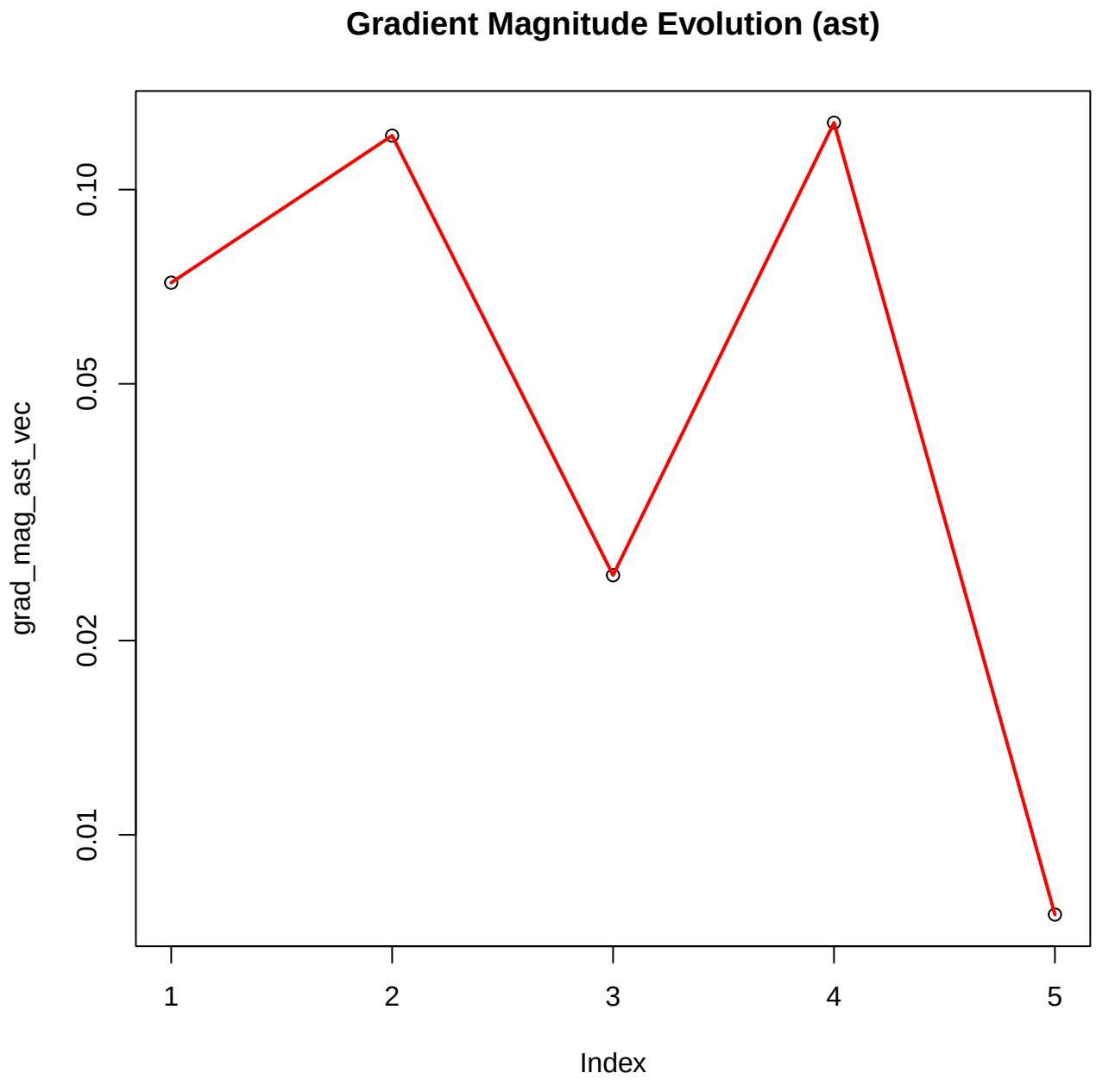

Value (dag)
loss_dag_vec
0.490
0.492
0.494
0.496
0.498
0.500
0.502
2
4
6
8
10
Index

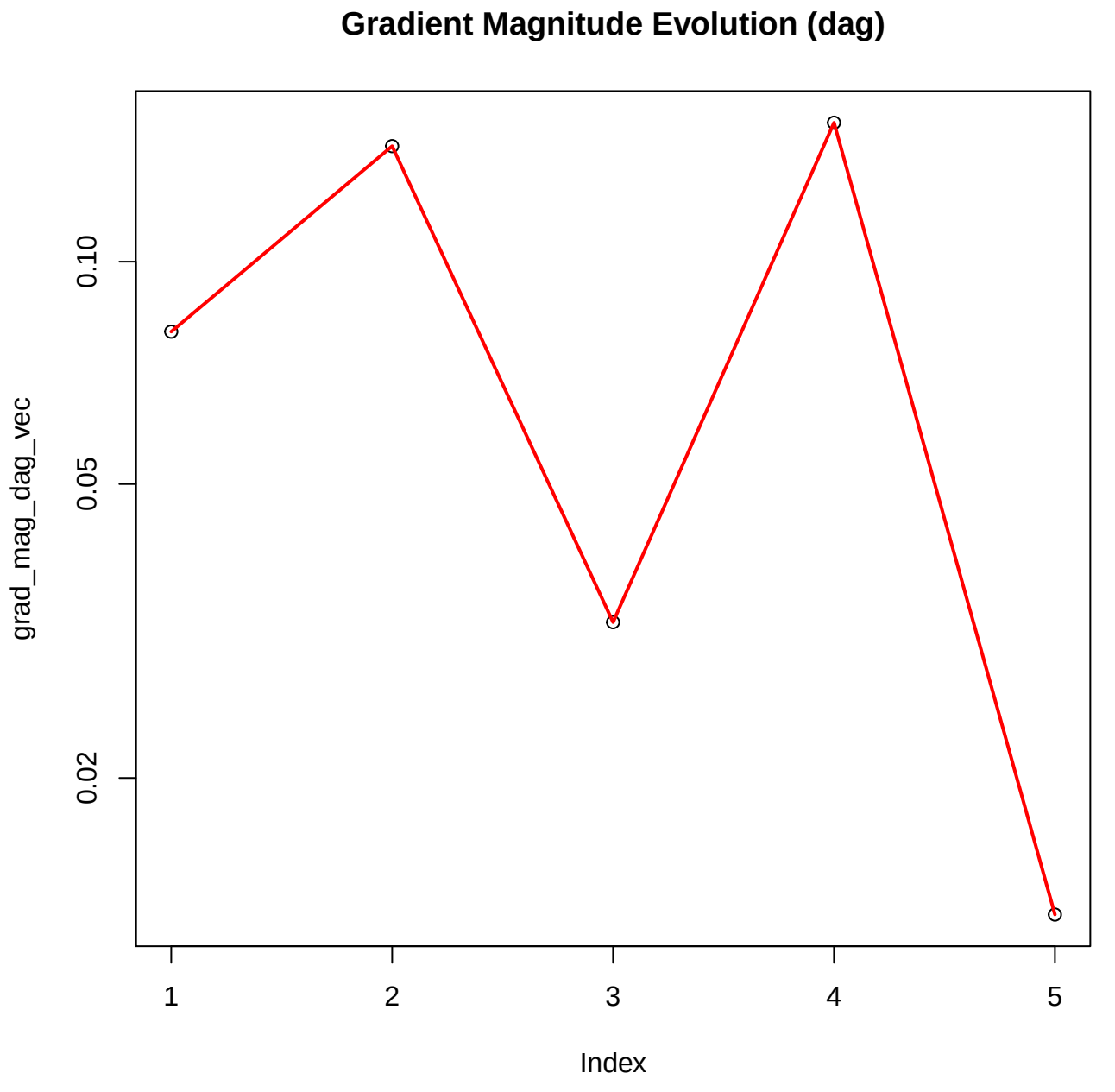

Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
2e−03
5e−04
2e−04
5e−05
1e−05
0
50
100
150
200
Index

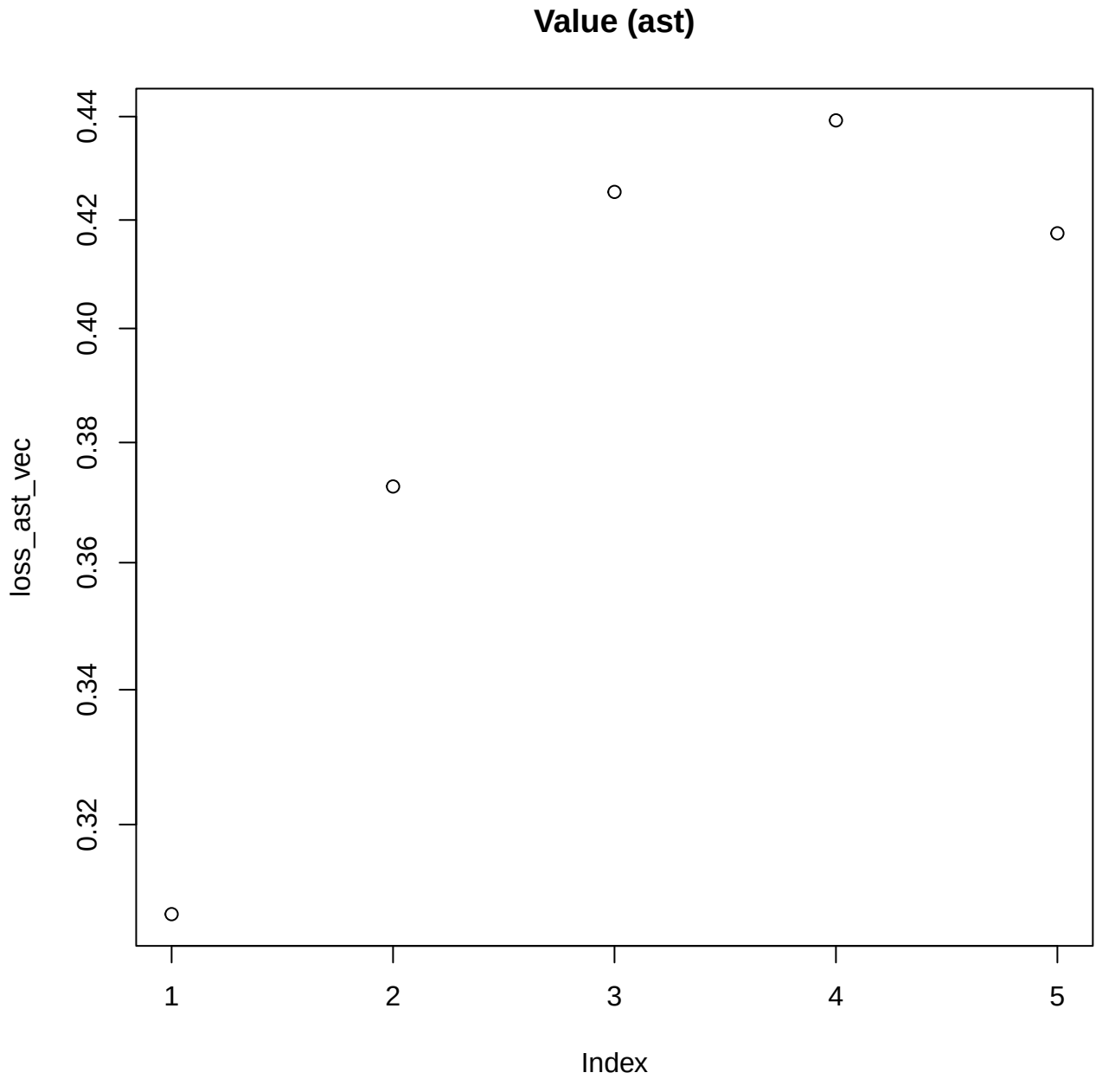

Gradient Magnitude Evolution (dag)
grad_mag_dag_vec
2e−03
5e−04
2e−04
5e−05
1e−05
0
50
100
150
200
Index

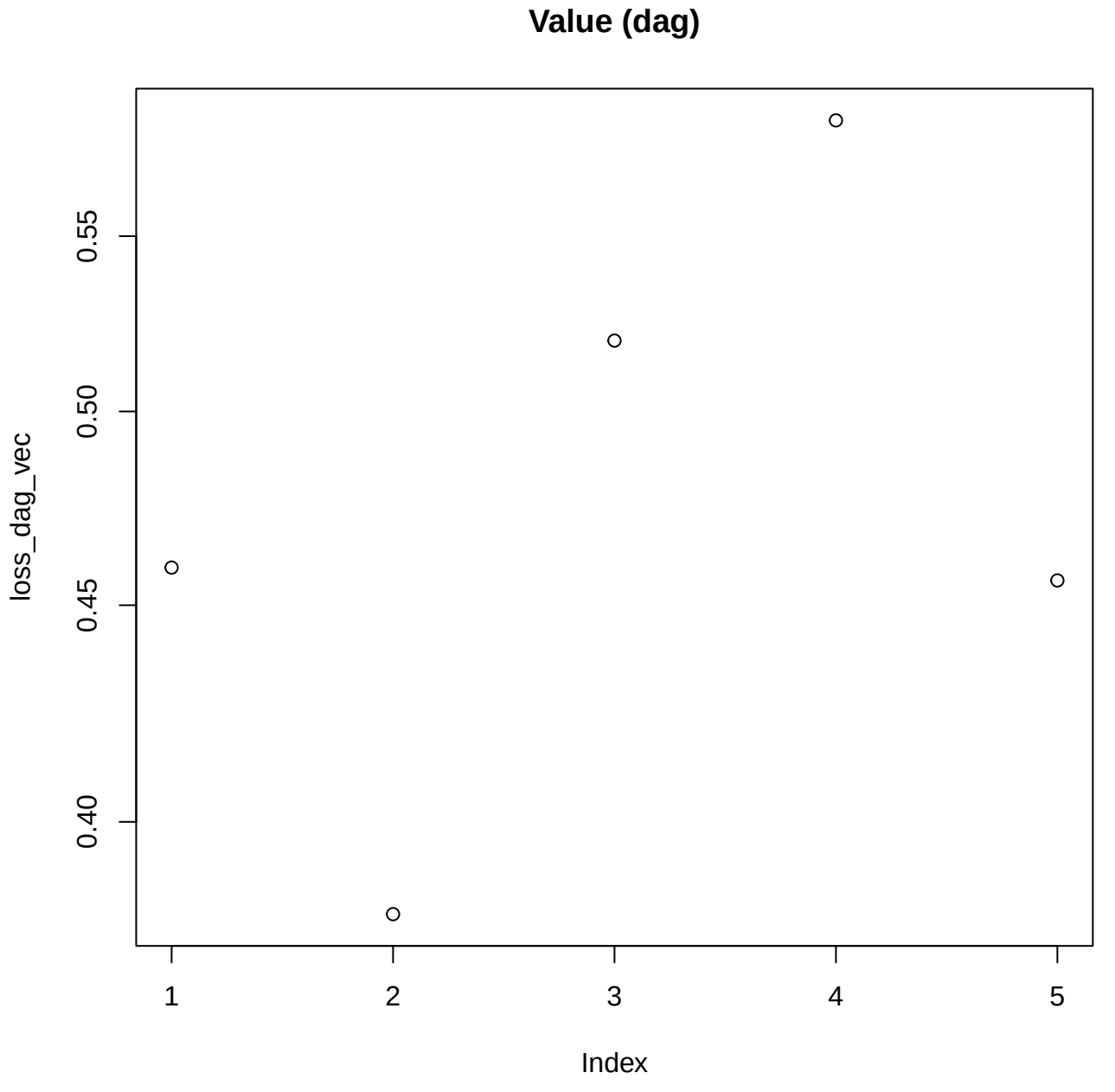

Value (ast)
loss_ast_vec
0.540
0.542
0.544
0.546
0.548
0
50
100
150
200
Index

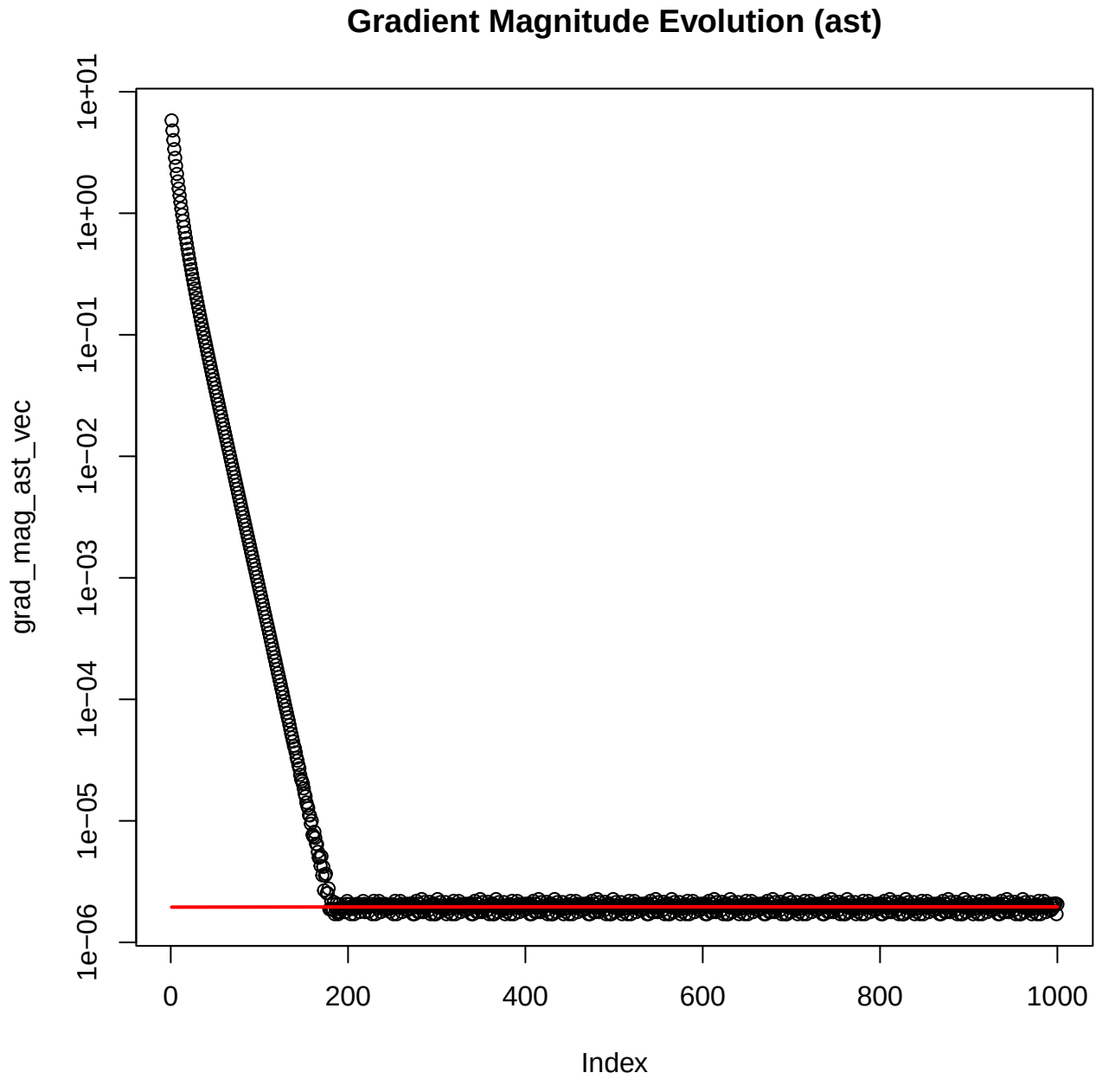

Value (dag)
loss_dag_vec
0.452
0.454
0.456
0.458
0.460
0
50
100
150
200
Index

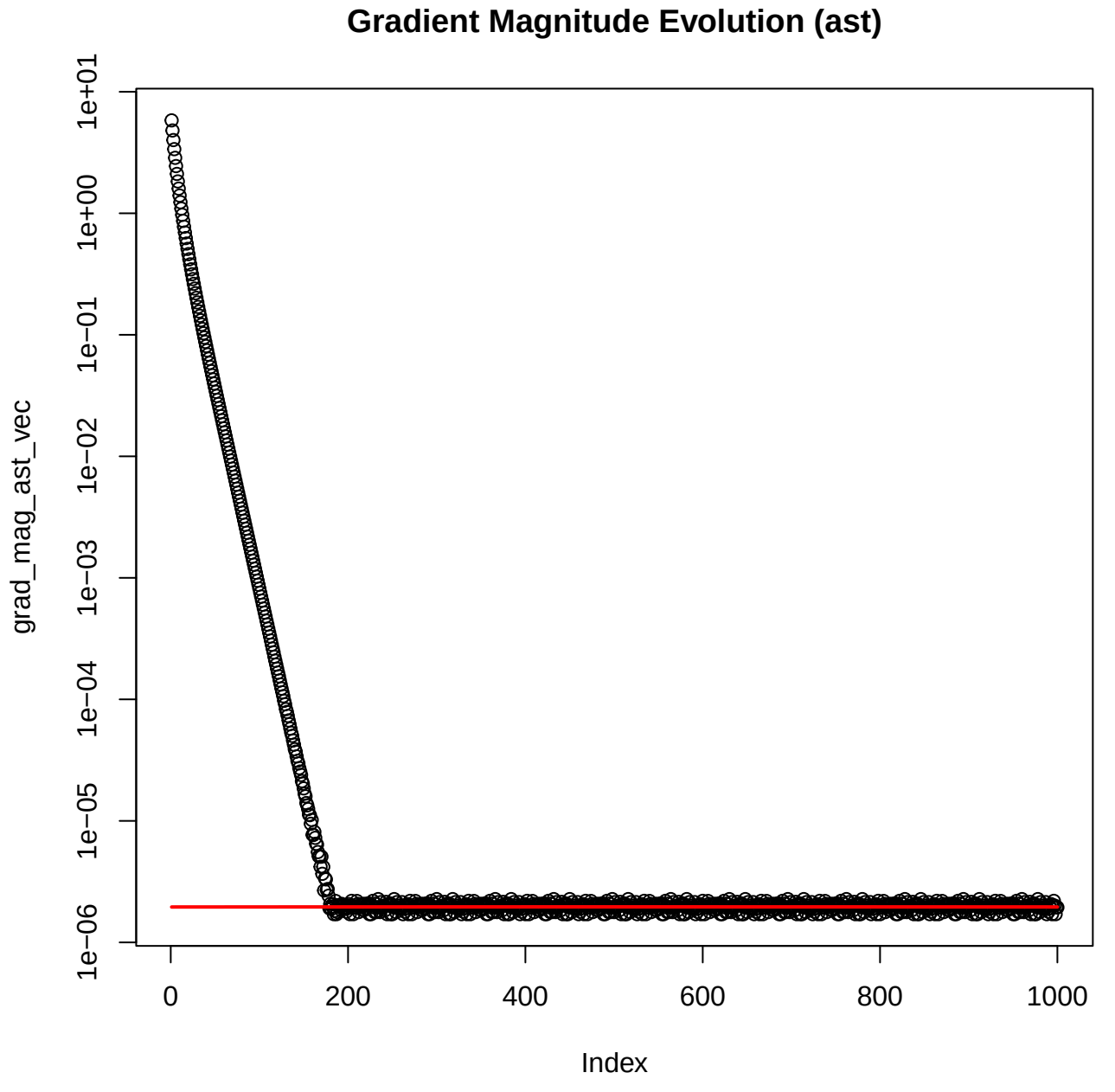

Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
0.01
0.02
0.05
0.10
1
2
3
4
5
Index

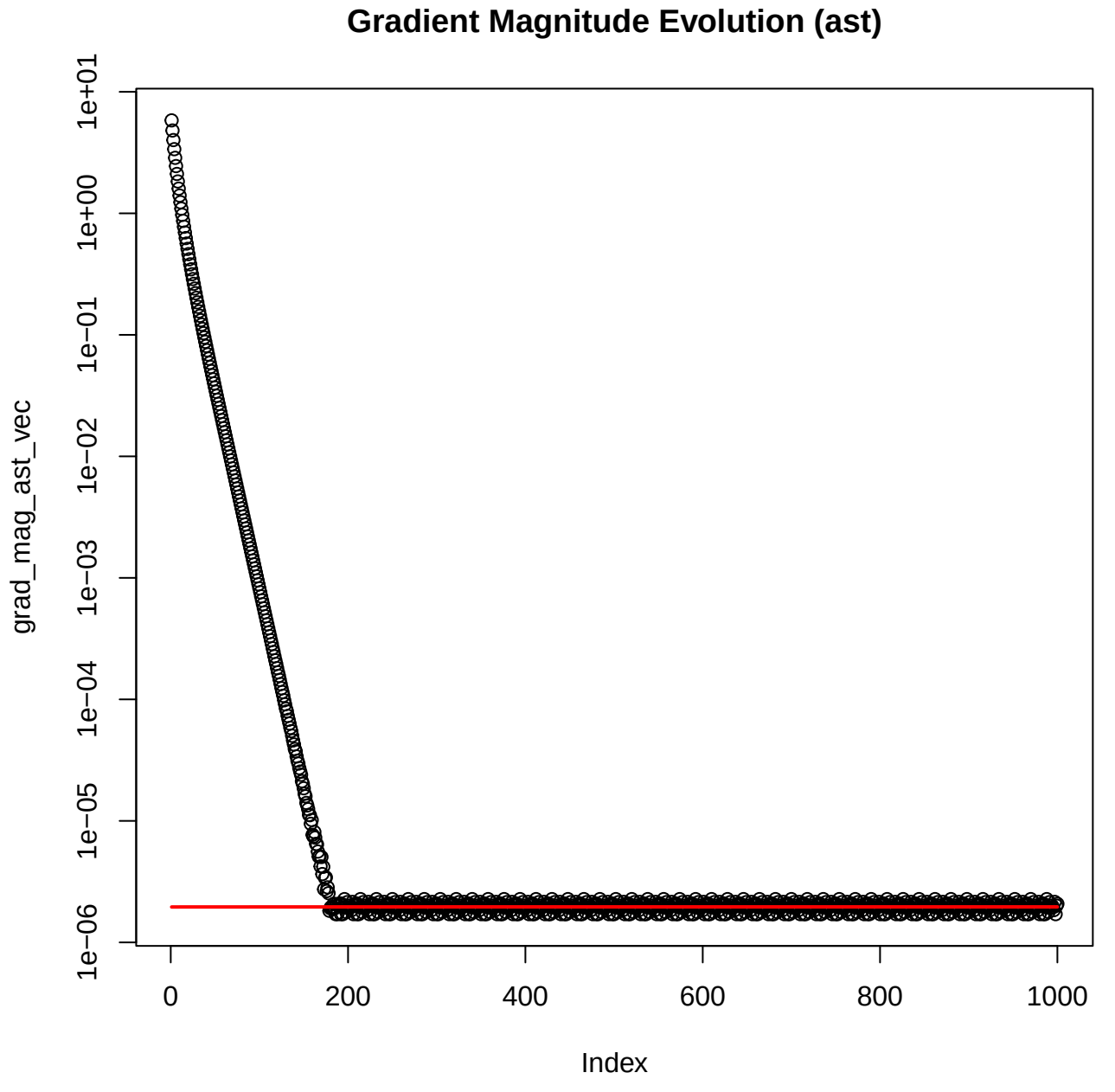

Gradient Magnitude Evolution (dag)
grad_mag_dag_vec
0.02
0.05
0.10
1
2
3
4
5
Index

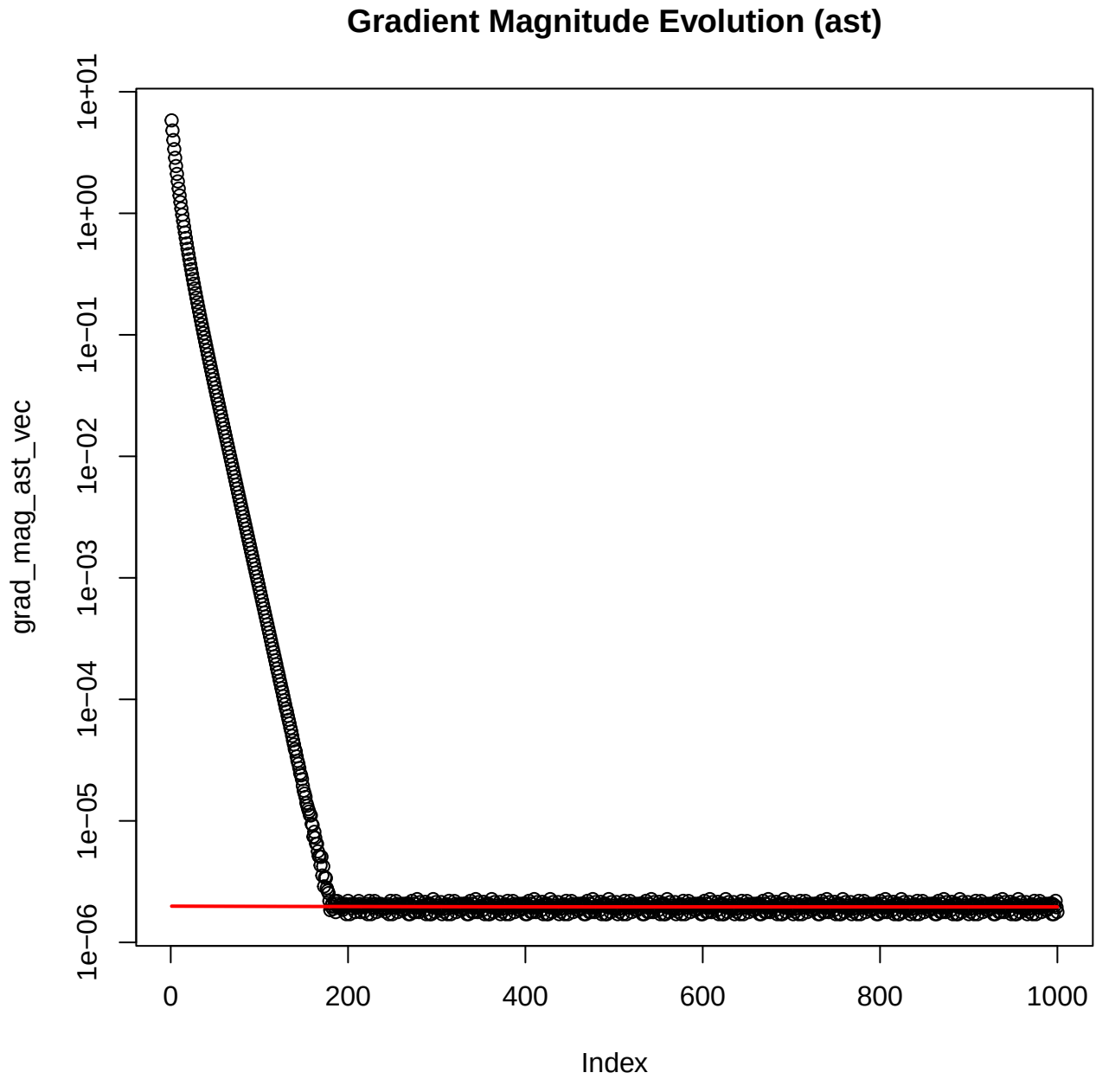

Value (ast)
loss_ast_vec
0.32
0.34
0.36
0.38
0.40
0.42
0.44
1
2
3
4
5
Index

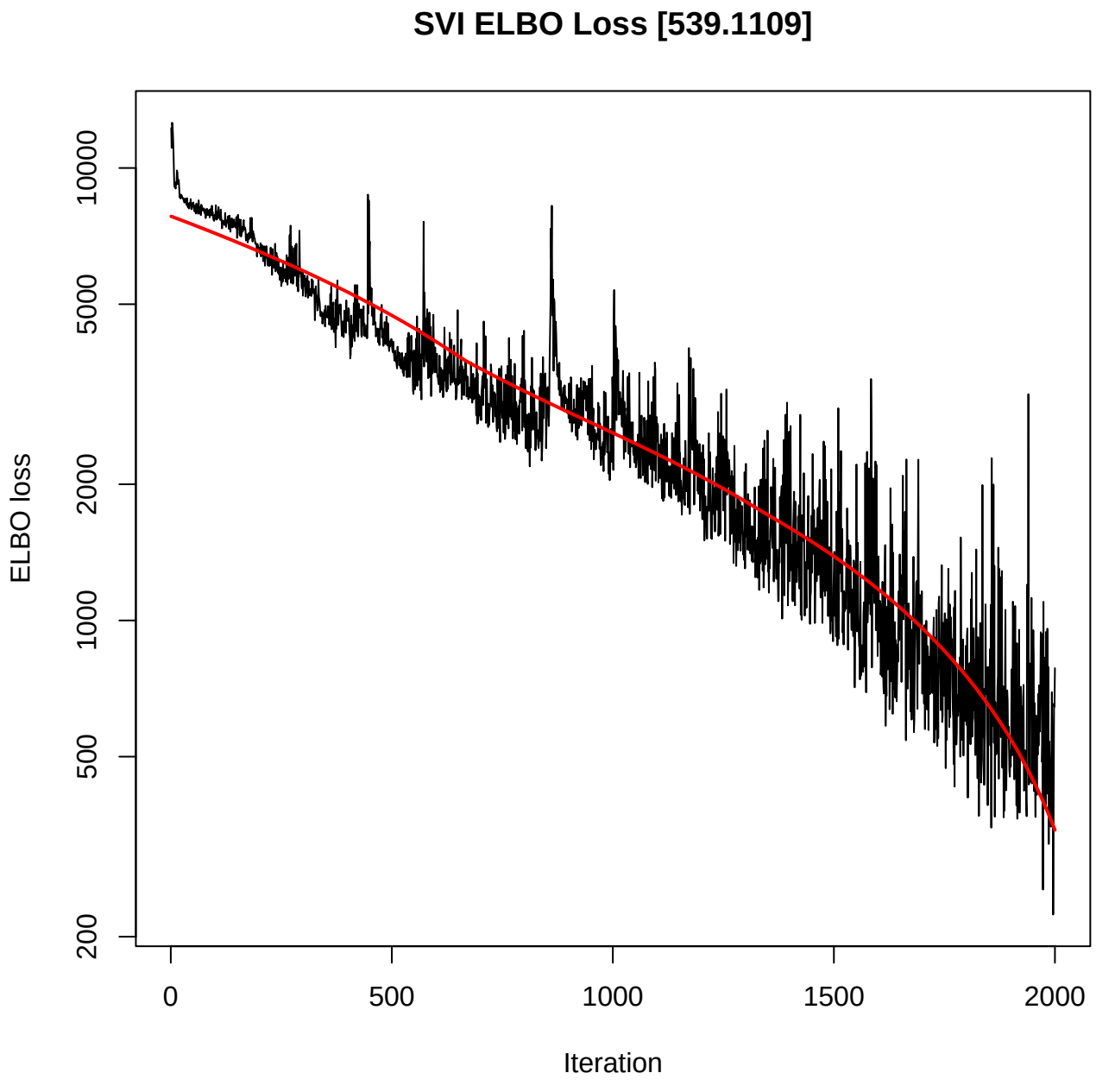

Value (dag)
loss_dag_vec
0.40
0.45
0.50
0.55
1
2
3
4
5
Index

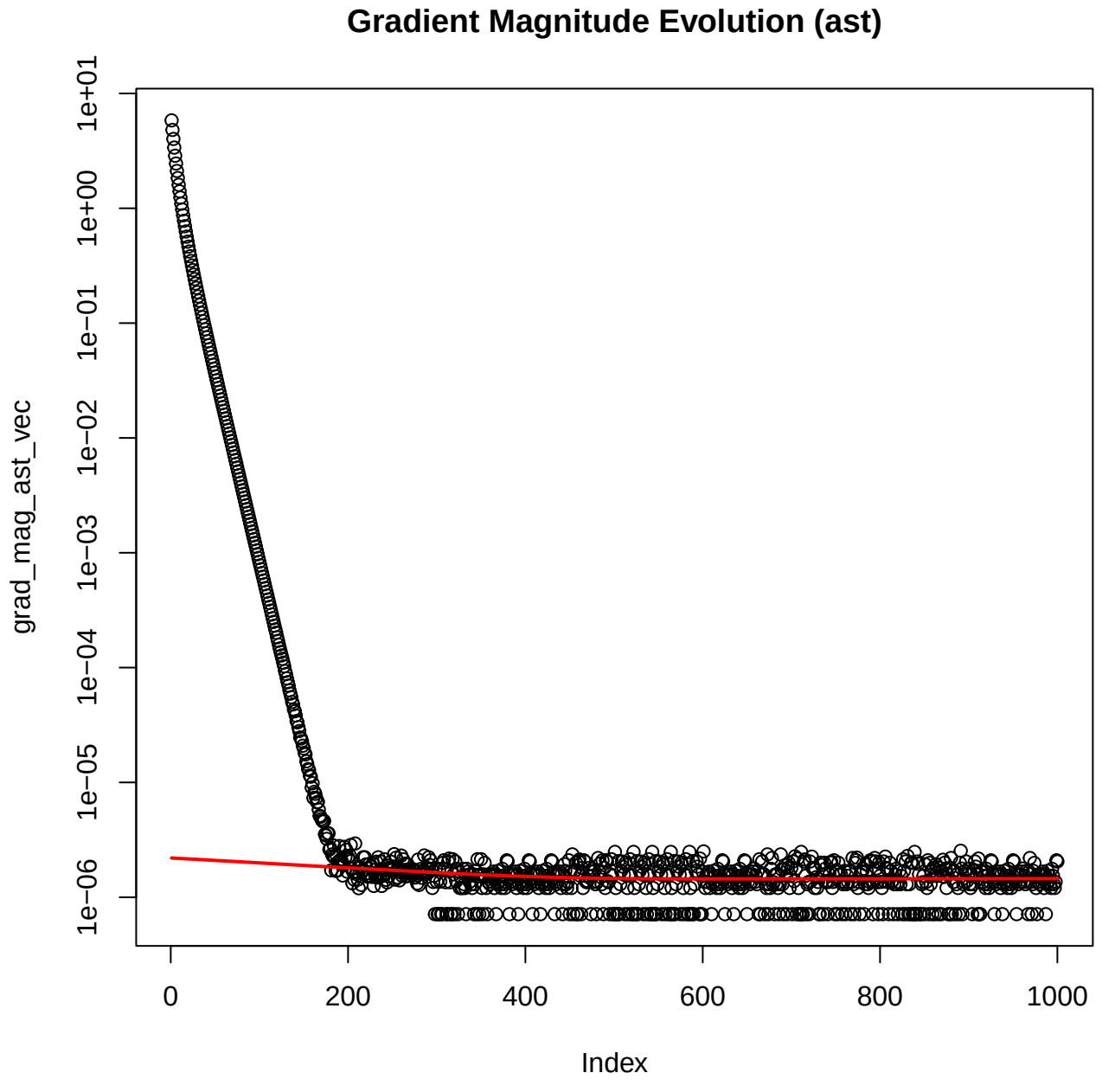

Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
1e+01
1e+00
1e−01
1e−02
1e−03
1e−04
1e−05
1e−06
0
200
400
600
800
1000
Index

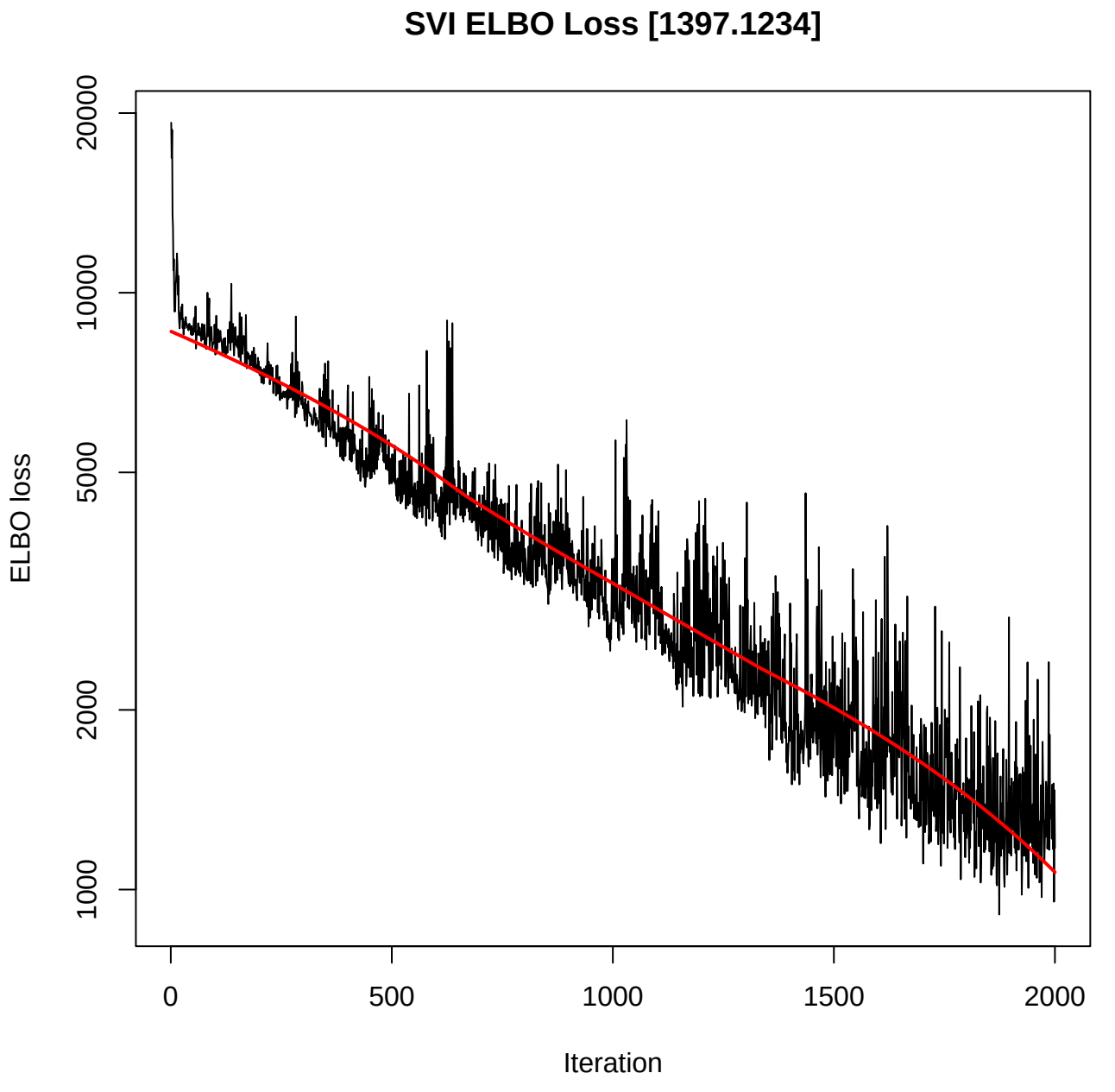

Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
1e+01
1e+00
1e−01
1e−02
1e−03
1e−04
1e−05
1e−06
0
200
400
600
800
1000
Index

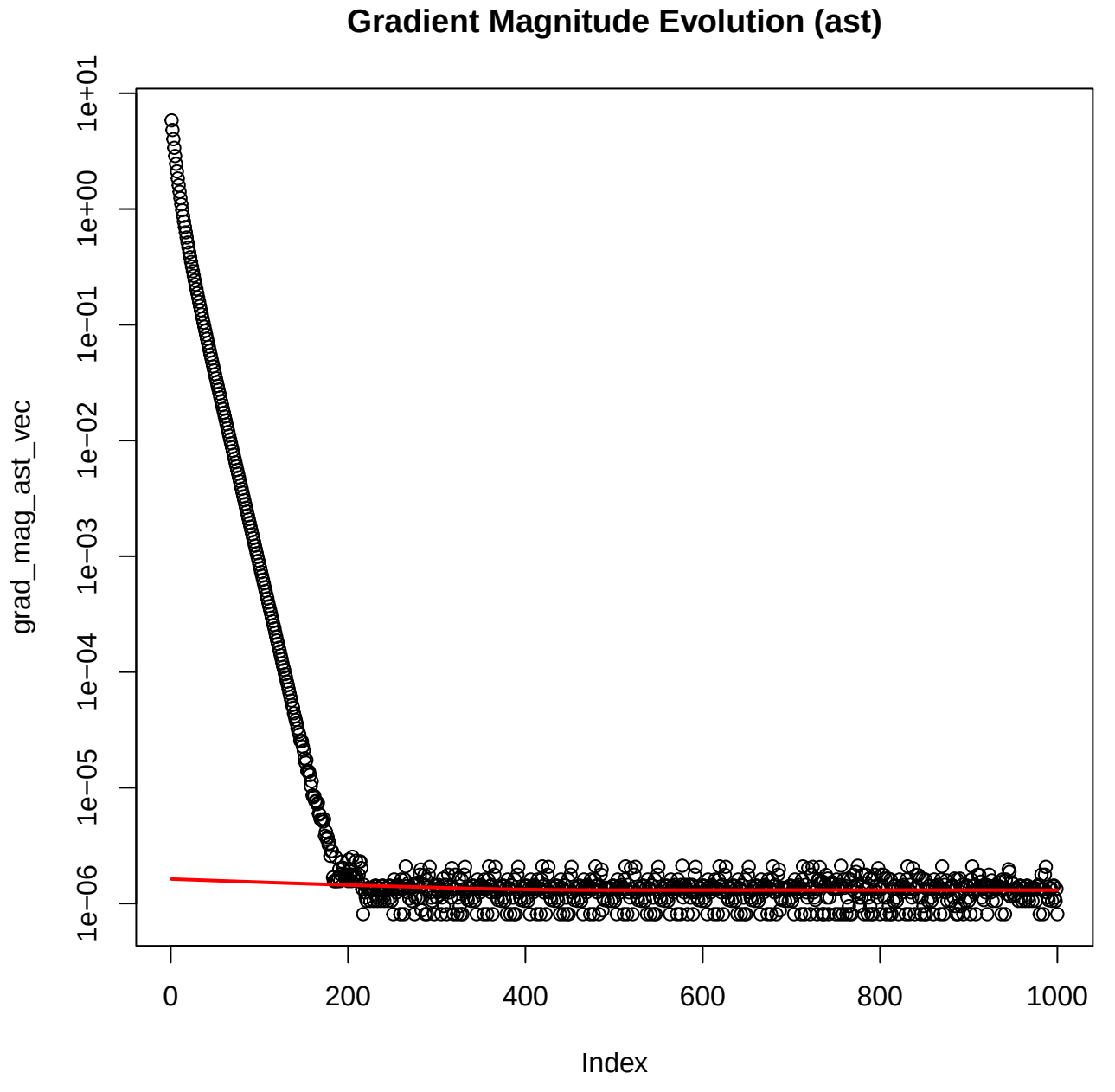

Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
1e+01
1e+00
1e−01
1e−02
1e−03
1e−04
1e−05
1e−06
0
200
400
600
800
1000
Index

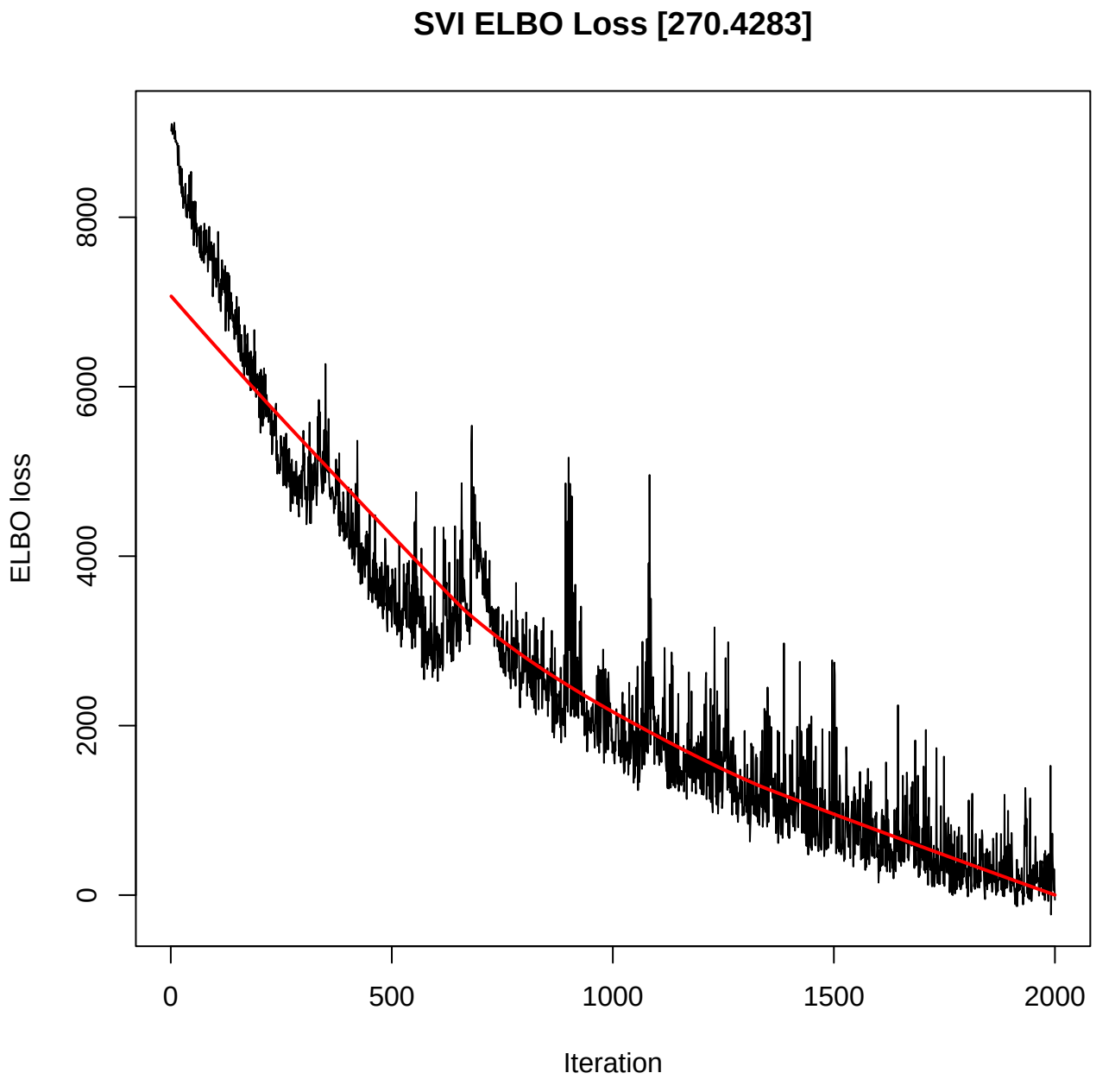

Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
1e+01
1e+00
1e−01
1e−02
1e−03
1e−04
1e−05
1e−06
0
200
400
600
800
1000
Index

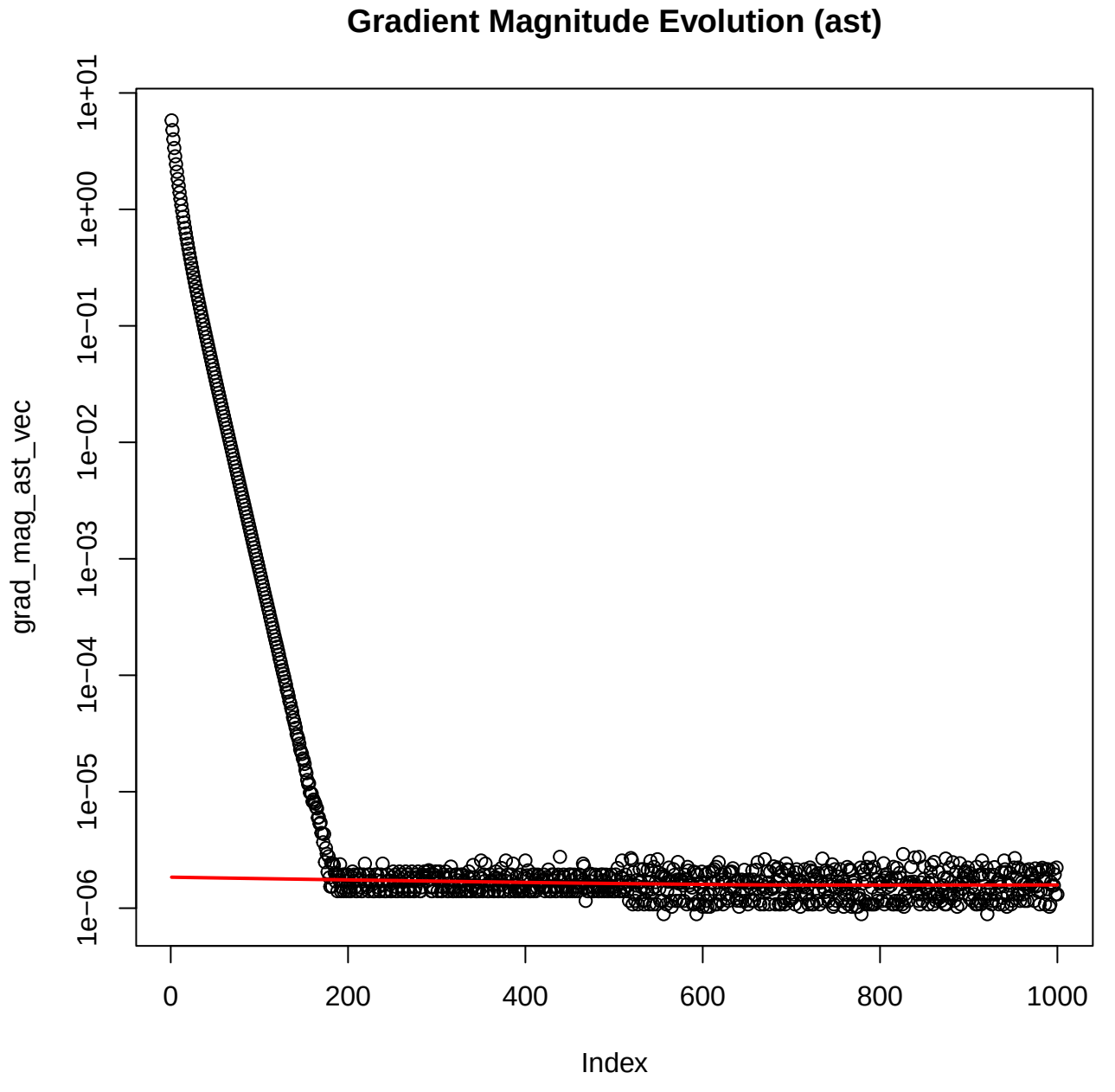

SVI ELBO Loss [539.1109]
ELBO loss
10000
5000
2000
1000
500
200
0
500
1000
1500
2000
Iteration

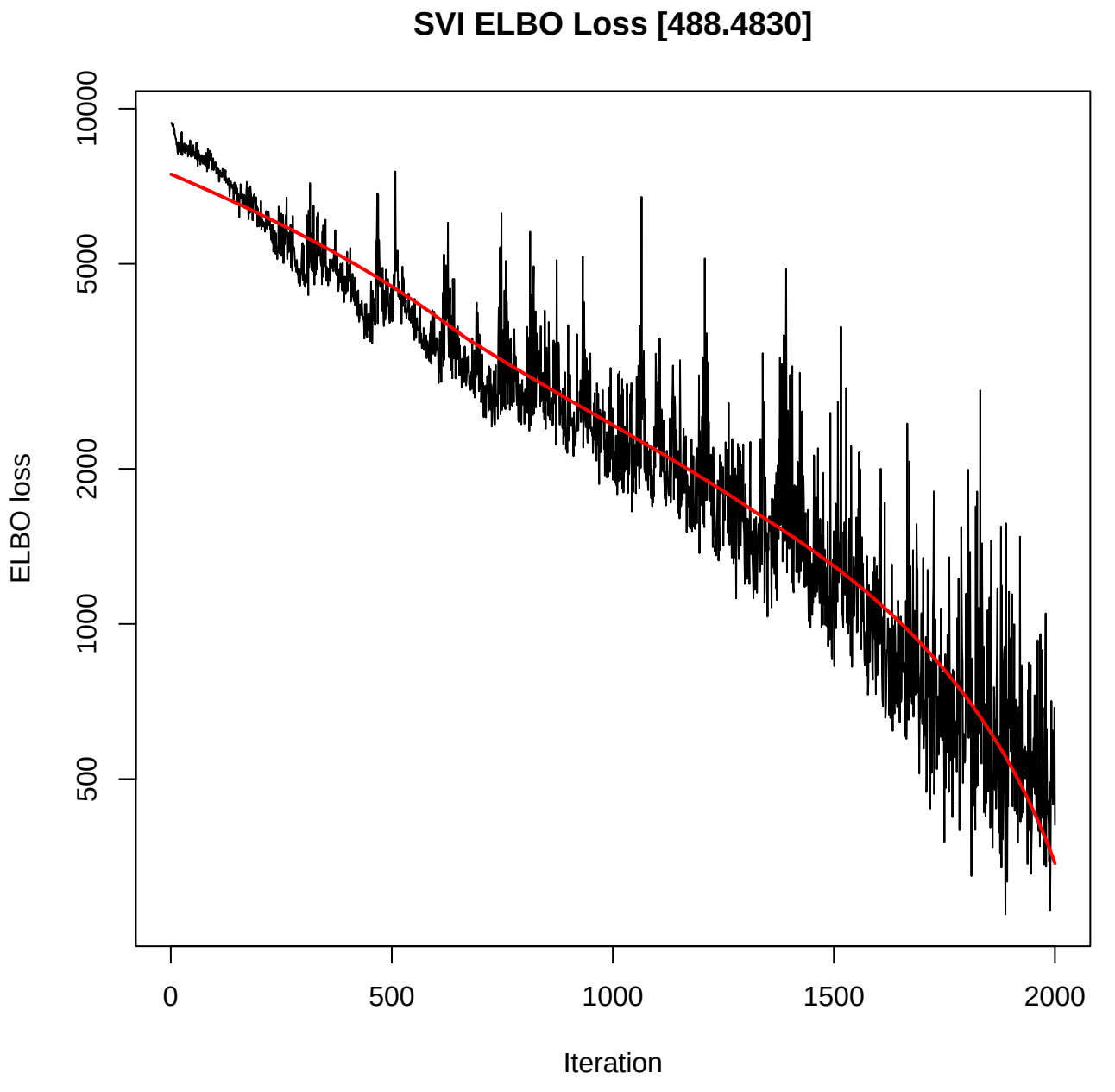

Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
1e+01
1e+00
1e−01
1e−02
1e−03
1e−04
1e−05
1e−06
0
200
400
600
800
1000
Index

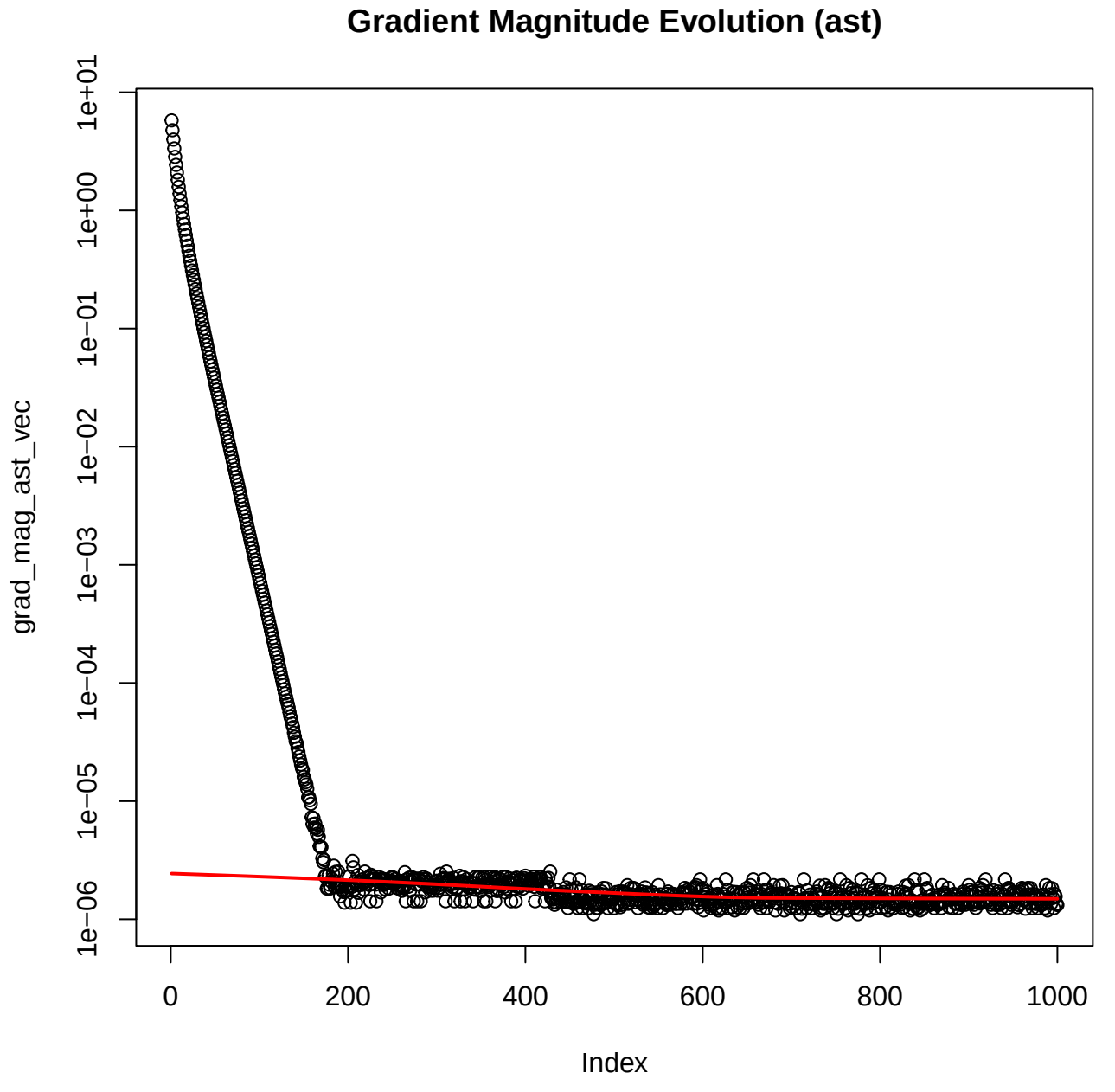

SVI ELBO Loss [1397.1234]
ELBO loss
20000
10000
5000
2000
1000
0
500
1000
1500
2000
Iteration

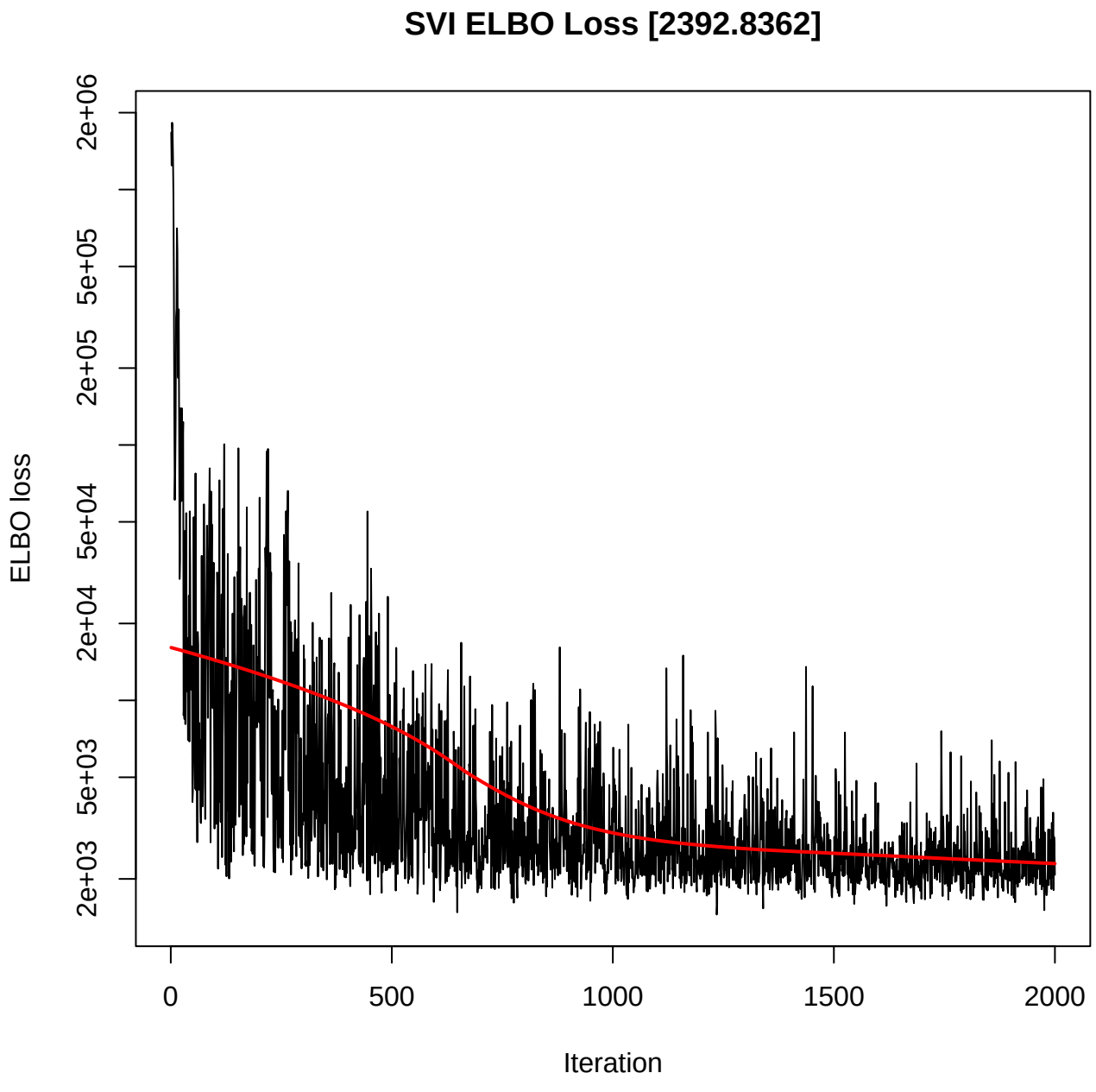

Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
1e+01
1e+00
1e−01
1e−02
1e−03
1e−04
1e−05
1e−06
0
200
400
600
800
1000
Index

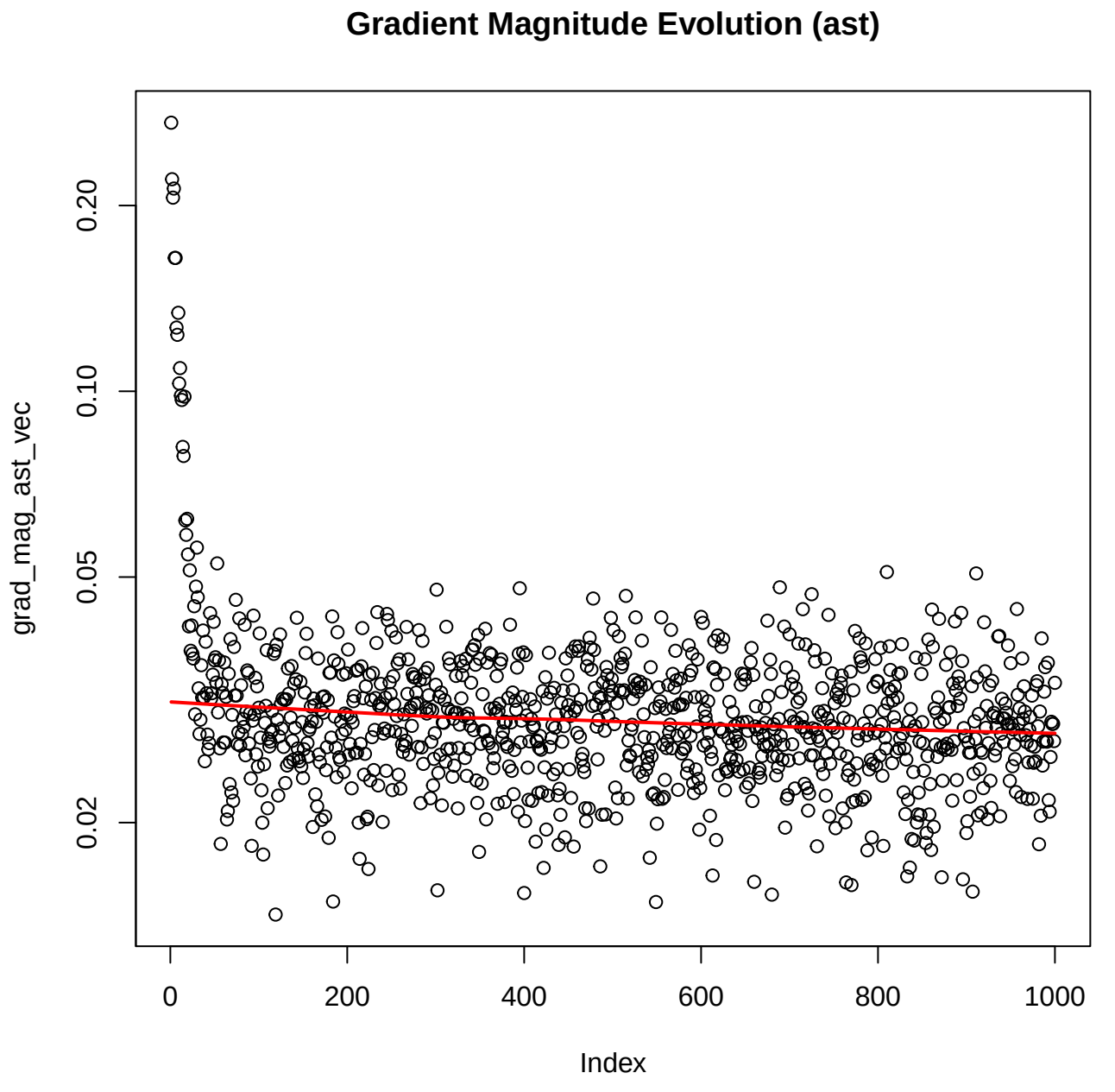

SVI ELBO Loss [270.4283]
ELBO loss
0
2000
4000
6000
8000
0
500
1000
1500
2000
Iteration

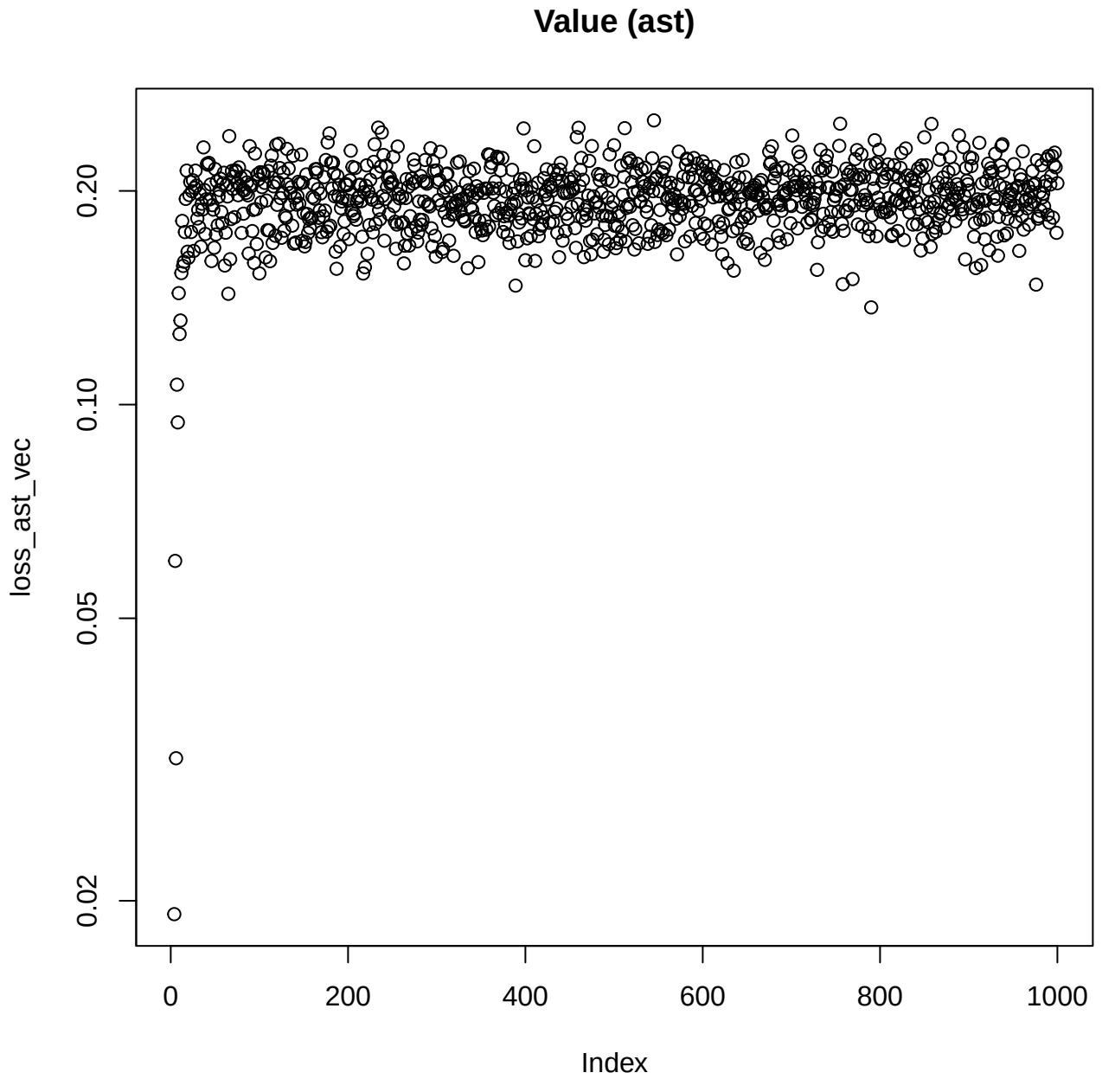

Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
1e+01
1e+00
1e−01
1e−02
1e−03
1e−04
1e−05
1e−06
0
200
400
600
800
1000
Index

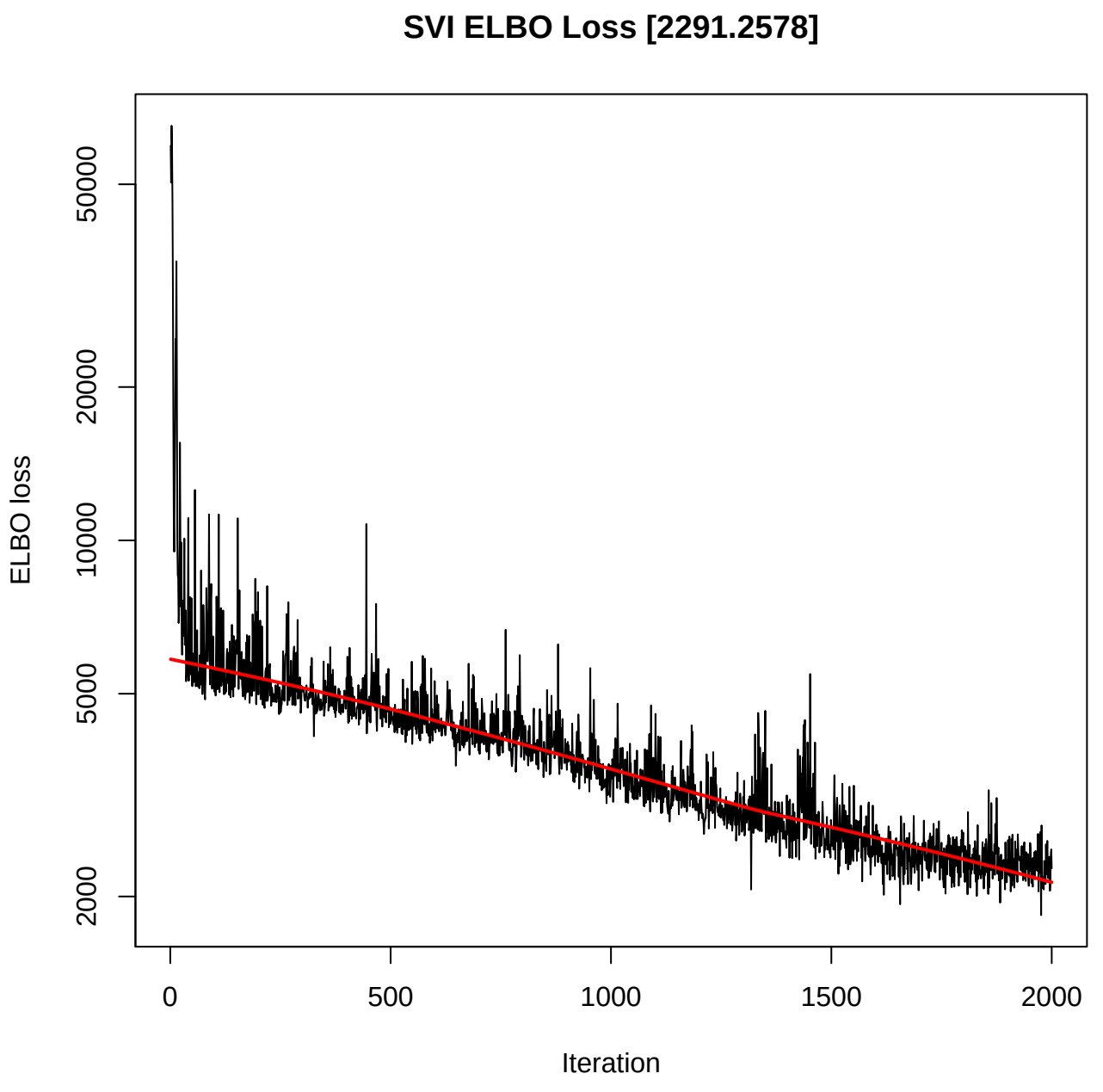

SVI ELBO Loss [488.4830]
ELBO loss
10000
5000
2000
1000
500
0
500
1000
1500
2000
Iteration

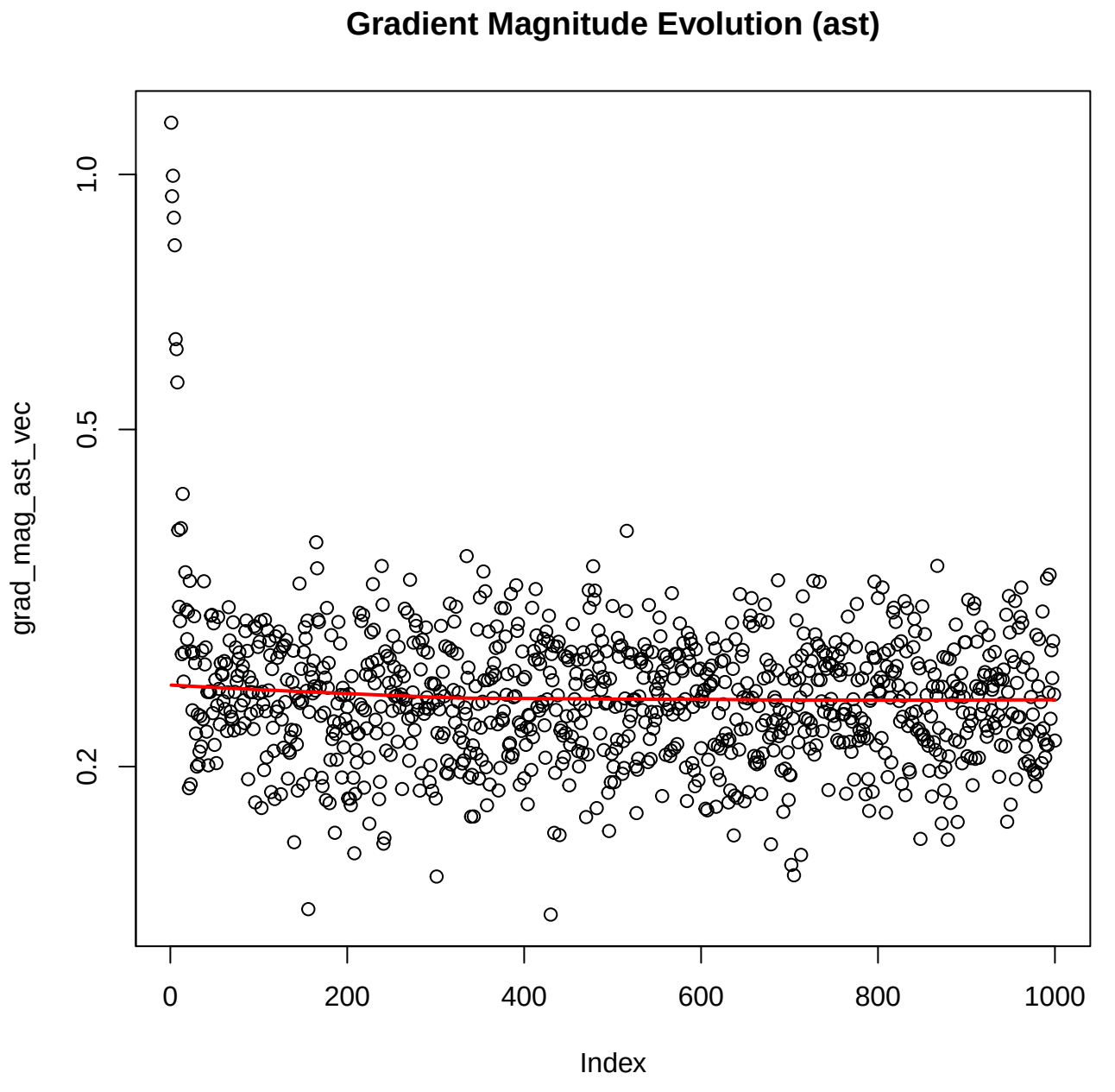

Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
1e+01
1e+00
1e−01
1e−02
1e−03
1e−04
1e−05
1e−06
0
200
400
600
800
1000
Index

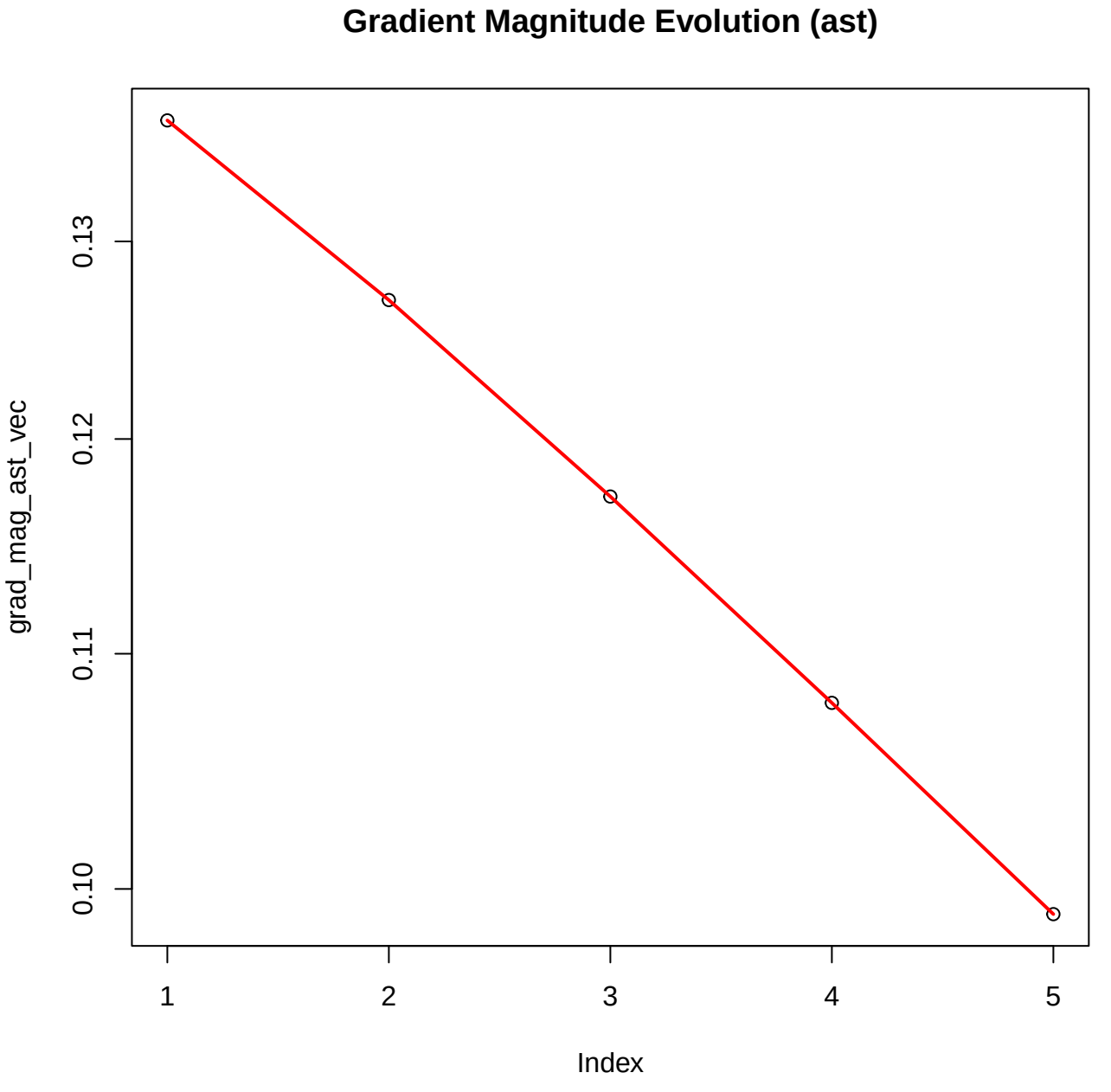

SVI ELBO Loss [2392.8362]
ELBO loss
2e+06
5e+05
2e+05
5e+04
2e+04
5e+03
2e+03
0
500
1000
1500
2000
Iteration

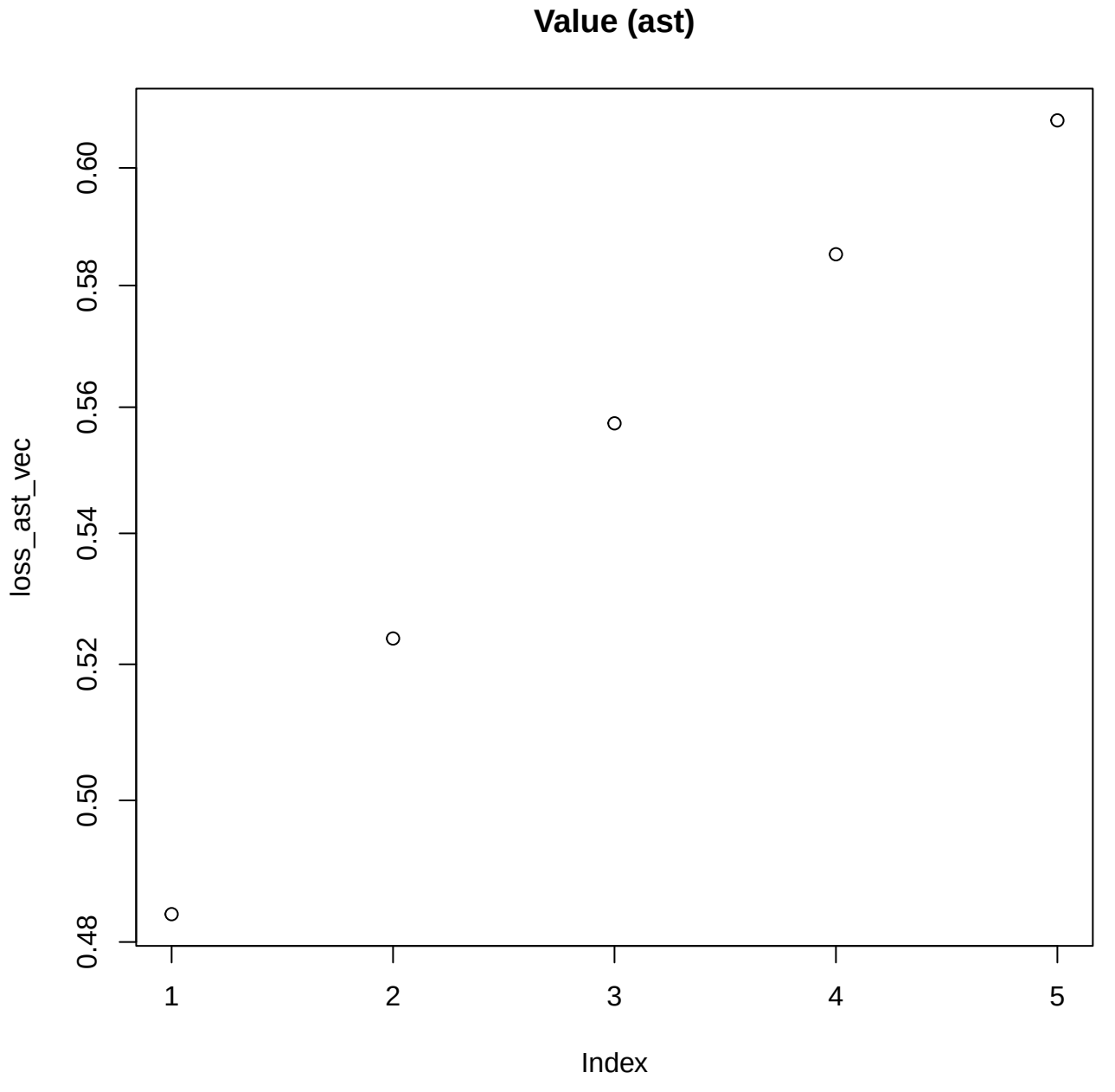

Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
0.02
0.05
0.10
0.20
0
200
400
600
800
1000
Index


Value (ast)
loss_ast_vec
0.02
0.05
0.10
0.20
0
200
400
600
800
1000
Index


SVI ELBO Loss [2291.2578]
ELBO loss
50000
20000
10000
5000
2000
0
500
1000
1500
2000
Iteration


Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
1.0
0.5
0.2
0
200
400
600
800
1000
Index


Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
0.10
0.11
0.12
0.13
1
2
3
4
5
Index


Value (ast)
loss_ast_vec
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0.50
0.52
0.54
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0.60
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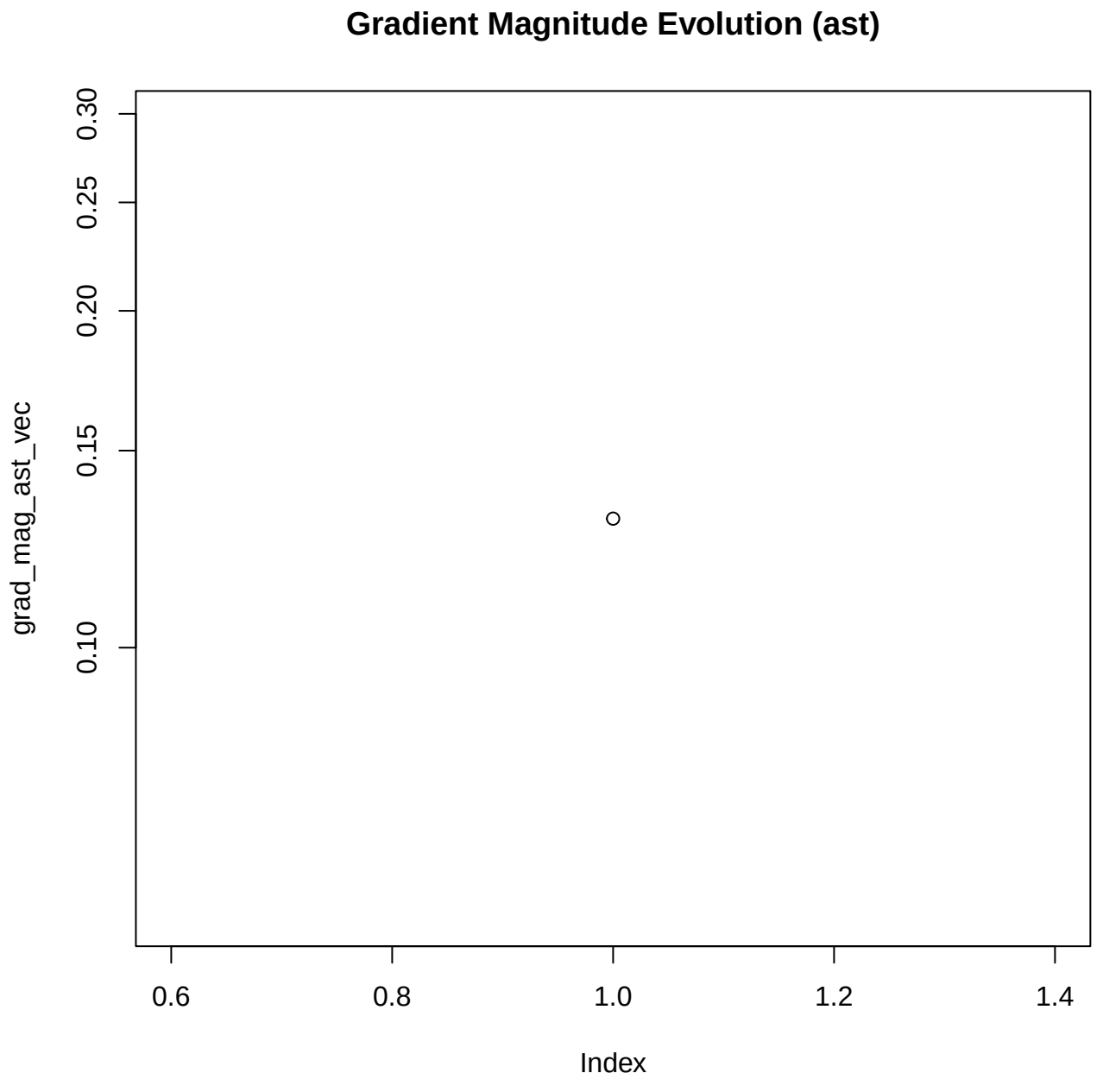
2
3
4
5
Index


Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
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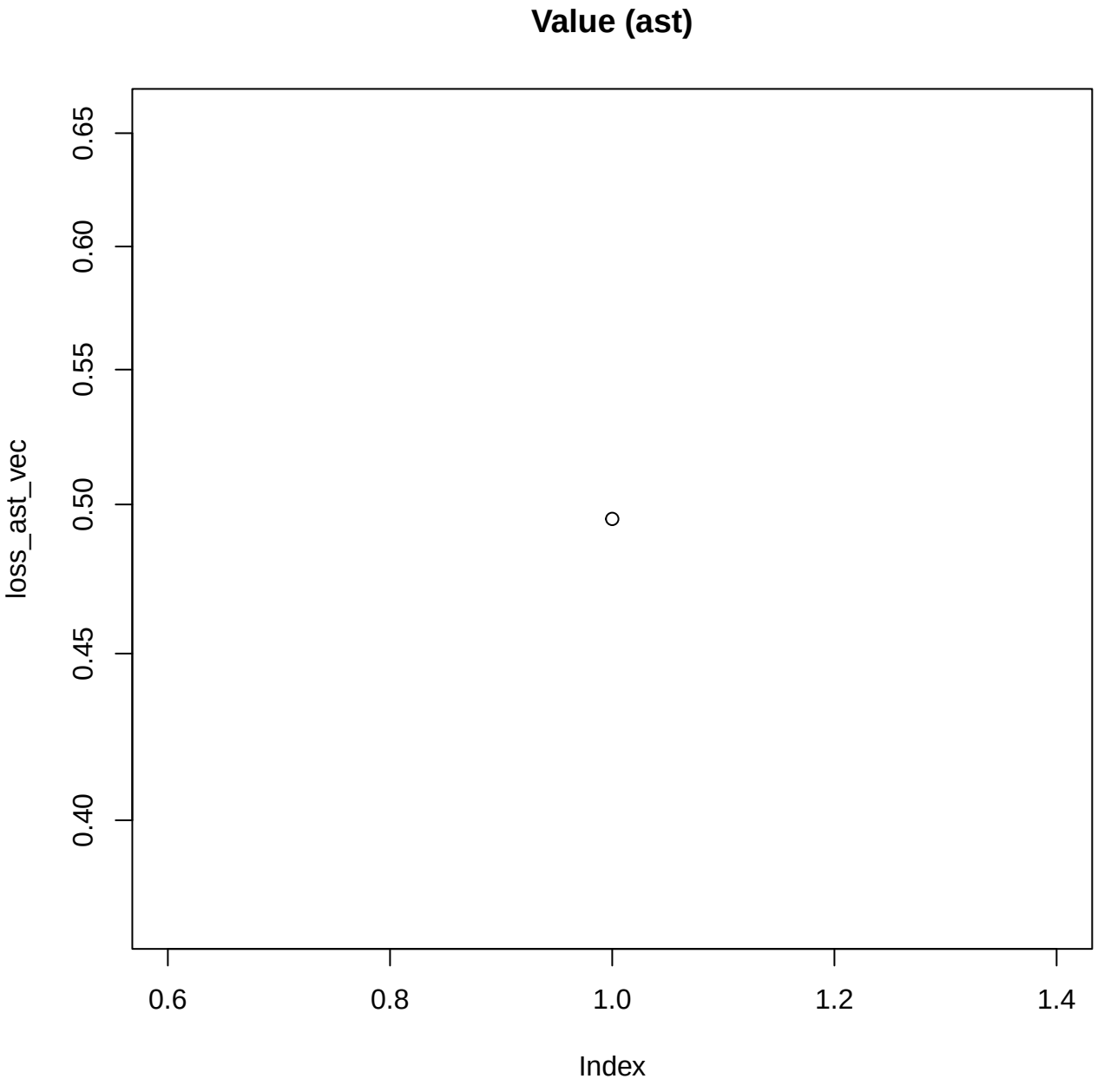
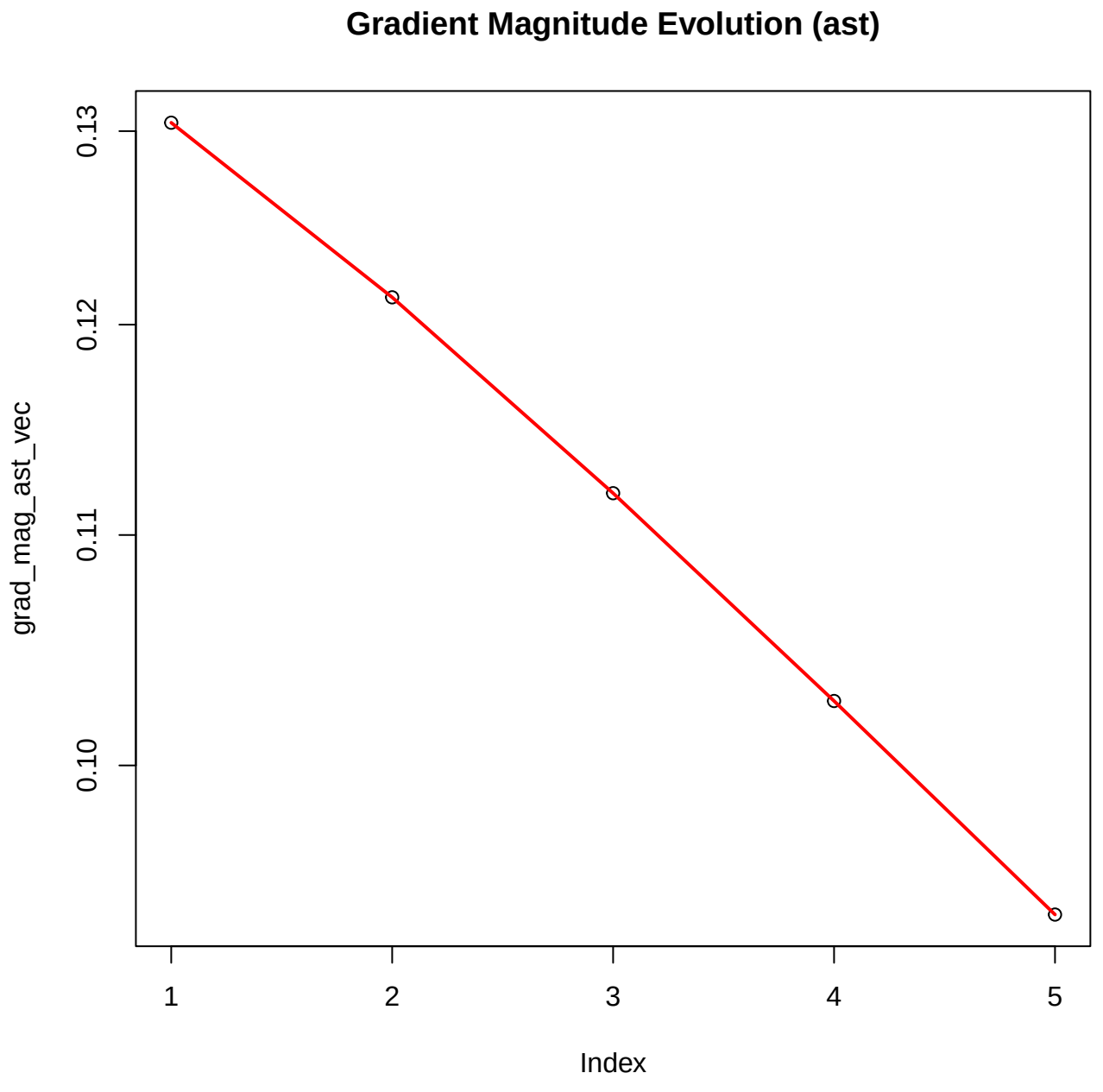
0.8
1.0
1.2
1.4
Index


Value (ast)
loss_ast_vec
0.65
0.60
0.55
0.50
0.45
0.40
0.6
0.8
1.0
1.2
1.4
Index


Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
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0.11
0.12
0.13
1
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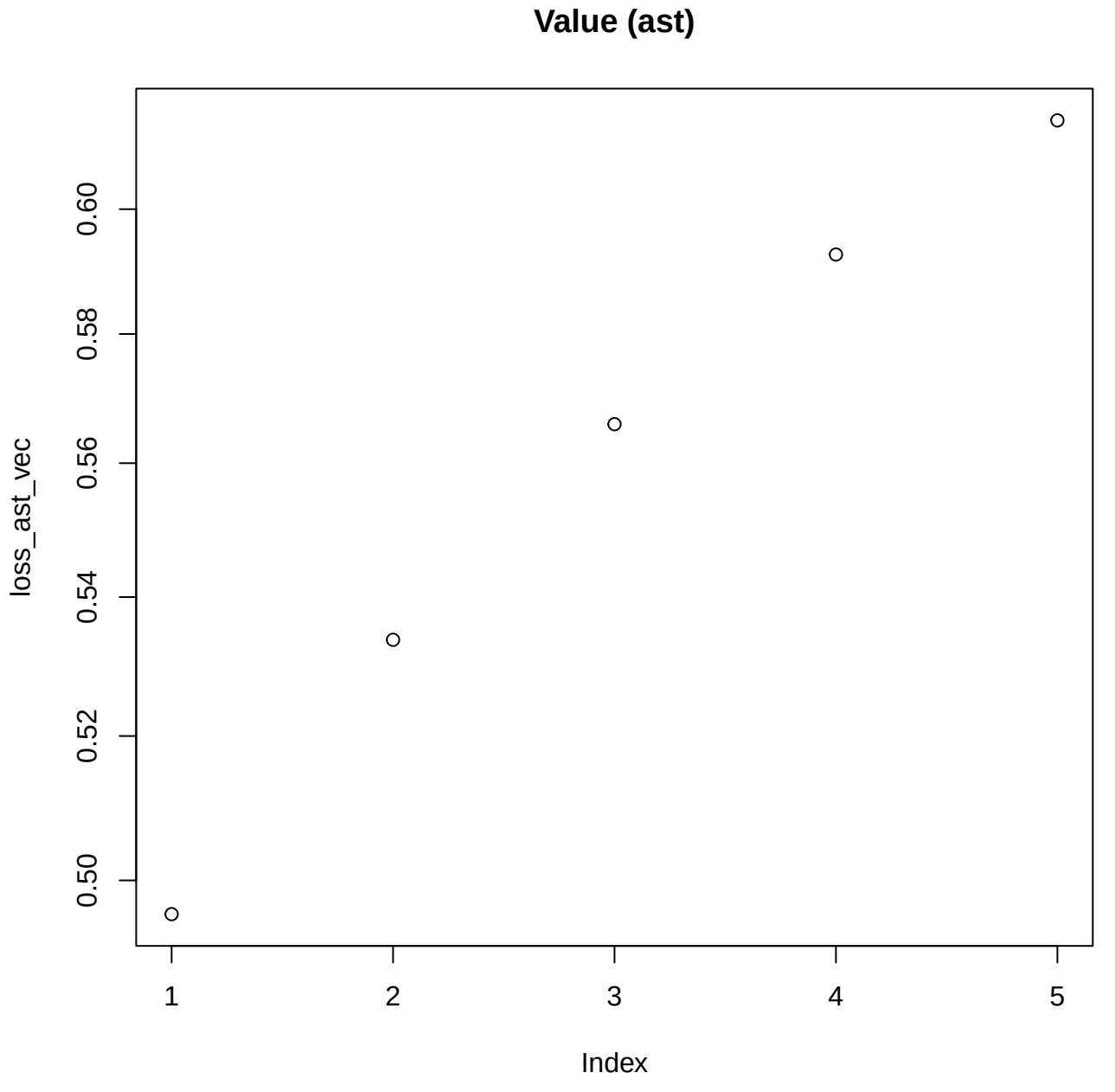
3
4
5
Index


Value (ast)
loss_ast_vec
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0.52
0.54
0.56
0.58
0.60
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2
3
4
5
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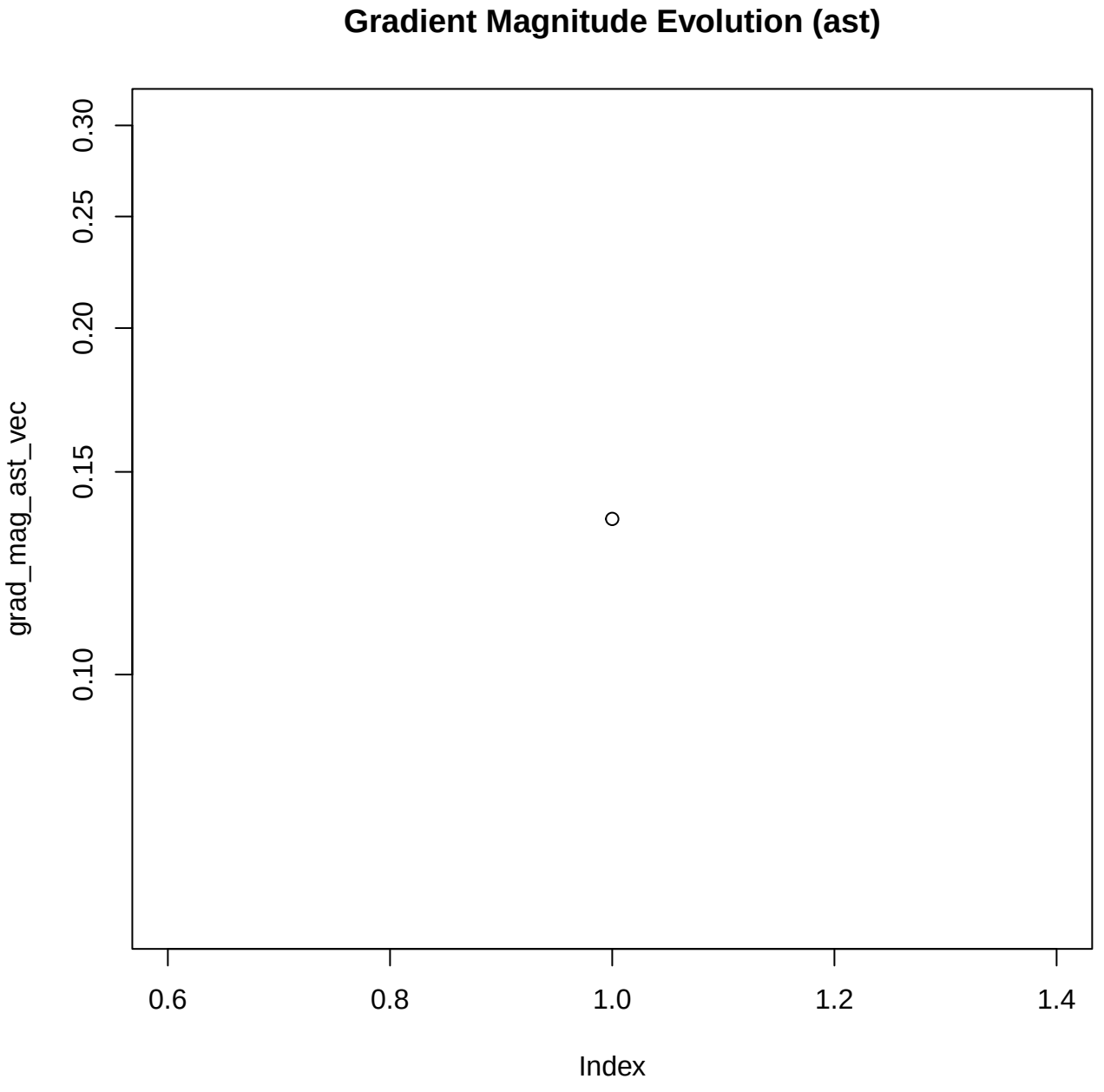
Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
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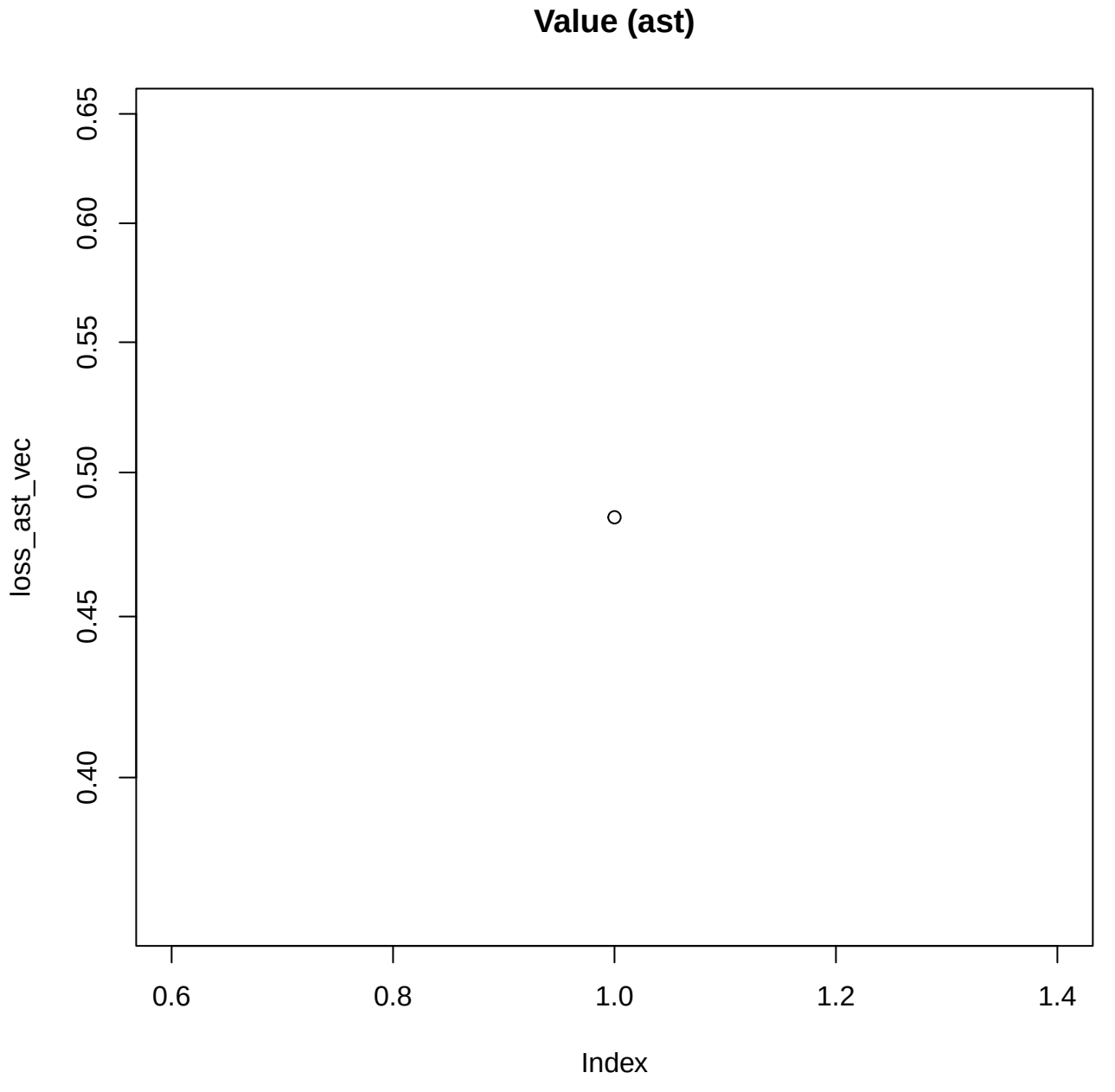
0.20
0.25
0.30
0.6
0.8
1.0
1.2
1.4
Index


Value (ast)
loss_ast_vec
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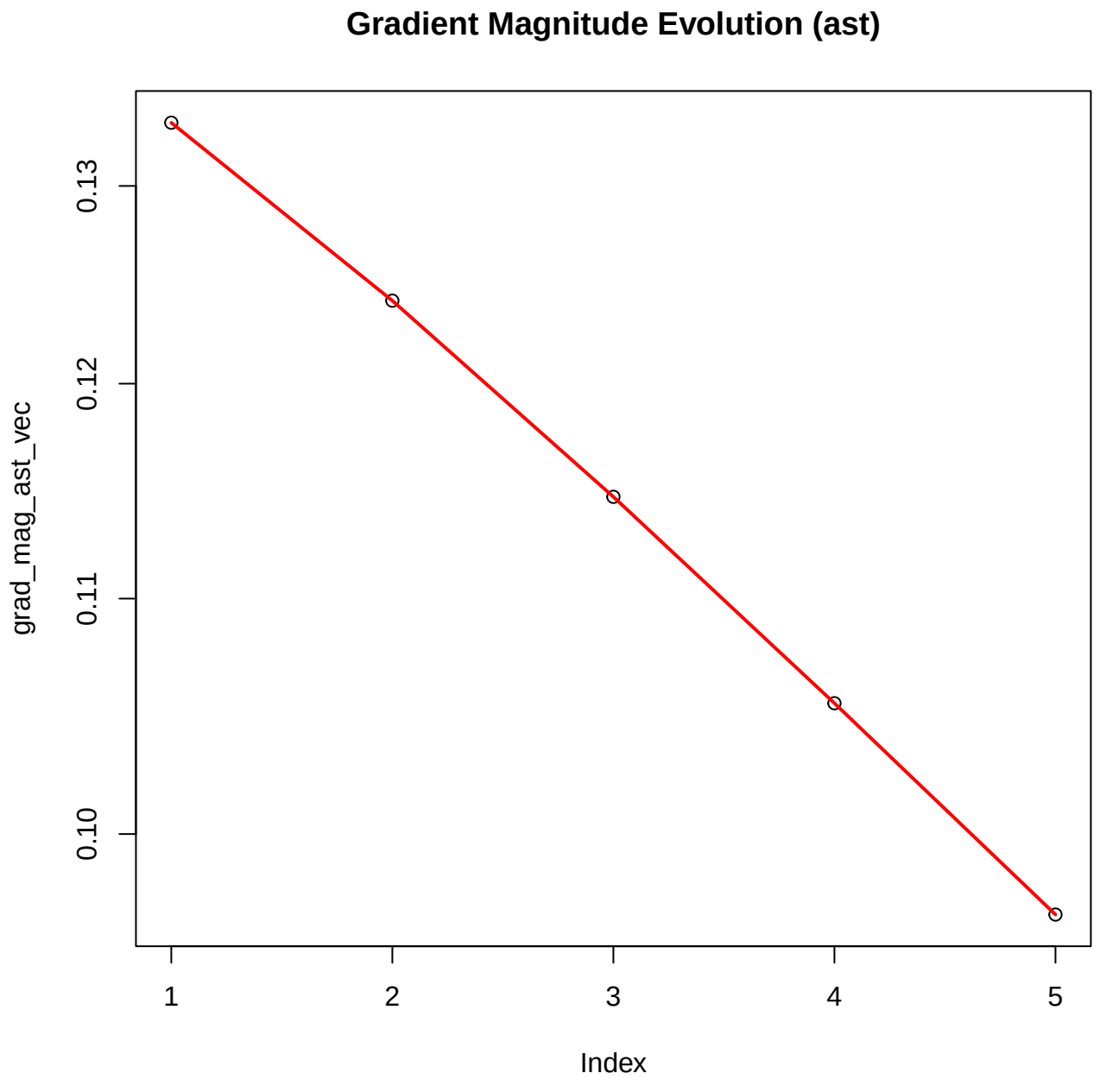
0.60
0.55
0.50
0.45
0.40
0.6
0.8
1.0
1.2
1.4
Index


Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
0.10
0.11
0.12
0.13
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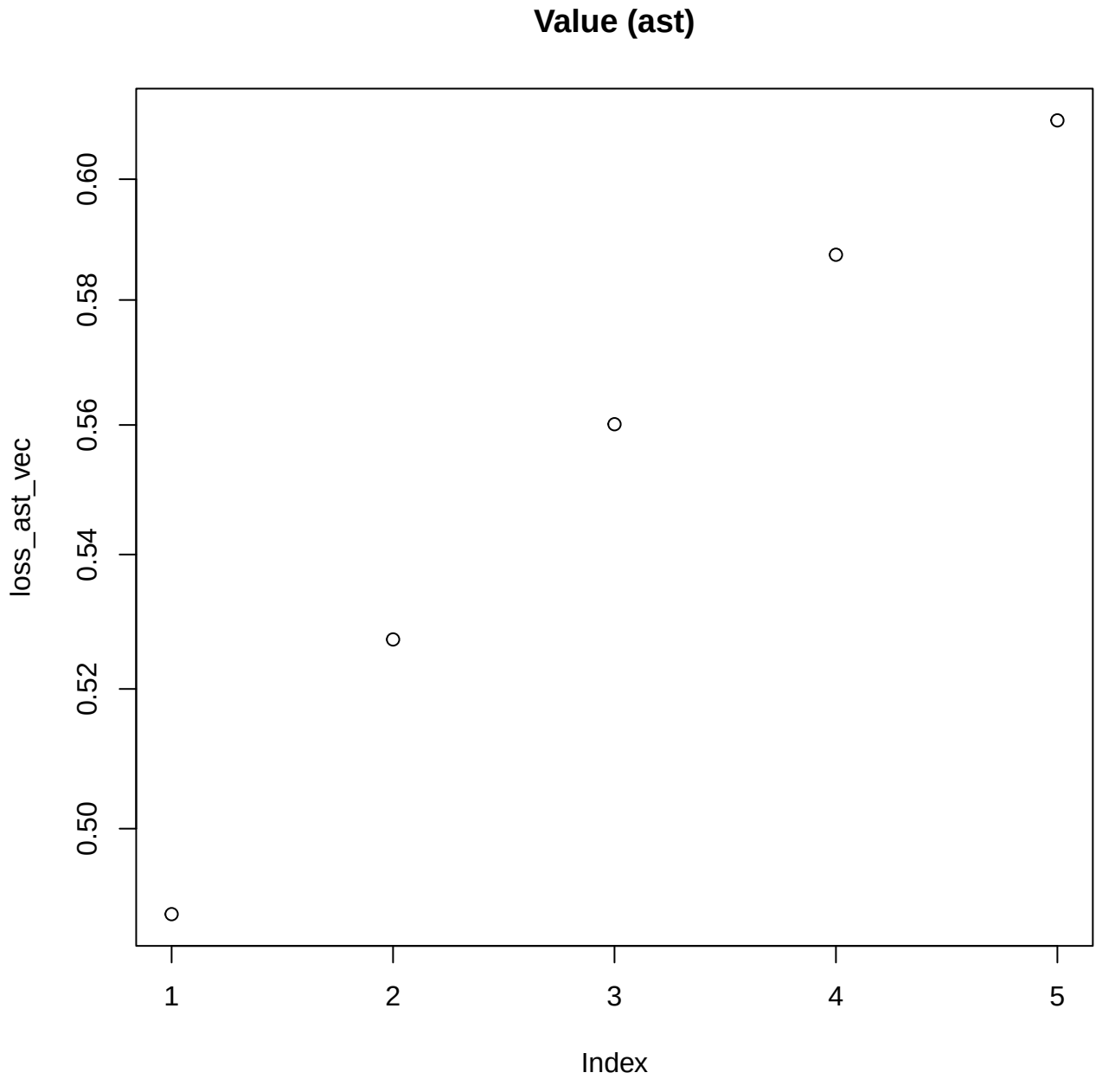
2
3
4
5
Index


Value (ast)
loss_ast_vec
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0.52
0.54
0.56
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0.60
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4
5
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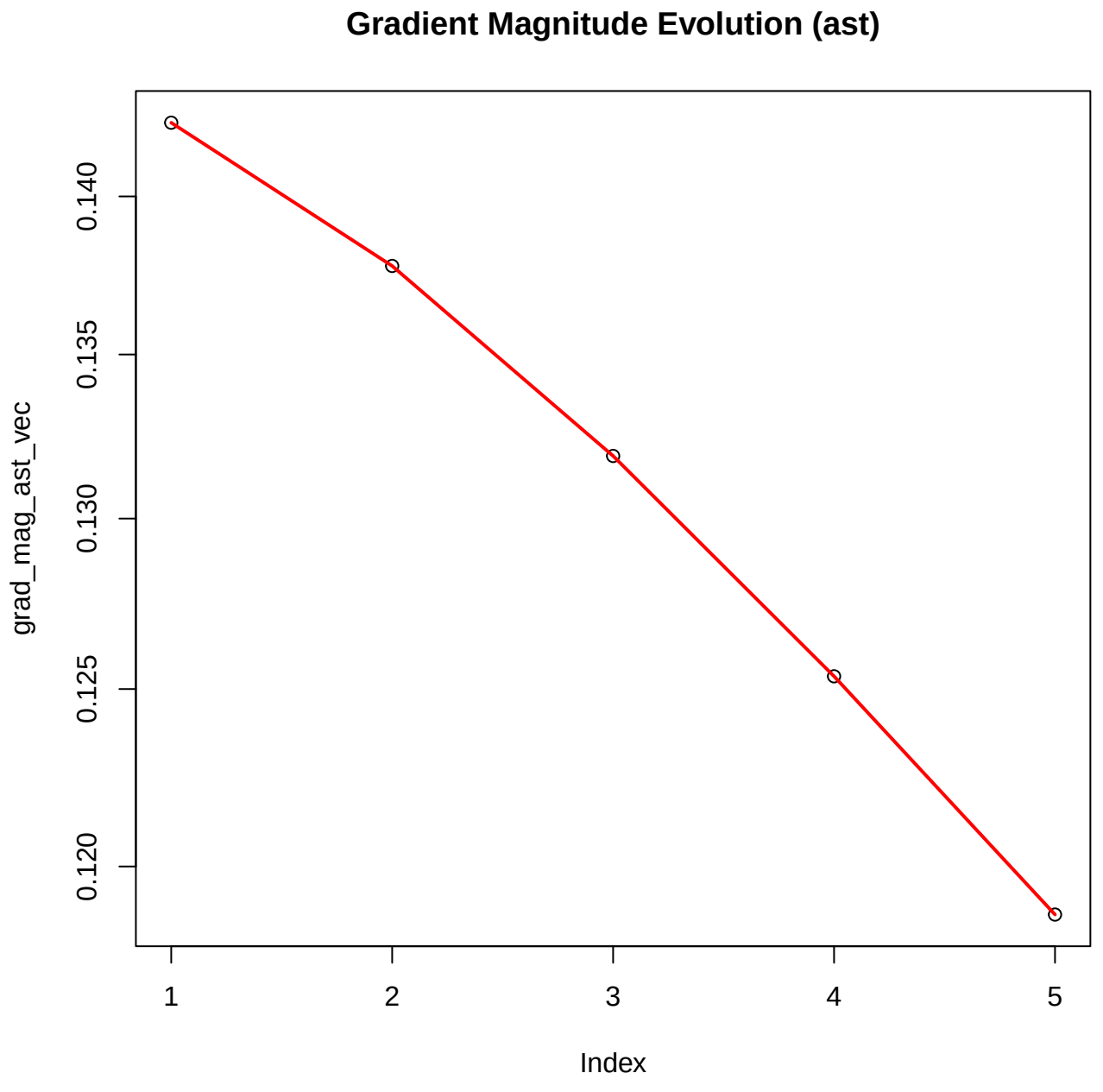
Gradient Magnitude Evolution (ast)
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0.140
1
2
3
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5
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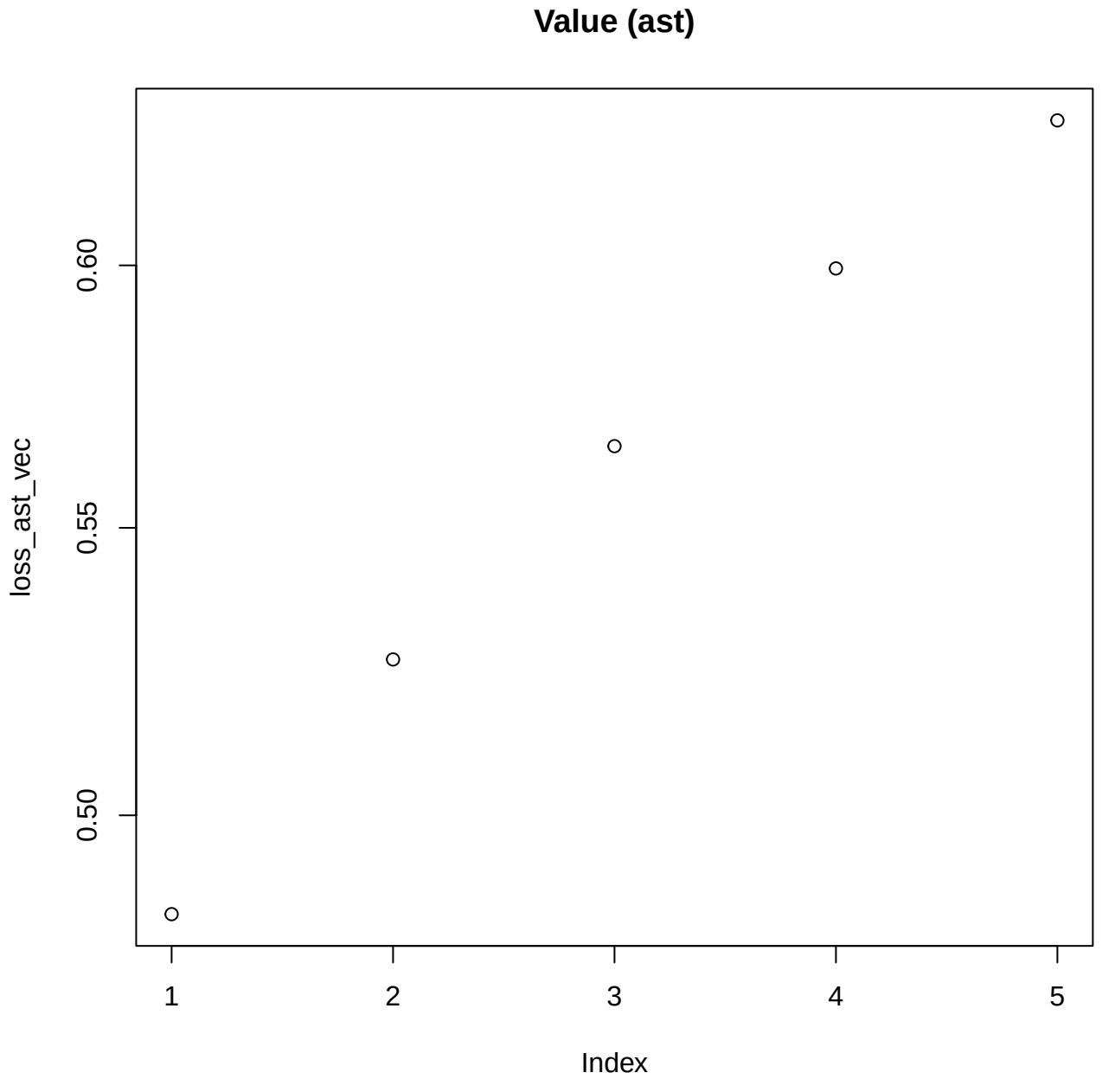
Value (ast)
loss_ast_vec
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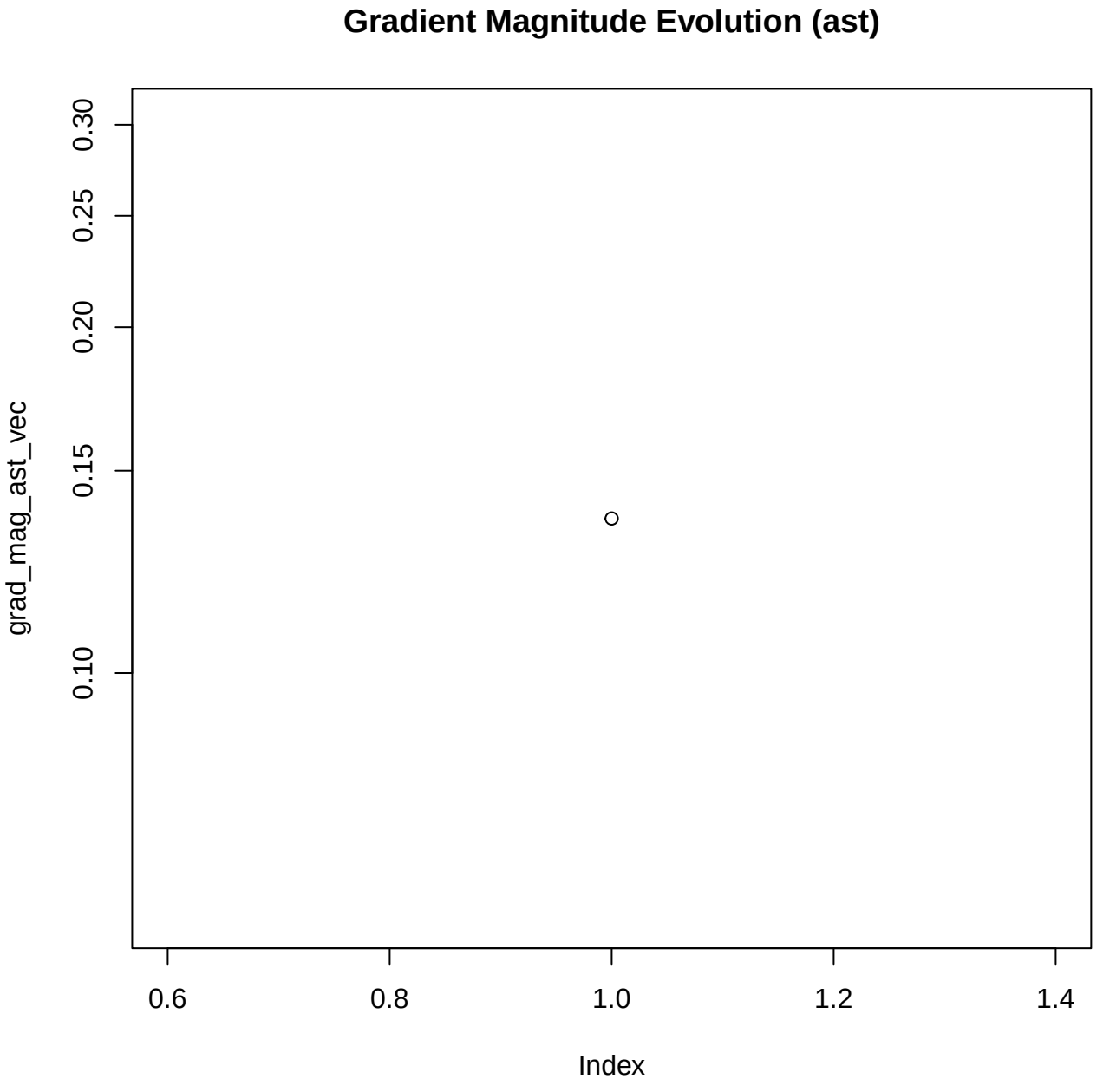
0.60
1
2
3
4
5
Index


Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
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0.15
0.20
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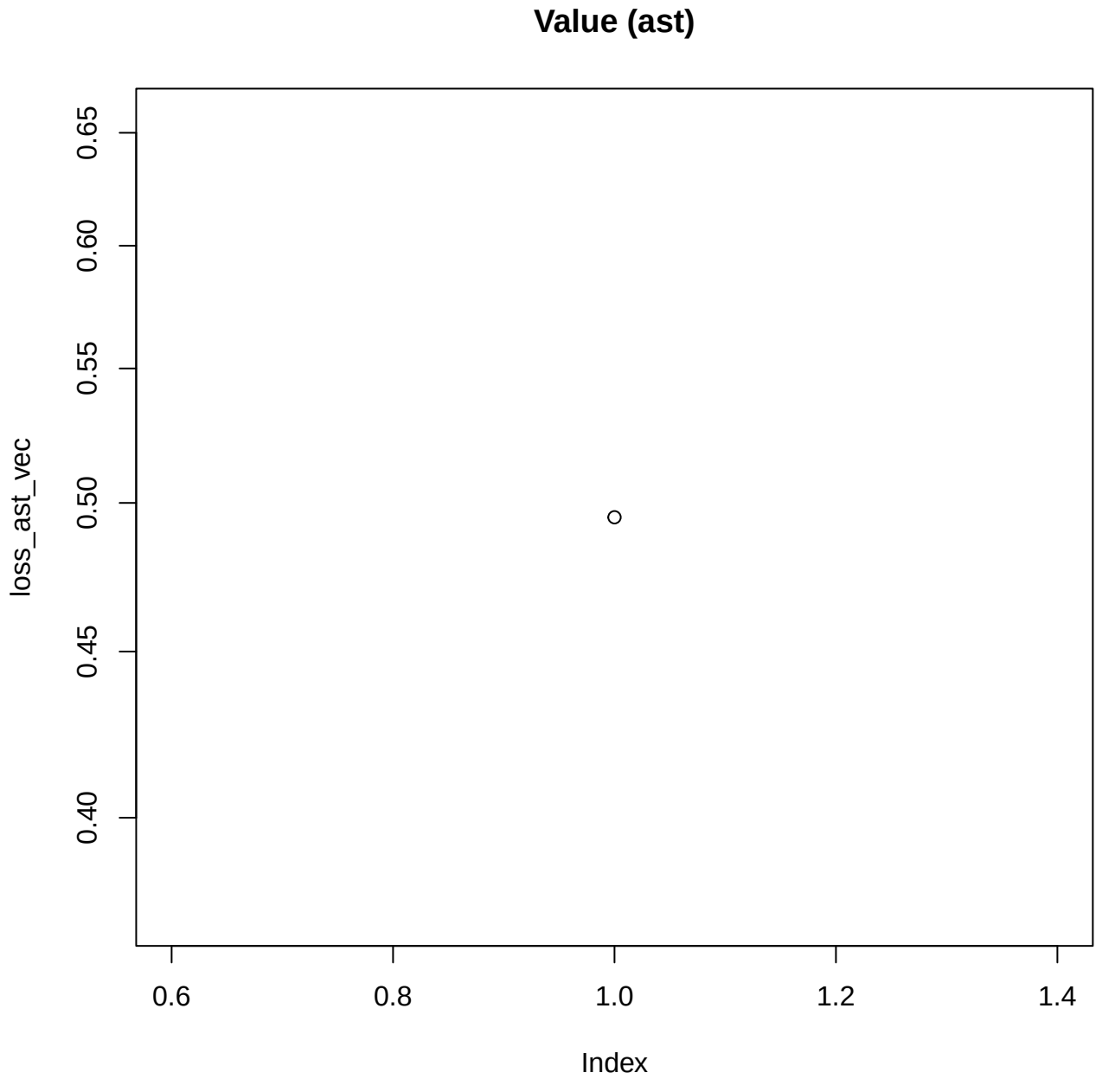
0.30
0.6
0.8
1.0
1.2
1.4
Index


Value (ast)
loss_ast_vec
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0.55
0.50
0.45
0.40
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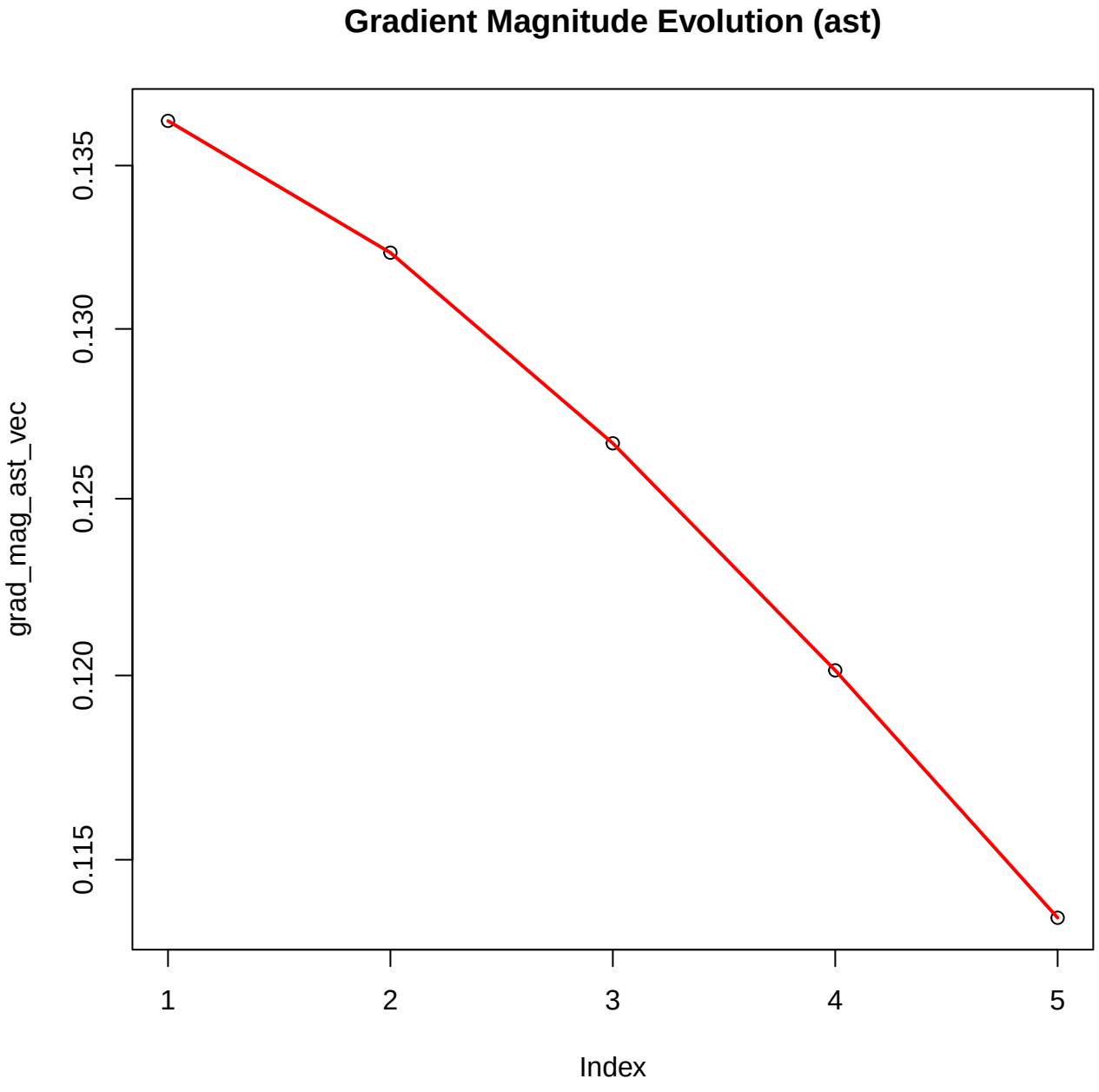
0.8
1.0
1.2
1.4
Index


Gradient Magnitude Evolution (ast)
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0.120
0.125
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0.135
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5
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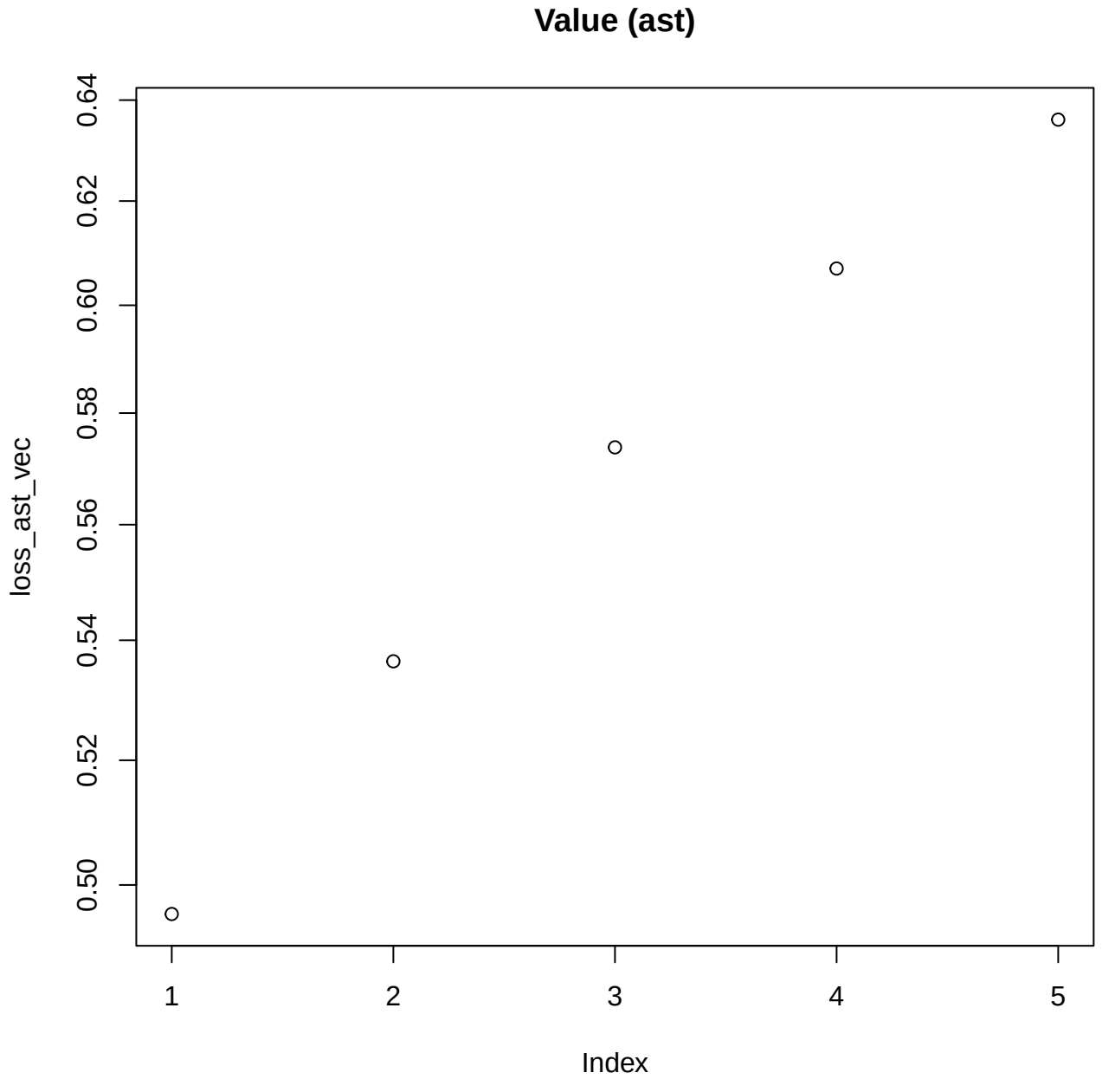
Value (ast)
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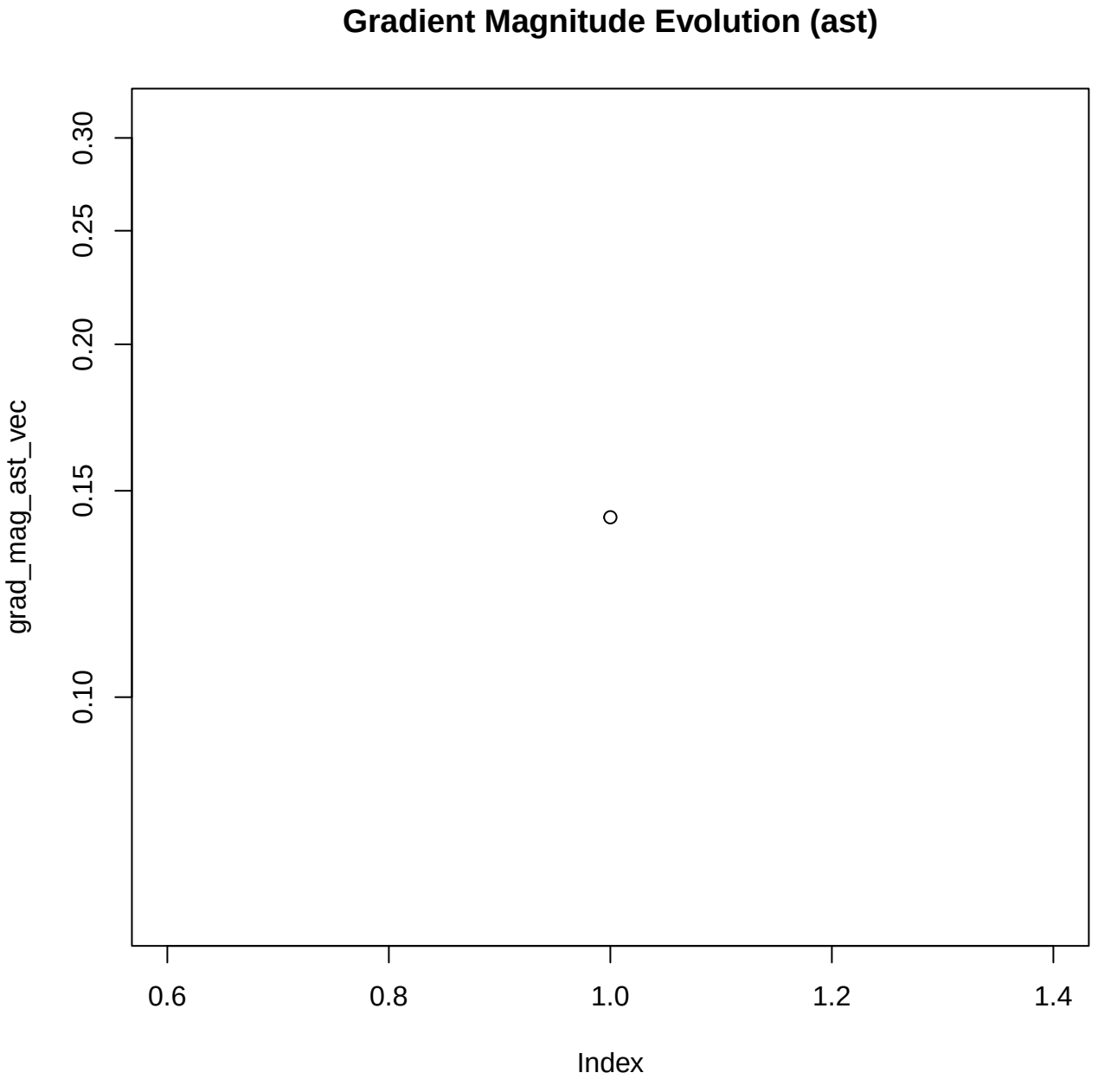
Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
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0.8
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1.2
1.4
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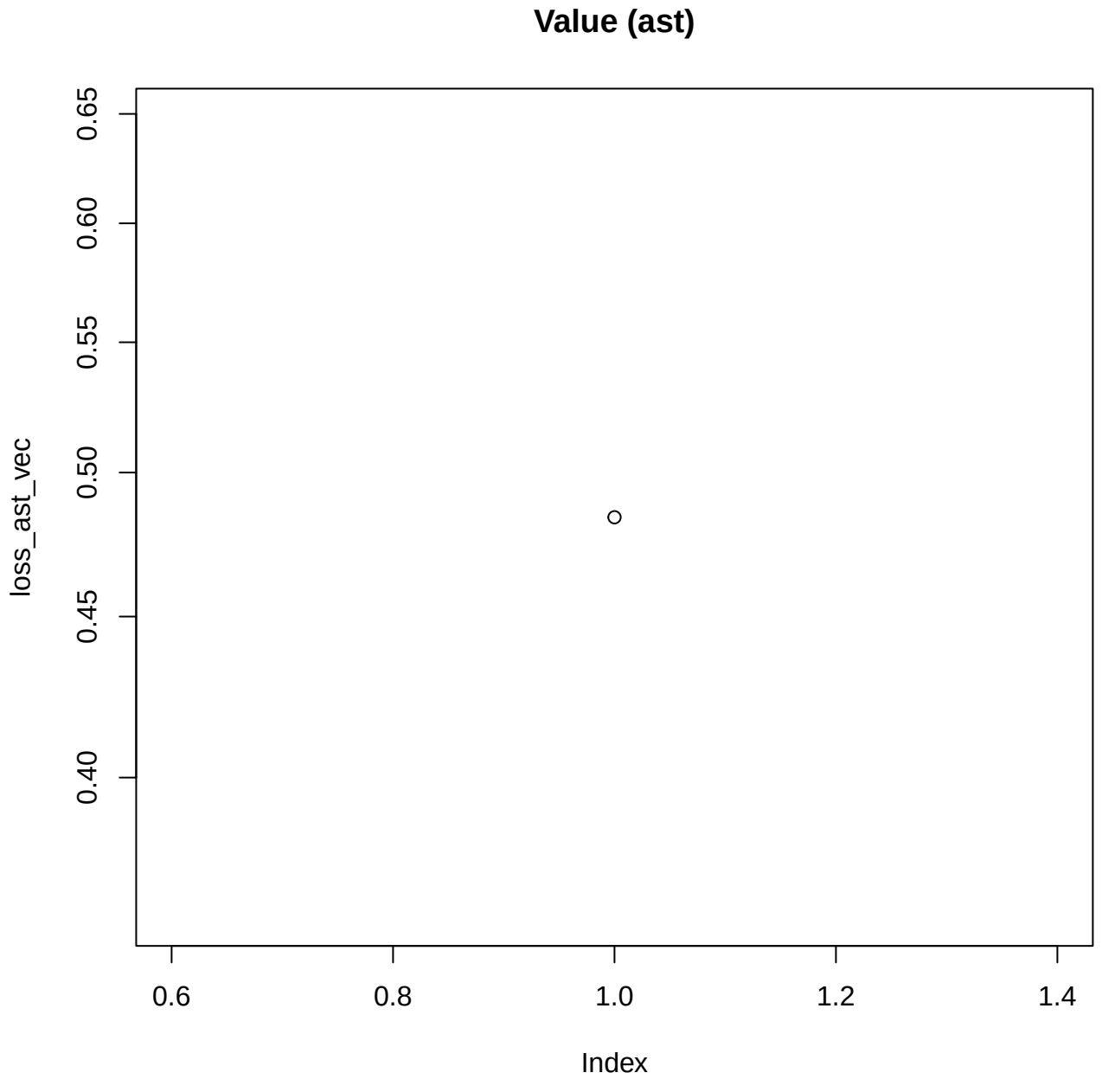
Value (ast)
loss_ast_vec
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1.4
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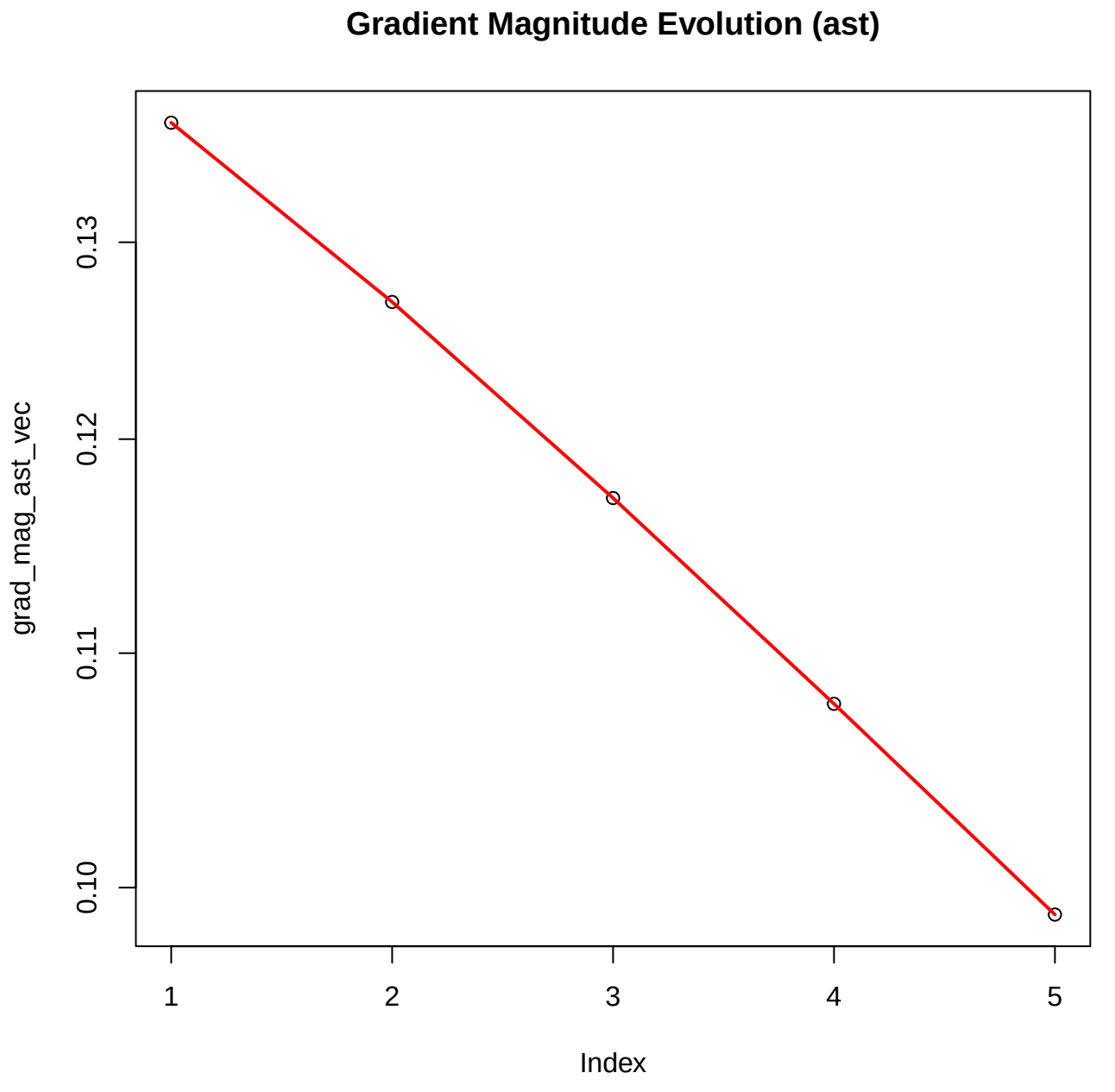
Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
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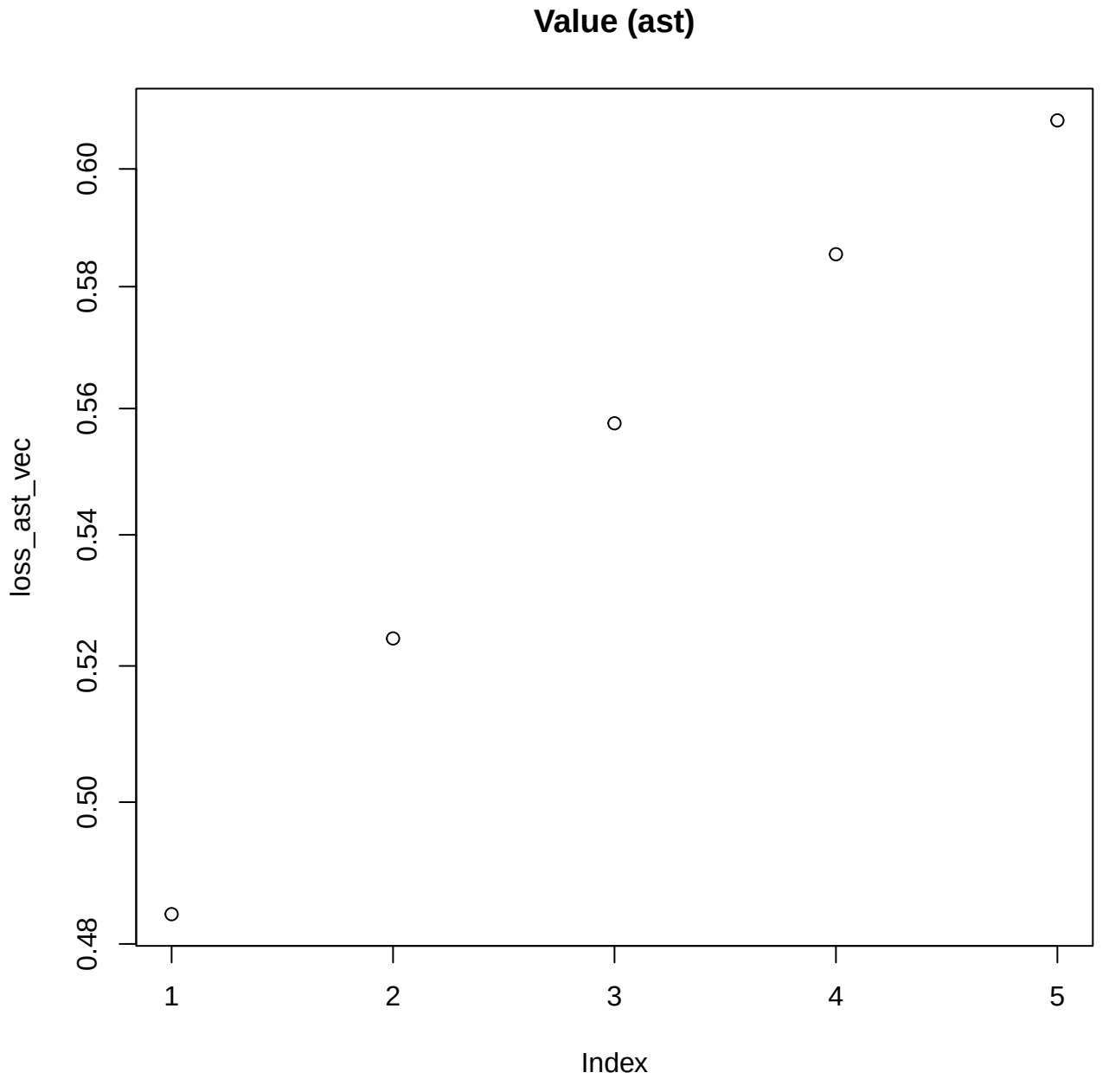
Value (ast)
loss_ast_vec
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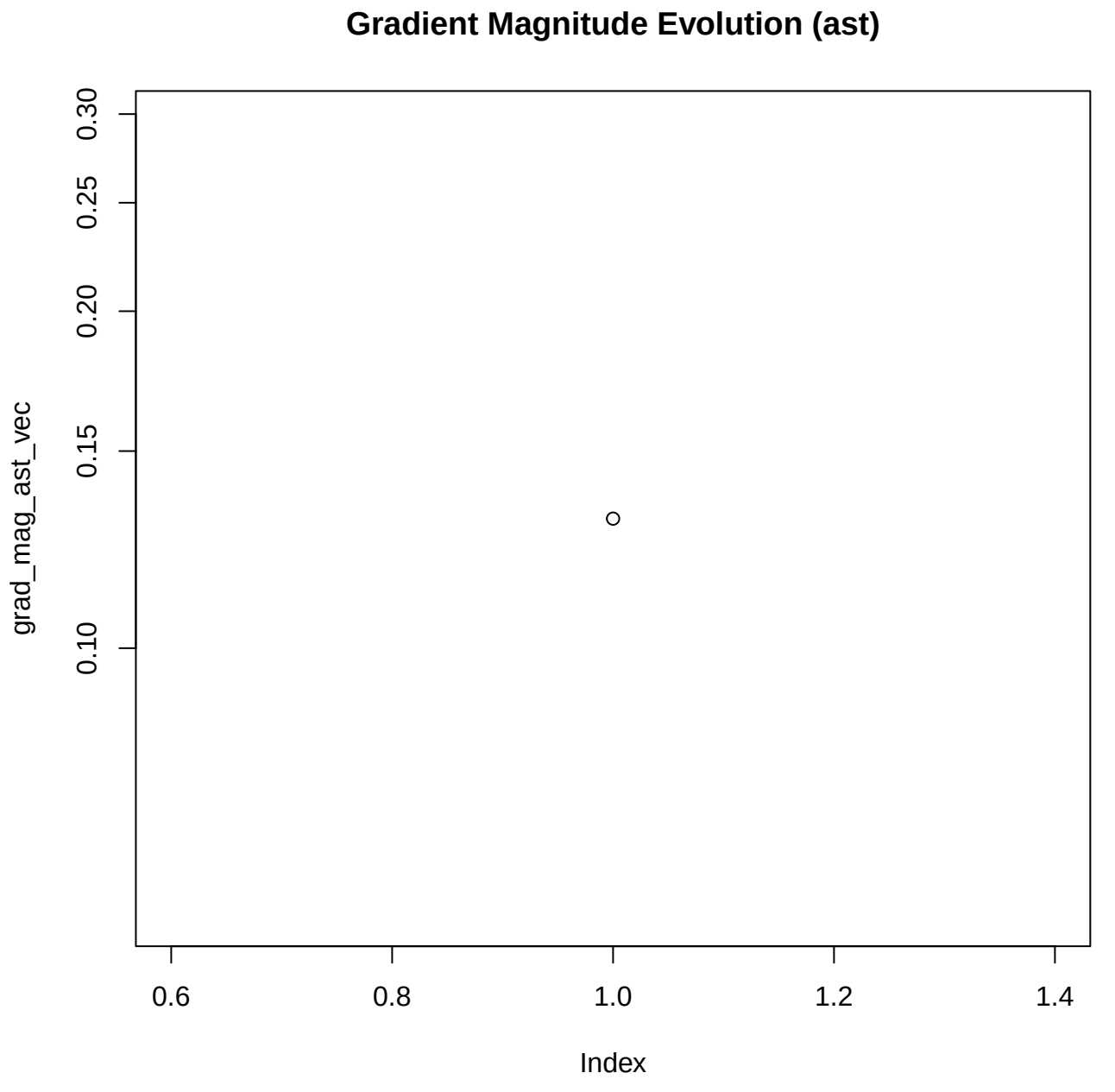
Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
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0.20
0.25
0.30
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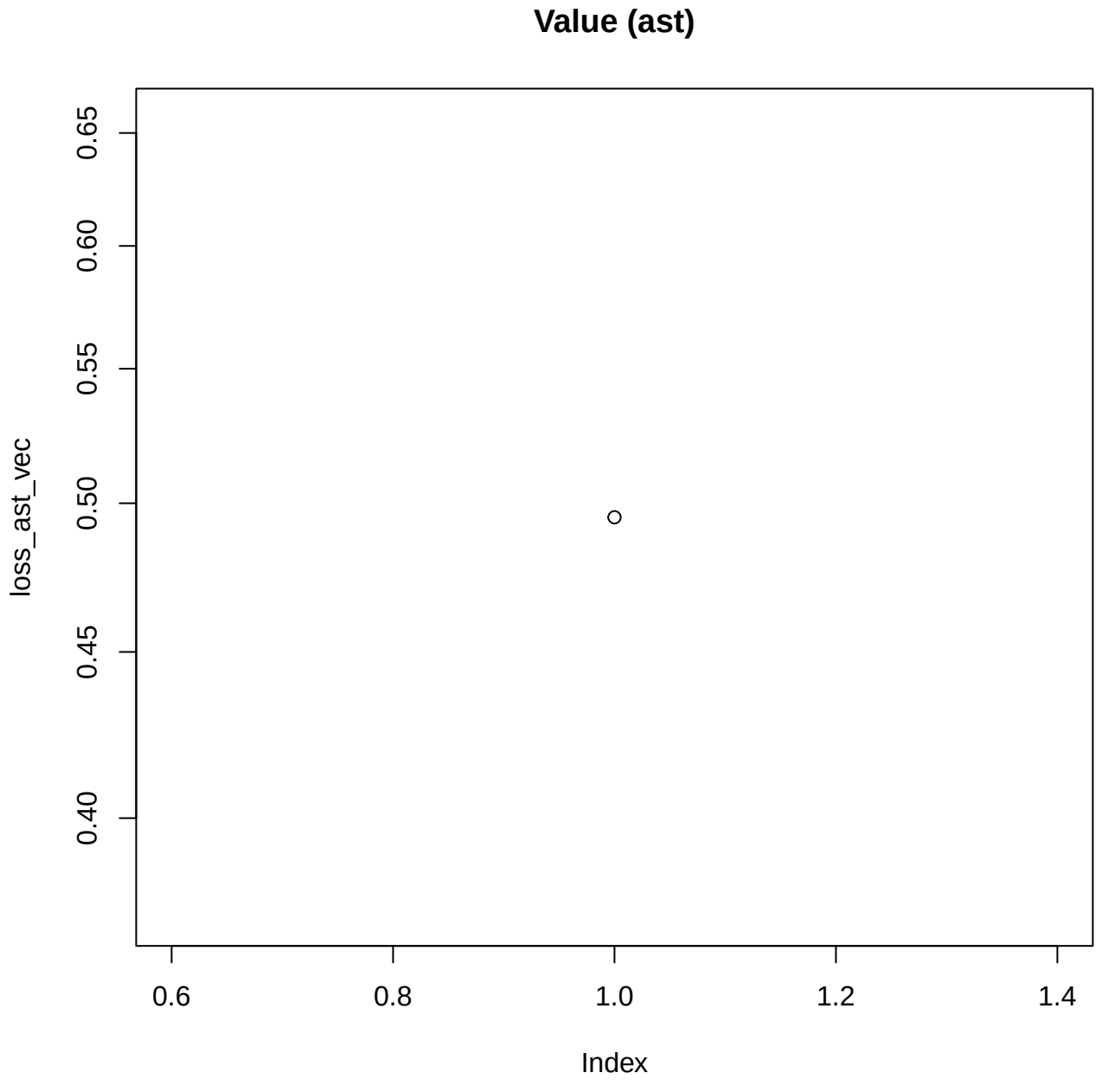
0.8
1.0
1.2
1.4
Index


Value (ast)
loss_ast_vec
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0.50
0.55
0.60
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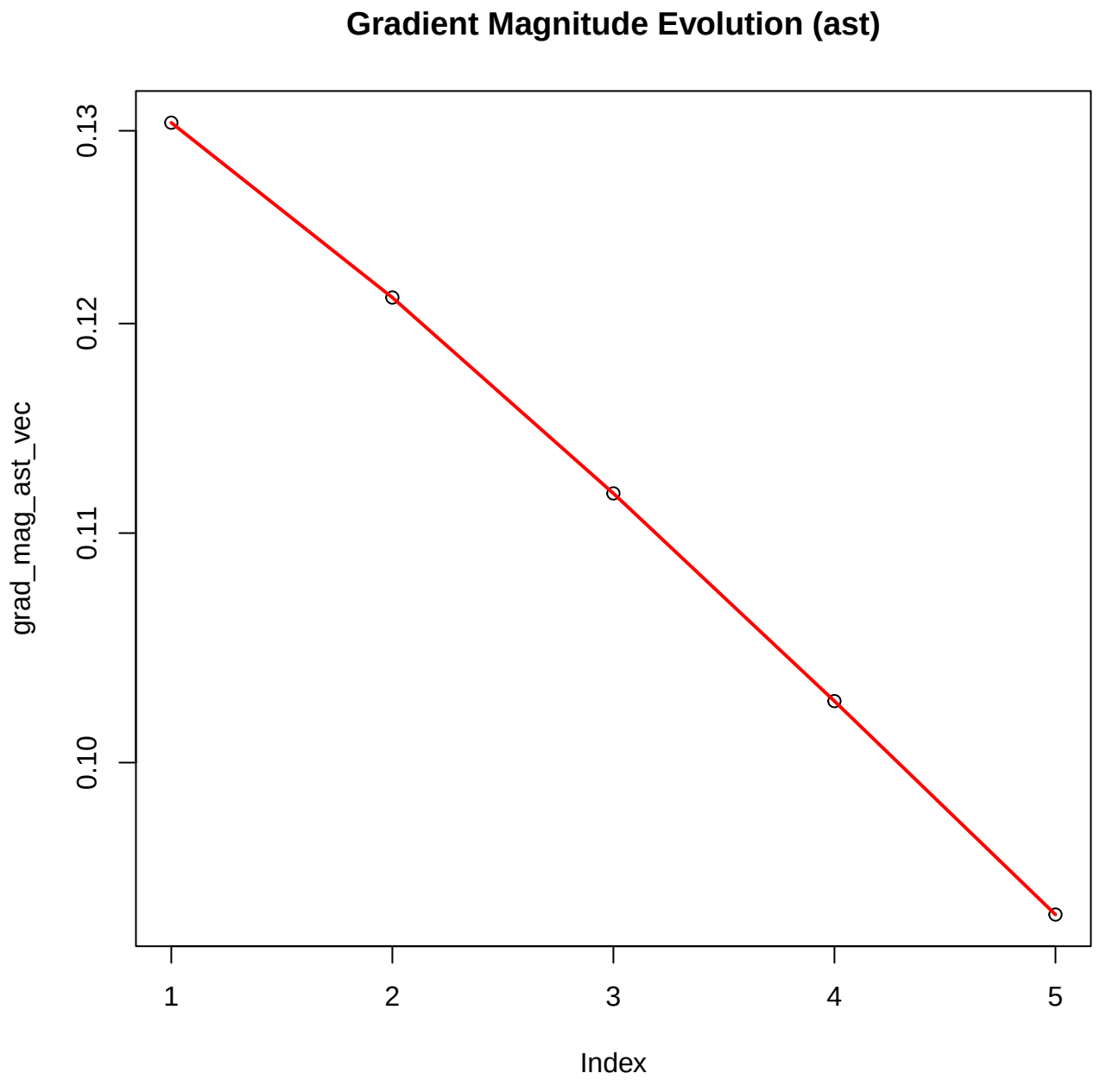
0.8
1.0
1.2
1.4
Index


Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
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0.11
0.12
0.13
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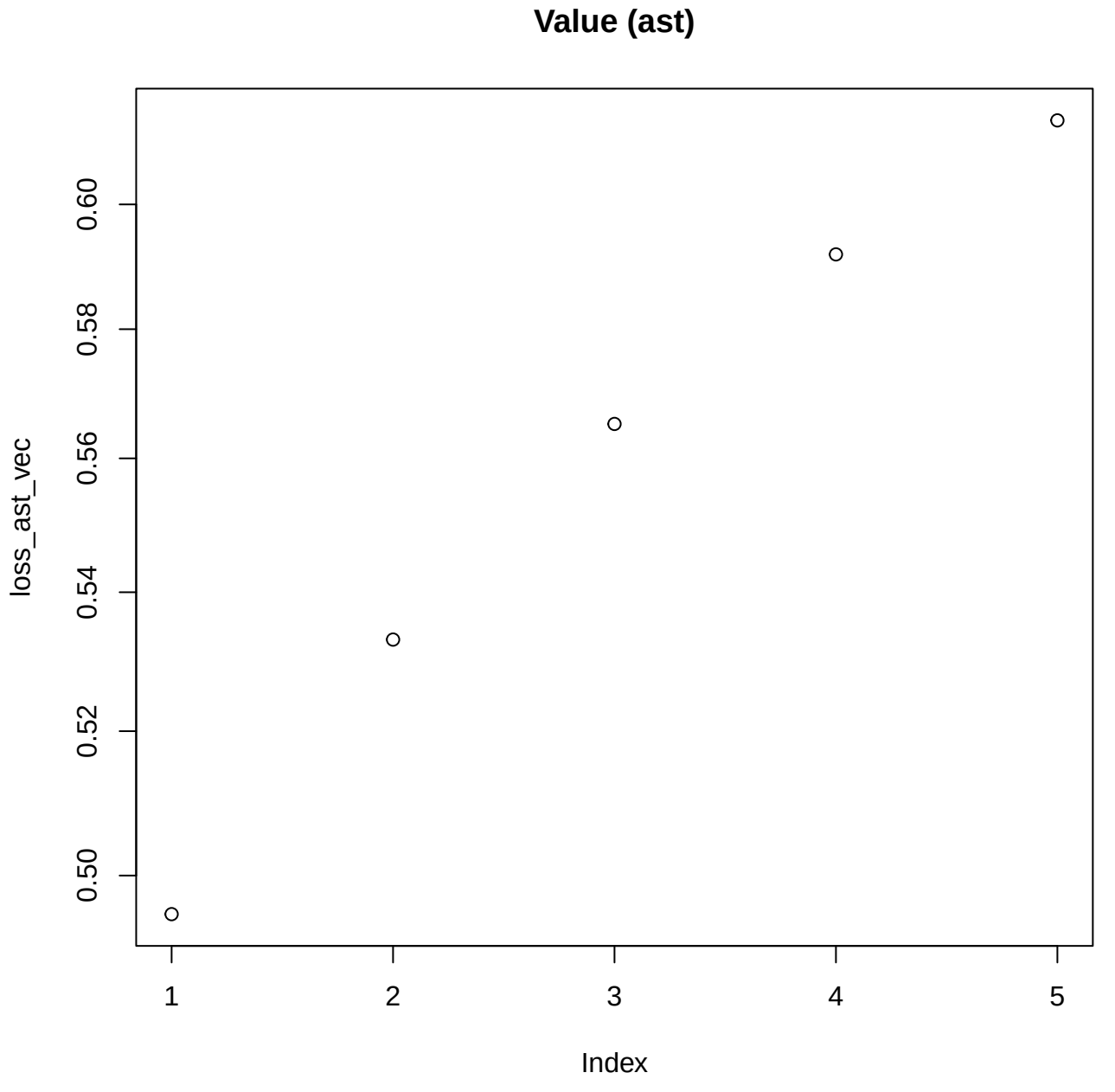
2
3
4
5
Index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loss_ast_vec
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0.54
0.56
0.58
0.60
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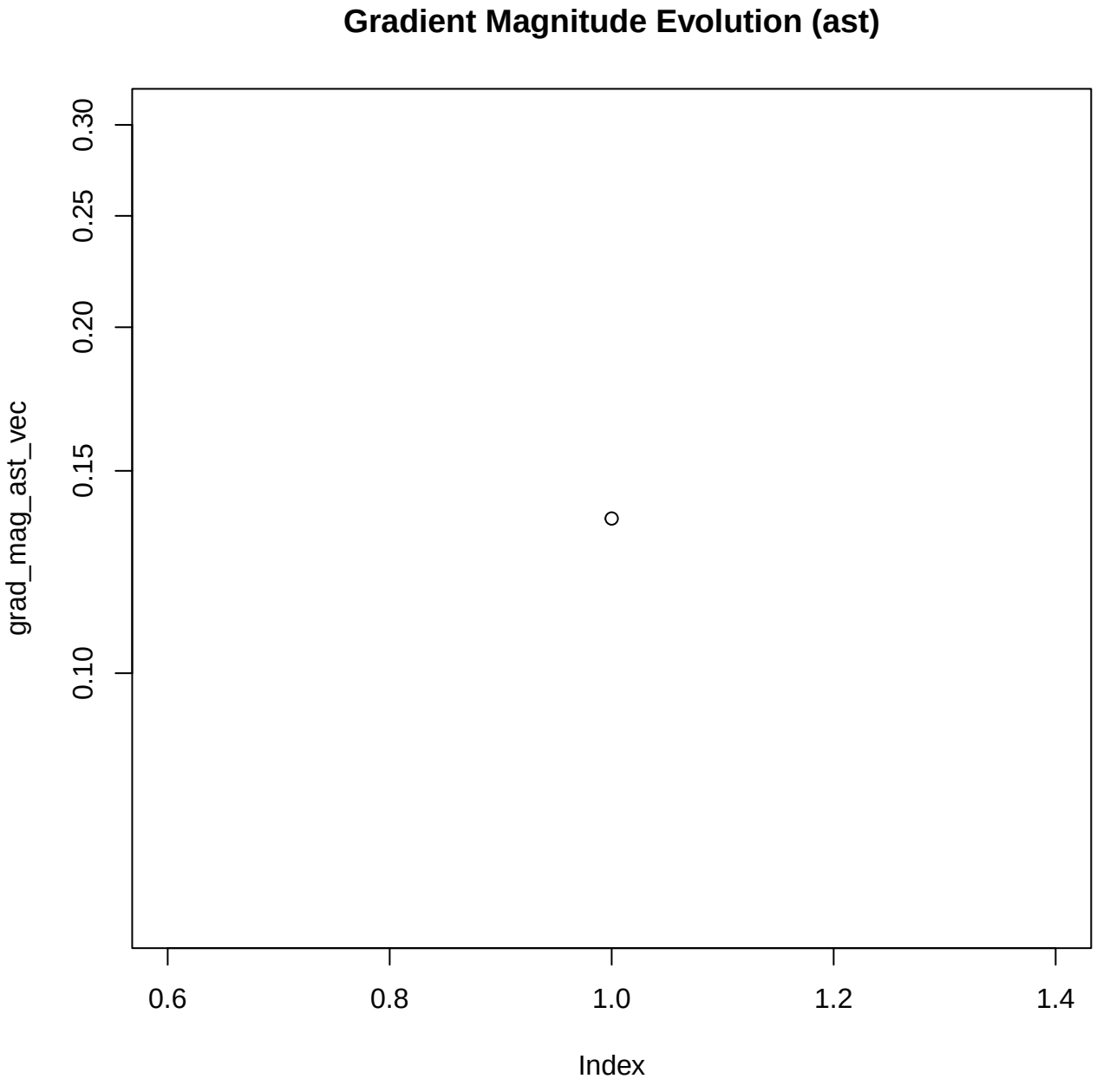
2
3
4
5
Index


Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
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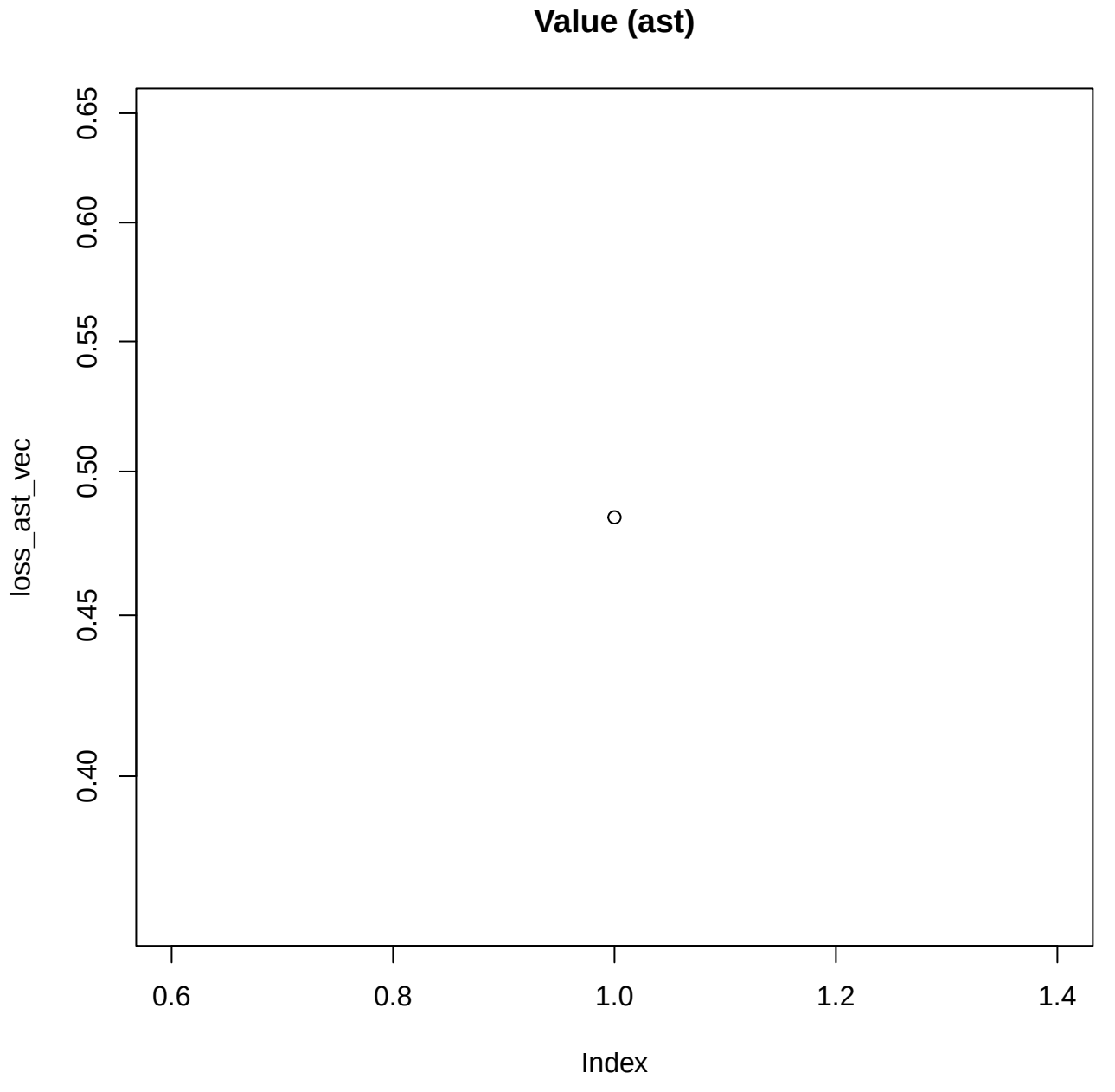
0.20
0.25
0.30
0.6
0.8
1.0
1.2
1.4
Index


Value (ast)
loss_ast_vec
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0.8
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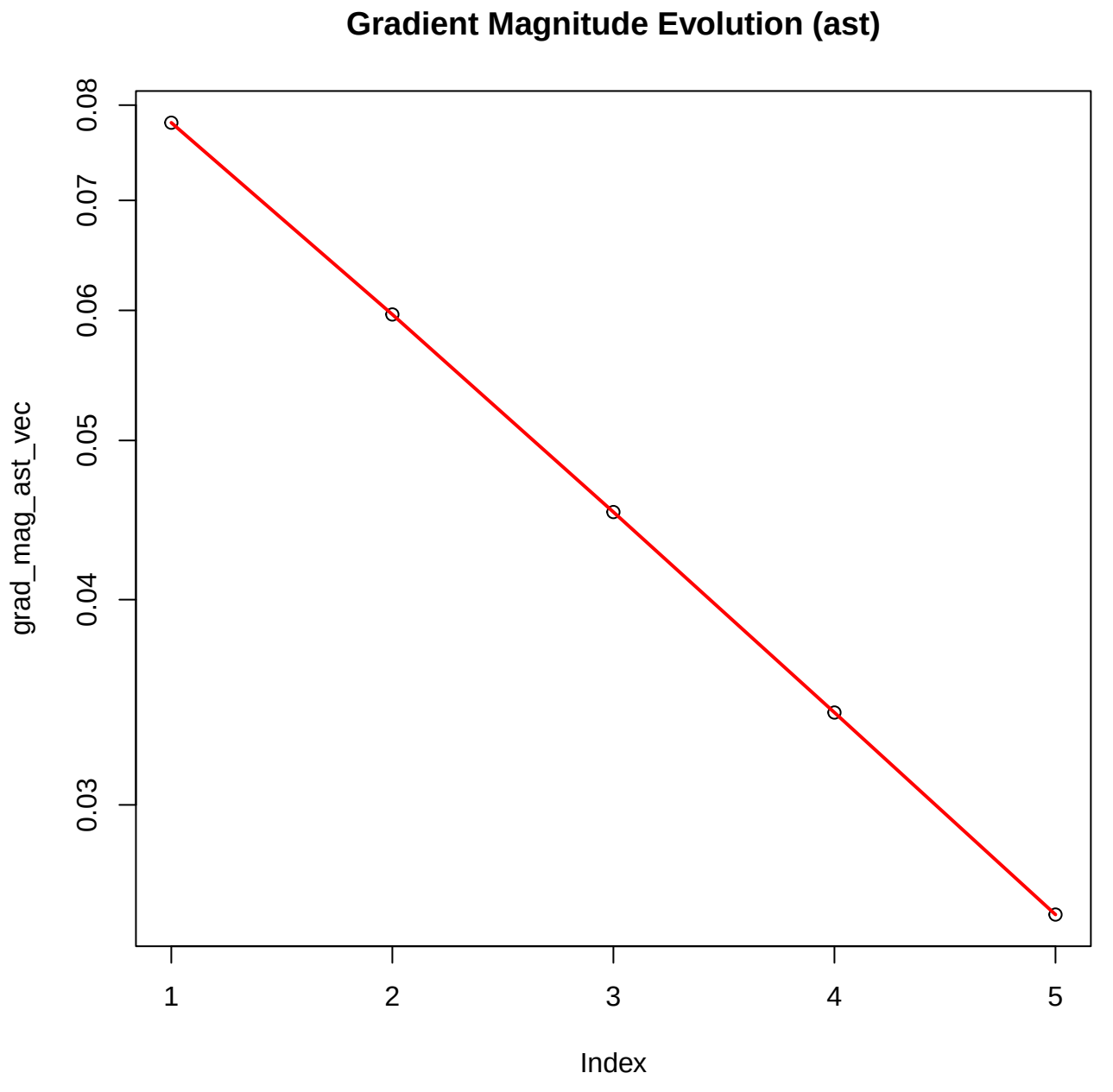
1.2
1.4
Index


Gradient Magnitude Evolution (ast)
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5
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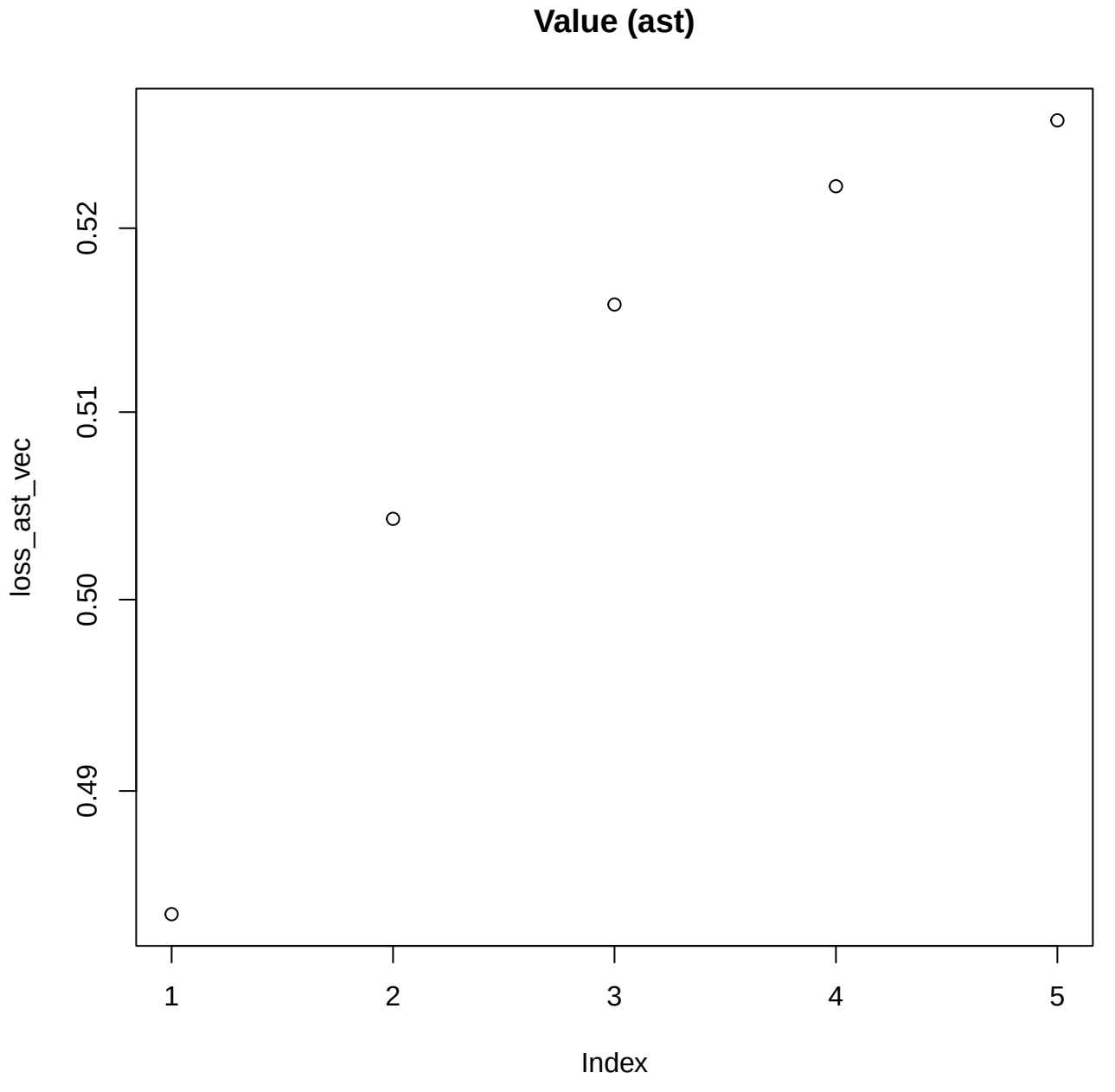
Value (ast)
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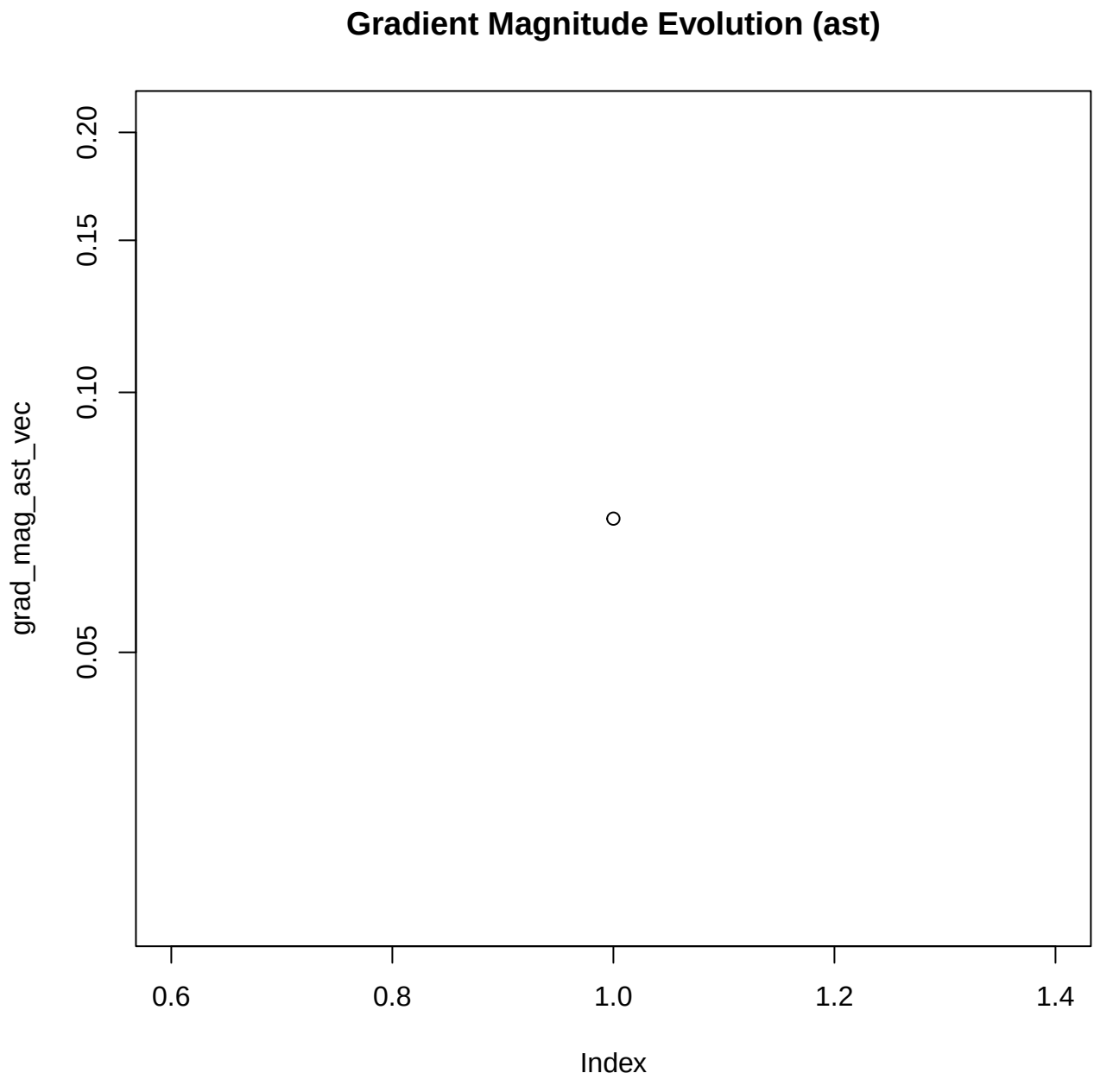
5
Index


Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
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0.10
0.15
0.20
0.6
0.8
1.0
1.2
1.4
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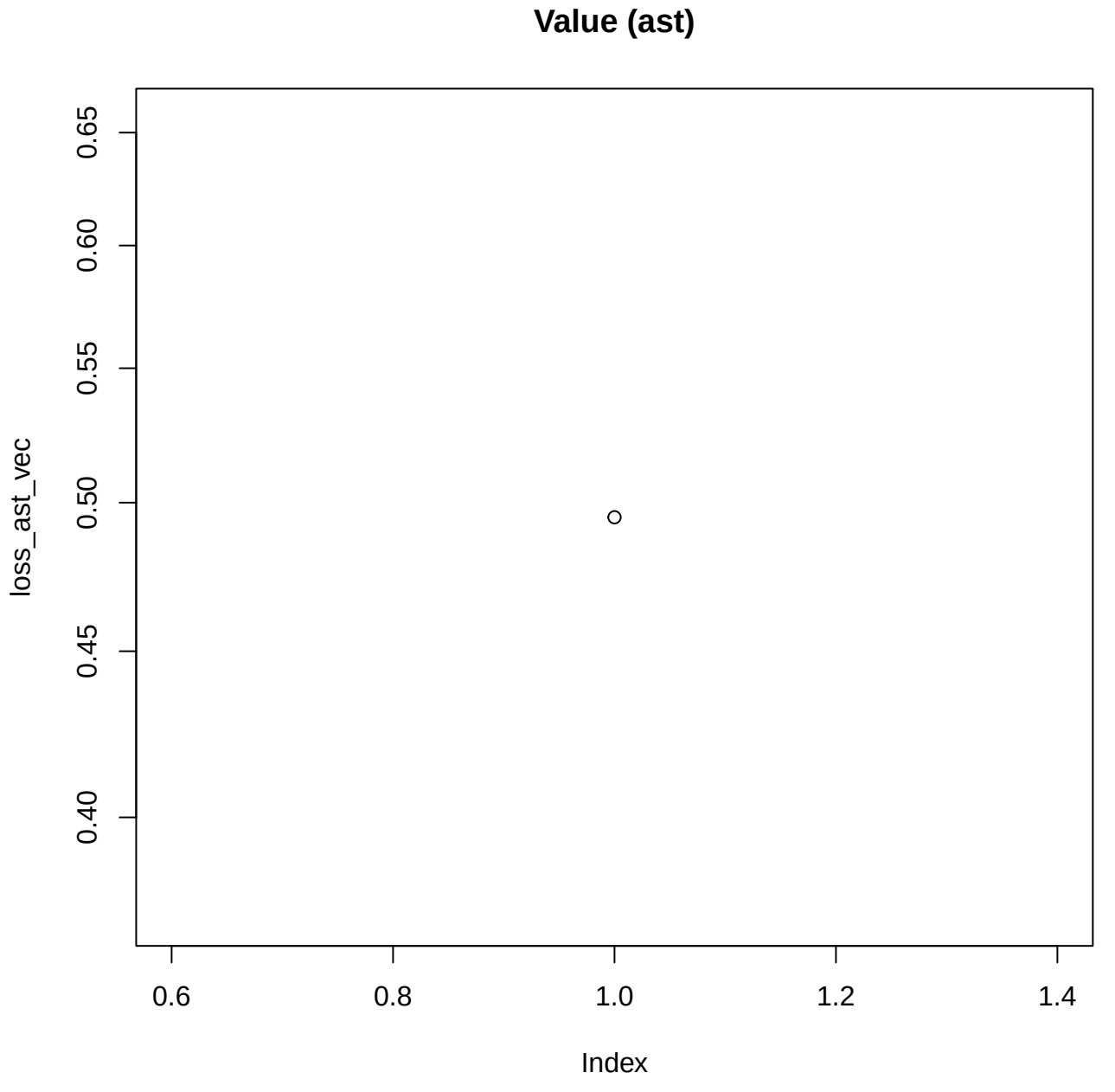
Value (ast)
loss_ast_vec
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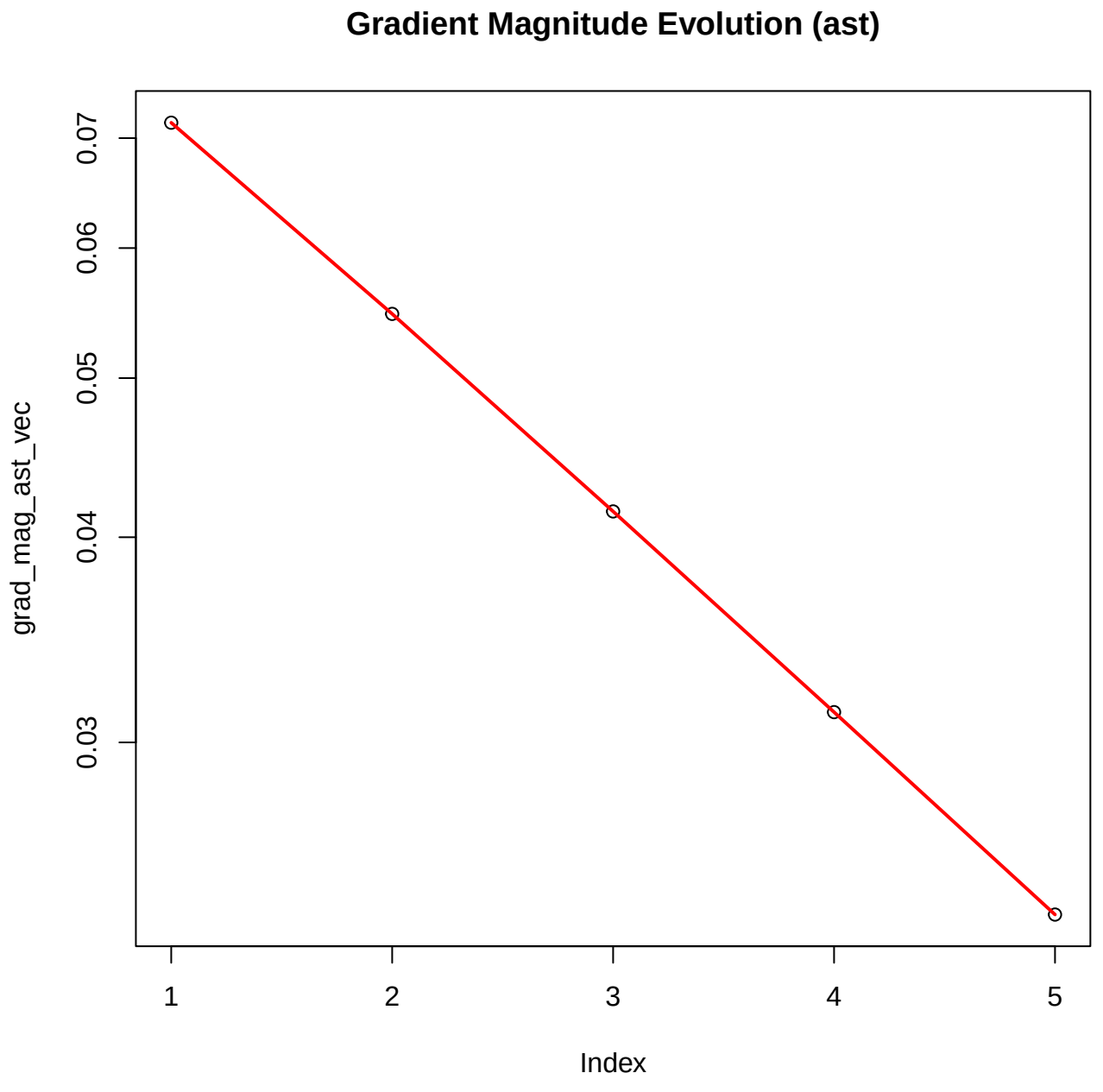
0.55
0.50
0.45
0.40
0.6
0.8
1.0
1.2
1.4
Index


Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
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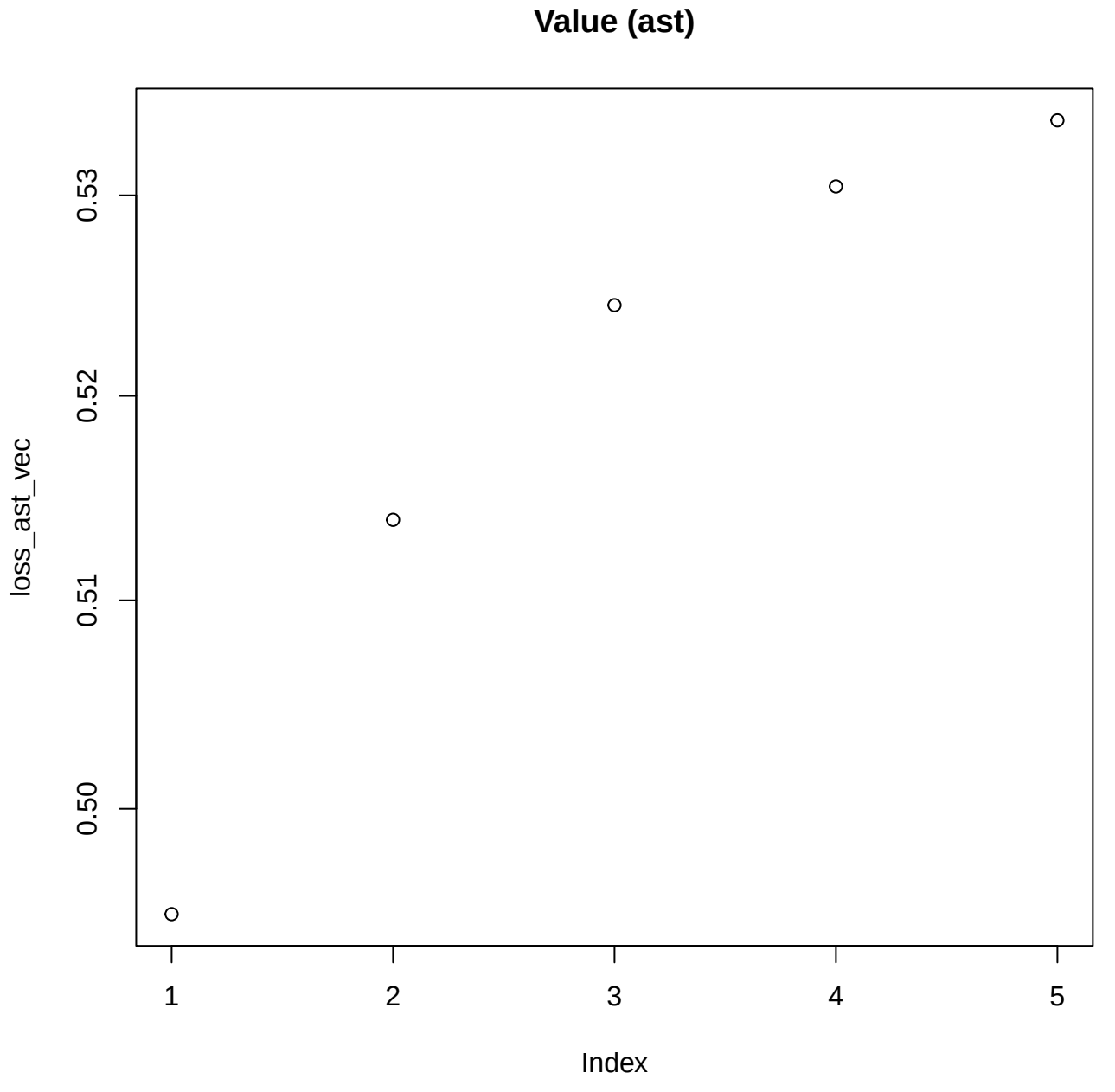
0.04
0.05
0.06
0.07
1
2
3
4
5
Index


Value (ast)
loss_ast_vec
0.50
0.51
0.52
0.53
1
2
3
4
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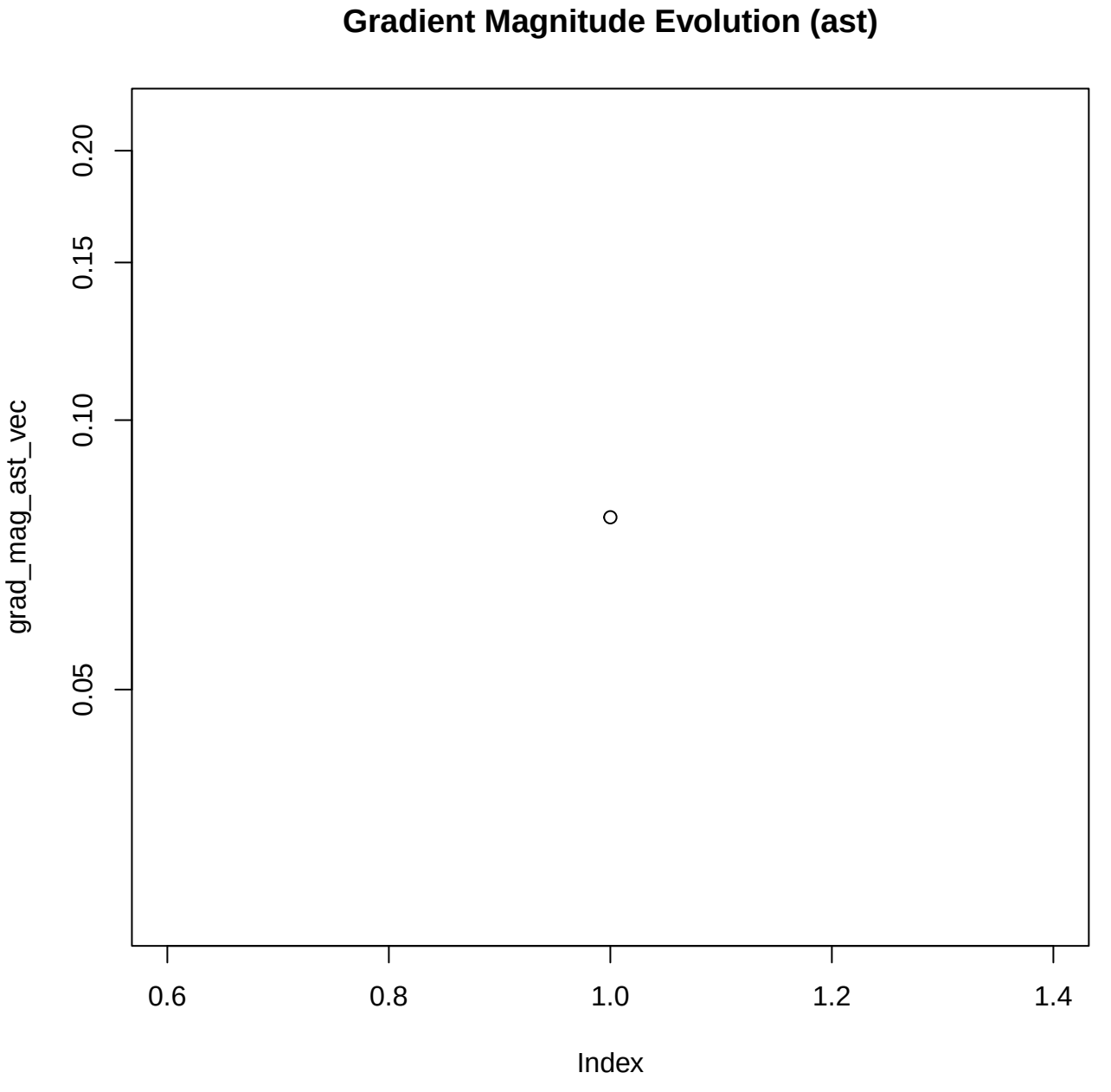
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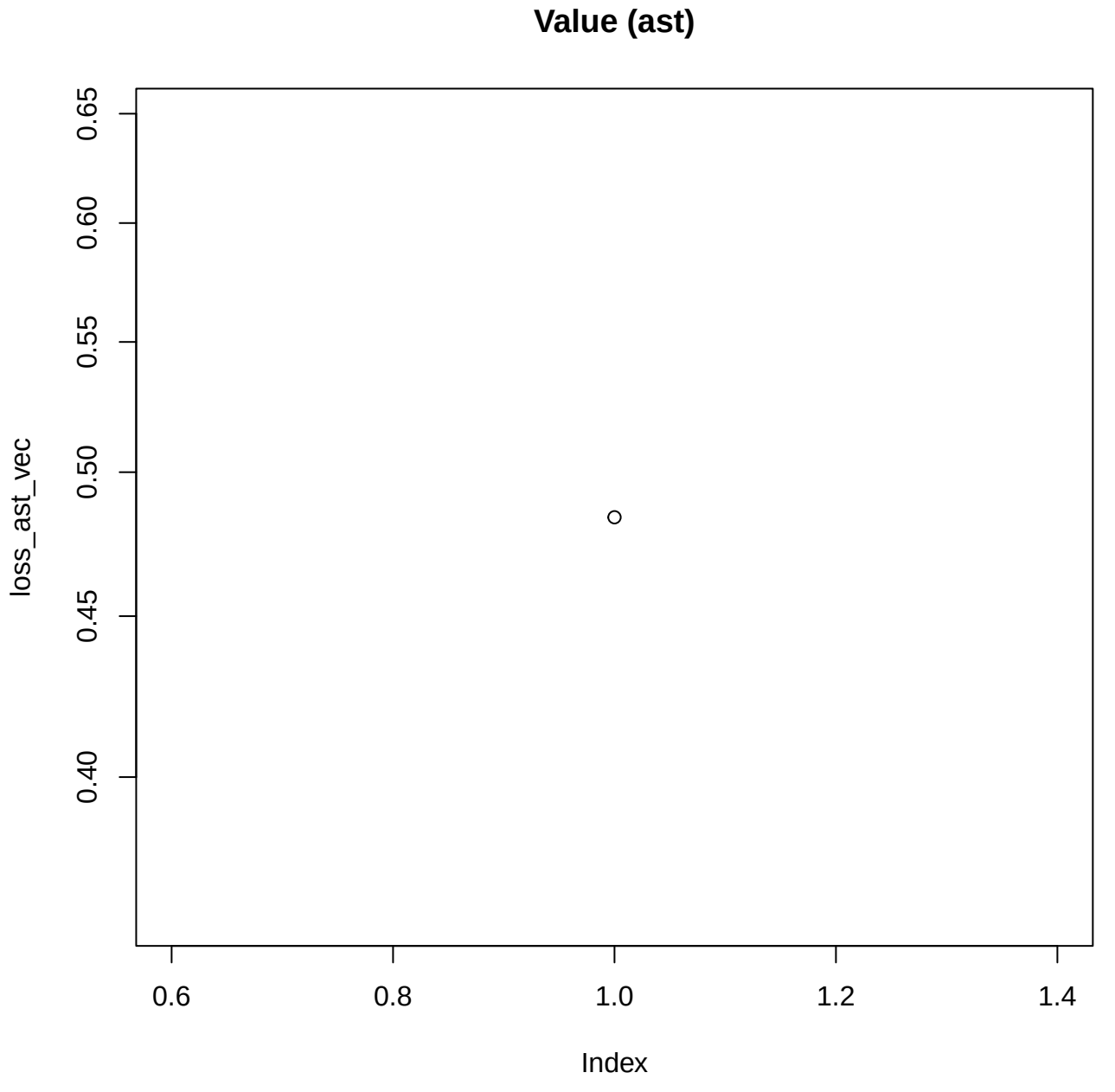
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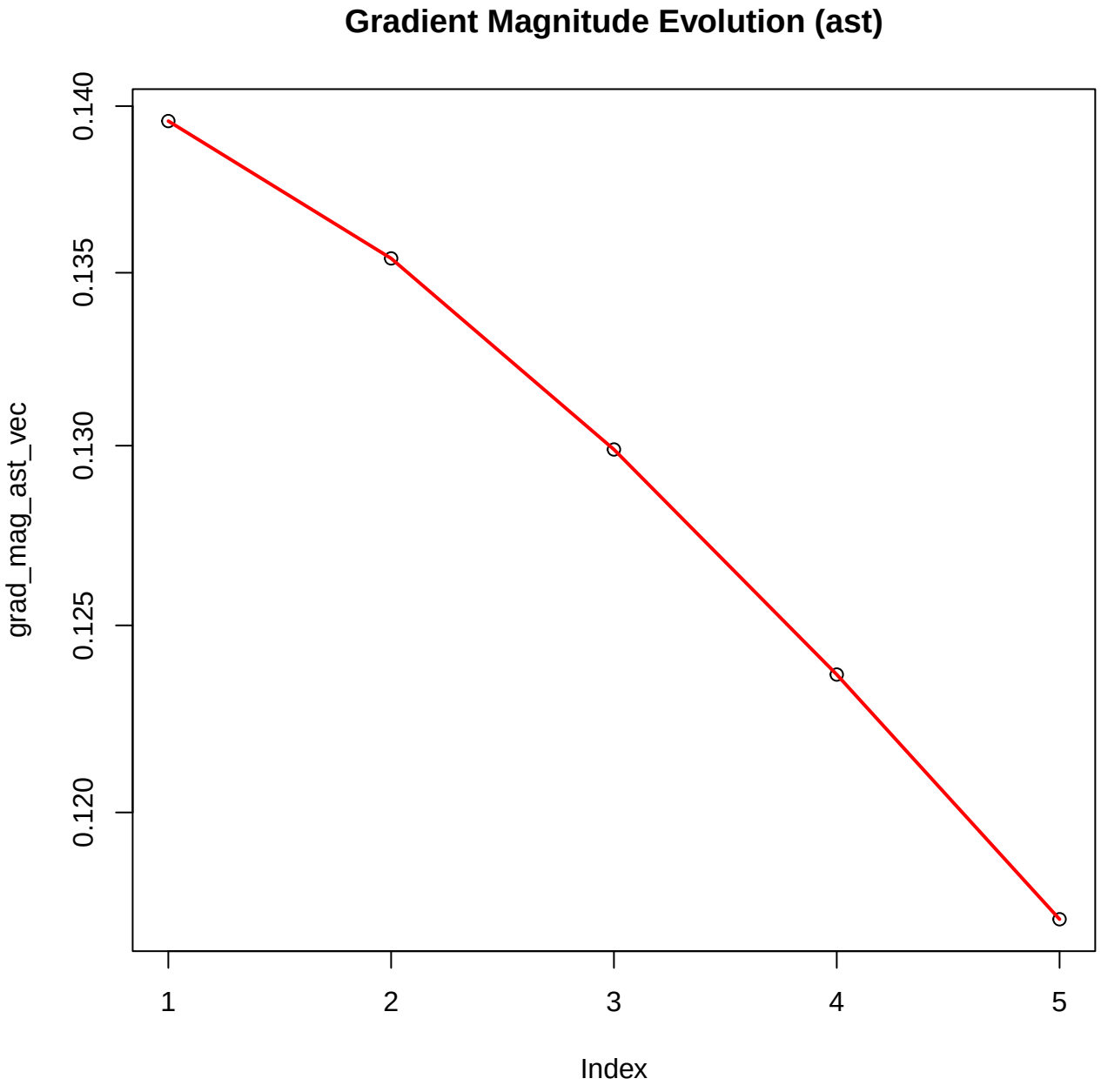
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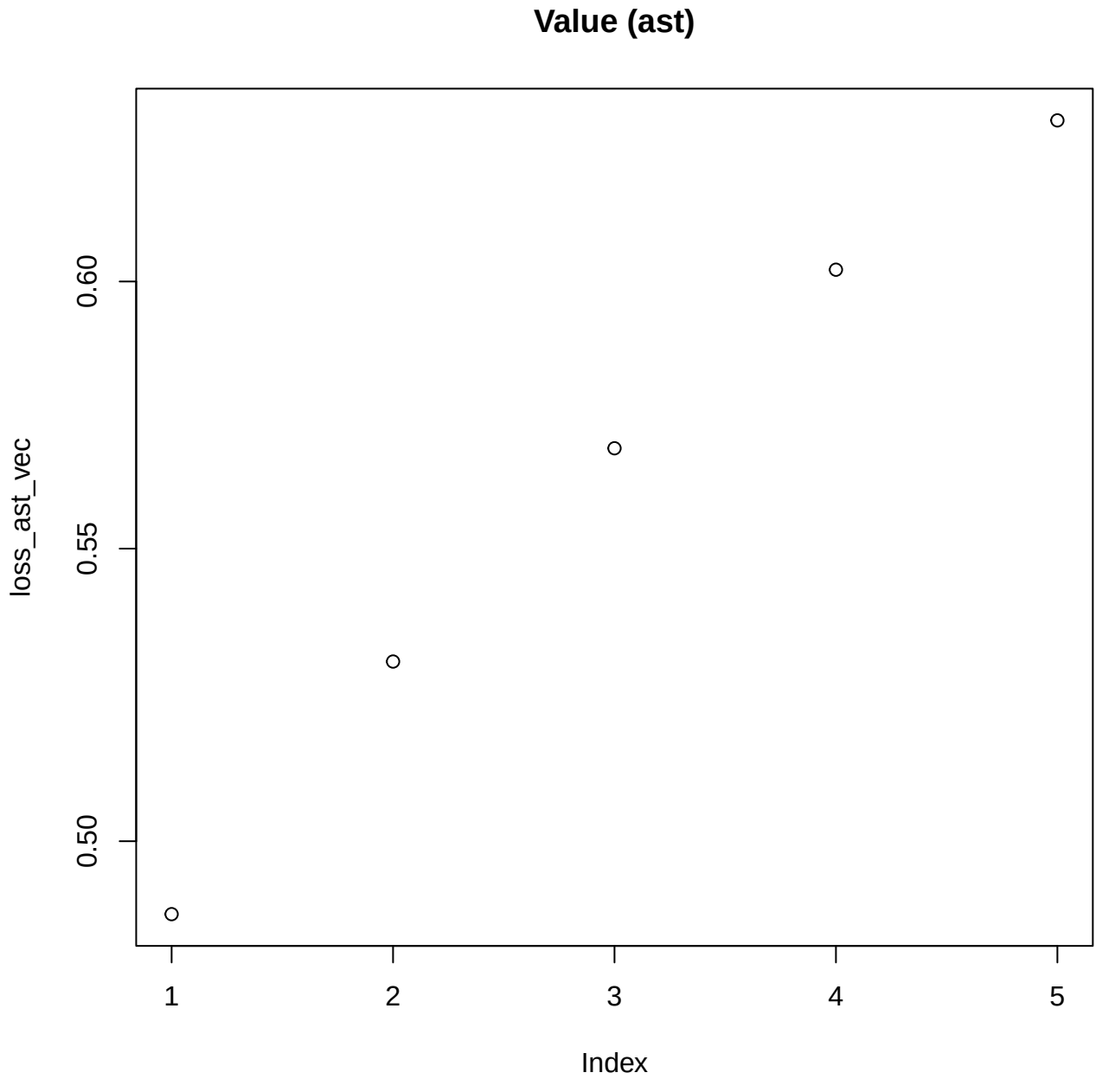
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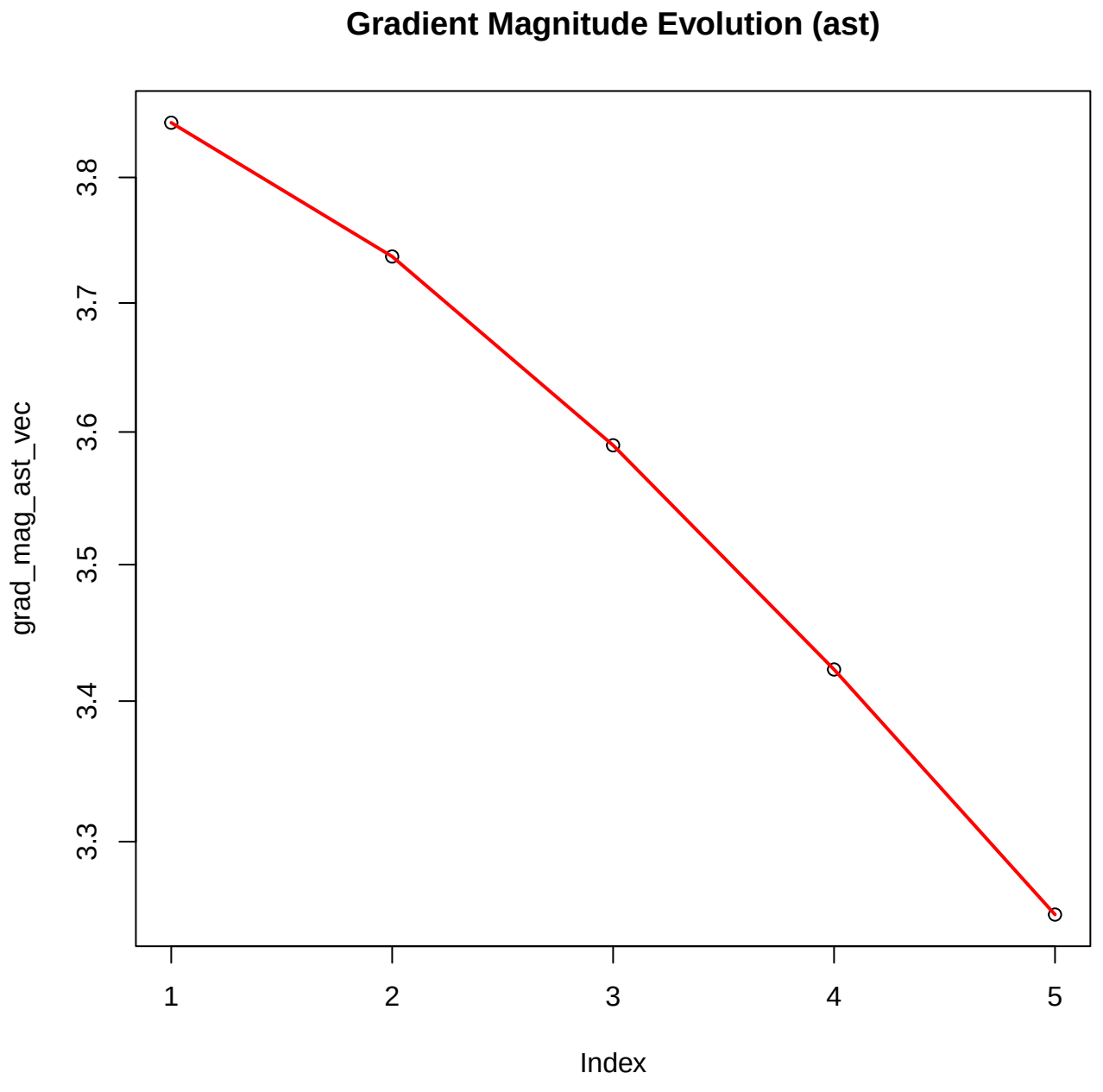
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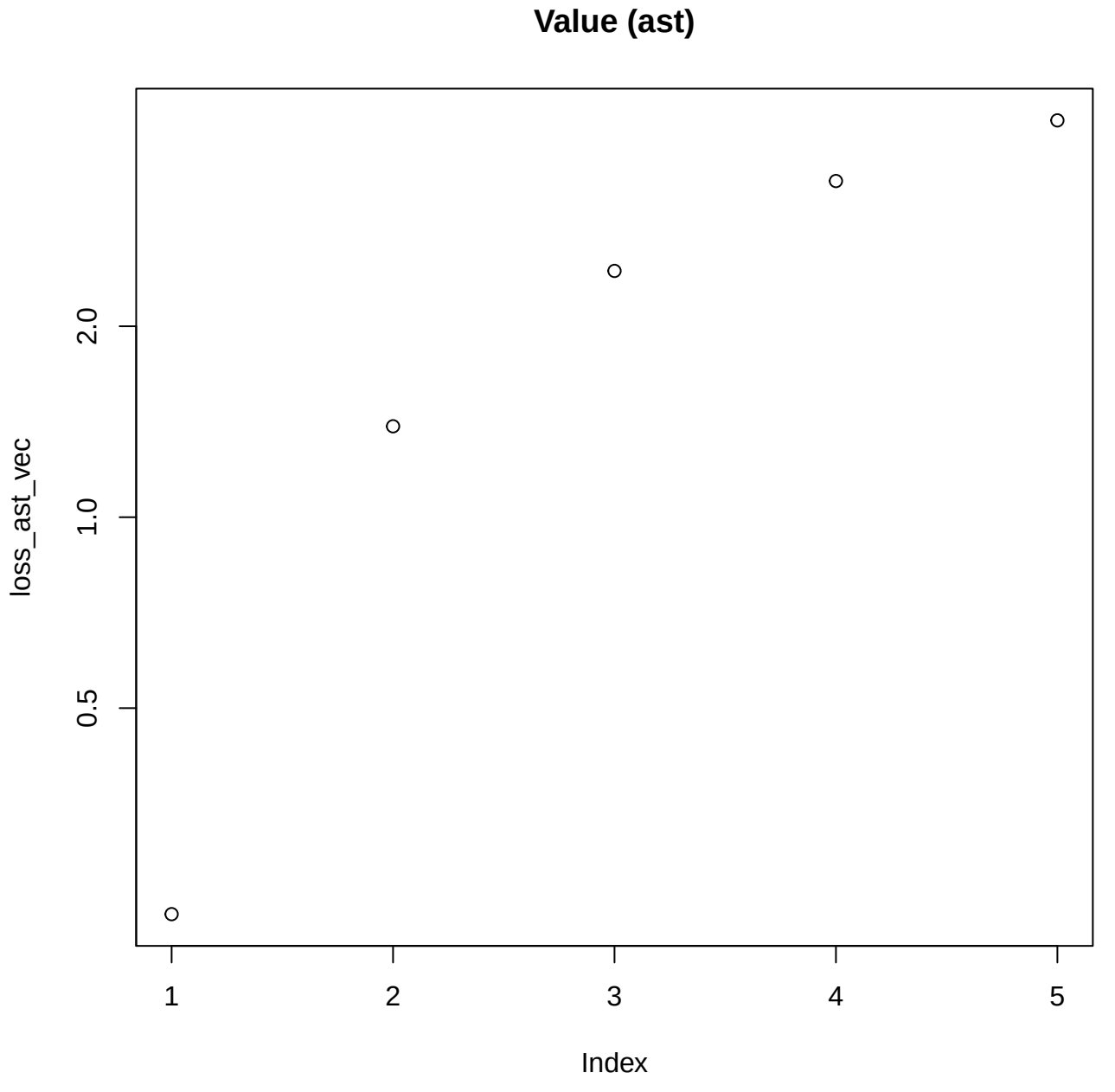
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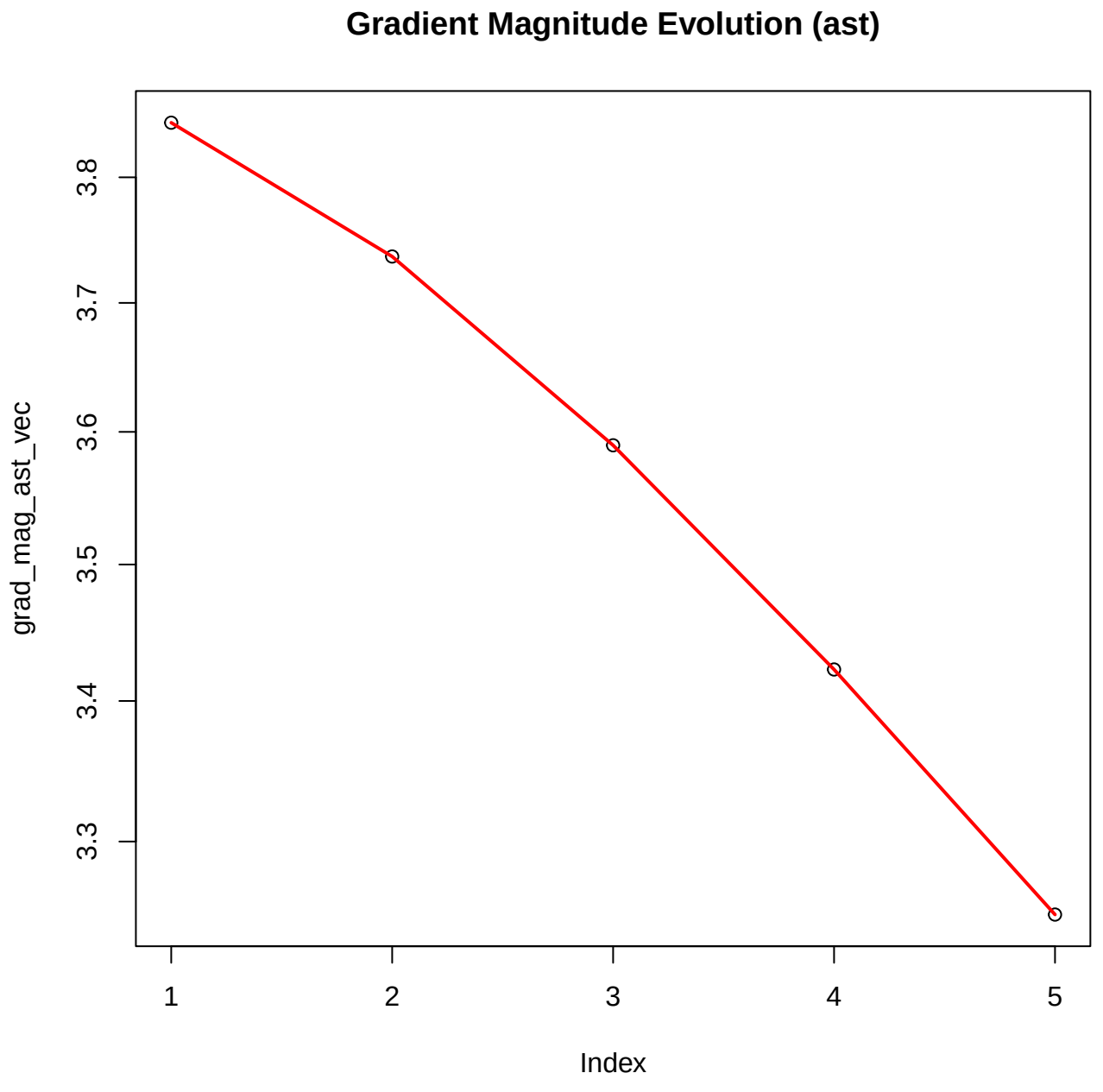
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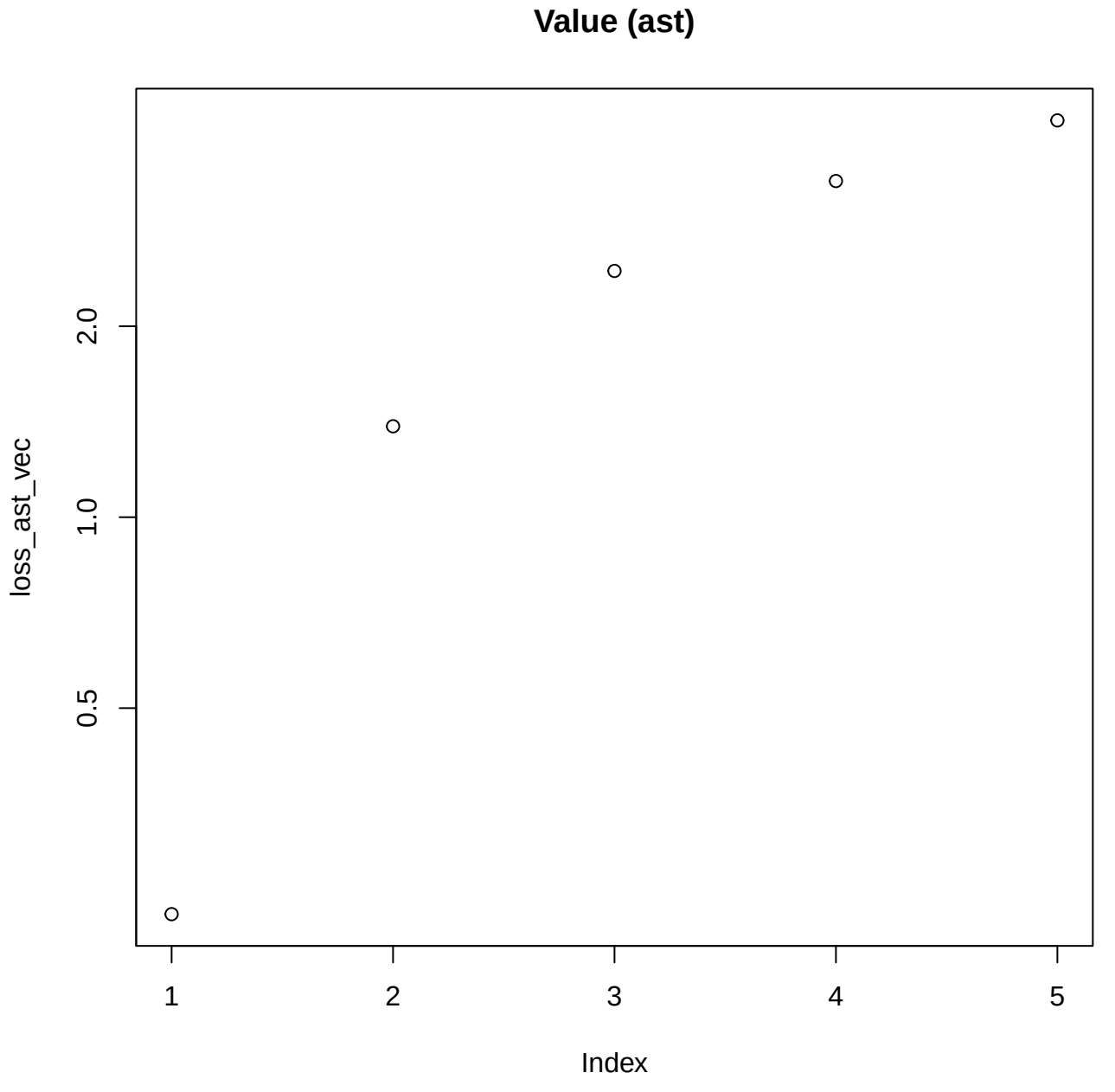
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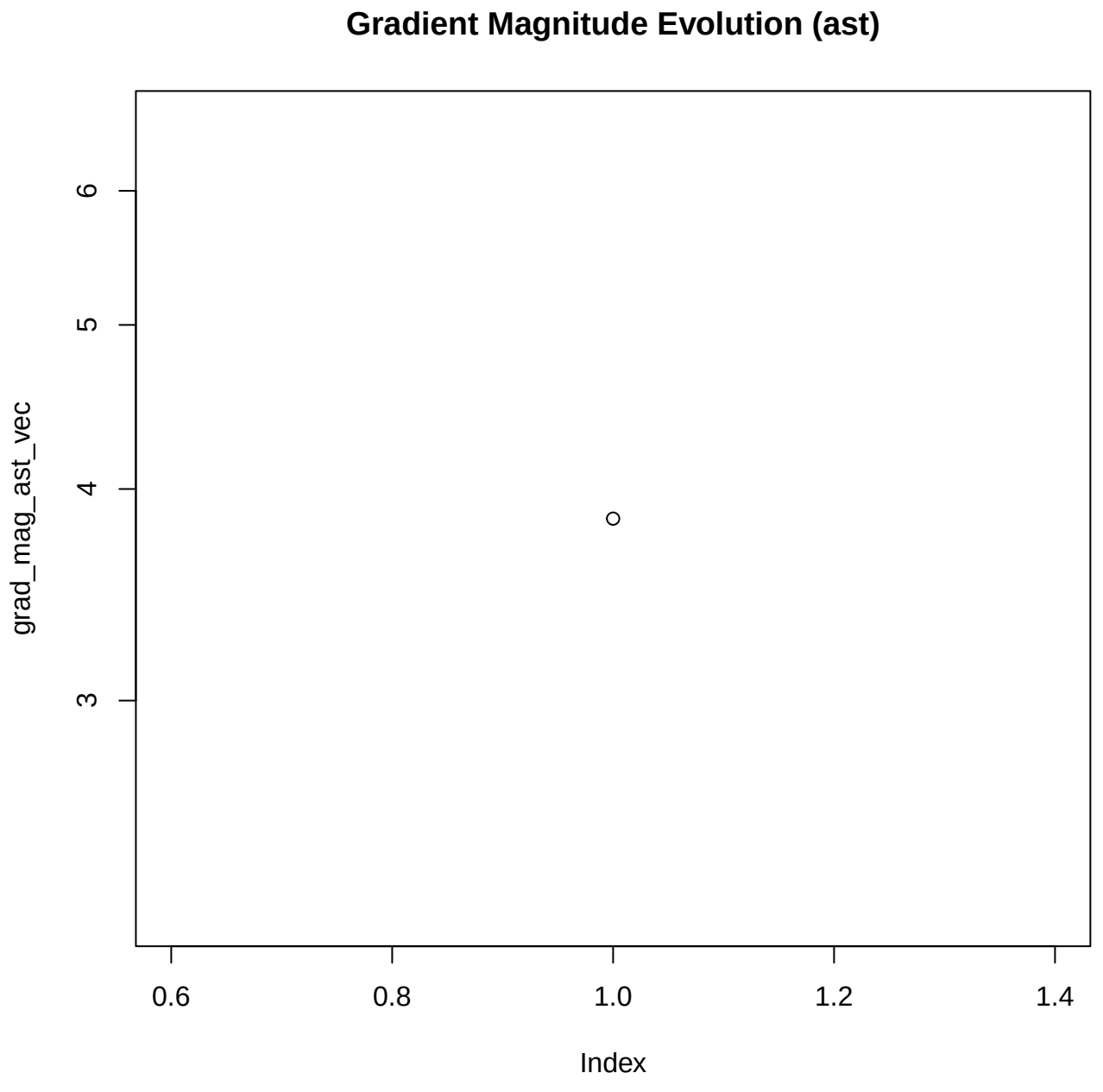
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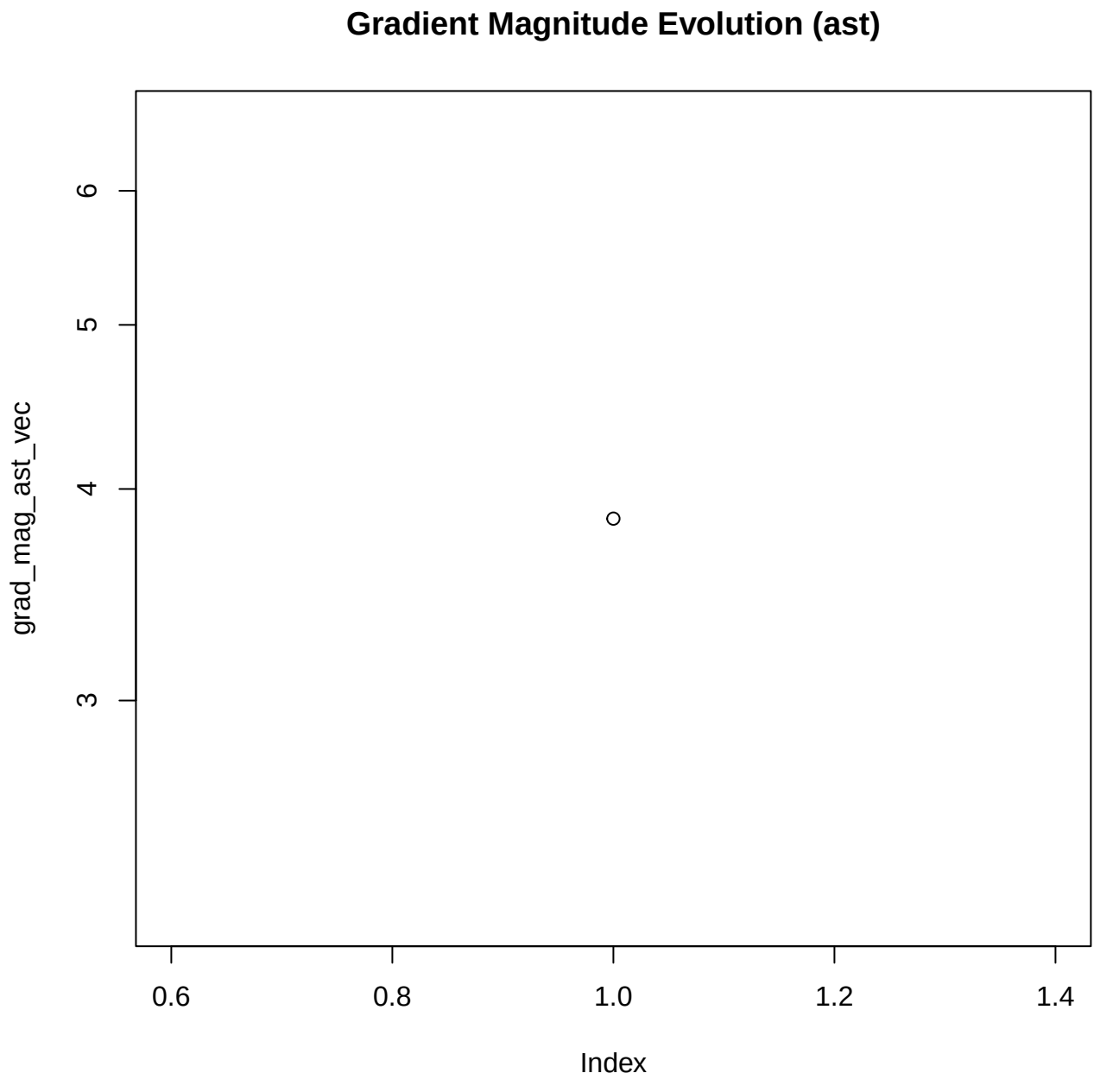
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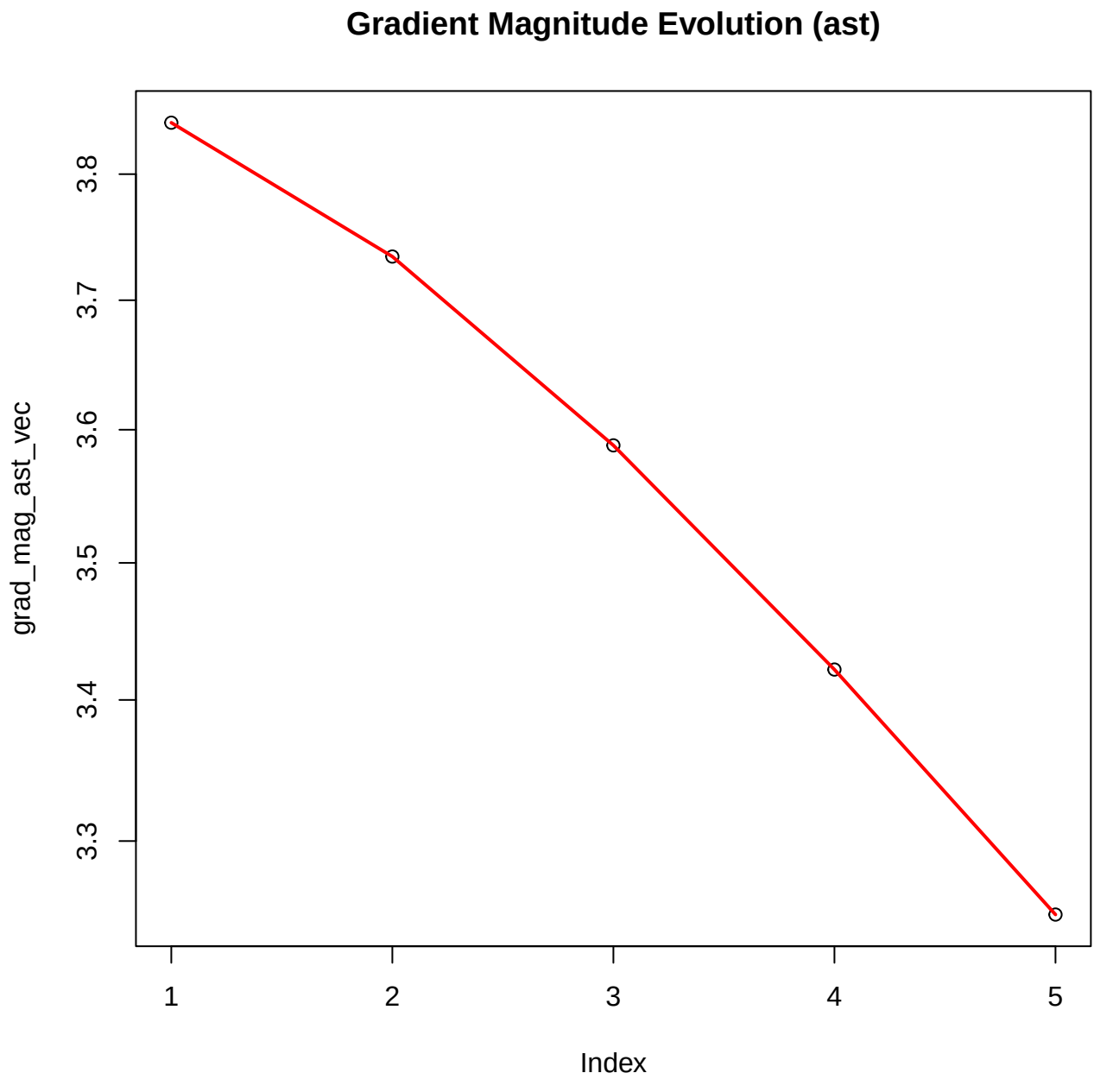
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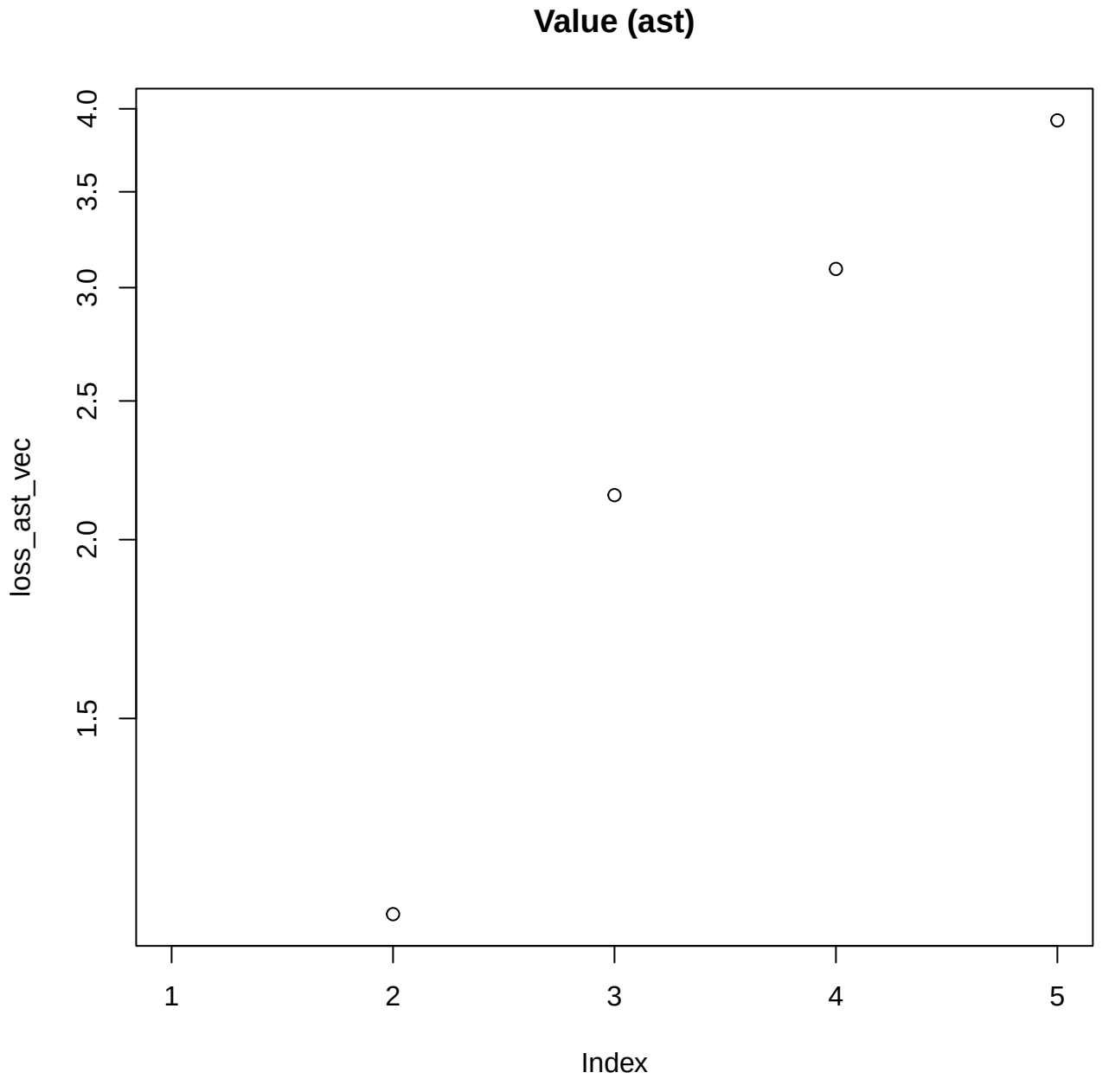
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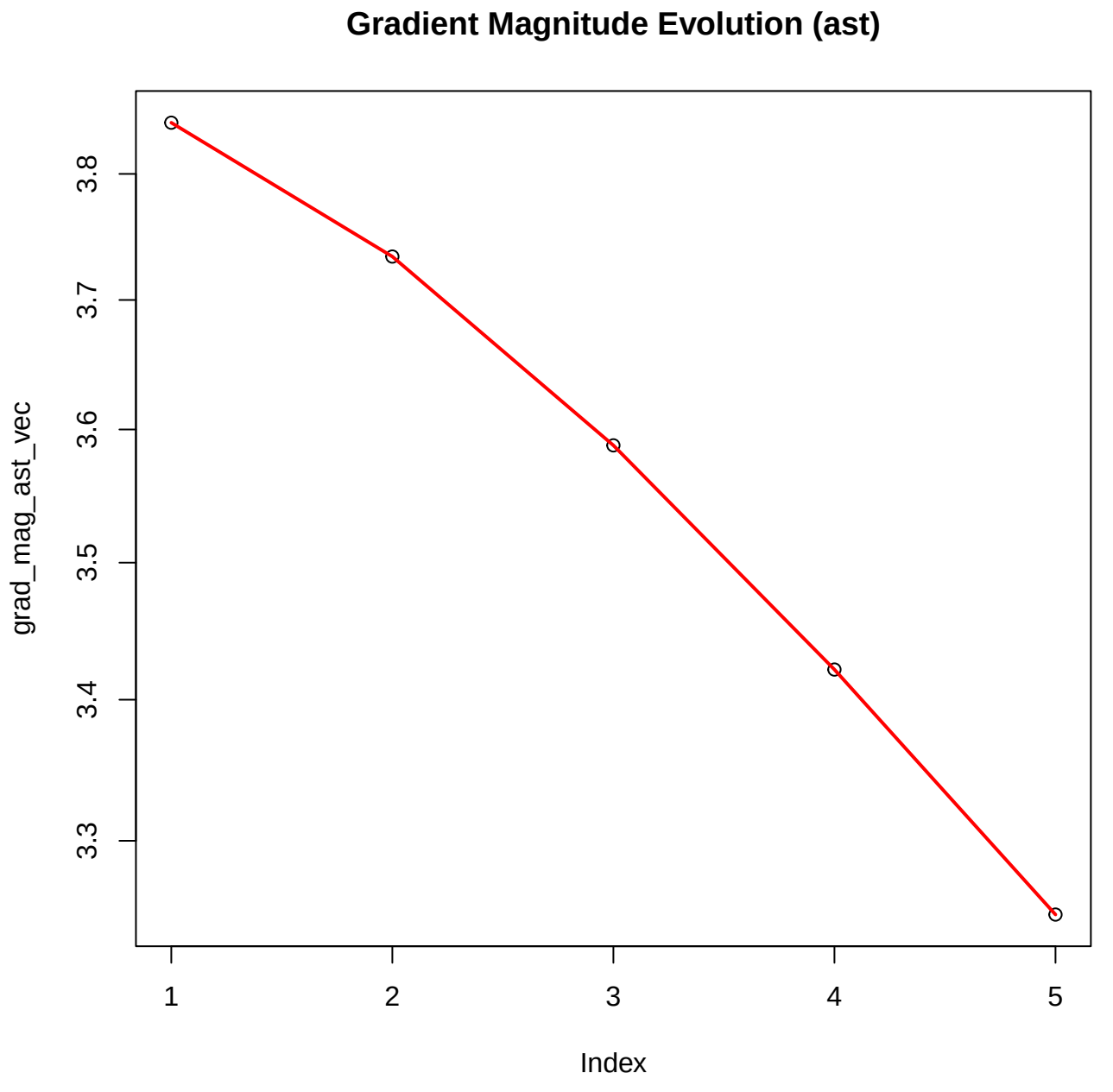
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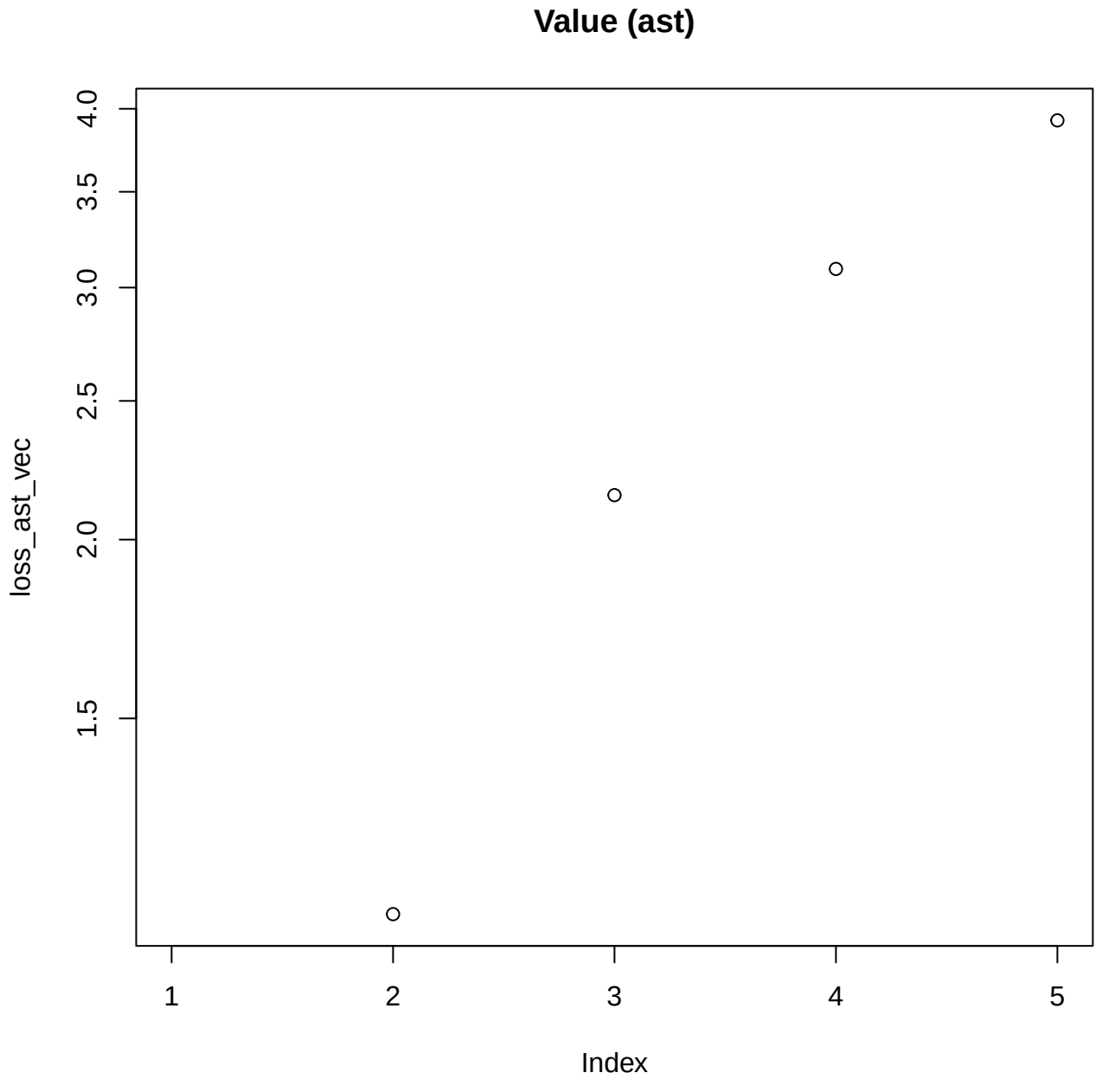
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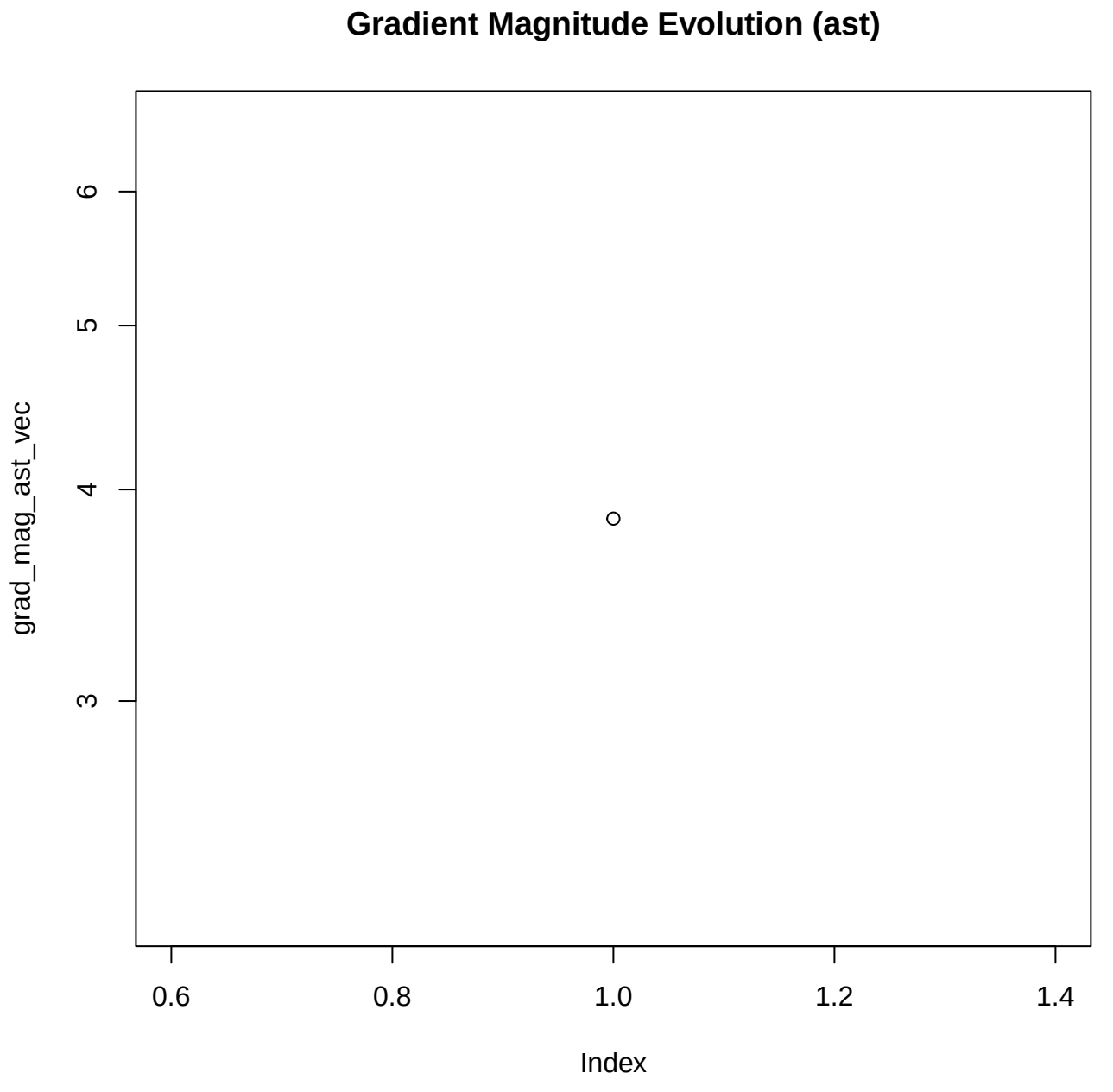
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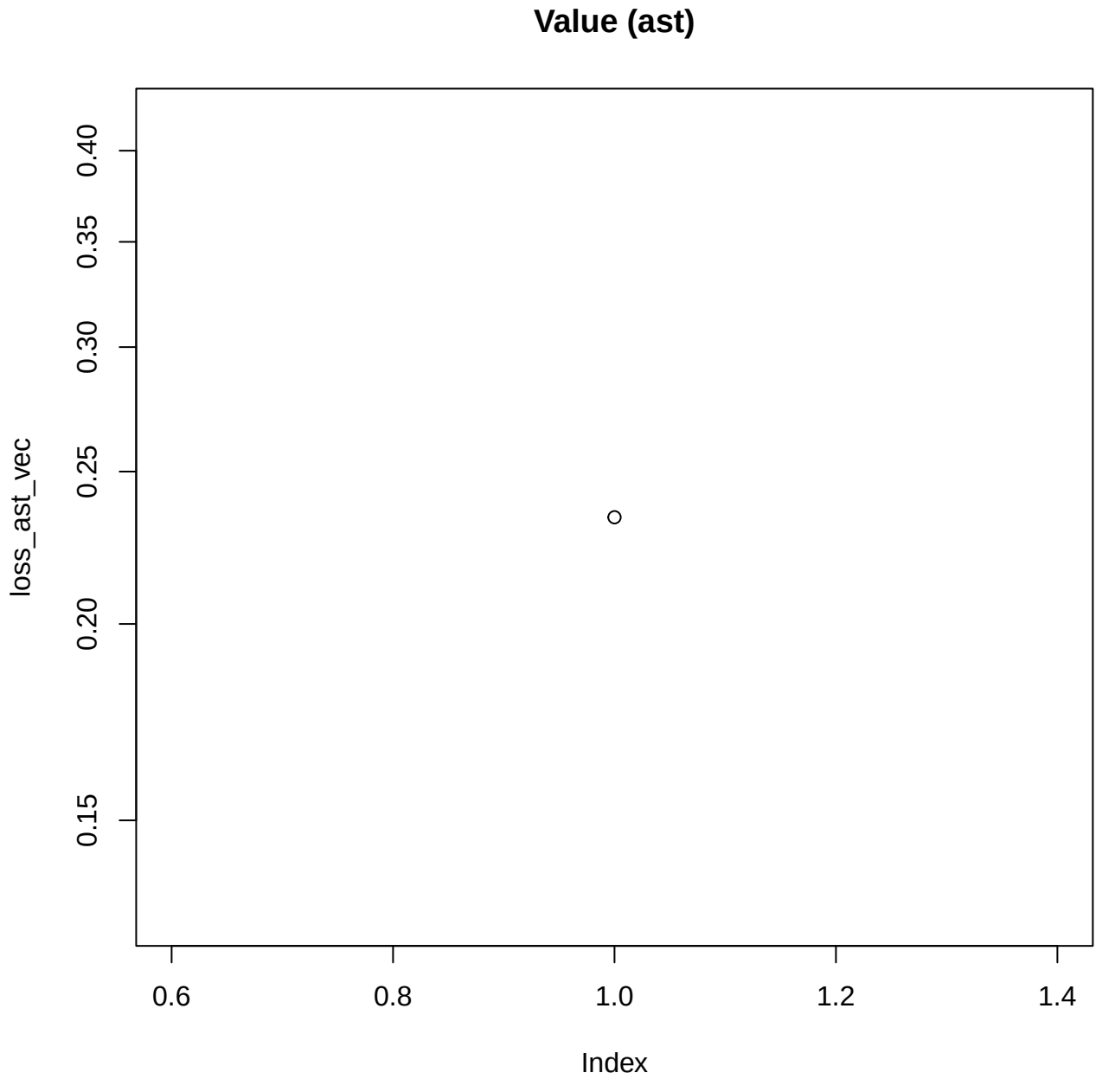
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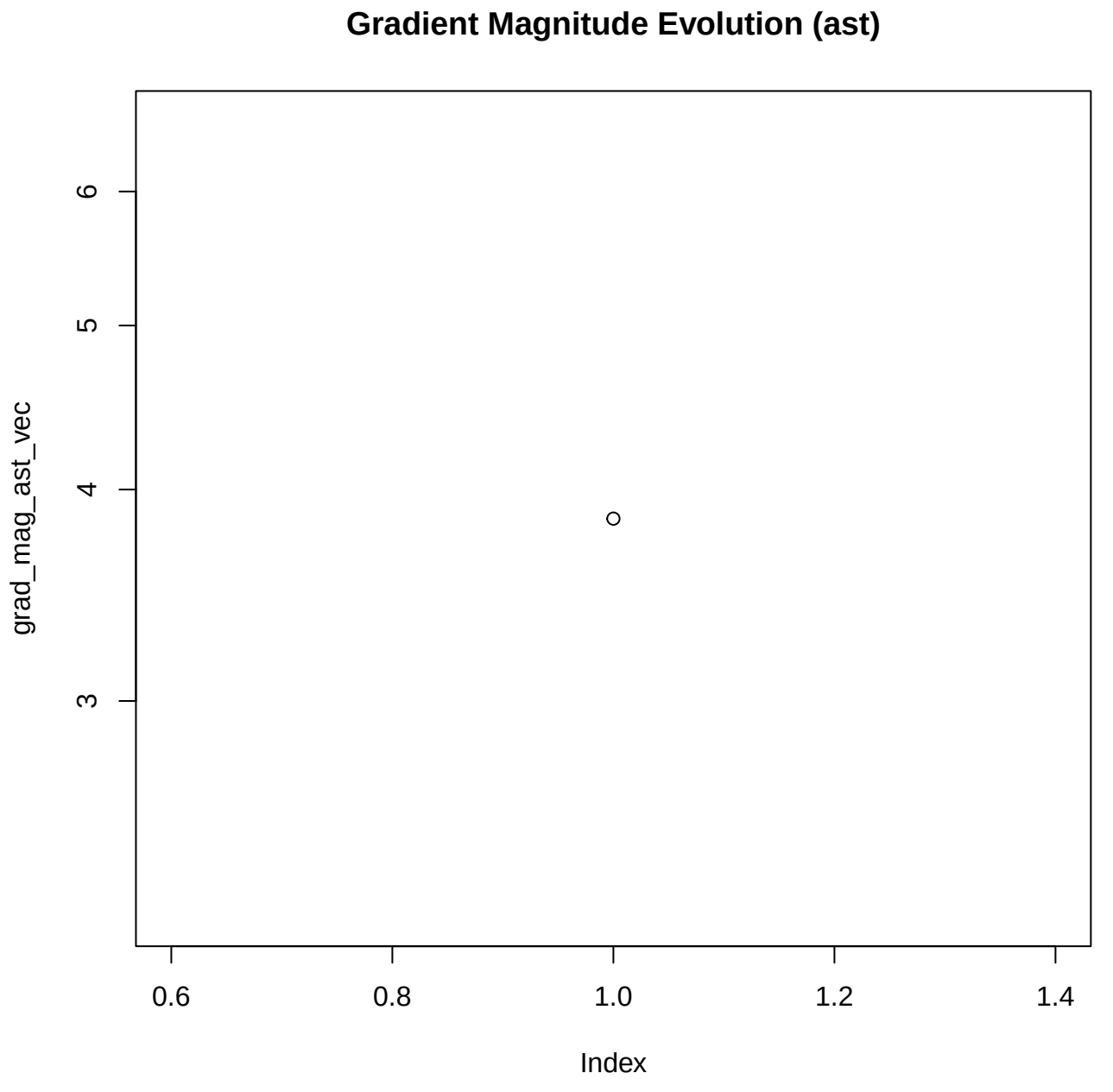
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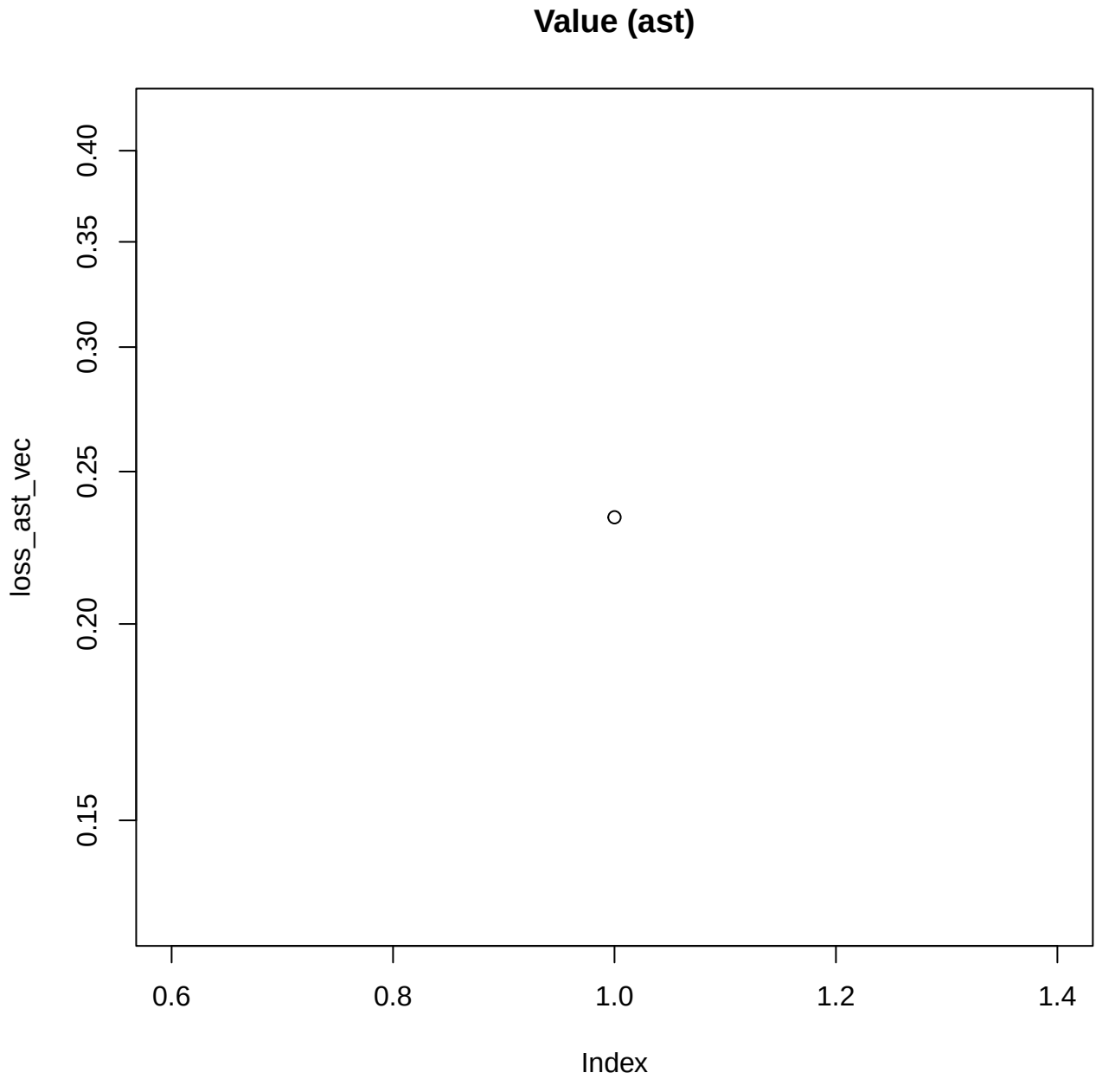
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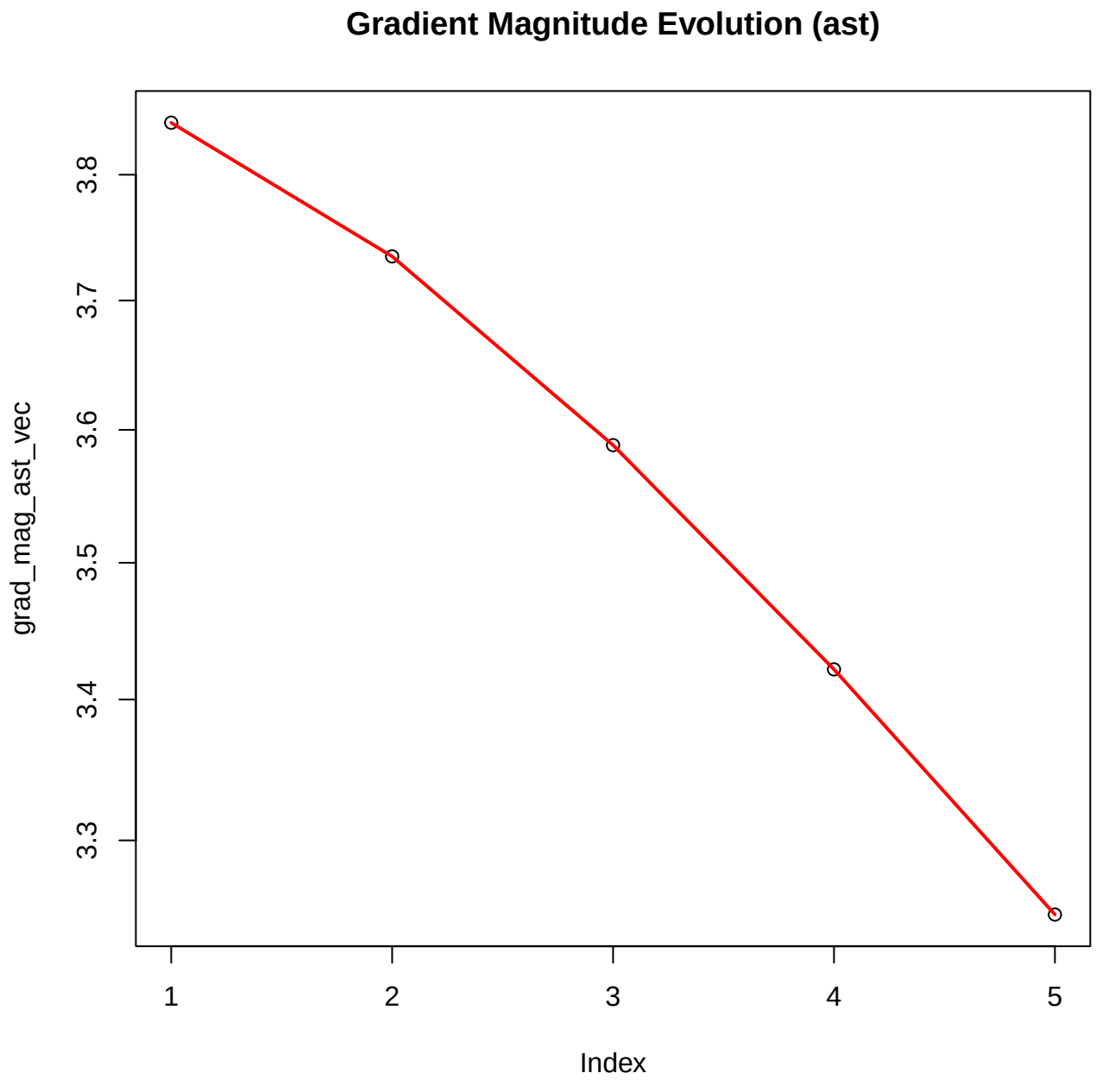
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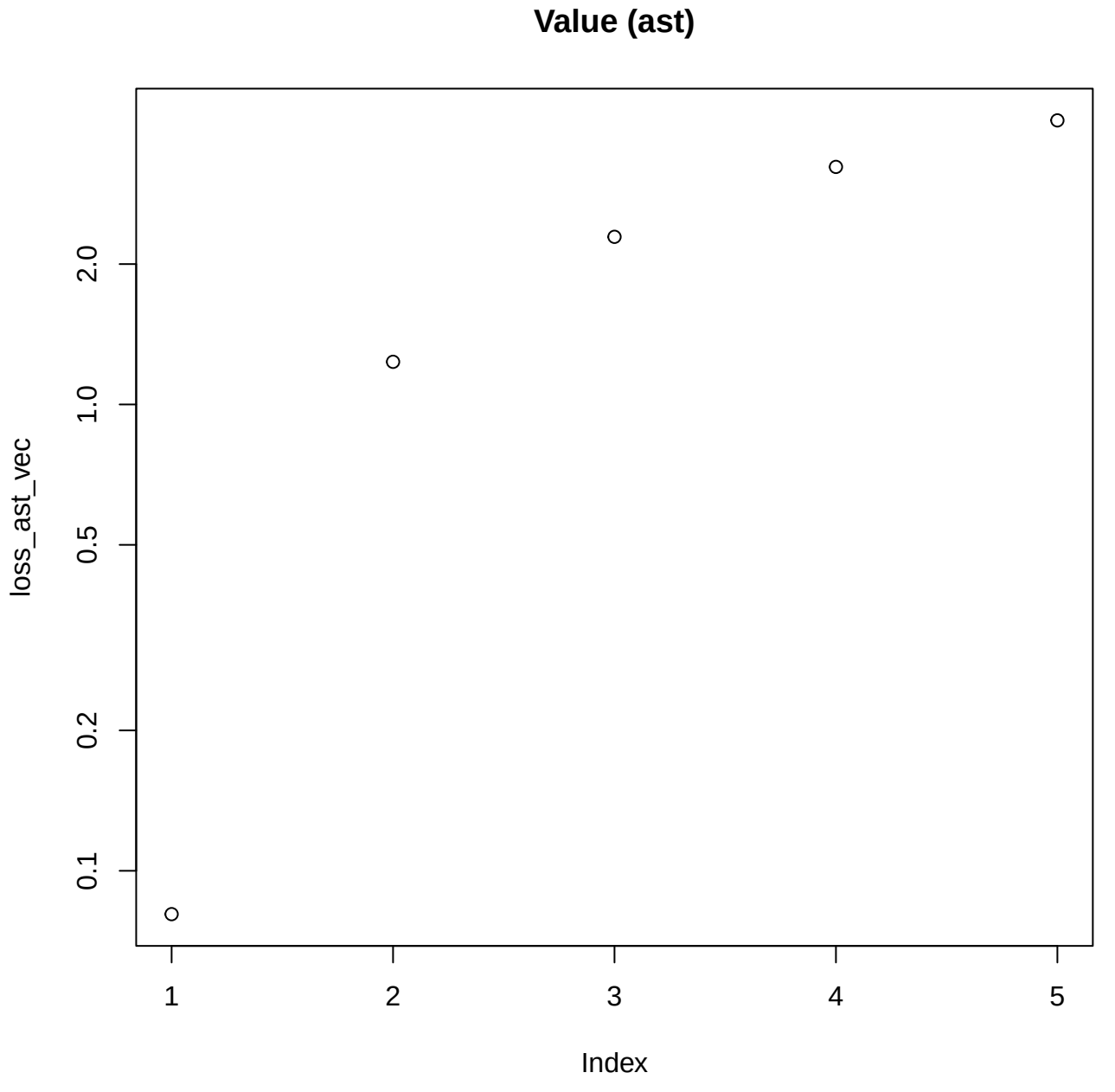
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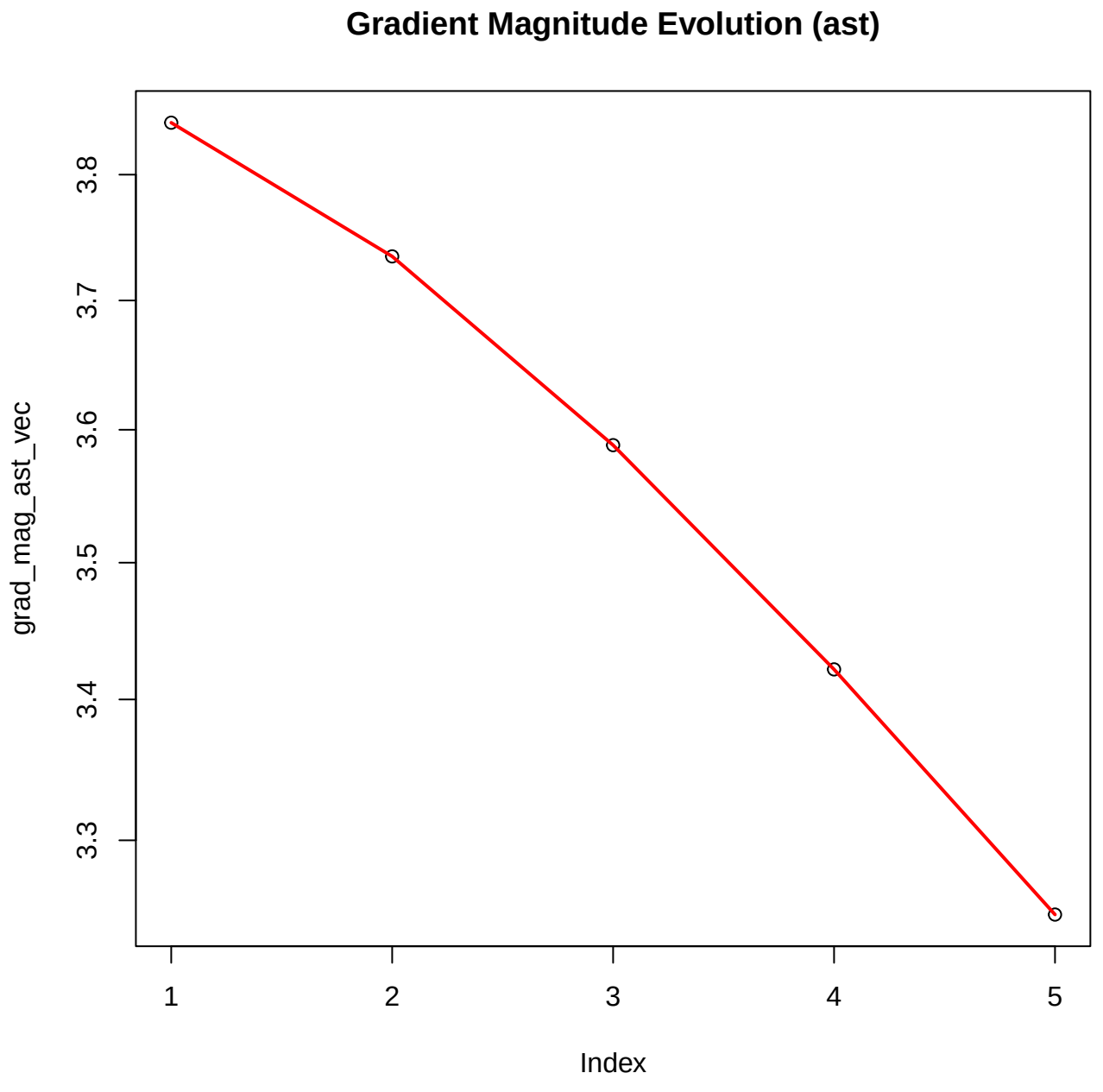
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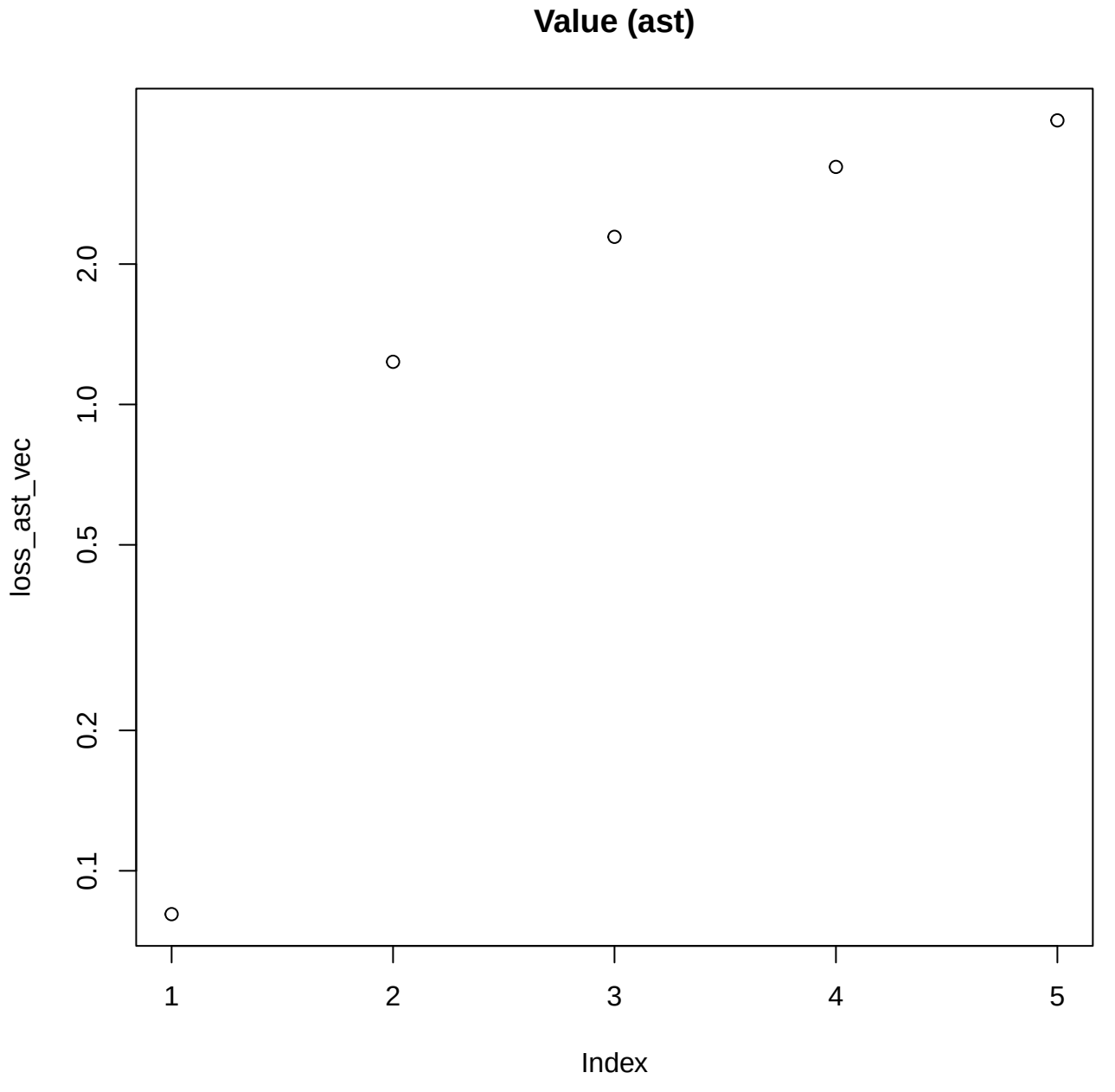
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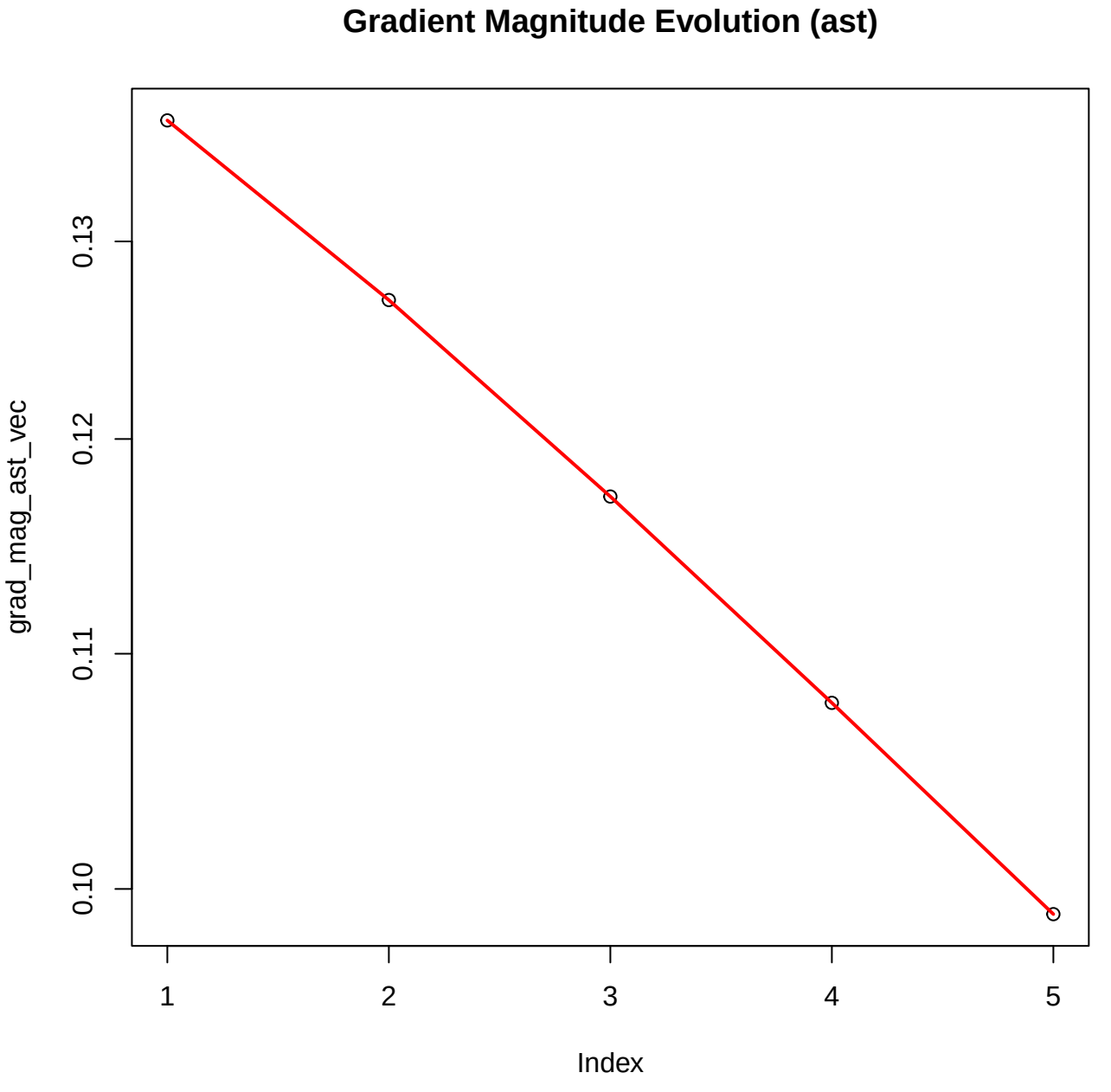
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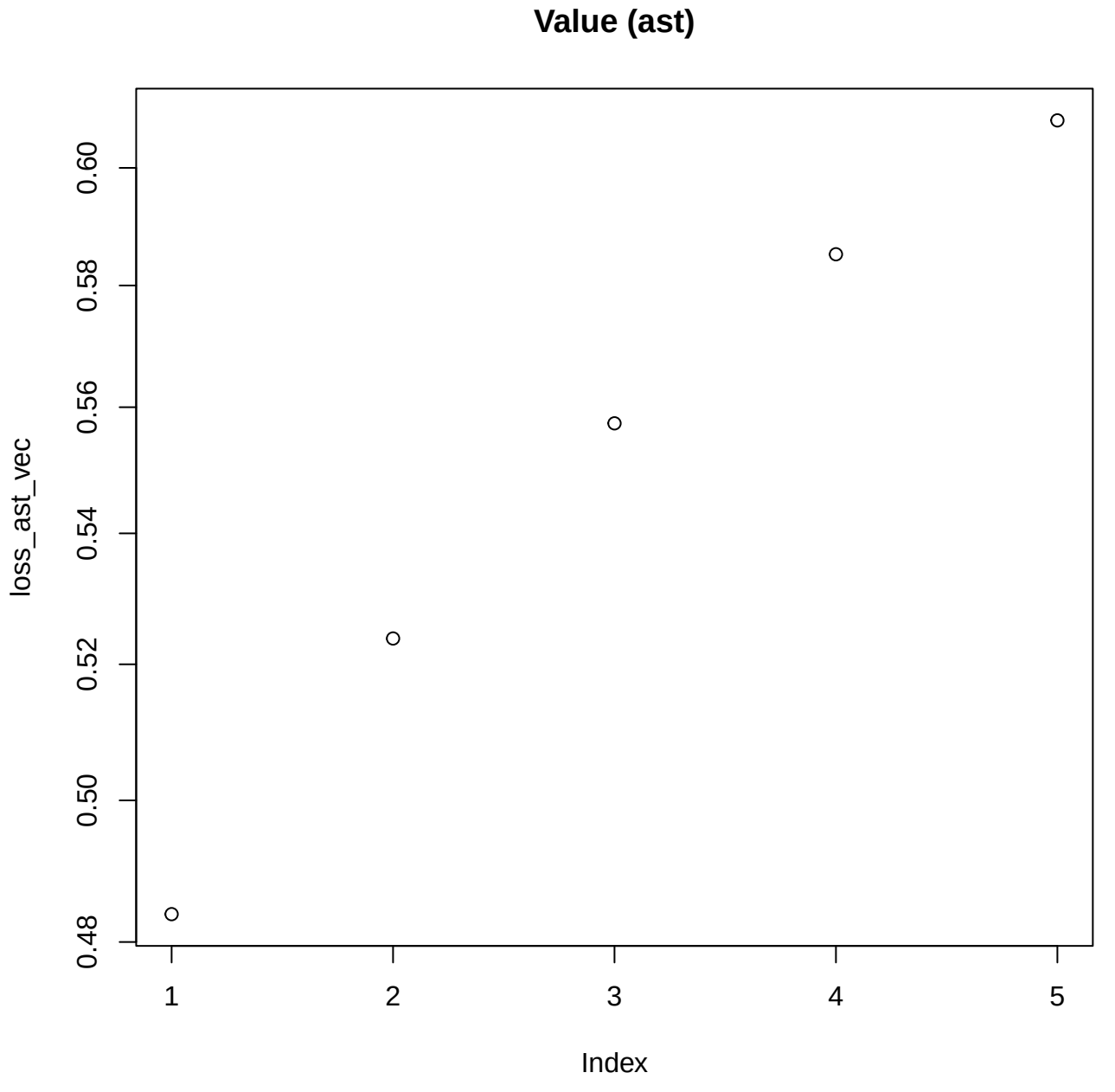
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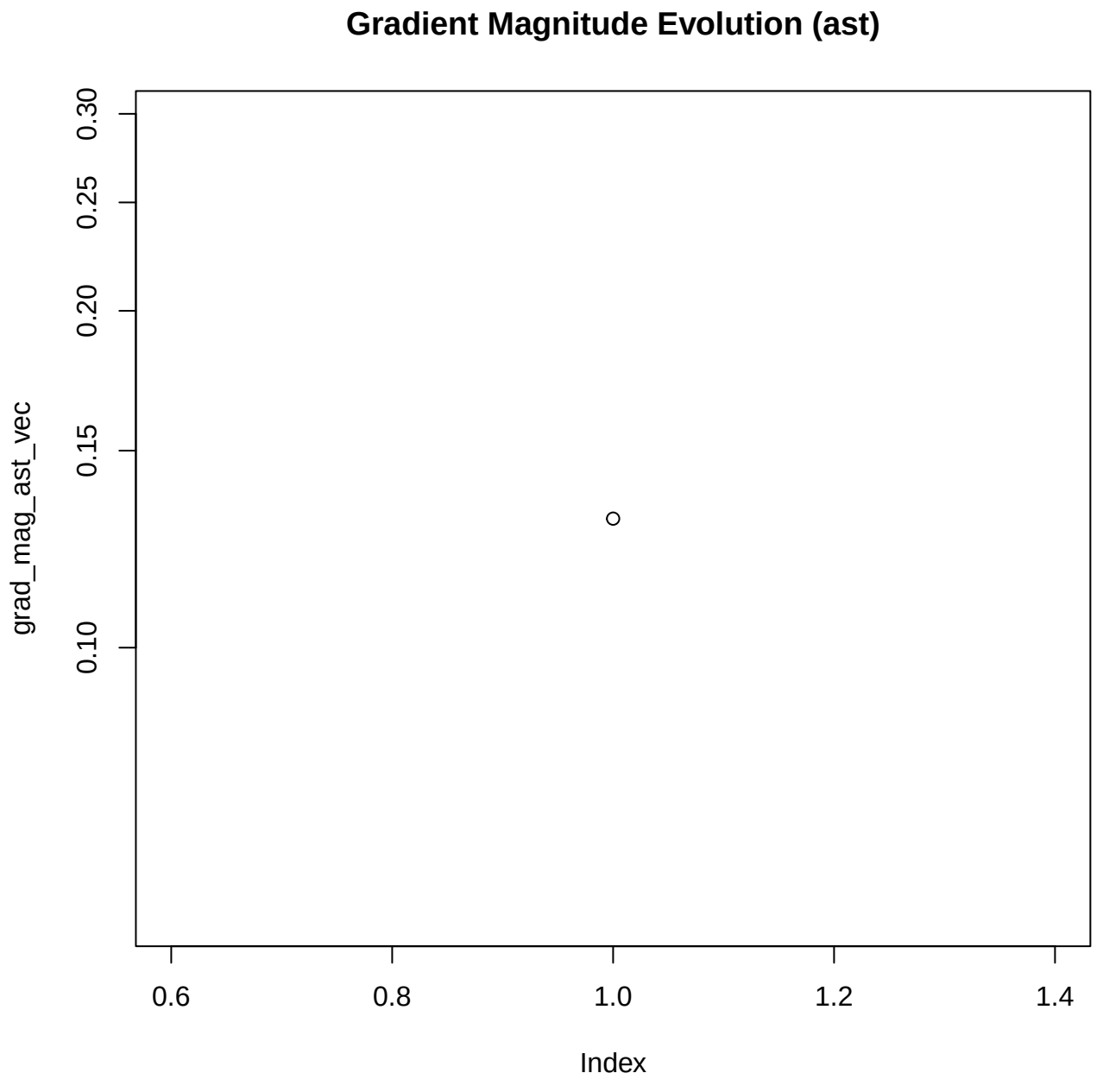
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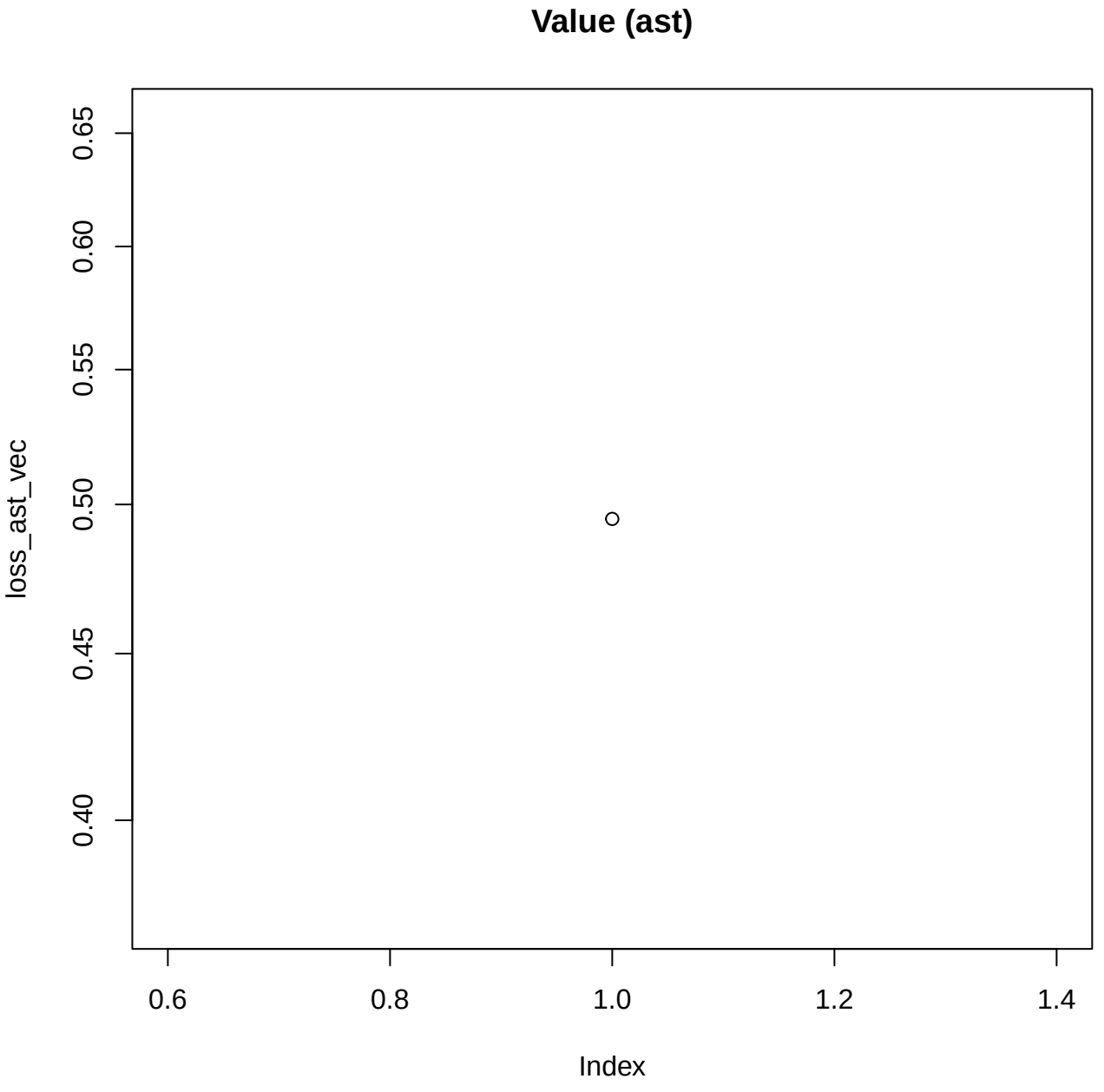
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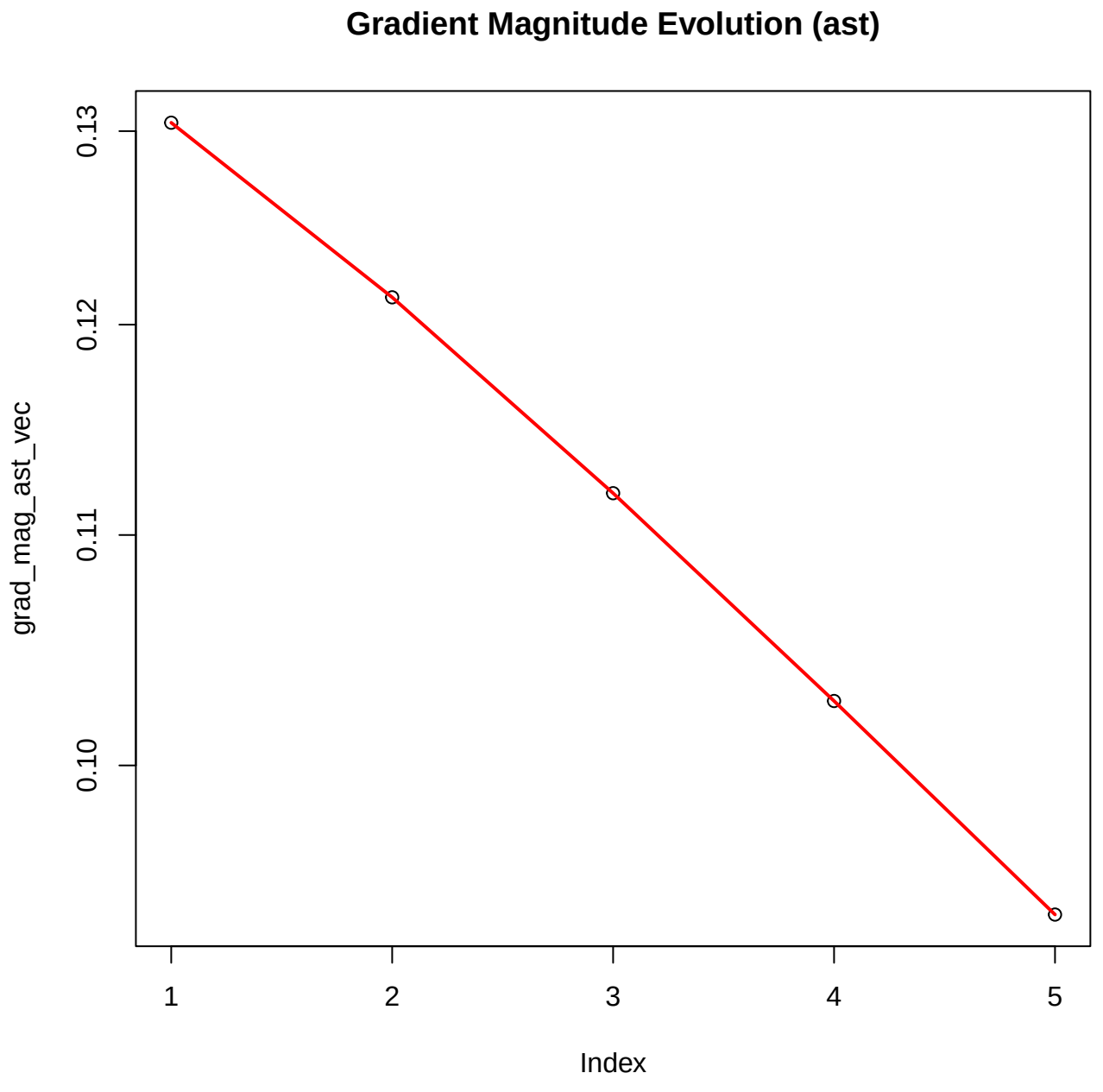
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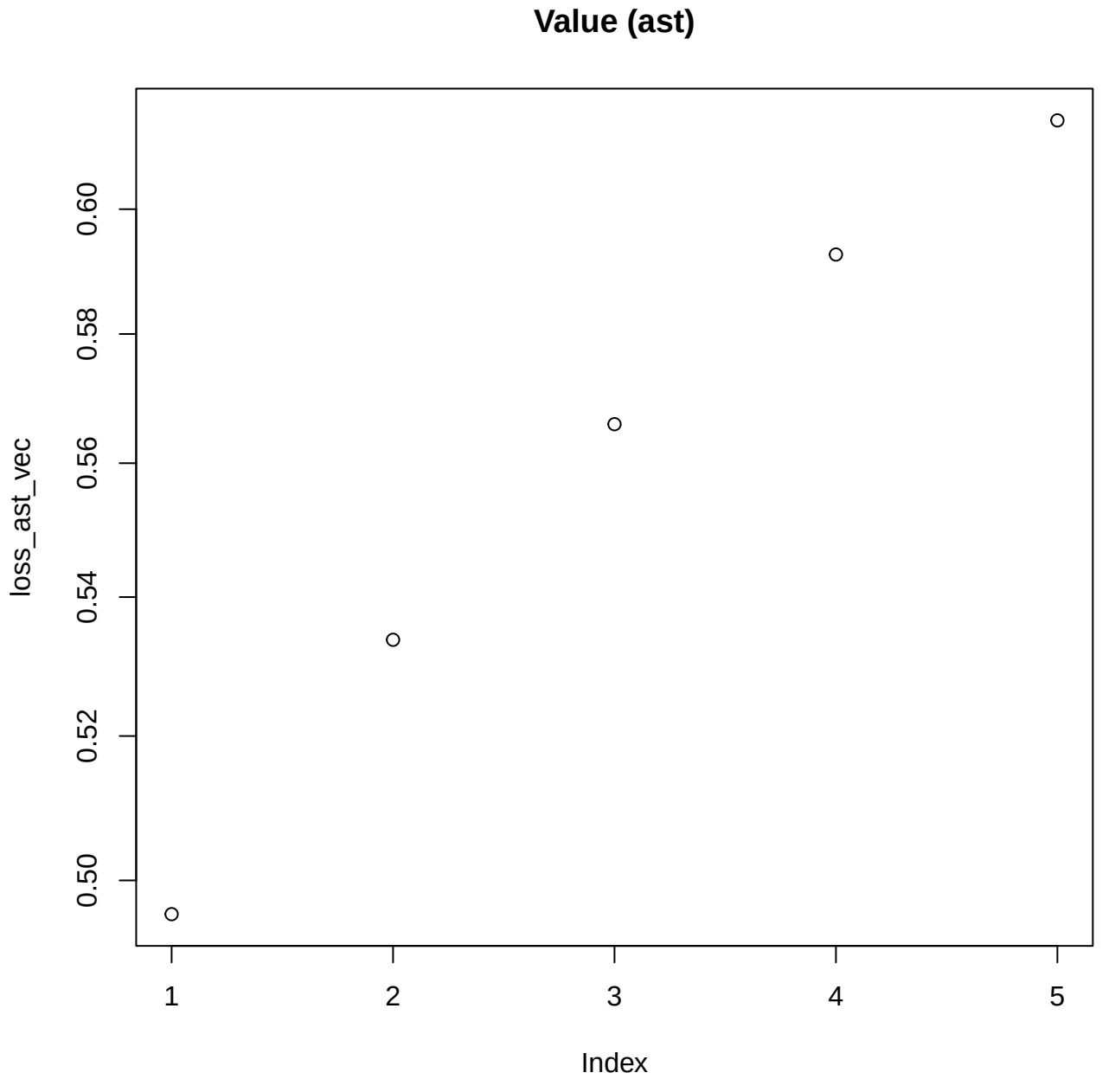
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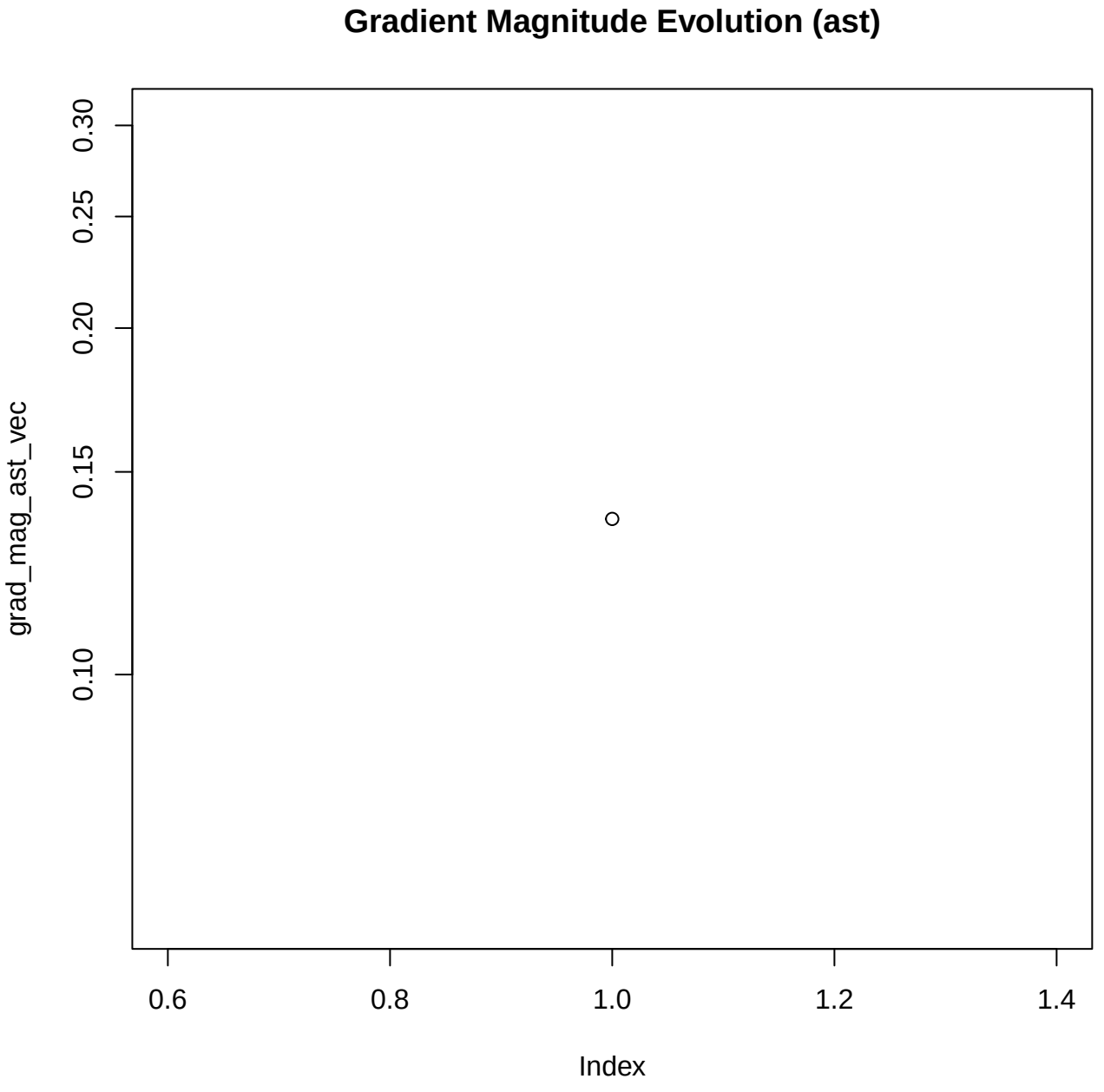
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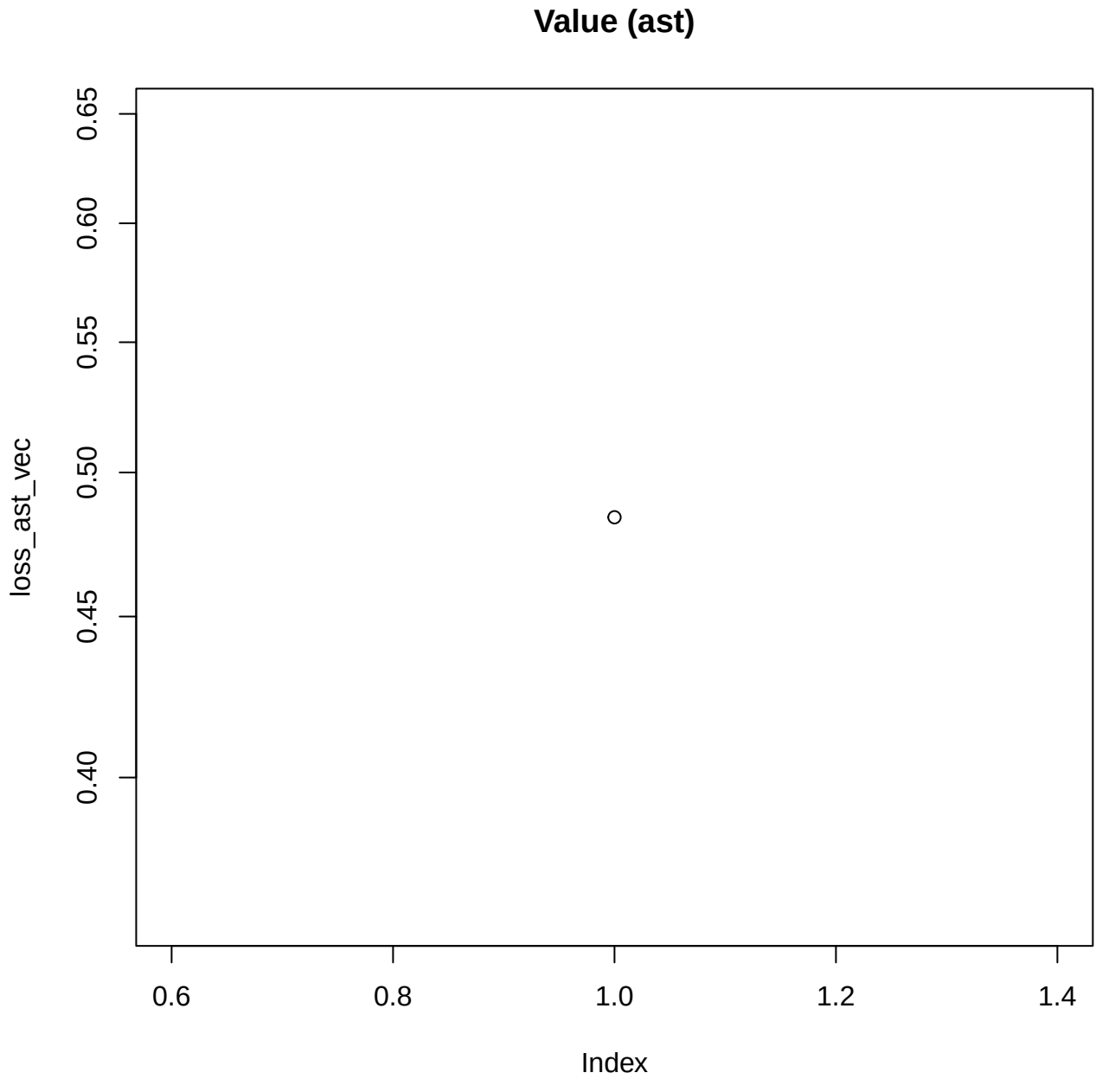
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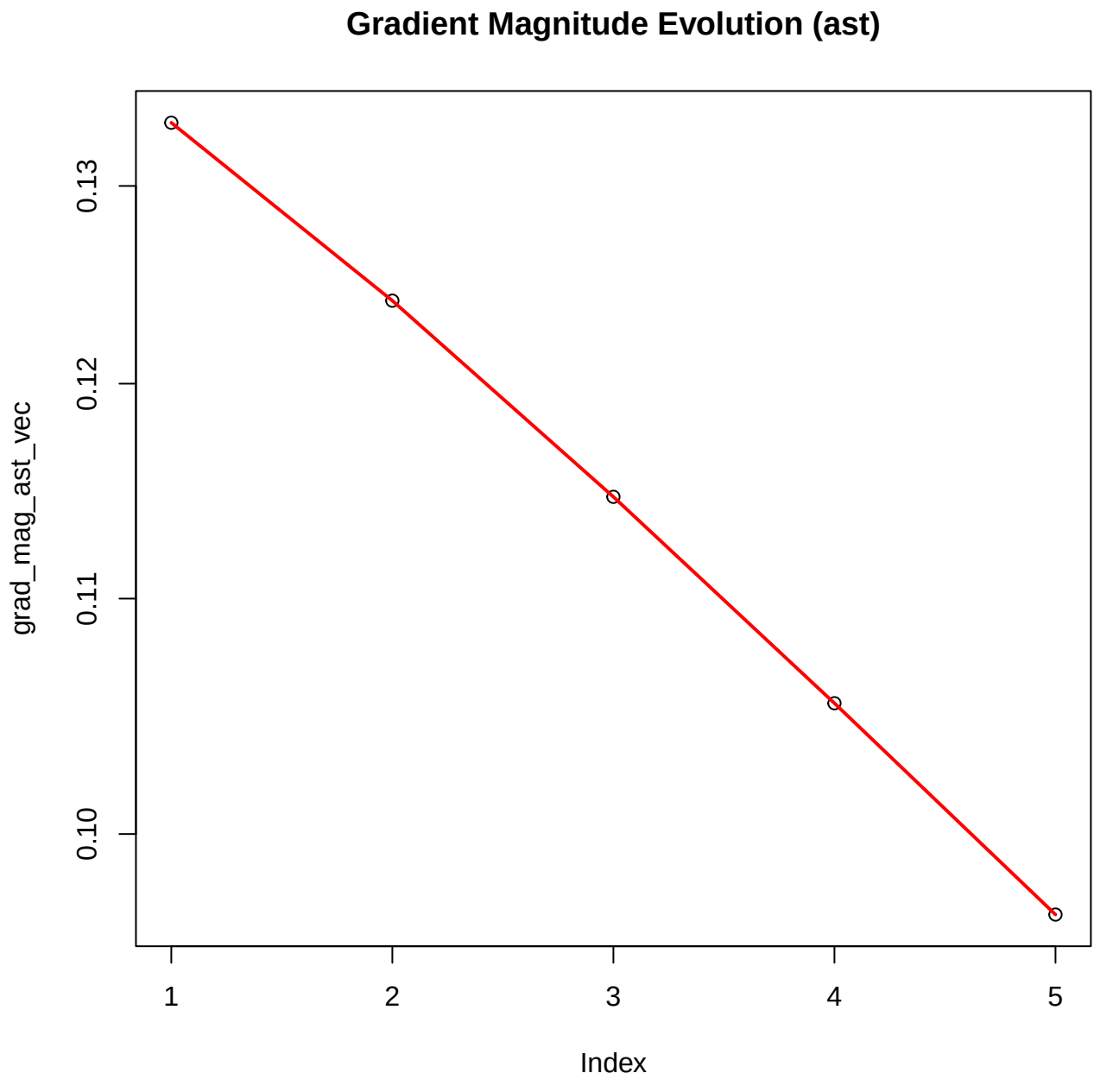
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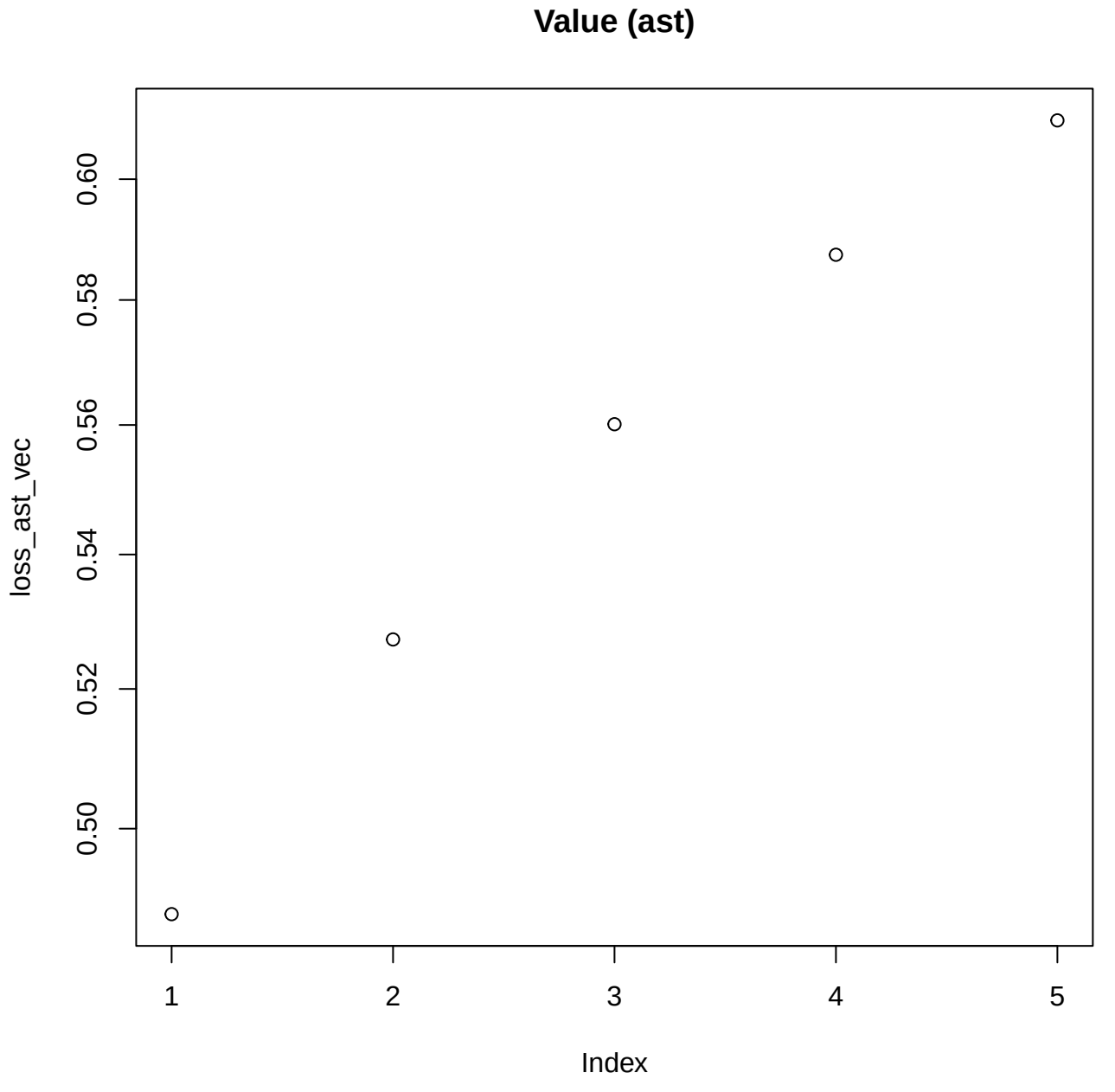
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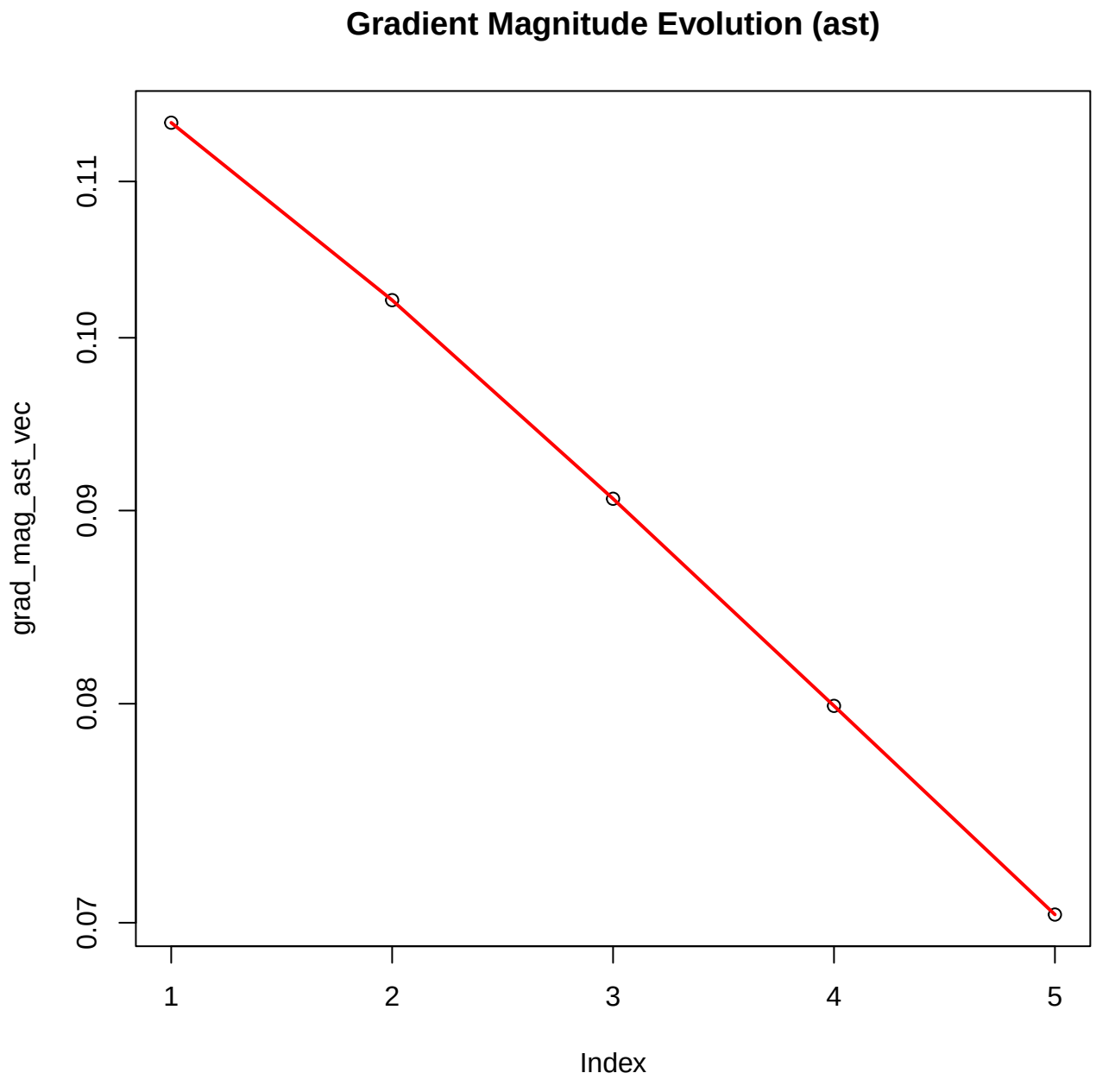
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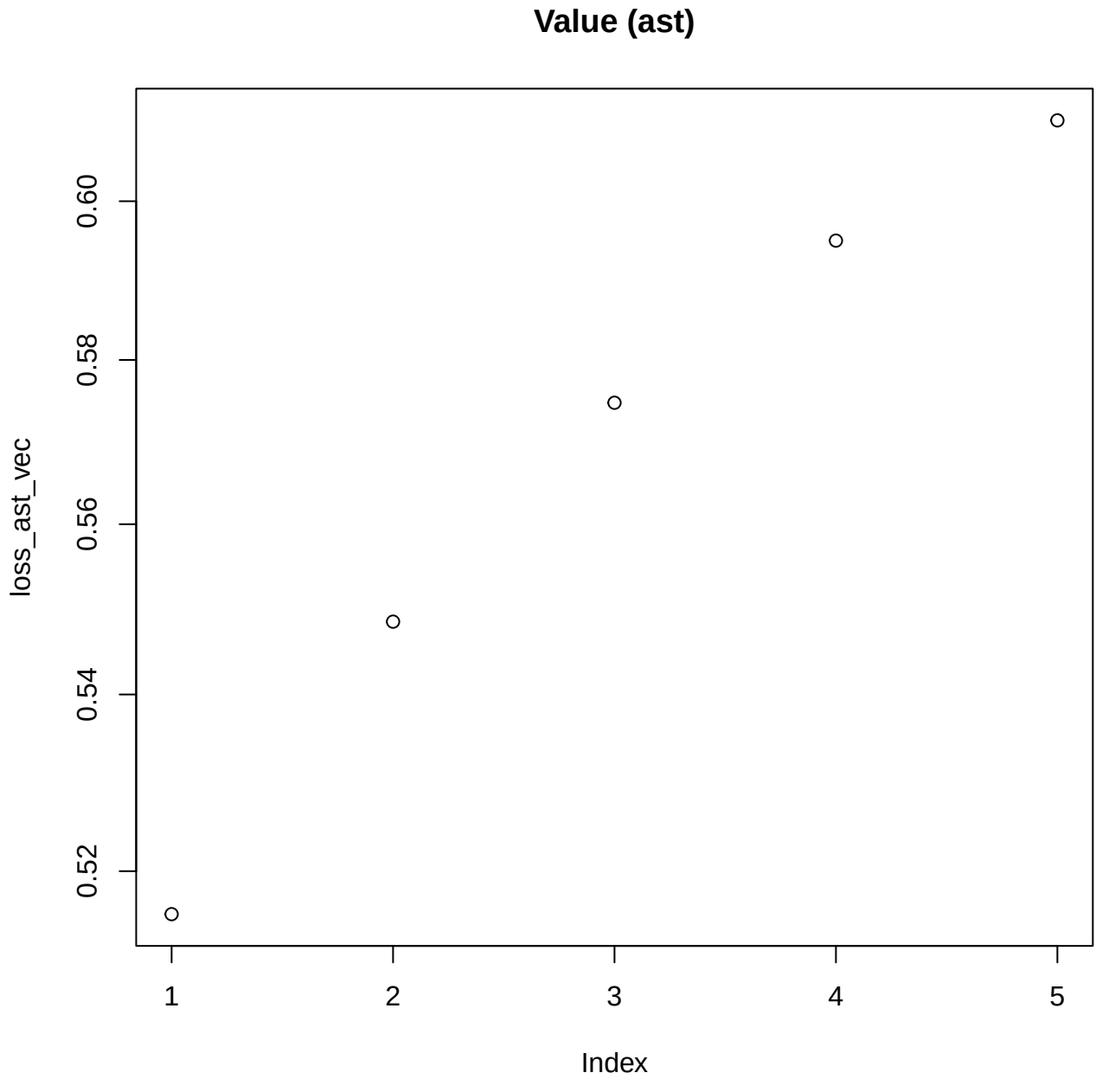
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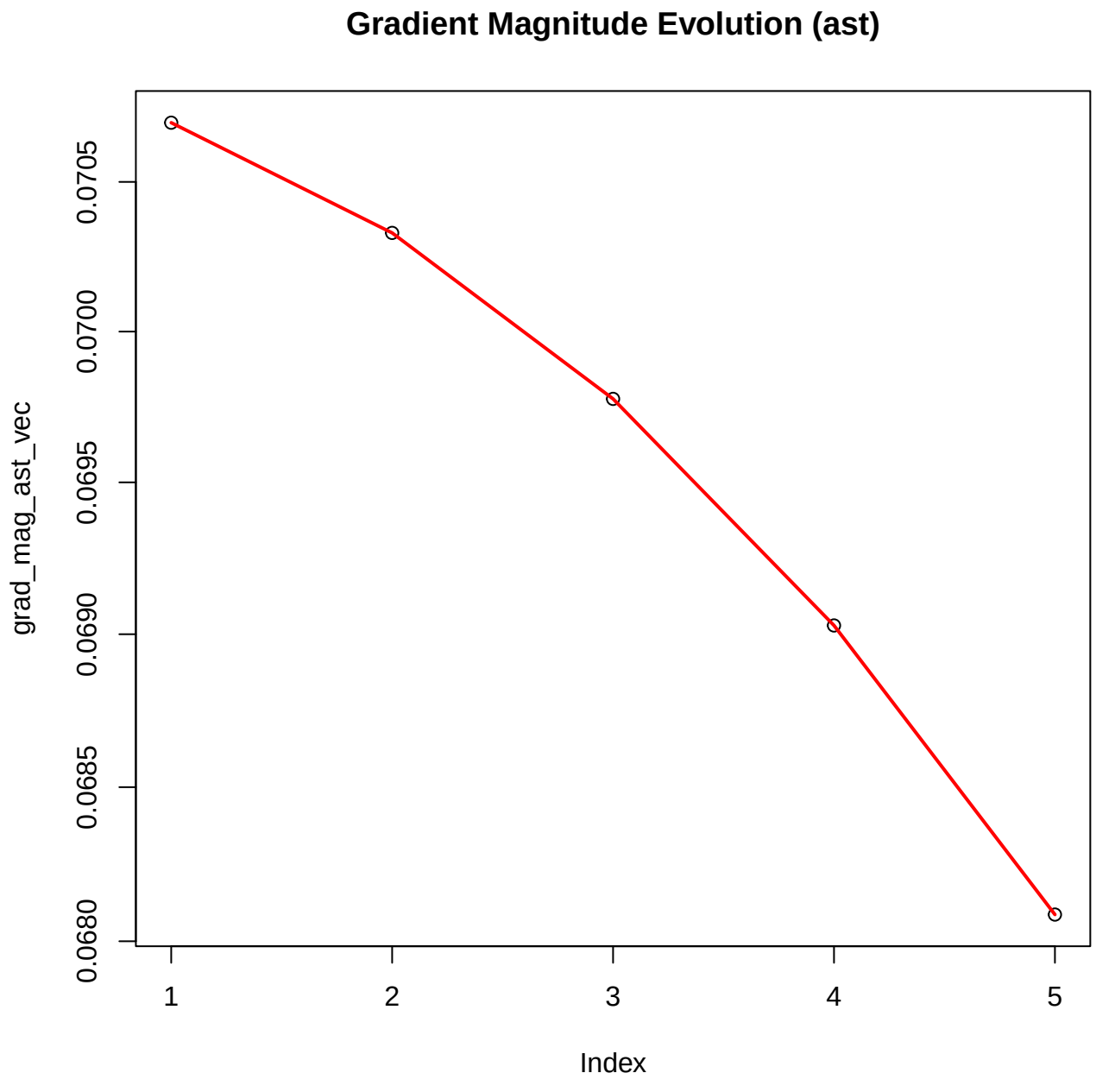
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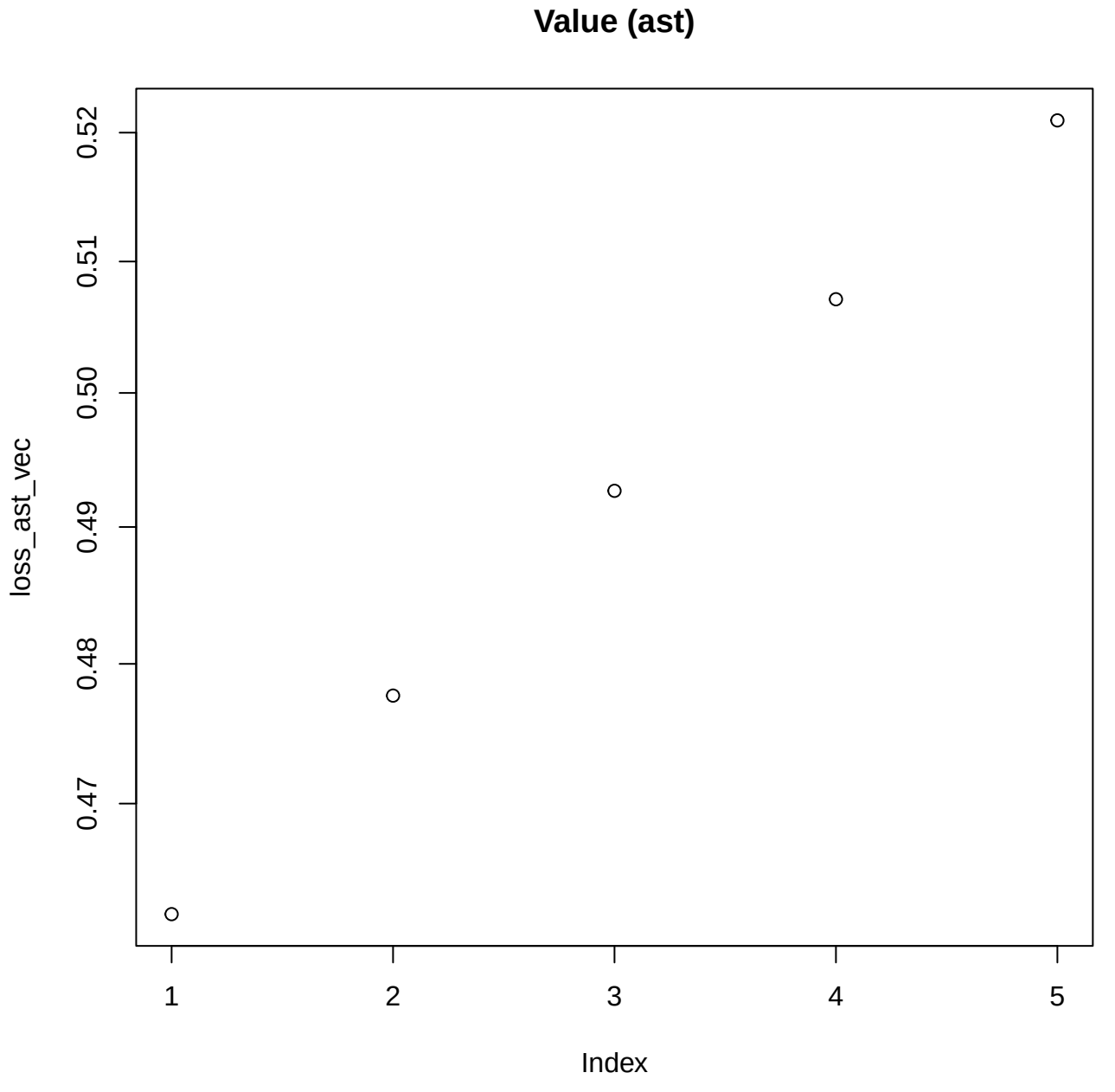
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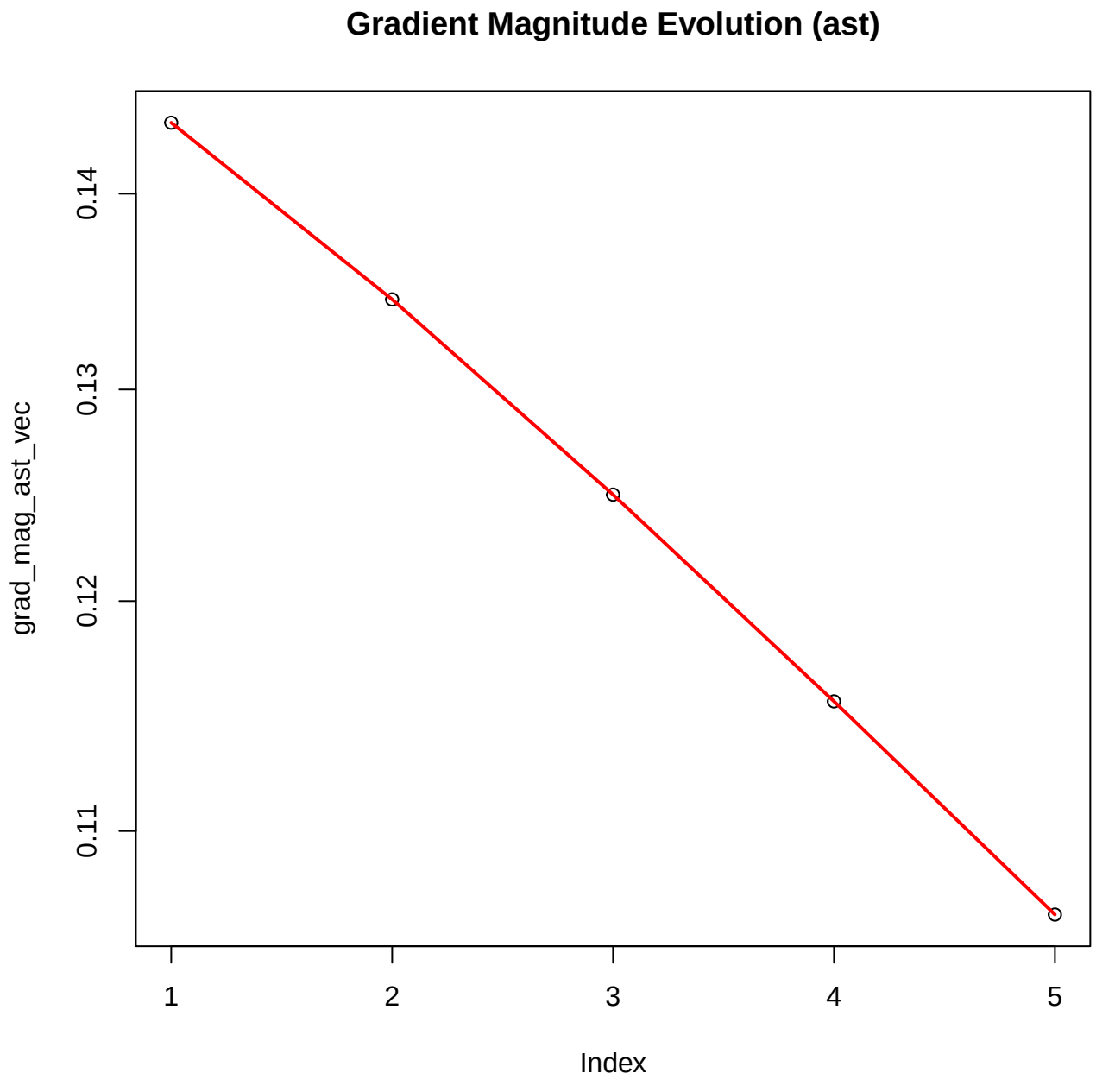
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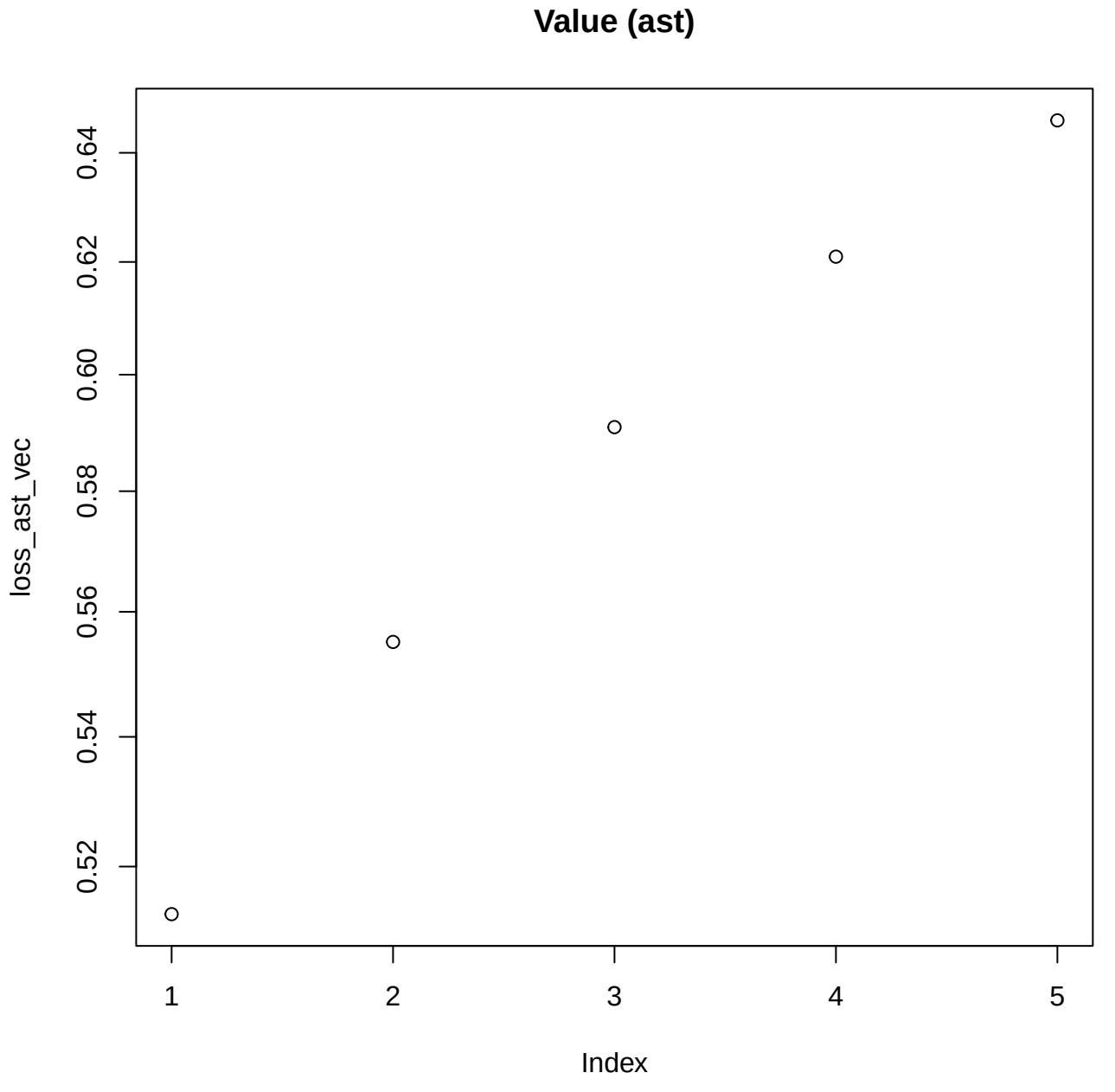
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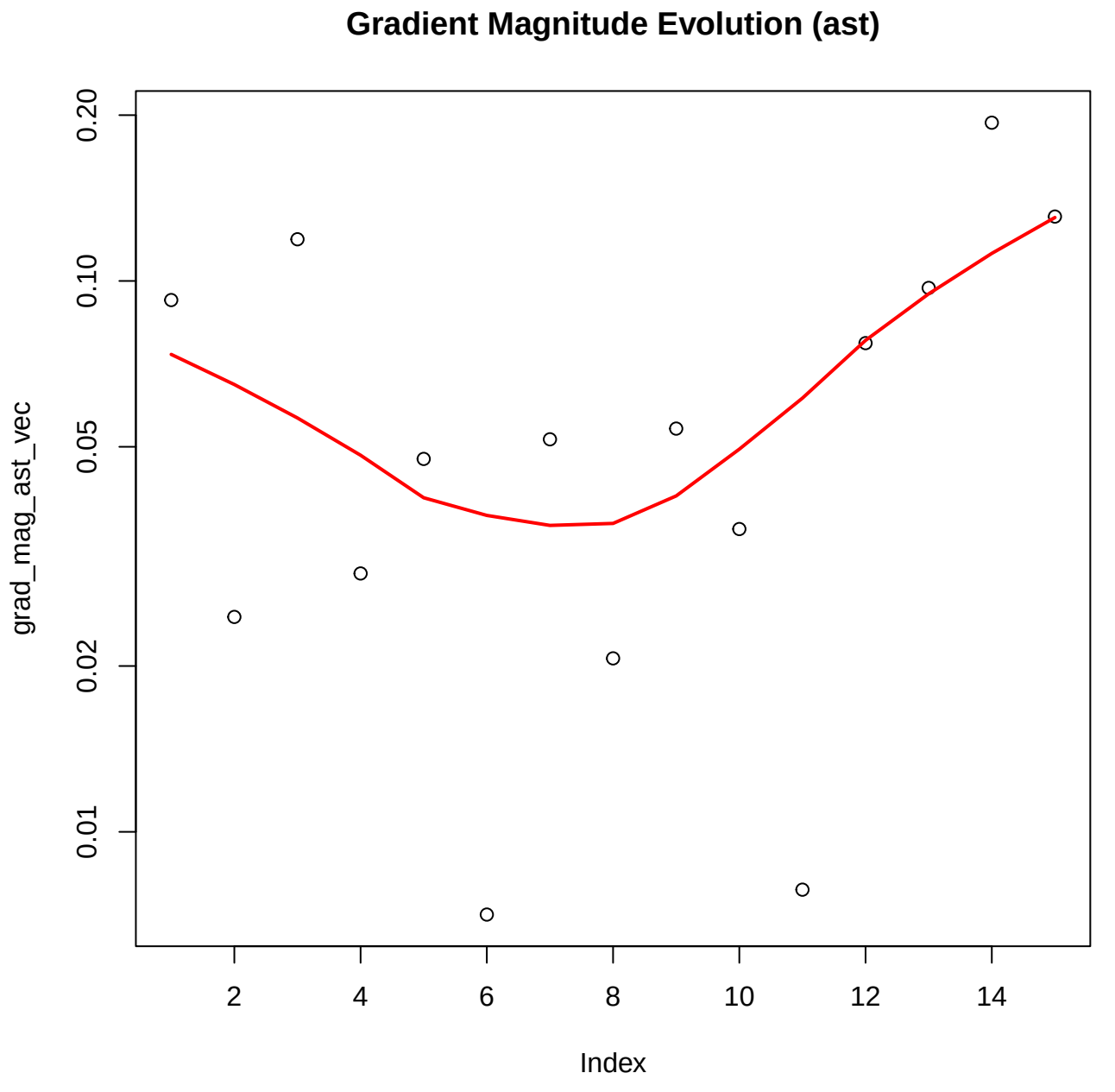
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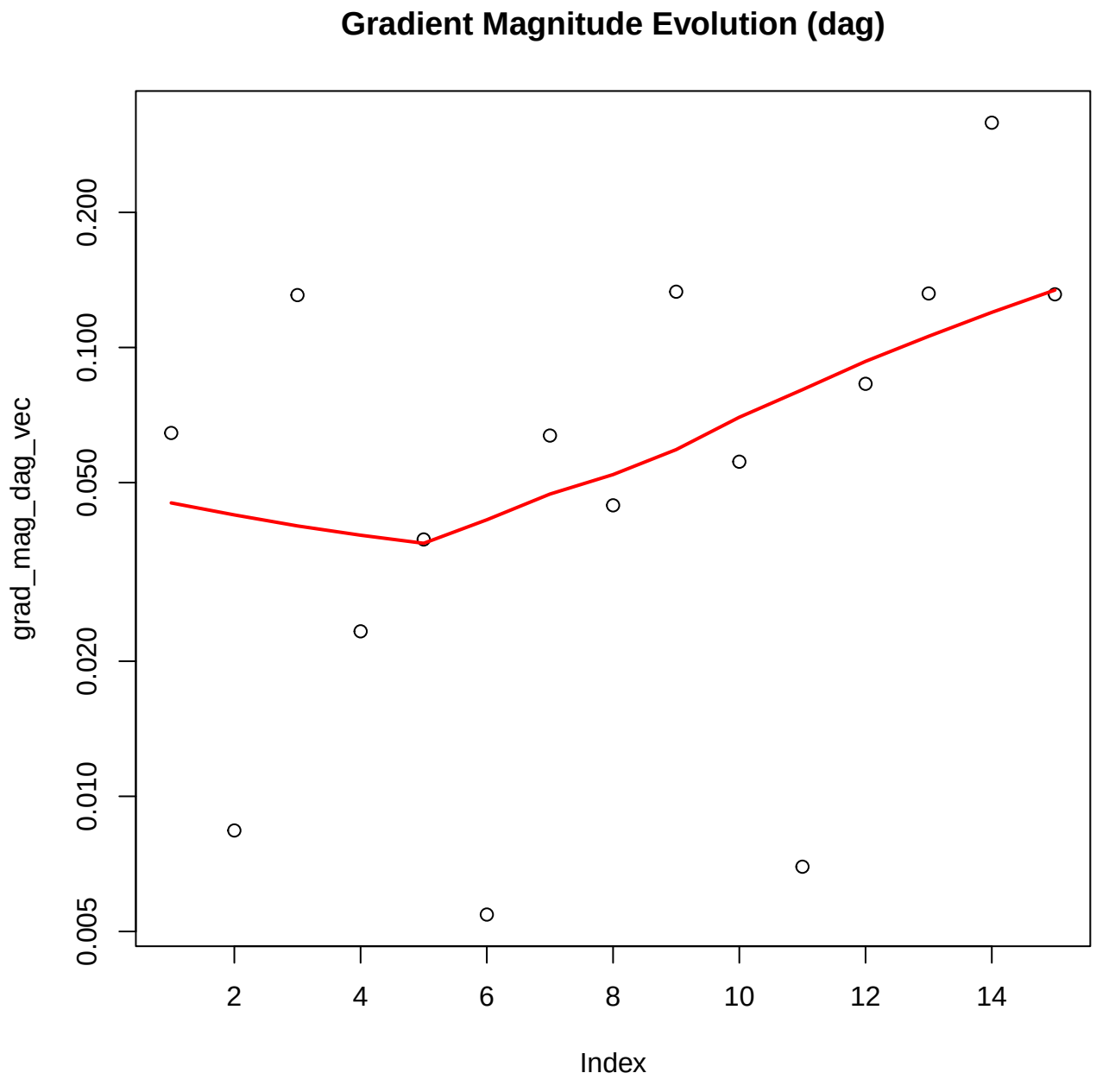
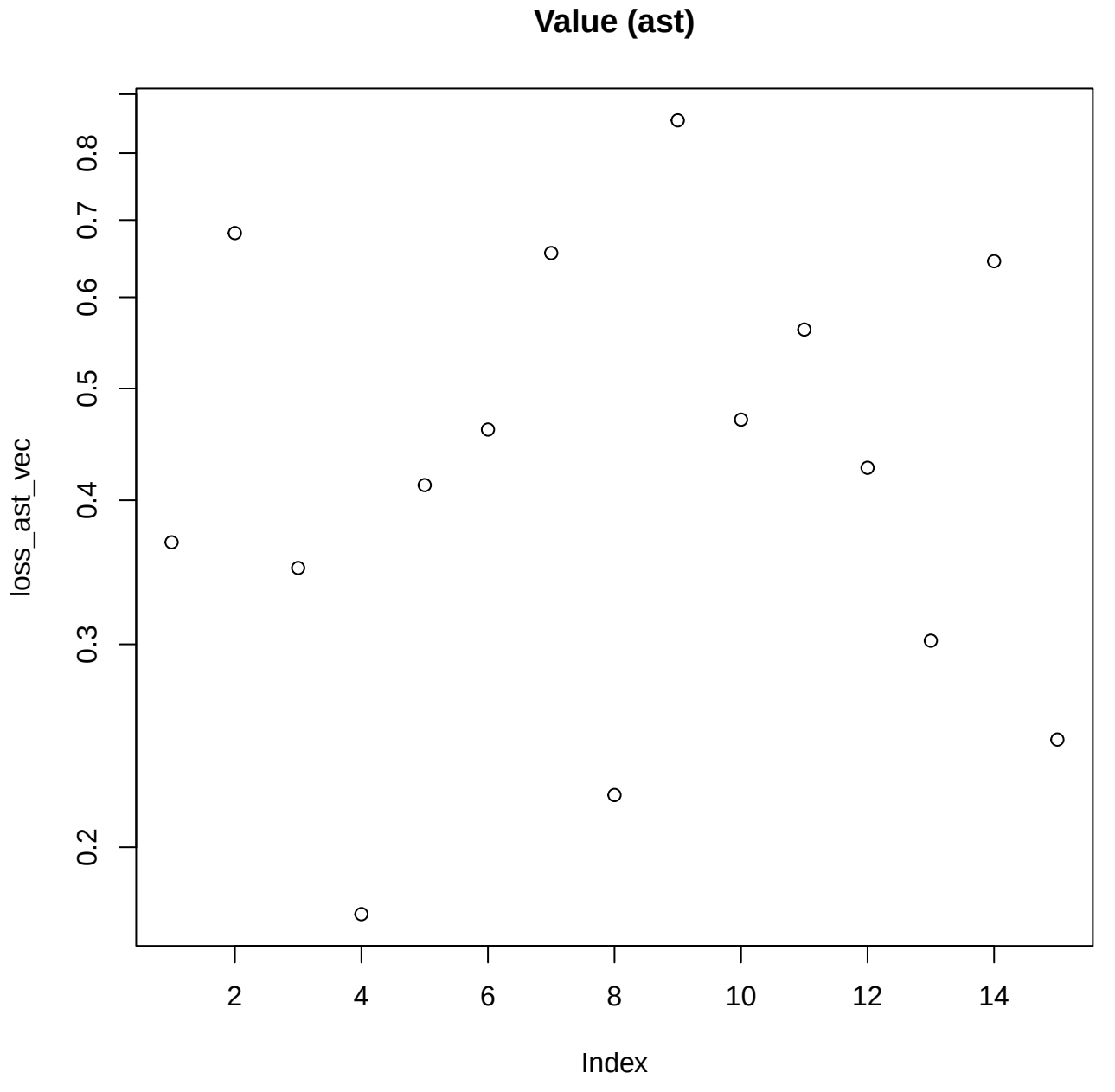
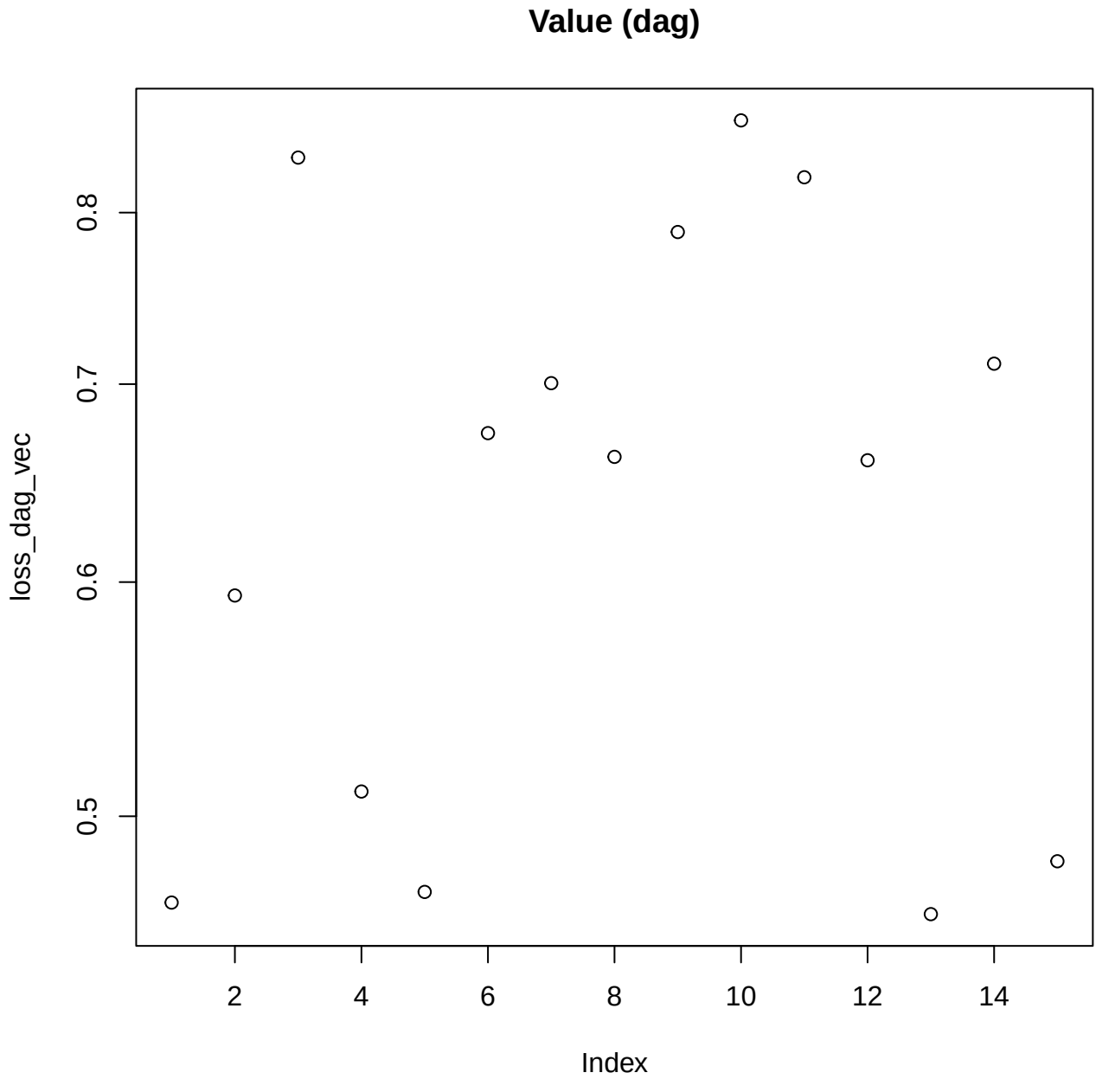
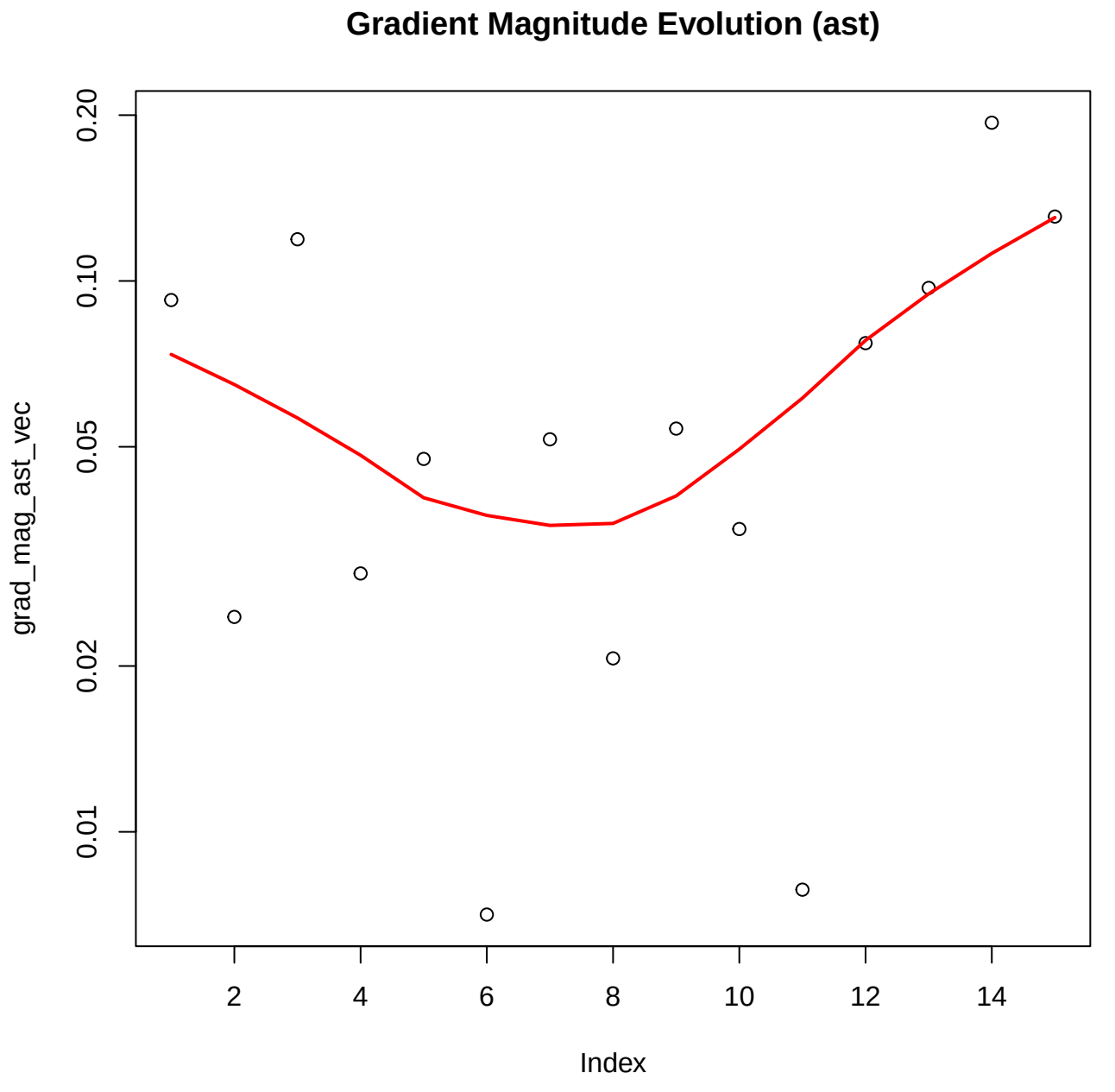
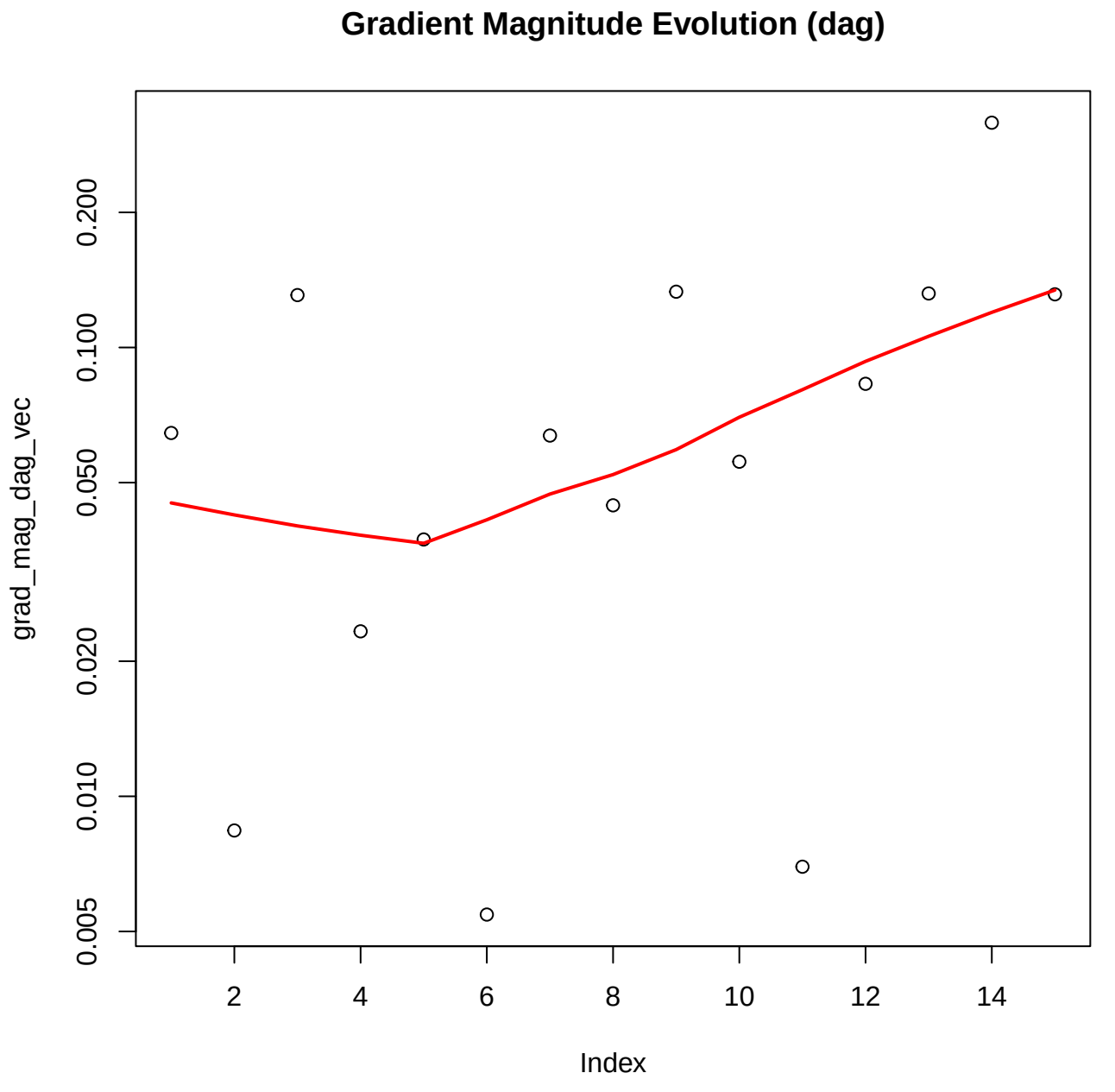
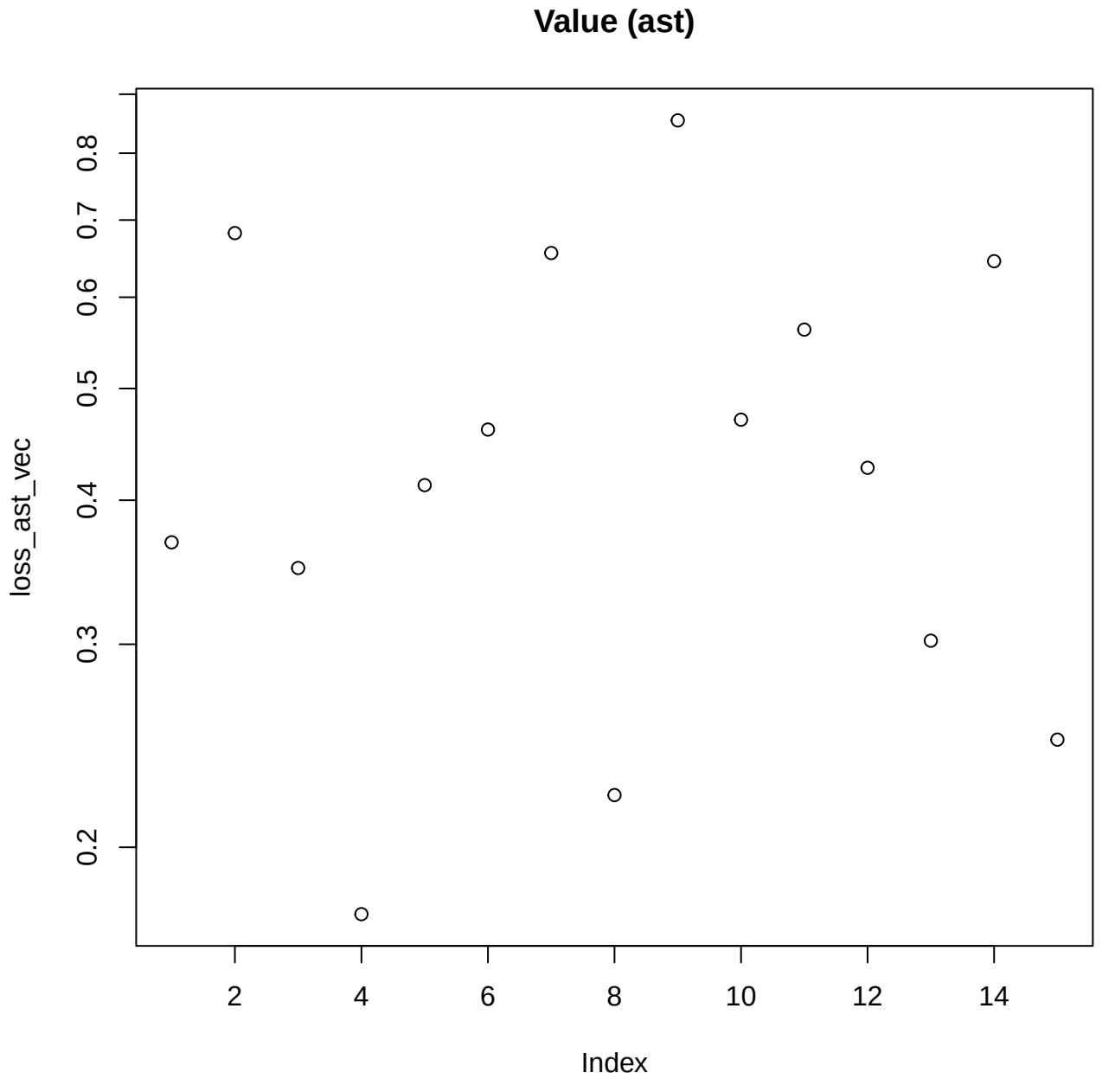

Gradient Magnitude Evolution (dag)
grad_mag_dag_vec
0.005
0.010
0.020
0.050
0.100
0.200
2
4
6
8
10
12
14
Index

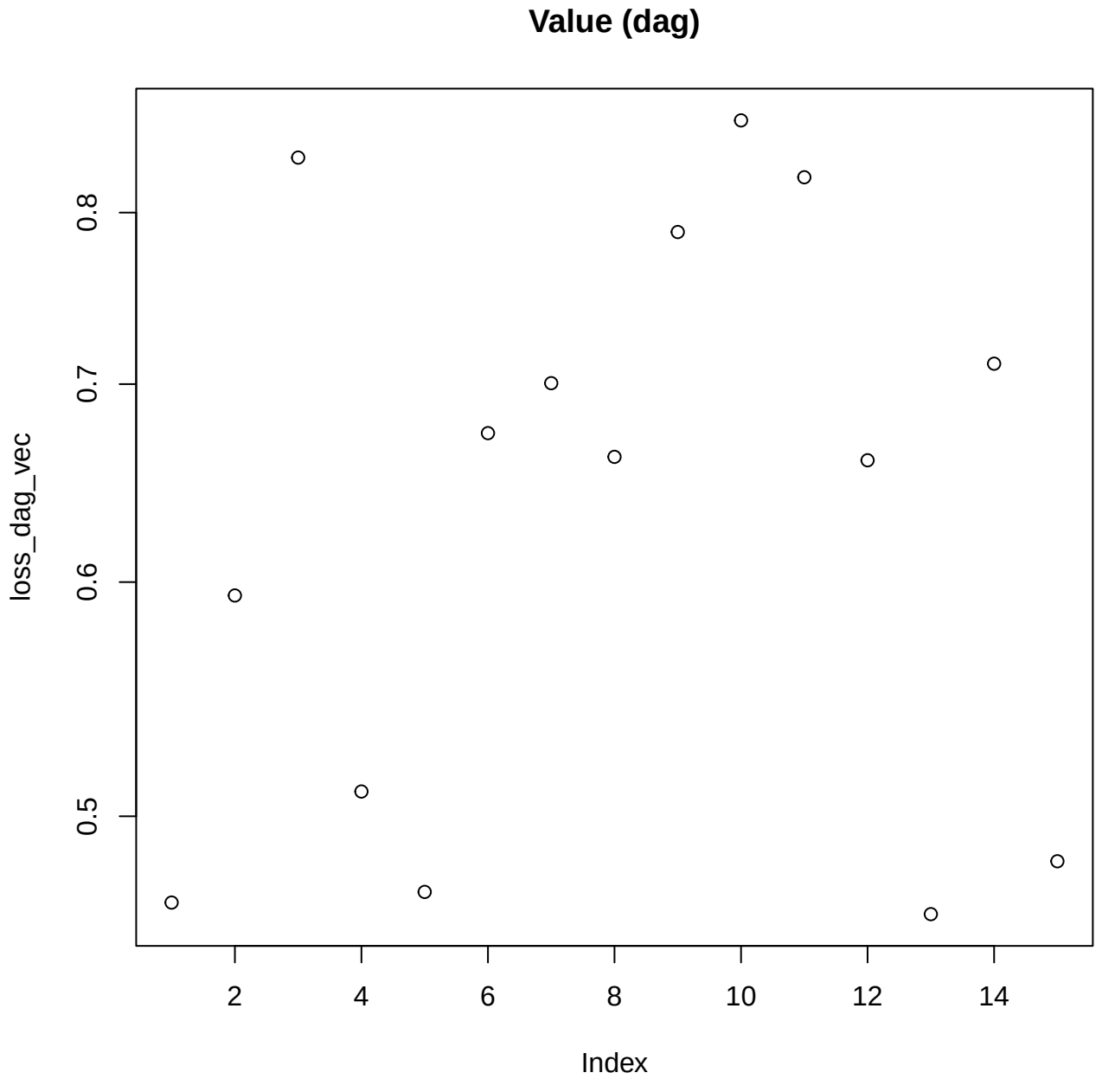

Value (ast)
loss_ast_vec
0.2
0.3
0.4
0.5
0.6
0.7
0.8
2
4
6
8
10
12
14
Index

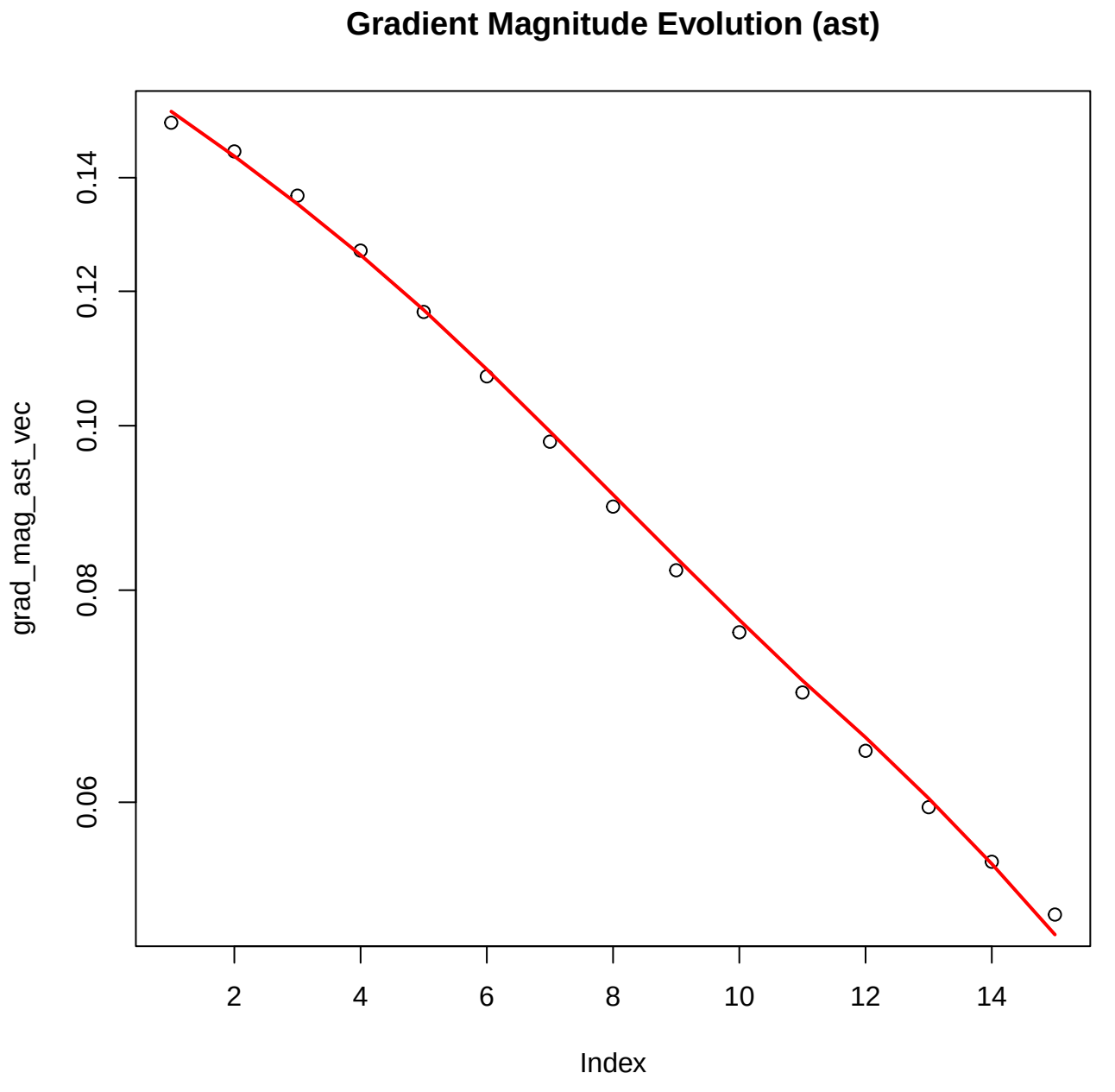

Value (dag)
loss_dag_vec
0.5
0.6
0.7
0.8
2
4
6
8
10
12
14
Index

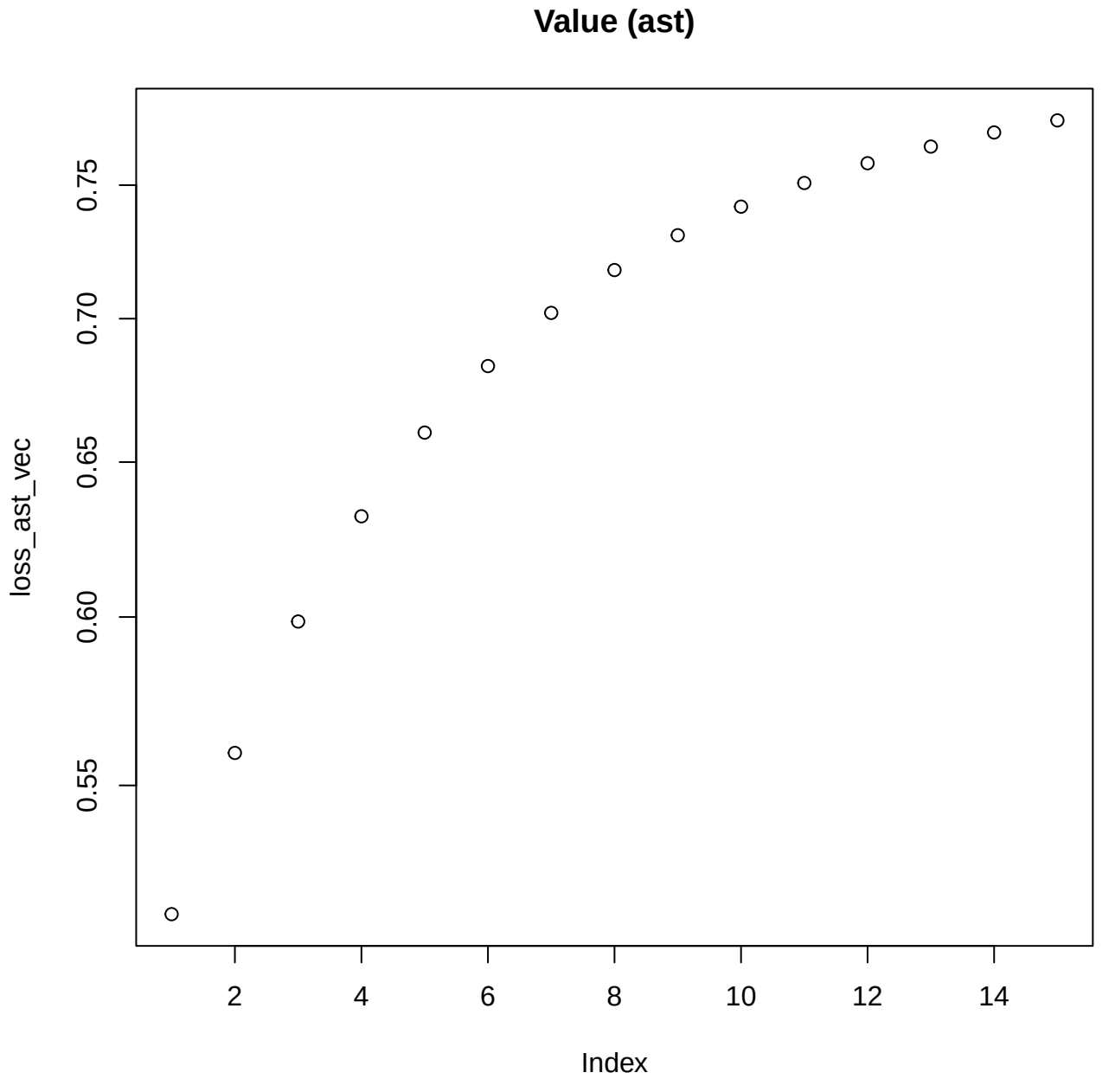

Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
0.01
0.02
0.05
0.10
0.20
2
4
6
8
10
12
14
Index

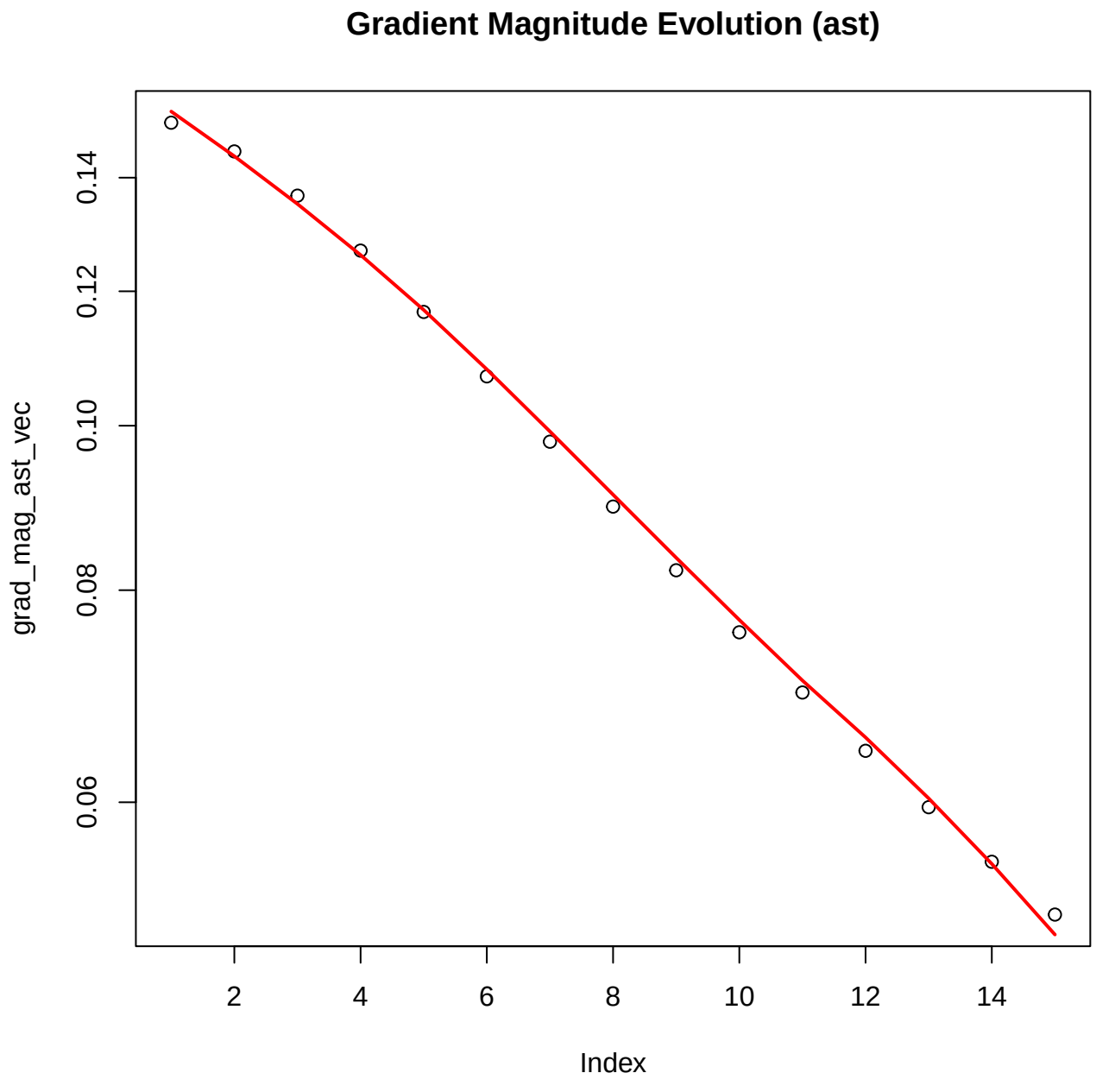

Gradient Magnitude Evolution (dag)
grad_mag_dag_vec
0.005
0.010
0.020
0.050
0.100
0.200
2
4
6
8
10
12
14
Index

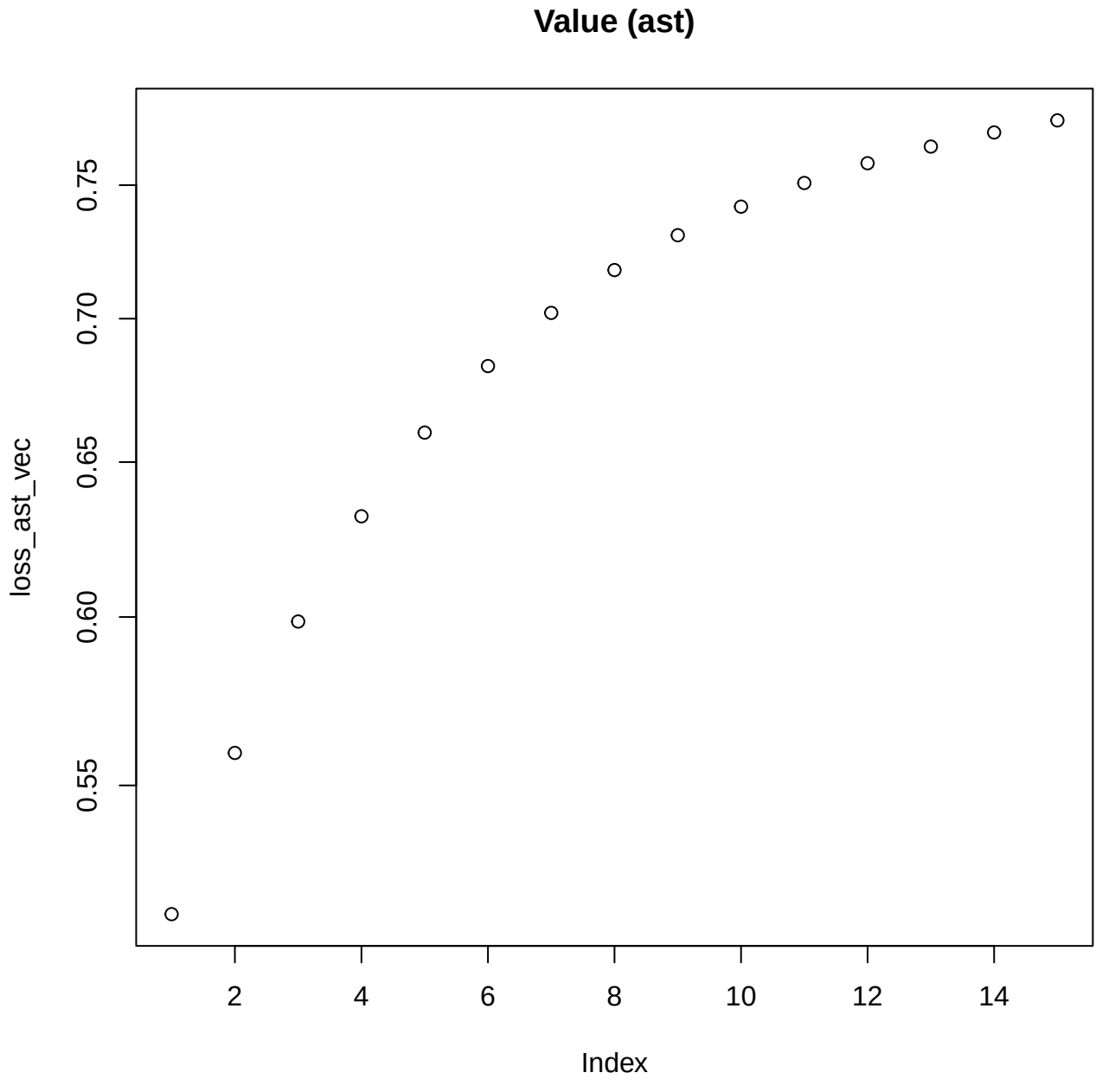

Value (ast)
loss_ast_vec
0.2
0.3
0.4
0.5
0.6
0.7
0.8
2
4
6
8
10
12
14
Index

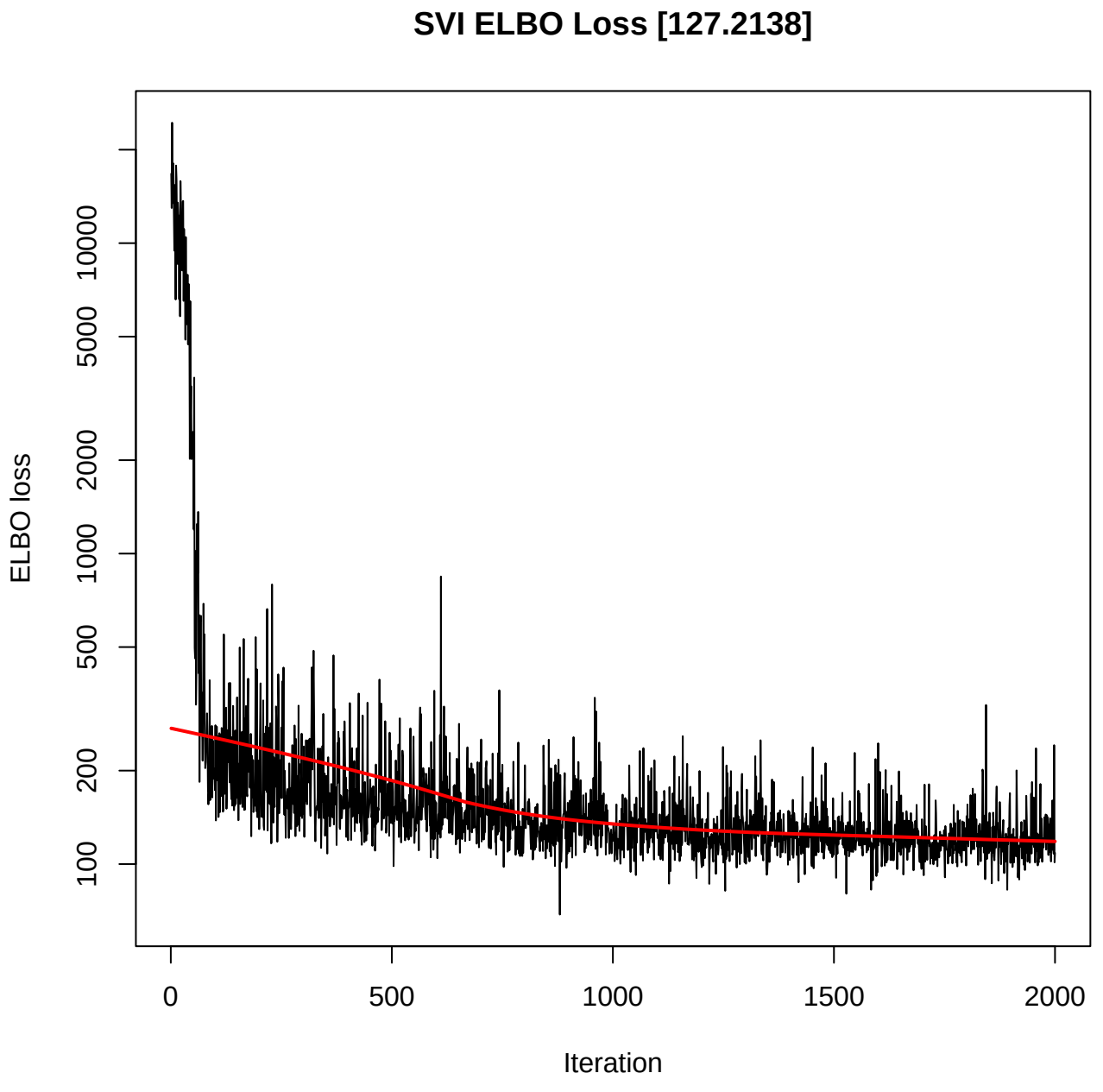

Value (dag)
loss_dag_vec
0.5
0.6
0.7
0.8
2
4
6
8
10
12
14
Index

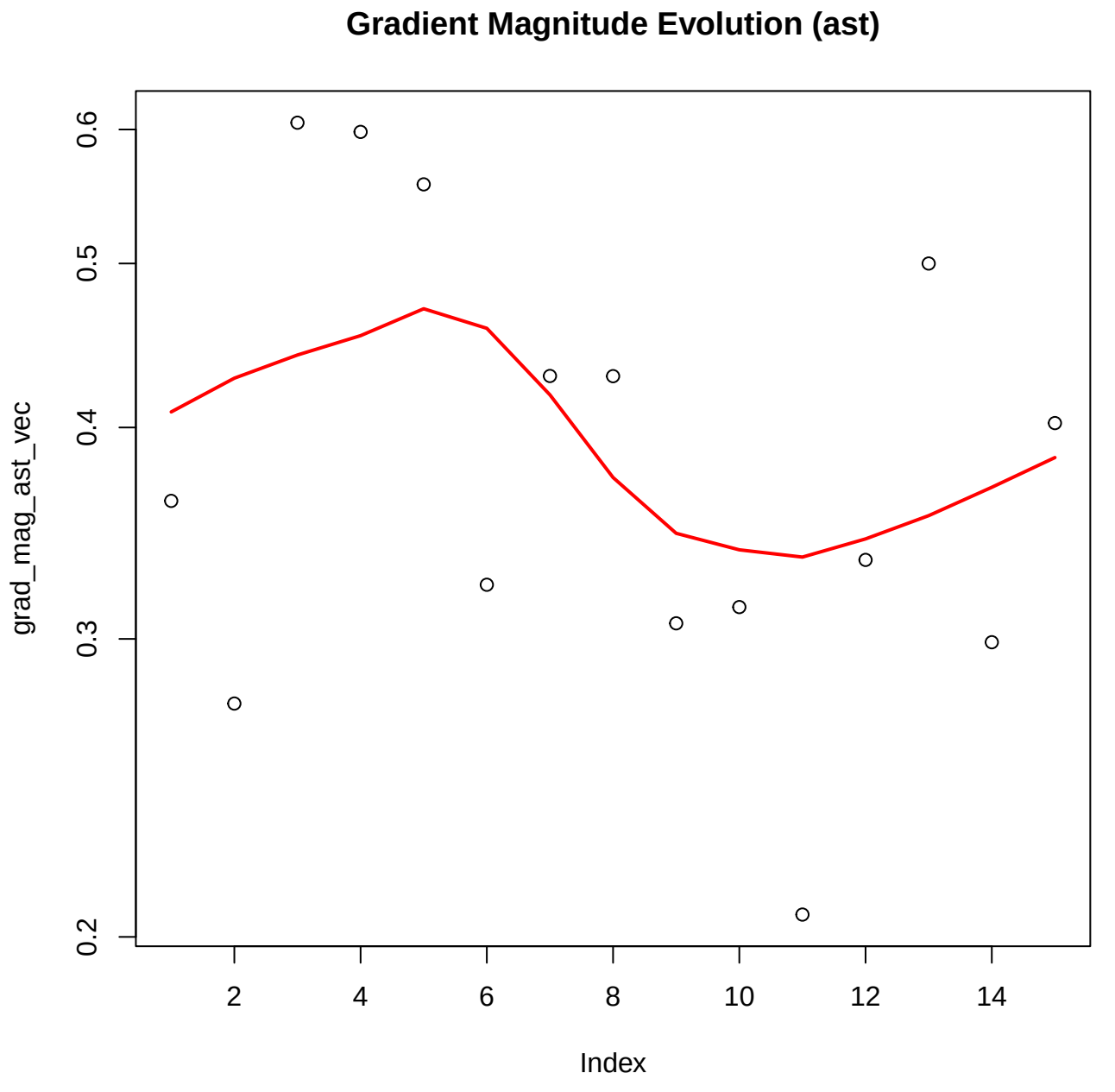

Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
0.06
0.08
0.10
0.12
0.14
2
4
6
8
10
12
14
Index

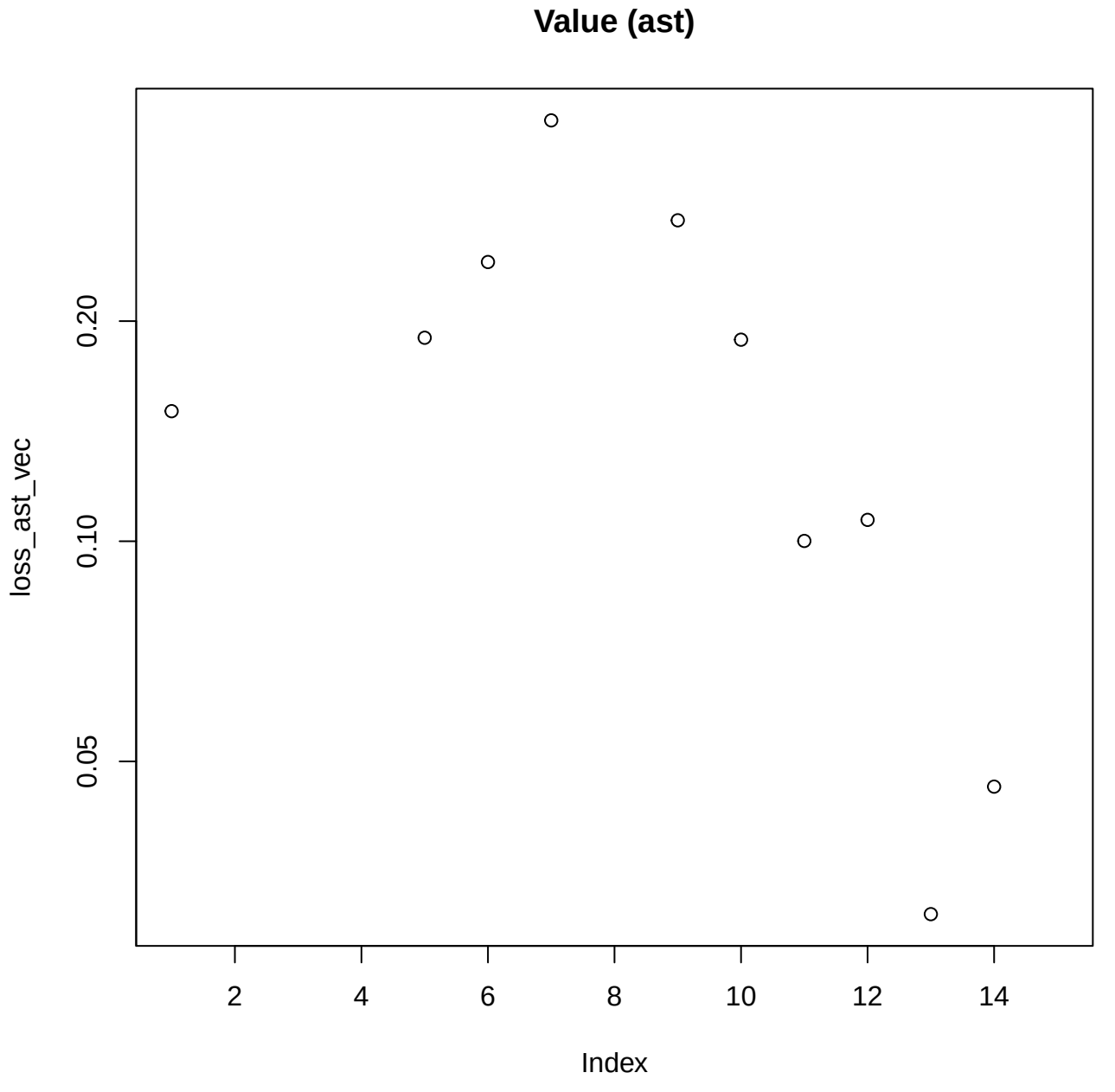

Value (ast)
loss_ast_vec
0.55
0.60
0.65
0.70
0.75
2
4
6
8
10
12
14
Index

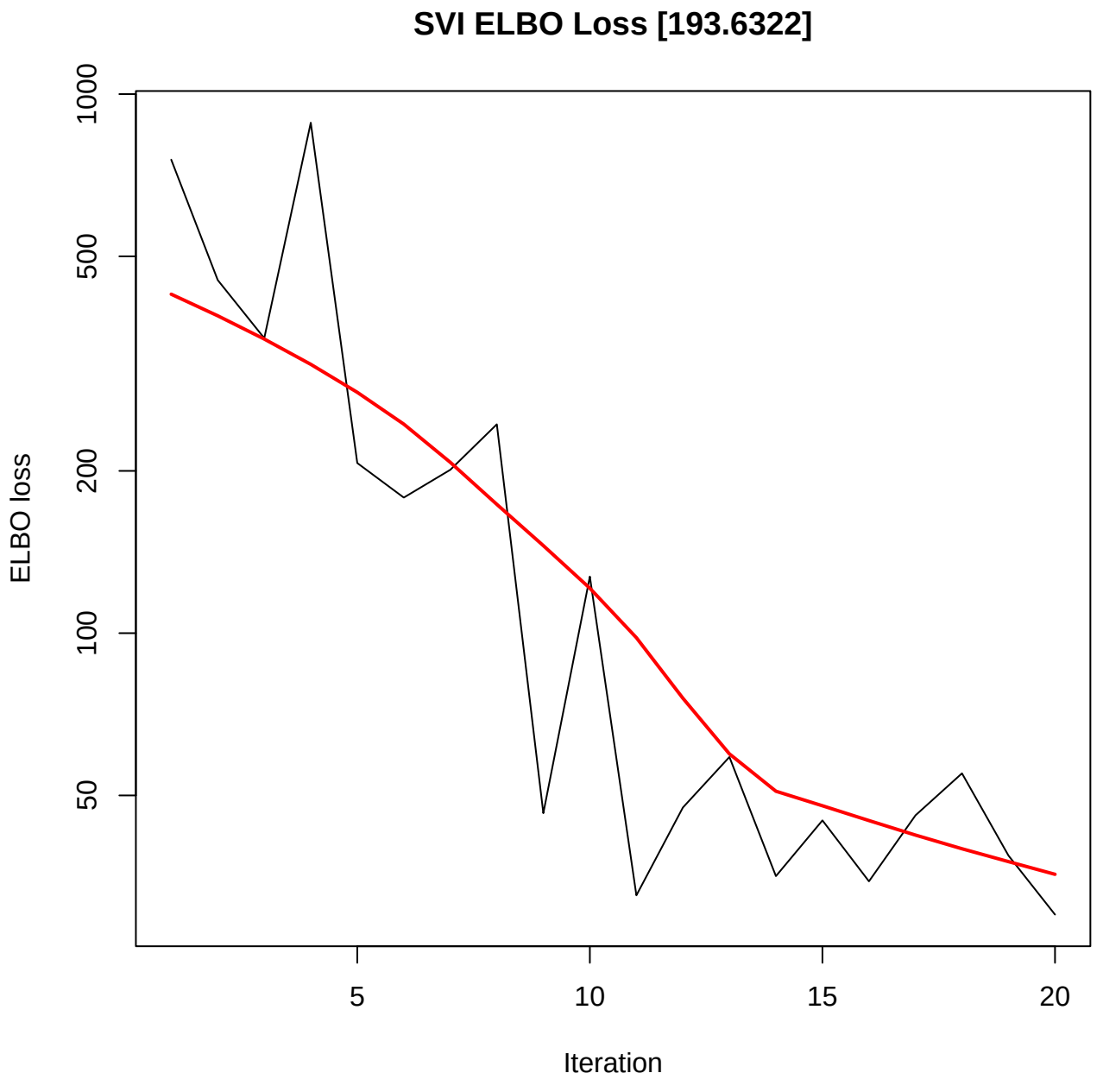

Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
0.06
0.08
0.10
0.12
0.14
2
4
6
8
10
12
14
Index

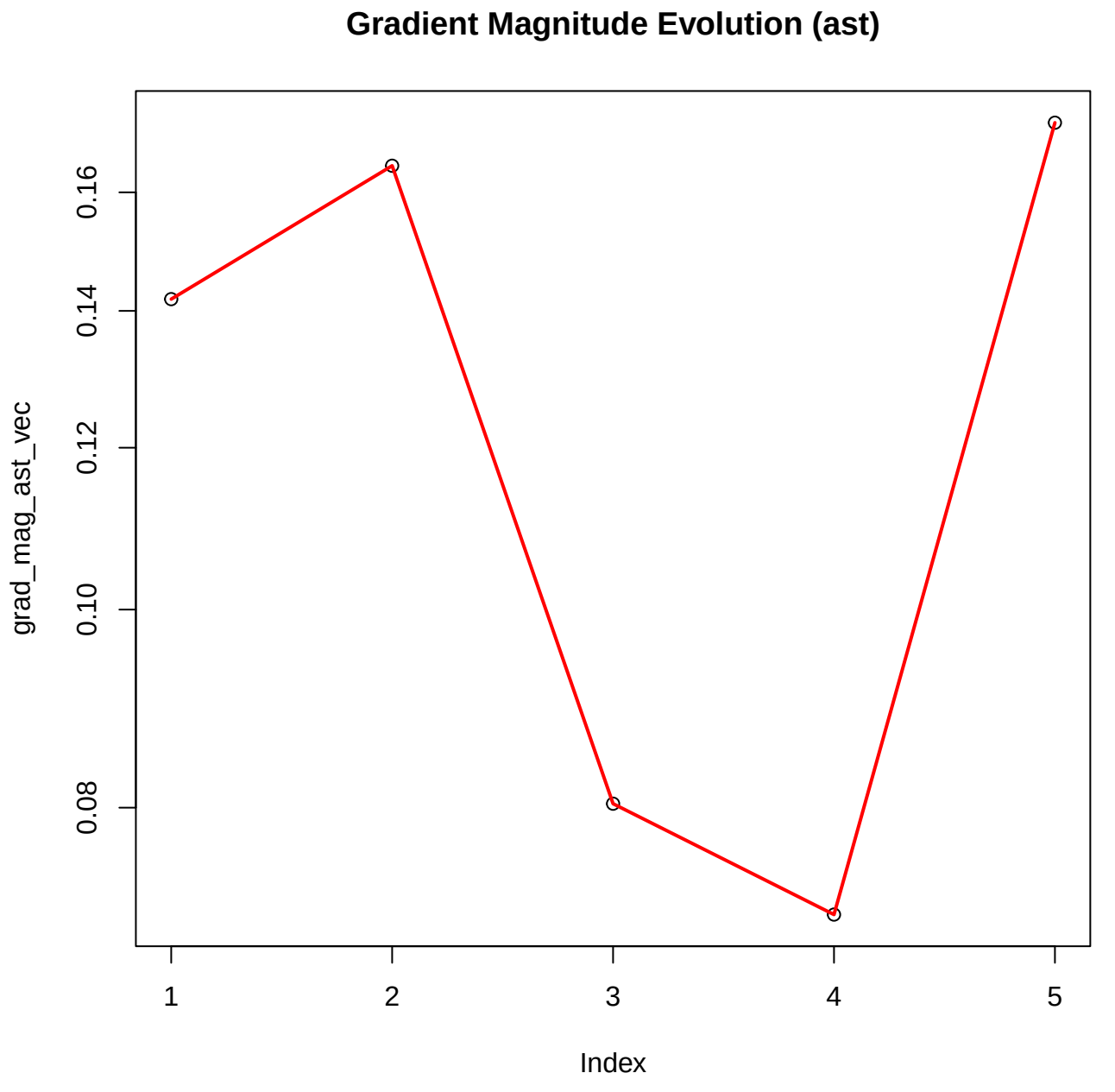

Value (ast)
loss_ast_vec
0.55
0.60
0.65
0.70
0.75
2
4
6
8
10
12
14
Index

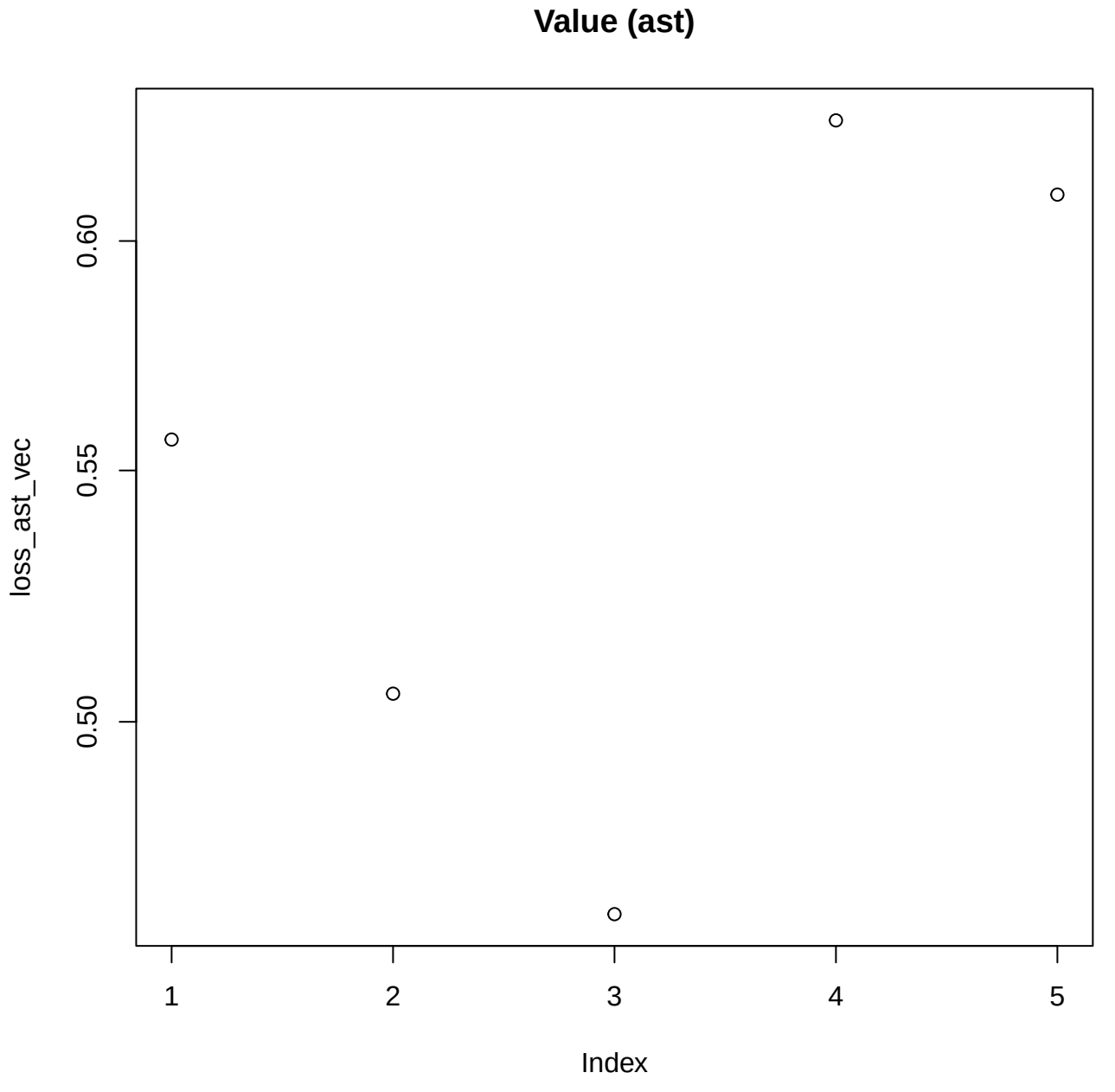

SVI ELBO Loss [127.2138]
ELBO loss
10000
5000
2000
1000
500
200
100
0
500
1000
1500
2000
Iteration

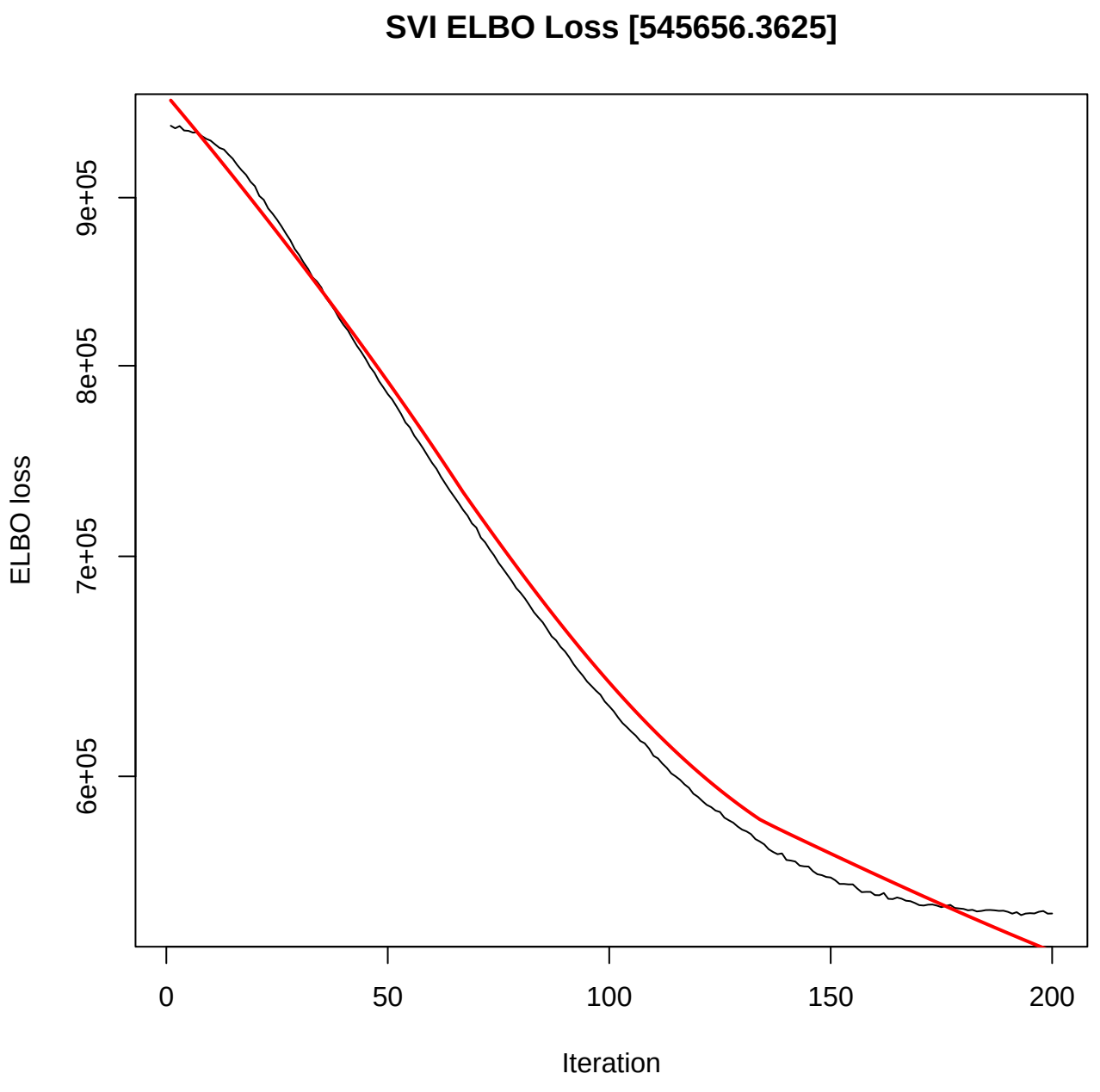

Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
0.2
0.3
0.4
0.5
0.6
2
4
6
8
10
12
14
Index

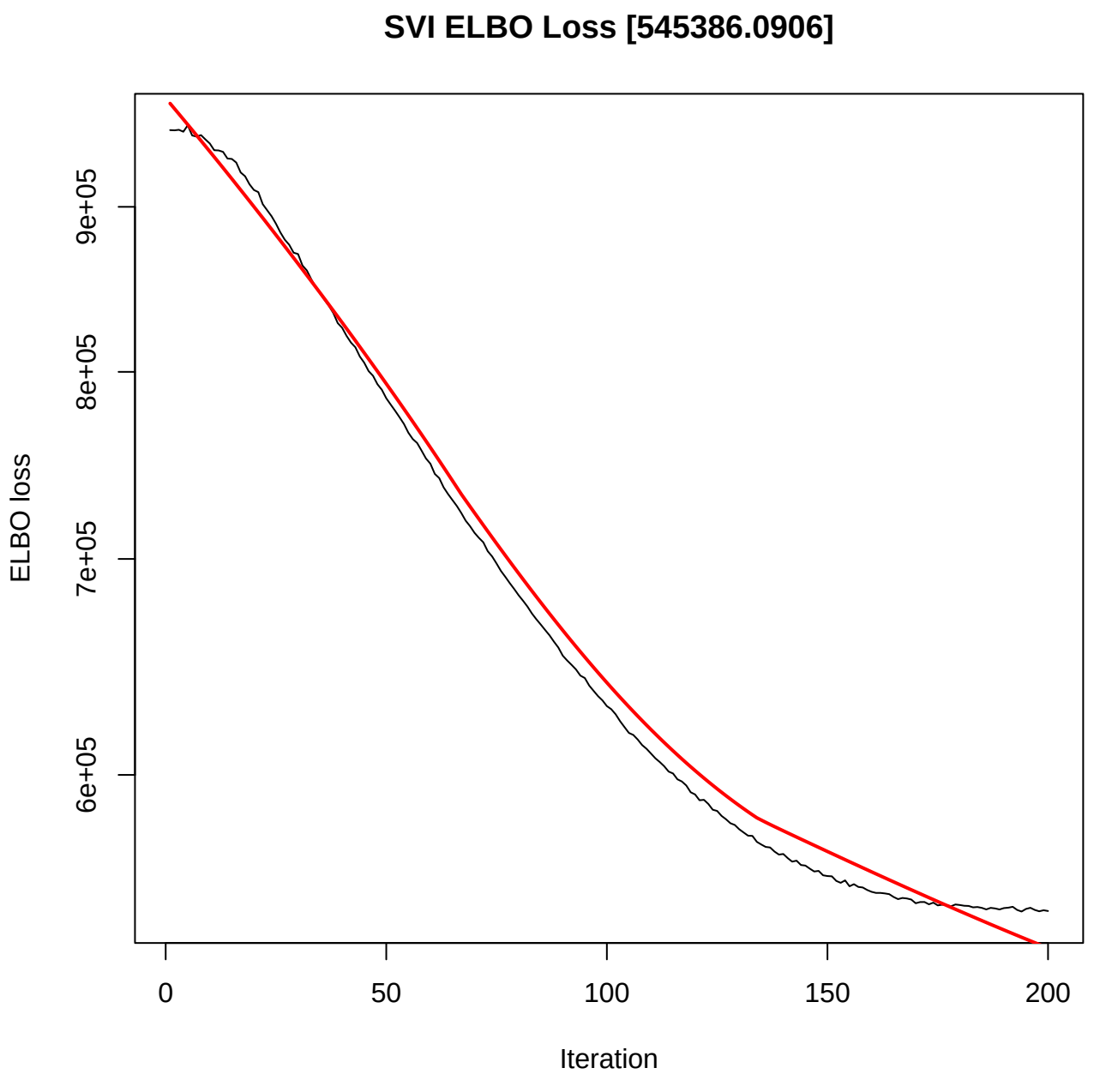

Value (ast)
loss_ast_vec
0.05
0.10
0.20
2
4
6
8
10
12
14
Index

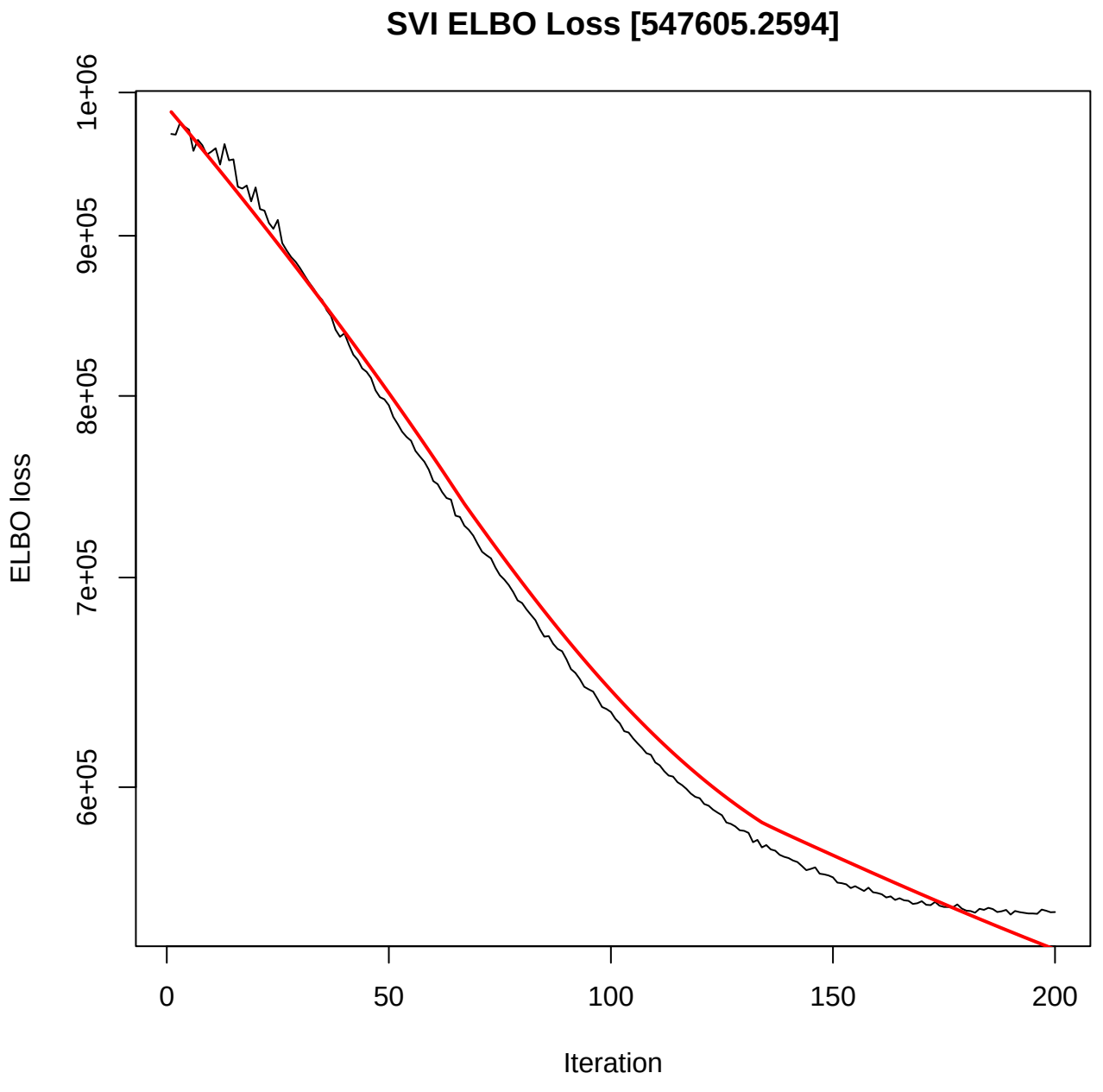

SVI ELBO Loss [193.6322]
ELBO loss
1000
500
200
100
50
5
10
15
20
Iteration

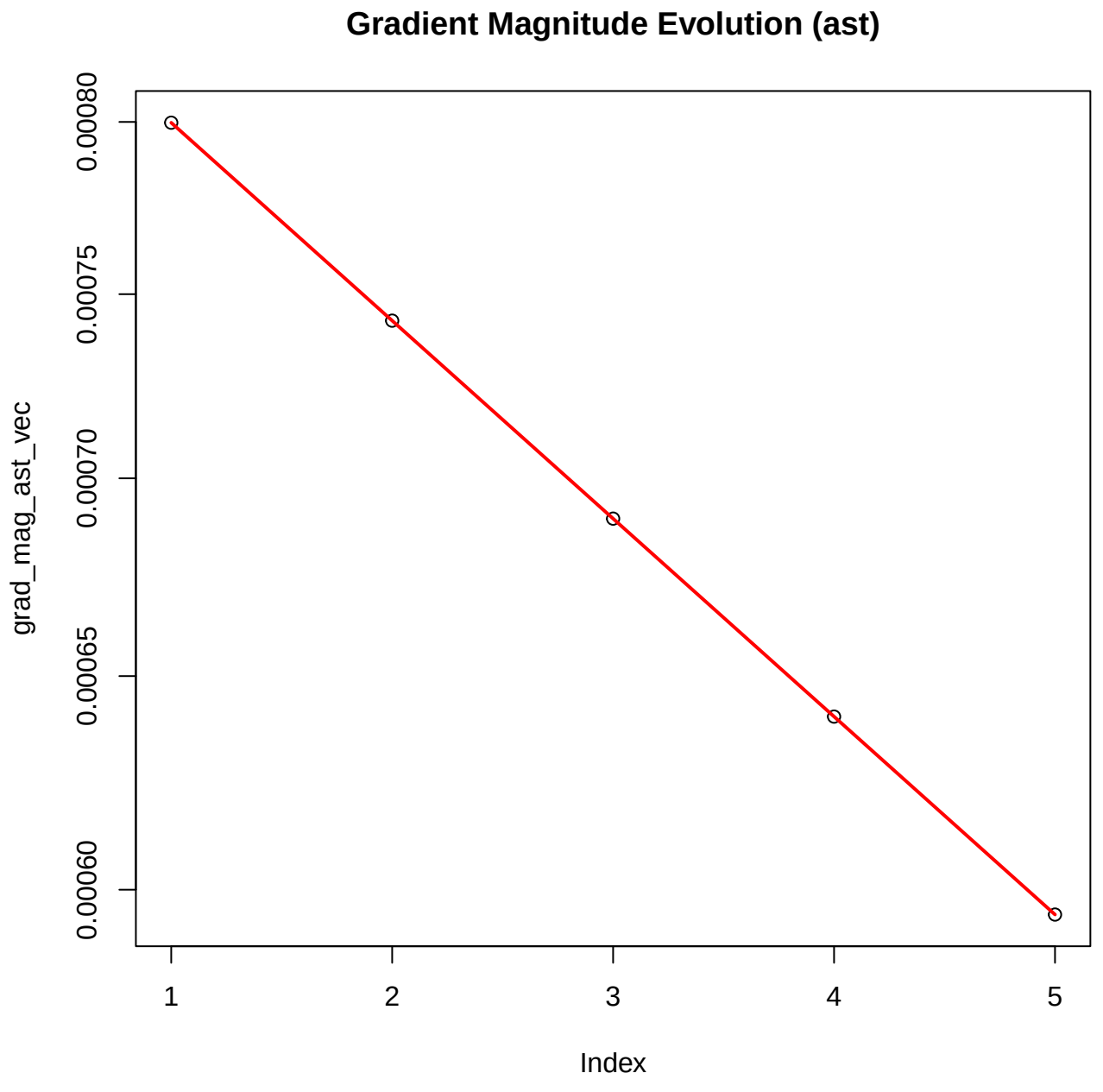

Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
0.08
0.10
0.12
0.14
0.16
1
2
3
4
5
Index

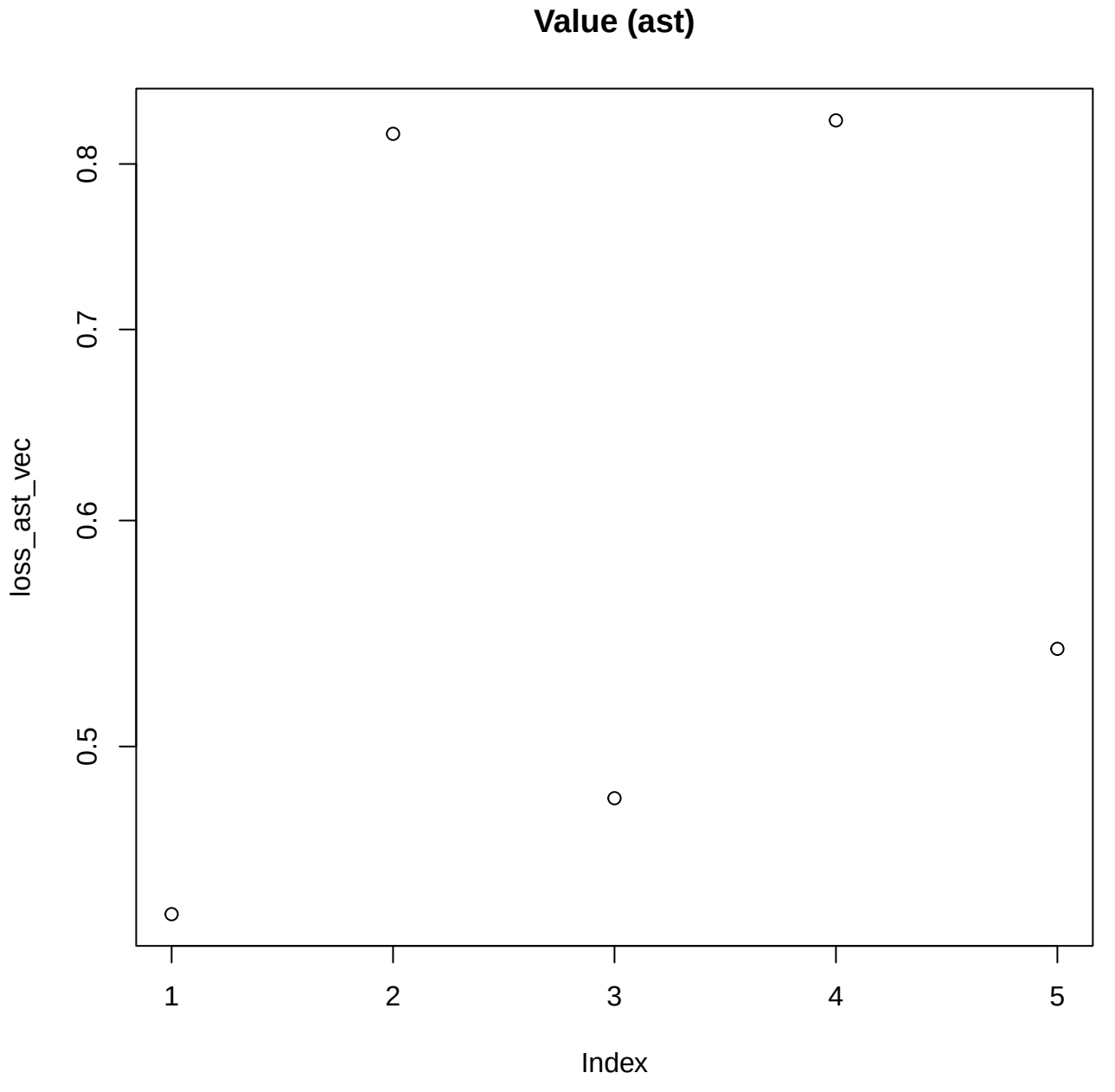

Value (ast)
loss_ast_vec
0.50
0.55
0.60
1
2
3
4
5
Index

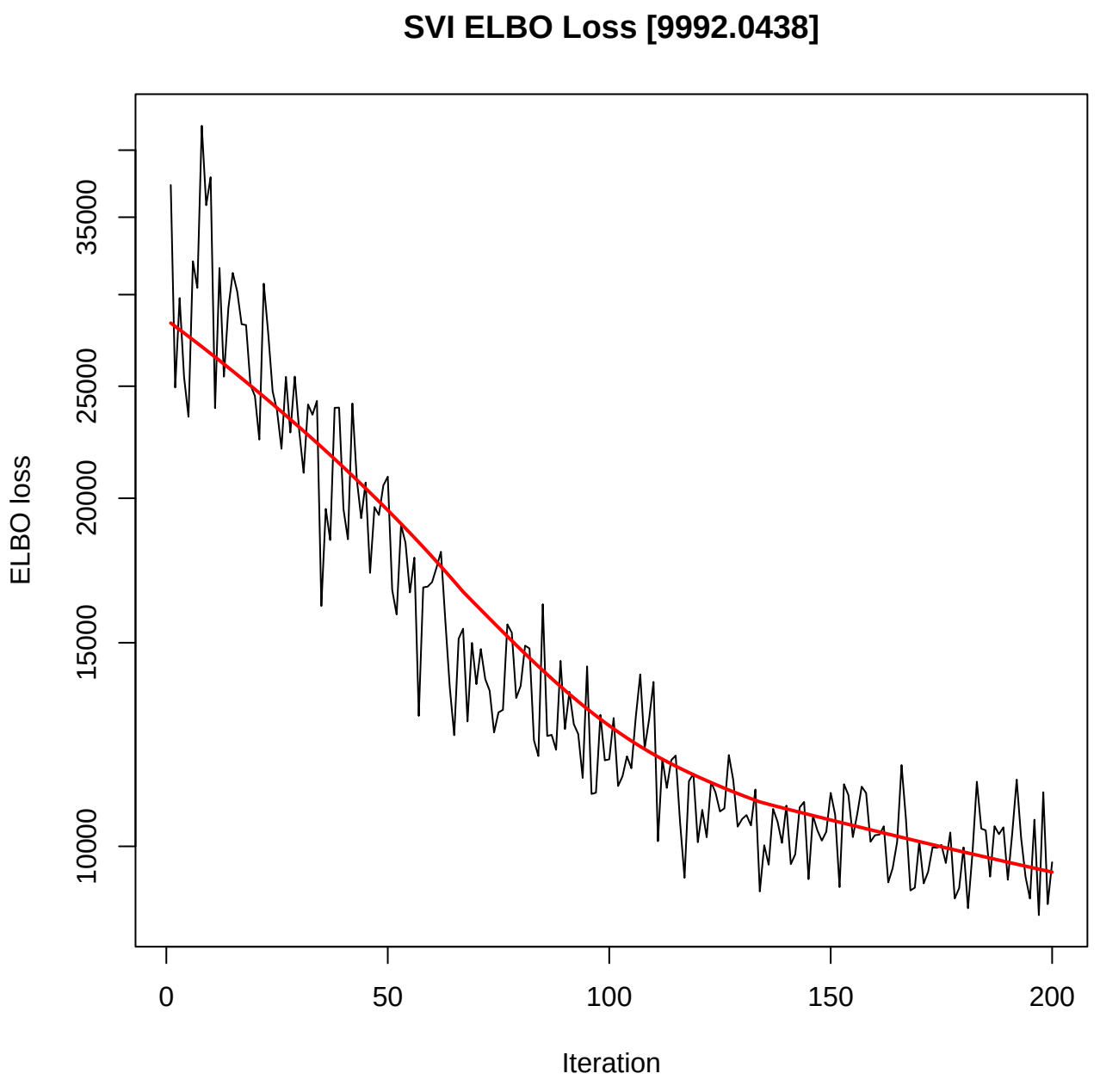

SVI ELBO Loss [545656.3625]
ELBO loss
9e+05
8e+05
7e+05
6e+05
0
50
100
150
200
Iteration

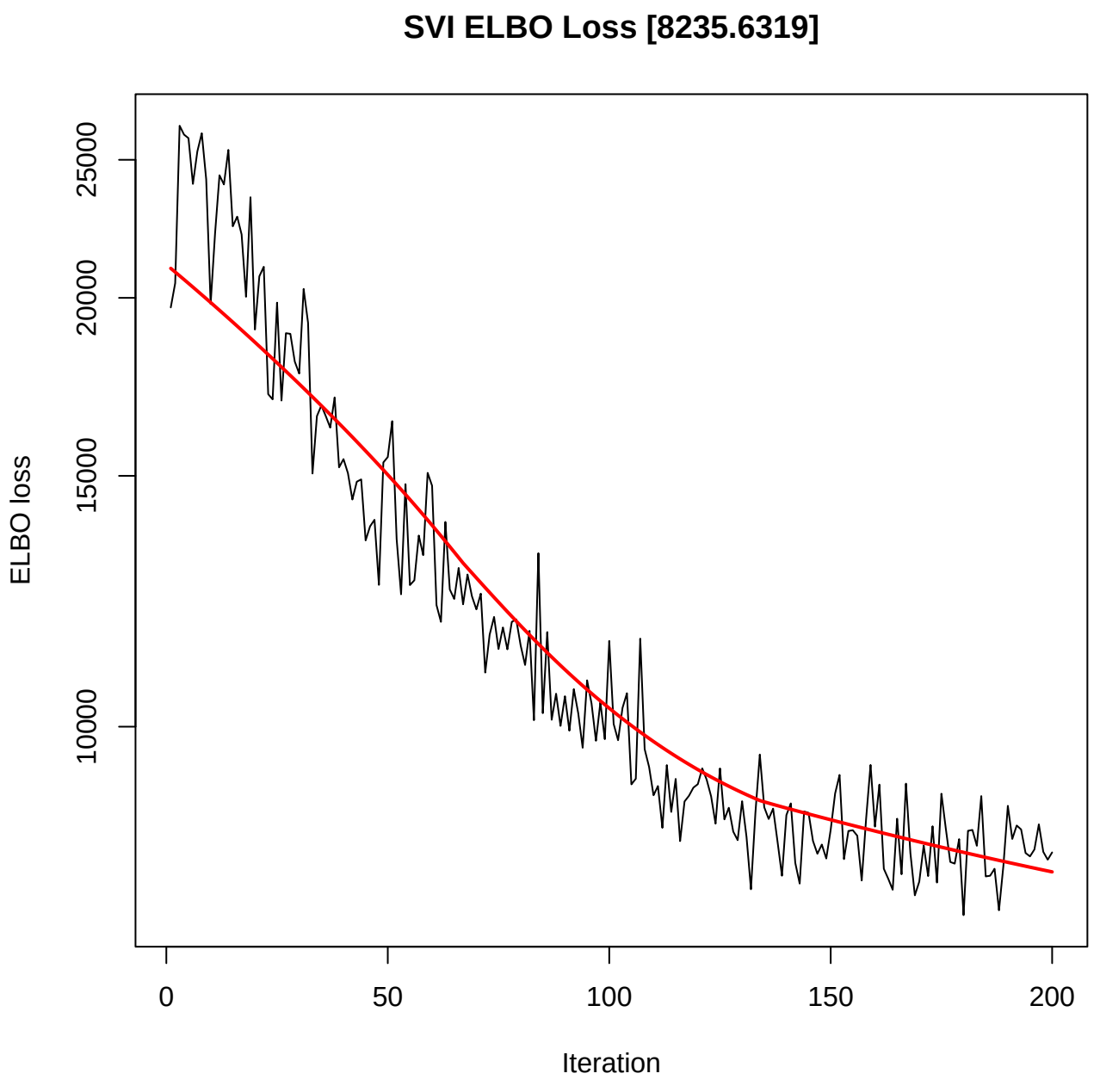

SVI ELBO Loss [545386.0906]
ELBO loss
9e+05
8e+05
7e+05
6e+05
0
50
100
150
200
Iteration

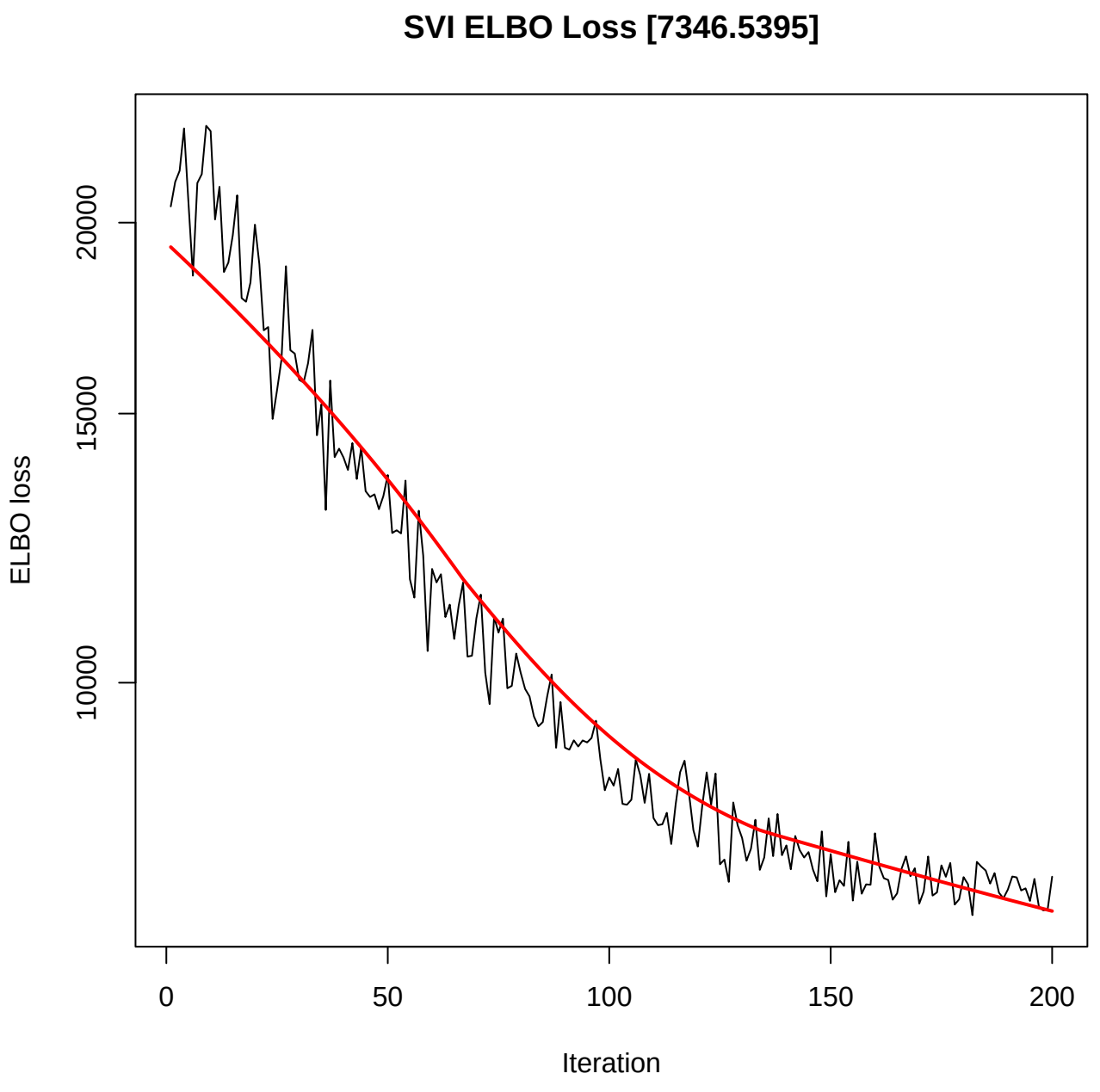

SVI ELBO Loss [547605.2594]
ELBO loss
1e+06
9e+05
8e+05
7e+05
6e+05
0
50
100
150
200
Iteration

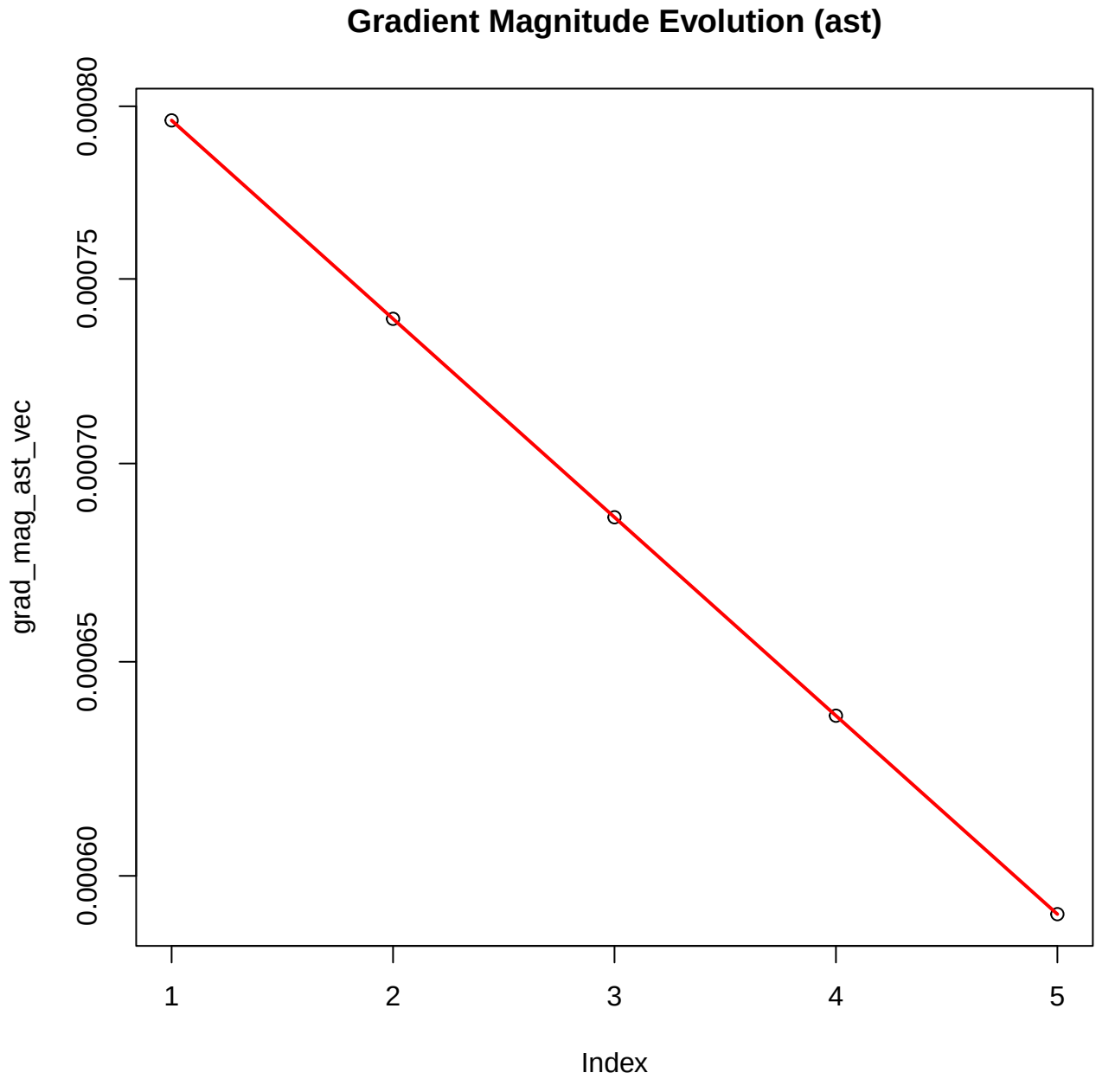

Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
0.00060
0.00065
0.00070
0.00075
0.00080
1
2
3
4
5
Index

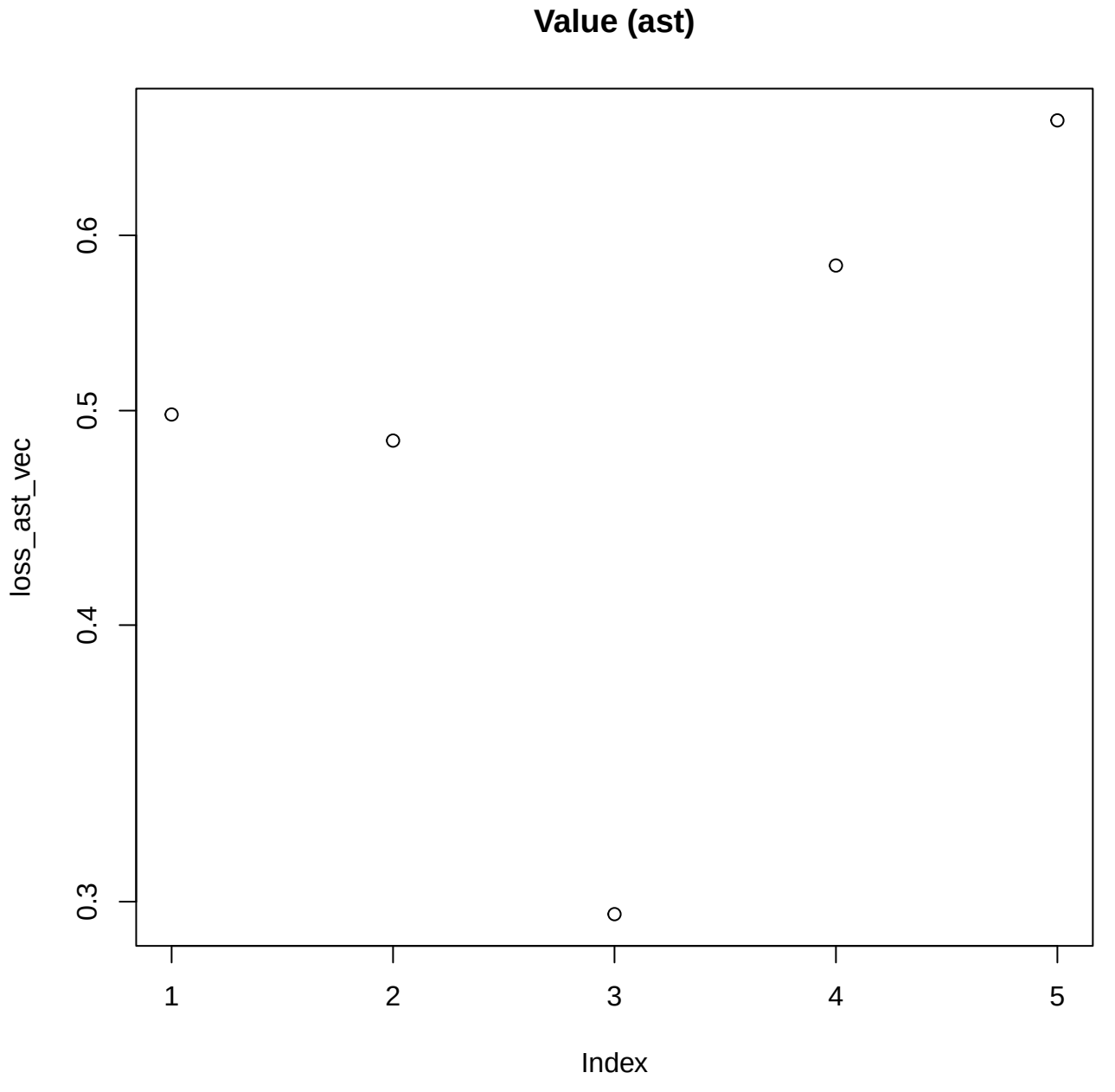

Value (ast)
loss_ast_vec
0.5
0.6
0.7
0.8
1
2
3
4
5
Index

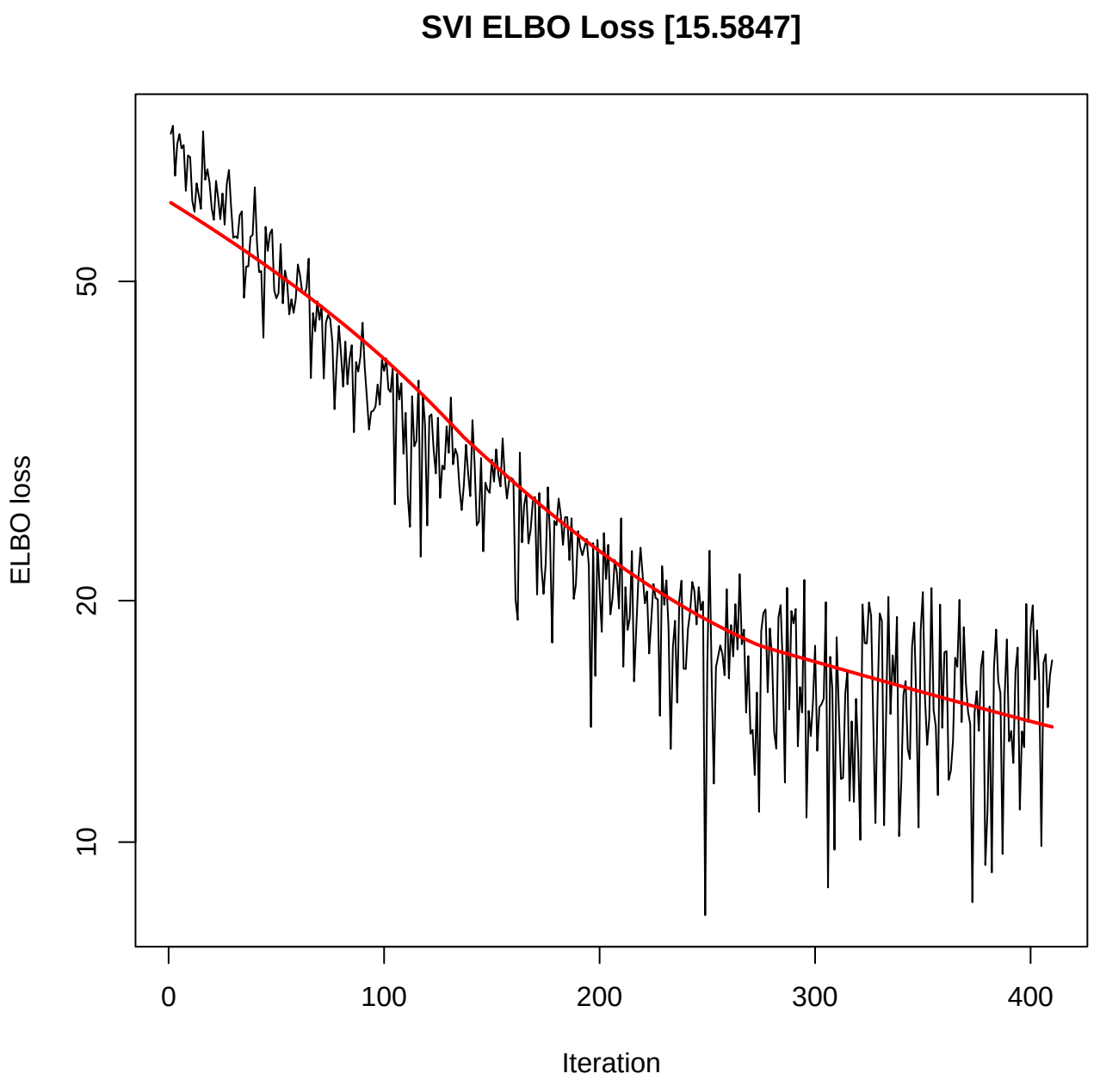

SVI ELBO Loss [9992.0438]
ELBO loss
10000
15000
20000
25000
35000
0
50
100
150
200
Iteration

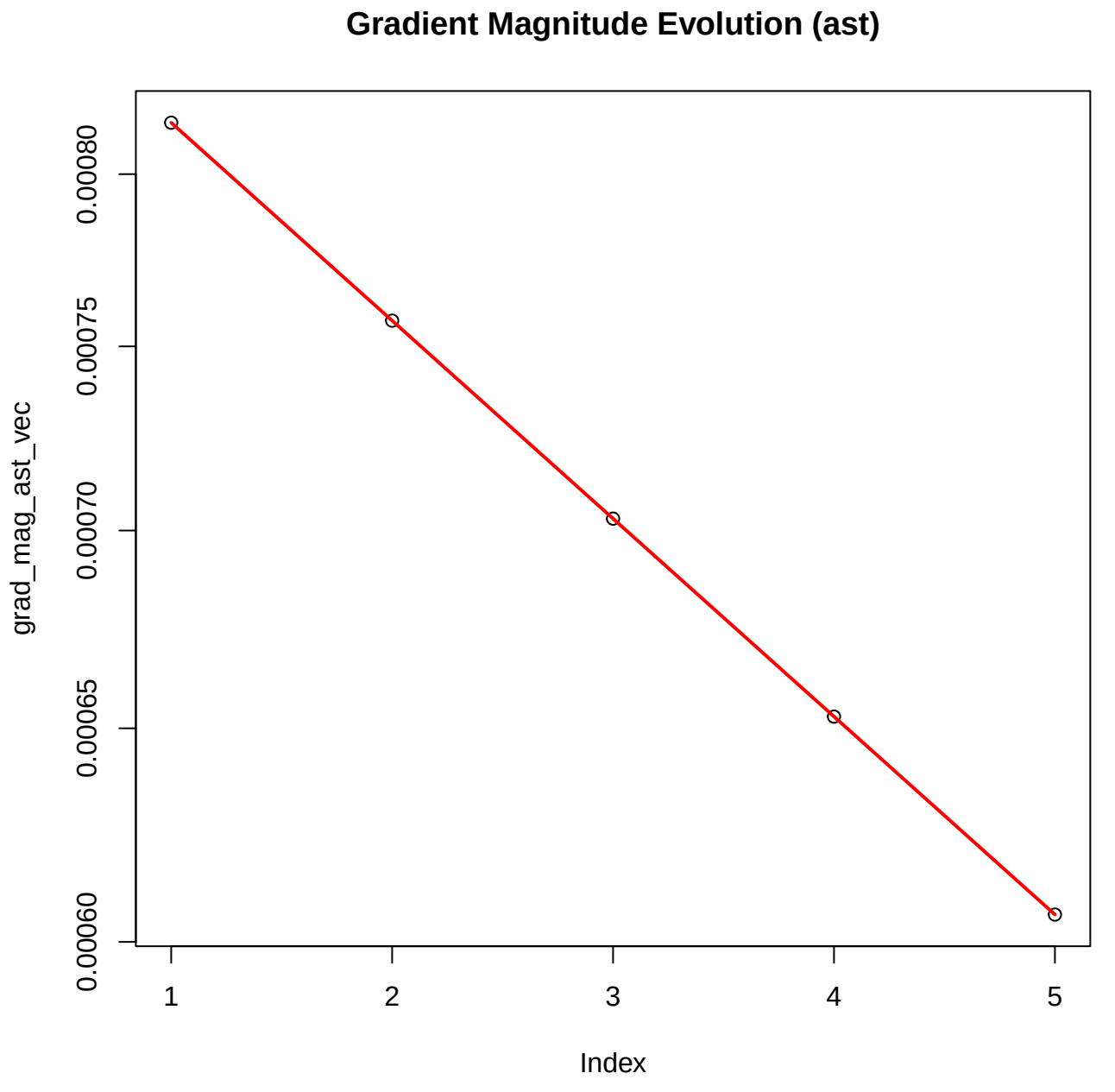

SVI ELBO Loss [8235.6319]
ELBO loss
25000
20000
15000
10000
0
50
100
150
200
Iteration

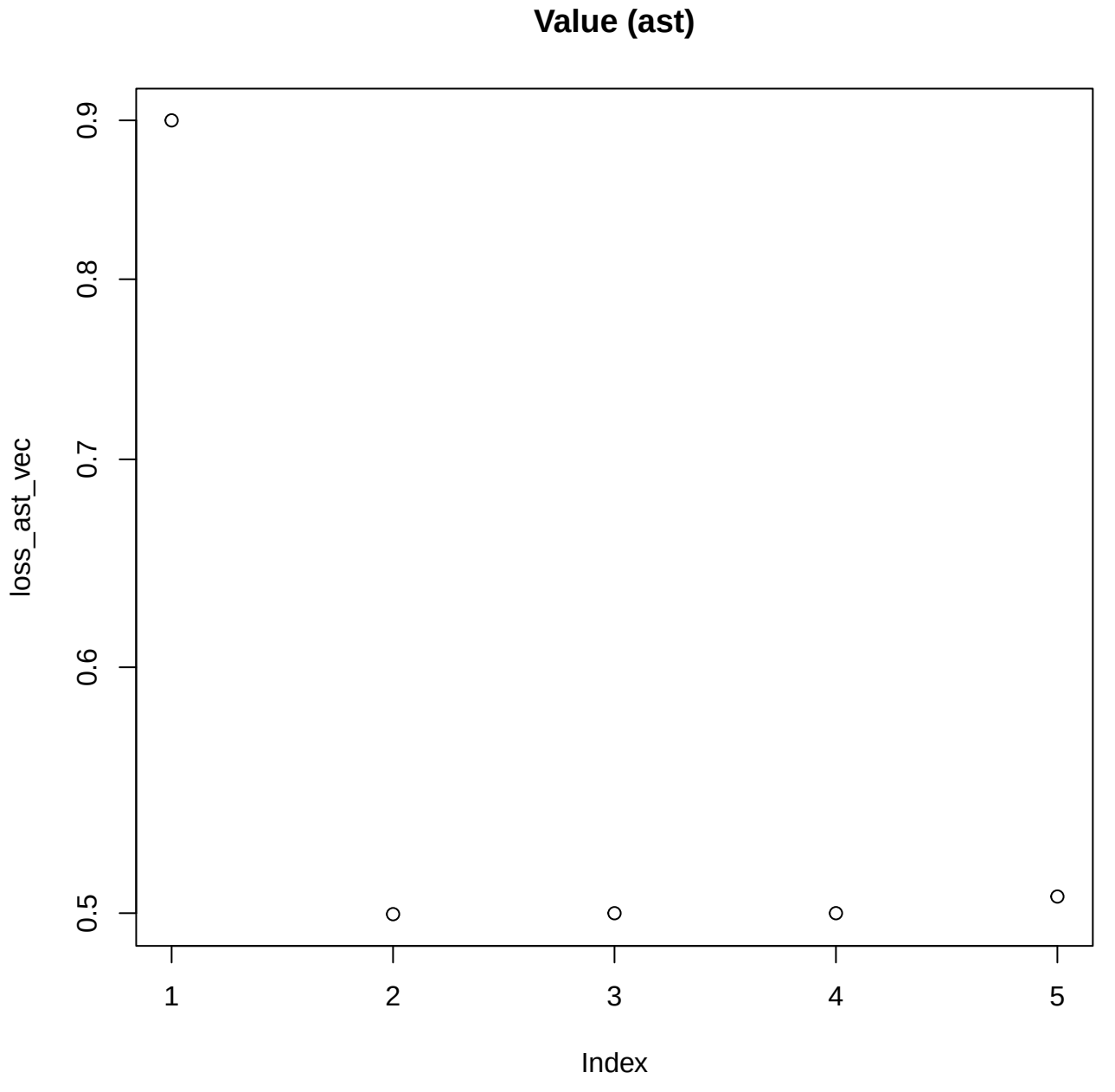

SVI ELBO Loss [7346.5395]
ELBO loss
20000
15000
10000
0
50
100
150
200
Iteration

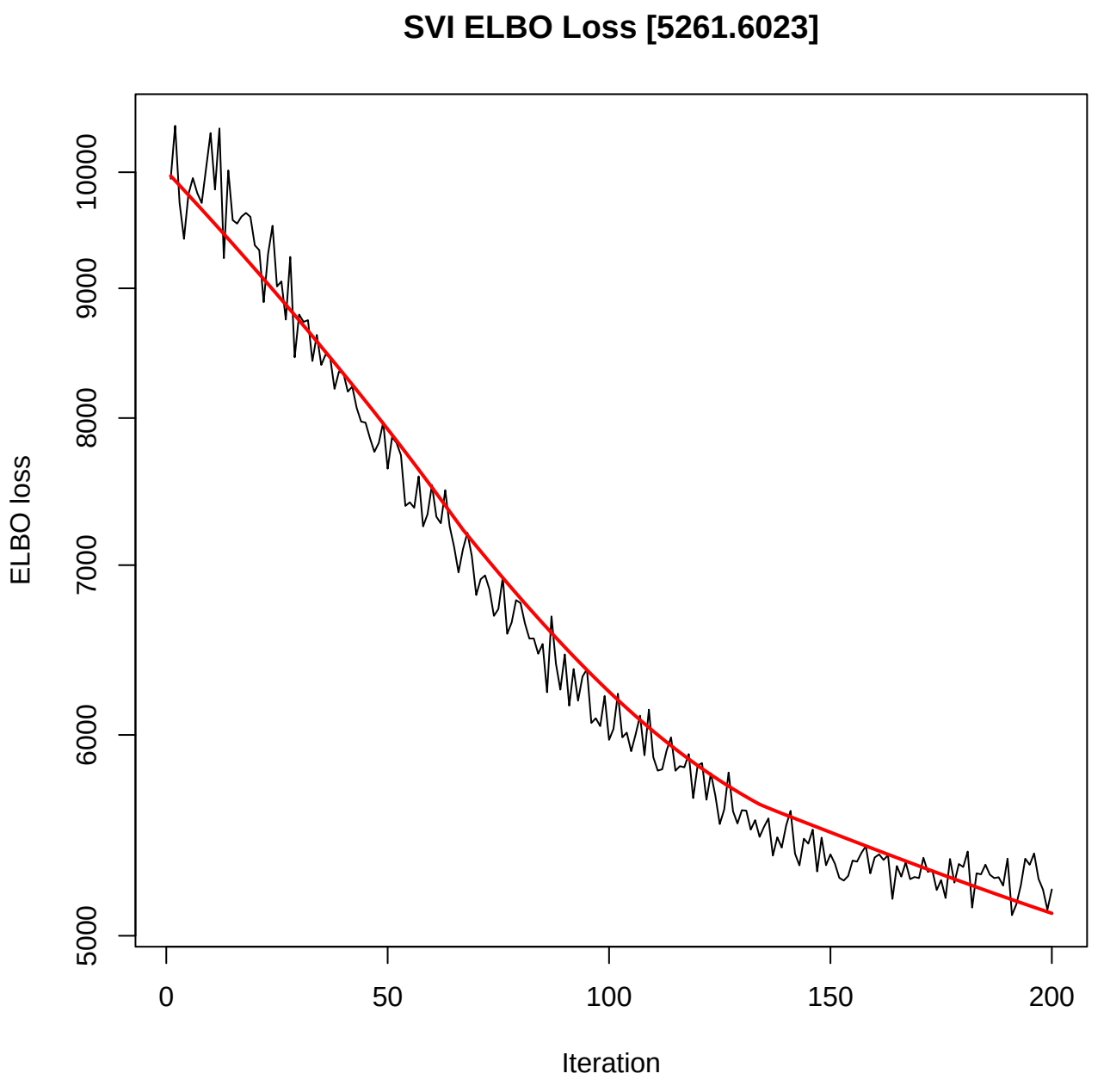

Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
0.00060
0.00065
0.00070
0.00075
0.00080
1
2
3
4
5
Index

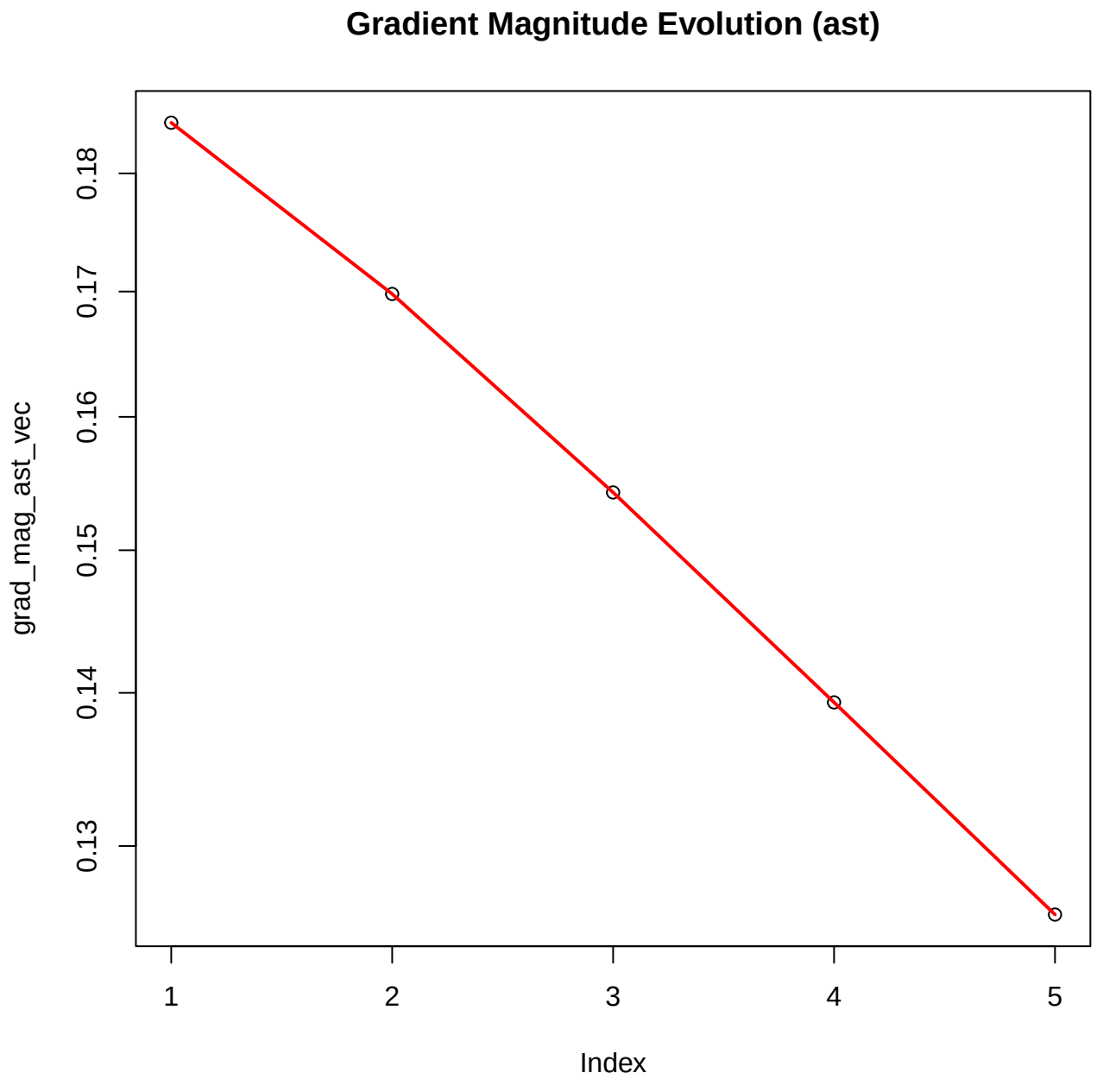

Value (ast)
loss_ast_vec
0.3
0.4
0.5
0.6
1
2
3
4
5
Index

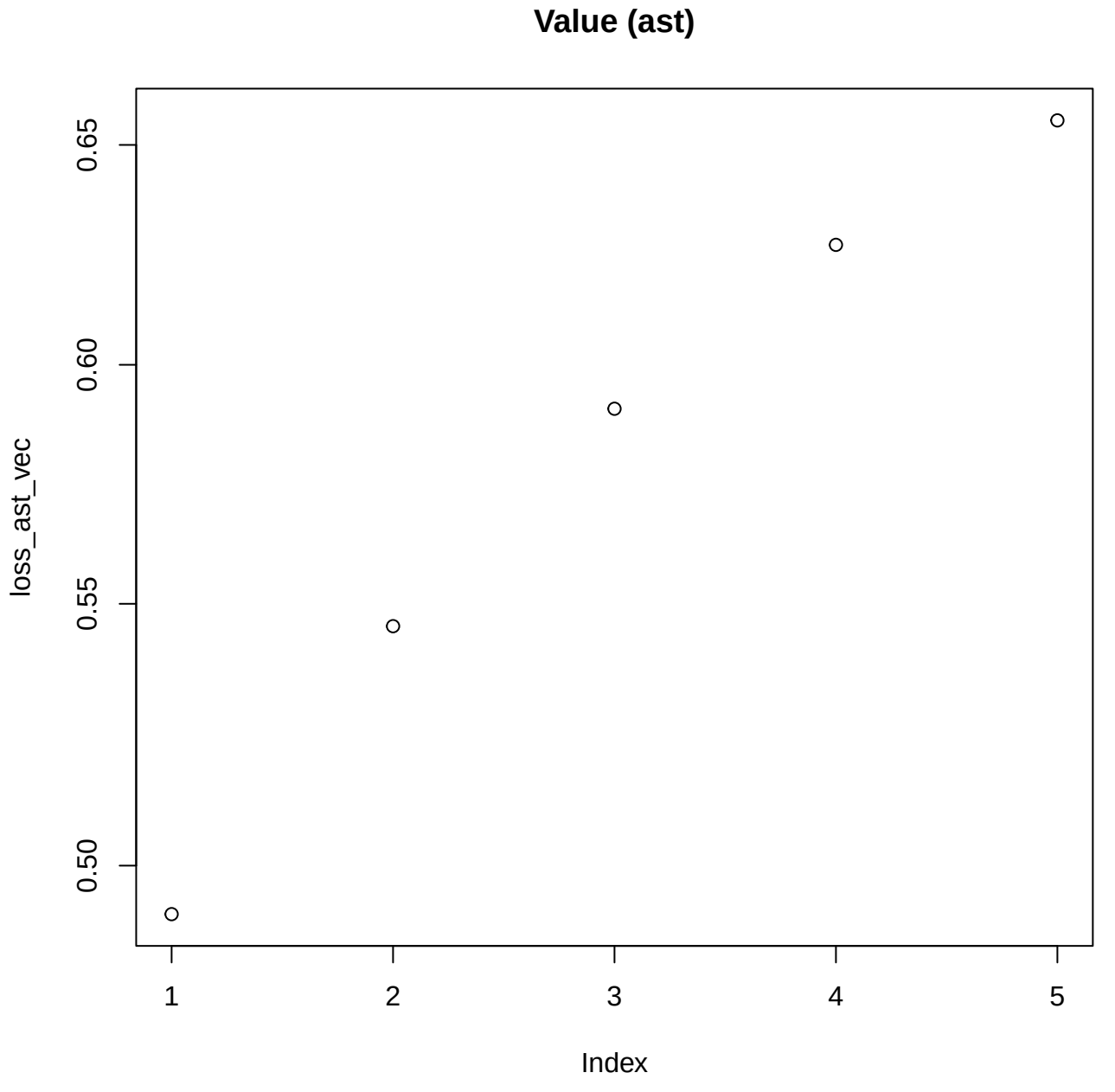

SVI ELBO Loss [15.5847]
ELBO loss
10
20
50
0
100
200
300
400
Iteration

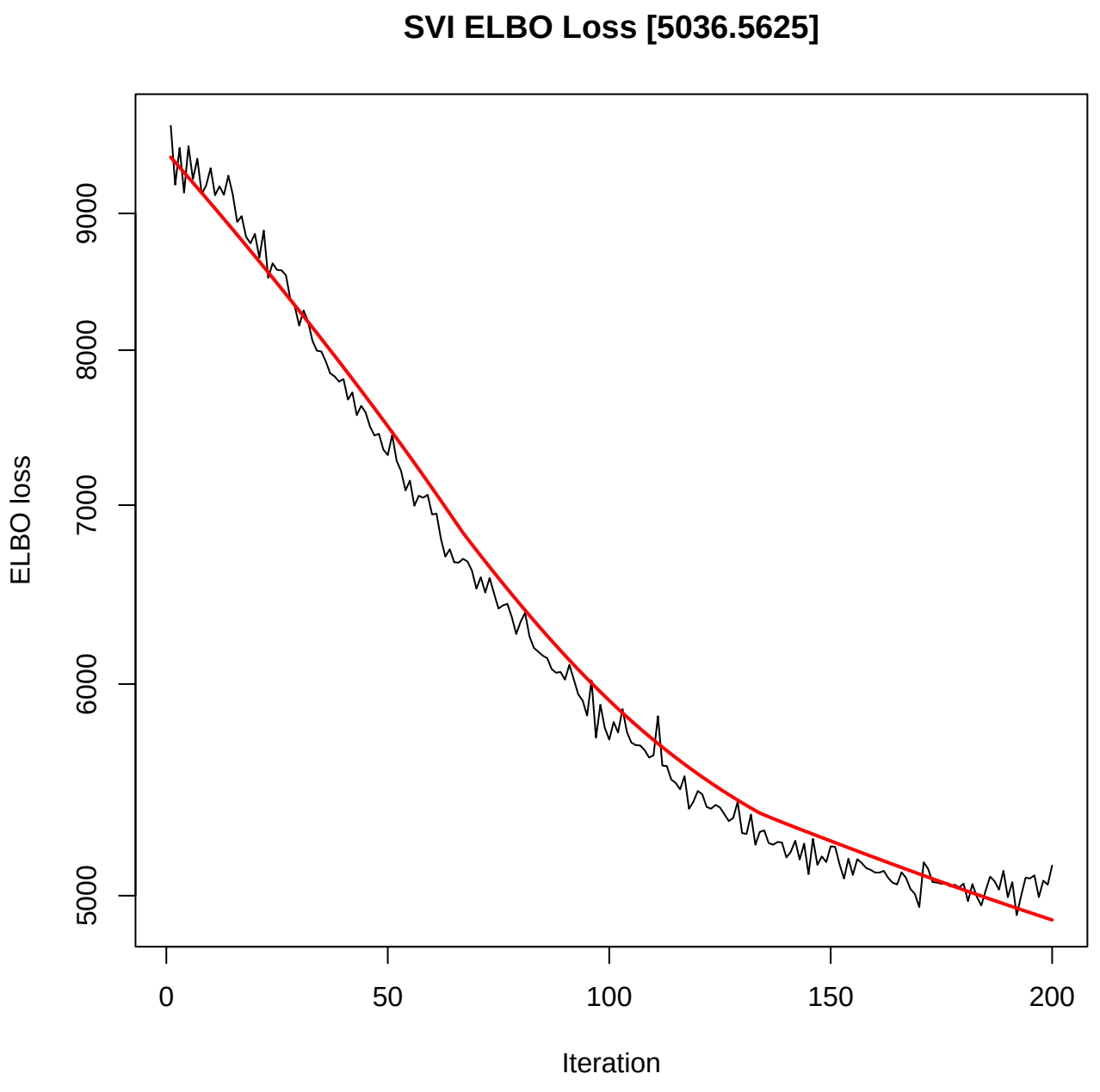

Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
0.00060
0.00065
0.00070
0.00075
0.00080
1
2
3
4
5
Index

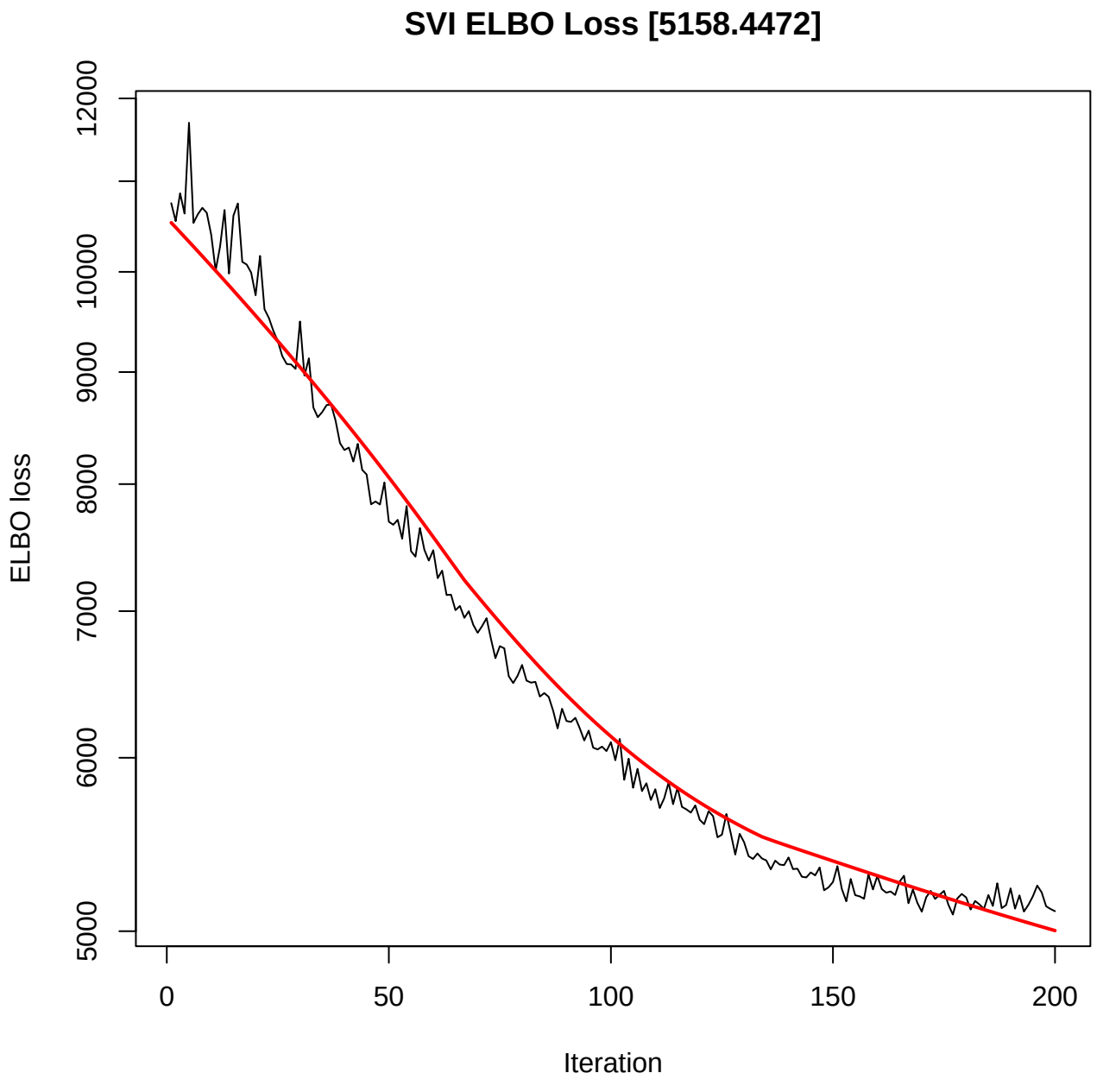

Value (ast)
loss_ast_vec
0.5
0.6
0.7
0.8
0.9
1
2
3
4
5
Index

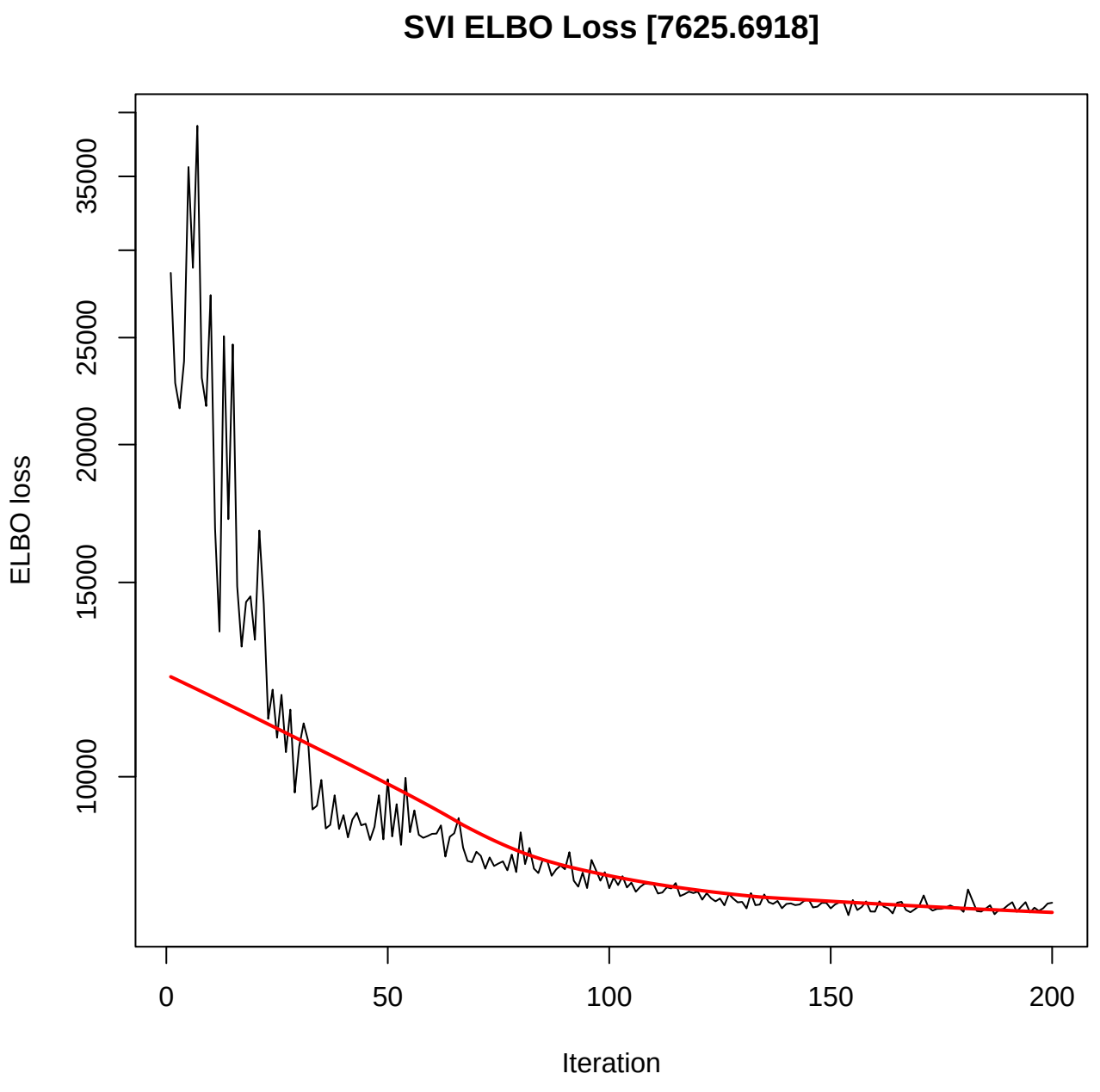

SVI ELBO Loss [5261.6023]
ELBO loss
10000
9000
8000
7000
6000
5000
0
50
100
150
200
Iteration

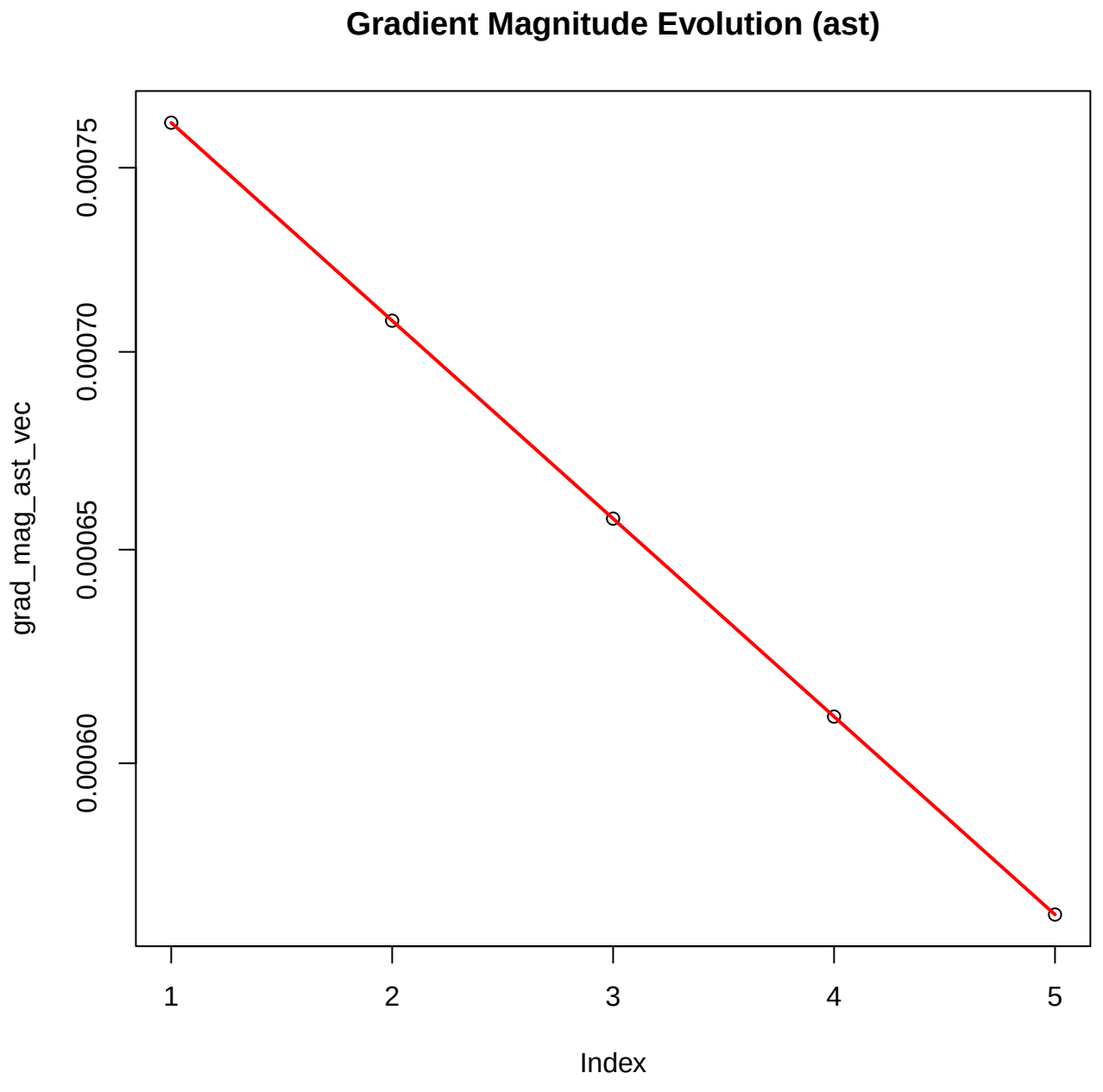

Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
0.13
0.14
0.15
0.16
0.17
0.18
1
2
3
4
5
Index

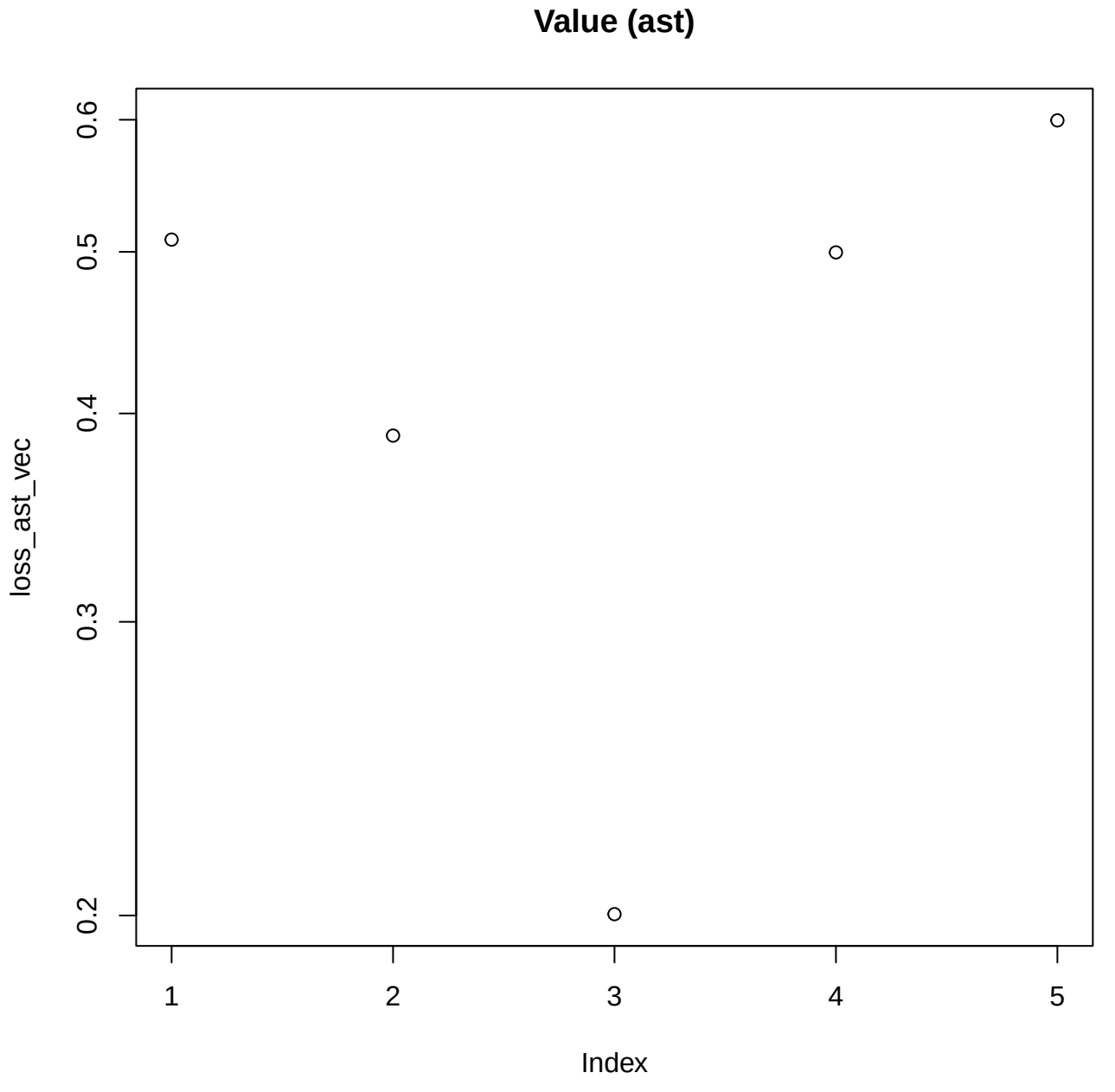

Value (ast)
loss_ast_vec
0.50
0.55
0.60
0.65
1
2
3
4
5
Index

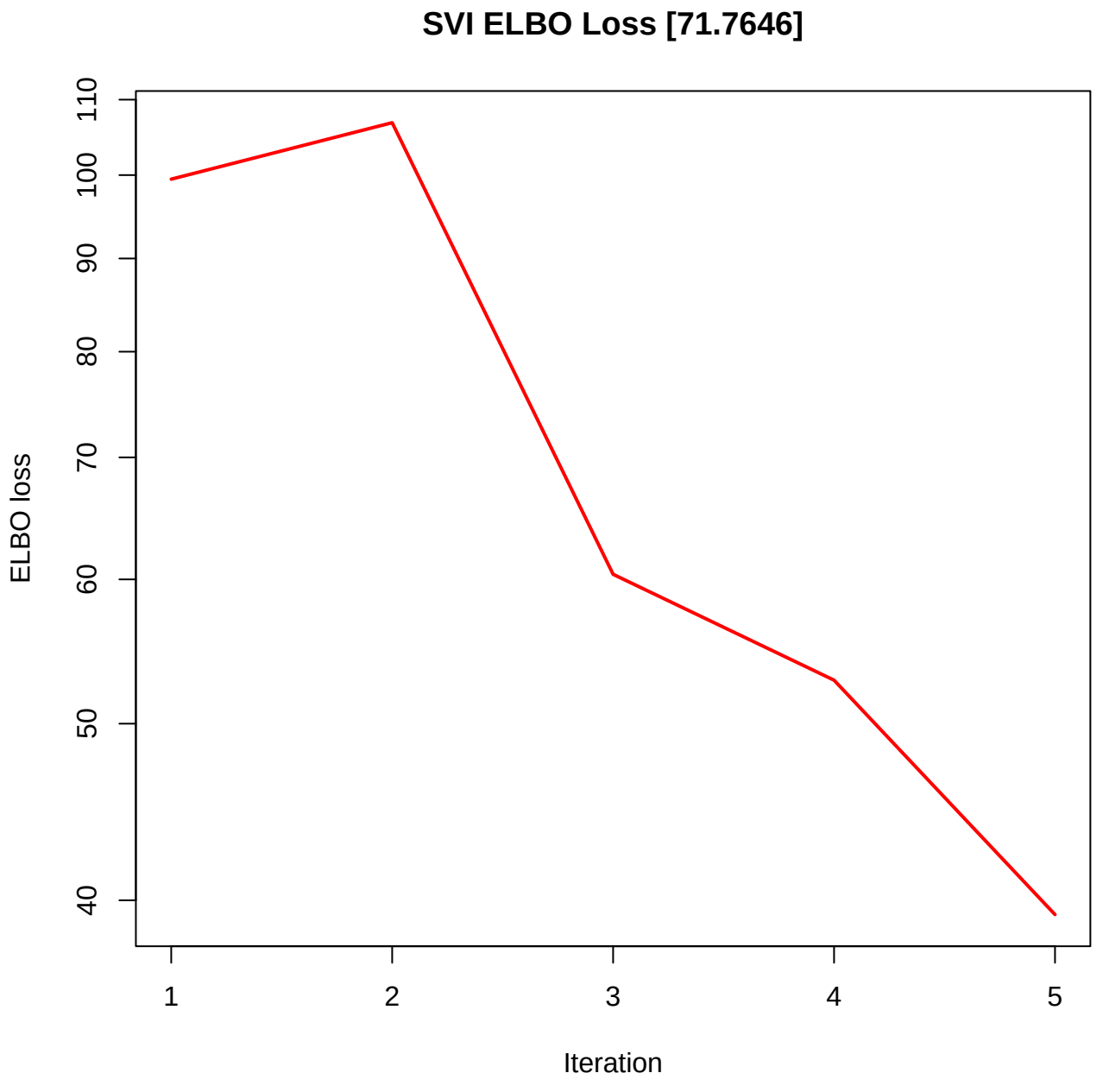

SVI ELBO Loss [5036.5625]
ELBO loss
9000
8000
7000
6000
5000
0
50
100
150
200
Iteration

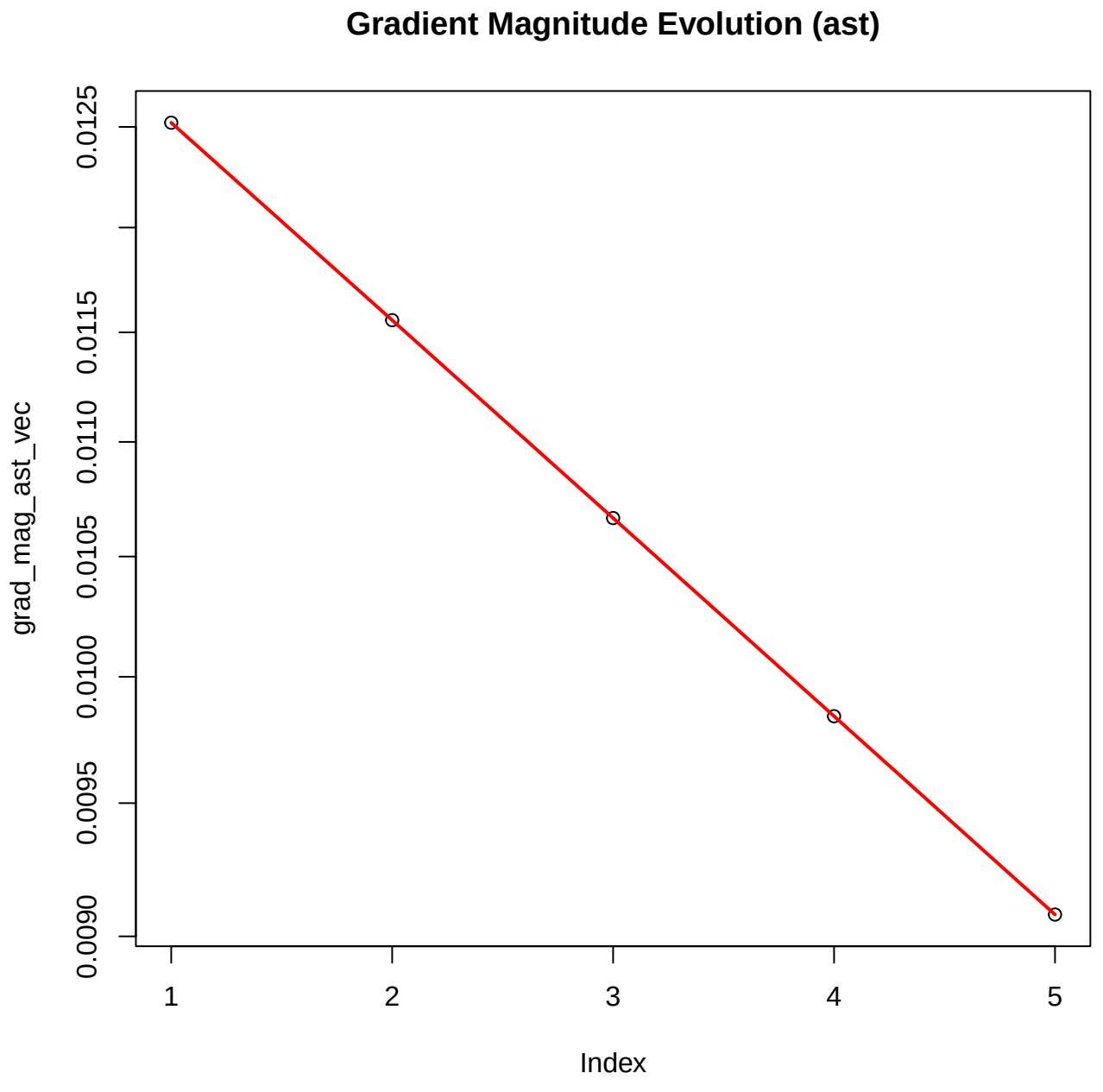

SVI ELBO Loss [5158.4472]
ELBO loss
Iteration
12000
10000
9000
8000
7000
6000
5000
0
50
100
150
200

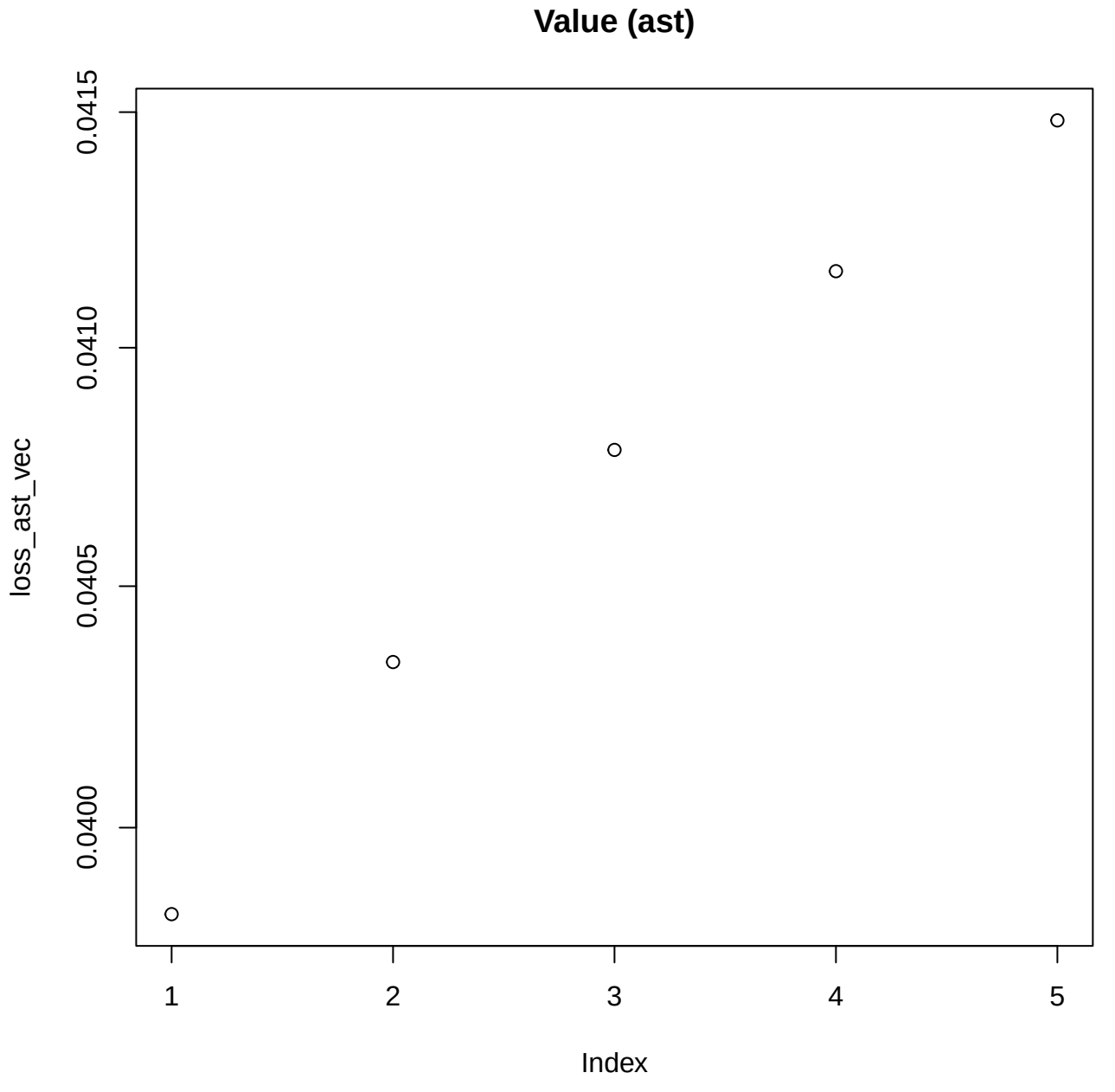

SVI ELBO Loss [7625.6918]
ELBO loss
35000
25000
20000
15000
10000
0
50
100
150
200
Iteration

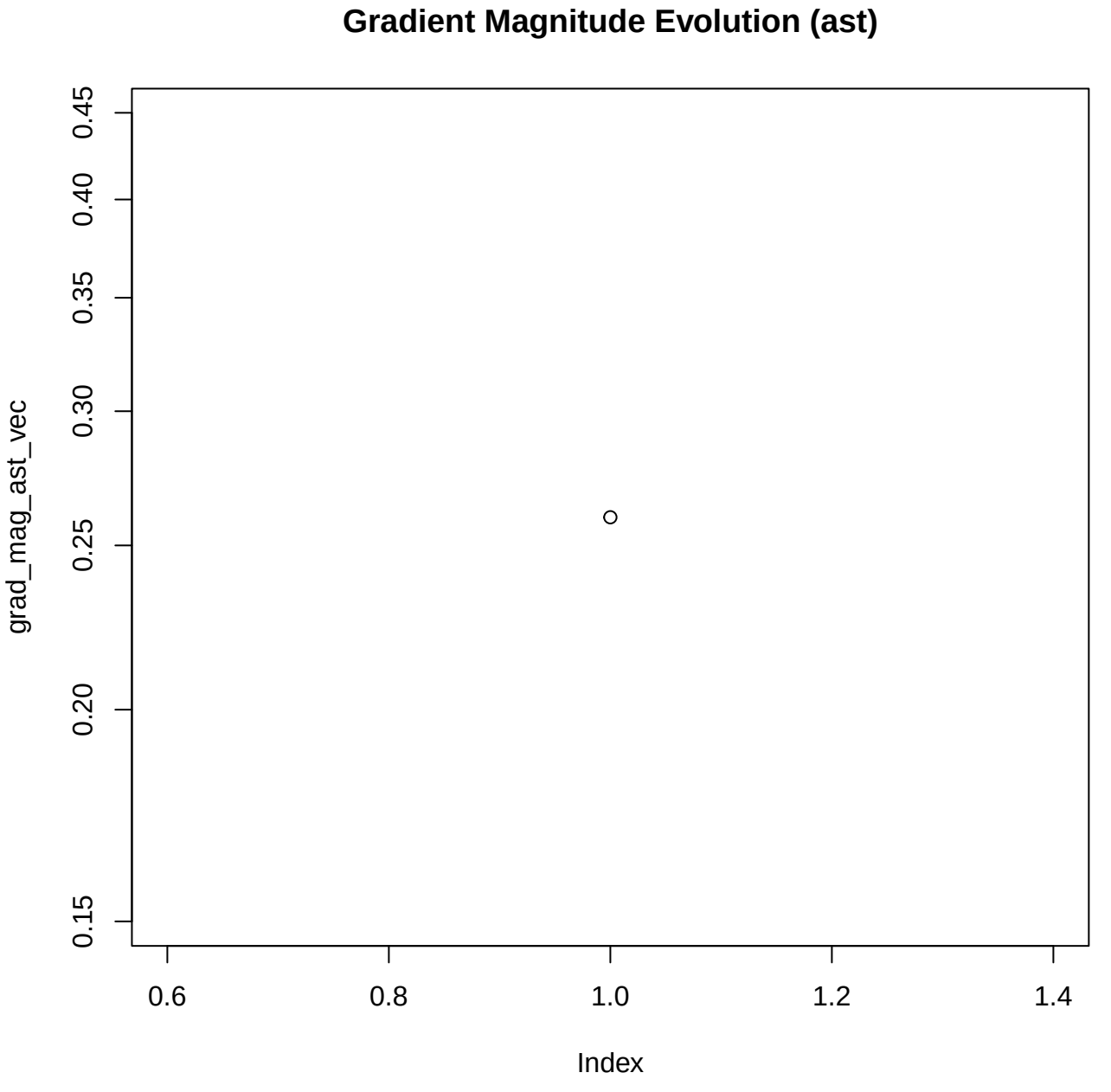

Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
0.00060
0.00065
0.00070
0.00075
1
2
3
4
5
Index

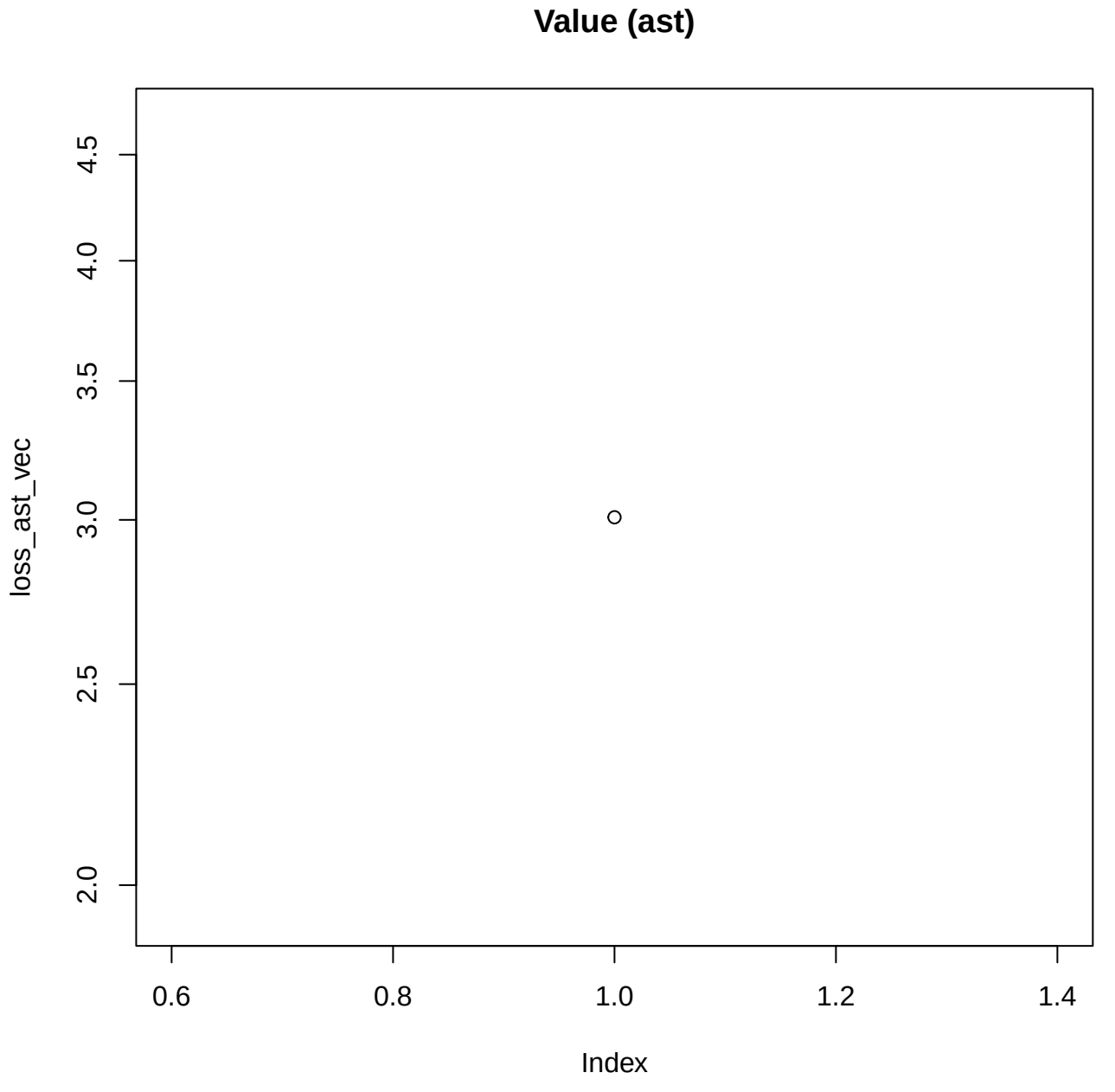

Value (ast)
loss_ast_vec
0.2
0.3
0.4
0.5
0.6
1
2
3
4
5
Index

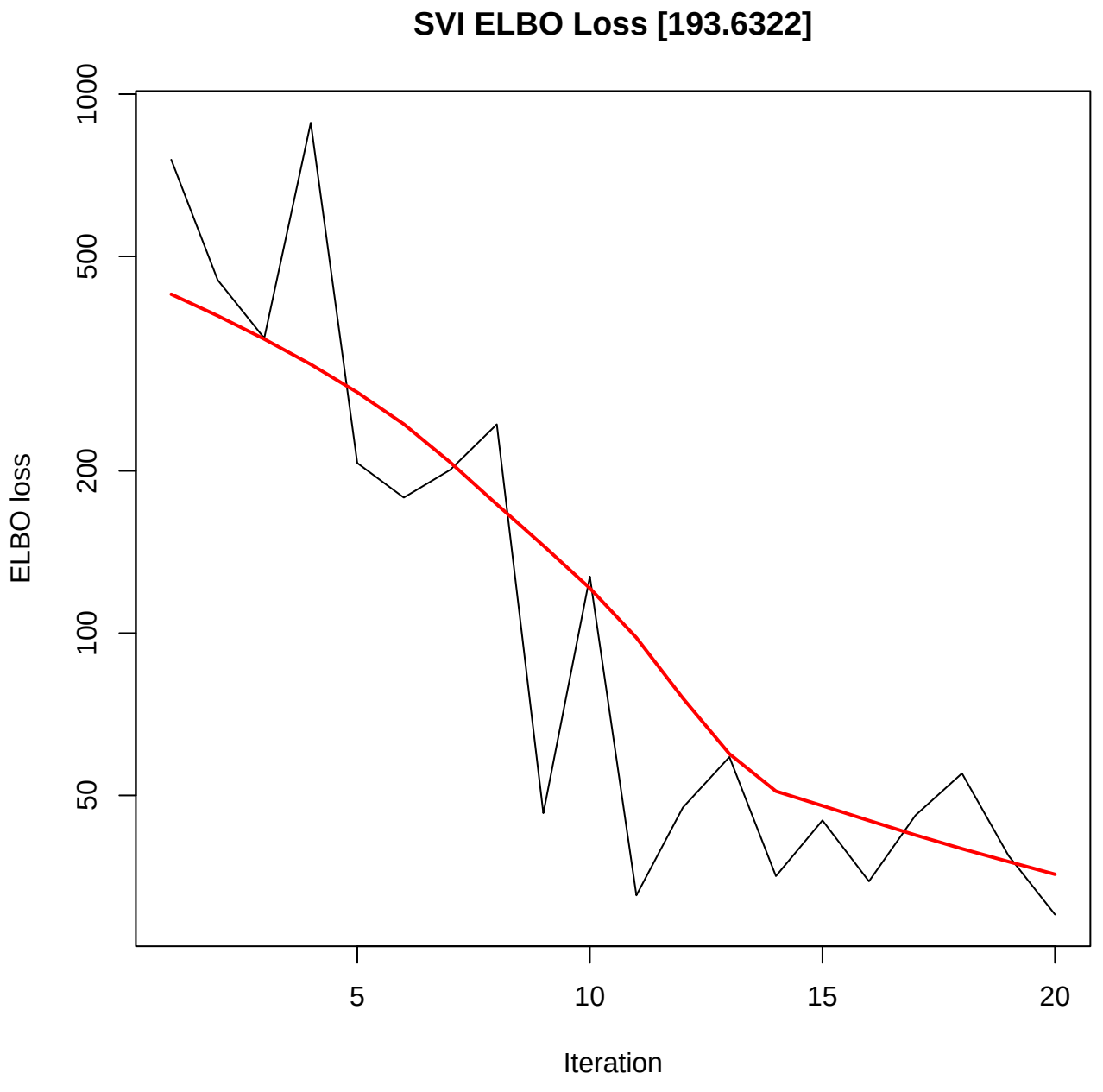

SVI ELBO Loss [71.7646]
ELBO loss
110
100
90
80
70
60
50
40
1
2
3
4
5
Iteration

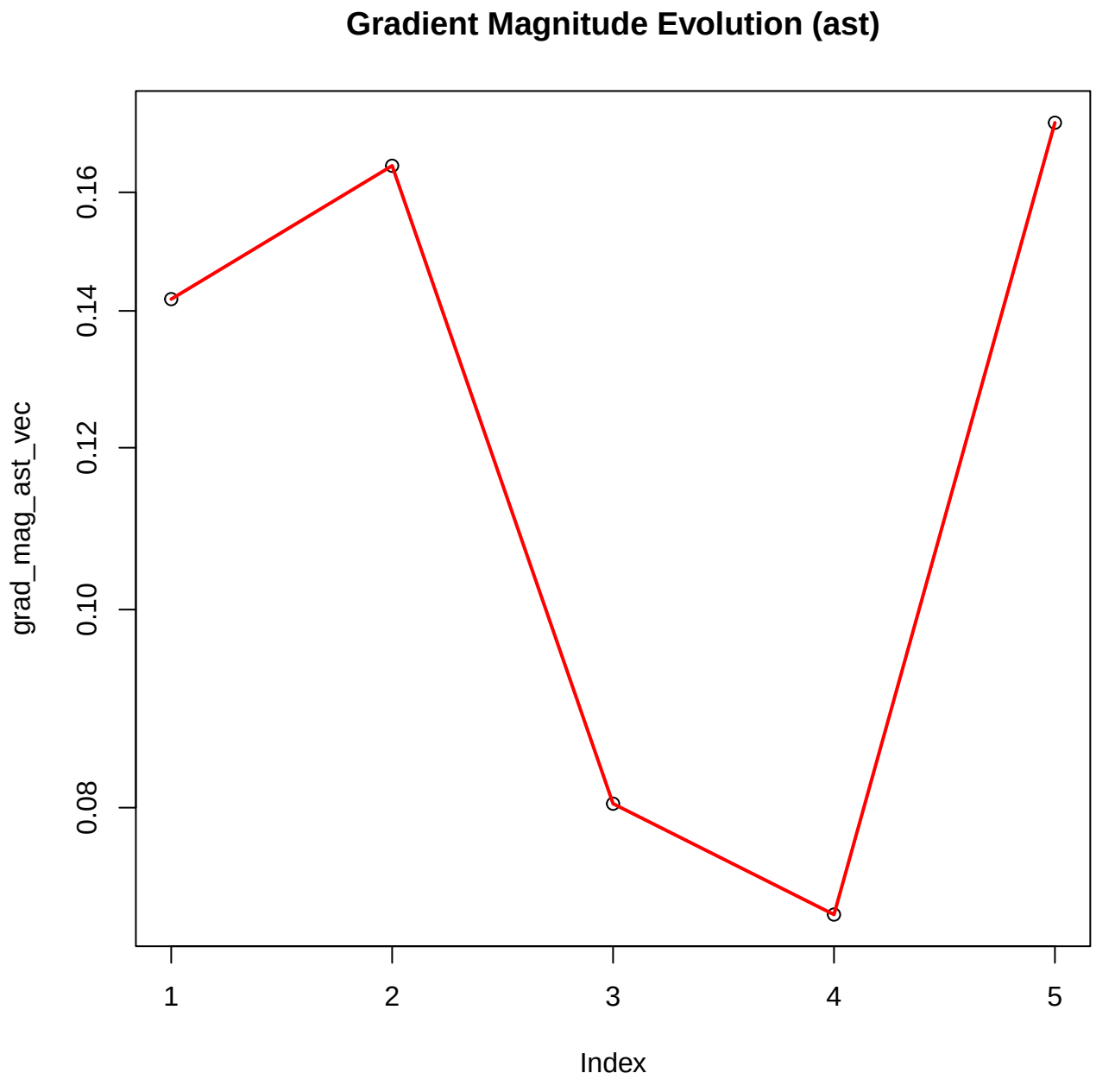

Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
0.0090
0.0095
0.0100
0.0105
0.0110
0.0115
0.0125
1
2
3
4
5
Index

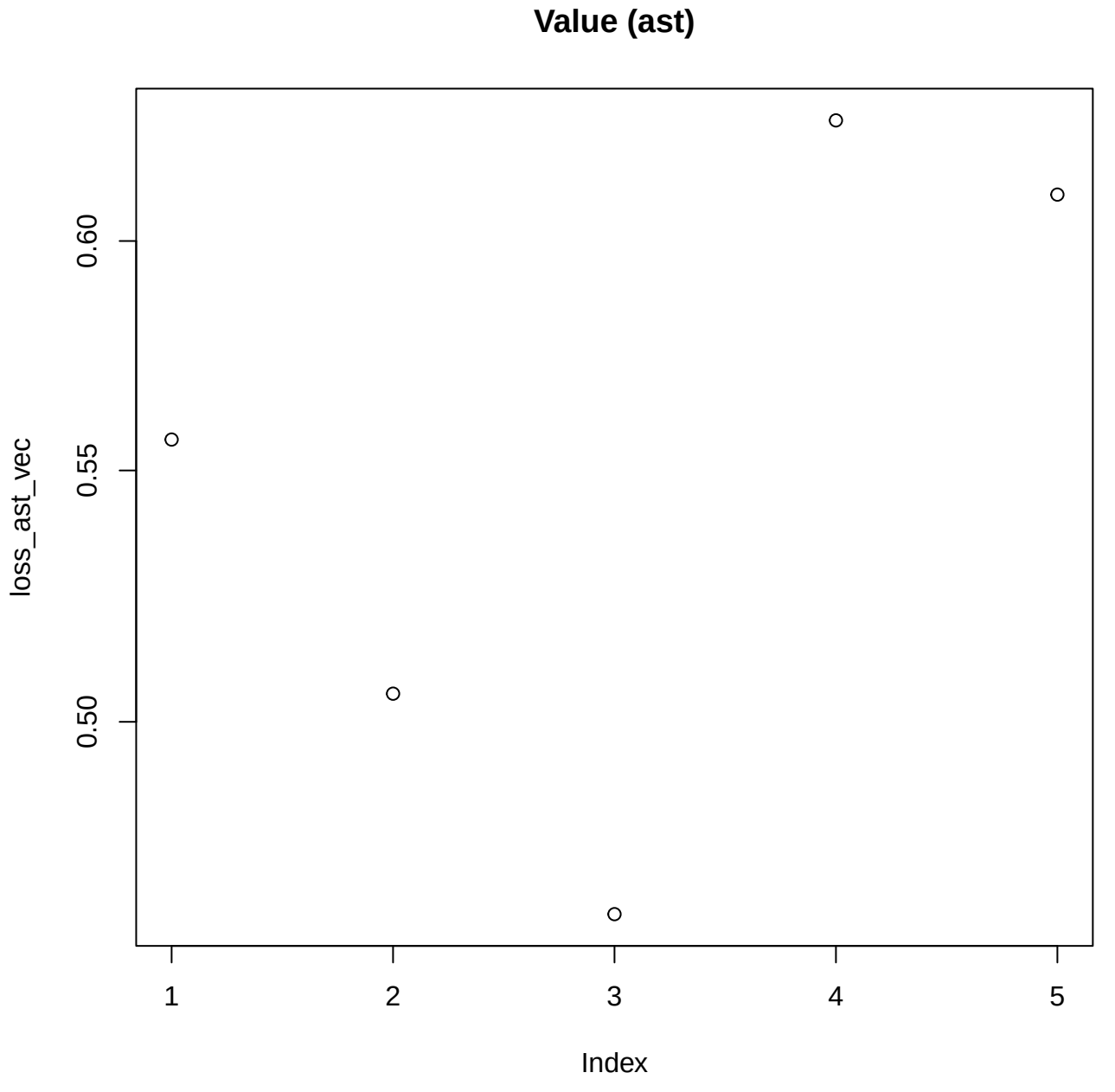

Value (ast)
loss_ast_vec
0.0400
0.0405
0.0410
0.0415
1
2
3
4
5
Index

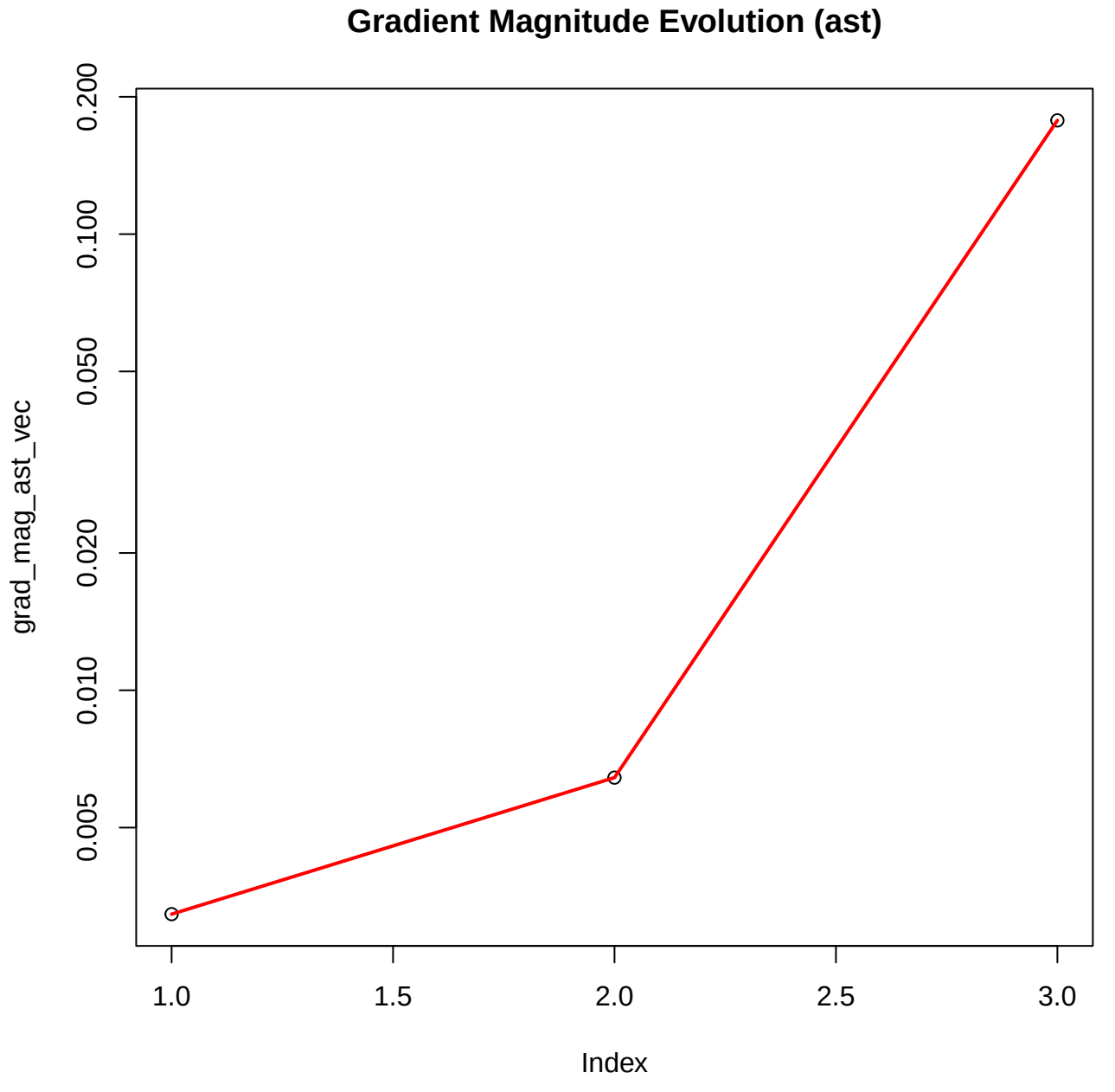

Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
0.15
0.20
0.25
0.30
0.35
0.40
0.45
0.6
0.8
1.0
1.2
1.4
Index

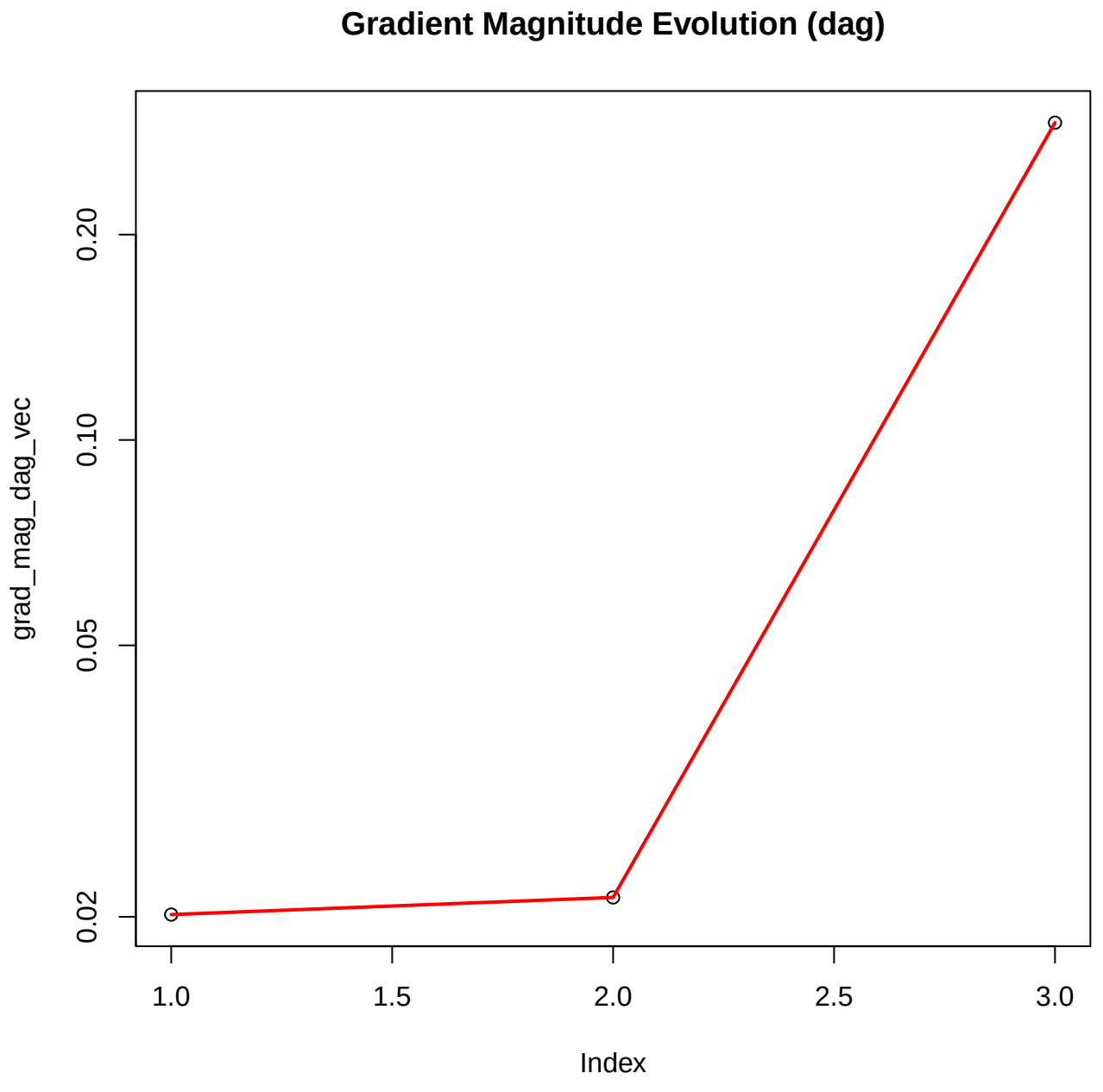

Value (ast)
loss_ast_vec
4.5
4.0
3.5
3.0
2.5
2.0
0.6
0.8
1.0
1.2
1.4
Index

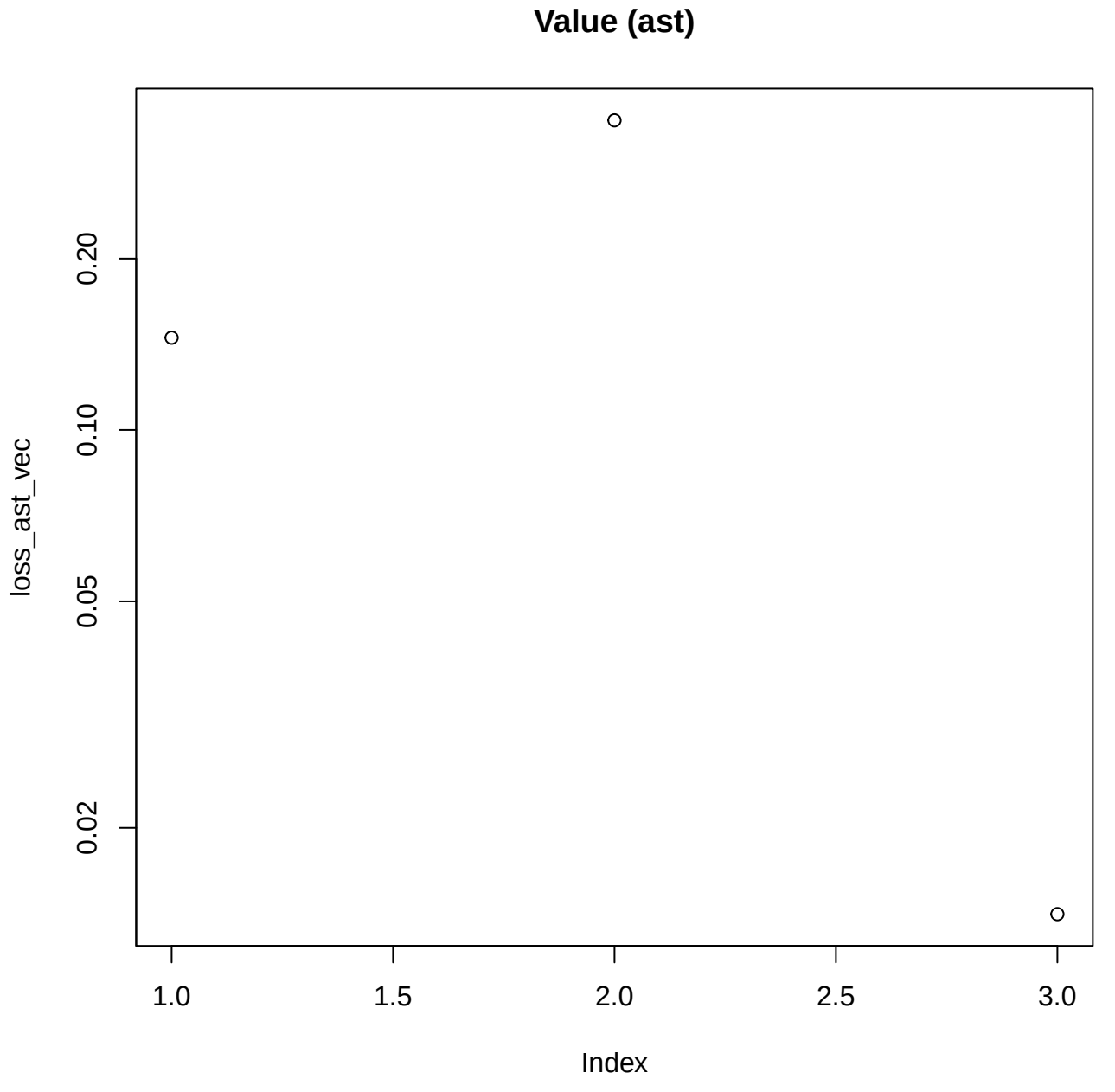

SVI ELBO Loss [193.6322]
ELBO loss
1000
500
200
100
50
5
10
15
20
Iteration

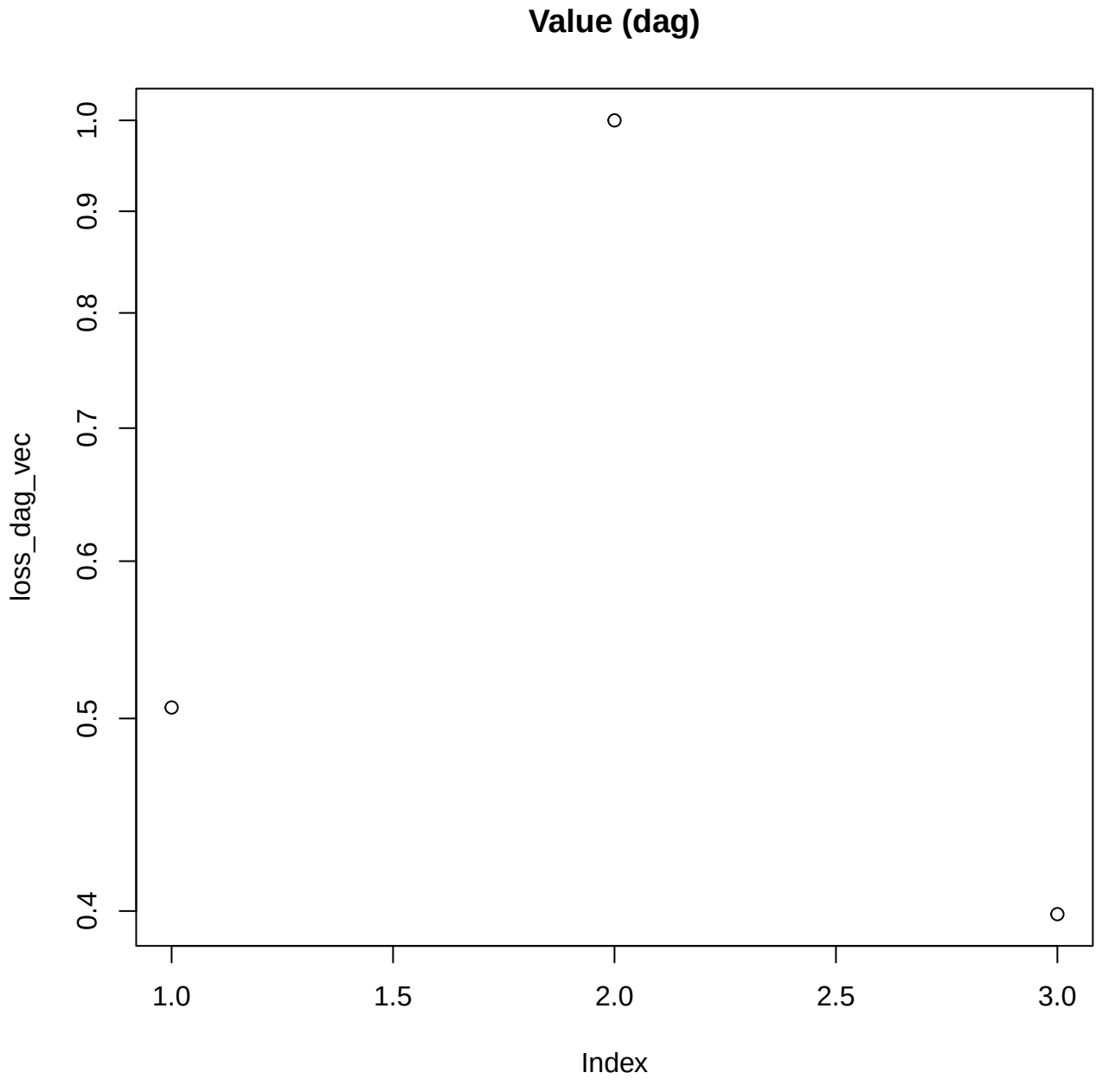

Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
0.08
0.10
0.12
0.14
0.16
1
2
3
4
5
Index

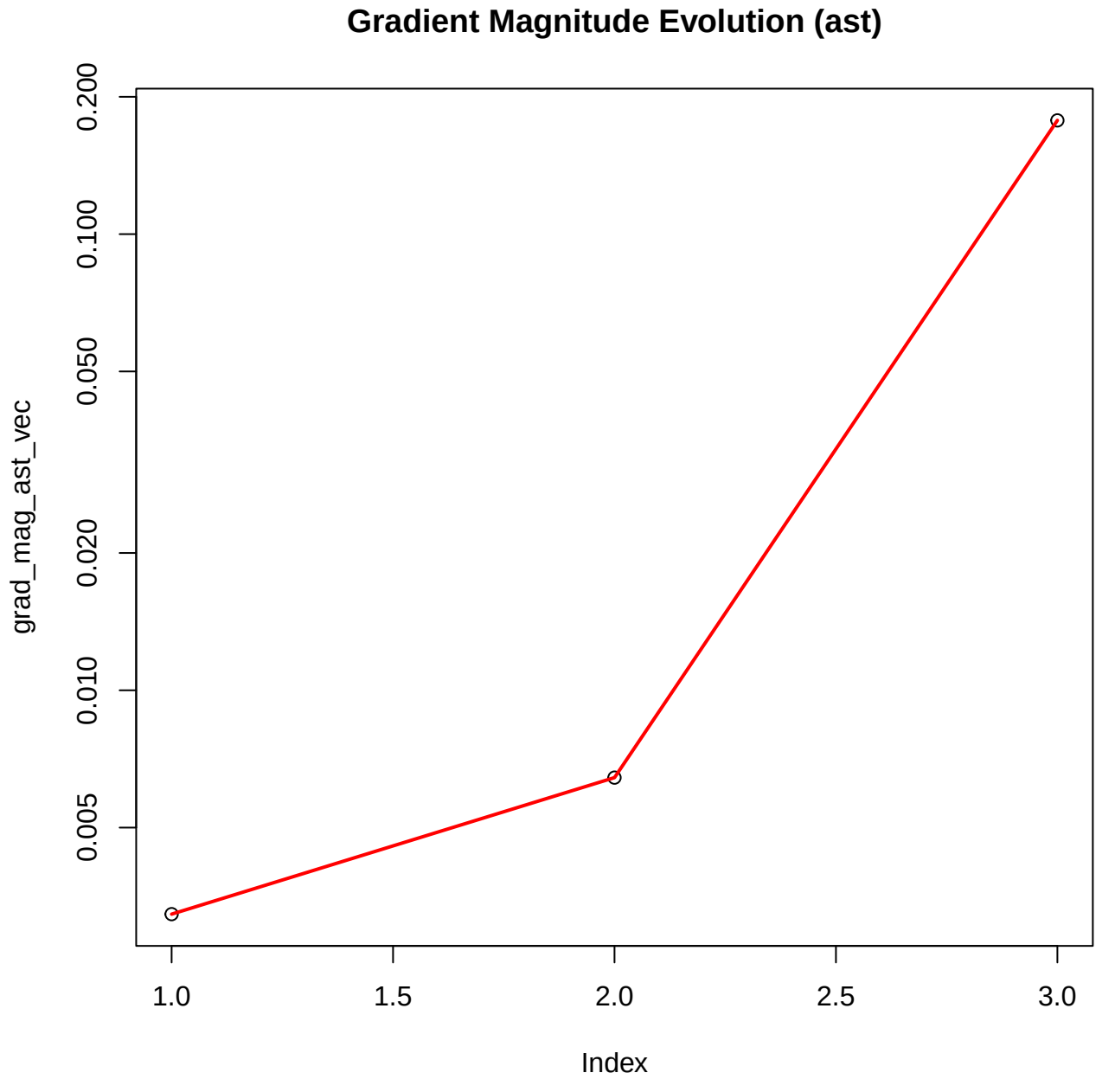

Value (ast)
loss_ast_vec
0.50
0.55
0.60
1
2
3
4
5
Index

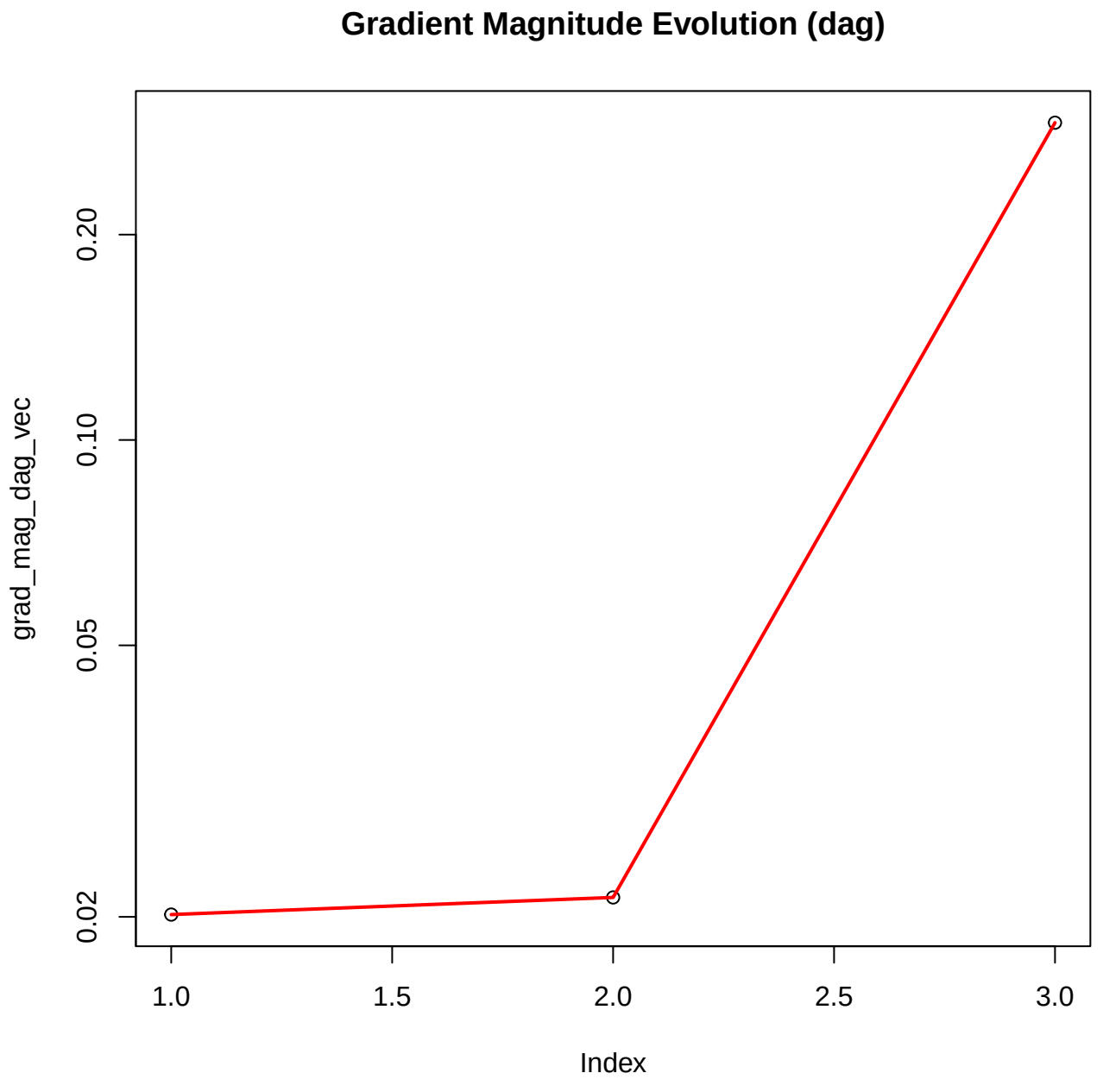

Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
0.005
0.010
0.020
0.050
0.100
0.200
1.0
1.5
2.0
2.5
3.0
Index

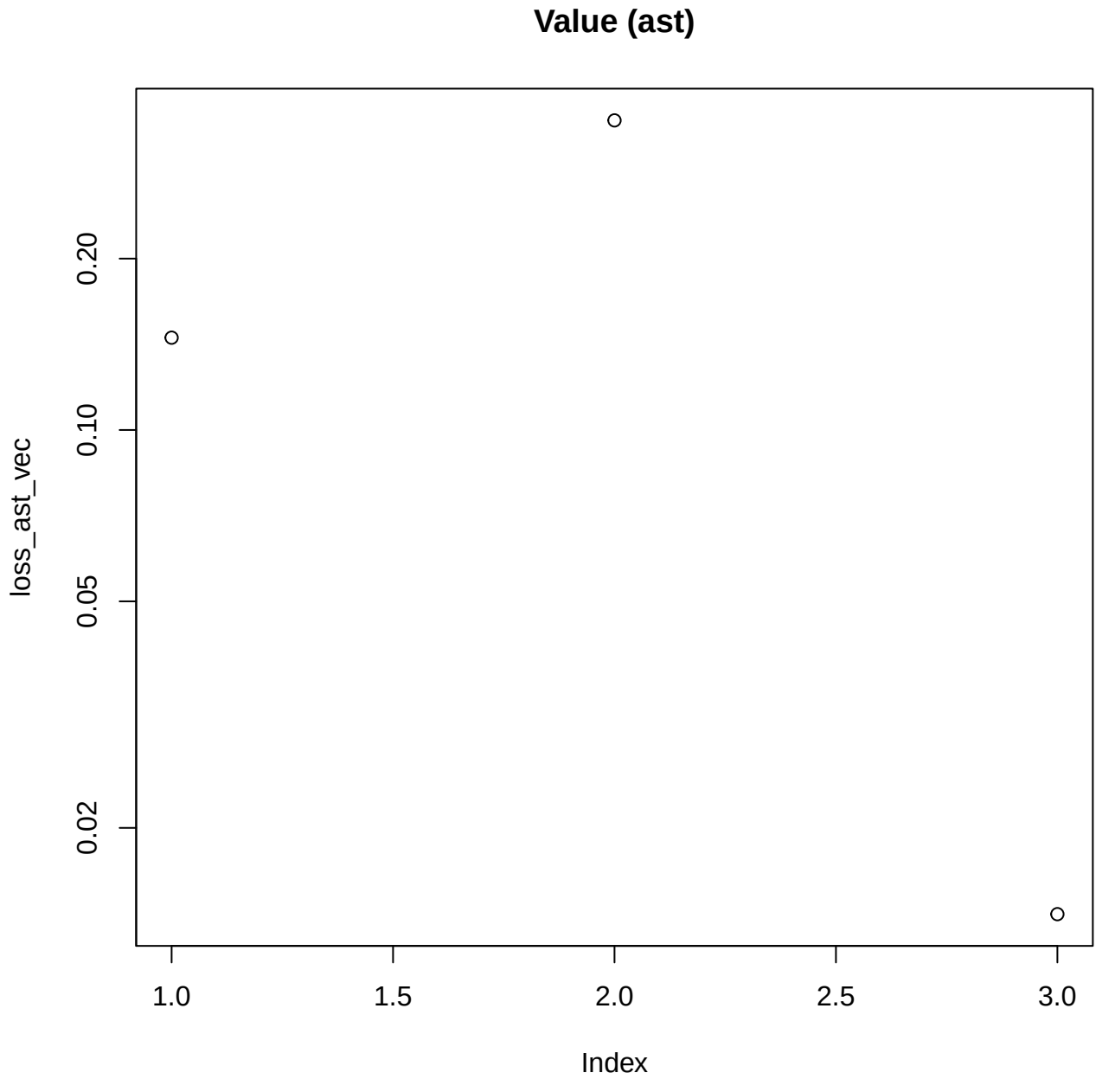

Gradient Magnitude Evolution (dag)
grad_mag_dag_vec
0.02
0.05
0.10
0.20
1.0
1.5
2.0
2.5
3.0
Index

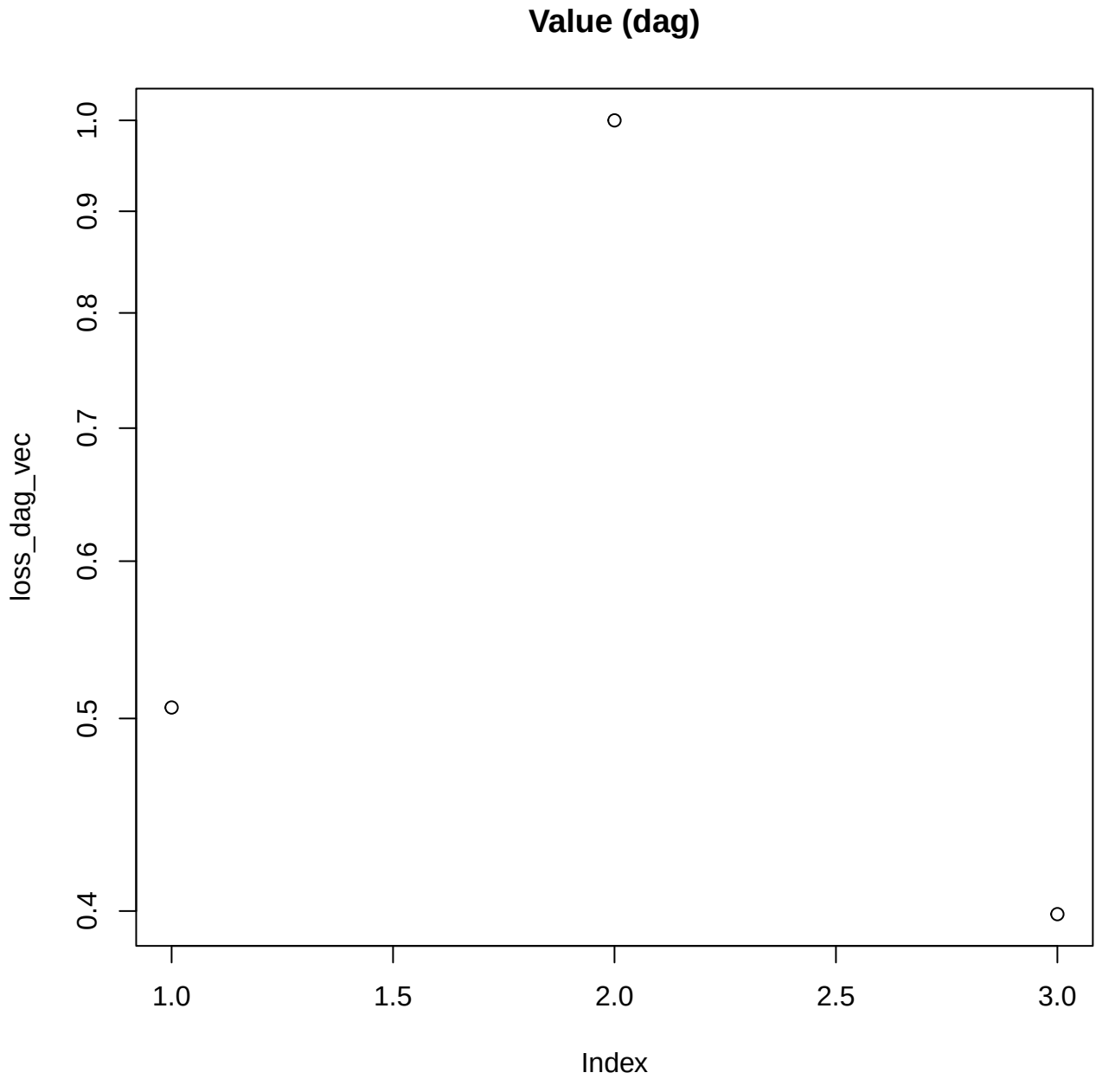

Value (ast)
loss_ast_vec
0.02
0.05
0.10
0.20
1.0
1.5
2.0
2.5
3.0
Index

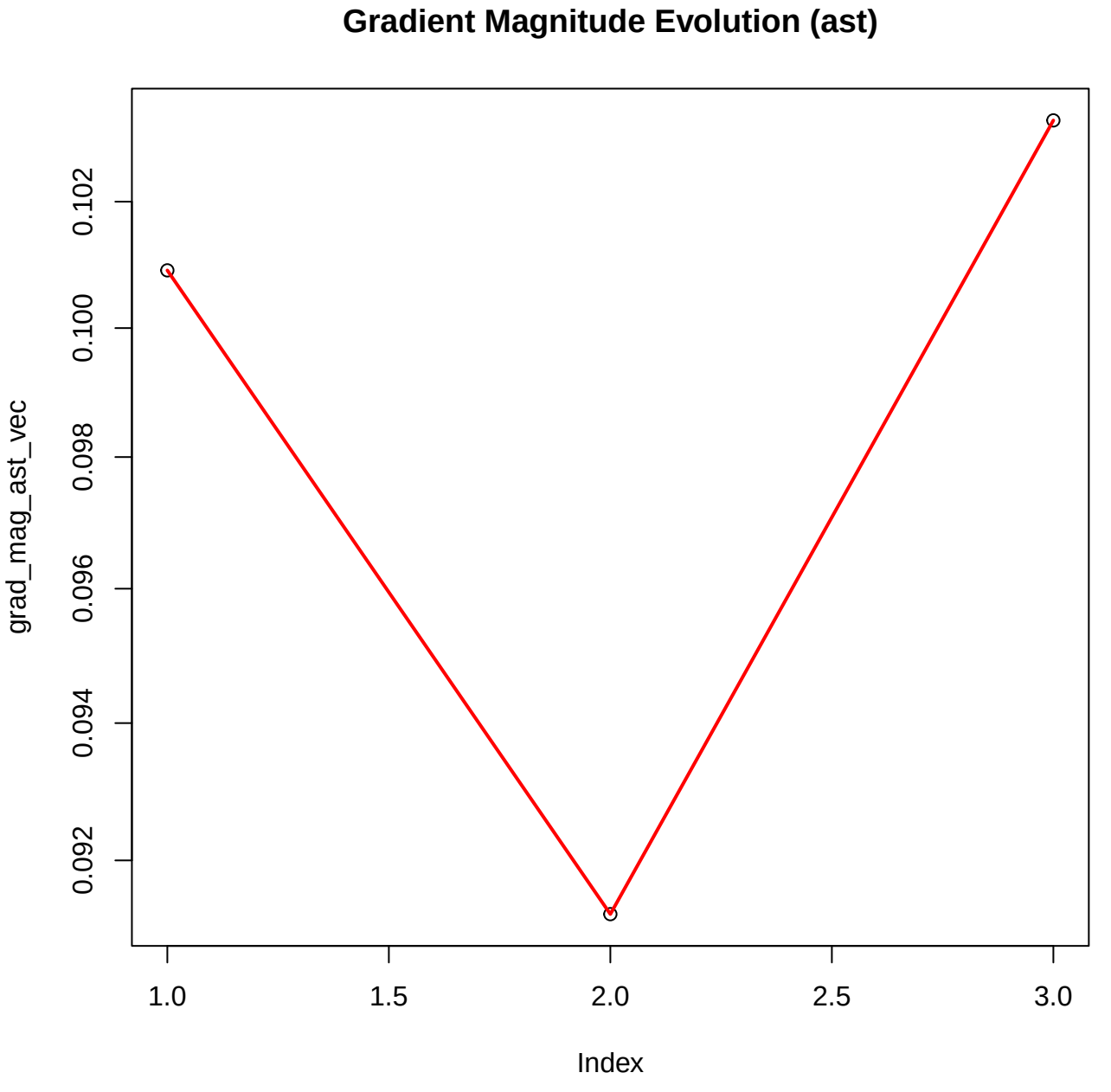

Value (dag)
loss_dag_vec
1.0
0.9
0.8
0.7
0.6
0.5
0.4
1.0
1.5
2.0
2.5
3.0
Index

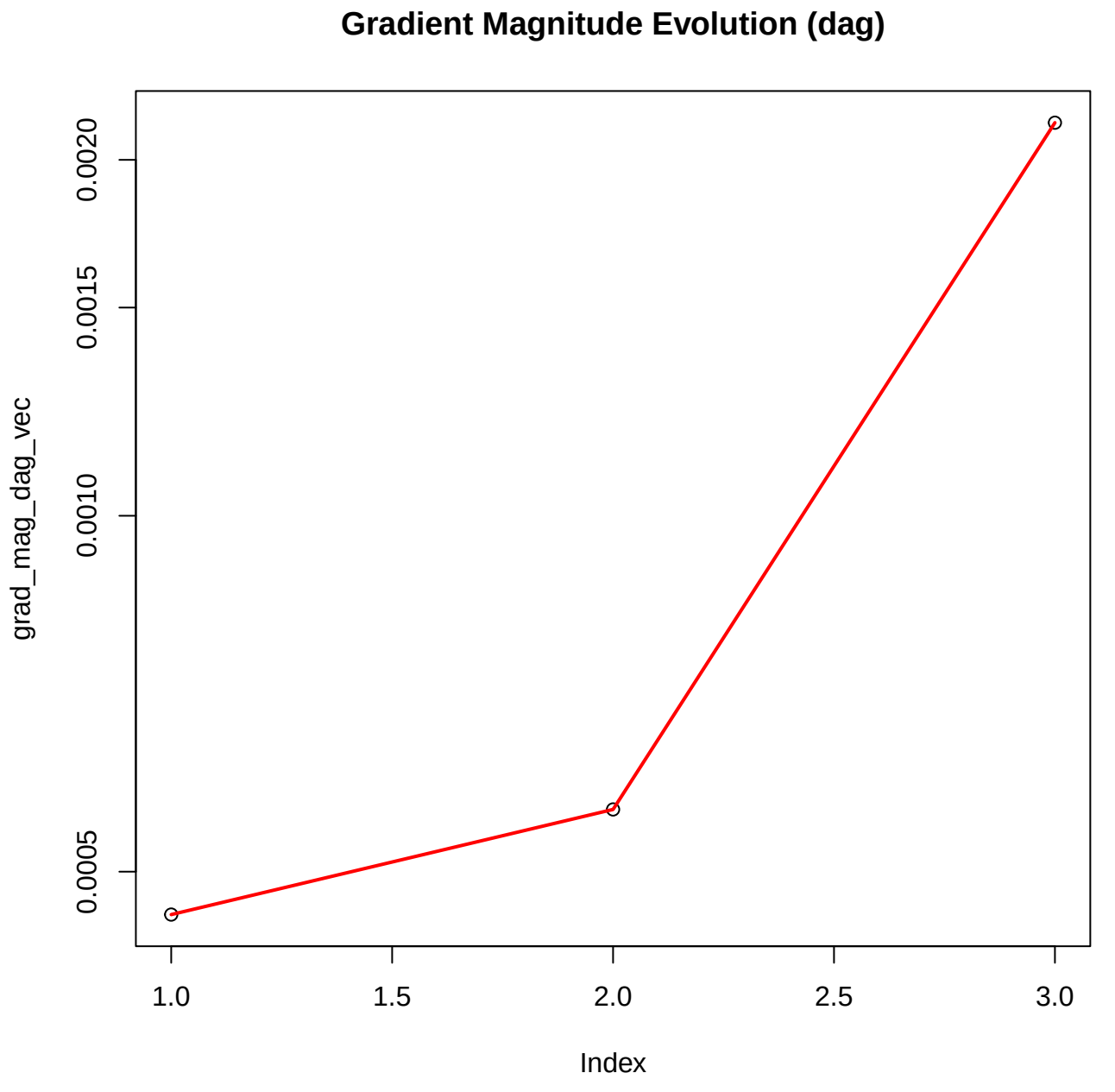

Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
0.005
0.010
0.020
0.050
0.100
0.200
1.0
1.5
2.0
2.5
3.0
Index

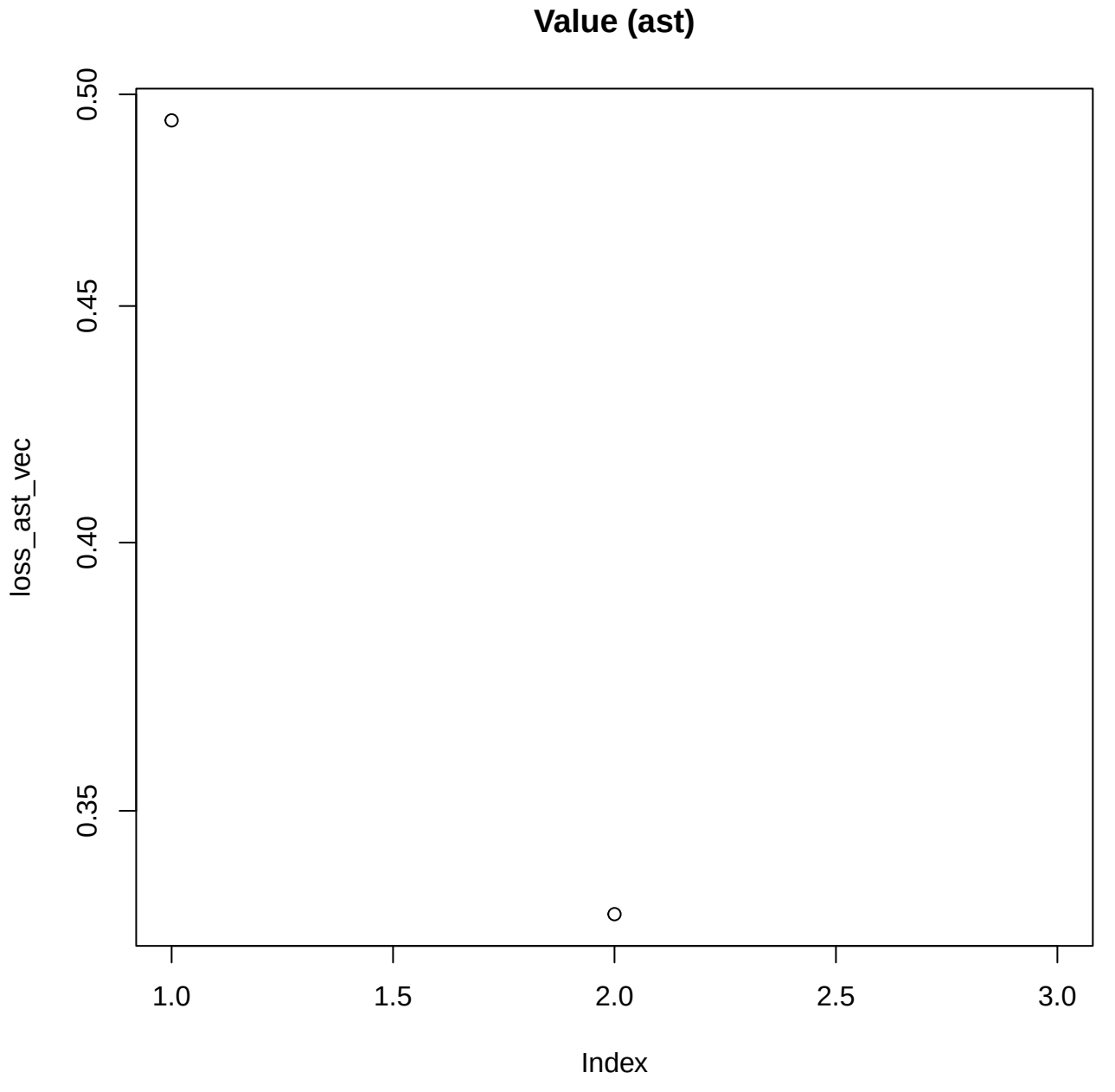

Gradient Magnitude Evolution (dag)
grad_mag_dag_vec
0.02
0.05
0.10
0.20
1.0
1.5
2.0
2.5
3.0
Index

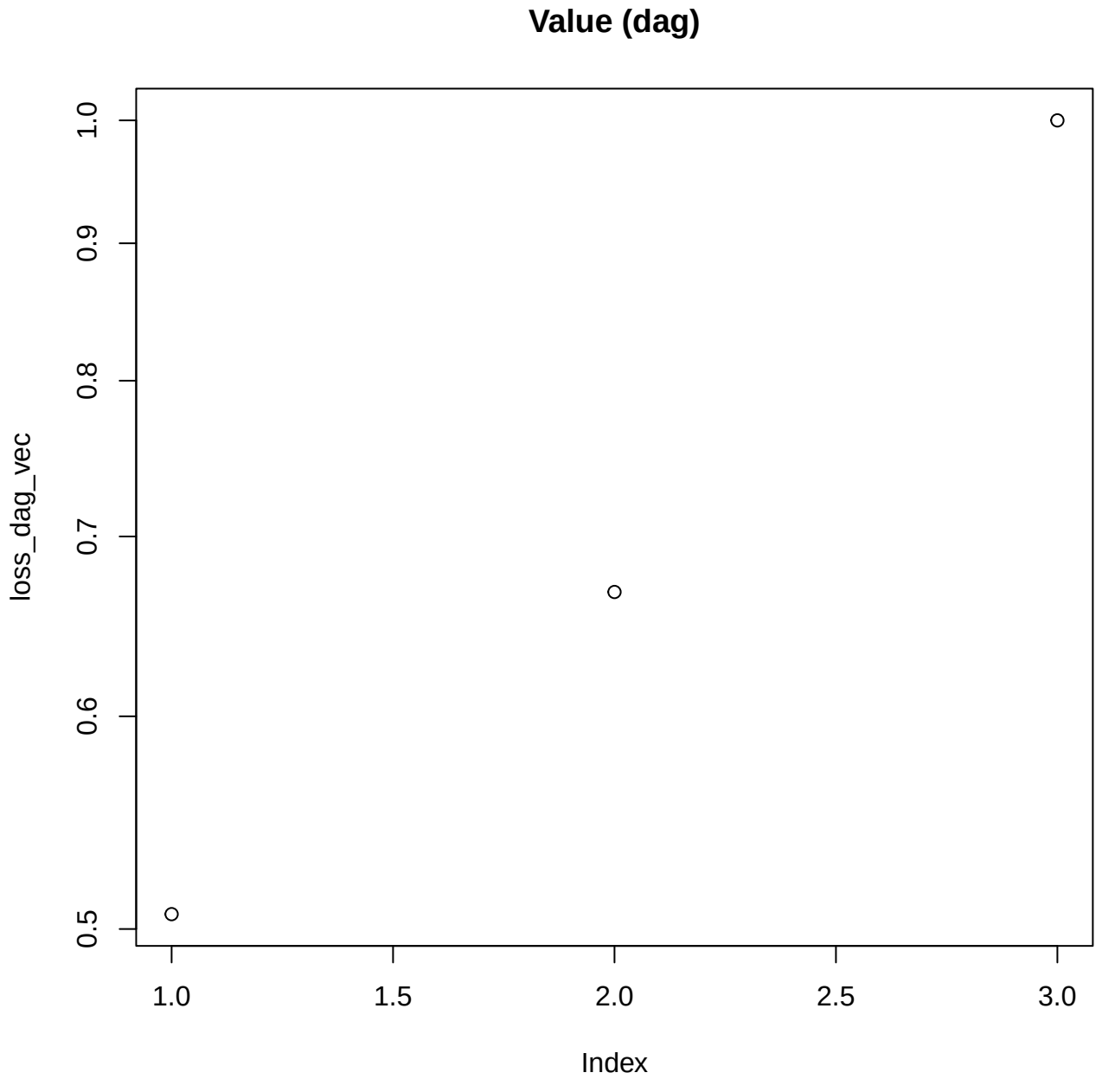

Value (ast)
loss_ast_vec
0.02
0.05
0.10
0.20
1.0
1.5
2.0
2.5
3.0
Index

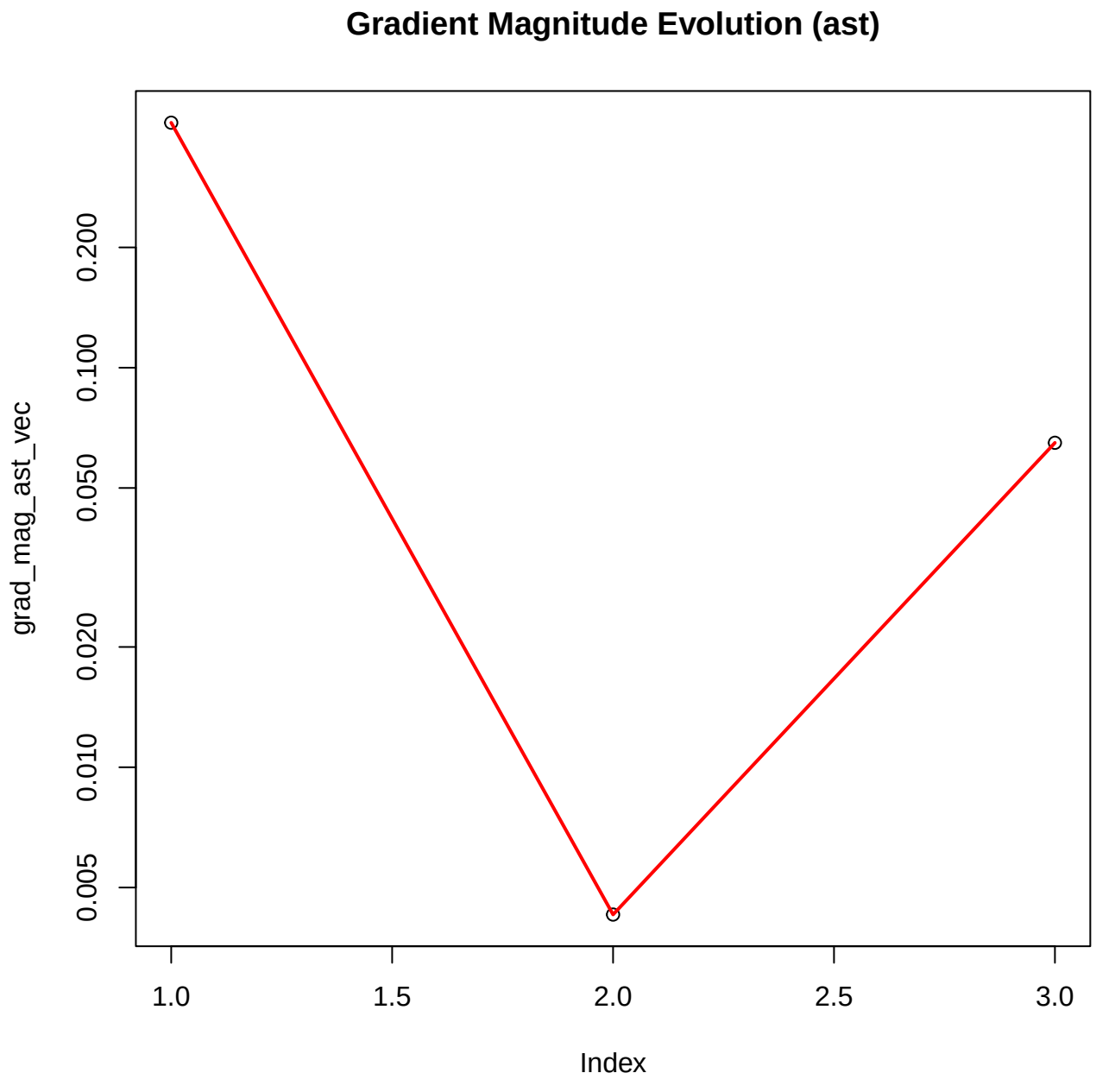

Value (dag)
loss_dag_vec
1.0
0.9
0.8
0.7
0.6
0.5
0.4
1.0
1.5
2.0
2.5
3.0
Index

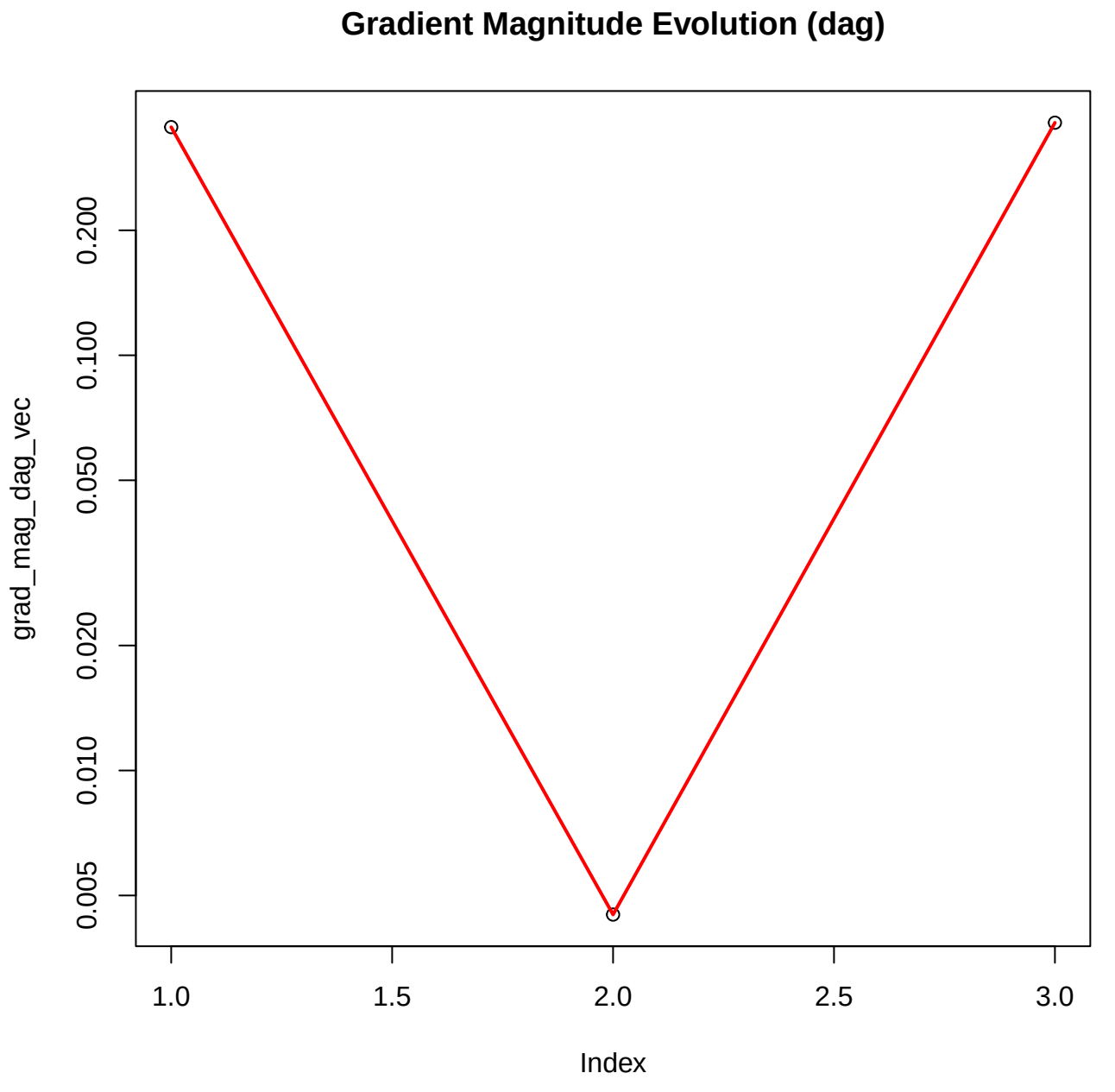

Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
0.092
0.094
0.096
0.098
0.100
0.102
1.0
1.5
2.0
2.5
3.0
Index

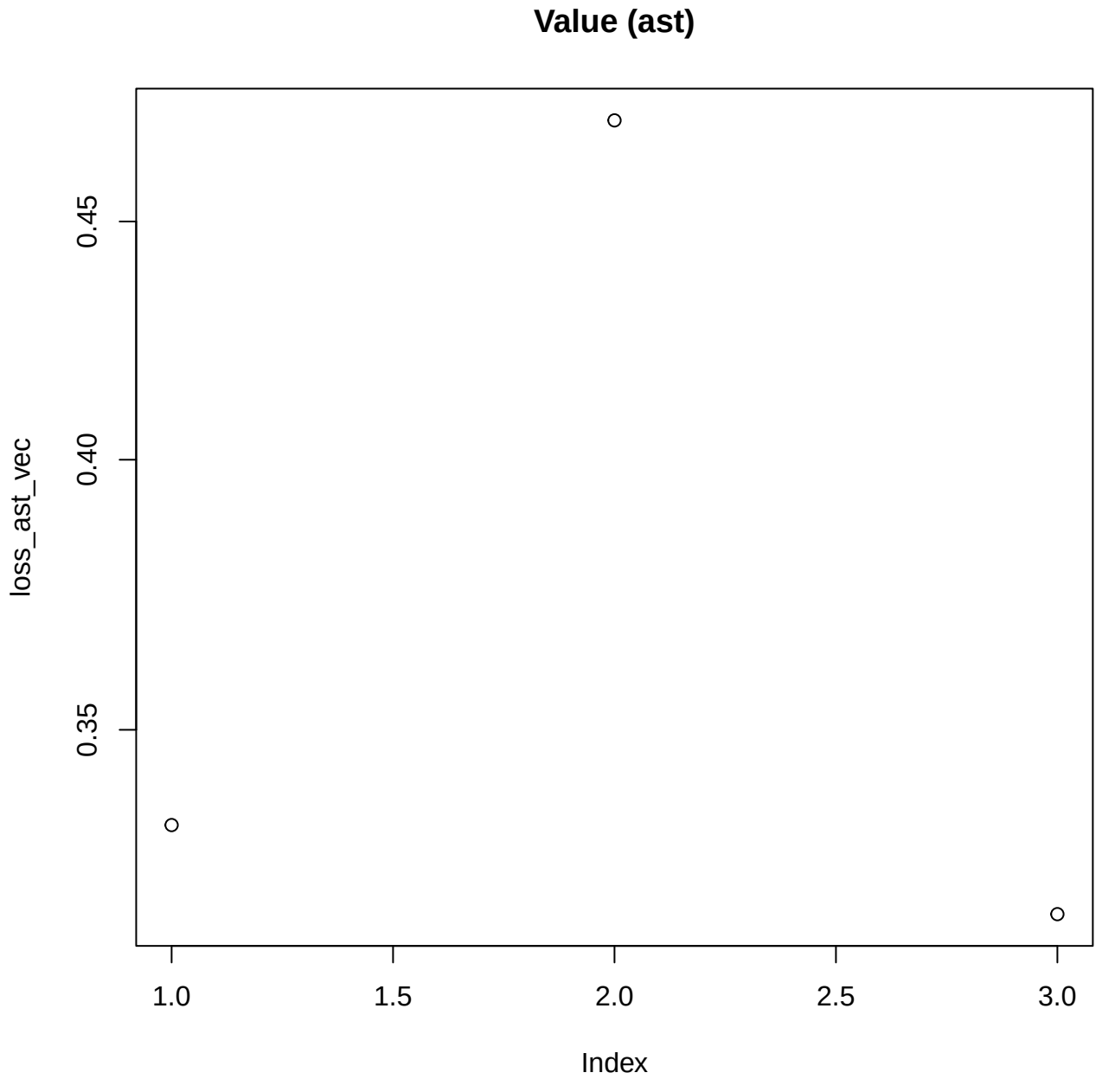

Gradient Magnitude Evolution (dag)
grad_mag_dag_vec
0.0005
0.0010
0.0015
0.0020
1.0
1.5
2.0
2.5
3.0
Index

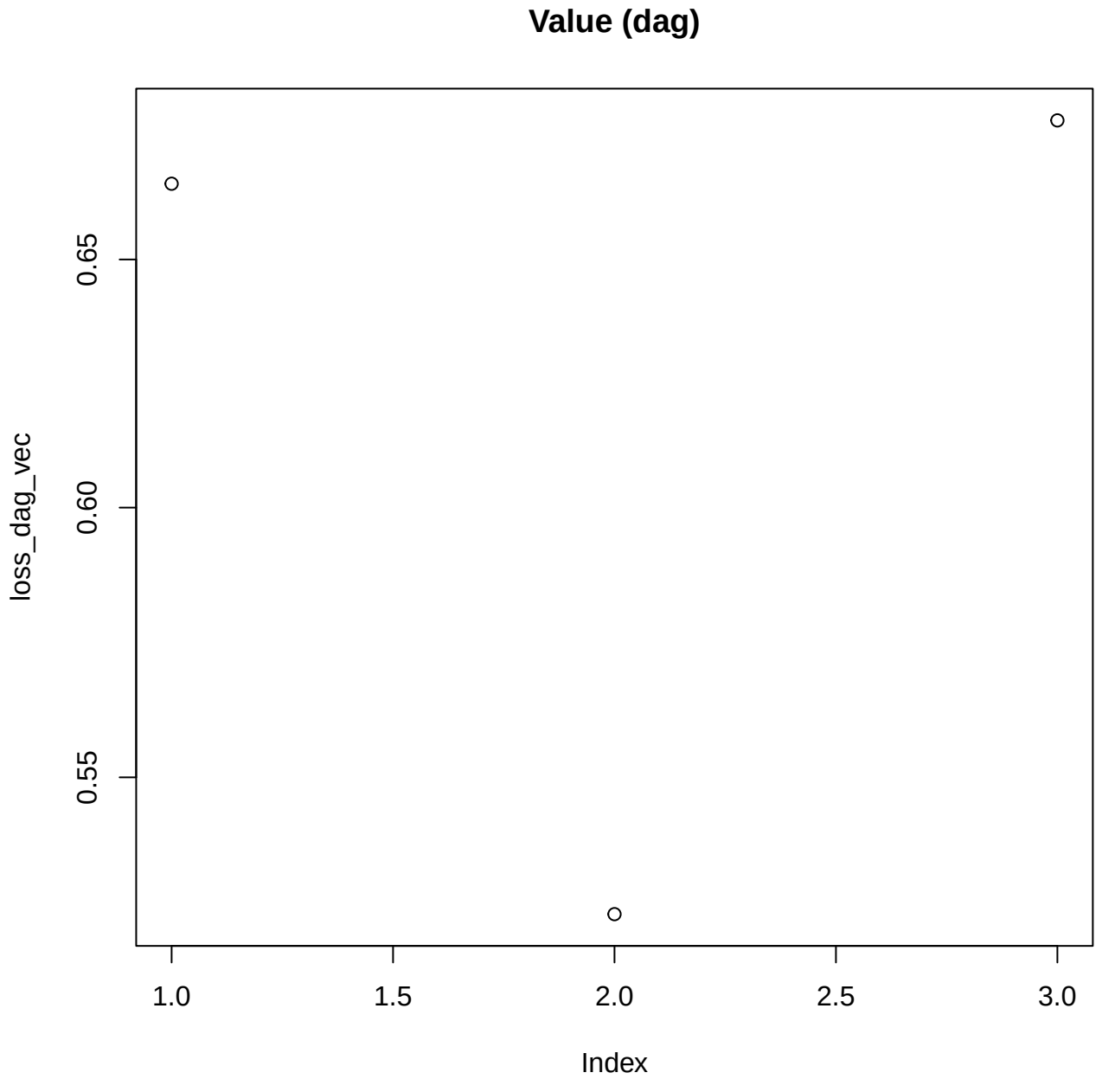

Value (ast)
loss_ast_vec
0.50
0.45
0.40
0.35
1.0
1.5
2.0
2.5
3.0
Index

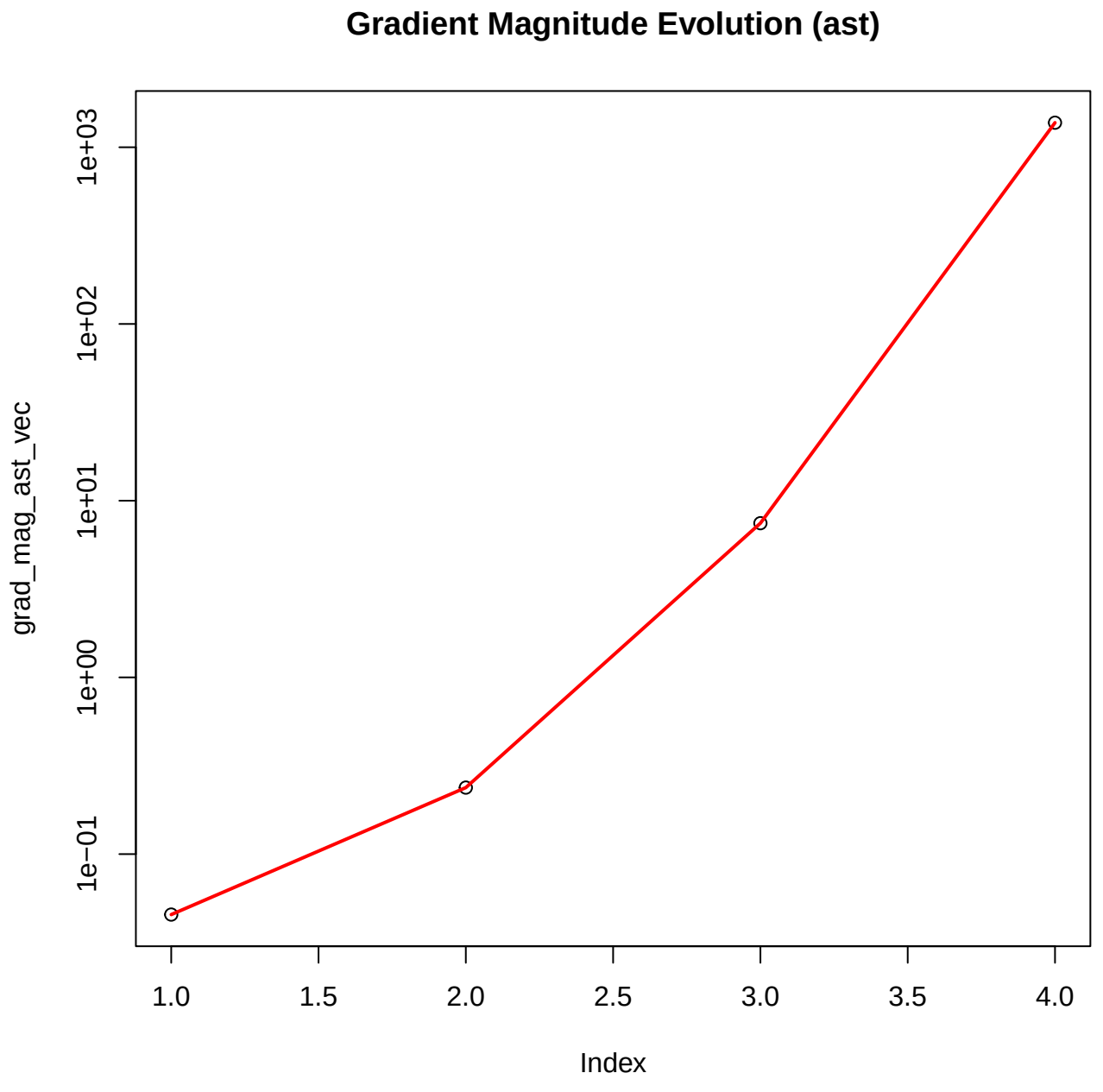

Value (dag)
loss_dag_vec
0.5
0.6
0.7
0.8
0.9
1.0
1.0
1.5
2.0
2.5
3.0
Index

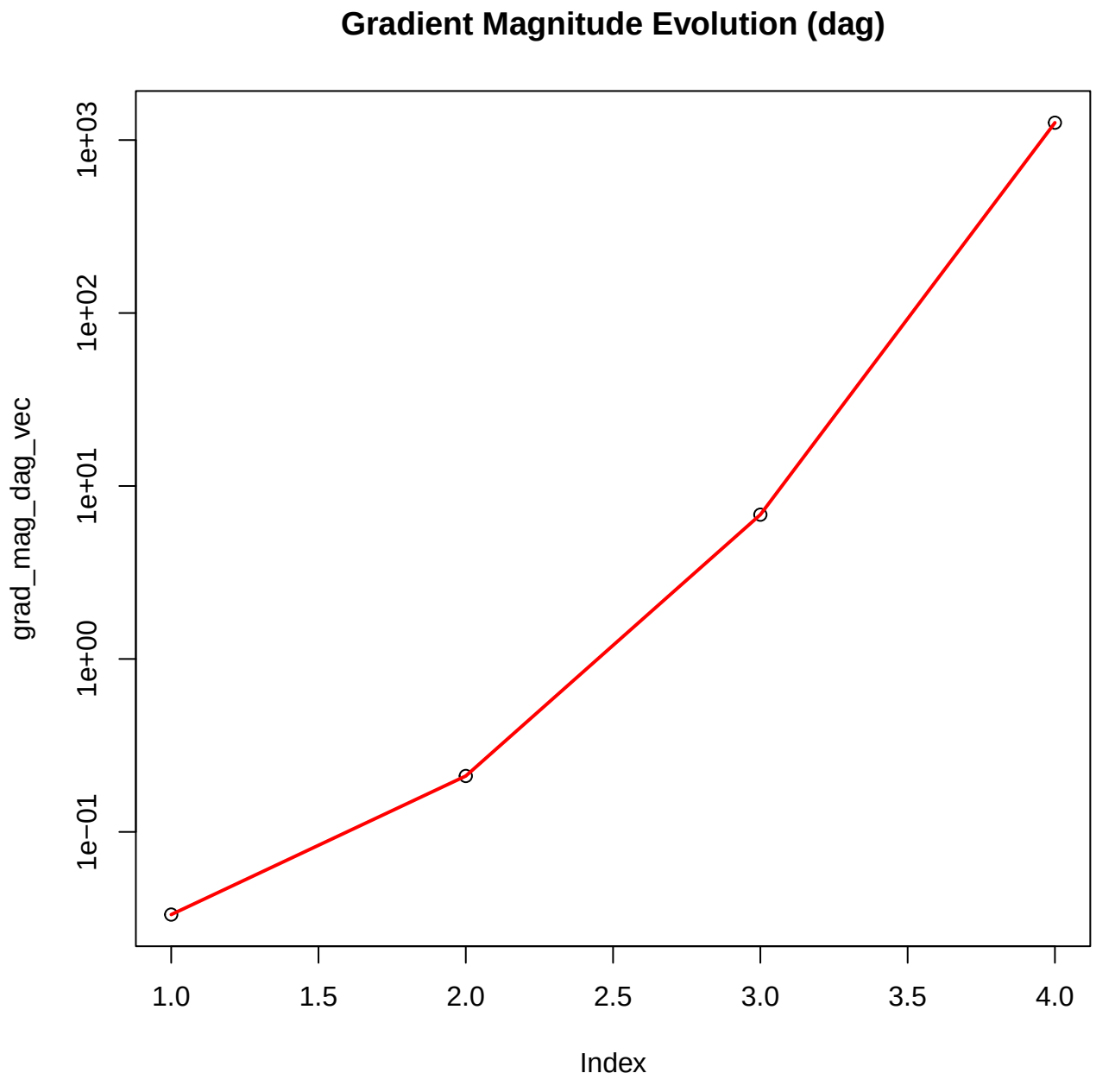

Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
0.005
0.010
0.020
0.050
0.100
0.200
1.0
1.5
2.0
2.5
3.0
Index

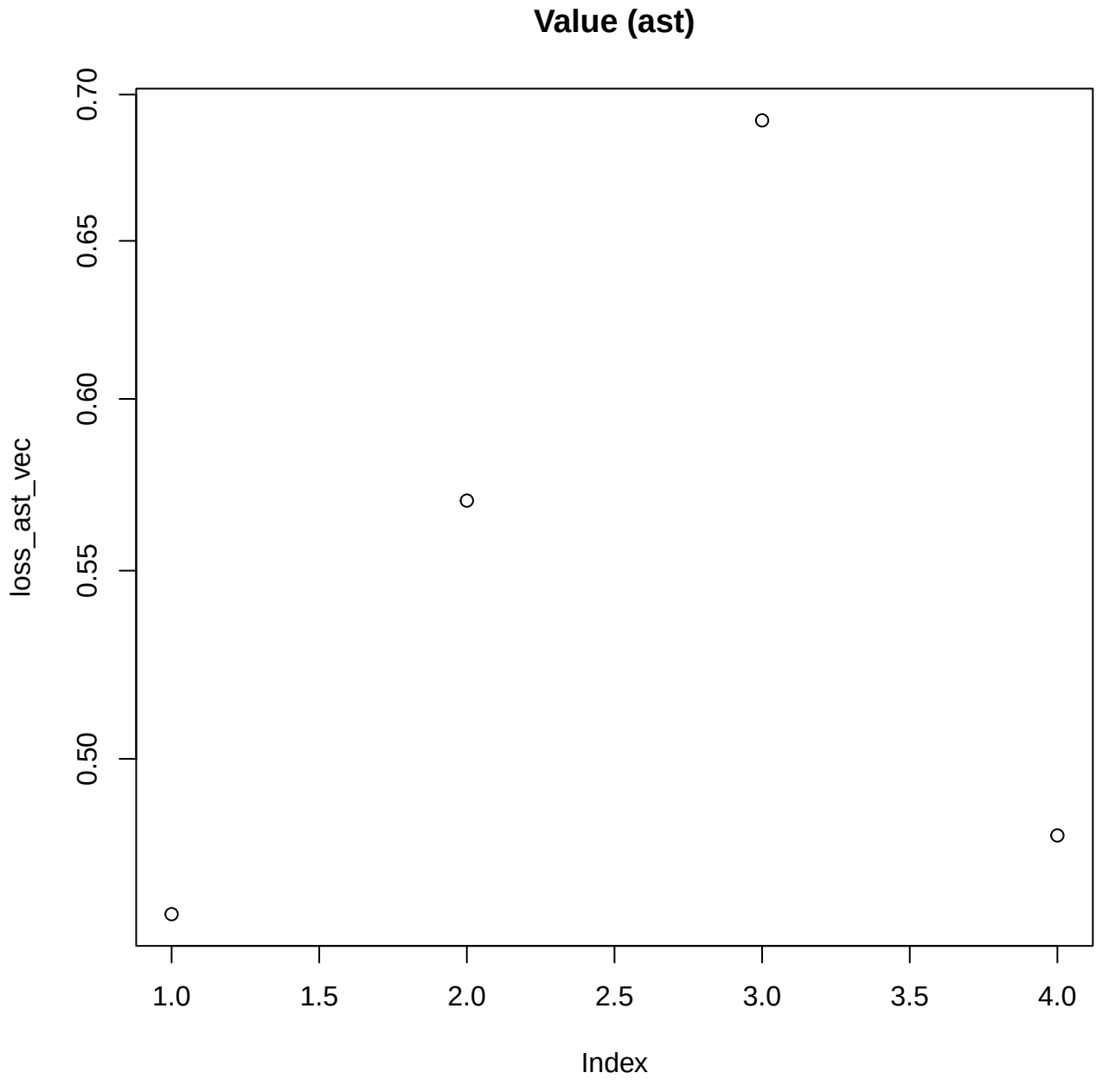

Gradient Magnitude Evolution (dag)
grad_mag_dag_vec
0.005
0.010
0.020
0.050
0.100
0.200
1.0
1.5
2.0
2.5
3.0
Index

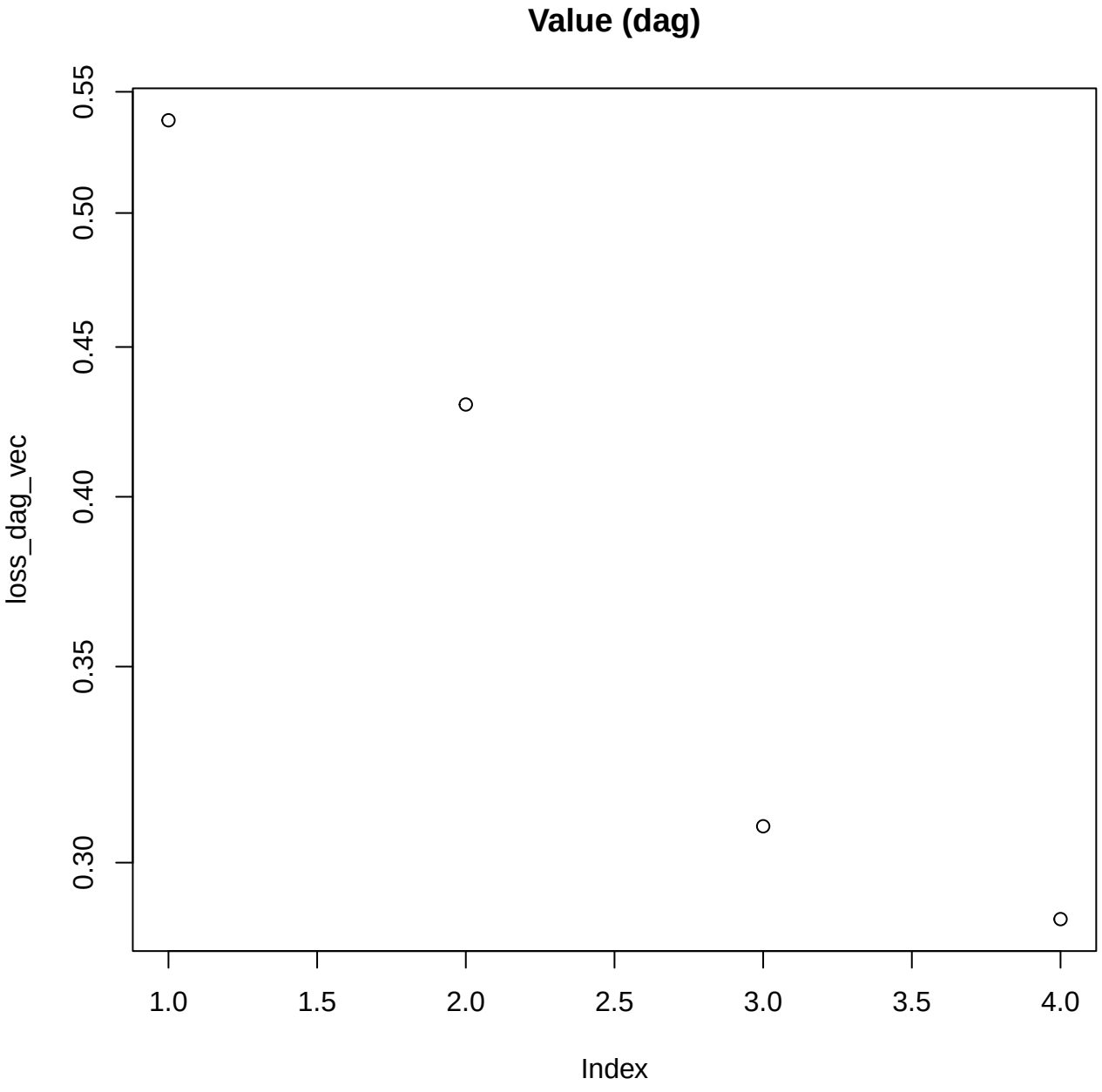

Value (ast)
loss_ast_vec
0.35
0.40
0.45
1.0
1.5
2.0
2.5
3.0
Index

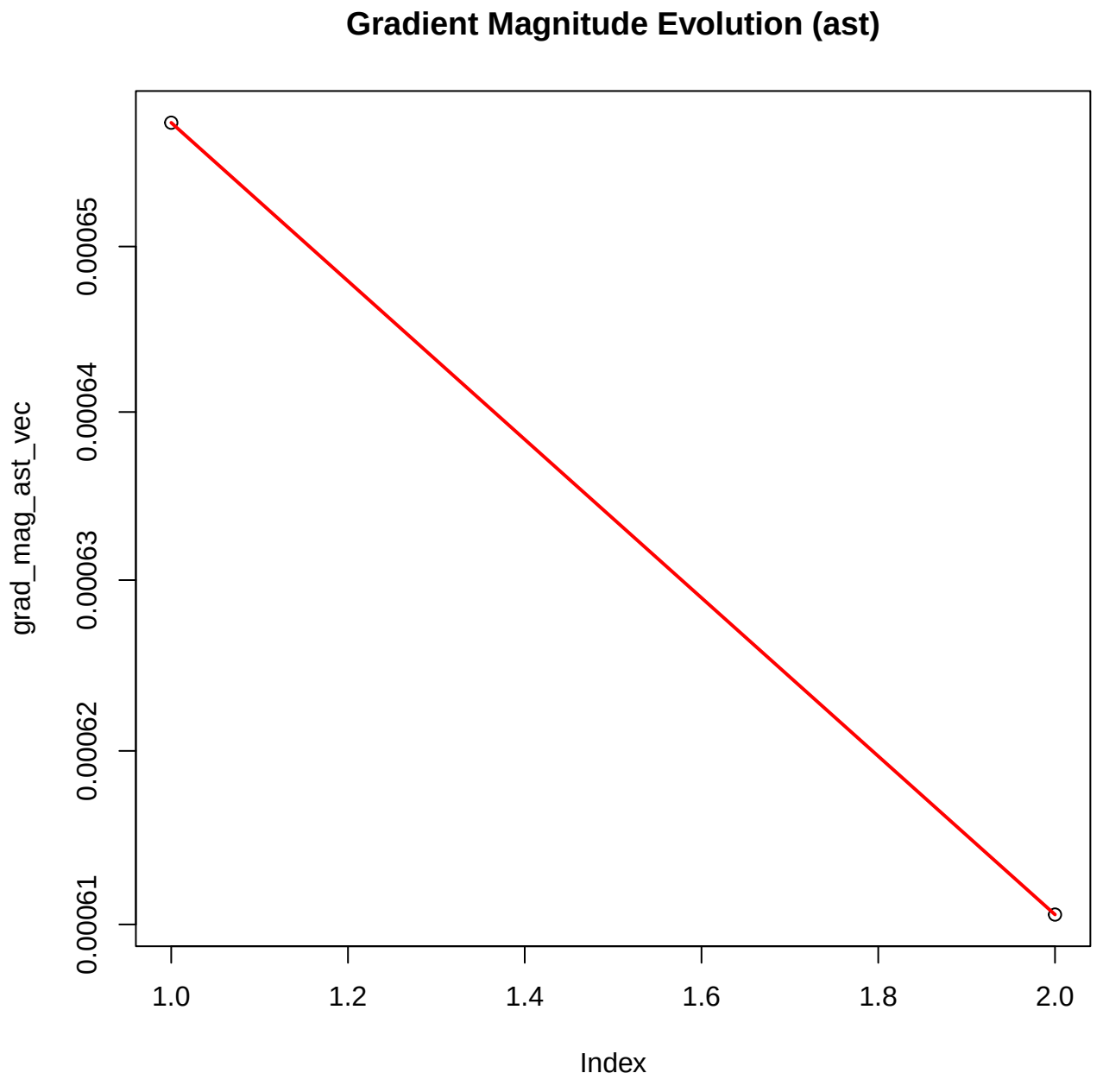

Value (dag)
loss_dag_vec
0.55
0.60
0.65
1.0
1.5
2.0
2.5
3.0
Index

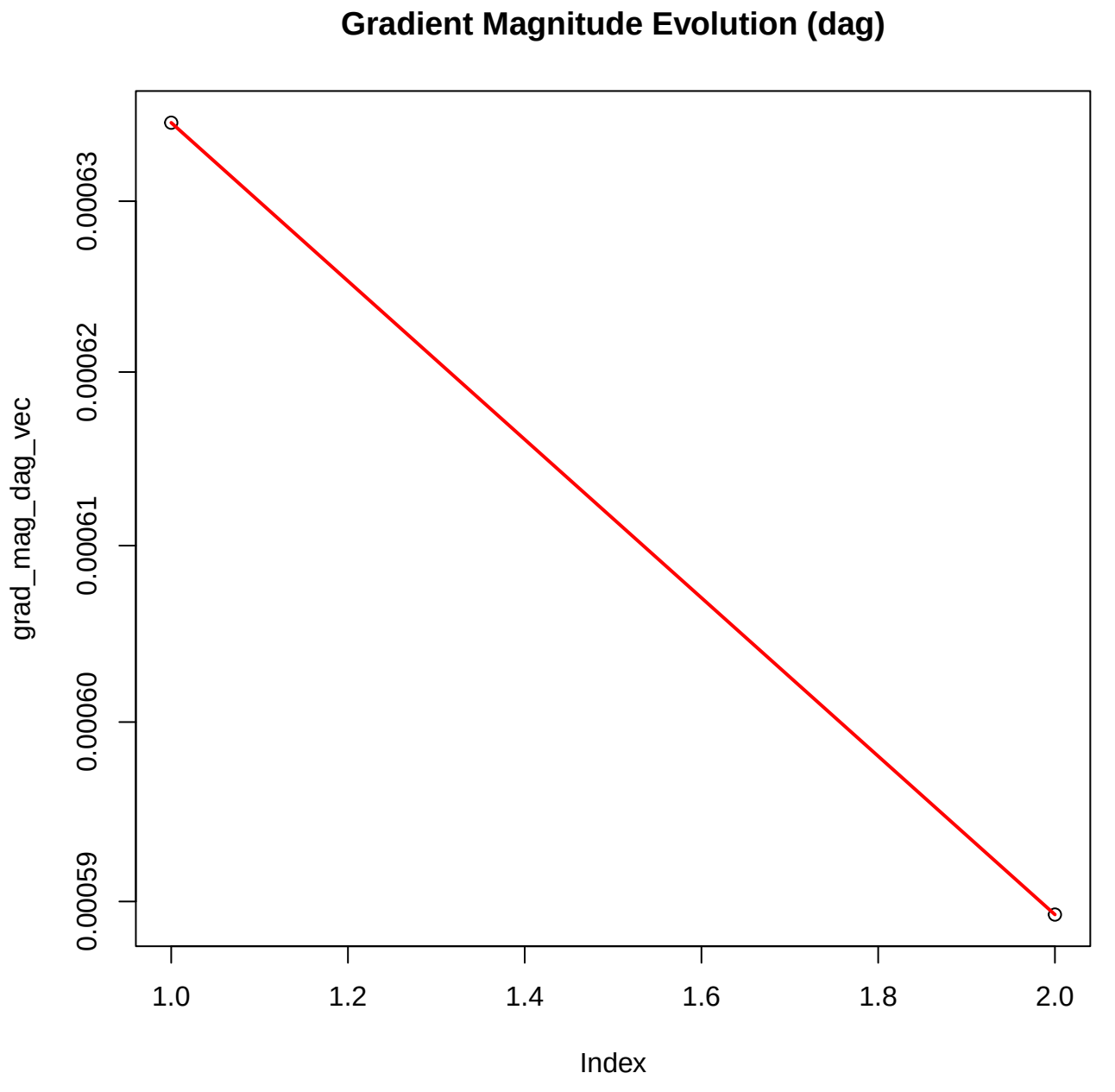

Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
1e−01
1e+00
1e+01
1e+02
1e+03
1.0
1.5
2.0
2.5
3.0
3.5
4.0
Index

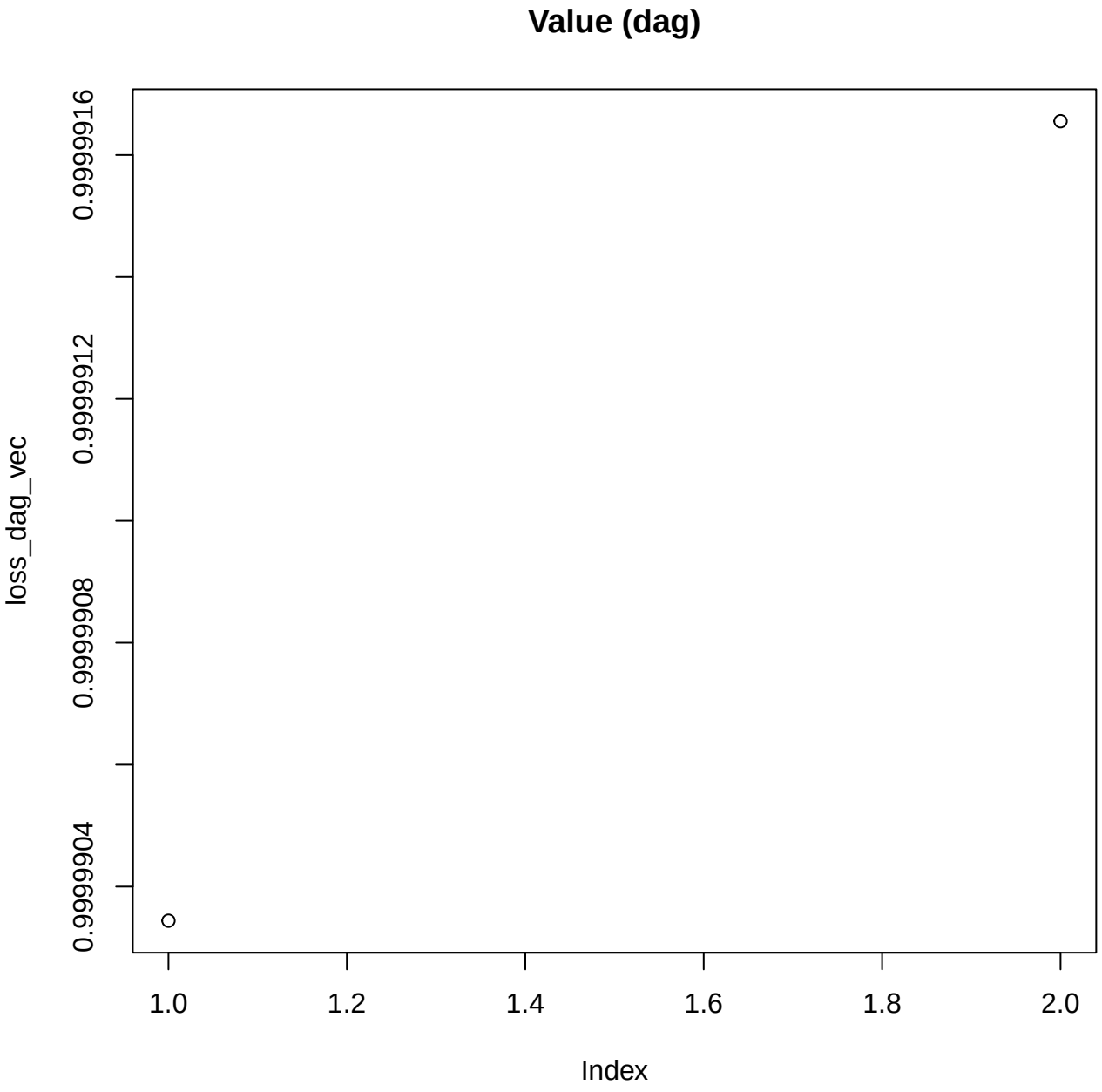

Gradient Magnitude Evolution (dag)
grad_mag_dag_vec
1e−01
1e+00
1e+01
1e+02
1e+03
1.0
1.5
2.0
2.5
3.0
3.5
4.0
Index

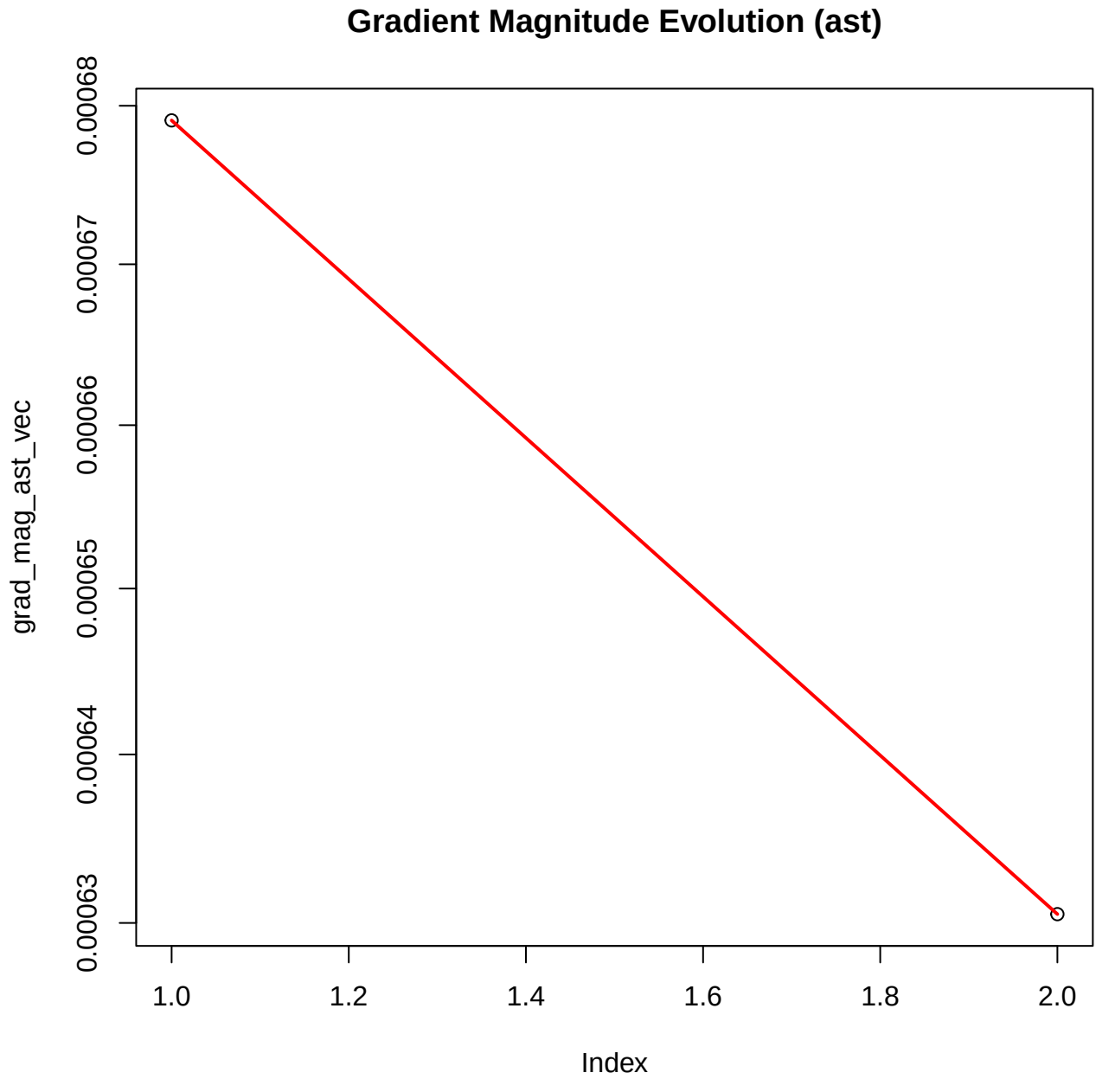

Value (ast)
loss_ast_vec
0.50
0.55
0.60
0.65
0.70
1.0
1.5
2.0
2.5
3.0
3.5
4.0
Index

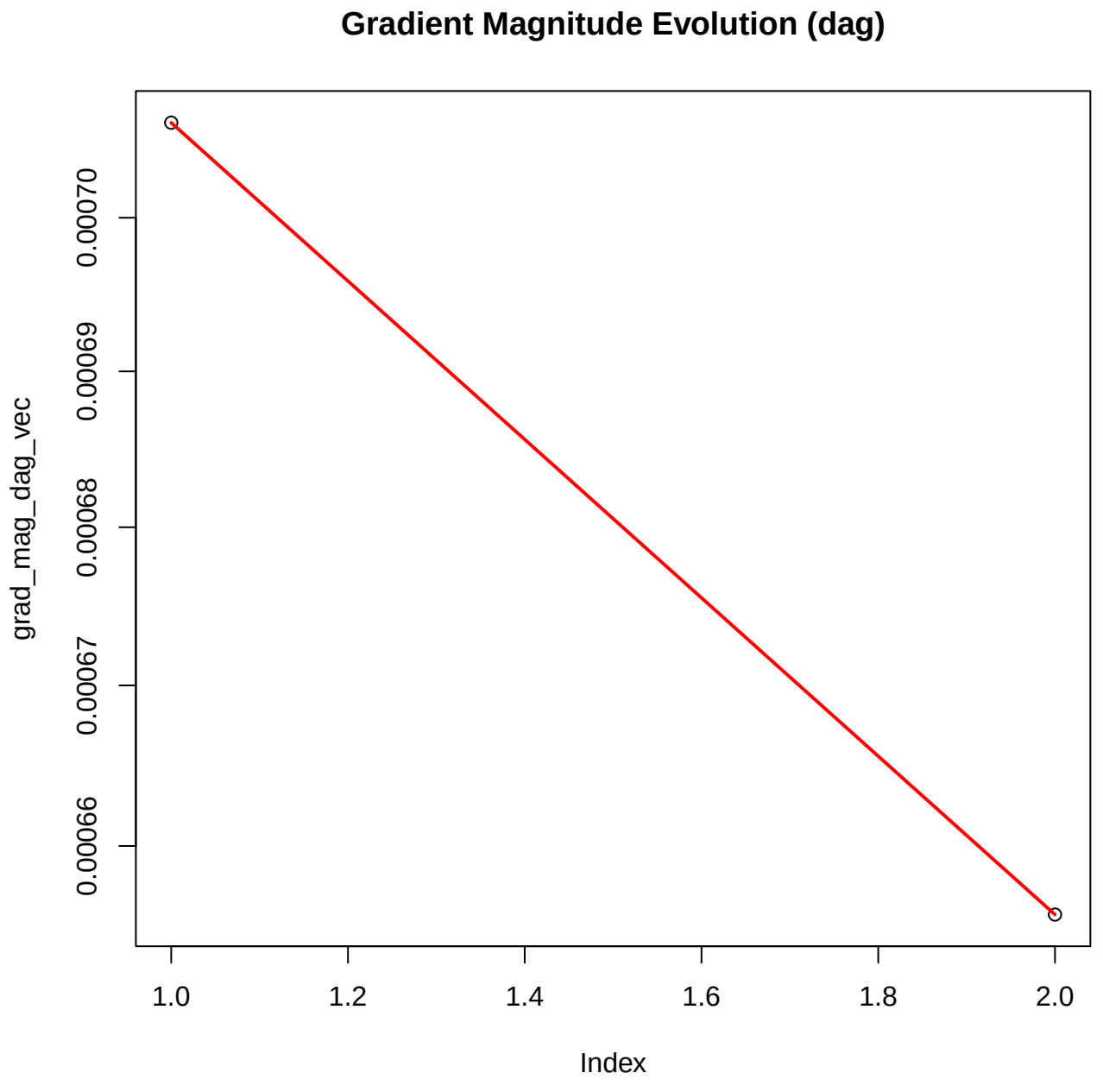

Value (dag)
loss_dag_vec
0.55
0.50
0.45
0.40
0.35
0.30
1.0
1.5
2.0
2.5
3.0
3.5
4.0
Index

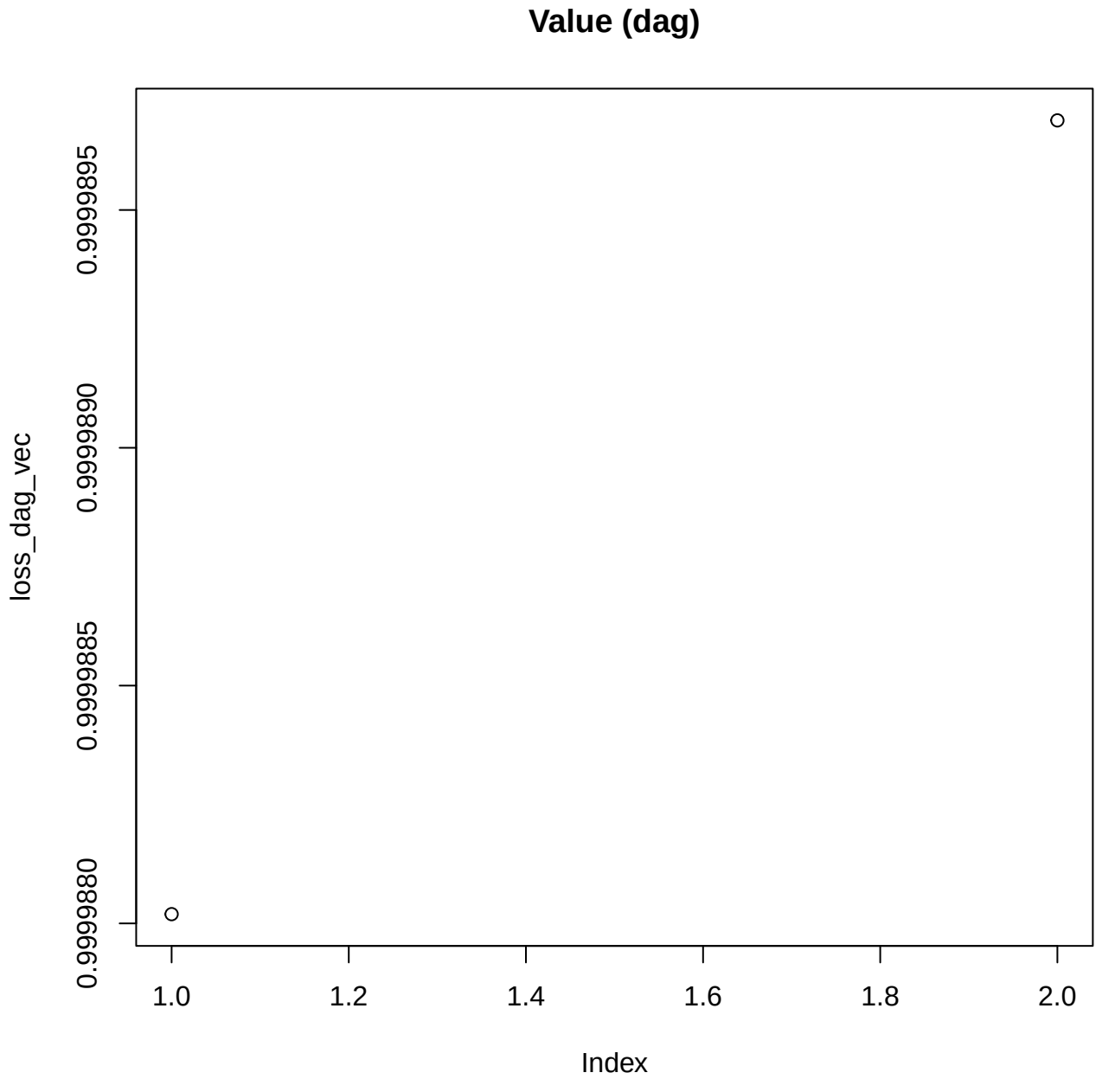

Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
0.00061
0.00062
0.00063
0.00064
0.00065
1.0
1.2
1.4
1.6
1.8
2.0
Index

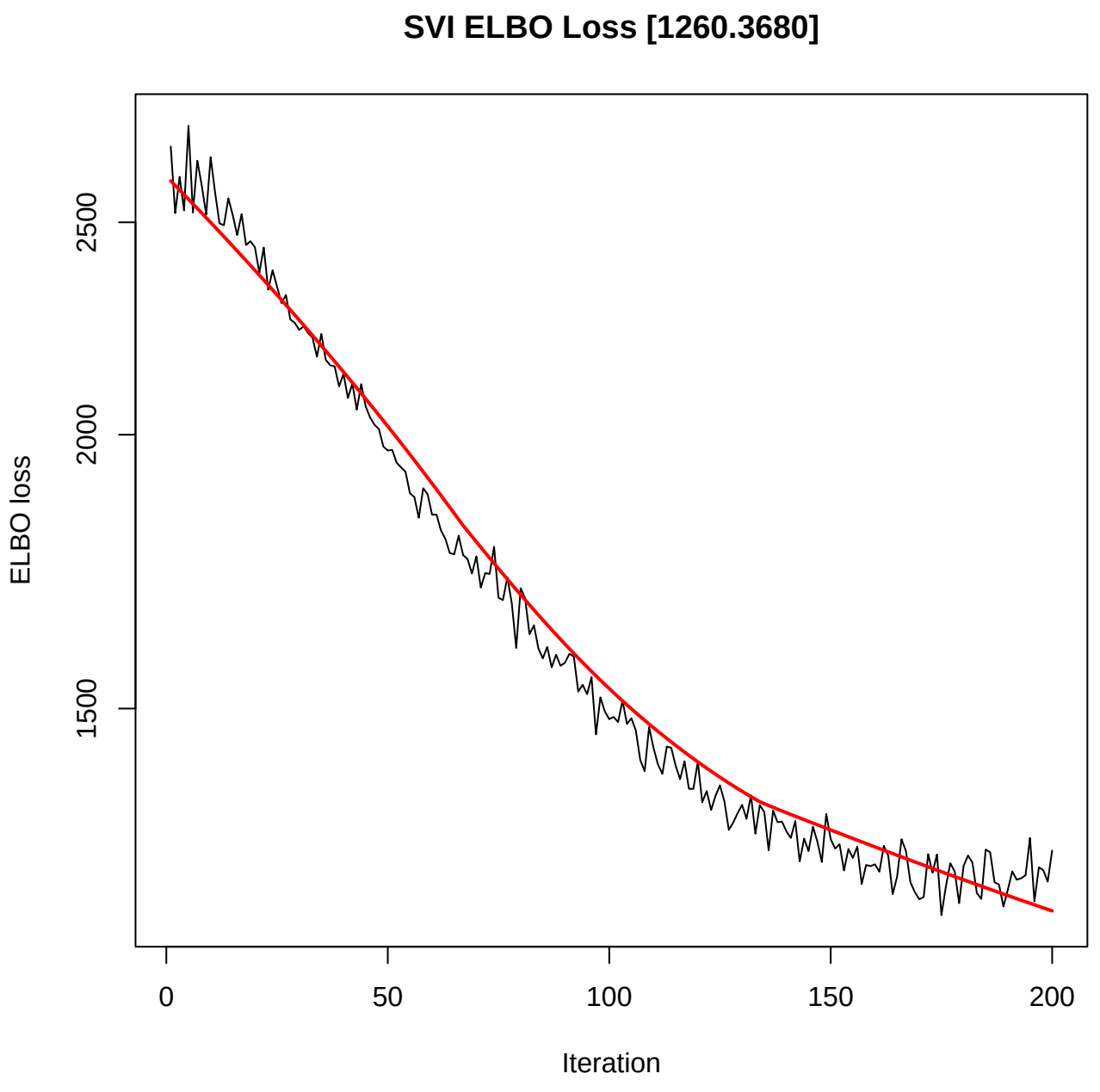

Gradient Magnitude Evolution (dag)
grad_mag_dag_vec
0.00059
0.00060
0.00061
0.00062
0.00063
1.0
1.2
1.4
1.6
1.8
2.0
Index

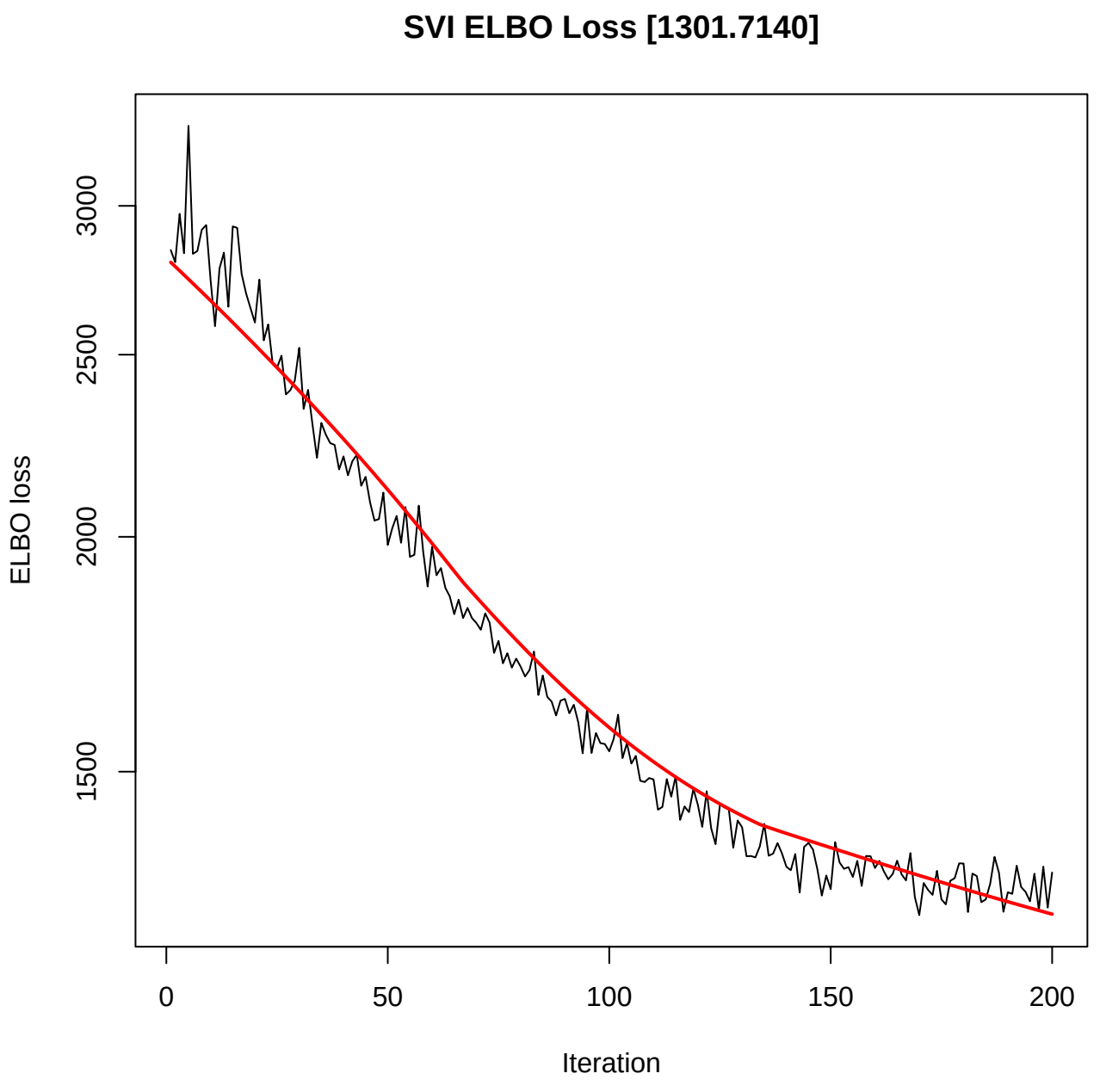

Value (dag)
loss_dag_vec
0.9999904
0.9999908
0.9999912
0.9999916
1.0
1.2
1.4
1.6
1.8
2.0
Index

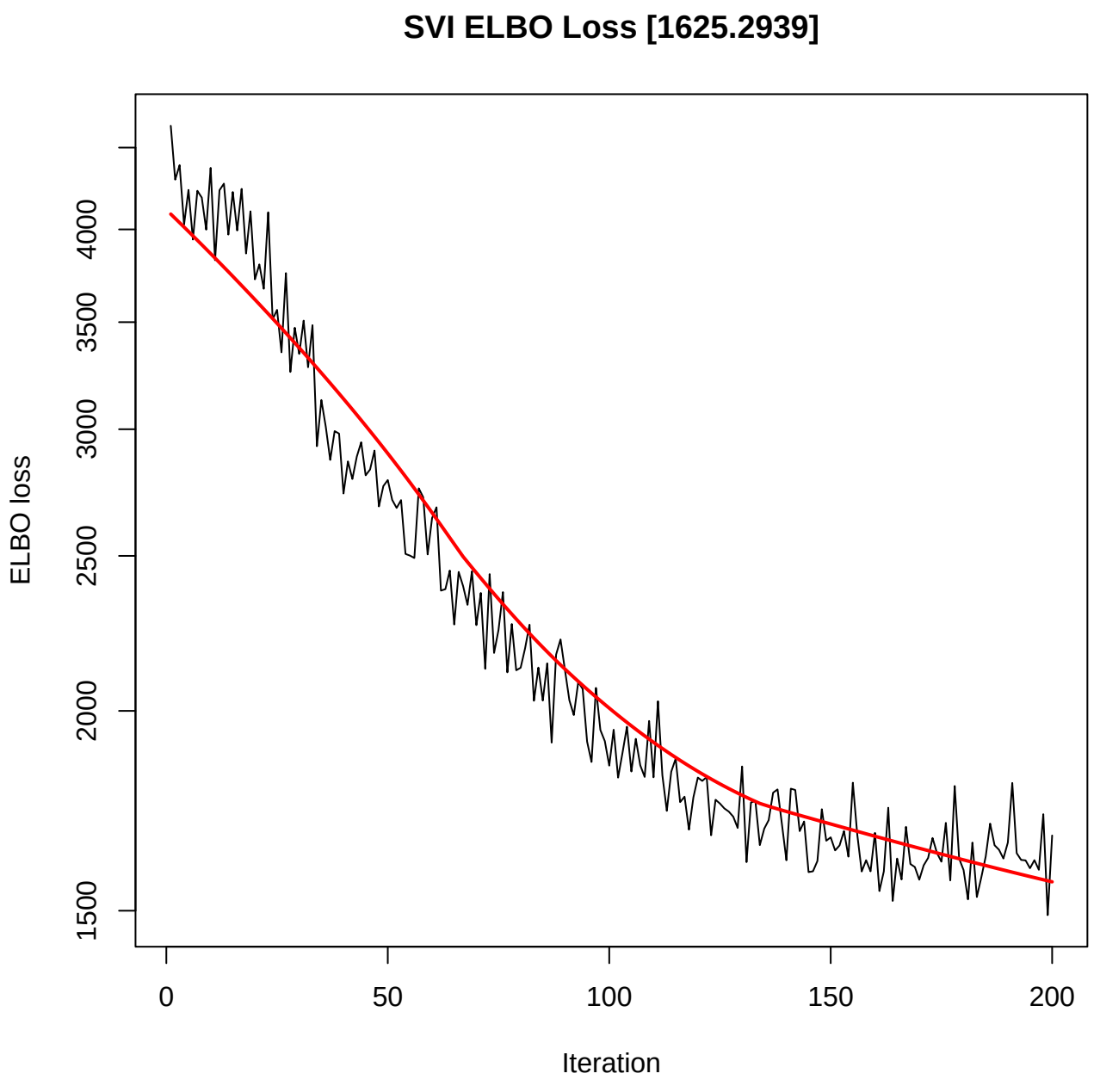

Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
0.00063
0.00064
0.00065
0.00066
0.00067
0.00068
1.0
1.2
1.4
1.6
1.8
2.0
Index

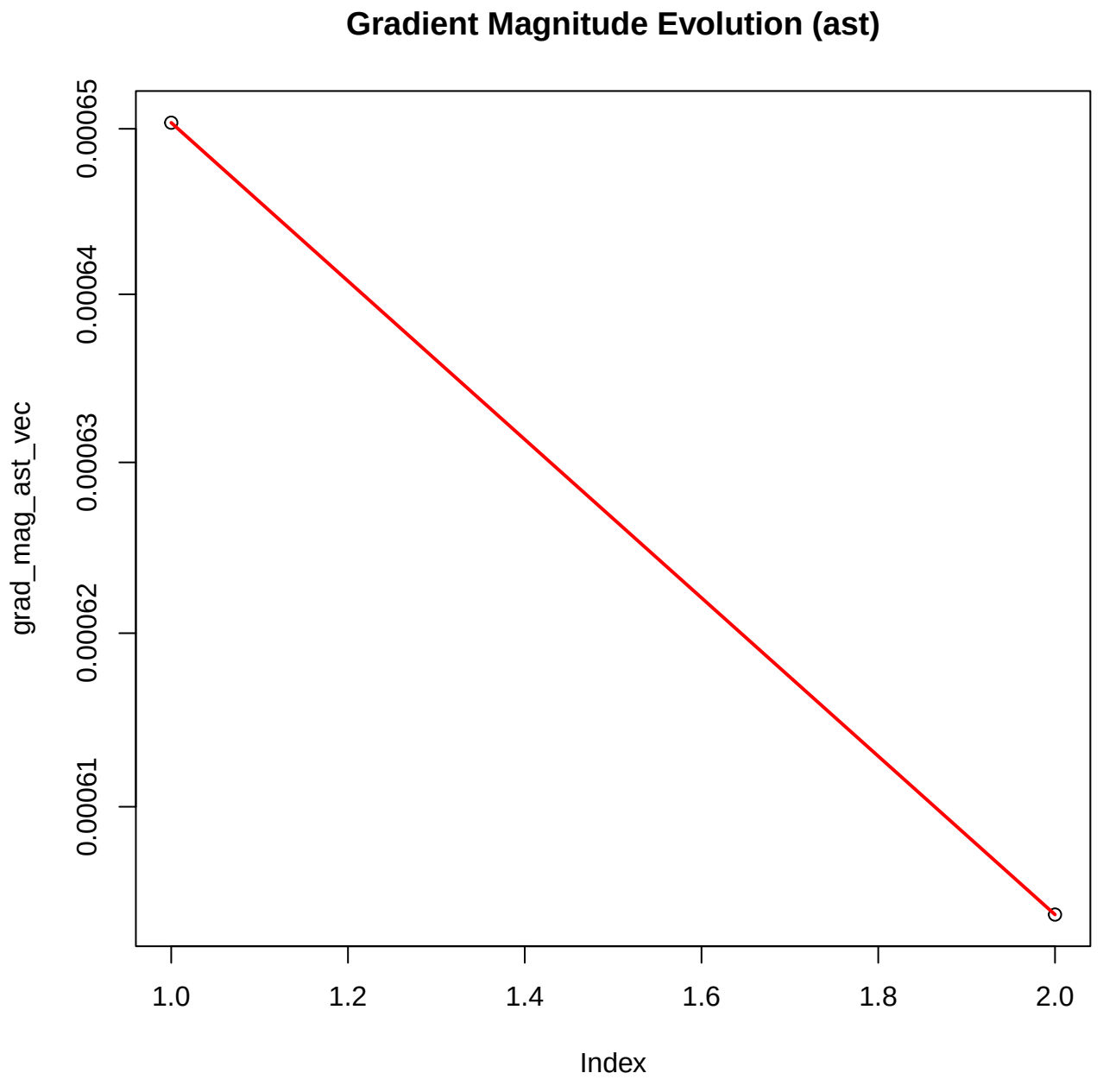

Gradient Magnitude Evolution (dag)
grad_mag_dag_vec
0.00066
0.00067
0.00068
0.00069
0.00070
1.0
1.2
1.4
1.6
1.8
2.0
Index

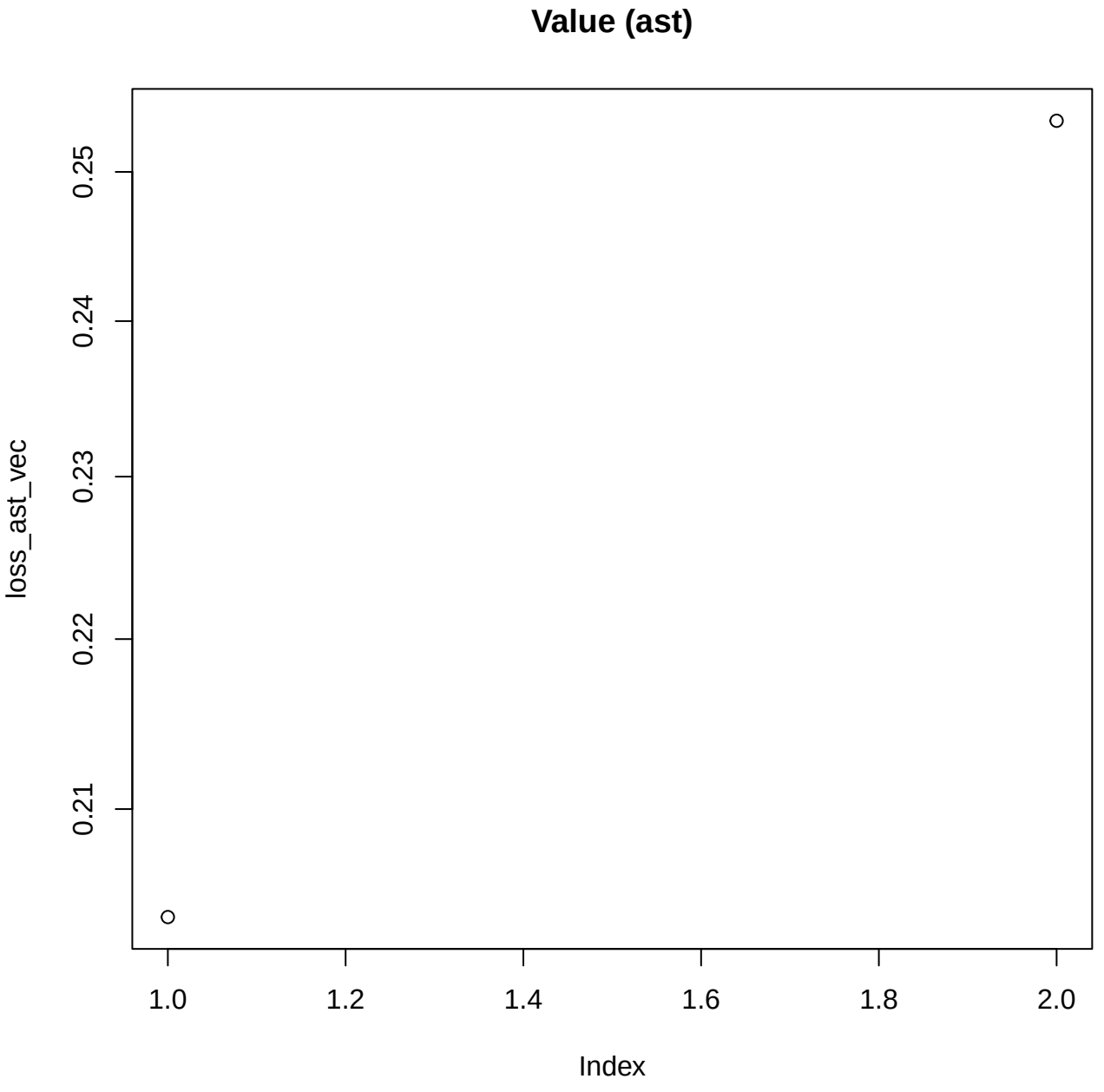

Value (dag)
loss_dag_vec
0.9999880
0.9999885
0.9999890
0.9999895
1.0
1.2
1.4
1.6
1.8
2.0
Index

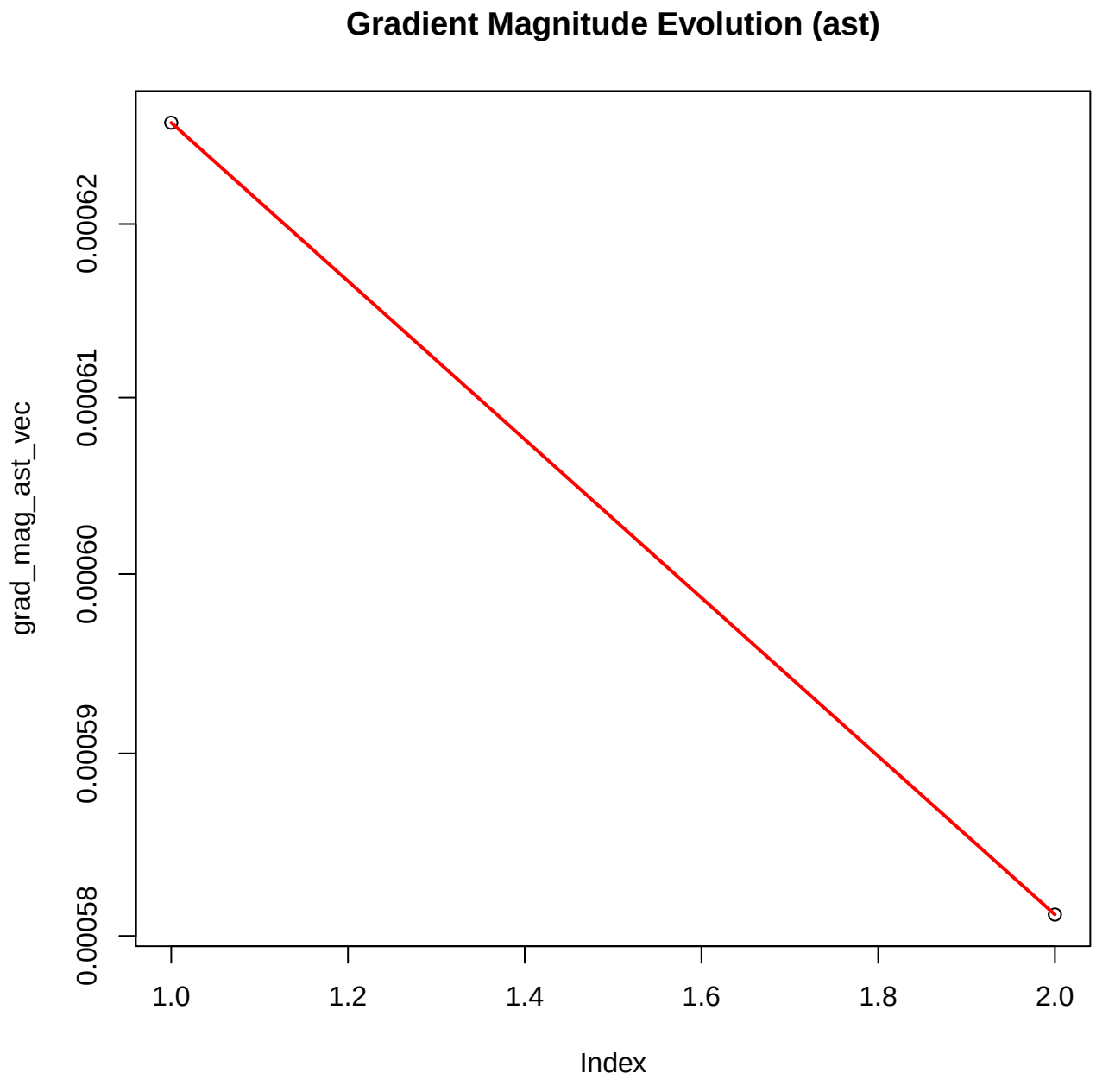

SVI ELBO Loss [1260.3680]
ELBO loss
2500
2000
1500
0
50
100
150
200
Iteration

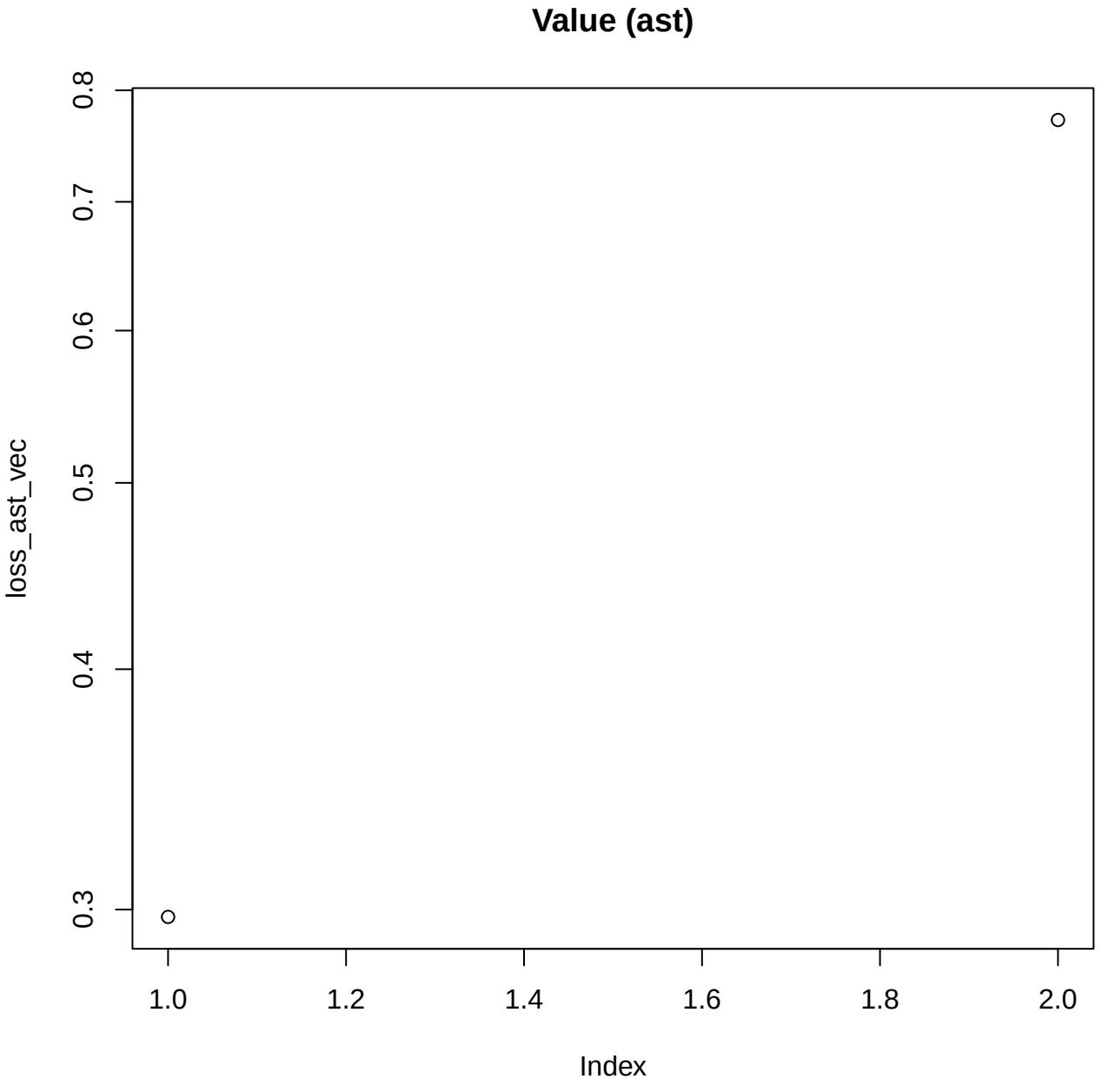

SVI ELBO Loss [1301.7140]
ELBO loss
3000
2500
2000
1500
0
50
100
150
200
Iteration


SVI ELBO Loss [1625.2939]
ELBO loss
1500
2000
2500
3000
3500
4000
0
50
100
150
200
Iteration


Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
0.00061
0.00062
0.00063
0.00064
0.00065
1.0
1.2
1.4
1.6
1.8
2.0
Index


Value (ast)
loss_ast_vec
0.21
0.22
0.23
0.24
0.25
1.0
1.2
1.4
1.6
1.8
2.0
Index


Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
0.00058
0.00059
0.00060
0.00061
0.00062
1.0
1.2
1.4
1.6
1.8
2.0
Index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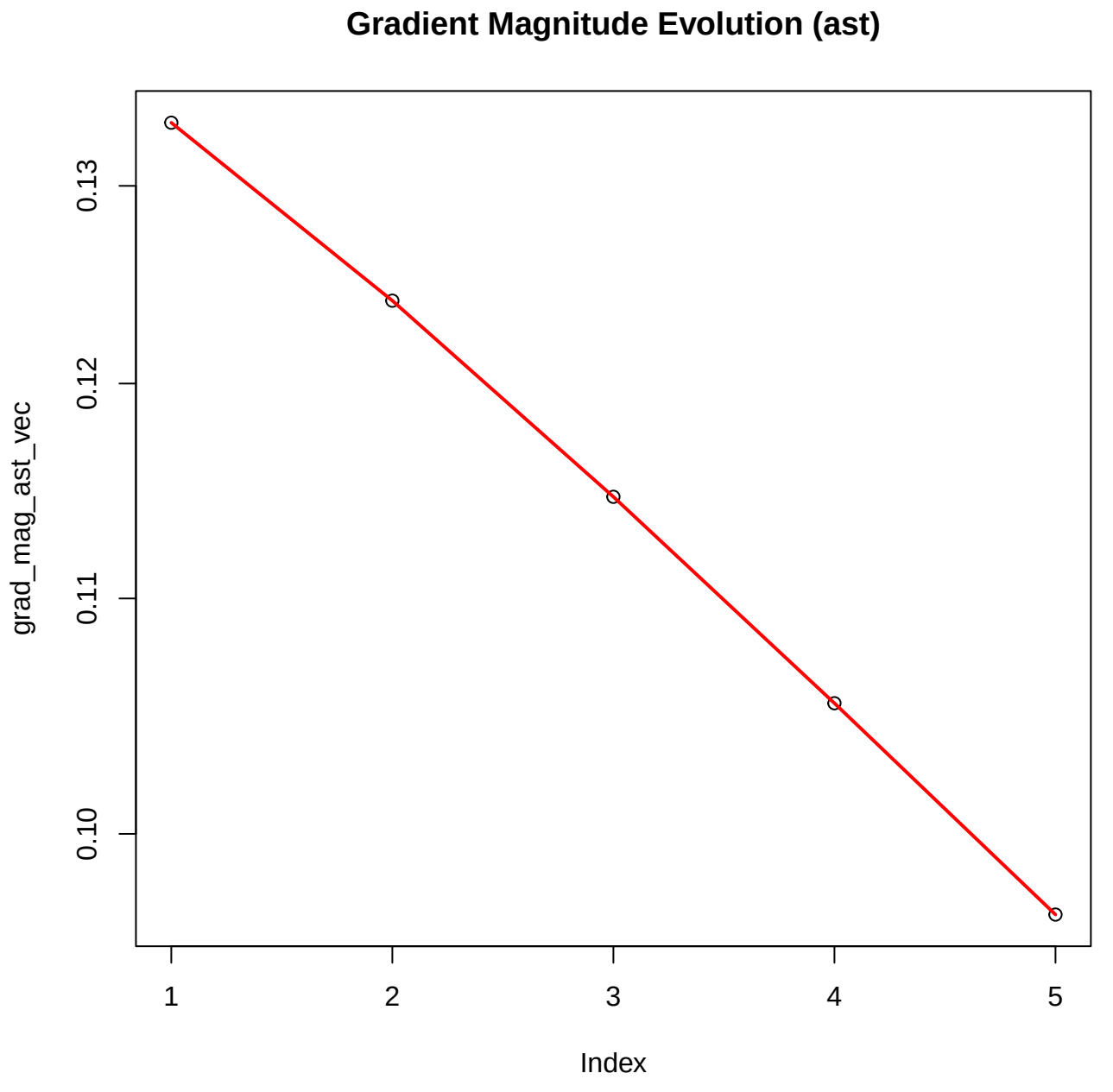
Gradient Magnitude Evolution (ast)
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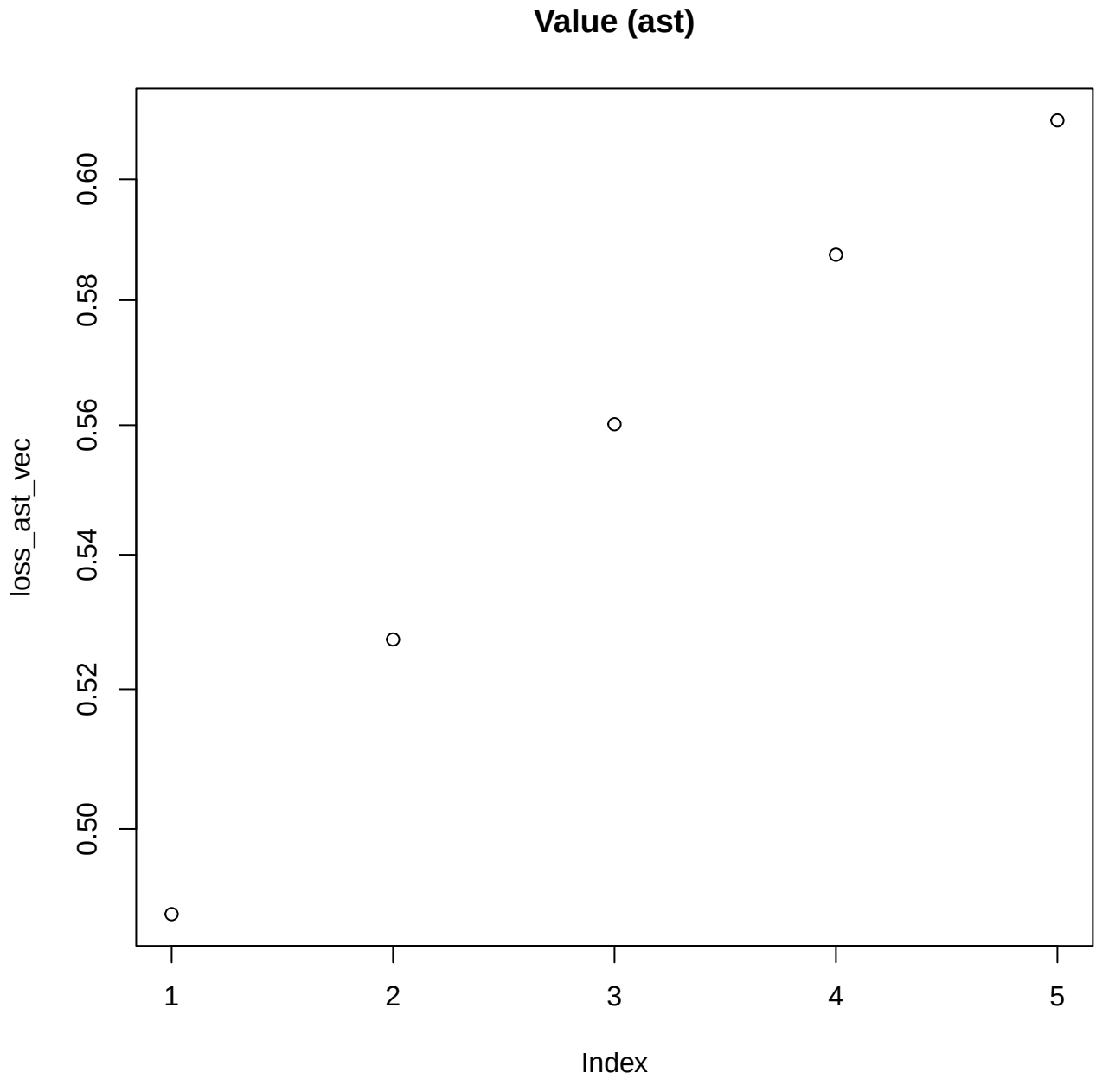
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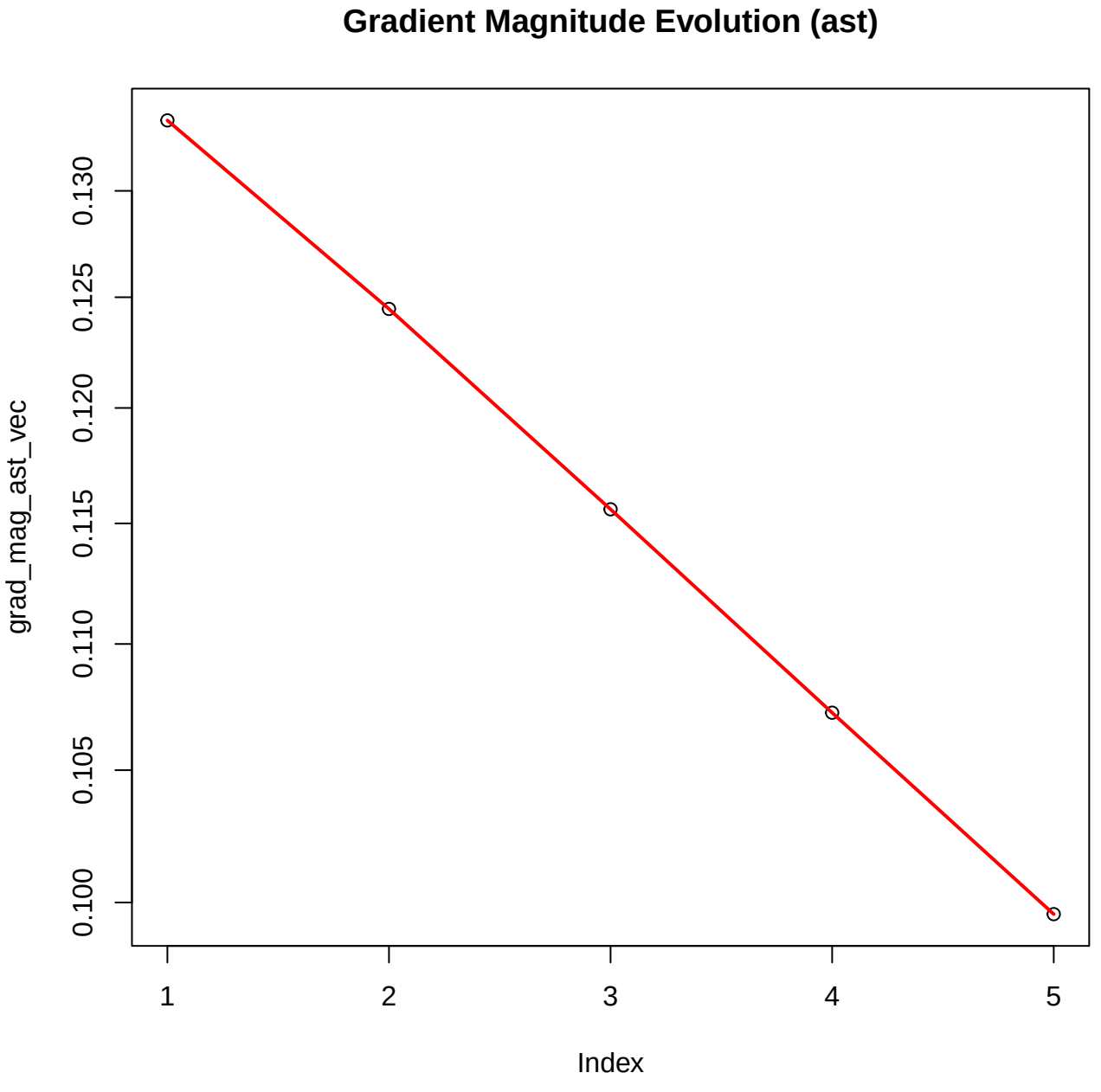
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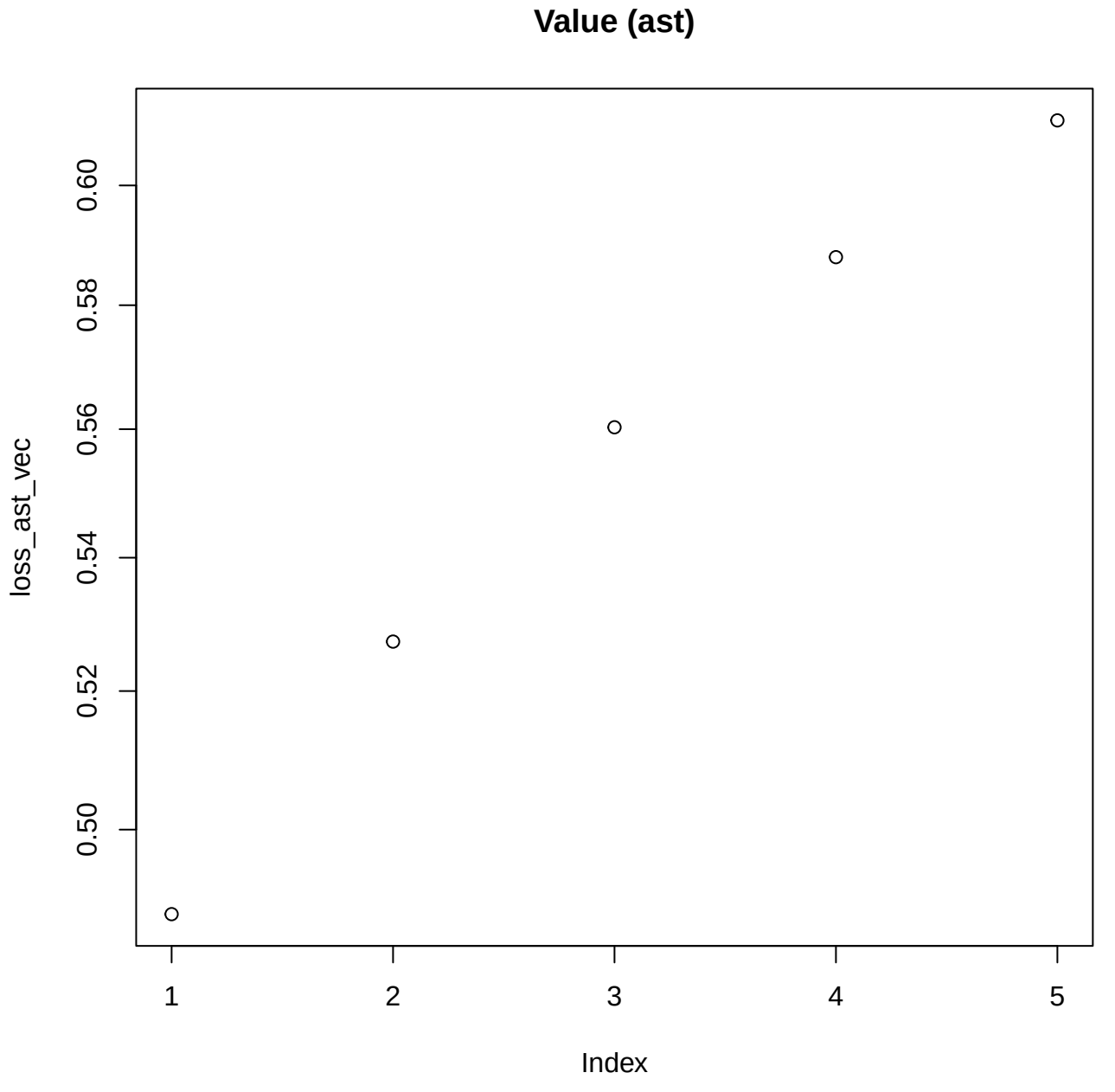
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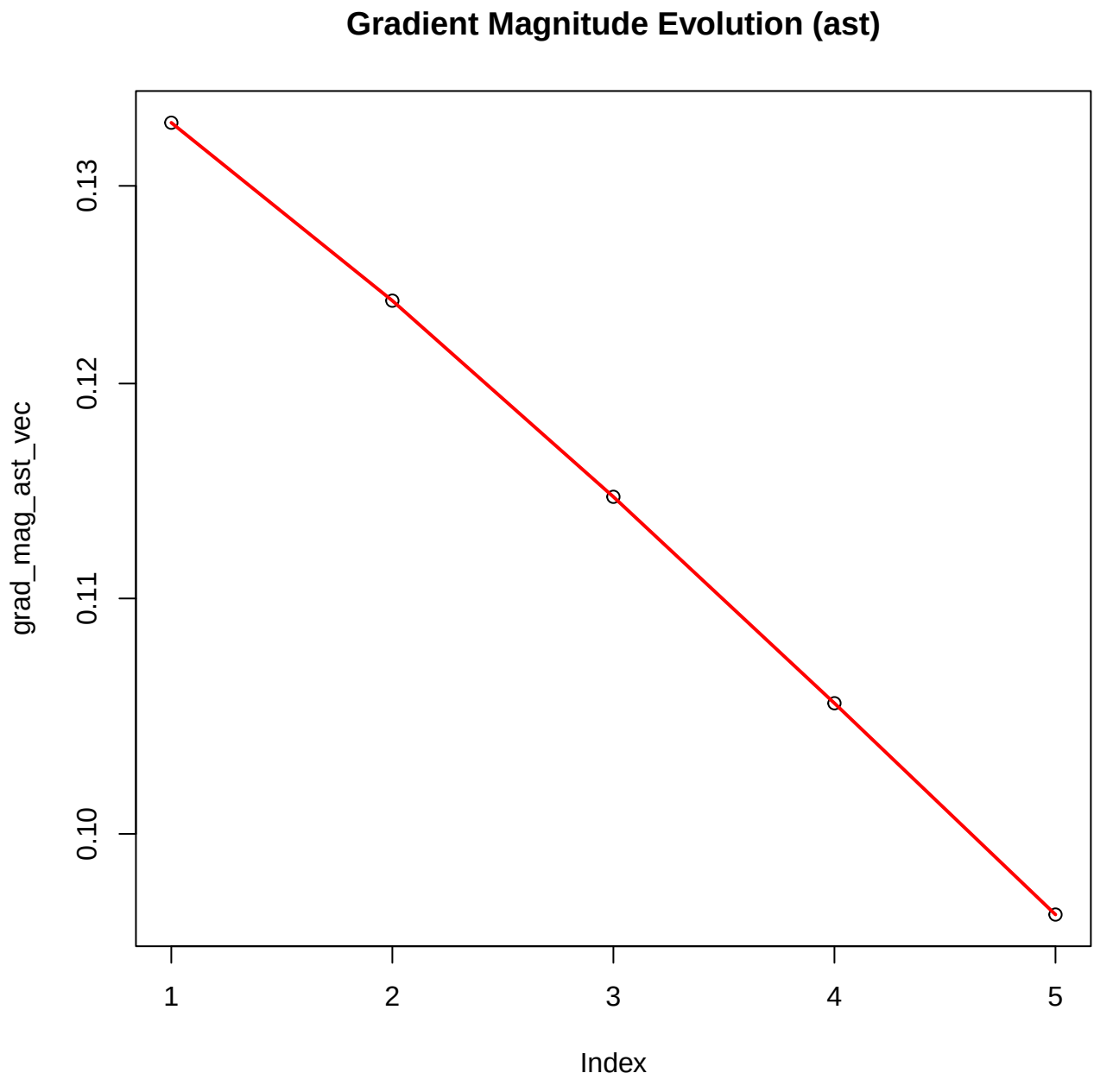
Gradient Magnitude Evolution (ast)
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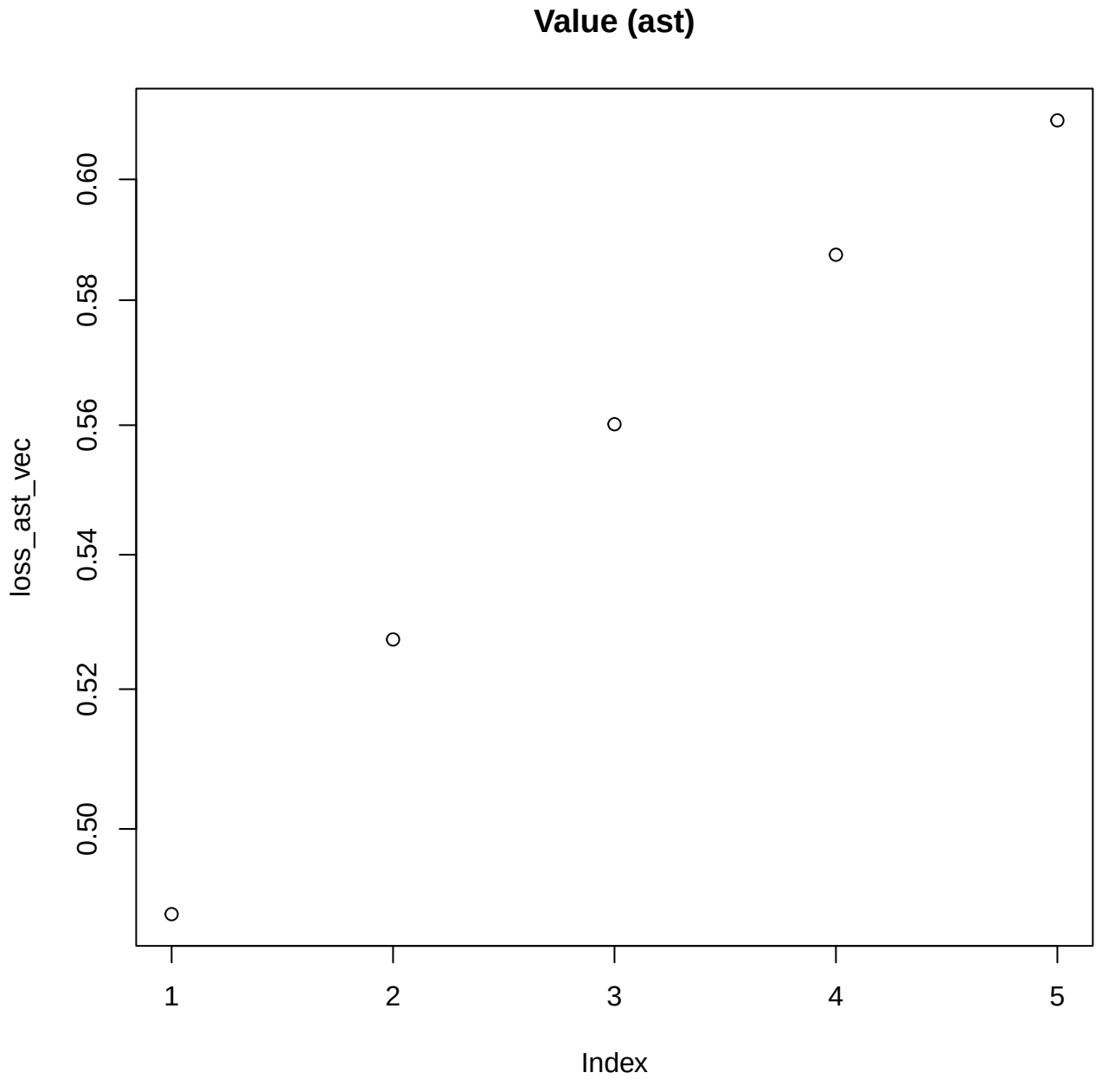
Value (ast)
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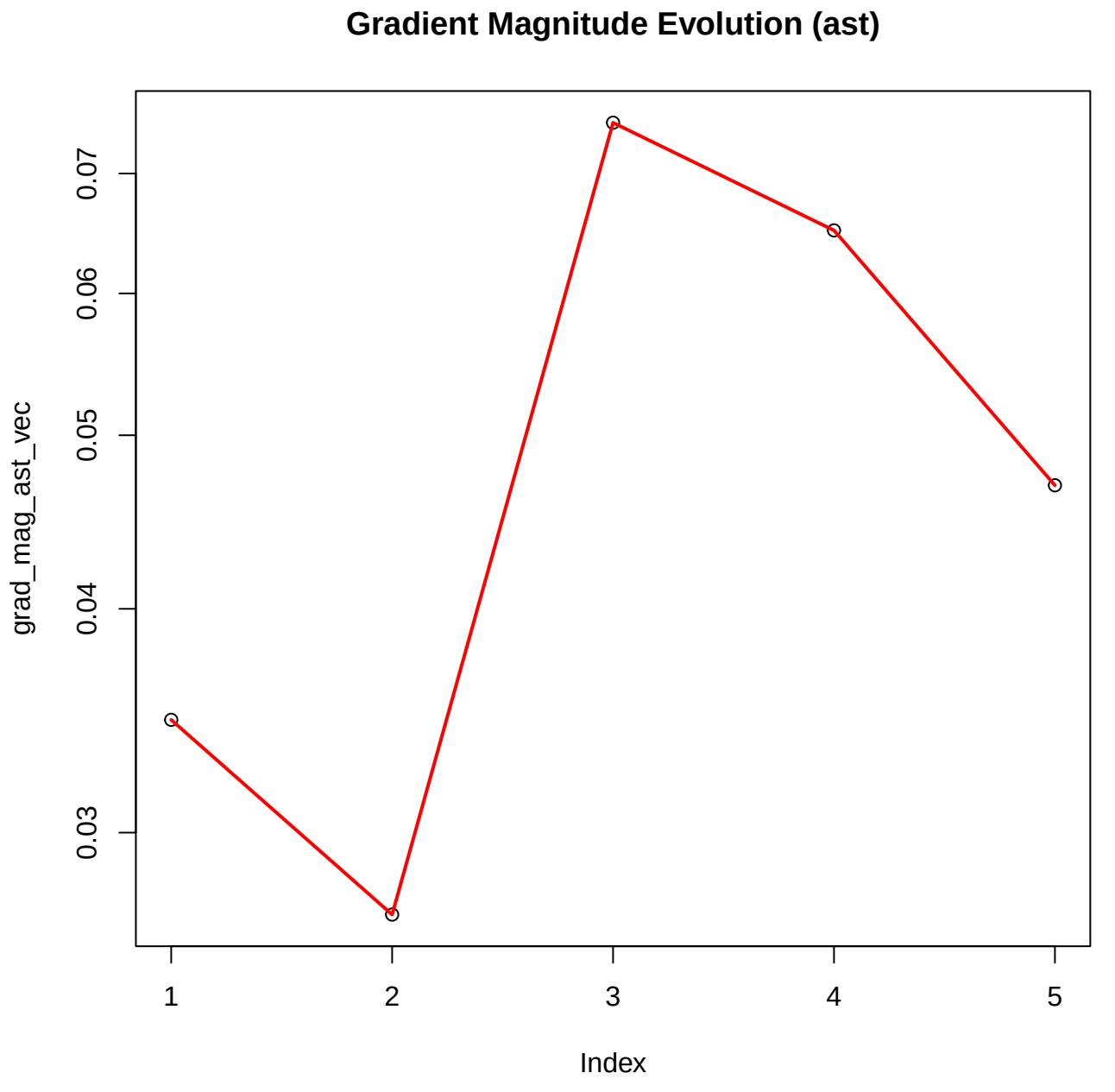
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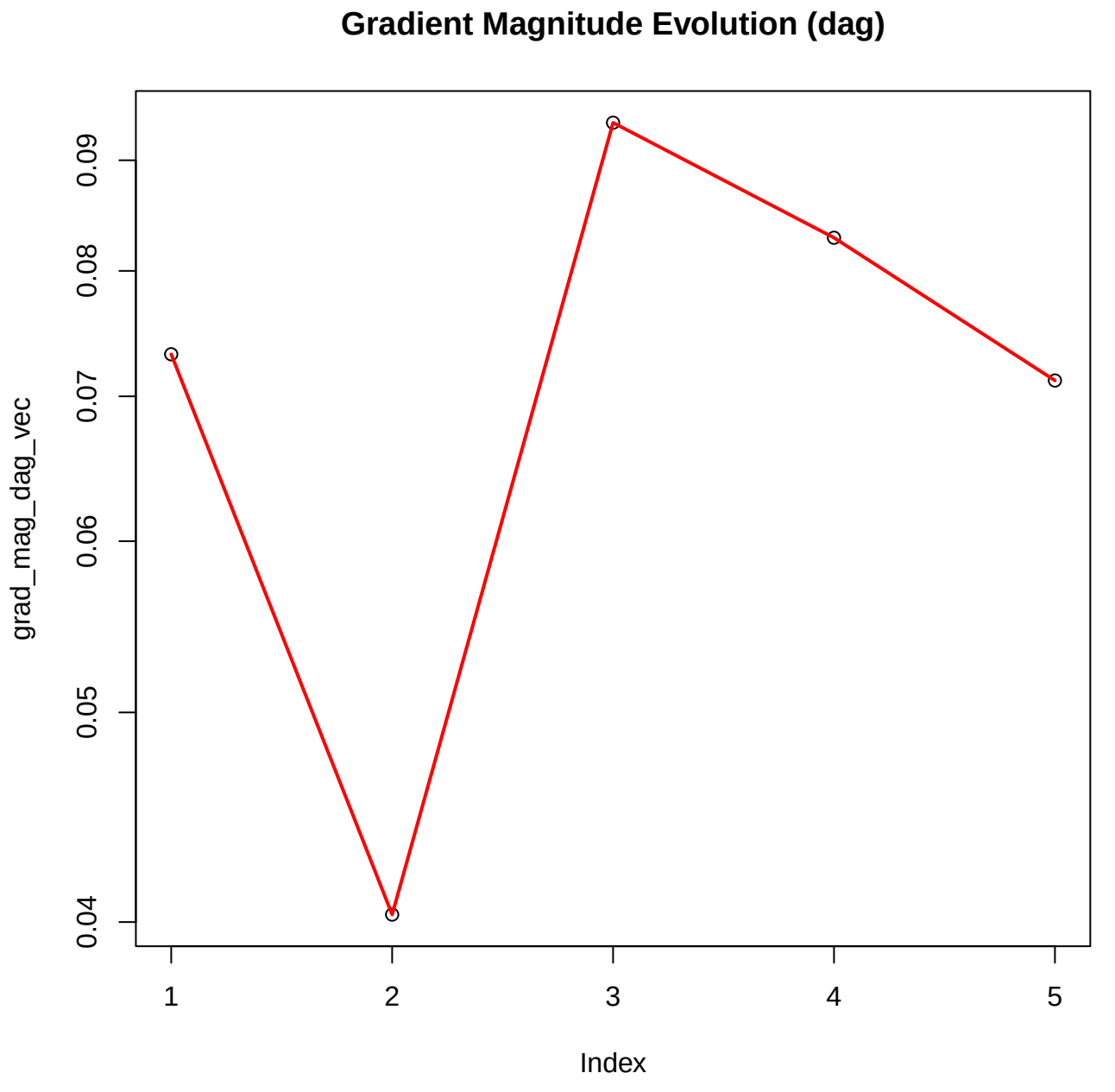
Gradient Magnitude Evolution (dag)
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0.09
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3
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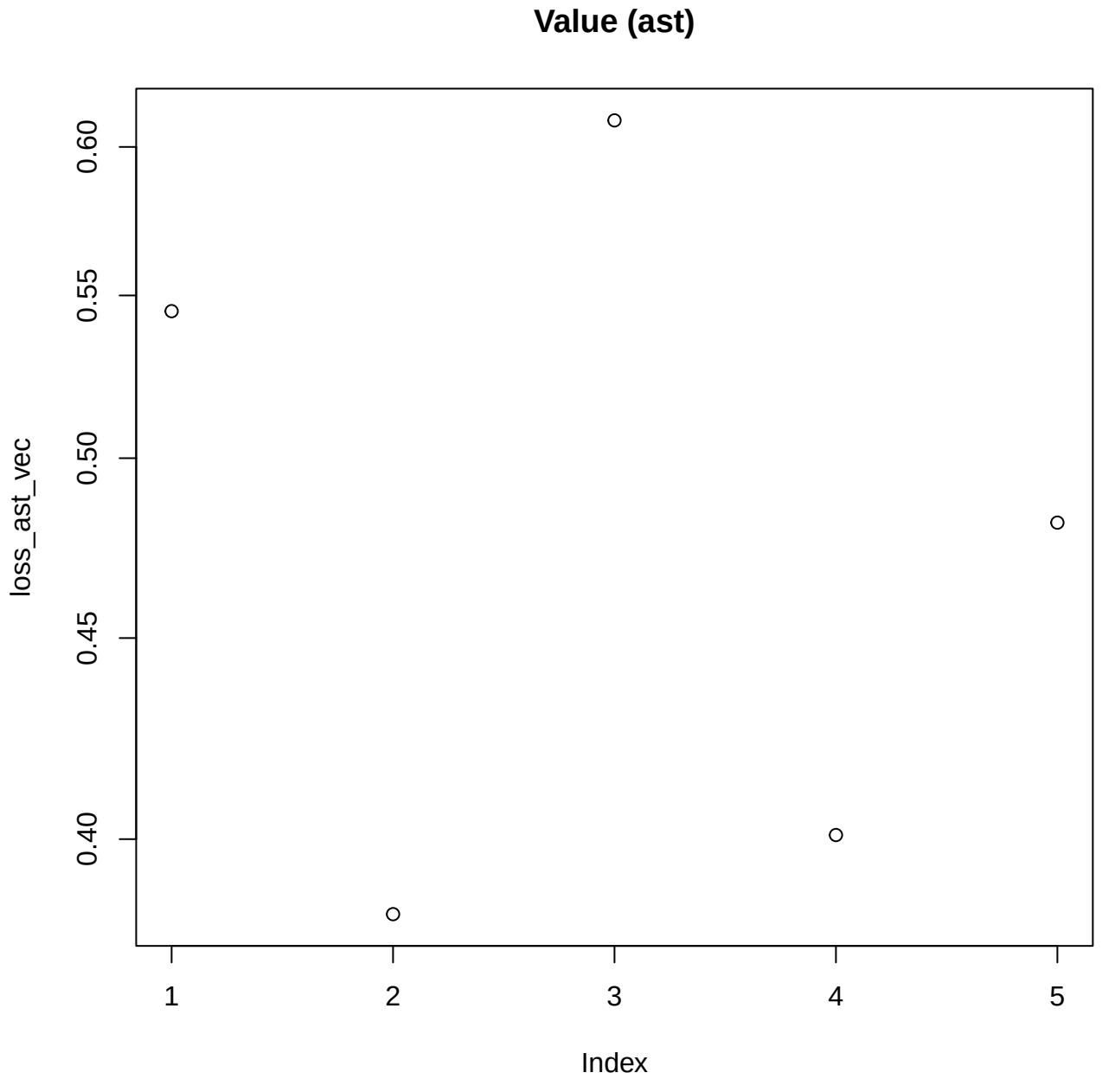
5
Index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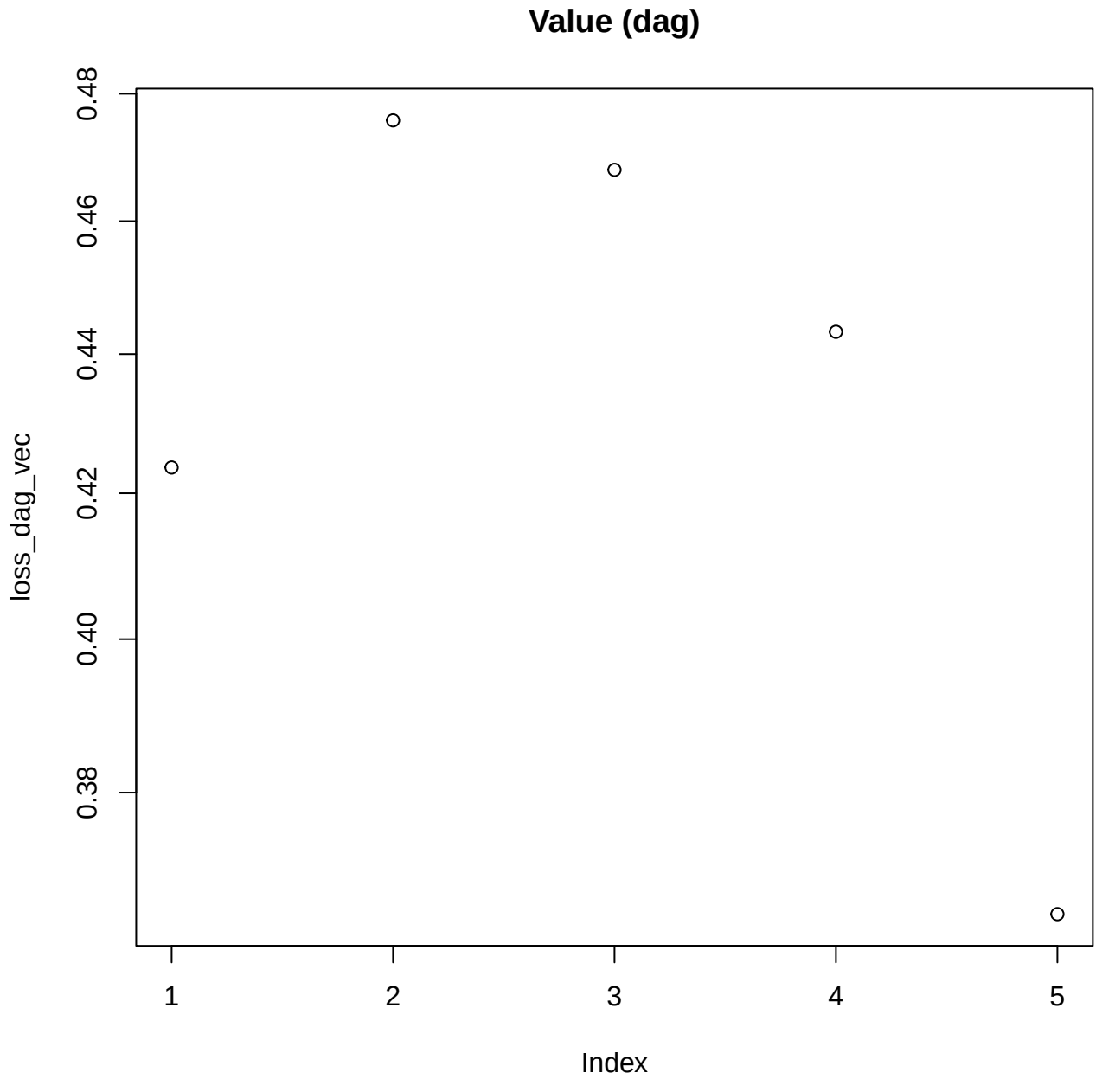
4
5
Index


Value (dag)
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0.44
0.46
0.48
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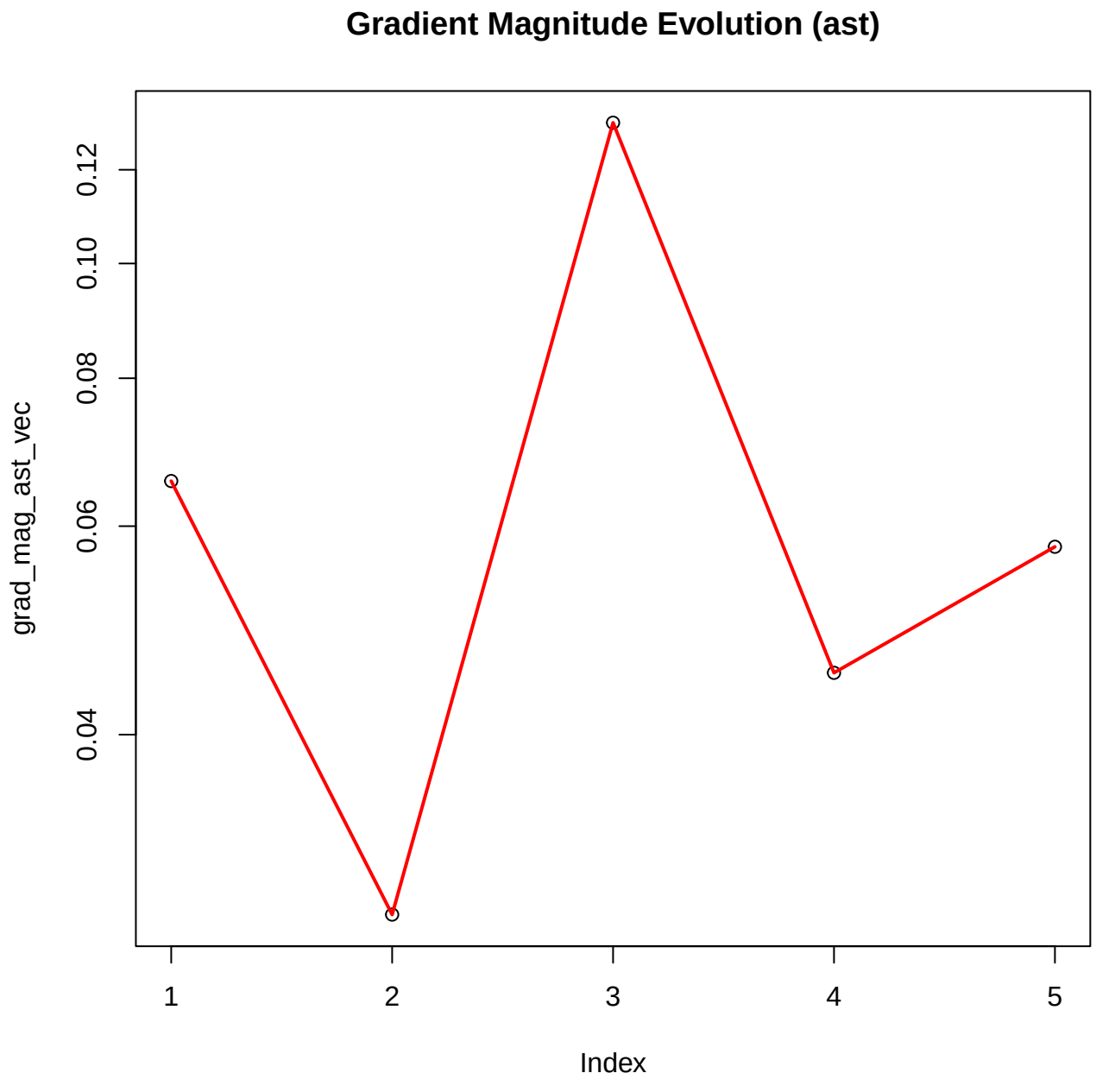
3
4
5
Index


Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
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0.06
0.08
0.10
0.12
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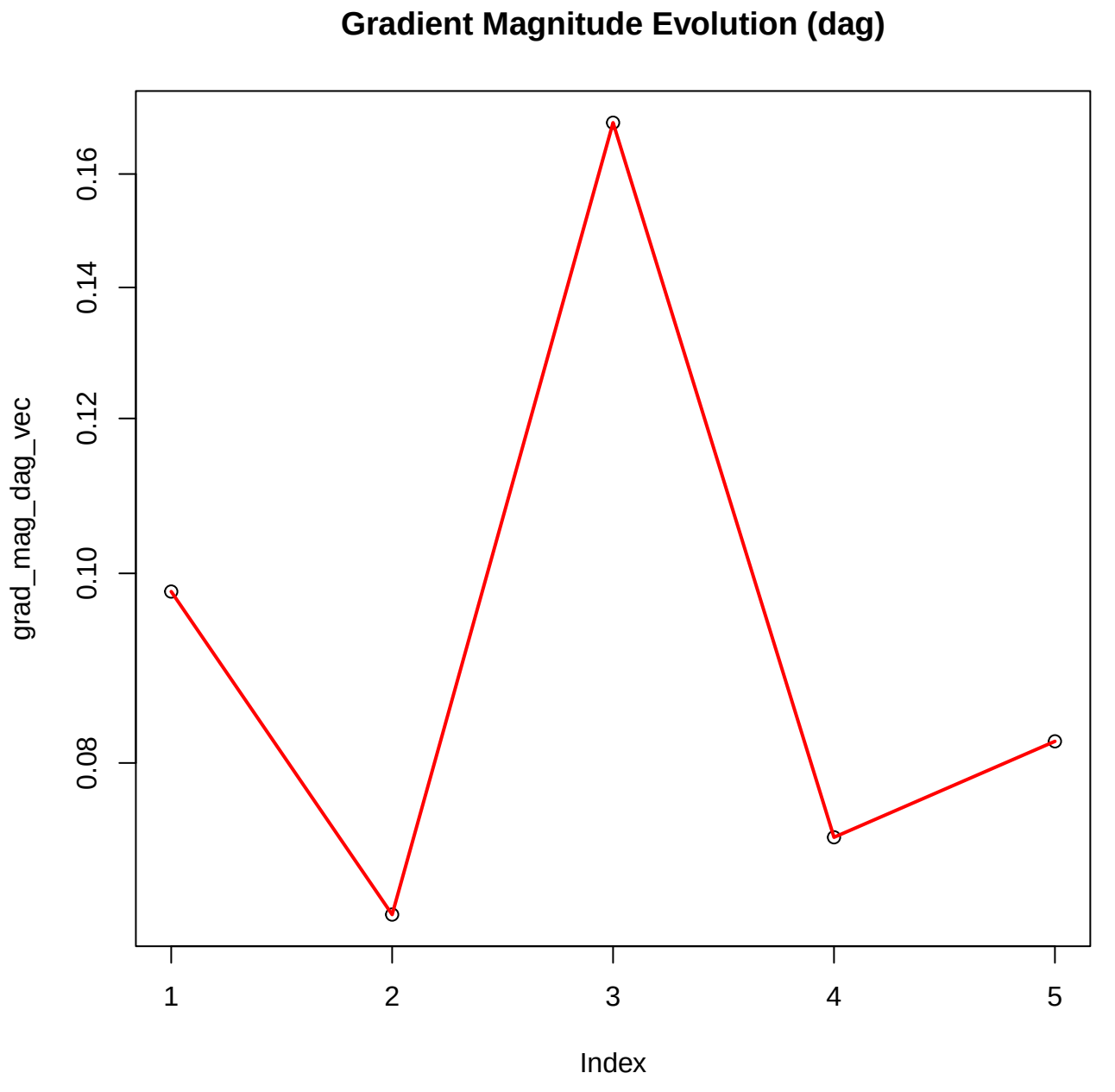
Index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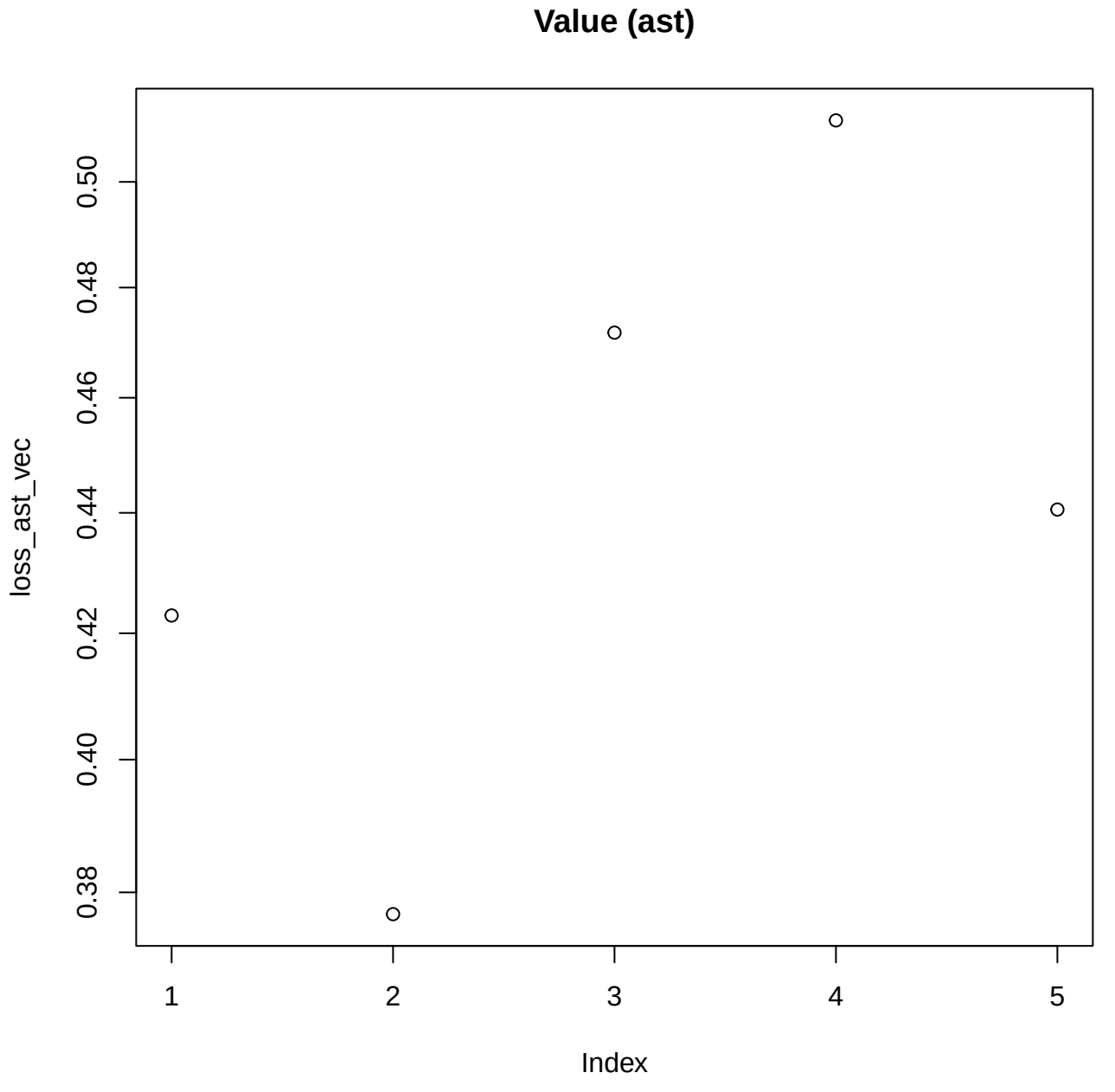
Value (ast)
loss_ast_vec
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0.50
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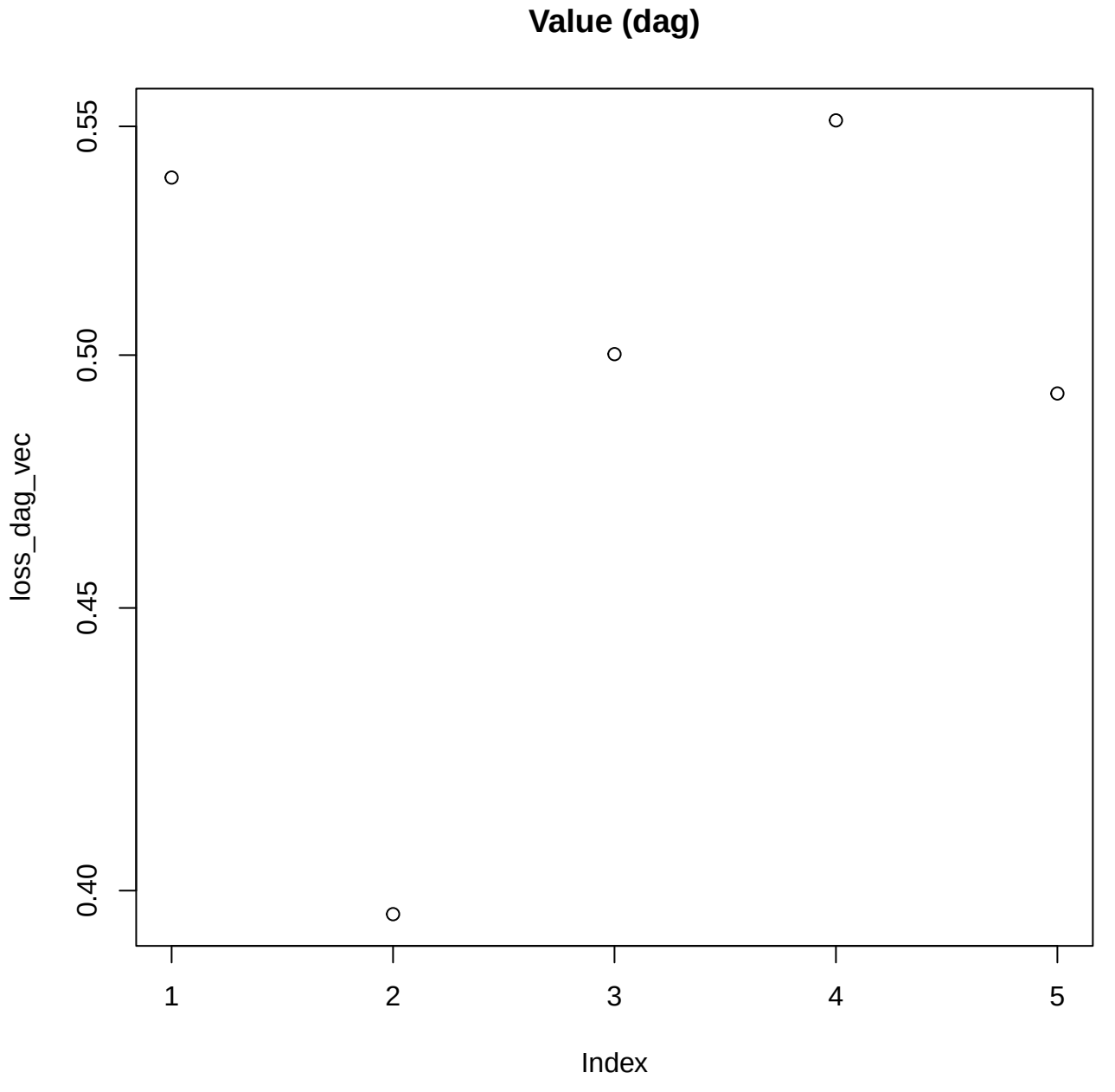
3
4
5
Index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3
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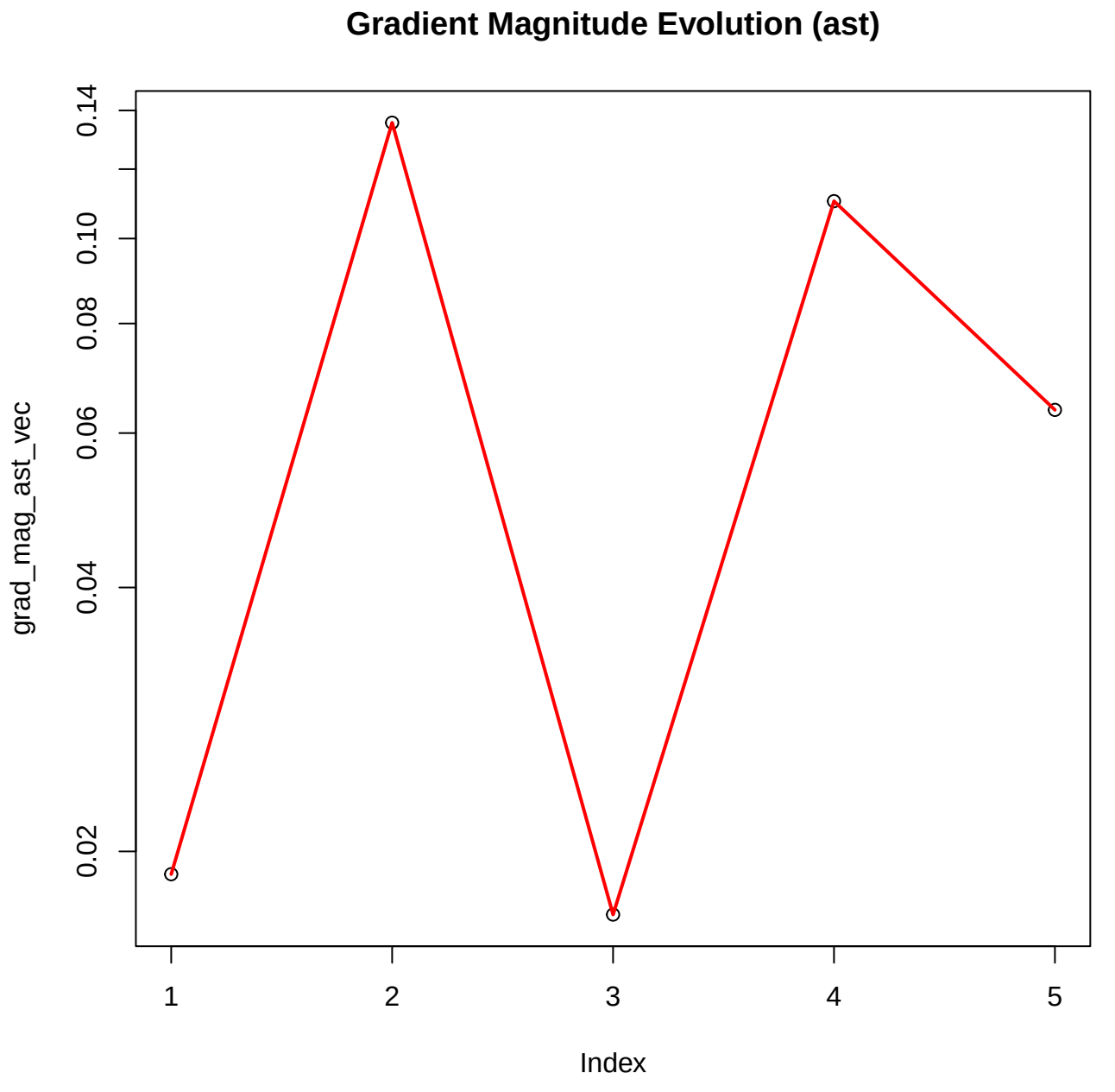
5
Index


Gradient Magnitude Evolution (ast)
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0.14
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3
4
5
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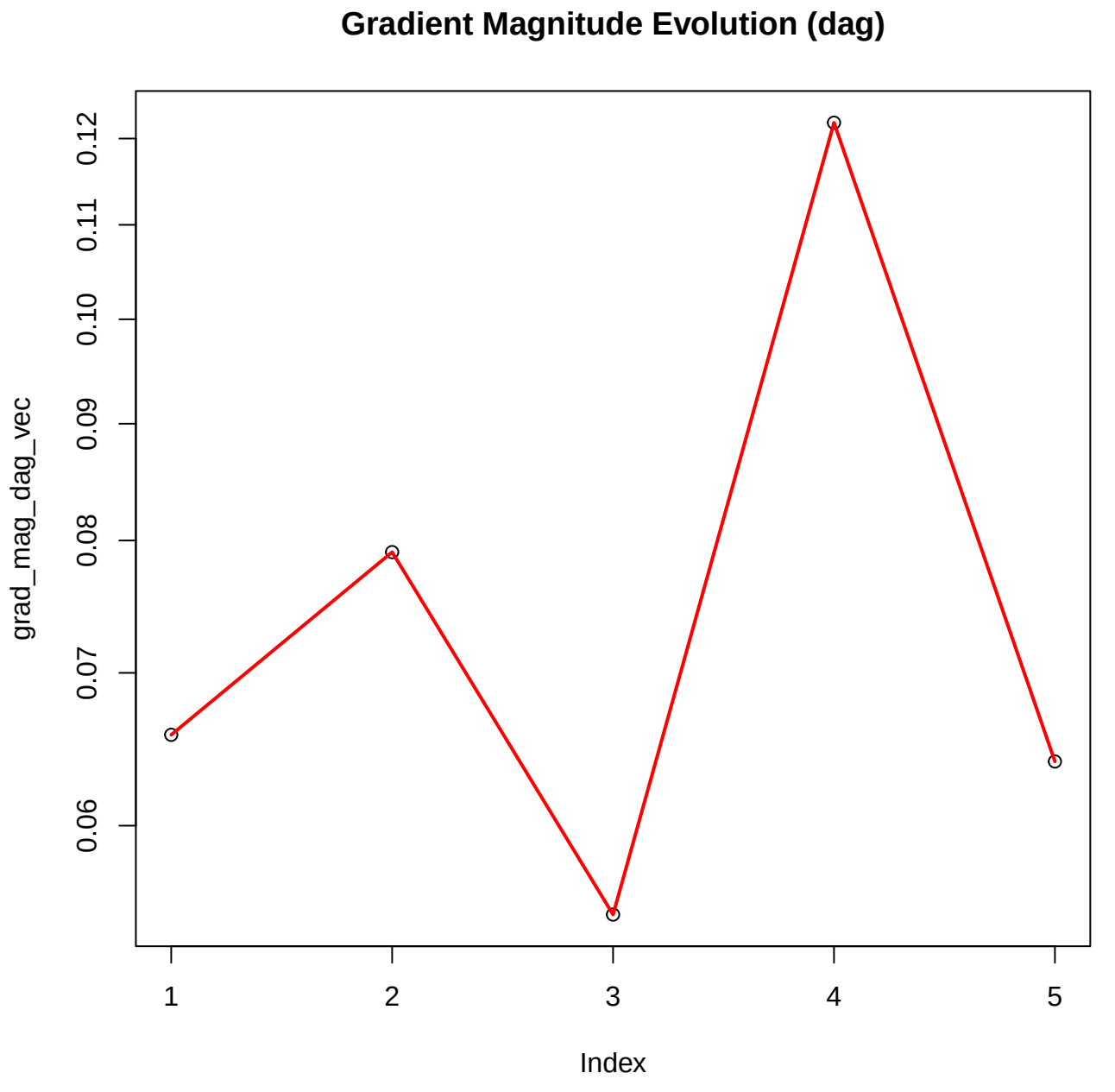
Gradient Magnitude Evolution (dag)
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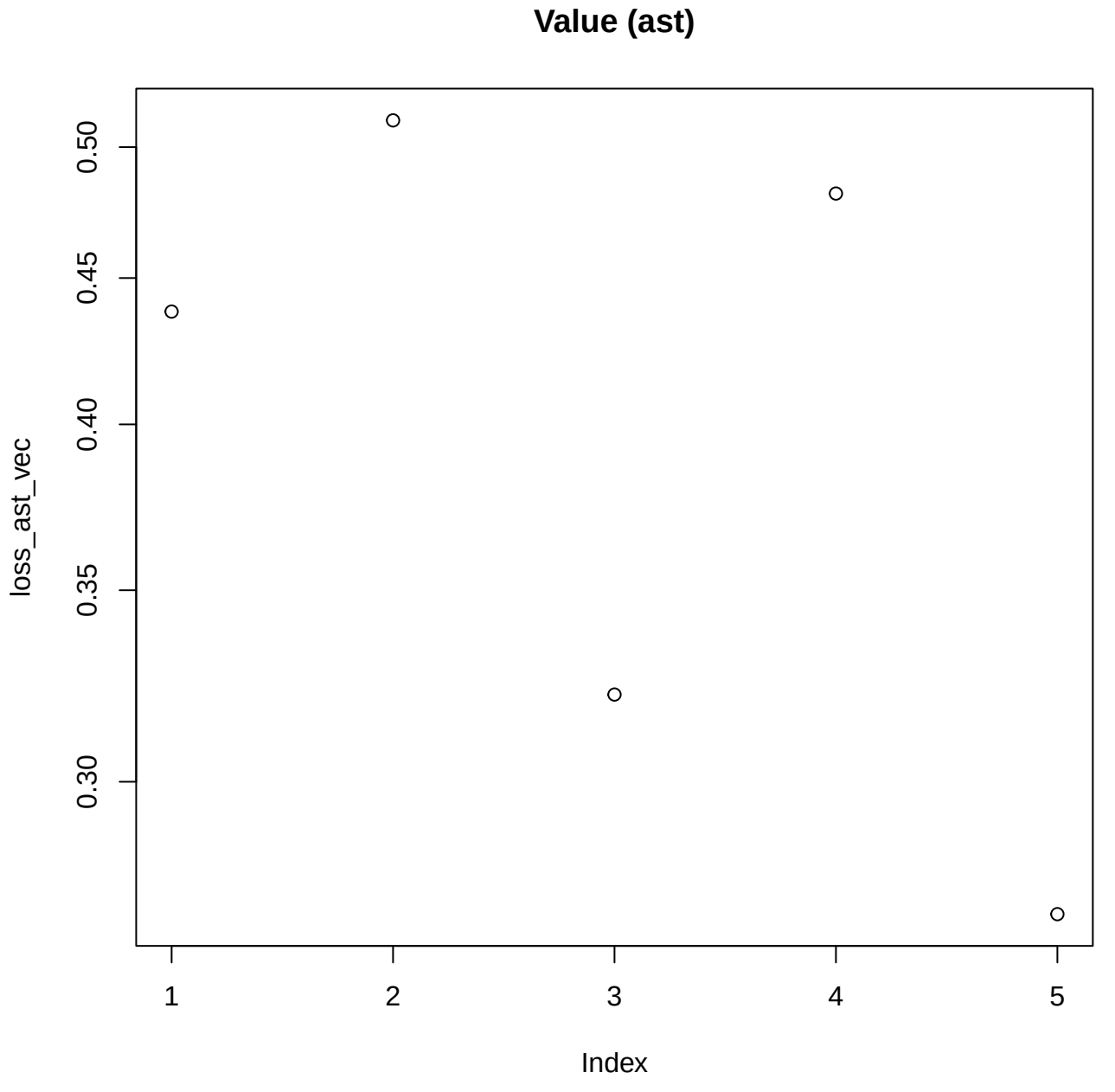
5
Index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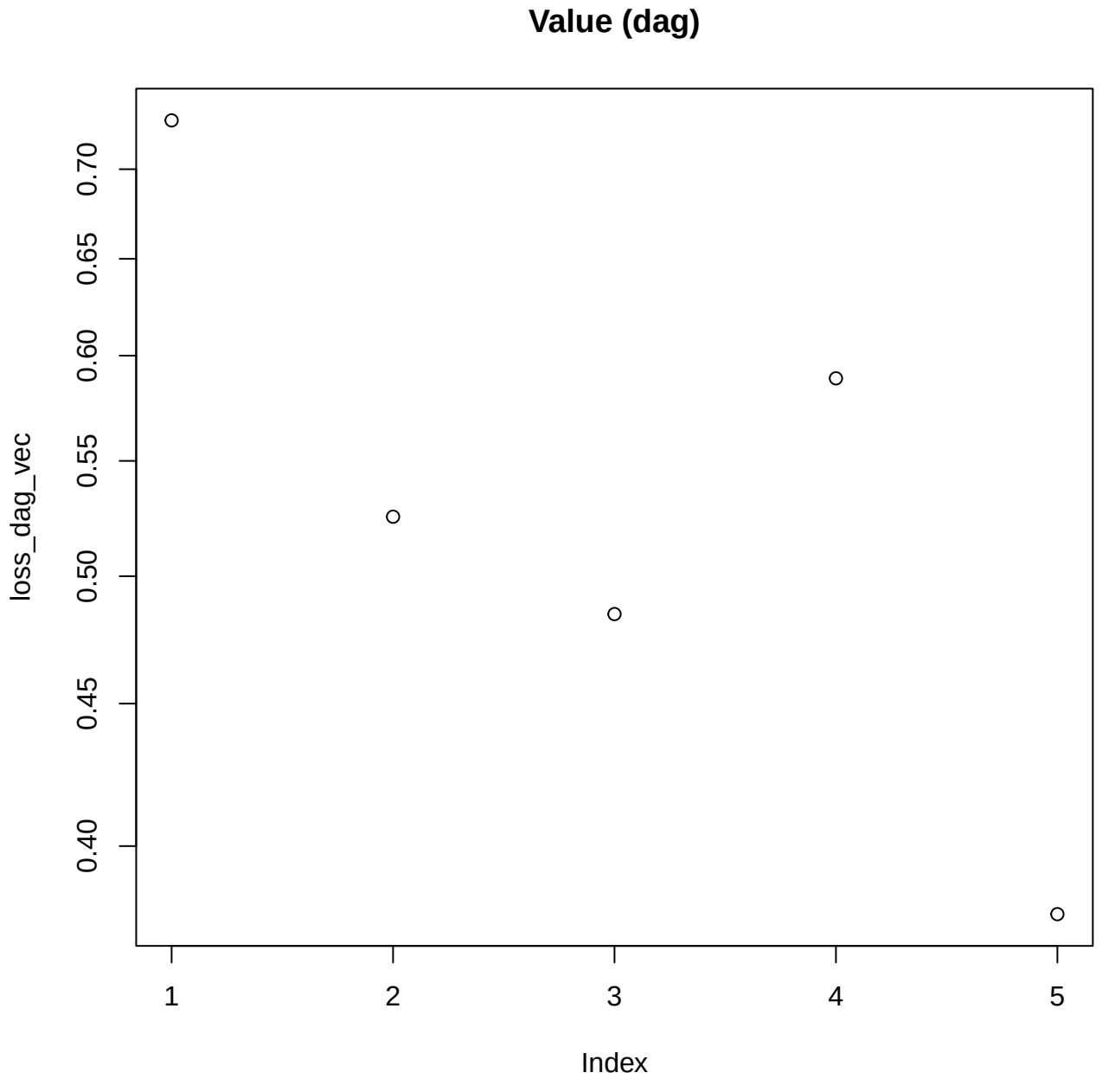
Index


Value (dag)
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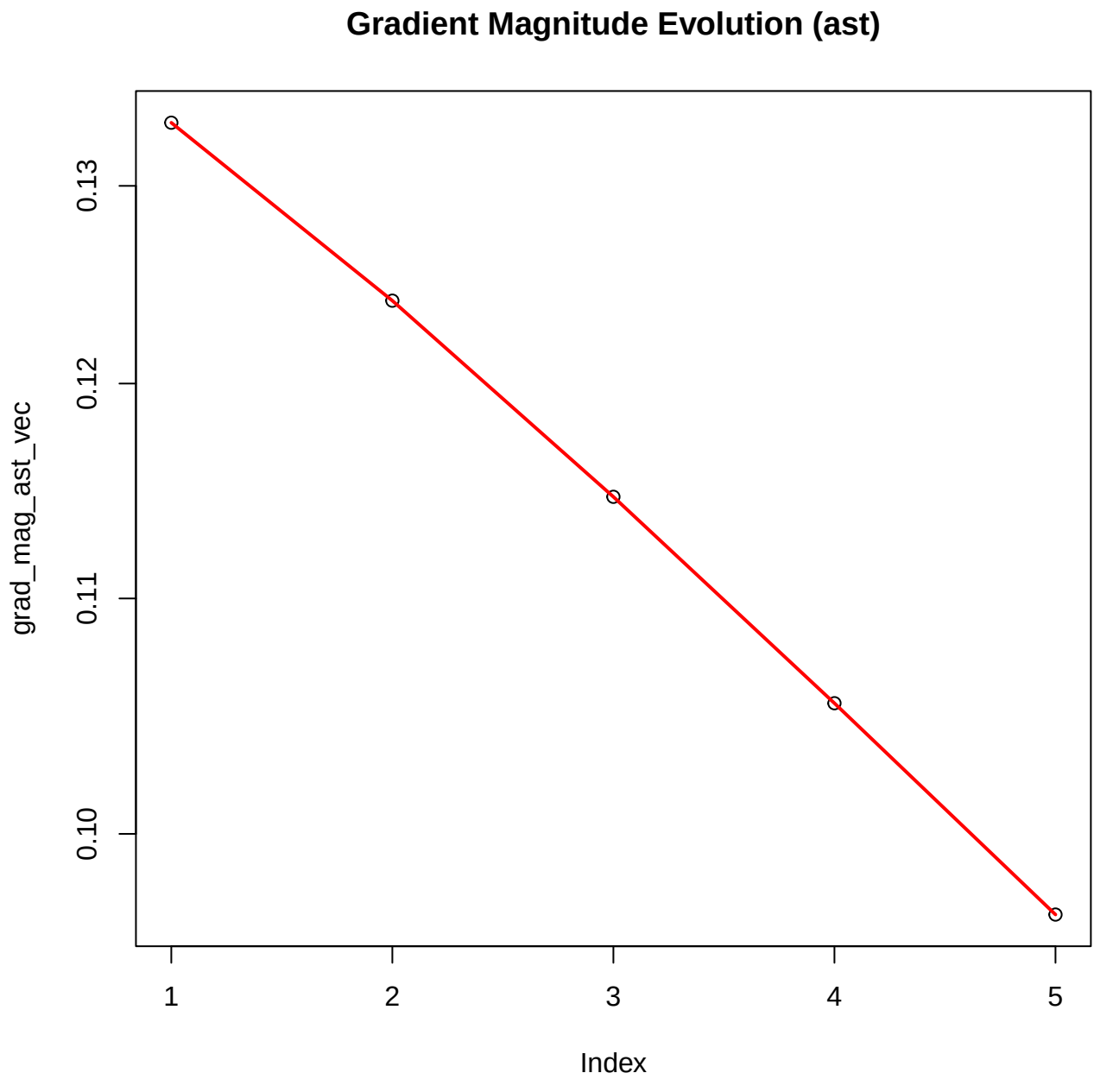
5
Index


Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
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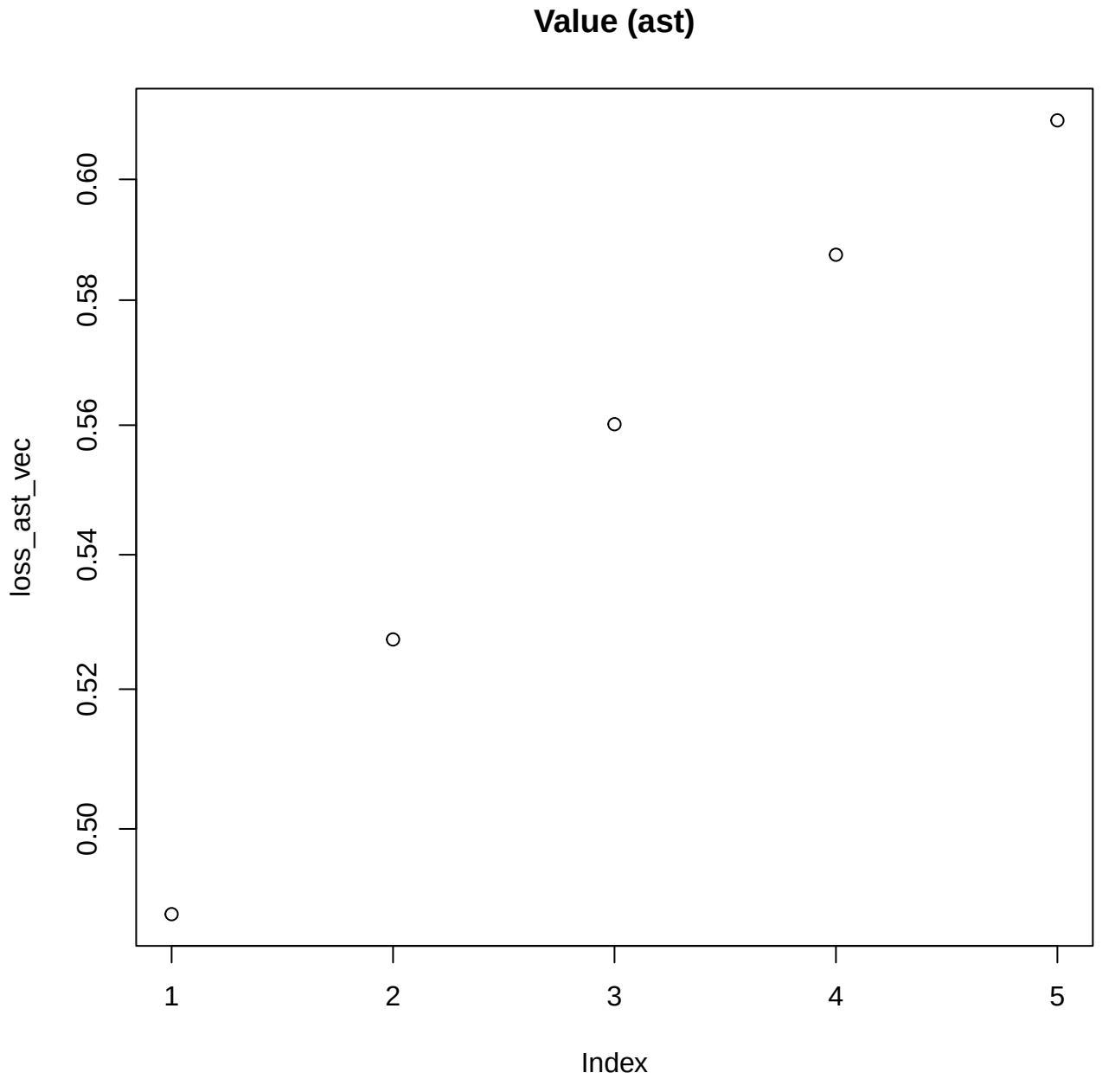
Value (ast)
loss_ast_vec
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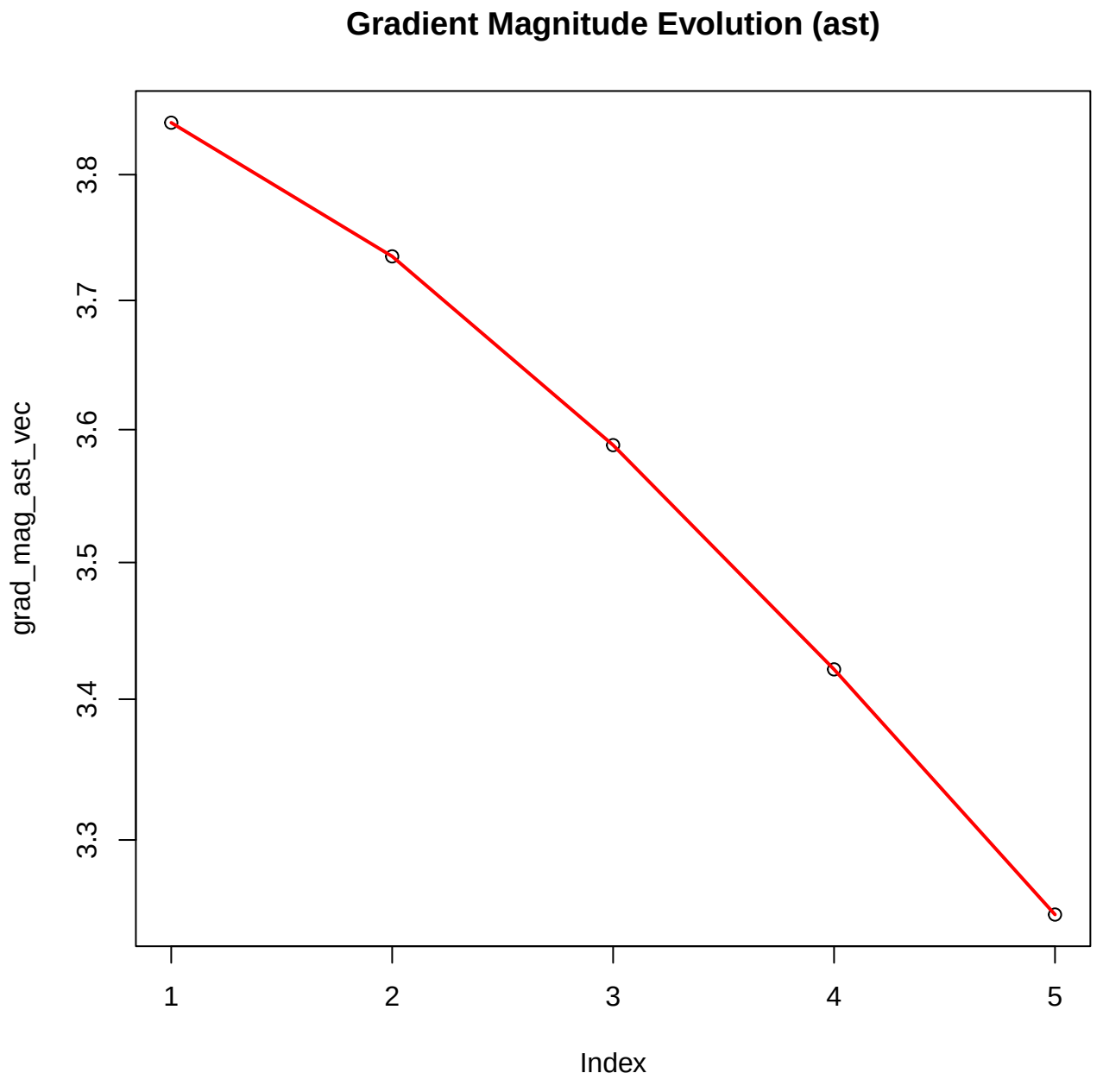
4
5
Index


Gradient Magnitude Evolution (ast)
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3.4
3.5
3.6
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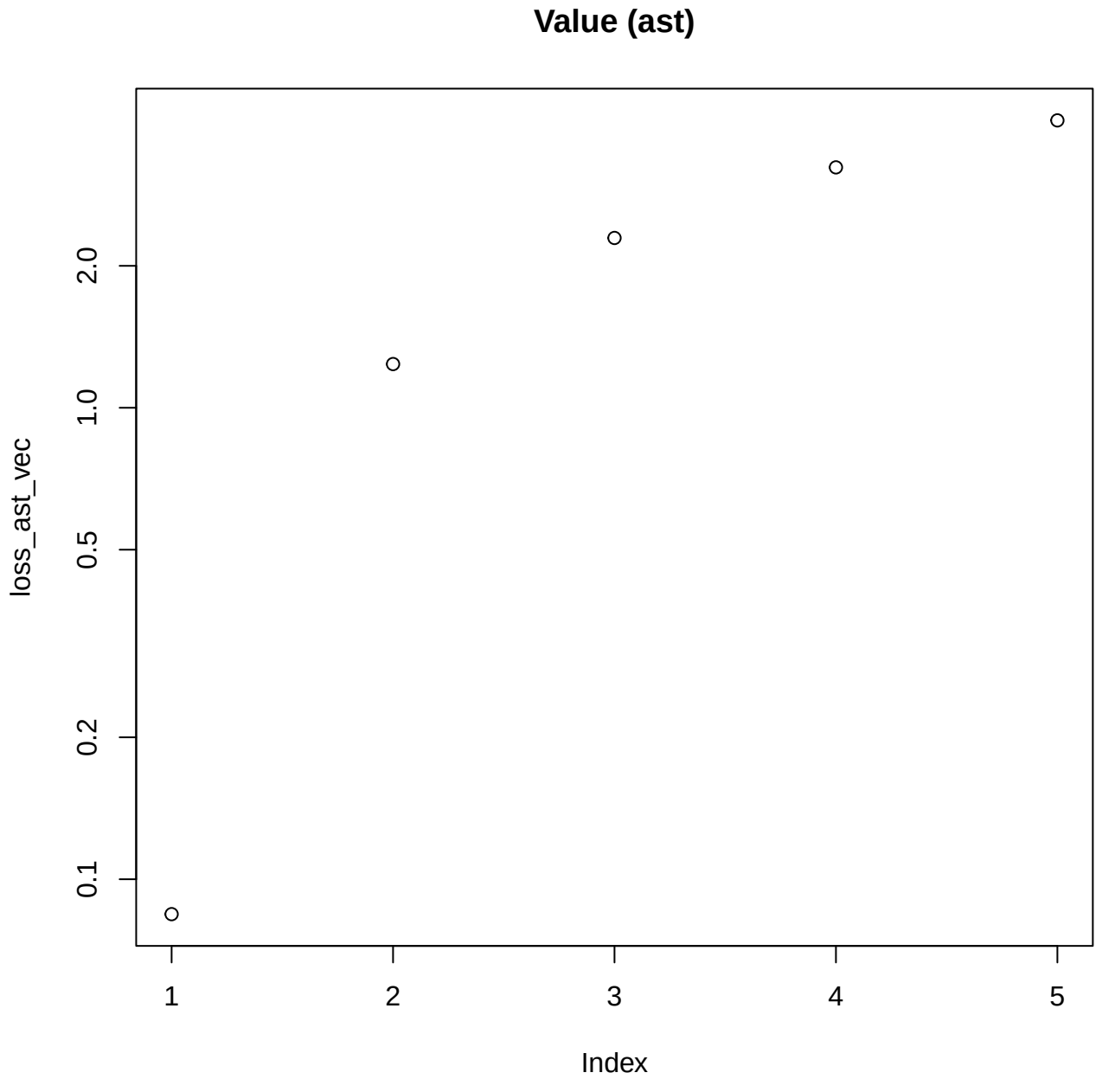
3.8
1
2
3
4
5
Index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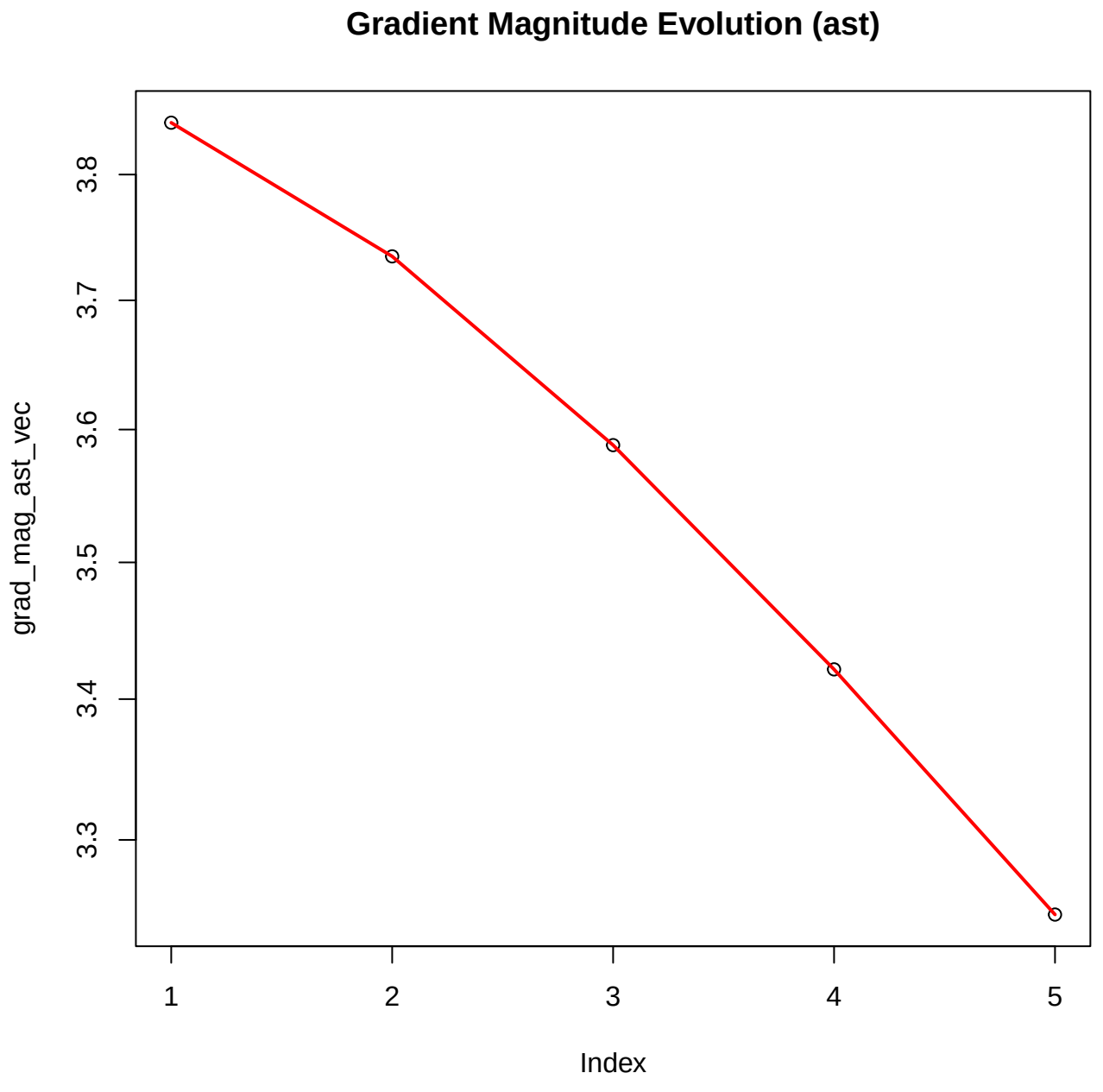
Index


Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
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3.5
3.6
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3.8
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2
3
4
5
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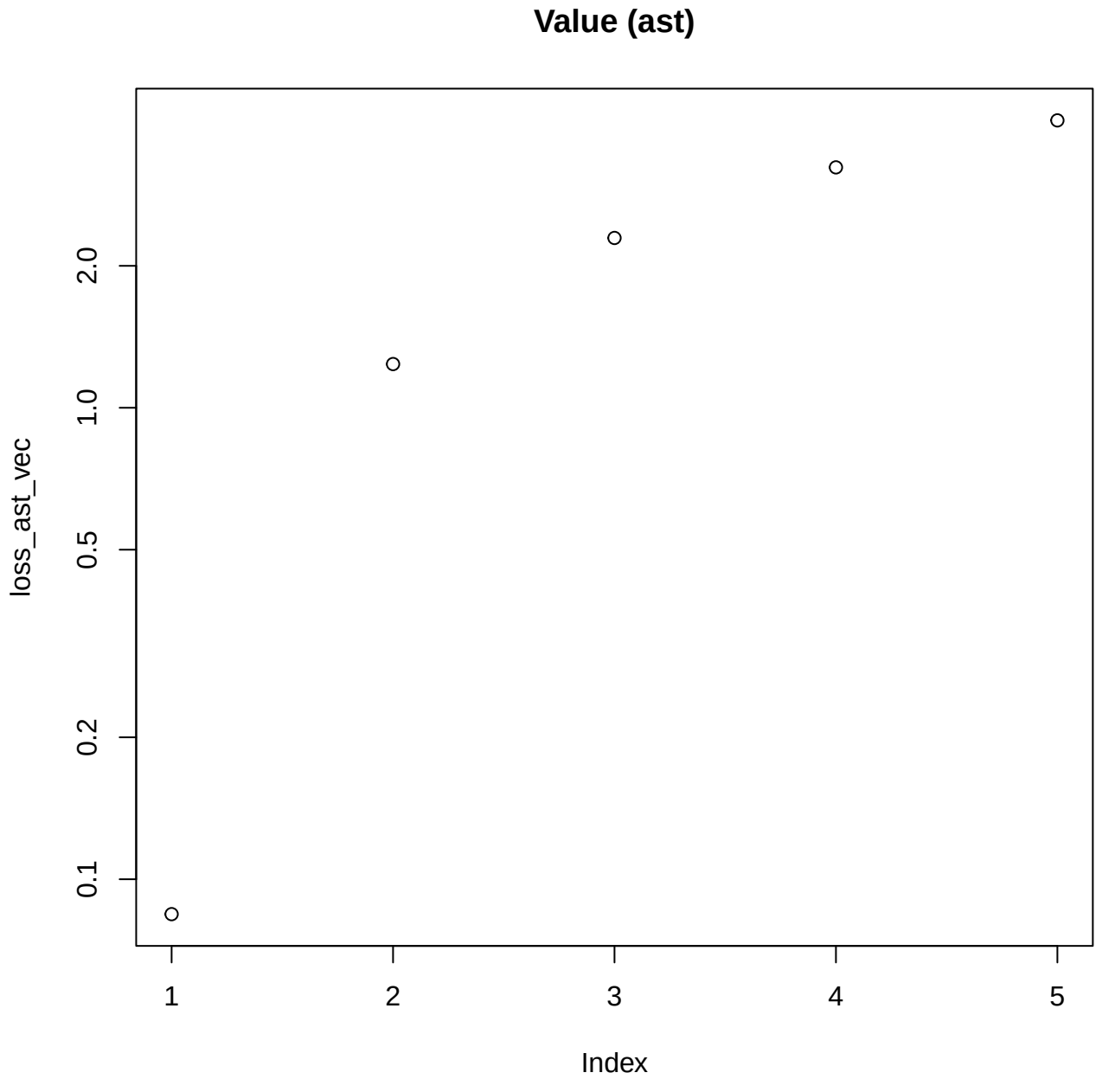
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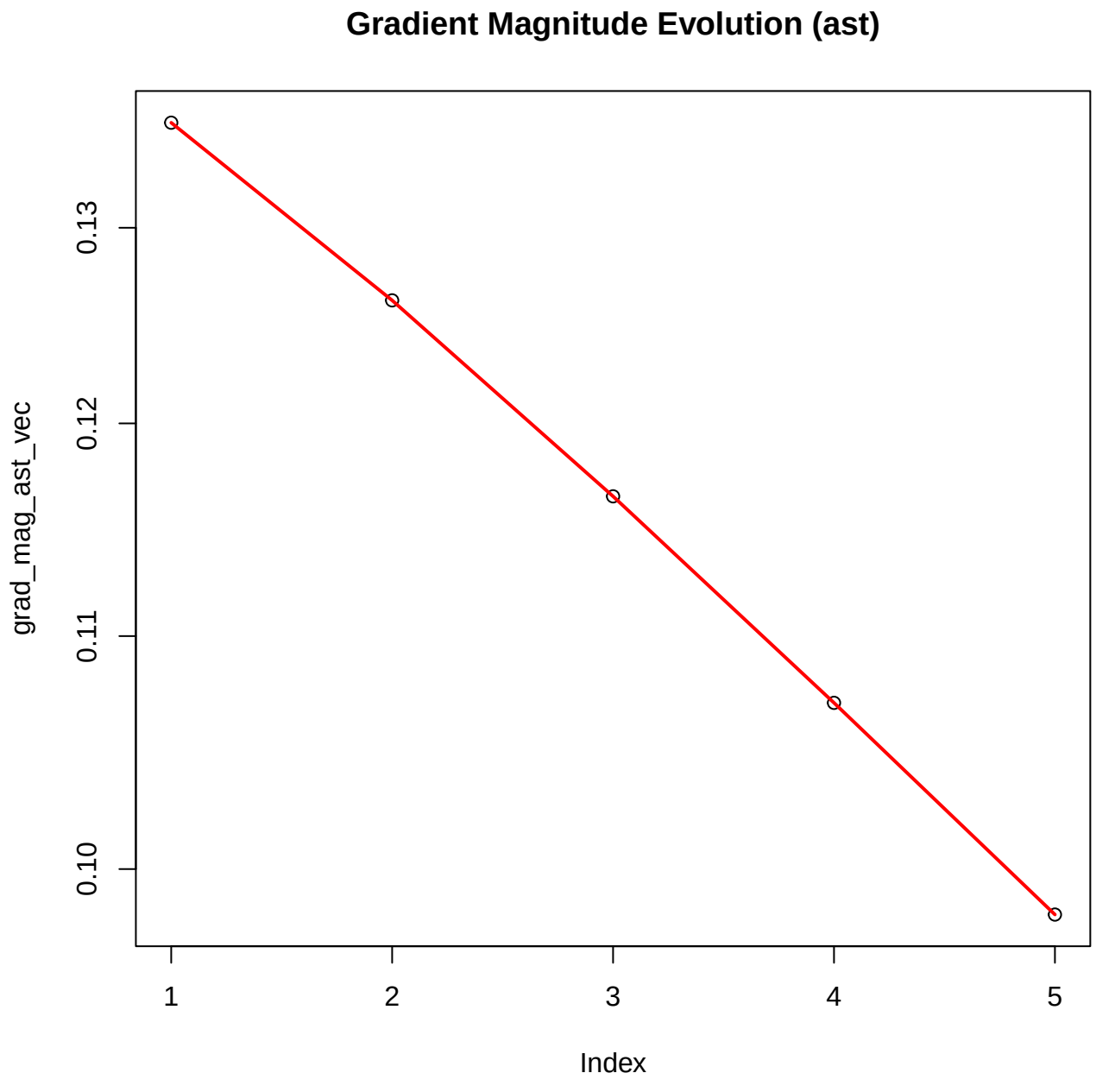
Gradient Magnitude Evolution (ast)
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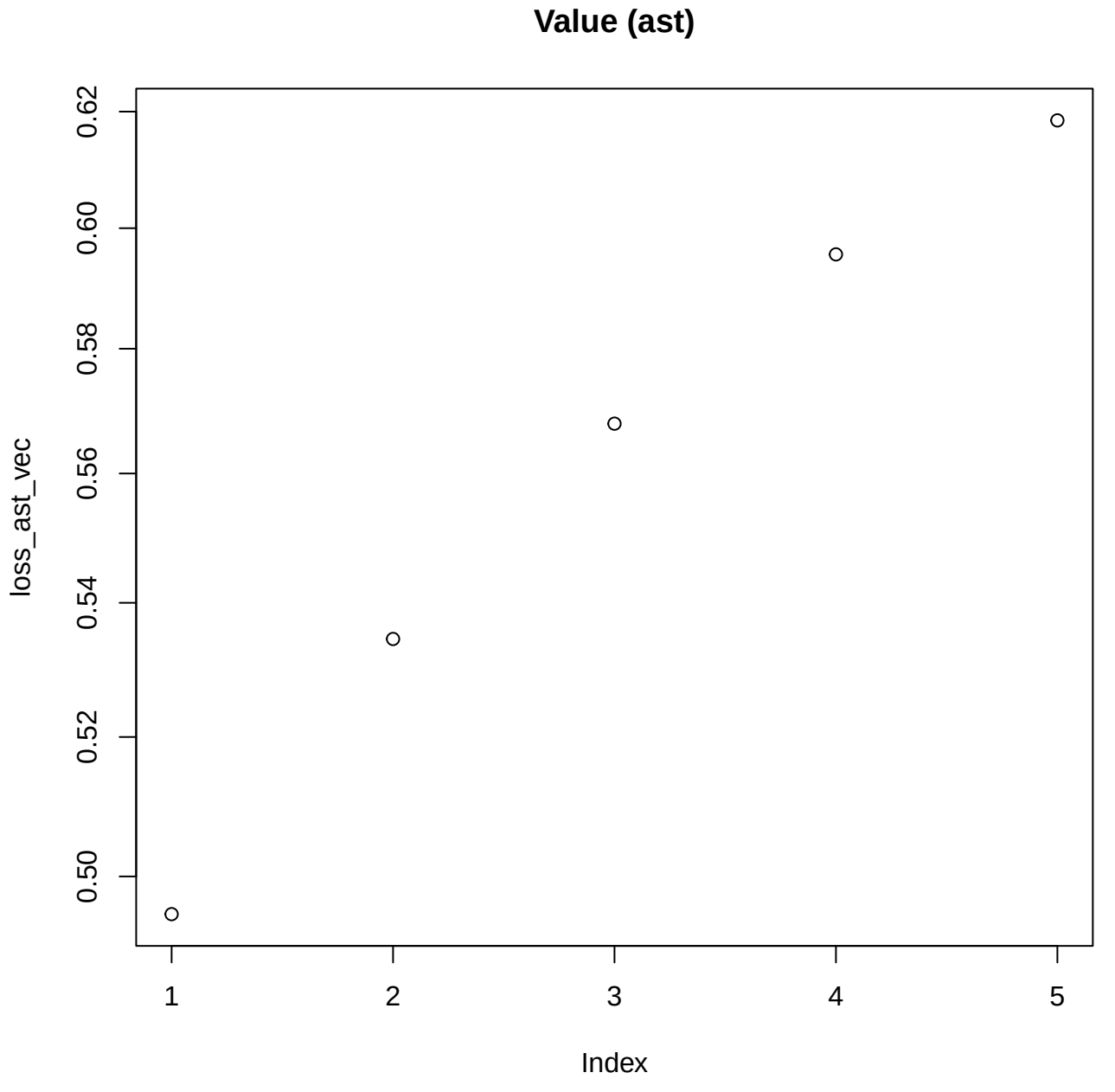
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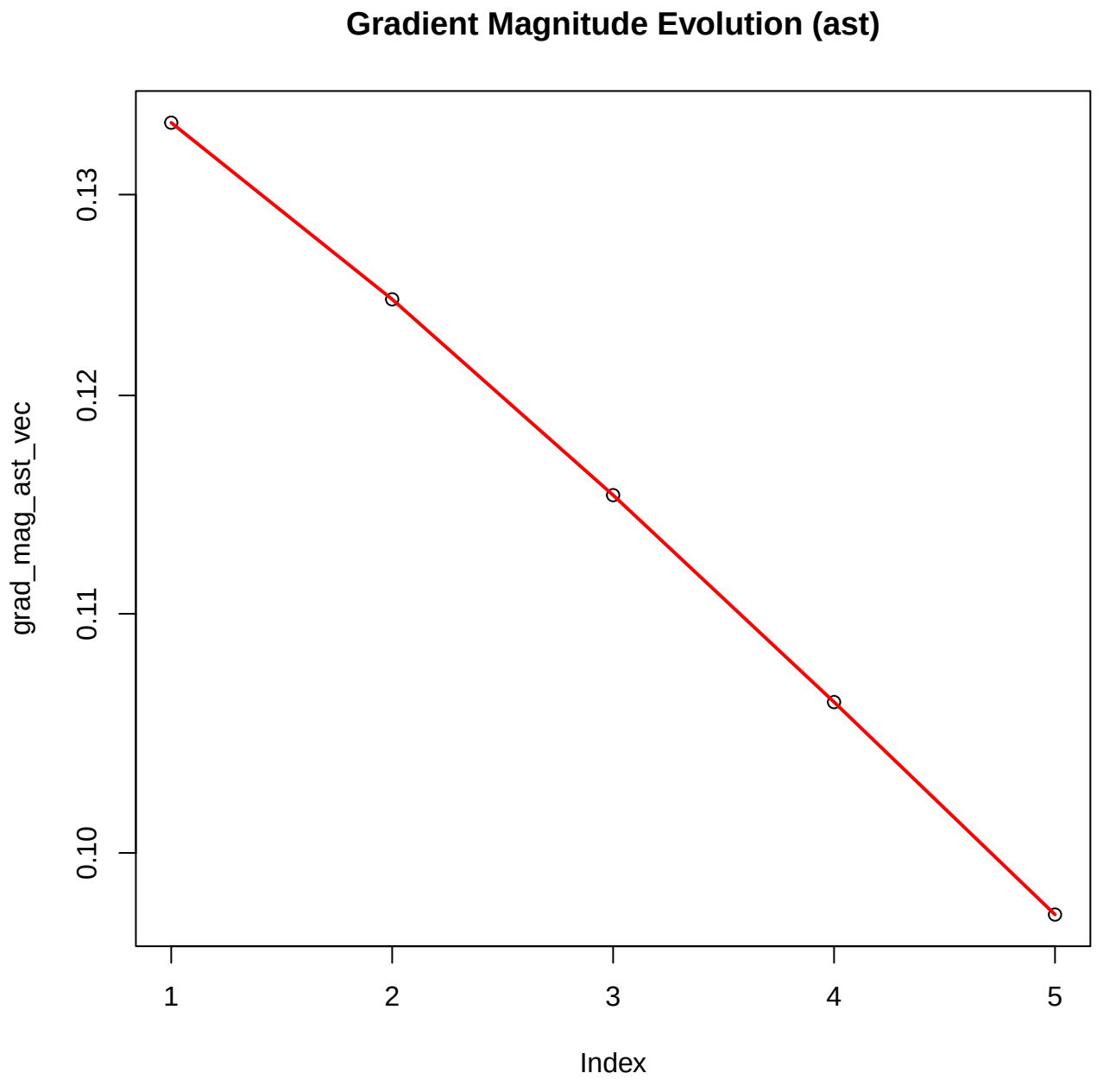
Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
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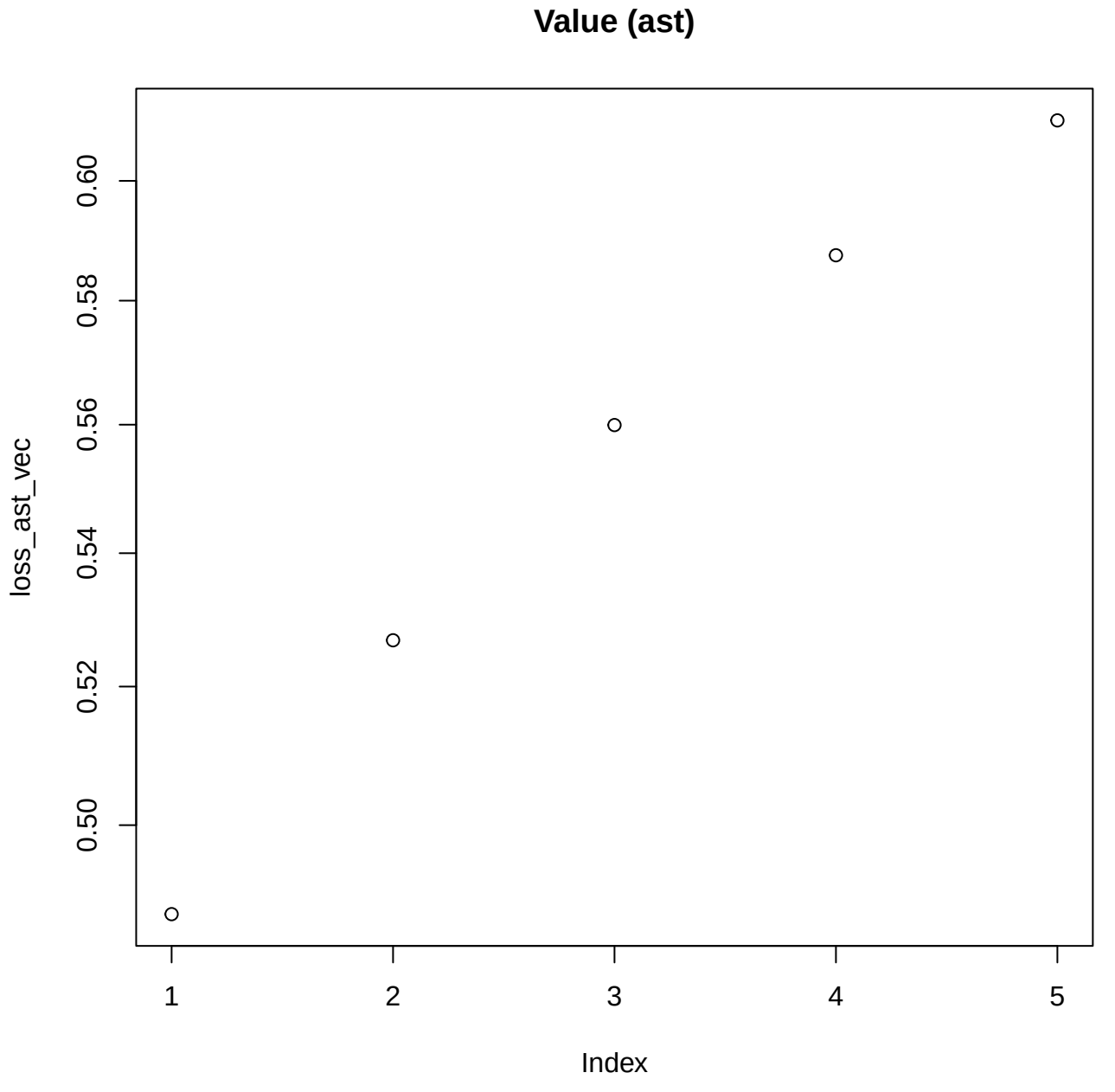
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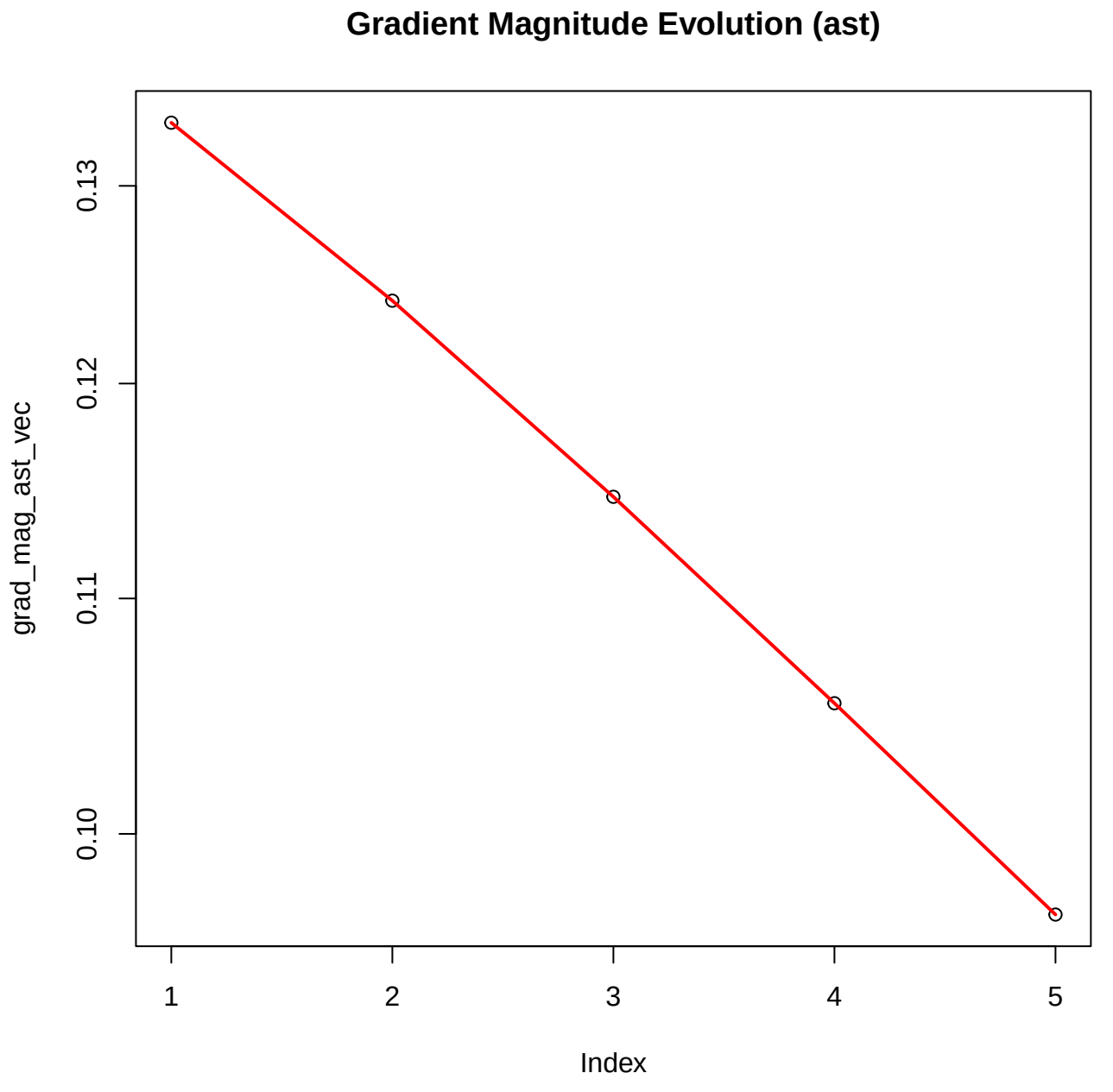
Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
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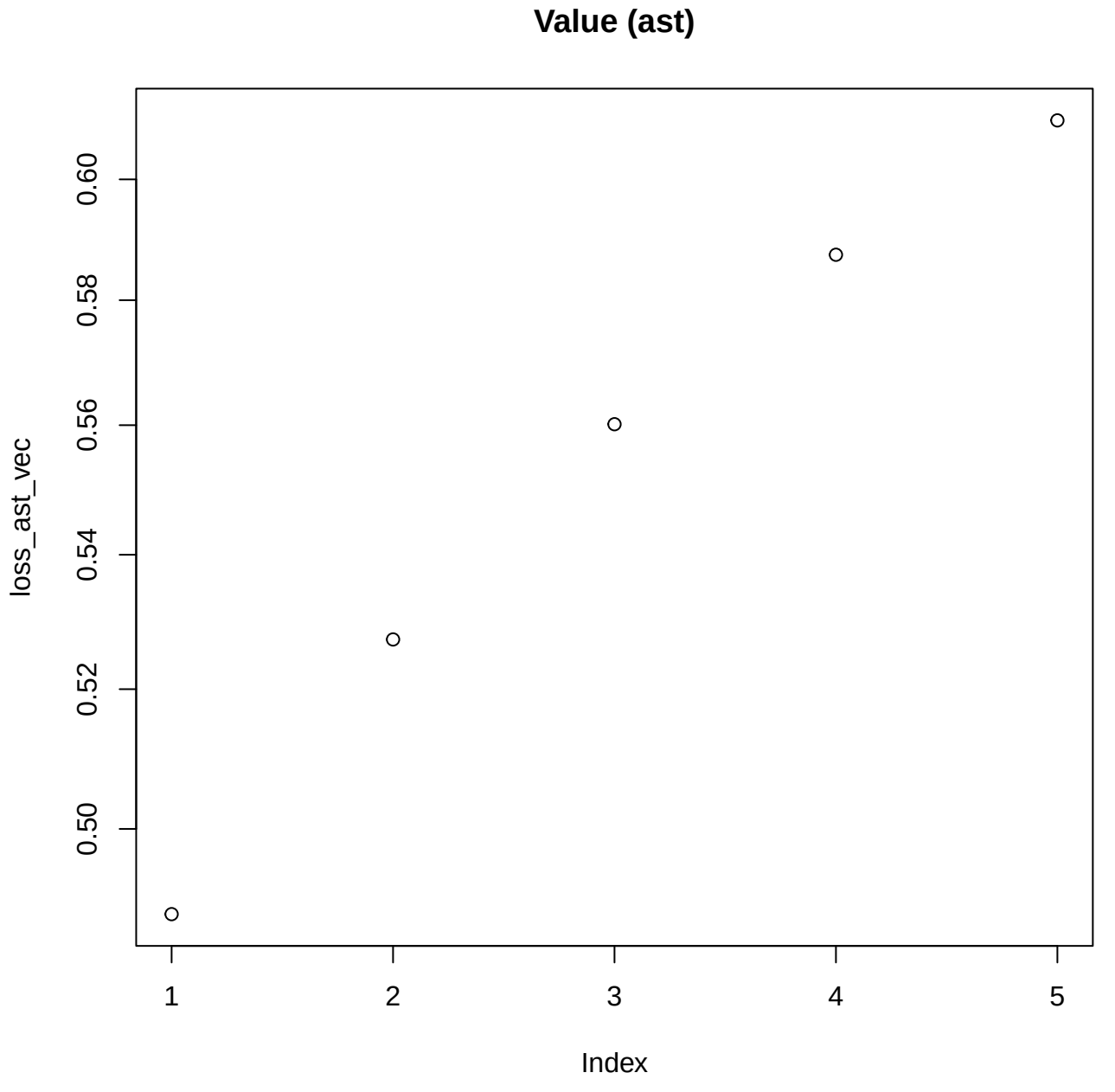
Value (ast)
loss_ast_vec
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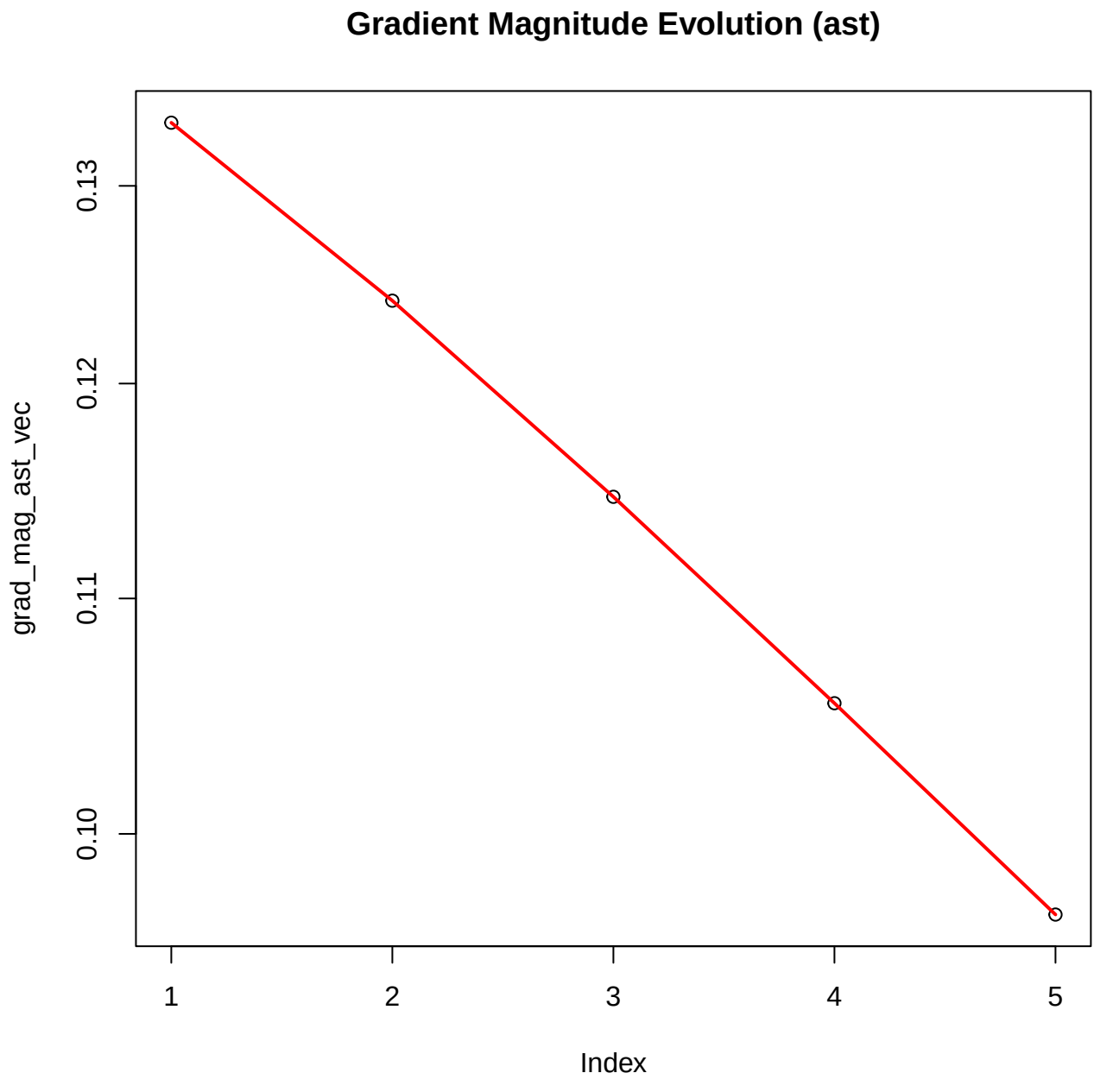
Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
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0.12
0.13
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2
3
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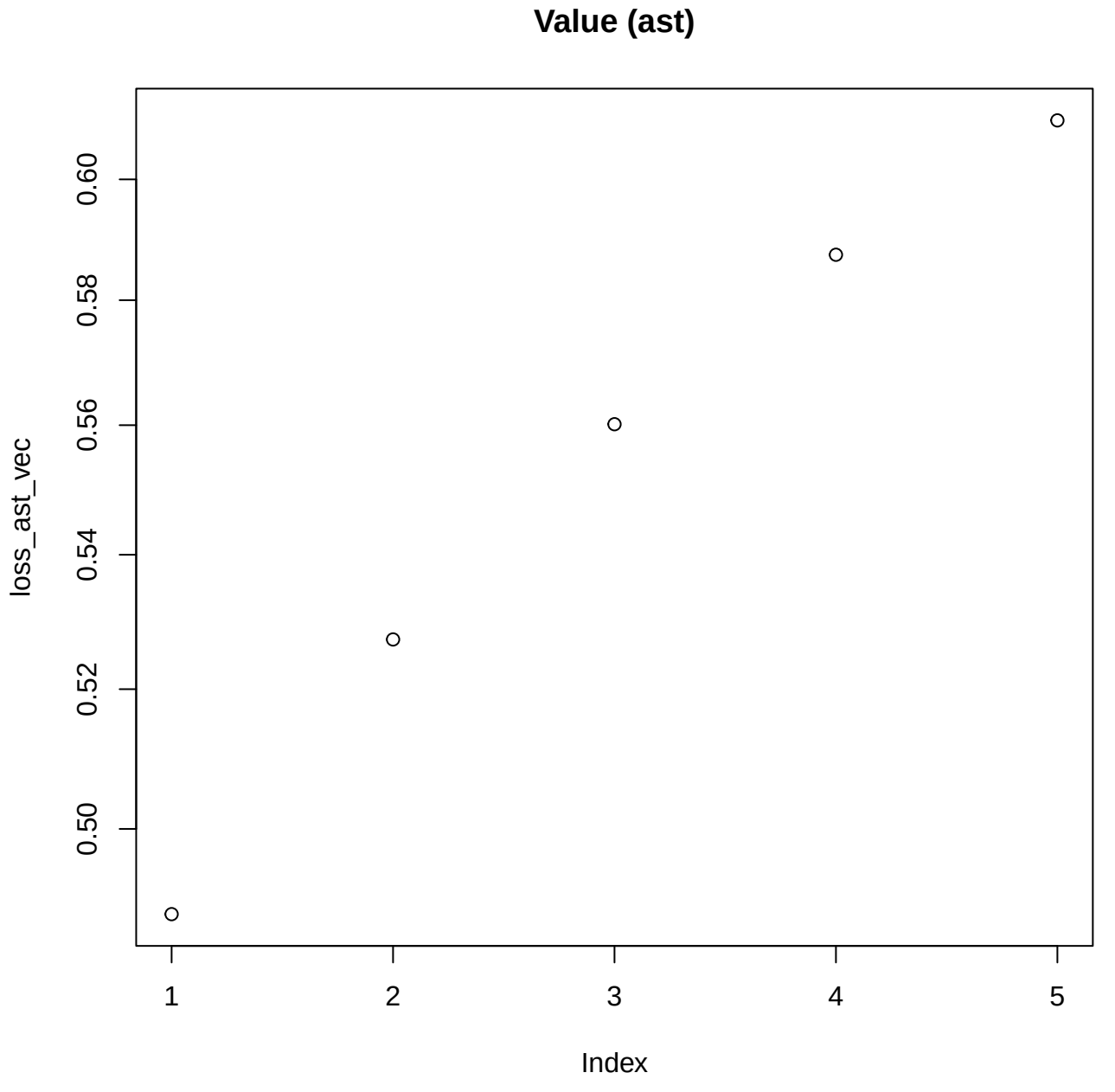
5
Index


Value (ast)
loss_ast_vec
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0.56
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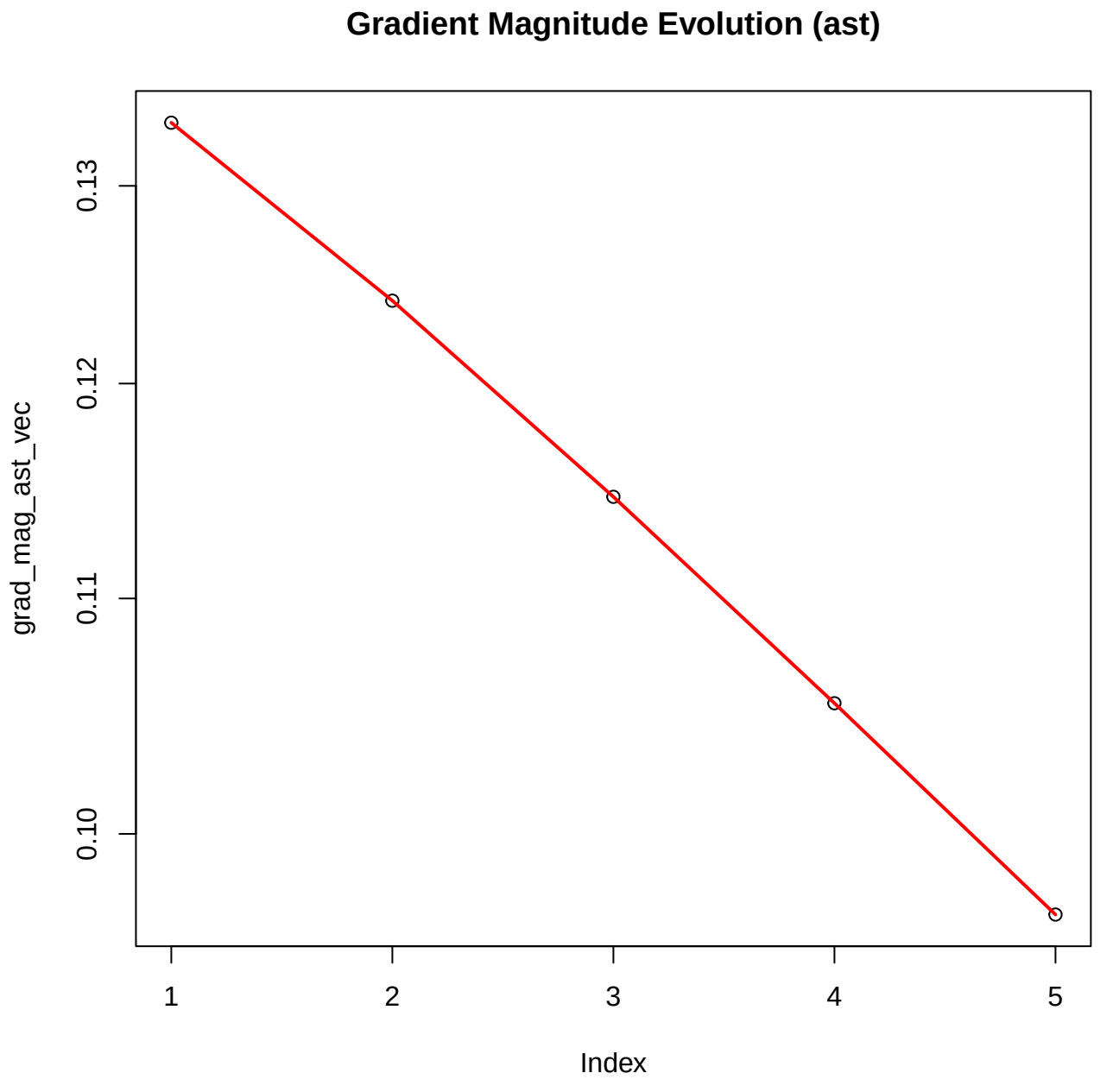
Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
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0.13
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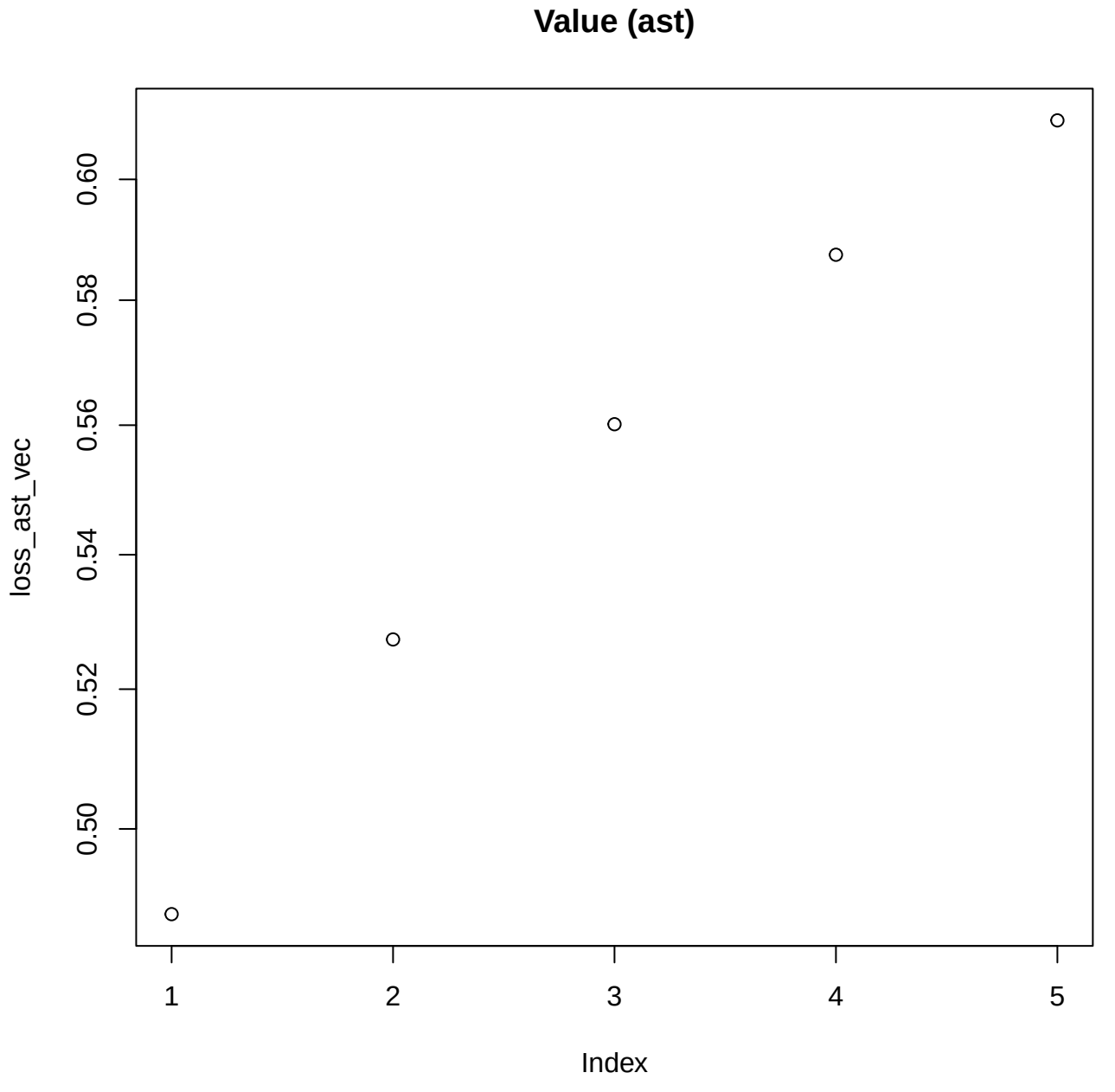
Value (ast)
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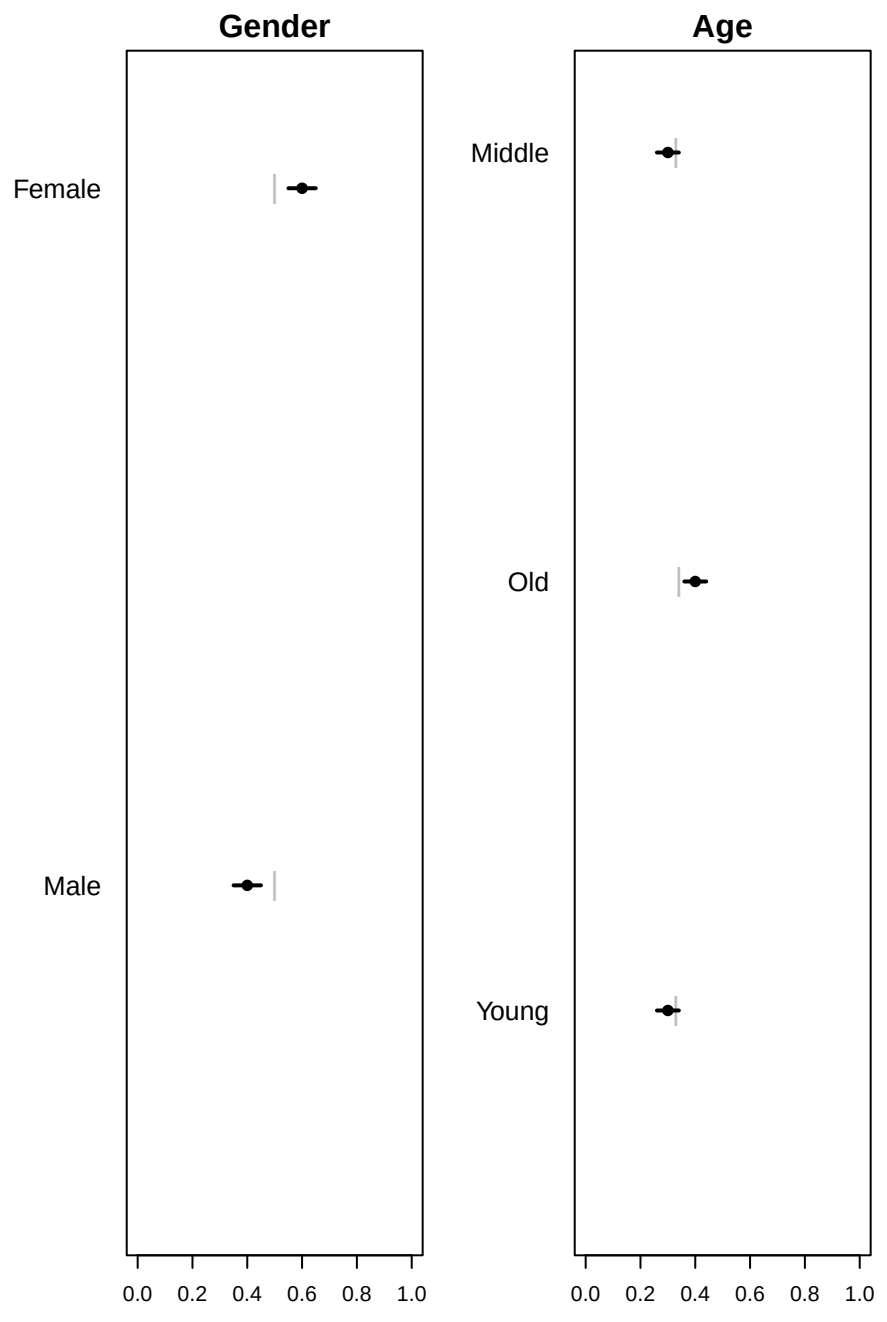
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Male
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0.6
0.8
1.0
Age
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Old
Young
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0.2
0.4
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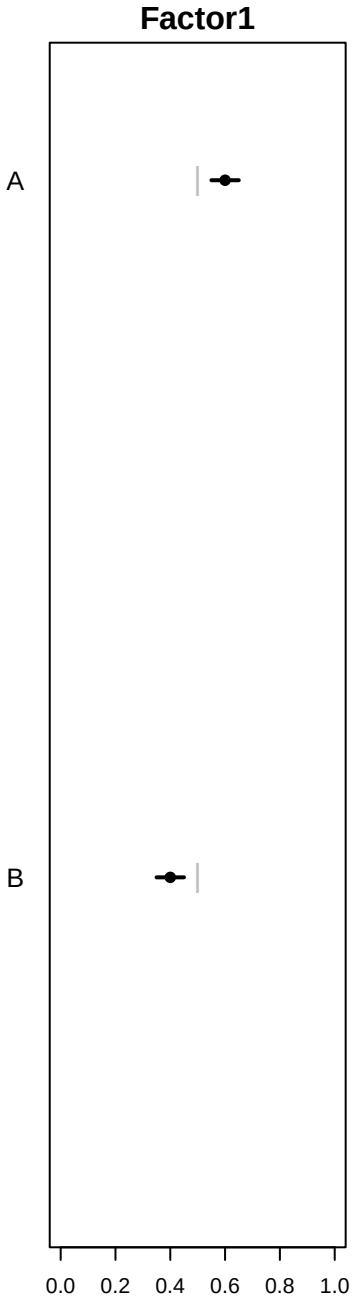
0.8
1.0


Factor1
A
B
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0.2
0.4
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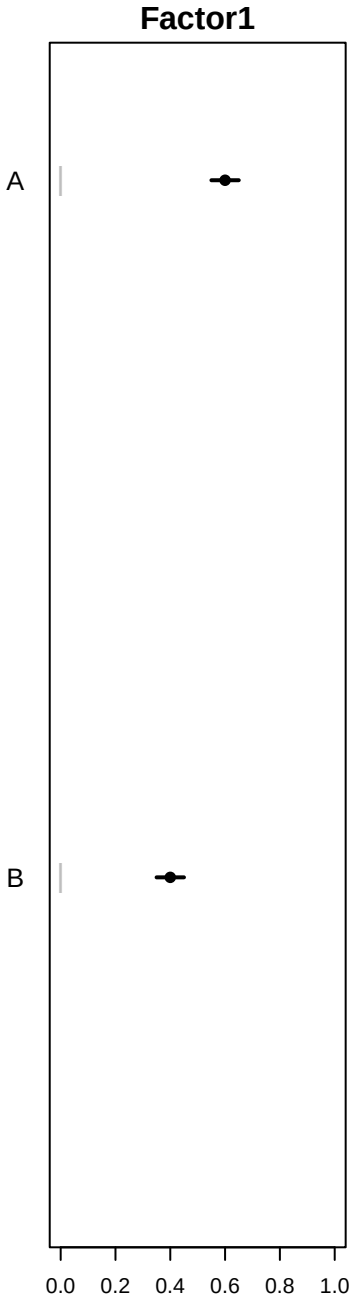
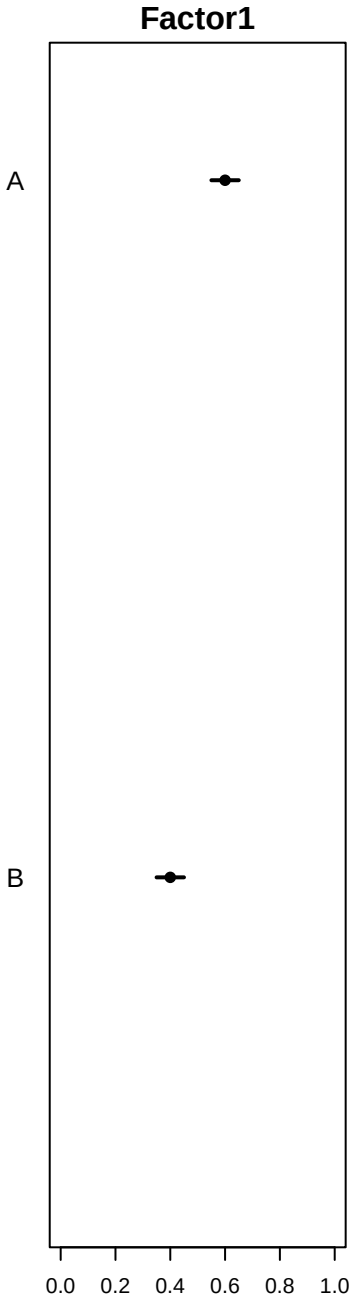
0.8
1.0


Factor1
A
B
0.0
0.2
0.4
0.6
0.8
1.0


Factor1
A
B
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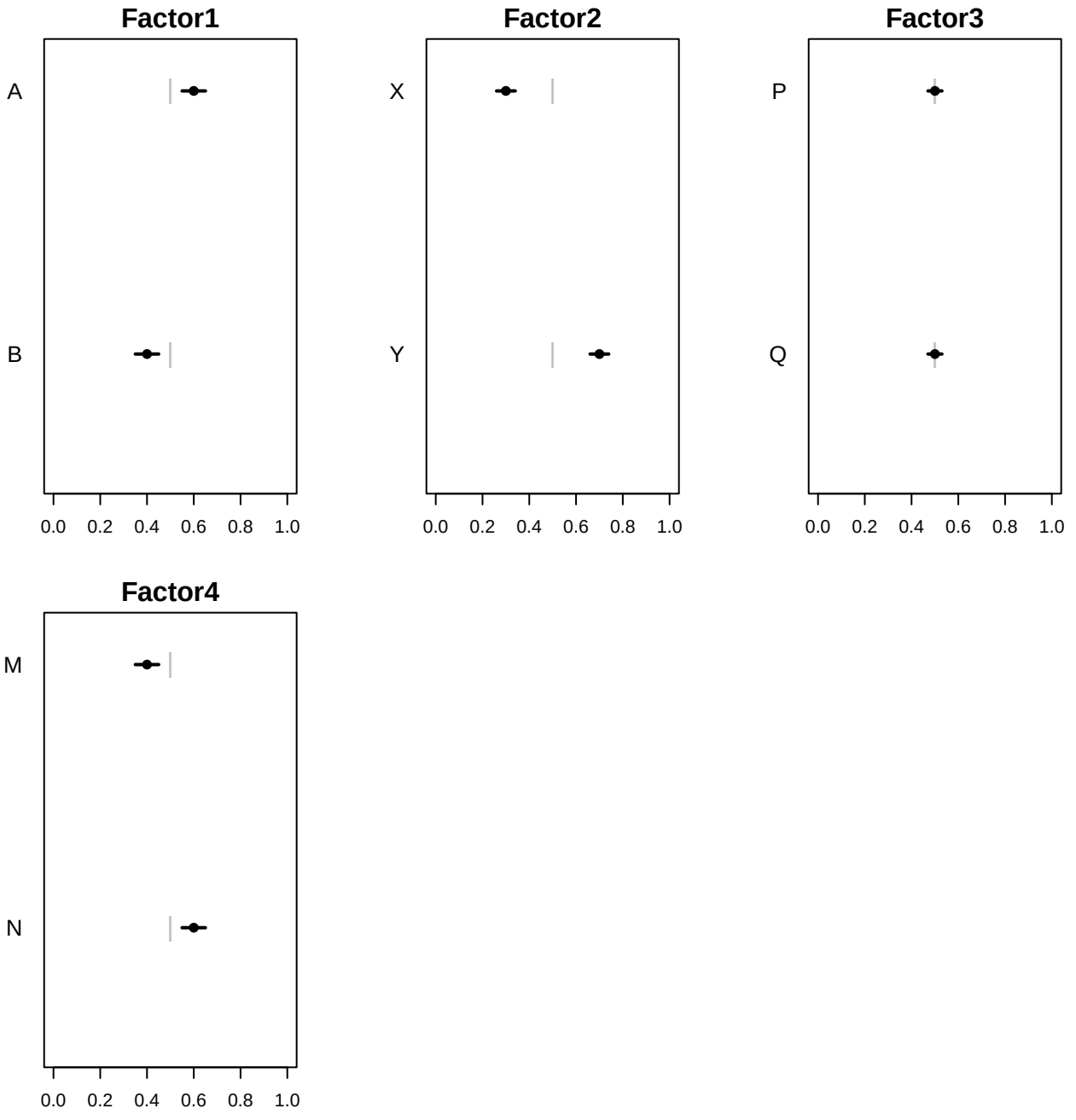
0.2
0.4
0.6
0.8
1.0


Factor1
A
B
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0.2
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0.6
0.8
1.0
Factor2
X
Y
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0.2
0.4
0.6
0.8
1.0
Factor3
P
Q
0.0
0.2
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M
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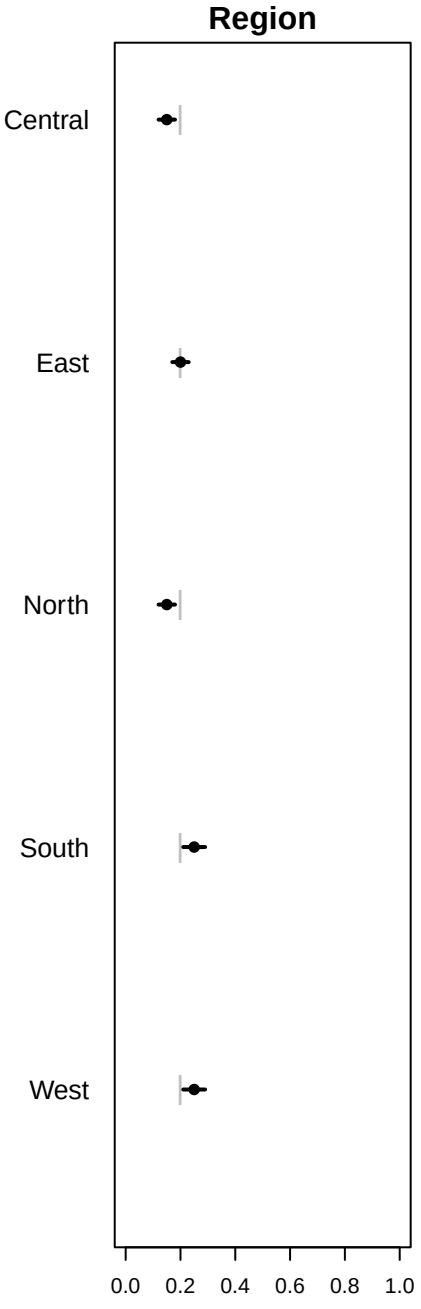
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East
North
South
West
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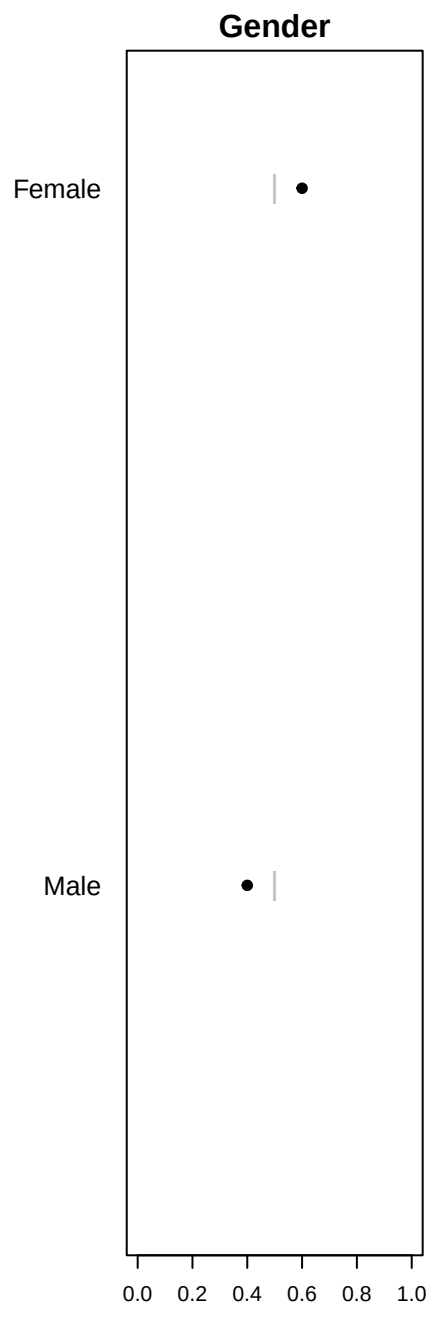
Gender
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Male
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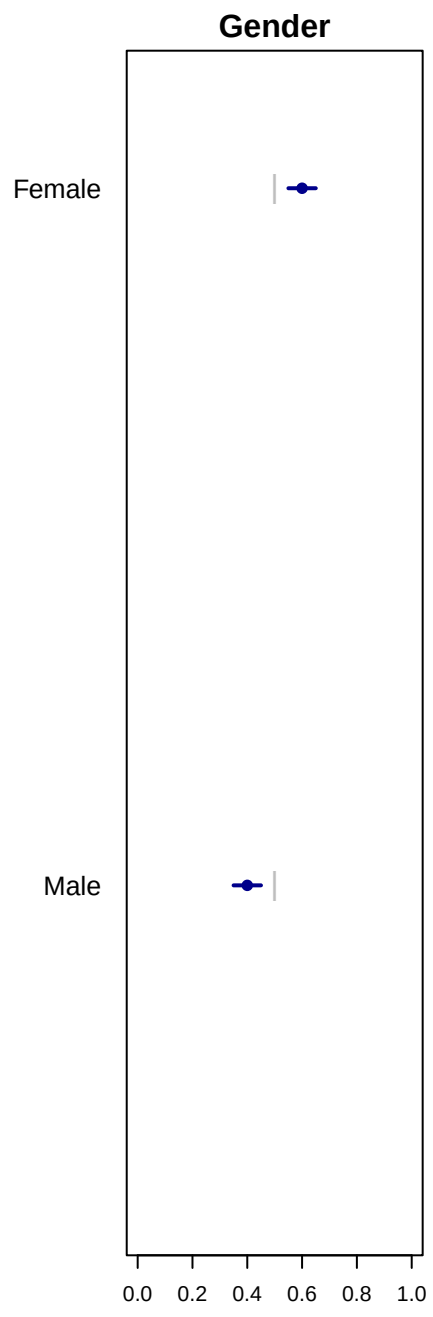
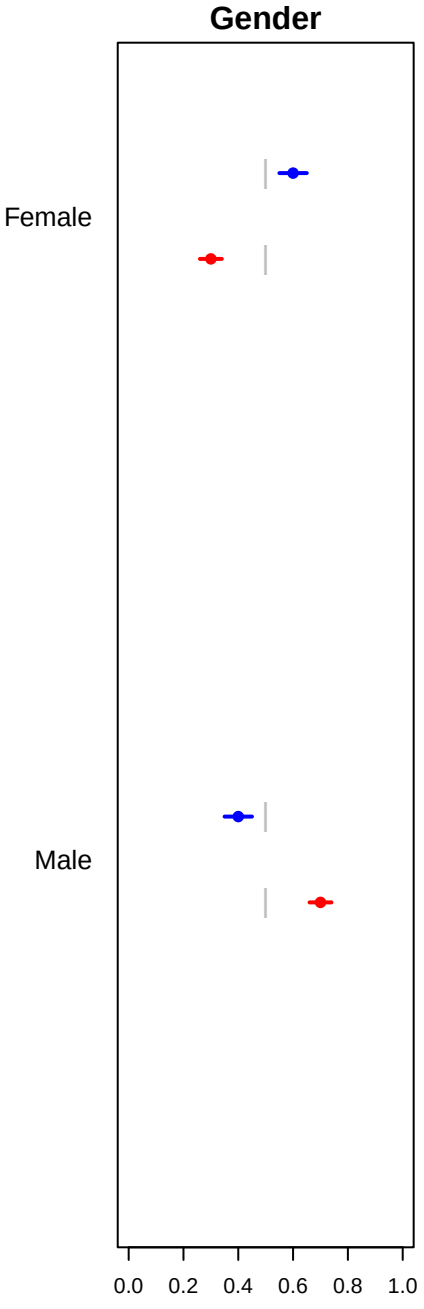
0.4
0.6
0.8
1.0


Gender
Female
Male
0.0
0.2
0.4
0.6
0.8
1.0


Gender
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Male
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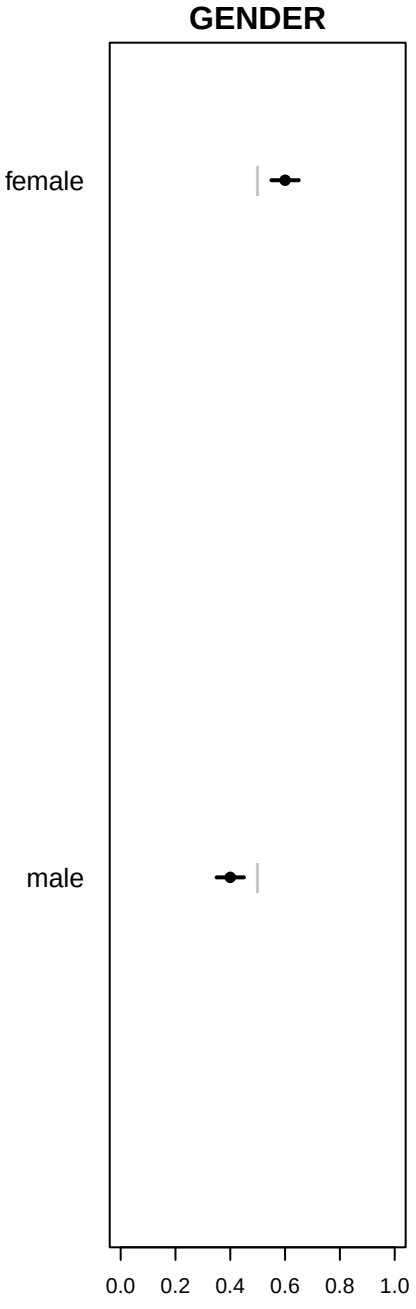
0.2
0.4
0.6
0.8
1.0


GENDER
female
male
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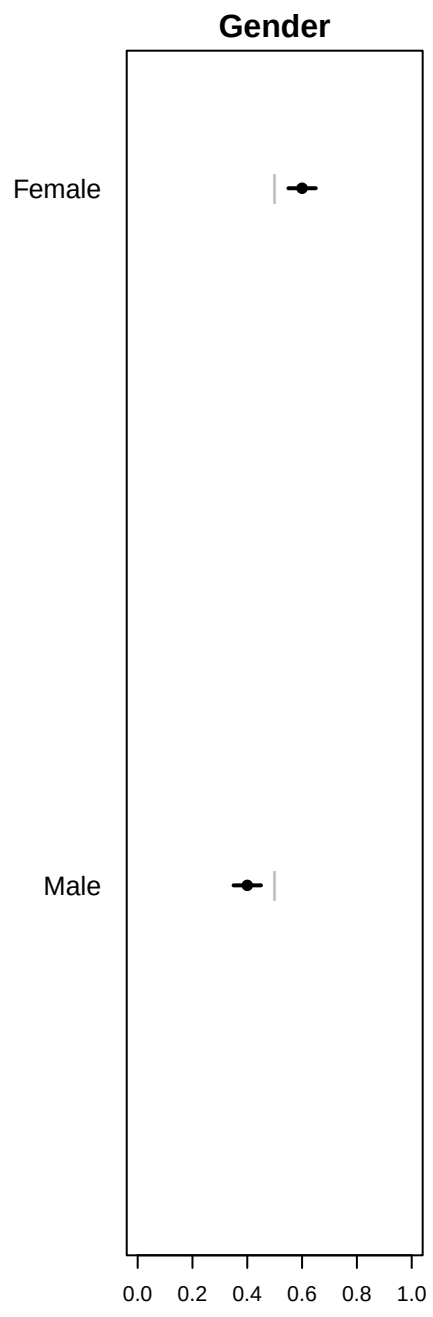
0.2
0.4
0.6
0.8
1.0


Gender
Female
Male
0.0
0.2
0.4
0.6
0.8
1.0